AIR+SEA+LAND+COMBAT

DCS WORLD

EAGLE DYNAMICS



DCS USER MANUAL

このマニュアルは、コンピュータゲーム、コンバットシミュレータ Digital Combat Simulator World (DCS World) 2.5のマニュアルです。このマニュアルでは、ユーザーインターフェース、ミッションエディタ、設定ウィンドウ、他のウィンドウやパネルの機能について説明しています。

ユーザーとのコミュニケーションを図るために、フォーラム ： http://forums.eagle.ru/
ここでは、ゲームやゲームプレイについての質問や追加情報などを得ることができます。

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Website: http://www.digitalcombatsimulator.com

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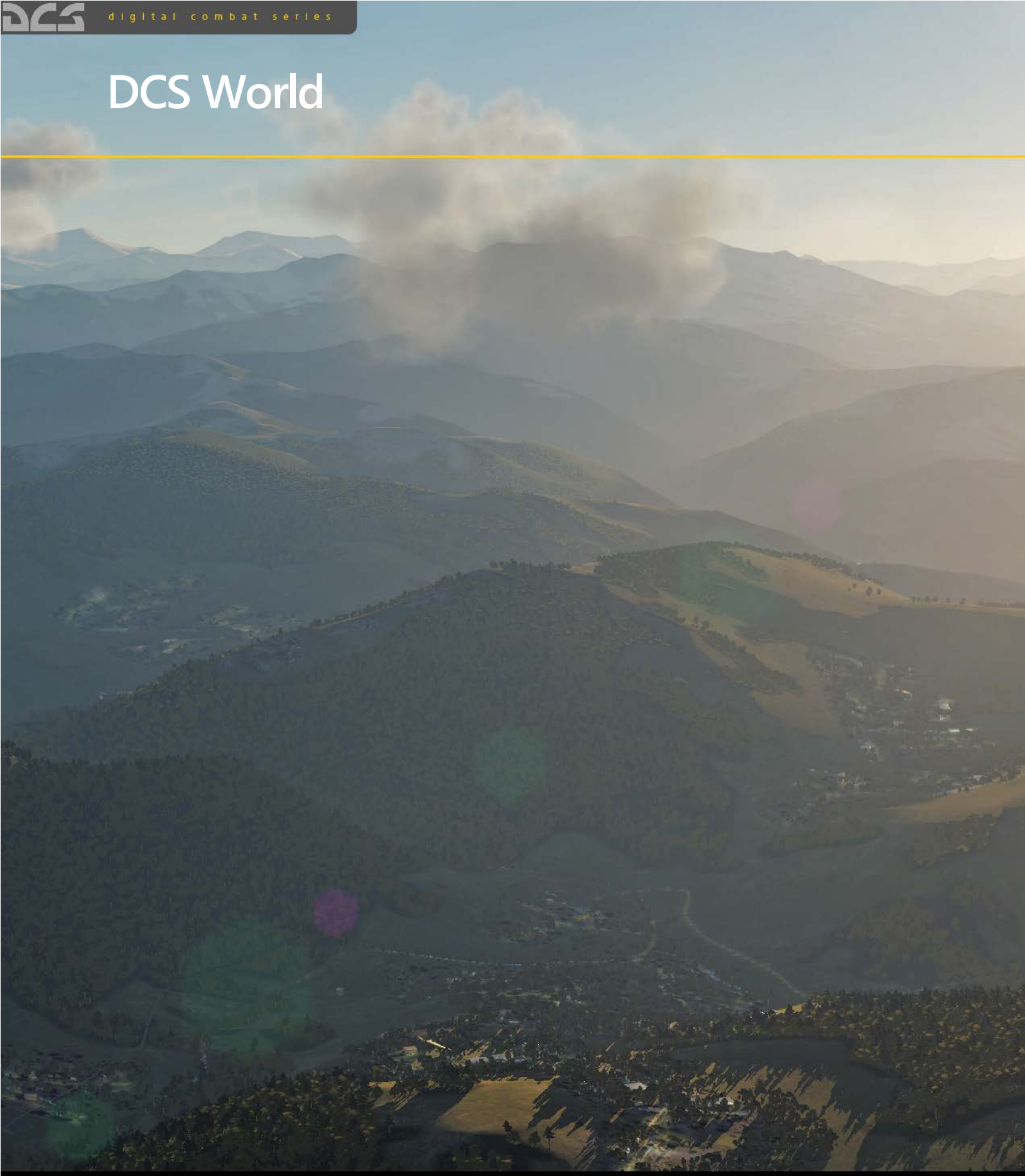
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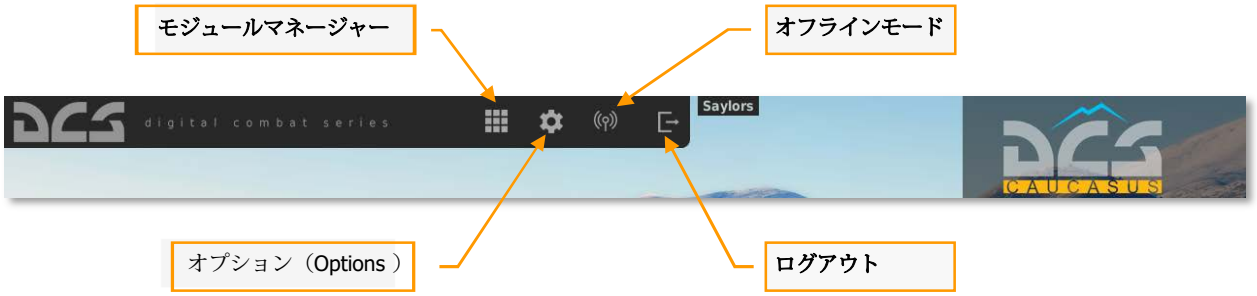
DCS World



MAIN MENU

メインメニュー（MAIN MENU）

DCS を起動すると、プログラムローディング画面のあと、、メインメニュー画面が表示されます。上段パネルにはボタンが配置されています。



- **モジュールマネージャー（MODULE MANAGER）**. モジュールマネージャでは、新しいDCSモジュールの購入や、購入したモジュールの表示、管理を行います。
- **オプション（OPTIONS）**. オプション画面はいくつかのタブで構成されています。グラフィック、オーディオ、入力、ゲームプレイのオプションをカスタマイズすることができます。
- **オフラインモード（OFFLINE-MODE）**. 長時間インターネットに接続できないことが予想される場合は、このモードを使用します。承認（authorization）が成功している場合のみ、 このモードを有効にしてDCS はオフライン専用モードに切り替えることができます。DCS はこのモードを無期限に維持することができますが、すべてのネットワークサービス(モジュールマネージャー、マルチプレイヤー、ニュース)は利用できません。**オフラインモードの解除は、最初に アクティベートしたコンピュータと同じコンピュータからのみ行えます。＊ これができない場合は、テクニカルサポートに連絡して、アクチベーション解除を行って下さい。**

  ＊ *BIOS と Windows のバージョンは、必ず同じでなければなりません。例えば、DCSがオフラインモードの時にBIOSを変更すると、ユーザーはログインできなくなります。*
- **LOG OUT**. ログイン/ログアウト/アカウントの変更 . DCSアプリケーションは、 http://www.digitalcombatsimulator.com で登録したアカウントを使用します。テクニカルサポートでも同じアカウントを使用します。Eagle Dynamics フォーラム： http://forums.eagle.ru/は別のアカウントが必要になります。

メインメニューでは、右のリストからゲームモードを選択することができます。これらはゲームの各モードに移動したり、ゲームを終了したりできます。キャプションの上にマウスを置くと、キャプションが点灯しまので、マウスを左クリックしてそのモードに移動して下さい。



左側には開閉式のニュースパネルがあります。EDからの最新ニュースやイベントの情報が表示されます。ニュースはパネル右上にある2つの「<>」の矢印で記事の替えができます

メインメニューボタンは以下の通りです。:

- **INSTANT ACTION**. ブリーフィング画面を省略して、あらかじめ作成されたミッションを素早く起動させることができます。 このボタンをクリックするとポップアップ画面が表示され、飛行する機体とミッションシナリオを選択することができます。 ミッションを選択せずにこの画面を終了するには、右上の黄色のXをクリックします。
- **CREATE FAST MISSION**. ミッションジェネレーターを使用して、指定したパラメータに応じたミッションを素早く作成することができます。 ミッションはシンプルモードとアドバンスモード（ミッションオプションの数が増えます）のいずれかで作成できます。
- **MISSION**. シングルミッションを開きます。シングルミッションはミッションエディタで自分で作成したり、インターネットからダウンロードしたりすることができます。この画面から、そのようなミッションにアクセスすることができます。
- **CAMPAIGN**. 新たなキャンペーンを選択したり、既存のキャンペーンを継続する場合には、これを選択します。ミッションとは異なり、キャンペーンとは、ミッションの結果に基づいて論理的にリンクされた一連のミッション群です。
- **MULTIPLAYER.** マルチプレイヤーインターフェースを開きます。オンラインゲームをホストしたり、参加したりすることができます。
- **LOGBOOK**. この画面から、パイロットのプロファイルを作成し、その実績や統計情報、飛行時間などを記録することができます。

- **ENCYCLOPEDIA**. エンサイクロペディアは、ゲーム内のすべての空、陸、海のユニットのデータと画像を提供する辞典です。
- **TRAINING**. 豊富な訓練ミッションが用意されており、飛行機を実際に操縦して操縦方法を学習します。
- **REPLAY**. 各ミッションをフライトした後、そのミッションをリプレイファイル（トラックファイル）として保存するオプションがあります。保存したリプレイファイルはここからアクセスできます。
- **MISSION EDITOR**. ミッションエディタでは、大小さまざまなミッションを作成することができます。これらのミッションは、シングルプレイヤー用ミッション、マルチプレイヤー用ミッション、トレーニング用ミッション、またはキャンペーンを作成するために使用することができます。 これは強力なツールで、ゲームに含まれているミッションやキャンペーンの作成に使用されているものと同じものです。
- **CAMPAIGN BUILDER**. キャンペーンビルダーでは、エディターで作成したミッションを使って、キャンペーンを作成することができます。
- **EXIT**. 赤いEXITボタンを押すと、ゲームを終了してデスクトップに戻ります。

画面の下端に沿って機種アイコンが表示されています。これらのアイコンは、DCSモジュールを表しています。インストールされていて利用可能なものはカラーで表示されます。アイコンをクリックすると、各モジュールの情報を読むことができ、背景のテーマに選択することができます。



モジュールマネージャー（MODULE MANAGER）

DCSモジュールマネージャーは、新しいDCSモジュールの購入、購入したモジュールの更新、インストールしたモジュールのレビュー、DCSのスペシャルオファー（セール）についての情報を読むことができるゲーム内ツールです。
モジュールマネージャーの右下にある更新ボタンをクリックして、いつでもモジュールとモジュールの状態を更新することができます。
モジュールマネージャは、4つのタブに分かれています。

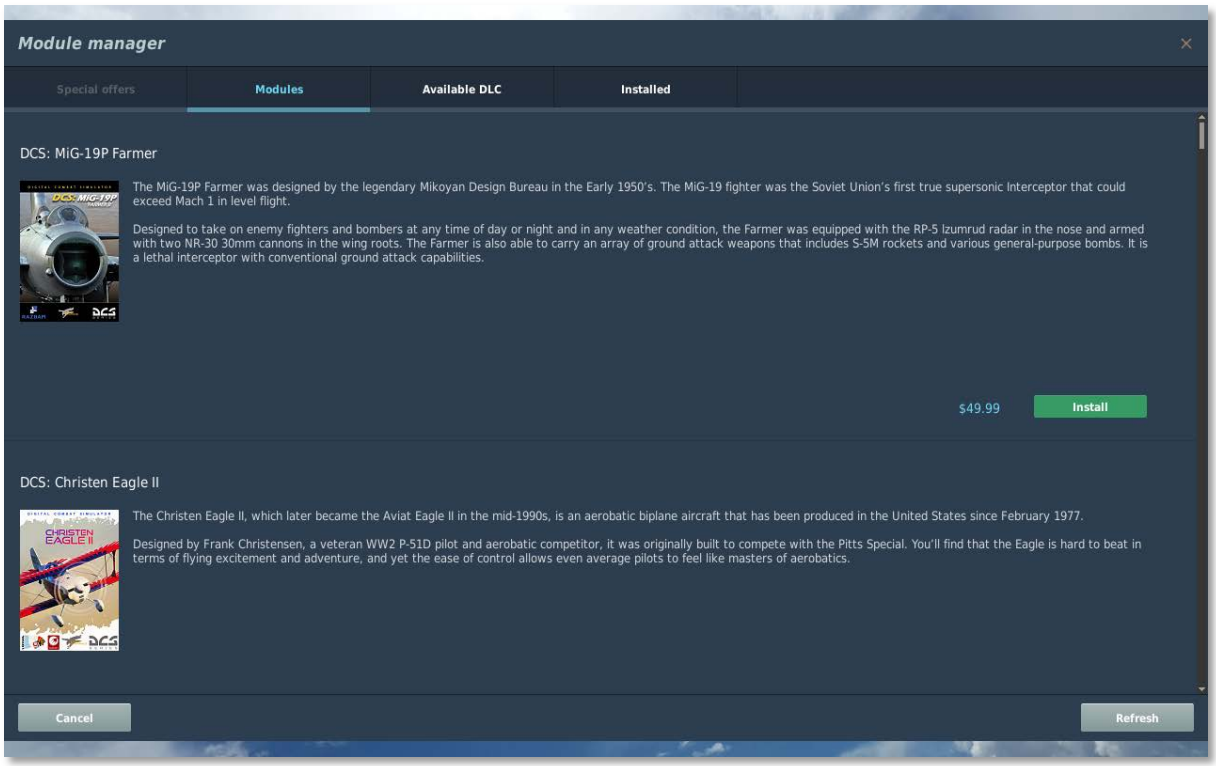
- Special Offers
- Modules
- Available Add-Ons
- Installed

Special Offers（スペシャルオファー）タブ

セールなどの特典がある場合は、スペシャルオファータブを選択すると表示されます。

Modules （モジュール）タブ

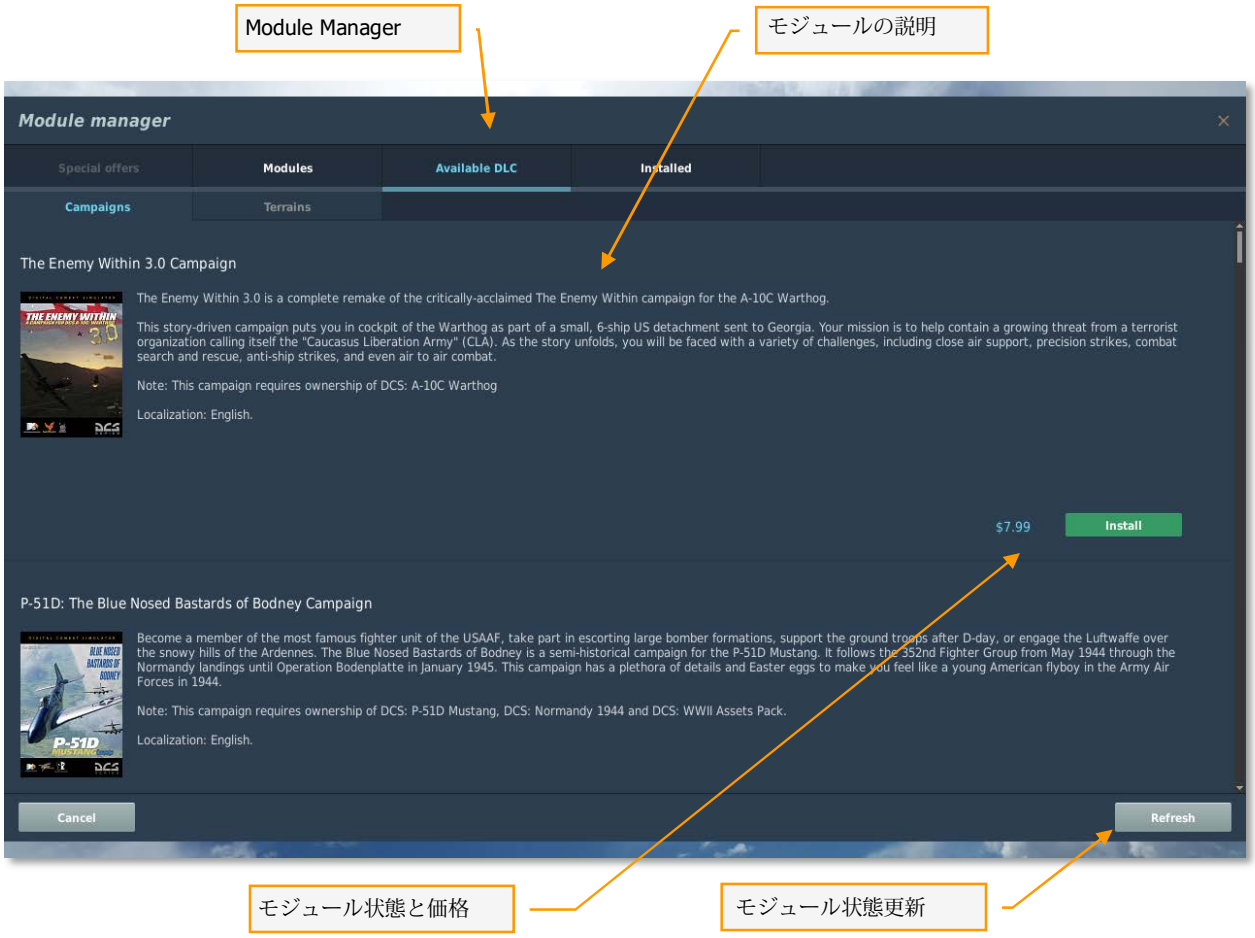
このページでは、利用可能なDCSモジュールがすべて表示され、すでに購入されているかどうかを確認することができます。



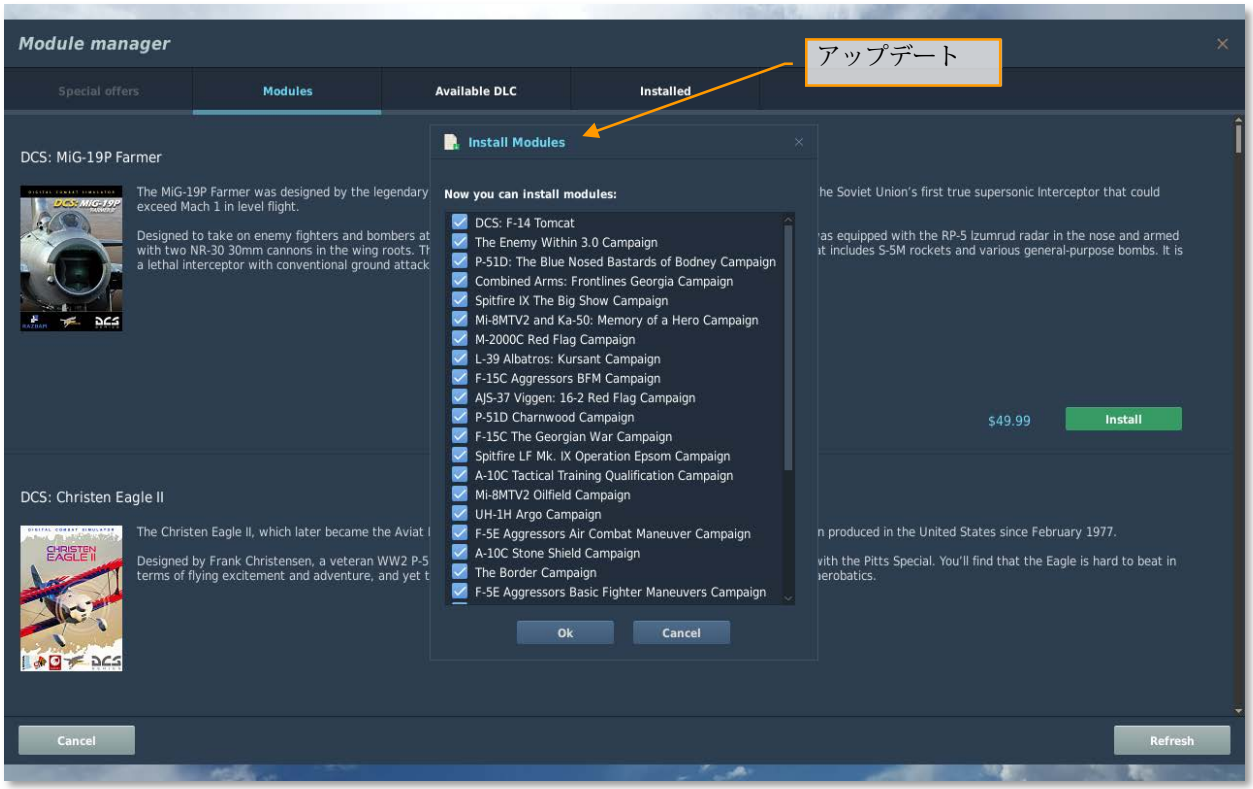
Available DLC（利用可能DLC）タブ

このページでは、利用可能なすべてのDLCと、すでに購入済みかどうかが表示されます。すでに購入済みの場合は、モジュール一覧の右端に「インストール」と表示されます。インストールボタンを押すと、モジュールをインストールまたは再インストールすることができます。「Buy」が表示されている場合、ボタンを押すと選択したモジュールの購入をすることができます。

Install/Buyの左に、そのモジュールの価格とモジュールの説明があります。



モジュールマネージャを起動した時に、購入したモジュールの新しいアップデートが検出されると、利用可能なアップデートを一覧表示するポップアップウィンドウが表示されます。アップデートをインストールするかキャンセルするかを選択します。



インストール済（Installed）タブ

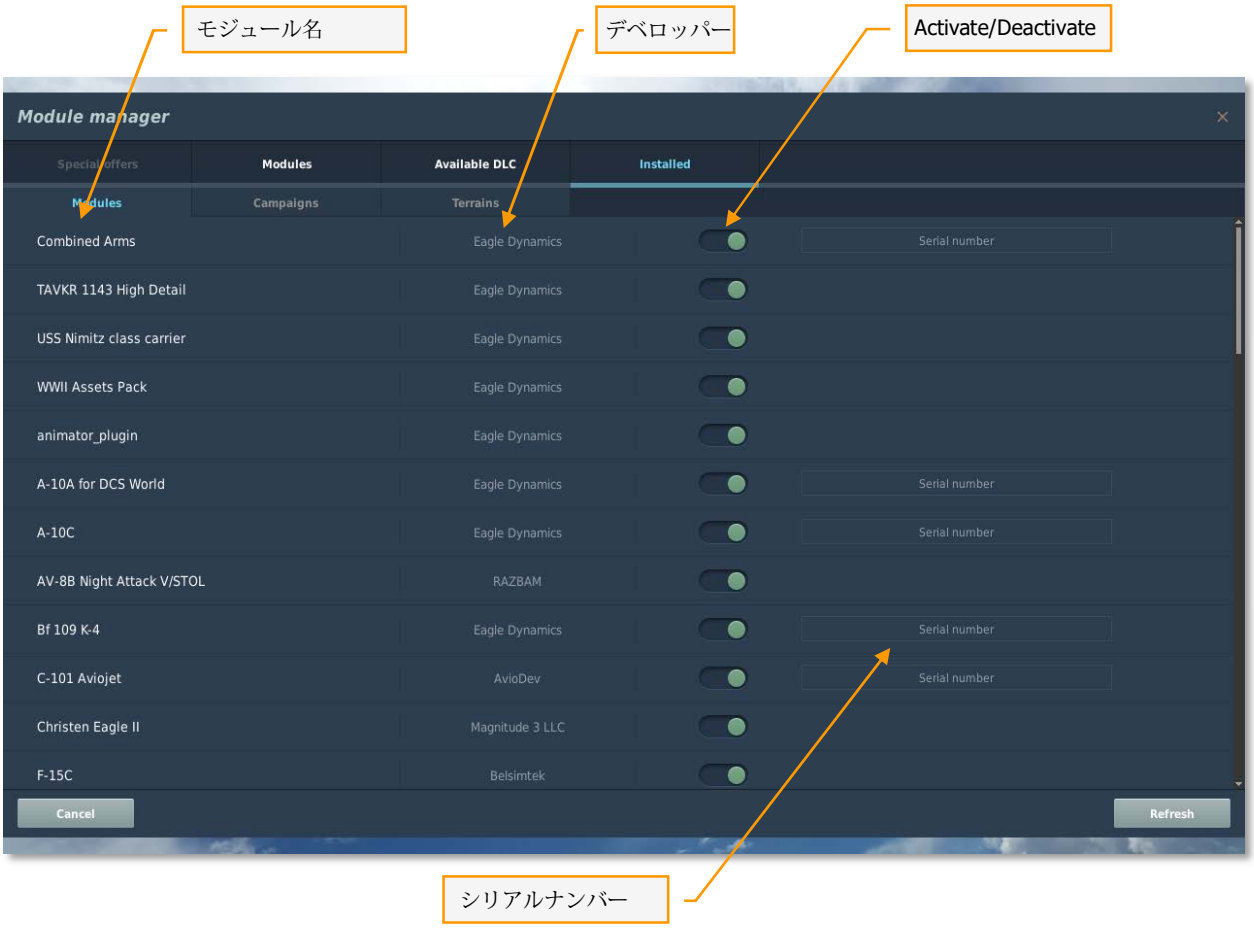
このタブページには、インストールされているすべての DCS モジュールが表示されます。各行にはインストールされているモジュールが表示され、各列にはモジュールに関する情報が表示されます。

Module Name. インストールされているDCSモジュールの名前

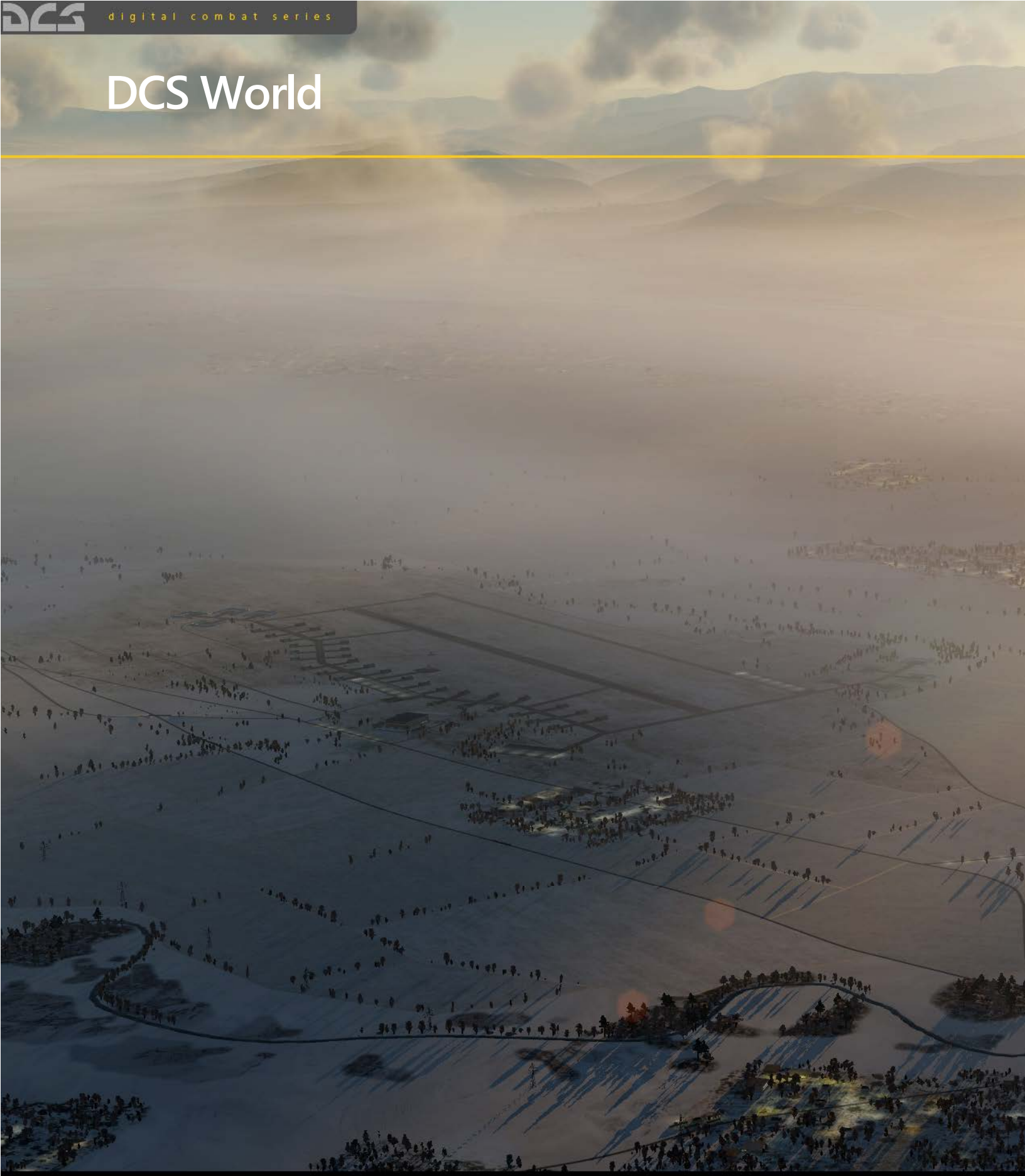
Developer. モジュールの発売元

Activate/Deactivate. このボックスのチェックを外すと、選択したモジュールはアクティベート解除（deactivate）します。チェックボックスをオンにすると、モジュールがアクティブになります。

Serial Number. この列には、選択したモジュールのシリアル番号が表示されます。シリアル番号フィールドをクリックすると、シリアル番号が表示され、シリアル番号をクリップボードにコピーするか、シリアル番号を非表示にするかを選択できます。


モジュール名
デベロッパー
Activate/Deactivate
Module manager
Special offers
Modules
Available DLC
Installed
Modules
Campaigns
Terrains
Combined Arms
Eagle Dynamics
Serial number
TAVKR 1143 High Detail
Eagle Dynamics
USS Nimitz class carrier
Eagle Dynamics
WWII Assets Pack
Eagle Dynamics
animator_plugin
Eagle Dynamics
A-10A for DCS World
Eagle Dynamics
Serial number
A-10C
Eagle Dynamics
Serial number
AV-8B Night Attack V/STOL
RAZBAM
Bf 109 K-4
Eagle Dynamics
Serial number
C-101 Aviojet
AvioDev
Serial number
Christen Eagle II
Magnitude 3 LLC
F-15C
Belsimtek
Cancel
Refresh
シリアルナンバー

DCS World



OPTIONS

オプション（OPTIONS）

オプション画面では、グラフィックスやオーディオの設定、入力オプション、ゲームプレイの設定をカスタマイズすることができます。 オプション画面にアクセスするには、メインメニューの上にあるオプションボタン（歯車のアイコン）をマウスで左クリックします。



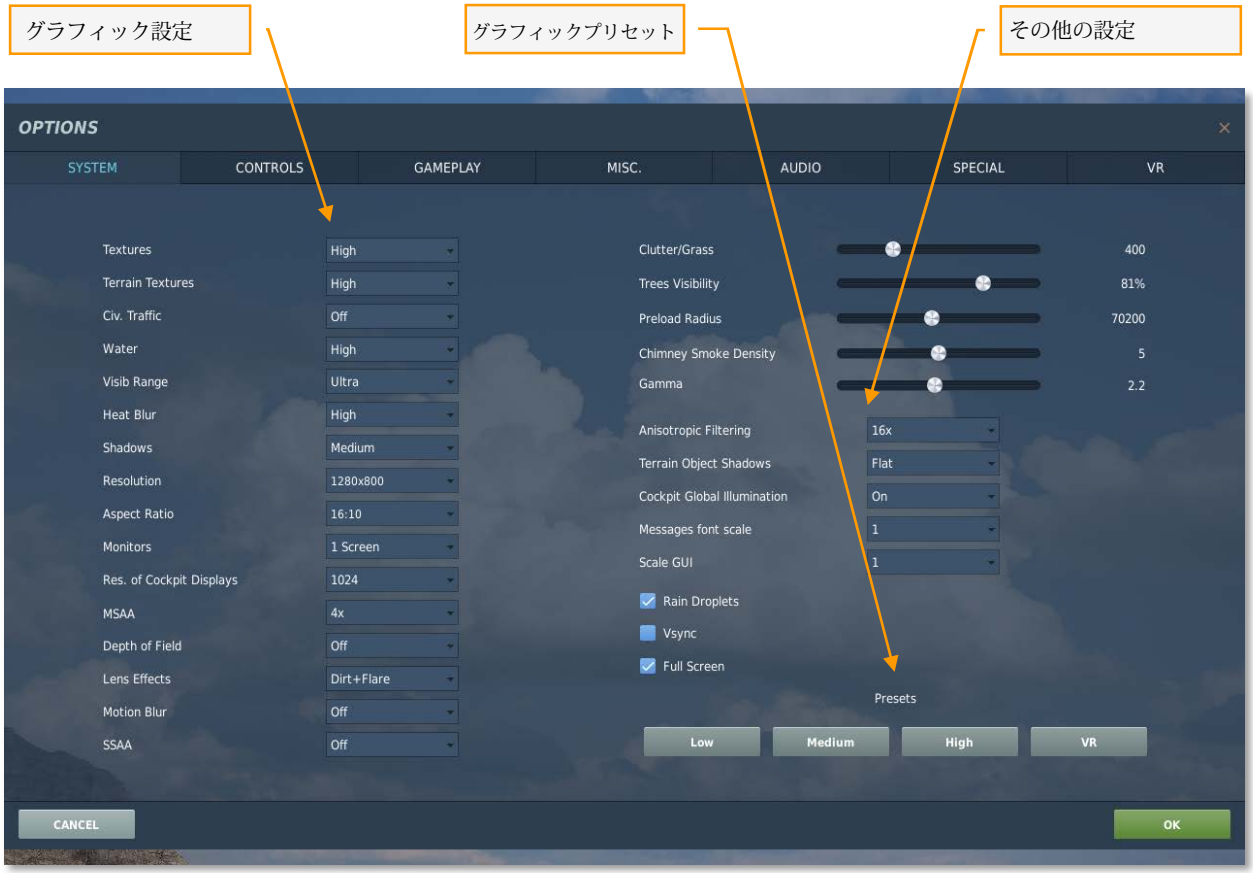
オプションページには、上部にいくつかのタブがあり、グラフィックス、入力、ゲームプレイ、その他、オーディオ、スペシャル、VRの設定を調整することができます。 タブをマウスで左クリックすると、目的のオプション設定ページが表示されます。

画面の一番下に、オプションへの変更を保存せずにメインメニューに戻るCANCELボタンと、変更を保存してメインメニューページに戻るOKボタンがあります。

また、画面右上の黄色のXボタンを押すことで、オプションページを終了することができます。

システムオプション（System Options）

システムオプション画面では、グラフィックスの設定を好みやハードウェア（CPU、RAM、グラフィックスカード）に合わせてカスタマイズすることができます。



画面下部には、オプションの変更を保存せずにメインメニューに戻るCANCELボタンと、変更内容は保存してメインメニューに戻るOKボタンがあります。

また、画面右上の黄色のXボタンを押すことで、オプションページを終了することができます。

グラフィック設定（Graphics Settings）

画面の左側には、いくつかのグラフィック設定が用意されています。 選択は、各項目のドロップダウンリストを使用して、希望の設定を選択することによって行います。

Textures（テクスチャ）

- **Low** – 低解像度テクスチャ。地形、オブジェクト、コックピット
- **Medium** – 中解像度テクスチャ。地形、オブジェクト、コックピット
- **High** – 高解像度テクスチャ。地形、オブジェクト、コックピット

Terrain Textures （地形テクスチャ）

このオプションは地形と建物のテクスチャの詳細レベルに影響を与えます。DCSがすでに地形をロードした（ミッションまたはミッションエディタでロードしている）後にこのオプションを変更した場合、変更を有効にするためには再起動が必要です。ゲームインターフェースをロードした直後にオプションメニューに入った場合は、再起動の必要はありません。

- **Low** – 選択したマップの地形表面のテクスチャは低解像度がロードされます。
- **High** – 選択したマップの地形表面のテクスチャは高解像度がロードされます。

Civ. Traffic（民間車両）

- **Off** – 民間交通(車両・鉄道)の通行無し
- **Low, Medium, High** – 民間交通(車両・鉄道)の頻度

Water（水面）

- **Low** – 立体的な表面（3D surface.）を持たない基本的な水面
- **Medium** – 立体的な水面と大陸棚（shelf）を表現
- **High** – 立体的な水面と大陸棚（shelf）を水面反射を表現

Visib Range （レンダリング距離）

プレイヤーの航空機/ユニットから地上オブジェクト（都市、建物、森林）を描画（レンダリング）する際の距離を変更します。PCの能力によっては、最大値に設定するとフリーズや、パフォーマンスの低下を起こします。

- **Low** – プレイヤーユニットから半径7km以内の地上物をレンダリングします。
- **Medium** – プレイヤーユニットから半径12km以内の地上物をレンダリングします。
- **High** – プレイヤーユニットから半径17km以内の地上物をレンダリングします。
- **Ultra** – プレイヤーユニットから半径24km以内の地上物をレンダリングします。
- **Extreme** – マップ上に存在する地上オブジェクトの最大レンダリング半径。

Heat Blur（ヒートブラー：エンジンの排気によって周辺や背景がぼやけて見える効果）

- **Off** – ヒートブラー無し

- **Low** – エンジンノズルの後方に部分的に描画
- **High** – エンジンノズルの後方に完全に描画

Shadows（シャドウ：影）

- **Off** – すべてのオブジェクトの影を無し
- **Flat Only** – 航空機/ユニットから地面へのフラットな影のみを表示、コクピット内や、航空機への影は表示されません。
- **Low** – 航空機/ユニットからの地上への低品質なプロジェクションシャドウ（シャドウマップ）を表示、コクピット内、航空機への影を表示
- **Medium** – 航空機/ユニットからの地上への中品質なプロジェクションシャドウ（シャドウマップ）を表示、コクピット内、航空機への影を表示
- **High** – 航空機/ユニットからの地上への高品質なプロジェクションシャドウ（シャドウマップ）を表示、コクピット内、航空機への影を表示

Resolution（解像度）

ゲームの解像度を選択します。アスペクト比は、選択した解像度に合わせて最適になるよう自動的に設定されます。マルチモニターを使用している場合は、合計解像度を手動で入力することができます。

Aspect Ratio（アスペクト比）

ディスプレイのアスペクト比は、現在の解像度に自動的に設定されますが、手動で入力することもできます。デフォルトは4:3(1280x768)です。

Monitors（モニター）

このゲームでは、複数のモニターに映像を出力することができます。10種類のデフォルト設定が事前に用意されています。(自分で作成することも可能です。）：

- **1 Screen** – すべての映像は、1 台のモニターに表示されます（デフォルト）。
- **3 Screen** – ビデオは3つのモニターに渡って分割表示されます。
- **Camera + LMFCD (on right side)** – 映像は２つのモニターに分割されます。左がメインカメラ用で右側が左MFCD用です。
- **Camera + LMFCD** – 映像は２つのモニターに分割されます。右がメインカメラ用で左側が左MFCD用です。
- **Camera + LMFCD (automatic placement)** – 映像は２つのモニターに分割されますが、左MFCDの位置は実際の航空機の位置に準じます。
- **Camera + RMFCD (on left side)** – 映像は２つのモニターに分割されます。右がメインカメラ用で左側が右MFCD用です。
- **Camera + RMFCD** – 映像は２つのモニターに分割されます。左がメインカメラ用で右側が右MFCD用です。

- **Camera + RMFCD (automatic placement)** – 映像は２つのモニターに分割されますが、右MFCDの位置は実際の航空機の位置に準じます。
- **LMFCD + Camera + RMFCD** – 映像は、左MFCD用、メインカメラ用、右MFCD用の3つのモニターに分割されます。
- **Stereo** – ステレオイメージ映像

補足： これらの設定をカスタマイズしたり、独自の設定を作成するには、「\Saved Games\DCSConfig\MonitorSetup」を参照してください。それぞれの設定は、個別のluaファイルで定義されています。

Res. of Cockpit Displays （コクピット画像の解像度）

ミラー、ターゲティングビデオインジケーター、マルチファンクションディスプレイ、ムービングマップなどのコックピット画像の解像度を設定します。 解像度を高く設定すると、ゲームプレイのスムーズさに悪影響を及ぼす可能性があるので注意が必要です。

- **256** – 低画質です。256x256ピクセルのテクスチャにレンダリングされています。フレームを通して更新されます。（updated through the frame.）
- **512** – 標準画質。512x512ピクセルのテクスチャにレンダリングされています。フレームを通して更新されます。
- **512 Every Frame** – 標準画質。512x512ピクセルのテクスチャにレンダリングされます。フレーム毎（Each frame）に更新されますが、フレームを通して更新されるプリセットの選択に比べてゲーム全体のパフォーマンスが低下します。
- **1024** – 高画質。1024x1024ピクセルのテクスチャにレンダリングされます。フレームを通して更新されます。
- **1024 Every Frame** – 高画質。1024x1024ピクセルのテクスチャにレンダリングされます。フレーム毎に更新されますが、フレームを通して更新されるプリセットの選択に比べてゲーム全体のパフォーマンスが低下します。

MSAA

MSAA (MultiSample Anti-Aliasing) - SSAA のスーパーサンプリングの改良版で、リソース消費がより少なくなっています。スムージングは、SSAA のようにシーン全体ではなく、オブジェクトのエッジに沿ってのみ行われます。ゲームの解像度が上がると、グラフィックスカードのビデオメモリがより必要になります。

- **Off** – MSAA アンチエイリアスをオフにします。
- **2x, 4x** – 選択した値でのMSAAスムージングを行います。

Depth of Field （被写界深度）

被写界深度とは、ピントを合わせた部分の前後のピントが合っているように見える範囲（距離）のことです。

- **Off** – 被写界深度効果なし
- **Bokeh** – 背景のぼかしを強くする効果をオンにします。パフォーマンスを著しく低下させ、主に動画/スクリーンショット作成にのみ使用します。

- **Simple** – 標準的な被写界深度効果を得られます。

Lens Effects（レンズ効果）

- **None** – キャビンウインドウへの全効果を無効にします。
- **Dirt** – キャビンウインドウの 汚れを表現します。
- **Dirt+Flare** – 現実のようにコックピット内の汚れやフレアを再現します。ただし、コンタクトを発見するのは困難になります。

Motion Blur （モーションブラー）

カメラの動きが速いときに画像をぼかす効果があります。シーンによりダイナミックさとスピード感を与える必要がある場合に有効です。モーションブラーを適切に使用すると、画面上で起こっていることに映画的な演出を加えることができます。また、必要に応じて低フレームレートをごまかしたり、ゲーム体験に滑らかさを加えたりするのに役立ちます。

- **Off** – モーションブラーをオフ
- **On** – モーションブラー効果をオン。 画面高速移動時に、ピントを合わせず画像全体をぼかしてスピード感を出します。

SSAA

SSAA（SuperSample Anti-Aliasing）：最も重い（高負荷）なだけでなく、 最も美しく、ビデオカードへの負荷も高くなります。シーン全体を滑らかにします。パワフルな最上位のグラフィックカード用。

- **Off** – SSAA anti-aliasingをオフにします。
- **x1.5, x2** – 選択した値で重いSSAAアンチエイリアスを有効にします。非常に強力なシステムのみ推奨。

同時に、マルチサンプリング（MSAA）とスーパーサンプリング（SSAA）は、競合と性能低下を避けるために同時に使用すべきではありません。（MSAA 、 またはSSAAのどちらかを使用します。）

注意： *フレームレートを上げたい場合、最も影響のあるグラフィック項目で低く設定スベき項目は、Terrain Textures、 Water、 Visib Range、 Shadows 、MSAAです。*

ページの右側には、その他のグラフィック設定があります。

Auxiliary Settings （その他の設定）

Clutter/Grass（雑草）

このスライダーは、草木が地面に描画される距離を調整します。

Trees Visibility （木々）

このスライダは、プレイヤーの機体/ユニットから選択したパーセンテージに応じて木の密度を調整します。値が大きいほど、森、野原、川沿い、道路、都市に描画される木が多くなります。値が小さくなると、木は森林帯にのみ、より少ない数で描かれます。

Preload Radius （プレロード半径）

ユーザーの位置を中心として事前読み込みする地形の半径をメートル単位で指定します。大きく設定すると、ハードディスクのスワップを減らすことができますが、より多くのシステムRAMを必要とします。

Chimney Smoke Density （煙突の煙の密度）

スライダーはマップ上に描画する煙突の数の密度を調整します。

Gamma （ガンマ）

ゲーム画像のガンマ値を調節します。

Anisotropic Filtering （異方性フィルタリング ）

最新のグラフィックカードではfpsにほとんど影響しません。フィルタリングファクター（2x, 4x, 8x, 16x）の設定しかありません。フィルタリング係数が高いほど、テクスチャはよりクリアで自然に見えます。通常、高い値では、小さなアーチファクト（artifacts ）が、傾いたテクスチャの最も離れたピクセルに表示されるぐらいです。原則として、視覚的な歪みの大部分を取り除くのには4x か 8x で十分です。

- **Off** – 異方性フィルタリングをオフ
- **2x, 4x, 8x**, **16x** – 選択された係数による異方性フィルタリングを行います。特定の角度で、消えるオブジェクトや表面のテクスチャの表示を改善します。

Terrain Object Shadows（地形オブジェクトの影）

- **Default** – 地上の樹木、建物、電柱、塔などのからのフラットな影＋シェーディング。
- **Flat** – 地上の樹木、建物、電柱、塔などのからのフラットな影
- **Off** – 樹木、建物、電柱、塔などのからの影をオフ

Cockpit Global Illumination （コクピット全体の照明）

光源からの直射光だけでなく、反射光（コックピット内での光の反射・屈折、dancing shadows）も考慮した照明方法で、よりリアルな映像を実現しています。

- **Off** – 直射光のみ
- **On** – よりリアルな光のシミュレーションに使用されるグローバルライティングアルゴリズムを有効にします。より多くのPCリソースを必要とし、フレームレートに大きな影響を与える可能性があります。

Messages font scale （メッセージフォントスケール）

高解像度の2K-4Kモニター用で、フライト中のプレイヤーのメッセージのフォントサイズを、選択した倍数に応じて拡大縮小します。

Scale GUI （スケールGUI）

高解像度の2K-4Kモニター用で、インターフェイス表示を、選択した倍数応じて拡大します。

Rain Droplets （水滴）

キャビンランタンの窓に落ちる雨滴を描画します。雨滴はヘリコプターの窓ガラスのワイパーには影響を受けません。

Vsync

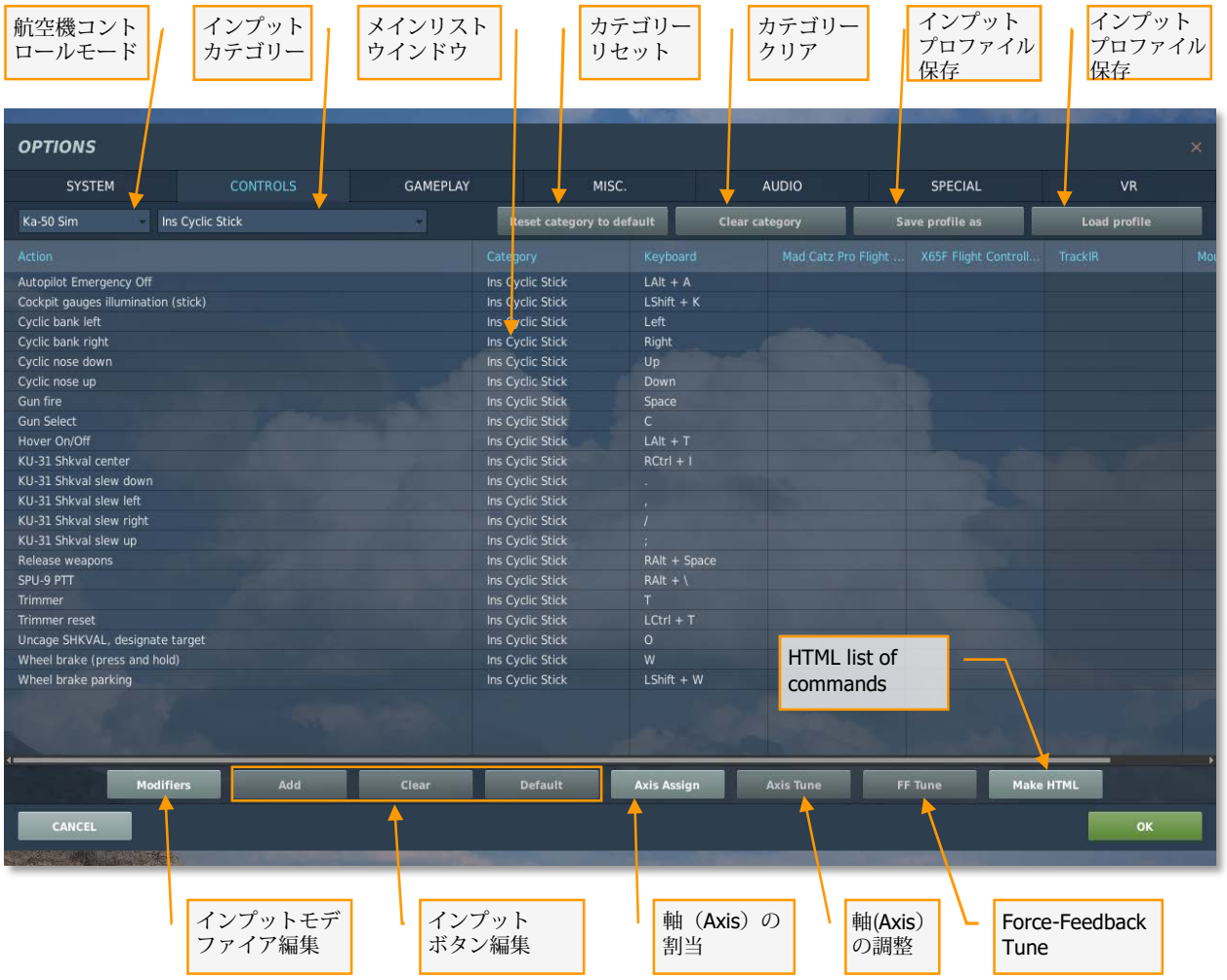
このボックスをオンにすると、モニターのリフレッシュレートとフレームレートの垂直同期が有効になります。

Full Screen （フルスクリーン）

このボックスにチェックを入れるとフルスクリーン表示になります。フルスクリーンがチェックされていない場合、ゲームはウィンドウモードで実行されます。デスクトップの解像度がゲームの解像度設定よりも高く、このボックスが選択されていない場合、ゲームウィンドウはモニターのディスプレイよりも小さくなります。

Controls Settings （コントロール設定）

コントロール設定タブでは、ジョイスティック、マウス、キーボード、ラダーなどの入力デバイスを各機能に割り当てることができます。キーに割り当てたり、軸（AXIS）への割当やカーブの調整が行えます。フォースフィードバックの調整も行えます。 また、複数の入力プロファイルを作成して保存することもできます。



画面下部には、オプションの変更を保存せずにメインメニューに戻るCANCELボタンと、変更内容は保存してメインメニューに戻るOKボタンがあります。

また、画面右上の黄色のXボタンを押すことで、オプションページを終了することができます。

Aircraft Control Mode（航空機コントロールモード）．ドロップダウンリストには、ミッションで利用可能なコントロールモードが表示されています。：

- Aircraft Game（ゲーム）．ゲームアビオニクスモードでミッションを開始するときの設定です。
- Aircraft Sim（シム）．シミュレーションモード（ゲームアビオニクスモードがオフ）の設定です。

- General（一般）. プレイヤーやクライアント制御の機体に関係のない設定です。

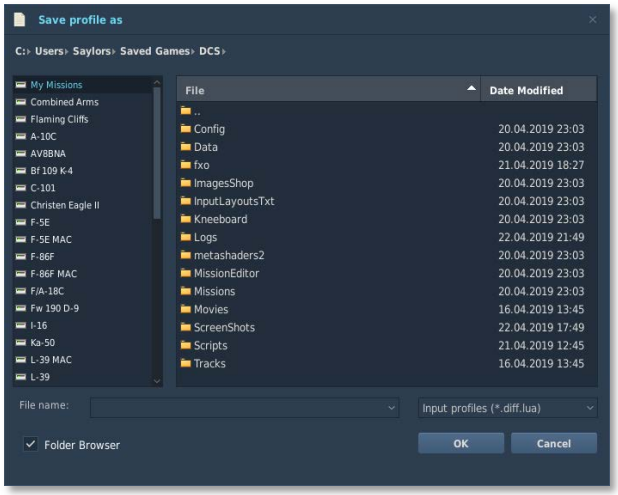
Input Category(インプットカテゴリ). インプットカテゴリのドロップダウンリストには、プレイヤー機の一般的な入力カテゴリがすべて表示されます。　メインリストからカテゴリを選択するには、そのカテゴリを左クリックします。そうすると、そのカテゴリの入力アクションがメインリストウィンドウに表示されます。

Main Listing Window（メインリストウインドウ）. このリストには、コンピュータに接続が検出された各入力デバイスに対して、割り当て可能なアクションが表示されます。 リストの左側列には、アサイン可能なアクションがリストアップされています。アクションのリストは、入力カテゴリーとプレイヤーモード（ゲーム、シム）によって決定されます。 アクションの列の右側には、検出された各入力デバイスの列があります。 アクションを入力デバイスにアサインするには、アクションの行とデバイスの列が交差するフィールドでマウスを左クリックします。 すると、白い割当ボックスが表示されます。注：コントローラの列がハイライト（選択）されているときにコマンド（キーやボタン）を押すと、その入力コマンドが特定のコントローラに割り当てられているかどうかを確認できます。

Reset category to default（カテゴリリセット）. このボタンを押すと、選択デバイスの選択されたカテゴリ内のすべてのキーボードまたはボタンの組み合わせがデフォル値にリセットされます。ハイライトされたセルが選択デバイスです。

Clear category(カテゴリクリアー). 入力デバイスのすべての入力割当をクリアしたい場合、入力デバイスの入力コマンドを左クリックしてから、CLEAR CATEGORY ボタンをクリックします。これにより、その入力デバイスのすべての入力割り当てが消去されます。

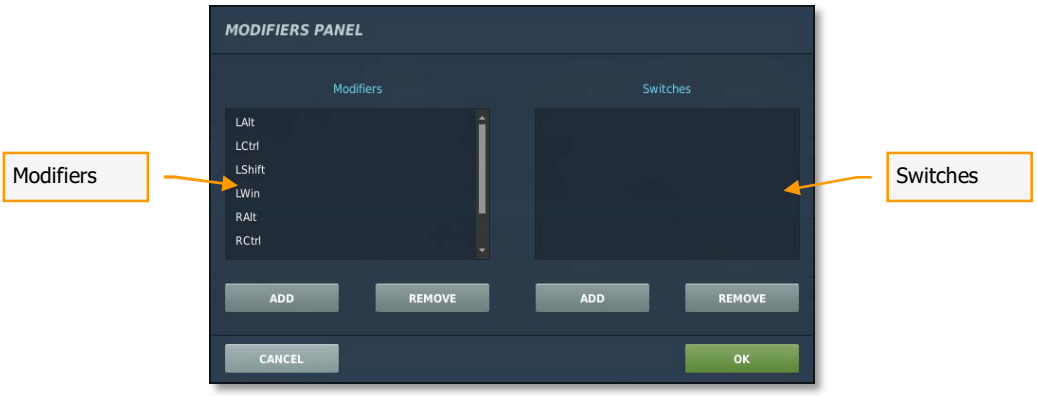
Save profile as（プロファイル保存）. プロファイルの入力オプションを変更（追加、削除、変更）した後、xSAVE PROFILE ASzボタンから保存することができます。 このボタンを押すと、「プロフィールを名前を付けて保存」ブラウザウィンドウが表示されます。 このウィンドウでは、コンピュータ上の任意の場所を参照してプロファイルを保存することができます。 プロファイルをデフォルトの名前で保存することもできますし、別の名前で新しいプロファイルを作成することもできます。 このように、Nつの航空機に対して複数のプロファイルを作成することができます。



Load profile（プロファイル読込）. 入力プロファイルが作成されると、LOAD PROFILE ボタンを選択してアクティブなプロファイルとしてロードすることができます。 このボタンを押すと、Load Profile ブラウザウィンドウが表示され、ロードするプロファイルを選択することができます。 目的のプロファイルを選択したら、OKボタンを押します。

注意：*ロード、セーブ機能はデバイス単位であり、機体プロファイル全体を保存するわけではありませ。*

Modifiers（モデファイア）. MODIFIERS ボタンを押すと MODIFIERS PANEL ウィンドウが表示されます。コントロールモディファイアとコントロールスイッチを設定して追加のコントロールアサインを行うことができます。



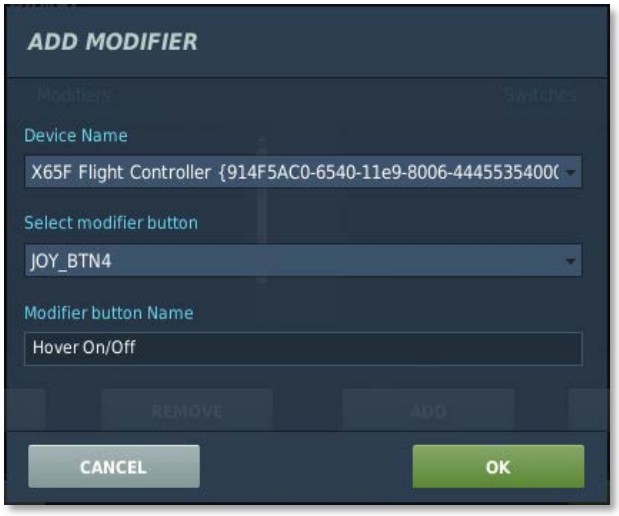
モディファイヤパネルは、左側のモディファイヤと右側のスイッチの 2 つの部分で構成されています。

モディファイヤは通常 [SHIFT], [CTRL] or [ALT] （デフォルト）かモデファイアとして登録したキーまたはボタンを使用します。を使用します。 例えば、ジョイスティックのボタンをモディファイヤとして割当、入力デバイスにアサインできるジョイスティックのコマンドを拡張することができます。

デフォルトで、いくつかのモデファイアがモデファイアウィンドウに表示されています。（LALT, LCTRL, LSHIFT, RALT, RCTRL, RSHIFT , RWIN). しかし、独自のモディファイヤを作成することもできます。モディファイヤリストウィンドウの下にあるADDボタンを押すことで、モディファイアを追加するウィンドウが表示されます。

ADD MODIFIERウインドウには３つのフィールドがあります。

- Device Name. このフィールドには、モデファイアを割り当てるデバイスを選択します。
- Select modifier button. このフィールドには、モデファイアとして押されたキーまたはボタンを表示します。
- Modifier button Name. このフィールドには、独自のモデファイア名、スイッチ名を入力することができます。



ADD MODIFIER パネルから、まずモディファイアを設定したいデバイスを選択します。 これはキーボード、マウス、ジョイスティック、スロットルなどです。 デバイスを定義したら、Select Modifier Button フィールドから上で指定したデバイスの特定のキーまたはボタンを選択します。 モデファイア名を入力して、OK ボタンを押し、新しいモディファイアを保存します。これでモディファイアリストウィンドウに表示されます。

モディファイアを削除するには、ウィンドウから選択して REMOVE ボタンを押します。

モディファイアが 1 つ以上リストにあれば、Assignment パネルの Add Assignment から任意のアクションにモディファイアを割り当てることができます。

Switches （スイッチ）

モディファイヤパネルの右側にあるスイッチウィンドウは、モディファイヤとは対照的にトグルとして機能します。モディファイヤでは、モディファイヤとキーを同時に押さなければなりません。 （例：CTRL+Eのように） スイッチを使用すると、まずスイッチを起動し、必要なキーを押して、それからスイッチをオフにすることができます。 これは、キーでモードを設定していると考えることができます。 例えば、キーボードの "E "ボタンをスイッチに、スティックの "Hat "ボタンをトリムのコントロールに割り当てておくと。飛行中に "E "ボタンを1回押すと、デフォルトのビューパンニング機能ではなく、ハットを使って機体のトリム調整を行うことができます。



ADD SWITCHES パネルから、まずスイッチを設定したいデバイスを選択します。 これには、キーボード、マウス、ジョイスティック、スロットルなどがあります。 デバイスを定義したら、Select Switch Button フィールドから上で選択したデバイスの特定のキーまたはボタンを選択します。 完了したら、名前をつけてOK ボタンを押して新しいスイッチを保存します。これでスイッチリストウィンドウに表示されます。
スイッチを削除するには、ウィンドウから選択して REMOVE ボタンを押します。
リストに少なくとも 1 つのスイッチがあれば、Add Assignment パネルから任意のアクションにスイッチを割り当てることができます。

Edit Input Buttons. メインリストウィンドウでアクションを作成、編集、削除する場合は、ADD、CLEAR、DEFAULT ボタンを使用します。

アクションにキーやボタンをadd(追加）、clear(クリア)、default(初期値）するには、まず、目的のアクションをリストアップしているカテゴリを選択する必要があります。 目的のアクションが表示されたら、アクション行とマッピングしたい入力デバイス列の交点のフィールドをマウスで左クリックします。 例えば、ランディングギア上げ下げのキーボードのキー割当を変更したい場合があります。 この例では、ランディングギアのトグルアクションの行を探し、キーボードの列と交差するフィールドでマウスを左クリックします。 選択すると、フィールドに青いボックスが表示されます。

アクションと入力デバイスが選択したら、3つのボタンのうちの1つを押します。

ADD. ADD ボタンを押すと、ADD ASSIGNMENT PANEL が表示されます。 このパネルには以下のフィールドがあります。：

- **Action**. メインリストウィンドウに表示されているアクションの名前を表示します。
- **Key / Button**. 目的のキーを押すか、または目的の入力軸を移動すると、キー、ボタン、または軸の名前がここに表示されます。 さらに、ドロップダウン矢印をクリックして、アクションに手動で割り当てることができるすべての入力を表示することができます。
- **Add Modifier**. ドロップダウンリストを使用して、アクションに割り当てることができるすべてのモディファイアとスイッチを表示します。 このリストは、モディファイアパネルから編集することができます。

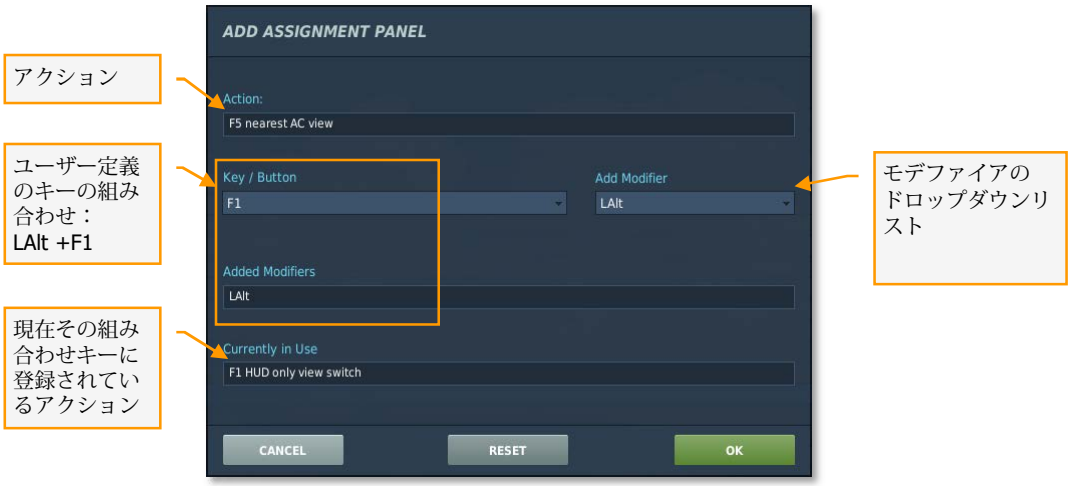
- **Added Modifiers**. モディファイアやスイッチを追加すると、自動的にこのフィールドに表示されます。
- **Currently in Use**. 割り当てられたキーまたはボタンが既に他のアクションで使用されている場合、競合はここに表示されます。
- **RESET**. RESETボタンを押すと、アクションはデフォルト設定に戻ります。

CLEAR. フィールドを選択してからCLEAR ボタンを押して割り当てを削除できます。

DEFAULT.　フィールドを選択して DEFAULT ボタンを押すと、アクションに元々割り当てられていたアサインが復元されます。

以下に、デフォルトの代わりに新しいキーボード/ボタンの組み合わせまたは軸を割り当てる方法例を示します。

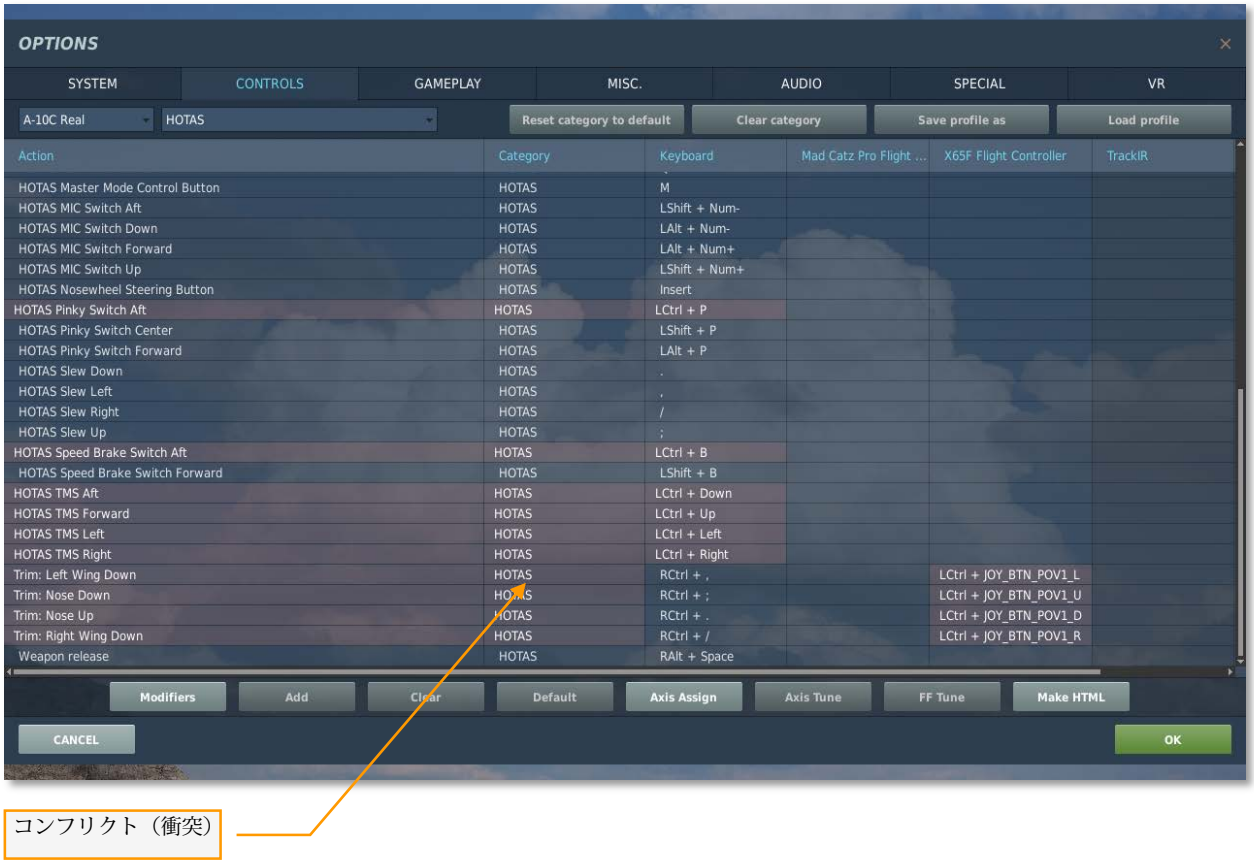
1. 必要なデバイスの列で、必要なアクションのセルを選択します。軸を割り当てる必要がある場合は、まず、アクションカテゴリのドロップダウンリストで AXIS COMMANDS カテゴリを選択します。
2. RESET をクリックして、セルに既に存在する組み合わせをクリアします。
3. ADD を押して[ADD ASSIGNMENT PANEL] ウィンドウを表示して、新しいキーの組み合わせを割り当てます。



- ACTIONフィールドには、組み合わせを割り当てるアクションが表示されます。
- KEY、BUTTON、AXIS フィールドには、割り当てられた（押された）キー、ボタン、または軸が表示されます。このフィールドでは、ドロップダウンリストからキー、ボタン、または軸を手動で選択もできます。
- ADD MODIFIER ドロップダウンリストでは、組み合わせに必要なモディファイアを手動で選択できます。
- ADDED MODIFIERS フィールドには、割り当てられたモディファイアの組み合わせが表示されます。

- CURRENTLY IN USEフィールドには、選択したキーやボタンの組み合わせがすでに他のアクションで使用されている場合、そのアクションの名前が表示されます（競合）

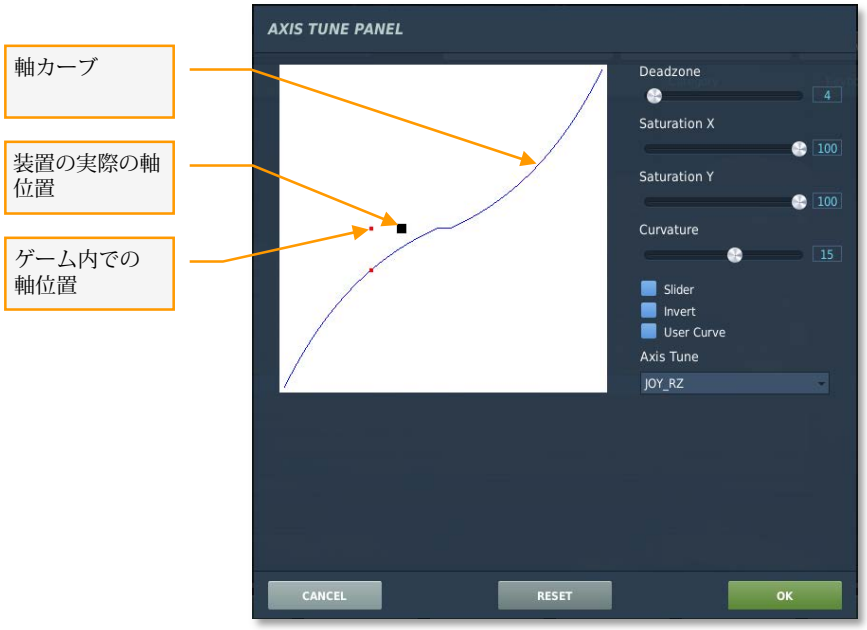
4. キーボードまたはジョイスティックで希望の組み合わせを押し、パネル上で入力が正しく設定されていることを確認します。
5. OK をクリックして、入力した組み合わせを確定します。
6. 割り当てに誤りがあった場合は、RESETボタンを押して割り当て処理をやり直して下さい。



誤ったモデファイアがある場合、赤いコンフリクト（競合）テキストが表示されます。例えば、現在使用中のモデファイアの一つを削除した場合、そのモデファイアを使うすべてのテキストが赤で表示されます

Axis Assign. Axis Commands カテゴリメニューを表示します。

Axis Tune. 入力軸をアクションに割り当てた場合、AXIS TUNE ボタンを押して、軸がどのように応答するかを調整することができます。ボタンを押すとAxis Tune Panel ウインドウが開きます。各軸ごとに調整が行なえます。このパネルには：



Deadzone. カーブの中心に「デッドゾーン」を作成することができます。お使いのジョイスティックの精度にもよりますが、デッドゾーンの値を5～10に設定すると良いでしょう。これは、キャリブレーションの問題で機体をトリミングする際の問題を回避するのに役立ちます。スライダーを右に動かすとデッドゾーンが増え、左に動かすと減ります。 実際の数値はスライダーの右側に表示されます。

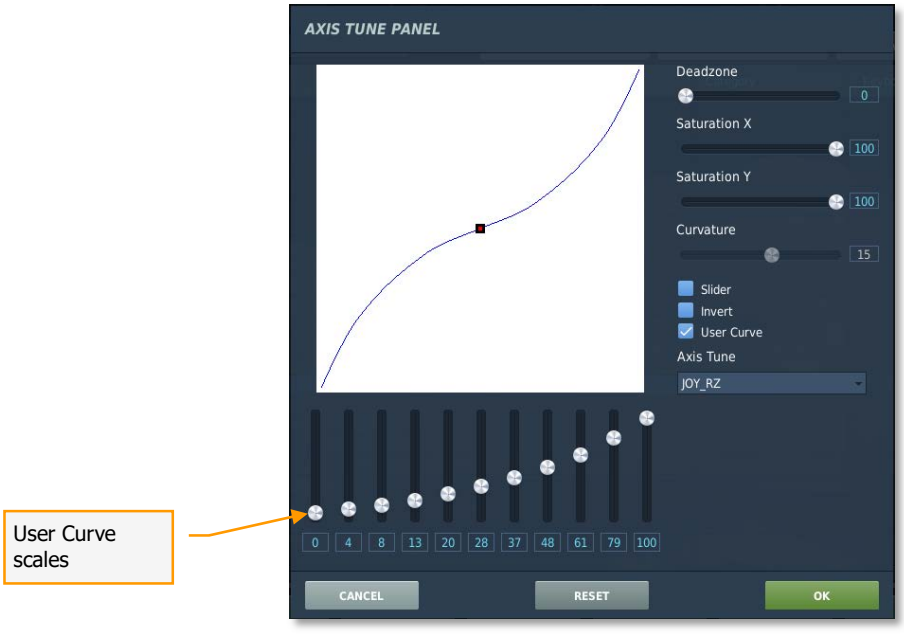
Saturation. これらの X と Y のスライダは、デバイスの入力値をゲーム内の入力値に変換する比率を調節できます。例えば、ジョイスティックの軸を全移動した時に、ゲーム内でも全移動にしたい時はXとYのリミットをウィンドウのコーナーに設定します。(100 のサチュレーションレベル)。ジョイスティックの入力を50%に減らしたい場合は、Yのサチュレーションレベルを50に設定します。

Curvature. このスライダを左右に動かすと、軸カーブの非直線性が増加したり減少したりします。 このカーブを使用して、軸の感度を調整することができます。 デッドゾーンの近くで浅いカーブを持つのが一般的で、微調整入力の精度を高めることができます。 実際の数値はスライダーの右側に表示されます。

Slider.　デバイスから何も入力がないときに中心に戻るデフォルトのカーブではなく、スライダー軸を設定することができます。例えば、スロットル軸にはスライダー機能を使用し、コントロールスティックには標準カーブ軸を使用したい場合などです。

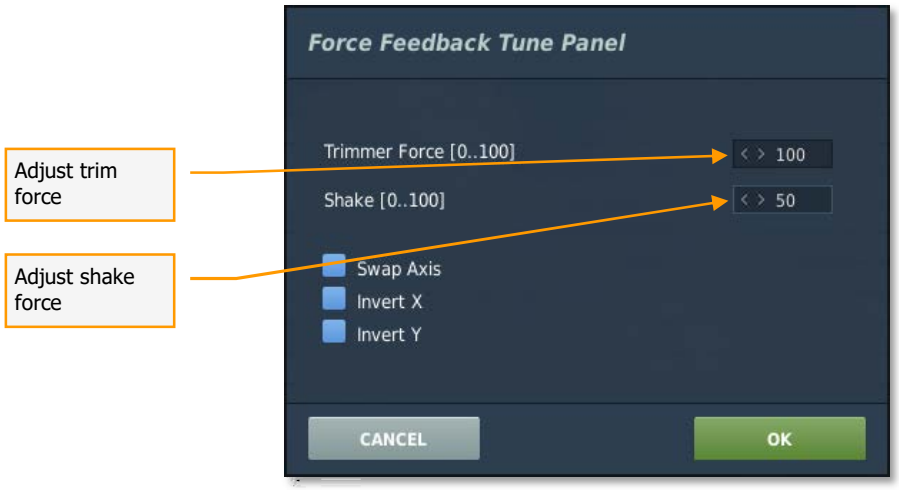
Invert. これにチェックを入れるとデバイスの軸を逆にします。

User Curve. Curvature スライダを使用するのではなく、User Curve を選択して、入力デバイスの応答を個別調整できる 11 の垂直スケールを表示することができます。 これらの範囲は、目盛りの左側に0%の入力から目盛りの右側に100%の入力までとなっています。



Axis Tune. このドロップダウンリストを使用して、チューニングする軸を選択します。

Force Feedback Tune. フォースフィードバックジョイスティックを使用している場合は、FF TUNE ボタンを押してForce Feedback Tuneパネルを表示させることができます。 このパネルでは、トリム(trim)の力とスティックの揺れを調整することができます。



フォースフィードバック軸を入れ替えるには、[Swap Axis] ラジオボタンをチェックします。

Swap Axis. このチェックボックスをオンにすると、選択した入力デバイスの軸を入れ替えることができます。これは、古いモデルのジョイスティックでは必要な場合があります。

Invert. このチェックボックスをオンにすると、X軸とY軸に沿った曲線を反転させることができます。

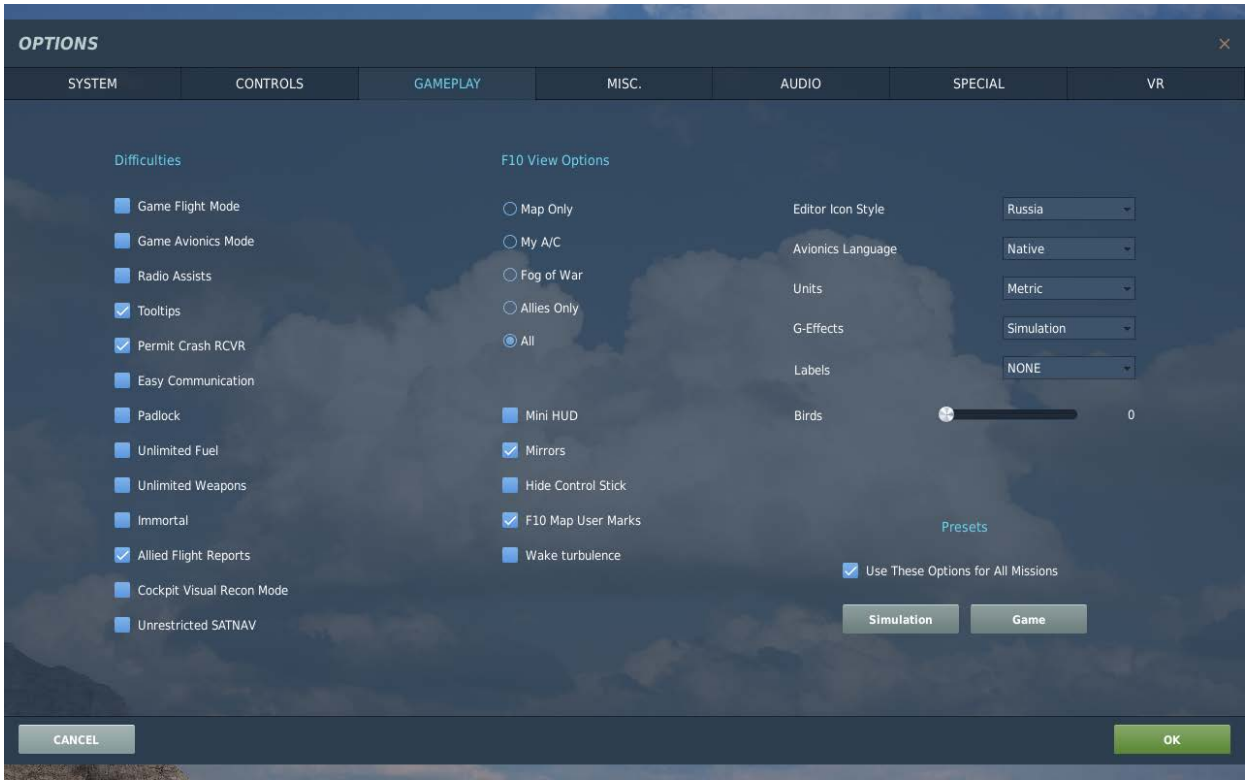
パネルの下部には、変更をキャンセルしてコントロールページに戻るCANCELボタンと、変更を保存してコントロールページに戻るOKボタンがあります。

Make HTML. 選択した航空機のコマンドのHTMLリストを作成することができます。このボタンを押すと、全てのデバイス用のHTMLファイルが入ったエクスプローラーウィンドウが開きます。

C:\Users\<User Name>\Saved Games\DCS\InputLayoutsTxt\

Gameplay Settings (ゲームプレイ設定)

ゲームプレイ設定タブでは、シミュレーション体験のリアルさを調節することができます。これらの設定を使用して、非常にリアルで難しいミッションをプレイすることもできますし、詳細なシステム知識や戦闘スキルを必要としない、リラックスしたカジュアルなプレイをすることもできます。



Difficulties （難易度）

difficulty オプションはチェックをしれて選択すると、オプションが有効になります。 difficulty オプションには次のようなものがあります。

Game Flight Mode. ゲームフライトモードは、航空機を操縦することの複雑さと難しさを大幅に軽減します。 選択されていない場合、航空機はより本格的な、より高度なフライトモデルシステムを使用しますが、初めて飛行機を操縦する人には難しいかもしれません。ゲームフライトモードは、航空機の飛行をはるかに簡単にし、気を使わずに遊ぶことができます。

Game Avionics Mode. ゲームアビオニクスモードは、画面上に3Dマーカーを表示することで、航空機の位置関係の把握を補助します。

Radio Assists. この機能を有効にすると、武器発射圏内である場合や、ミサイルの接近、敵ユニットの位置情報などを音声で知らせます。

Tooltips. コックピット内では、コントロールの上にマウスを置くと、コントロールの機能を示す簡単なメモが表示されます。

Permit Crash RCVR. クラッシュリカバリーを有効にすると、ミッション中に自機が破壊されたり、イジェクトした場合にリスポーンすることができます。

Easy Communication. イージーコミュニケーションをオンにすると、ラジオは自動的に選択された受信者の周波数に合わせて調整されます。さらに、ラジオメニューにはミッション中に存在するオブジェクトのみが表示され、連絡が可能な相手がハイライト表示されます。イージーコミュニケーションがオフの場合、プレーヤーは希望する連絡先に合わせて正しい無線機を正しい周波数にチューニングし、ラジオメニューから正しい通信相手を選択する必要があります。

Padlock. パドロックシステムは、コックピット内から見える範囲内のターゲットに対して視線をロック（パドロック）したままにします。パドロック対象は地上や空のユニットやミサイルに対して行えます。

Unlimited Fuel. これを有効にすると、燃料が消費されることはありません。 ただし、燃料を満タンにした場合、重量が増加するため、航空機の性能は低下することに注意してください。

Unlimited Weapons. これを有効にすると、武器を使用すると自動的に即座に補充されます。

Immortal. 有効にすると不死身になります。機体にダメージを与えたり、破壊したりすることができなくなります。

Allied Flight Reports. 割り当てられた無線周波数で味方のフライトの無線交信が有効になります。

Cockpit Visual Recon Mode. 地形上の任意の地上オブジェクトをマークし、味方の地図上にその位置を表示することができます。右マウスボタンで十字を表示し、左マウスボタンでオブジェクトを選択します。

Unrestricted SATNAV. 衛生ナビゲーション装置を持っている航空機であれば、時間と場所に関わらずこれを使用することが可能になります。

F10 View Options （F10ビューオプション）

ミッション中にF10キーを押すと、シミュレーションワールドのマップが表示されます。味方ユニットと敵ユニットの両方が表示され、コックピット外の戦場の全体像を把握することができます。 アイコンは一般的なユニットの種類に応じて描かれており、ブルーサイドのユニットには青色のアイコン、レッドサイドのユニットには赤色のアイコン、プレイヤーのユニットには白色のアイコンが表示されます。

F10ビューオプションでマップを表示する際に、表示される要素を設定することができます。

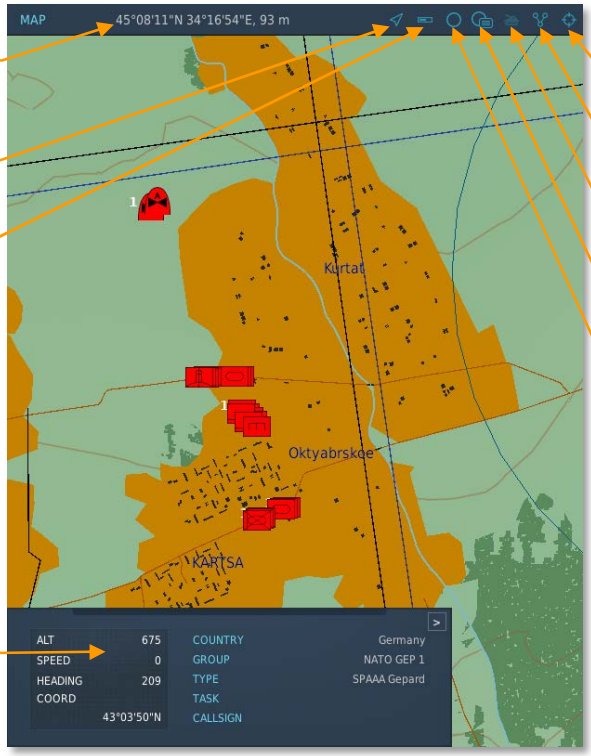
Map Only. 地図のみ表示され、オブジェクトアイコンは表示されません。

My A/C. マップ上には、プレイヤーの自機のアイコンのみが表示されます。

Fog of War. 味方ユニットによって発見された敵対ユニットのみが表示されます。

Allies Only. プレイヤーが所属する連合軍のユニットのみがマップ上に表示されます。

All. すべてのユニットがマップ上に表示されます。デフォルトの選択です。



カーソル位置の座標と標高

プレイヤー中心表示

ライフバー表示

ユニットデータ

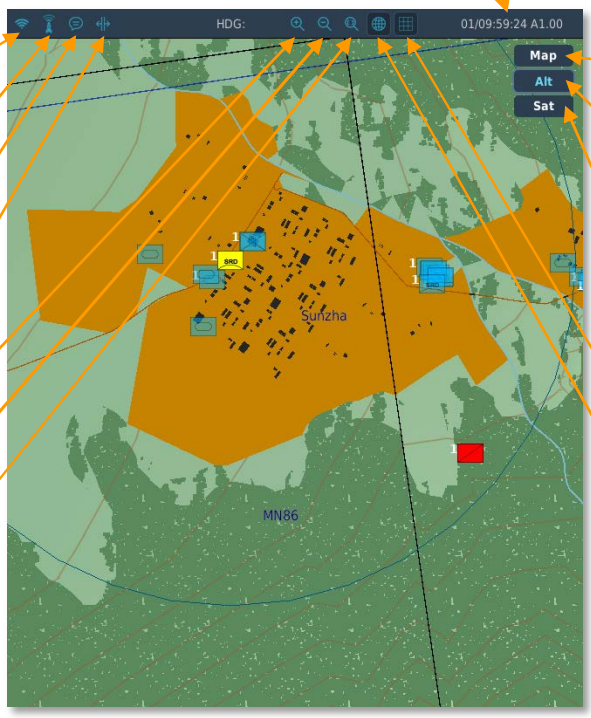
ターゲット

ルート

テイクコントロール

マーク表示

マーク追加



ゲームデータ/時間と加速倍率

Detection Zones

脅威ゾーン

ユニットラベル

Distance Tool

ズームイン

ズームアウト

Map 100%

Map Mode

標高マップモード

サテライトビュー

セクターナンバー

マップグリッド

F10マップでは画面上部に沿って、左から順に以下の機能を備えたツールバーがあります。

Unit Data（ユニットデータ）. ユニットのアイコンをクリックすると、画面左下にそのユニットのデータウィンドウが表示されます。ユニットのグループ名、陣営、コールサイン、タイプ、割り当てられたタスク、ユニットの高度、速度、方位、座標が表示されます。

Cursor Coordinate and elevation（カーソル位置の座標と標高）. 地図上でマウスを移動させると、カーソル位置の地理座標と標高がこのフィールドに表示されます。

Game data/time and acceleration rate of window（ゲーム内時刻と加速倍率）.現在のゲーム内（ミッション）時間とその時間加速倍率が表示されます。ゲーム内時間は日/時:分:秒で表示されます。加速倍率はAの文字で表示されます。例：A1.00は通常の時間の流れ[LShift +Z]を示し、加速倍率は1.00です。Pという文字は、時間が一時停止中であることを示します。

Player Center. このボタンを押すと、プレイヤーが中央になるよう、マップ移動します。

地図上に表示されているユニットの詳細情報を収集するには、そのユニットを左クリックして、ユニットデータウィンドウが表示せます。 選択したユニットアイコンは黄色くなります。 このウィンドウは情報のみのウィンドウで、ユニットの種類によっては以下のような情報を表示することができます。

- Unit type（ユニットの種類）
- Side（陣営）
- Unit callsign（ユニットのコールサイン）
- Group task（グループの任務）
- Unit altitude（ユニットの高度）
- Unit speed（ユニットの速度）
- Unit heading（ユニットの方位）
- Unit's current coordinates（ユニットの現在座標）

Show Lifebar（ライフバー表示）. このボタンをクリックすると、地上ユニットのライフバーが表示されます。

Add a mark（マーク追加））. マークを追加する。このボタンをクリックすると、F10 マップの敵ユニットに自分のマークを追加したり、サイン（署名）することができます。

Show marks（マーク表示）. このボタンをクリックすると、現在のユーザーと、セッション中の他のプレイヤーが追加したすべてのカスタムマーカーが表示されます。

Take control（テイクコントロール）. このボタン [又はRAlt+J] を押すと、選択したユニットや航空機を直接コントロールすることができます。

Show pathing. このボタンを押すと、地上ユニットと航空機に割り当てられたウェイポイントを表示します。

Detection Zones. 有効にすると、防空ユニットのレーダー範囲がユニットの周囲に円で表示されます。

Threat Zones（脅威ゾーン）. 有効にすると、防空ユニットの交戦範囲がユニットの周囲に円で表示されます。

Unit Labels（ユニットラベル）. 有効にすると、各ユニットの横にユニットの種類を示すラベルが表示されます。

Distance Tool（距離ツール）. マップ上でマウスを左クリックし、カーソルを新しい場所にドラッグすると、2点間に線が引かれます。 距離ツールボタンの右には線の距離が表示され、HDG（方位）ラベルの横には線の始点から終点までの方位が表示されます。

Zoom In Tool（ズームイン）. このボタンをクリックすると、カーソルが中央に(+)マークのついた虫眼鏡になります。 拡大したい地点で左クリックすると、クリックした地点をマップ中心としてマップを拡大します。 このモードを解除するには、このボタンをもう一度クリックします。する必要があります。

Zoom Out Tool（ズームアウト）. このボタンをクリックすると、カーソルが中央に(-)マークのついた虫眼鏡になります。 縮小したい地点で左クリックすると、クリックした地点をマップ中心として縮小します。 このモードを解除するには、このボタンをもう一度クリックします。する必要があります。

Map 100%. このボタンを押すと、マップを 100% のフルサイズに素早く戻します。

Map grid（マップグリッド）. 座標グリッドを表示します。

Square numbers. 座標グリッドのsquare numbersを表示します。

Map Type. 画面右上のボタンで、3種類から地図のタイプを選択できます。

- MAP. 標高データのない世界地図を表示します。
- ALT. 地図を表示しますが、標高データ（等高線）が入っています。
- SAT. 衛生写真による世界地図を表示します。

Additional Settings （追加設定）

画面の右側には、シミュレーションのゲームプレイオプションをカスタマイズするためのオプションがいくつかあります。 これらには以下になります。

- **Editor Icon Style.** ミッションエディタを使用してマップ上にユニットやポイントを配置する際には、西部スタイル（NATO）の軍事アイコンを使用するか、ロシアスタイルの軍事アイコンを使用することができます。 このドロップダウンリストを使用して、どちらかを選択します。





- **Avionics Language.** 航空機に英語とネイティブの両方のアドバイザリーメッセージがある場合、このオプションでどちらを聞くかを選択することができます。
- **Units（度量単位）.** ドロップダウンメニューでは、測定単位をインペリアル法またはメートル法から選択することができます。 この選択により、Mission Editor で使用する測定単位が決定されます。
- **G-Effects（G効果）.** 機体にかかるGに応じて、ドロップダウンリストから選択したレベルに応じて、Gフォースの効果が変わります。 オプションには、 None, Reduce, Realisticがあります。 Gモデルでは、パイロット（あなた）に負担がかかるような急激なGの立ち上がりをシミュレートしていることに注意してください。 そのため、リアリスティック設定では、急に操縦桿を引くのではなく、スムーズにGをかけていくようにしてください。
- **Labels.** この機能を有効にすると、シミュレーションワールドで見えるユニットの横にラベルが表示されます。 ラベルは、より簡単にユニットを見つけることができるように設計されており、ユニットとの距離に応じて詳細な情報を提供します。
  - **NONE** – マップマーカーを無効にします。
  - **FULL** – 完全なオブジェクト名を表示します。
  - **ABBREVIATED** – オブジェクト名を省略して表示します。
  - **DOT ONLY** – ドットのみ表示します。

自機とユニットが離れているほどラベルの情報量は少なくなります。 ラベル情報が提供される距離は、ユニットの種類によって異なります。 以下の表にまとめました。

航空機	ラベル
30 km	・
20 km	+ オブジェクトまでの距離
10 km	+ オブジェクトの名前
5 km	+ コールサイン
地上物	ラベル
20 km	・
10 km	+ オブジェクトまでの距離

5 km	+ オブジェクト名
艦船	**ラベル**
40 km	・
20 km	+ オブジェクトまでの距離
10 km	+ オブジェクト名
武器	**ラベル**
20 km	・
10 km	+ オブジェクトまでの距離
5 km	+ オブジェクト名

- **Birds（鳥）.** このスライダーを使用して、非常に低い高度で飛行しているときのバードストライクの可能性を調整します。100%に設定すると、バードストライクの確率が現実的になります。このシミュレーションでは、目に見える3Dの鳥が出現するわけでは有りません。損傷の可能性の高める効果（主にエンジンのフレームアウト）をシミュレートしています。
- **Mini HUD.** Flaming Cliffs 3の航空機を飛行する場合、このオプションでは、視野がどこに向いているかに関係なく常に表示されるミニHUDのON/OFFを設定できます。
- **Mirrors.** リアビューミラーを有効にします。 ミラーはシステムパフォーマンスに大きな影響を与える可能性があることに注意してください。
- **Hide Control Stick.** 操縦桿を非表示にして、コックピット下部の計器類や機器の視認性を向上させます。
- **F10 Map User Marks.** ミッションエディタのマップにユーザーマークの名前を配置したり、編集したりすることができます。
- **Wake turbulence（後方乱気流）.** 航空機が空を通過する際に後方に形成される大気の乱れを計算するようになります。空中給油をしたり、フォーメーションで飛んだり、敵機の６時に位置しているときに新しい感覚を得られます。

Presets（プリセット）

このウィンドウの下部には、すべてのミッションでここで設定したゲームプレイ設定を強制的に使用するオプションと、ゲームプレイ設定を簡単に設定できるようあらかじめ用意した2つのプリセットがあります。

Use These Options for All Missions. このオプションをチェックすると、全てのミッションにおいて、ここで設定したゲームプレイ設定を強制的に使用します。このボックスがチェックされていない場合は、ミッション作成者が設定したゲームプレイ設定が使用されます。

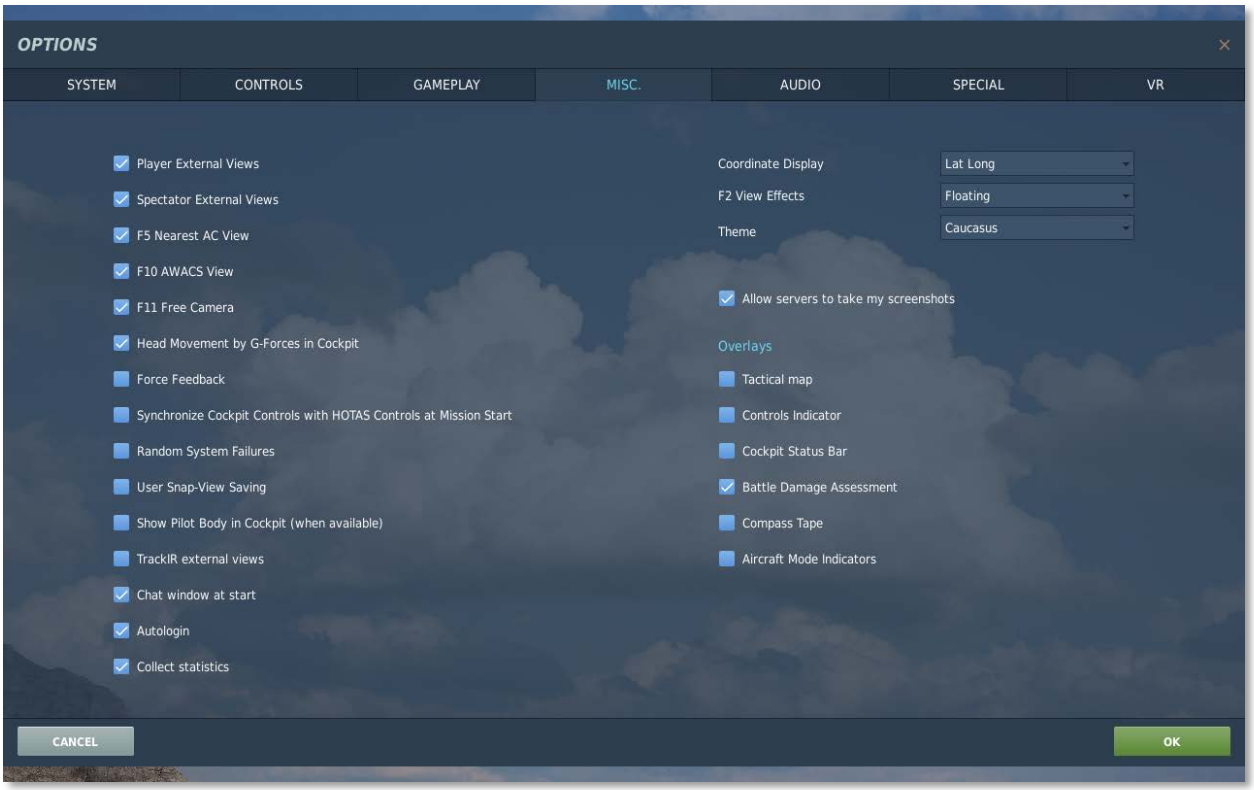
画面の右下には2つのプリセットボタンがあります。 これらのボタンでゲームプレイのオプションをリアルなシミュレーショ ンか簡単なゲームのどちらかで素早く設定できます。どちらかを押して基本の設定を作ってから、手動で自分の好みに合わせて行くのが良いでしょう。

画面の一番下には、オプションの変更を保存せずにメインメニューに戻るCANCELボタンと変更を保存してメインメニューページに戻るOKボタンがあります。

また、画面右上の黄色のXボタンを押すことで、GAMEPLAYページを終了することもできます。

Miscellaneous （その他）

Miscタブには、追加のゲームオプションが多数含まれています。



Player External Views. プレイヤーの外部ビューを許可します。

Spectators External Views. 観客（Spectator）の外部ビューを許可します。

F5 Nearest AC View. 外部ビューで「F5」最も近い航空機へのパッドロックを有効にします。

F10 AWACS View. 「F10」マップを有効にします。

F11 Free Camera. 「F11」フリーカメラビューを有効にします。

Head Movement by G-Forces in Cockpit. 飛行中の航空機の操縦に起因するGによるパイロットのわずかな頭の動きのシミュレートを行います。

Force Feedback. フォースフィードバック効果を有効にします。

Synchronize Cockpit Controls with HOTAS Controls at Mission Start. ミッション開始時にジョイスティックやスロットルのスイッチの状態に合わせて、コクピット内のスイッチ類を合わせます。

Random System Failures. ミッション中に航空機のシステム障害がランダムに発生するようになります。これらの故障は、個々の装置の故障平均時間をシミュレートして自動的に生成されます。

ミッションの一部として含まれるスクリプト化された機体の故障とは関係ありません。

User Snap-View Saving. 装置やHUD,MFD画面などへユーザー作成のカスタムビューを保存可能にします。

Show Pilot Body in Cockpit（パイロットの表示） (利用可能時). パイロットの腕がコックピットの特定のコントロールを見えなくすることがあるため、ユーザーはコックピットでのパイロットの体の表示をON/OFFすることができます。

TrackIR external views. TrackIR入力を介して外部カメラを操作できるようにします。

Chat window at start. 離陸開始時にチャットウィンドウを表示します。

Autologin. DCS起動時に、設定したログイン名とパスワードを使用して、自動的にアカウントにログインできるようにします。

Collect statistics（情報収集）. このチェックボックスのチェックを外さないでください - 私たちが収集したデータは、ゲームをより良いものにするのに役立ちます。これは私たちにあなたの銀行カードや個人的な連絡先リストへのアクセスを与えるものではありません。

Coordinate Display. ミッションエディタやゲーム内のパネルでの座標表示に、Lat/Long, Lat/Long Decimal, MGRS, Metric and Precise Lat/Long が選択できます。

- **Lat Long**. 座標を度-分-秒で表示します。
- **Lat Long Decimal**. 座標を度数形式で表示（分の10進数）
- **NATO Standard (MGRS)**. ミリタリーグリッドリファレンスシステム（Military Grid Reference System）は、NATOが目標地点の位置を計算するために広く使用している座標系です。MGRSはUTMに基づいており、各ゾーンを100×100kmの正方形に分割しています。これらの正方形は2文字のコードで識別され、最初の文字は経度ゾーンの東西の位置、2番目の文字は南北の位置です。
- 例えば、UTM座標の35 V 414668 6812844は、MGRS座標の35VMJ1466812844に相当します。MGRSにおける座標精度は、１メートルの精度レベルで設定されており、１５文字で表され、最後の１０文字は指定されたグリッド内のeastingとNorthingの値を表しています。
- **Metric**. Absolute X/Y coordinates.
- **Precise Lat/Long**. 座標を度数形式で表示します - 分の10進数 - 地理的秒数の100分の1まで正確に表示します。

F2 View Effects. 「F2」外部航空機カメラのカメラエフェクトを選択します。

- **None**. エフェクト無し
- **Floating**. フローティングカメラエフェクト
- **Shaking**. シェイキングカメラエフェクト

Theme（テーマ）ドロップダウンリストからメインメニューのバックグラウンドテーマを選択できます。

Allow servers to take my screenshots. マスターサーバーがゲームセッションのスクリーンショットを撮ることを許可します。

Overlays （オーバーレイ）

Tactical Map. 指定されたウェイポイントと方位を表示したミニマップを表示します。方向を示すためのマーカーも表示されます。

Controls Indicator. コクピットコントロールデバイスの現在状態を表示します。

Cockpit Status Bar（コクピットステータスバー）. 現在の飛行パラメータと座標を表示します。

Battle Damage Assessment（戦闘ダメージ評価）. プレイヤーが敵をヒットしたときに、与えたダメージをパーセンテージで表示します。ダメージとターゲットの破壊に関する情報は、リアルタイムで画面の左端に表示されます。破壊されたターゲットは赤で表示され、ダメージを受けたターゲットは黄色で表示され、ダメージのパーセンテージが表示されます。

Compass Tape. 画面上部にコンパステープを表示します。

Aircraft Mode Indicators（航空機のモード表示）. 画面の左側にあるオーバーレイを表示します。航空機のシステム（飛行経路、戦闘モード、選択武器、および対策システム（フレア/チャフ））に関する情報を表示します。

Audio Settings（オーディオ設定）



画面には、ゲーム内のオーディオ環境をコントロールするための設定があります。6つのスライダーと2つのラジオボタンの組み合わせて行います。

オーディオスライダーは、右に動かすと音量が上がり、左に動かすと音量が下がります。 それぞれのスライダーは、特定のオーディオチャンネルに特化しています。

Volume. これはオーディオ設定のマスターボリュームをコントロールし、すべてのチャンネルに等しく影響します。

GUI. ボタンクリックなどのGUI（Graphics User Interface）の音量は、このスライダーでコントロールされます。

World. コックピット外のオーディオソースの音量をコントロールします。

In-Cockpit. コックピット内のオーディオソースの音量レベルをコントロールします。例としては、スイッチやファンなどが挙げられます。

Helmet. ラジオメッセージや警告など、パイロットのヘッドフォンから聞こえる音量レベルを調整します。

Music. メニュー画面で流れる音楽の音量を調整します。

Switches. 航空機（AC）キャビン内のスイッチ音の音量を調整します。

Main audio out. ゲーム内のメインオーディオ出力の選択。

Headphones. ゲーム内のヘッドフォンからの出力を選択します。

スライダの下には、5つのラジオボタンがあります。

Hear like in Helmet. パイロットのヘルメットでサウンドシミュレーションを有効にします。外部の音がミュートされ、無線通信が大きくなります。

G-breath Effect. G過負荷時にパイロットの呼吸音を有効にする。

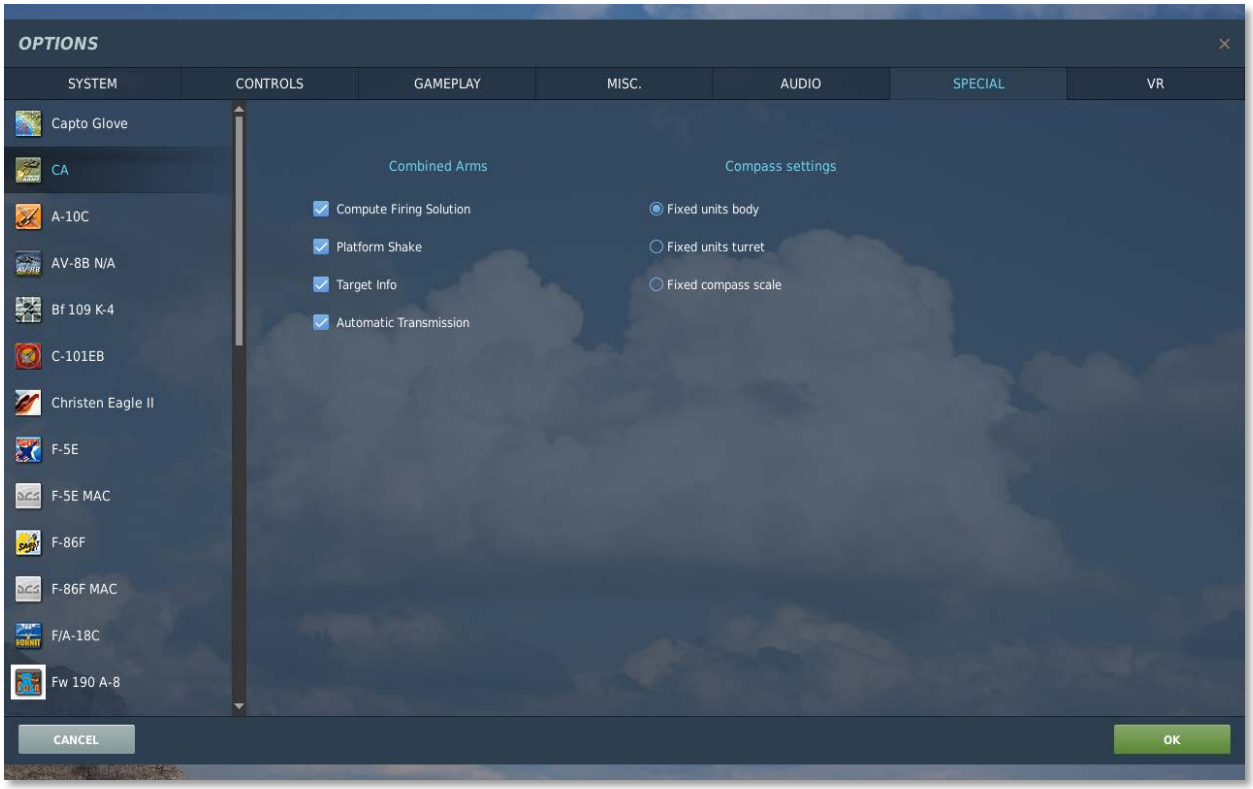
Radio Speech. すべての無線交信を聞くことができます。

Subtitles. 全てのテキスト字幕を表示します。

External View Cockpit Sounds and Radio Messages. 外部の航空機/ユニットビューに切り替えた際でも、コックピットからの音声と無線を聞くことができます。

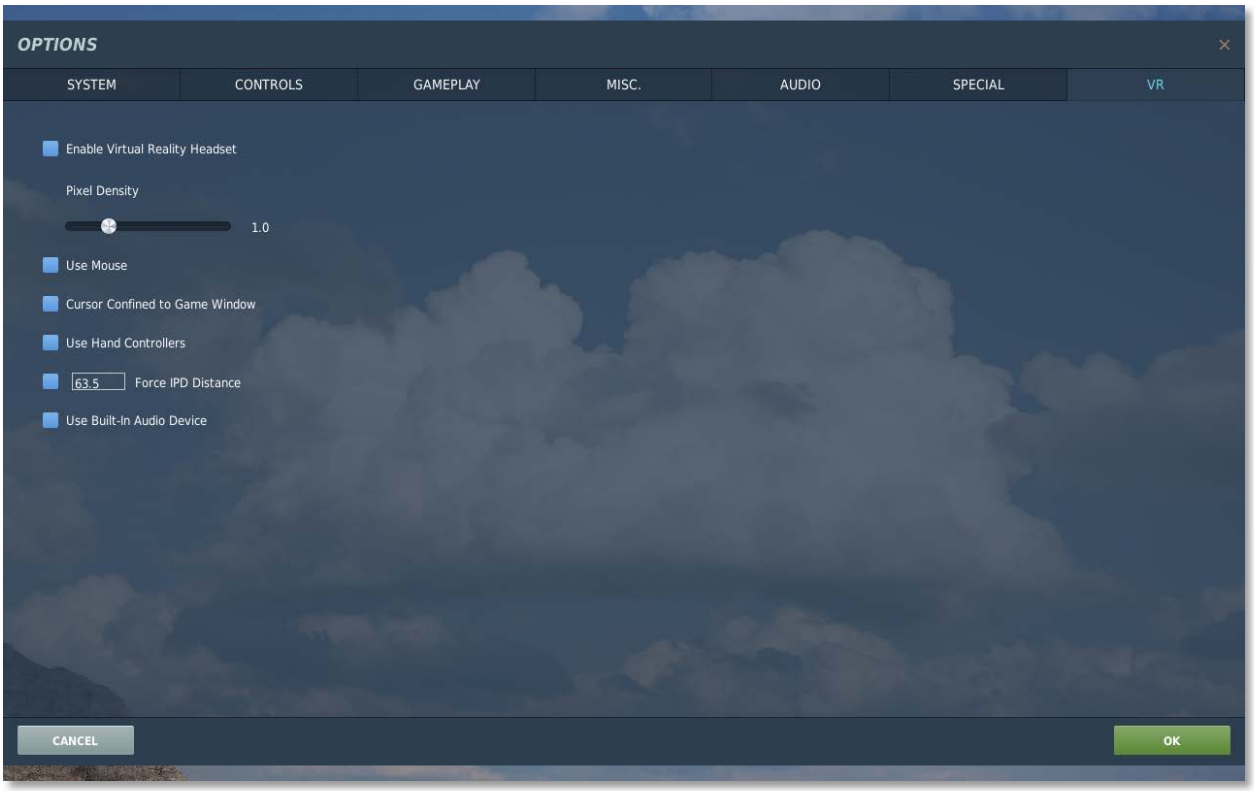
Special （スペシャル）

このページでは、左の列にインストールされたモジュールが表示されています。それを選択すると、各モジュールごとに設定グループがあります。これらの設定をオン/オフにすることで、各モジュールの操縦などが簡単になったり、複雑になったりします。



VR

このページでは、VR デバイスの設定を行います。



Enable Virtual Reality Headset. VRデバイスを使用を有効にします。

- **Pixel Density** – VRヘッドセットの画面のピクセル密度を変更し、解像度を上げることができます。この値を大きくするとパフォーマンスに大きく影響します。

Use Mouse. この設定を有効にすると、カーソルが現在のヘッド位置から外れ、マウス入力でコントロールできるようになります。

Cursor Confined to Game Window. このオプションを有効にすると、マウスカーソルがモニター上のゲームウィンドウからはみ出ないようになります。

Use Hand Controllers. 環境やインターフェイスの操作にハンドコントローラの仕様が可能になります。のサポートが含まれています。

Force IPD Distance. ユーザーが手動で瞳孔間距離を設定することができます。

Use Built-in Audio Device. VRヘッドセット内蔵のオーディオデバイスをオンにします。

digital combat series

DCS World



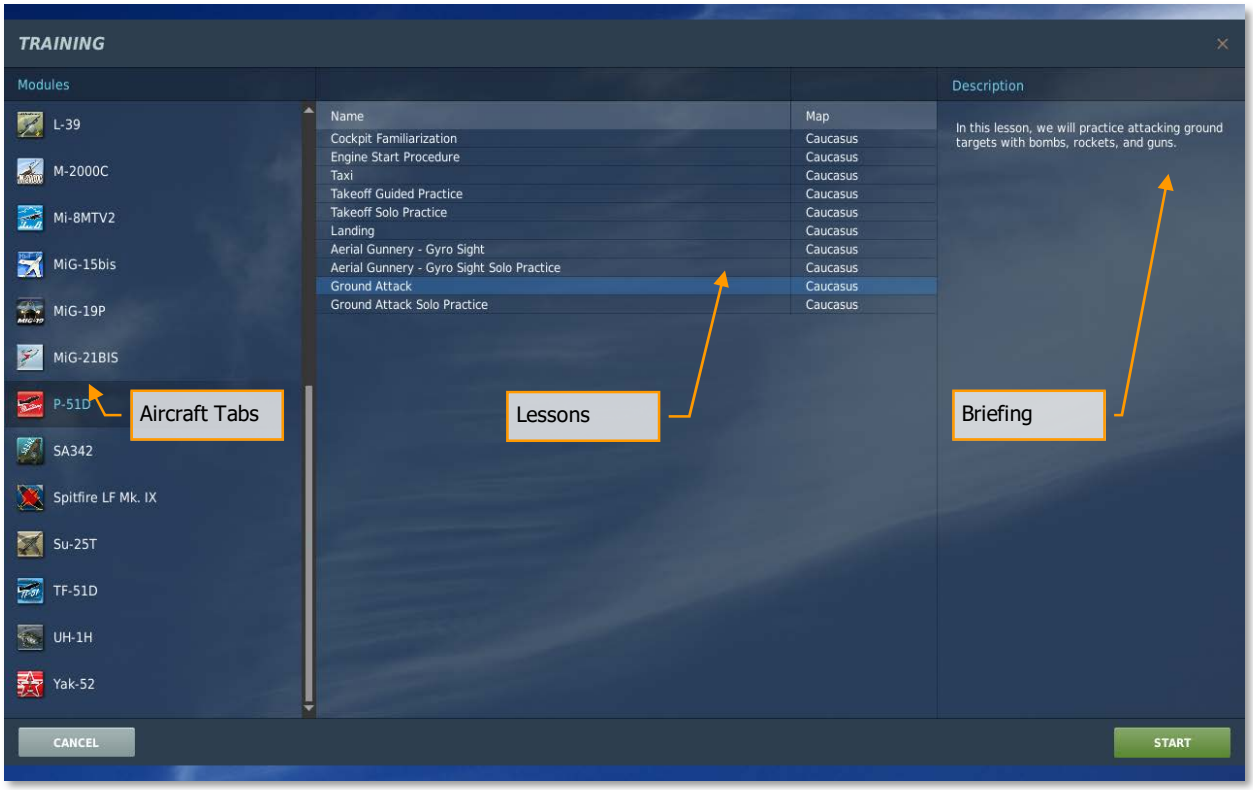
TRAINING

TRAINING （トレーニング）

機体の操縦方法を学ぶのに最適なように、インタラクティブなトレーニングミッションや録画済みのトレーニングビデオで構成されたトレーニングモジュールを用意しています。 トレーニングメニューに入るには、メインメニュー画面からTRAININGを選択してください。



トレーニングメニューは以下の3つのエリアに分かれています。



Aircraft Tabs. 練習したい航空機を選択します。

Lessons. 練習したい航空機を選択すると、その航空機でのトレーニングミッションがここにリスト表示されます。 これらのレッスンは、それぞれ特定の訓練を目的として設計されています。

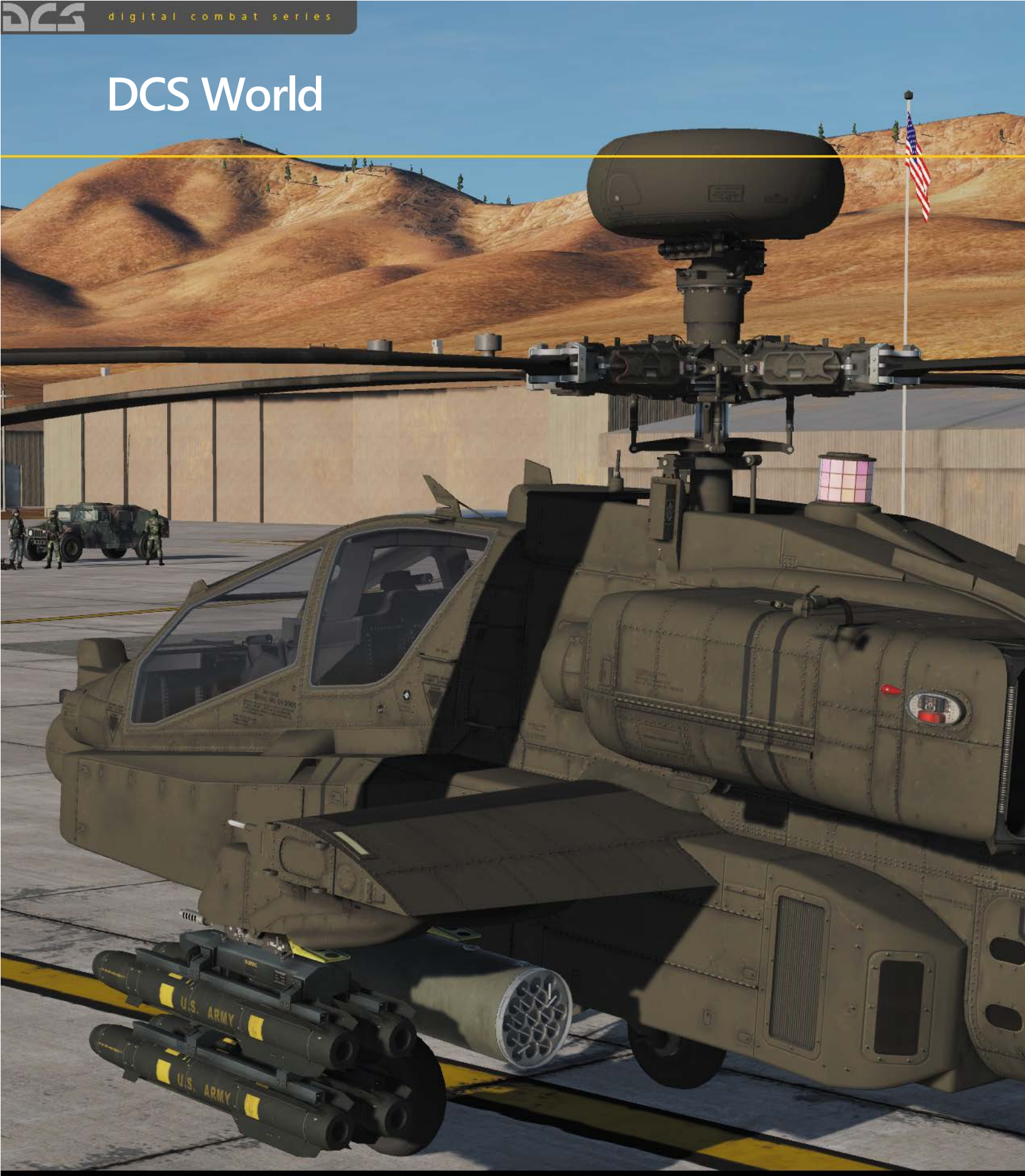
Briefing. レッスンを左クリックして選択すると、選択したレッスンの概要を説明するテキストが表示されます。

（日本語化されたミッションであってもここまでは英語です。）

画面の一番下には、メインメニューに戻るCANCELボタンと、レッスンを開始するSTARTボタンがあります。
また、画面右上の黄色のXボタンを押すことで、トレーニングページを終了することができます。

digital combat series

DCS World

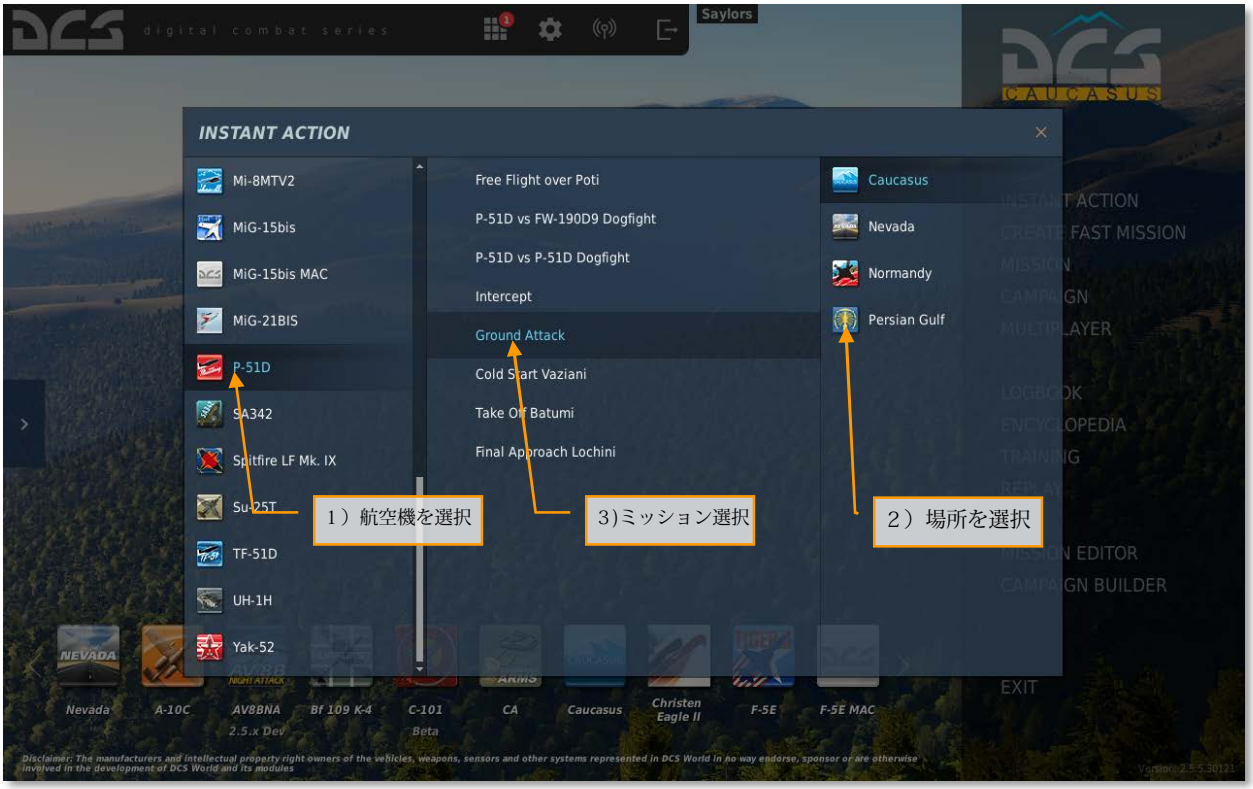


INSTANT ACTION

INSTANT ACTION （インスタントアクション）

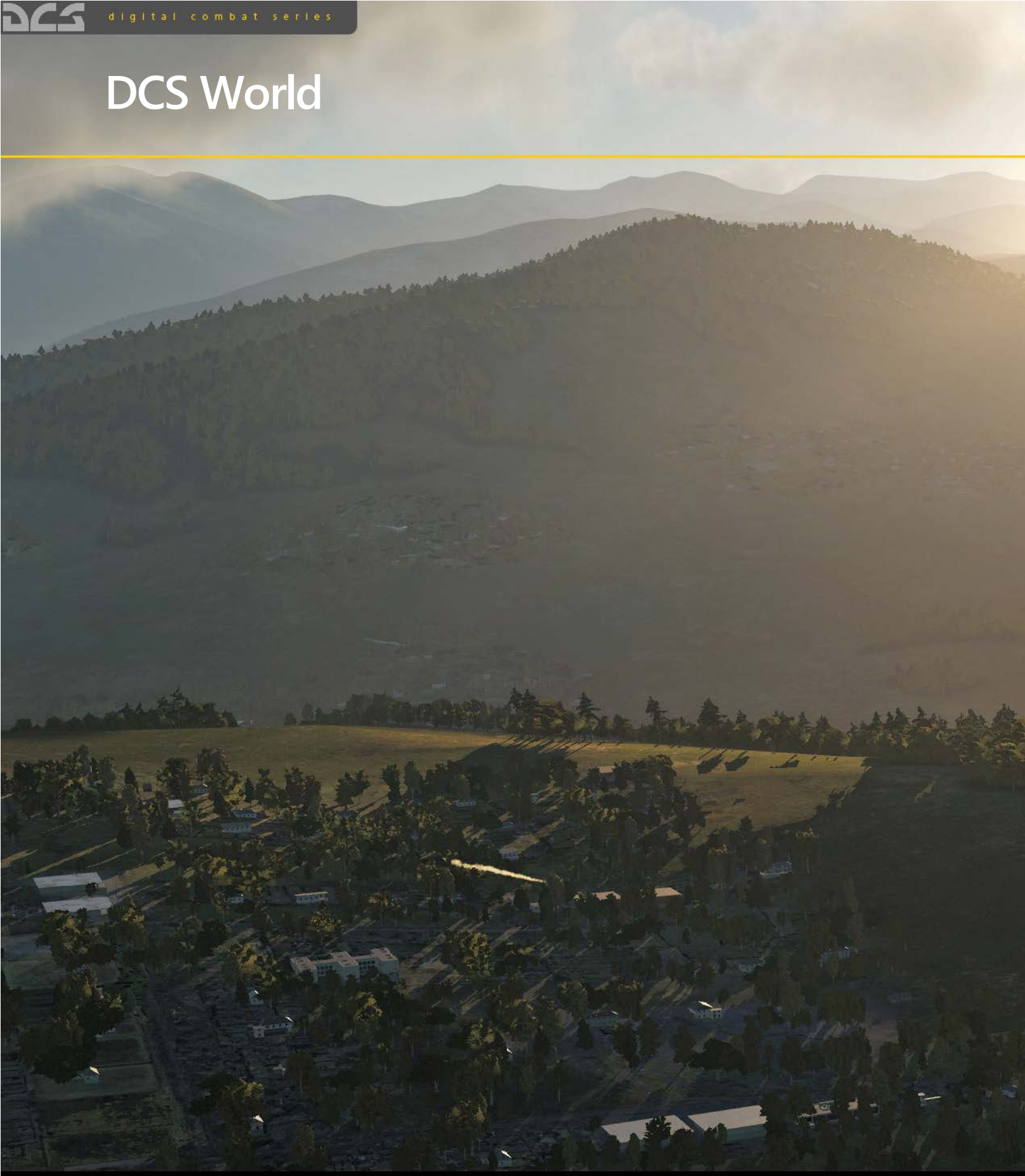
INSTANT ACTION （別名：クイックミッション）を押すと、航空機を選択できるポップアップウィンドウが開き、航空機ごとにいくつかのミッションを選択することができます。 航空機タブで航空機を選択し、場所を選んで、目的のミッションをクリックすると、事前に生成されたミッションを飛行することができます。インスタントアクションは、素早くミッションを飛ばすための便利な方法です。




INSTANT ACTION
Mi-8MTV2
MiG-15bis
MiG-15bis MAC
MiG-21BIS
P-51D
SA342
Spitfire LF Mk. IX
Su-25T
TF-51D
UH-1H
Yak-52
Free Flight over Poti
P-51D vs FW-190D9 Dogfight
P-51D vs P-51D Dogfight
Intercept
Ground Attack
Cold Start Vaziani
Take Off Batumi
Final Approach Lochini
Caucasus
Nevada
Normandy
Persian Gulf
1）航空機を選択
3)ミッション選択
2）場所を選択
Nevada
A-10C
AV8BNA
Bf 109 K-4
C-101
CA
Caucasus
Christen Eagle II
F-5E
F-5E MAC
EXIT

digital combat series

DCS World



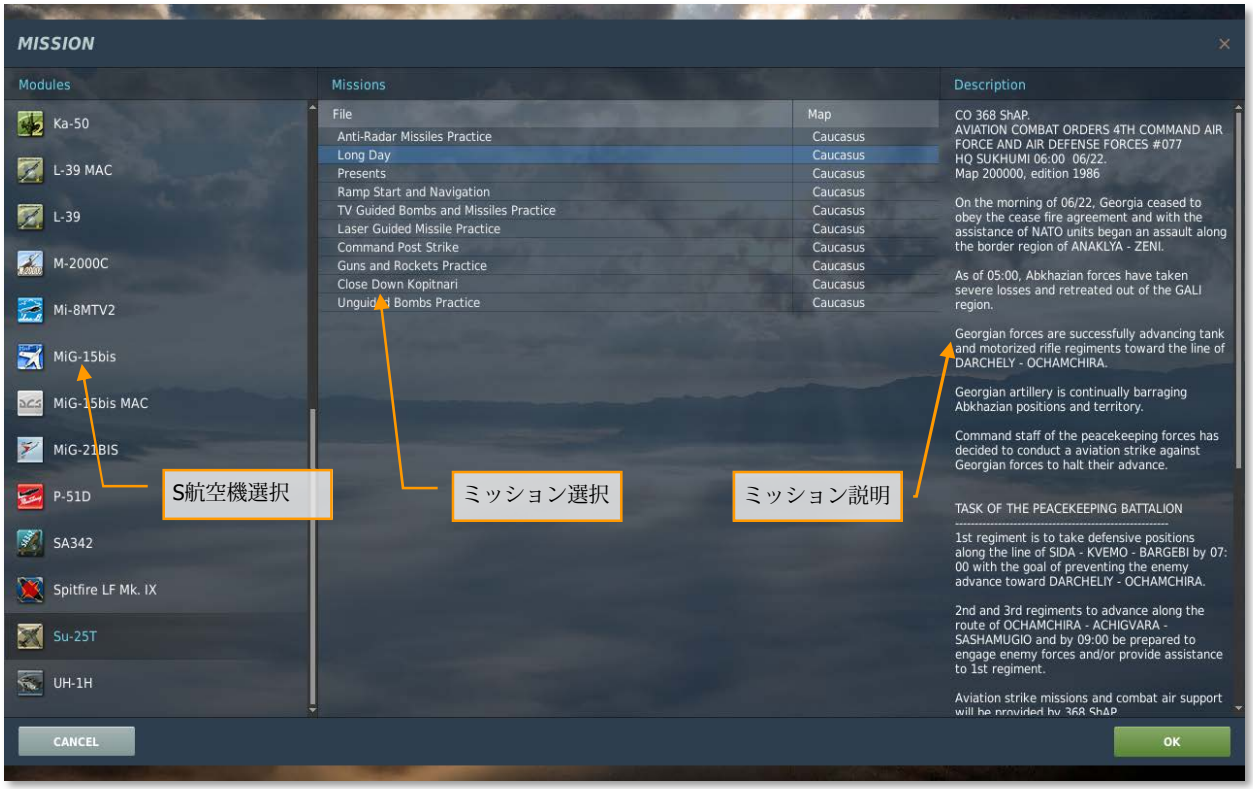
MISSION

MISSION （ミッション）

ミッションエディタを使用して、ミッションを作成し、ミッション画面から開くことでプレイすることができます。 ミッション画面に行くには、メインメニューページのMISSIONボタンをクリックします。



MISSIONを選択すると、ミッションの選択ページが表示されます。 このウィンドウでは、ローカルディスクを参照し、保存されたミッションを選択してロードすることができます。 ミッションは "*.miz "形式で保存されます。



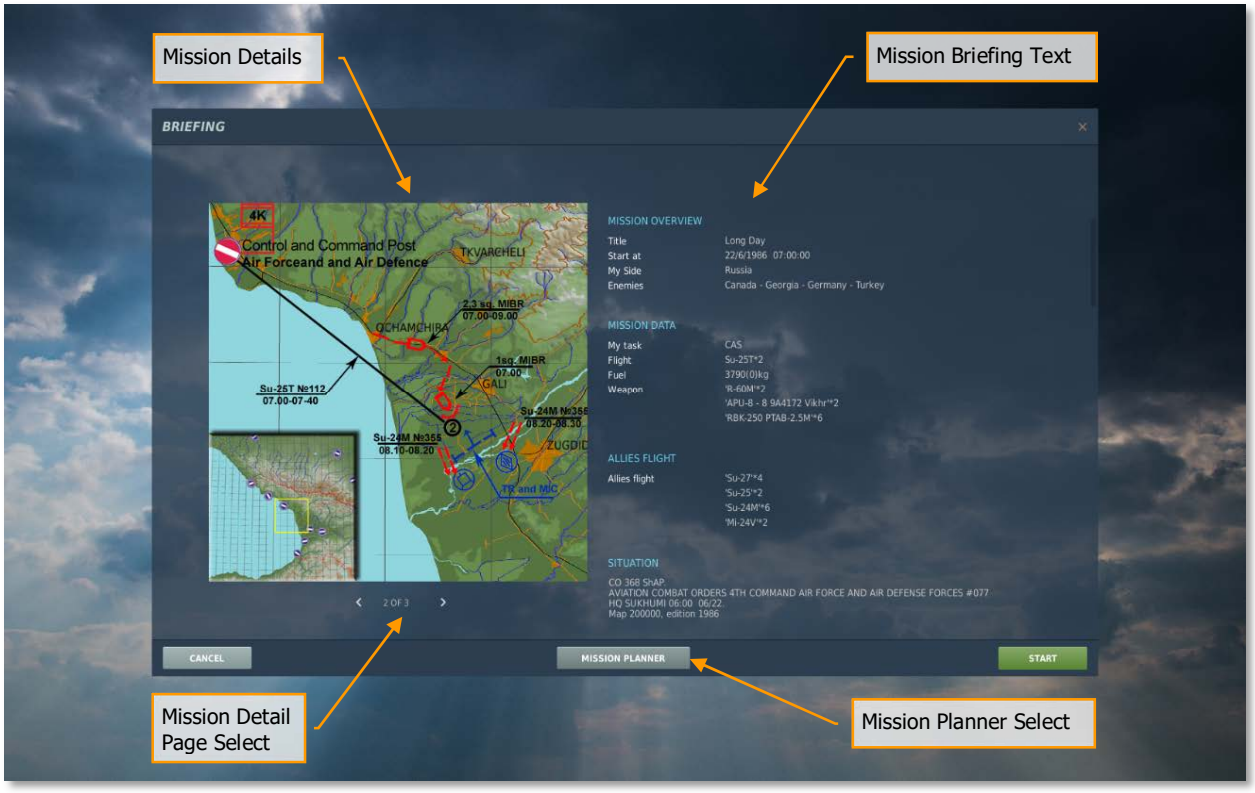
マウスの左ボタンで目的のモジュールとミッションを選択します。 ミッションが選択されると、そのミッションに書かれたブリーフィングが右側のMission Descriptionウィンドウに表示されます。（日本語化されたミッションではこのブリーフィングも日本語表示されます。）また、ウィンドウ下部のCANCELボタンを押すか、右上のXボタンを押すことで、ミッションをロードせずにウィンドウを終了することができます。

Briefing （ブリーフィング）

OKボタンを押すと、ミッションブリーフィング画面に移動します。 この画面では、1 つまたは複数のブリーフィング画像と、データとテキストのミッションブリーフィングが表示されます。

ミッション・ブリーフィング画面は以下の要素で構成されています。：

- **Mission Details（ミッション詳細）**. 各ミッションは、1つまたは複数のブリーフィング画像で構成されています。これらの画像はミッションデザイナーによって作成され、ミッションマップ、ターゲットエリアの写真、ターゲット車両の画像などのコンテンツで構成されています。
- **Mission Detail Page Select（ページ切り替え）**. 左右の矢印ボタンを使用して、ミッション詳細のブリーフィング画像を切り替えます。
- **Mission Briefing Text.** 画面右側のスクロール可能なテキストブロックには、開始時間、状況、目的、陣営、飛行情報や自動生成されたミッションデータなど詳細なミッションデータが表示されています。



画面の一番下には、３つのボタンがあります。

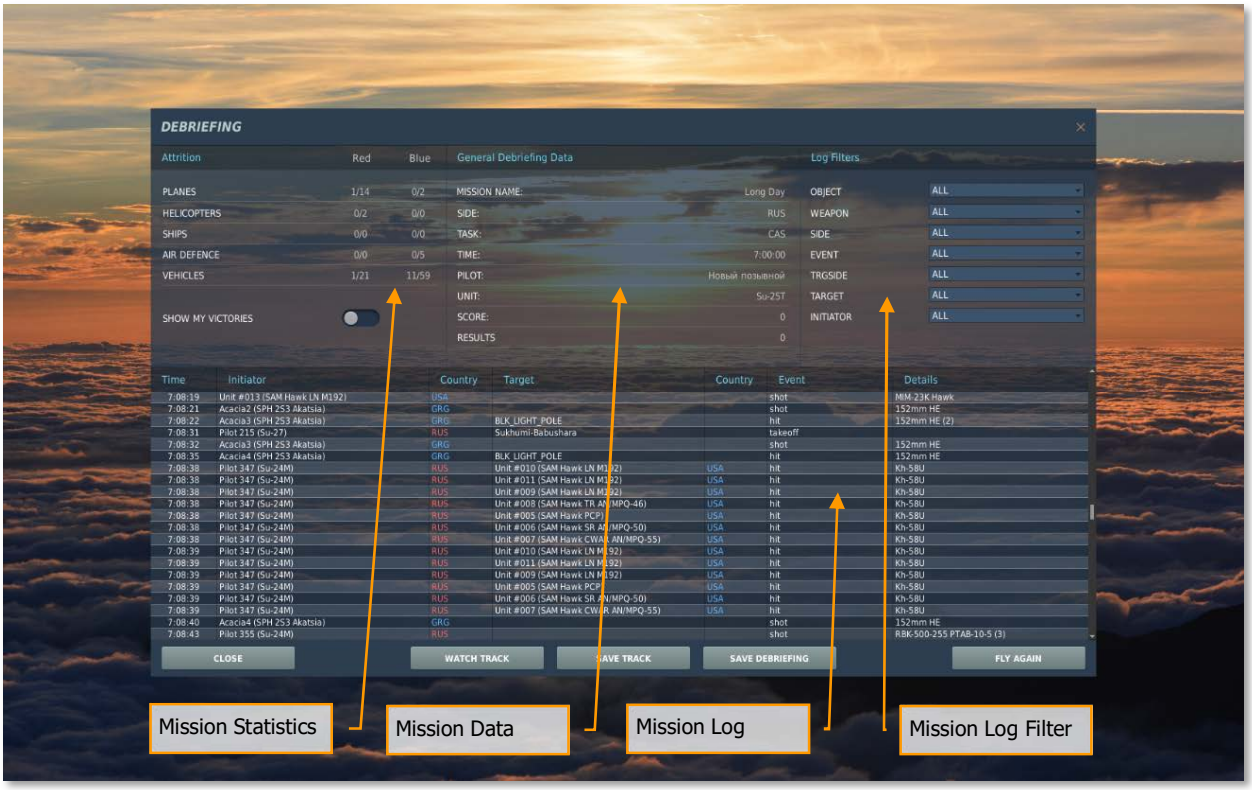
- **CANCEL**. キャンセルボタンを押すとミッションページに戻ります。 また、ページ右上の黄色の×を押してミッションをキャンセルすることもできます。

プランナーやエディタでミッションを編集する際に注意してください。自分が何をしているのか理解している場合にのみ、ミッションを編集してください。

- **MISSION PLANNER**. ミッションプランナーボタンを押すと、選択したミッションがミッションプランナーに表示され、フライトの詳細やルート、武器の編集が可能になります。
- **START**. STARTボタンを押すとミッションが開始され、シミュレーションに入ります。

View Debriefing （デブリーフィング）

ミッションを飛行し終えると、自動的にデブリーフィング画面が表示されます。



デブリーフィング画面は4つの主要エリアに分かれています：attrition、debriefing data,、 log filters、chronological log fileです。

ATTRITION. 画面のミッション統計の部分では、ミッション中に破損または破壊されたユニットの数がタイプ別、連合別に表示されます。表示されるユニットには次のものが含まれます。飛行機、ヘリコプター、艦船、航空防衛、車両。X/Y形式で表示されます。

X - 破壊されたユニットの数

Y - ミッションに参加したユニットの総数

- **SHOW MY VICTORIES**. SHOW MY VICTORIESのチェックボックスを使用して、すべてのユニットによる戦績を表示するか、自分だけの戦績を表示するかを選択することができます。

GENERAL DEBRIEFING DATA. ミッションデータを表示します。

- **MISSION NAME**. ミッションエディターで定義されているミッション名
- **SIDE**. プレイヤーの陣営

- **TASK**. プレイヤーの任務
- **TIME**. ミッション開始時間
- **PILOT**. パイロット名
- **UNIT**. ミッションで使用した機体
- **SCORE**. 獲得したスコア
- **RESULTS**. ミッションにおける目的達成度合い、この値が一定数を超えないと成功とみなされない。

LOG FILTERS. 画面右上のログフィルターセクションでは、ミッションログリストに表示されるログイベントをフィルタリングすることができます。 それぞれにドロップダウンメニューがあり、表示するデータを選別することができます。

- **OBJECT**. オブジェクトリスト
- **WEAPON**. ミッション中で使われた全兵器がリスト化されています。
- **SIDE**. All, Red, Blue.
- **EVENT（イベント）**. All, Dead, Hit, Shot.
- **TRGSIDE (ターゲットサイド)**. All, Red, Blue.
- **TARGET**. ミッション中にターゲットになったすべてのユニットとオブジェクトをリストアップします。
- **INITIATOR**. ミッションに出現した全ユニットがリスト化されています。

MISSION LOG. 画面の大きく下の部分には、ミッションログのフィルター設定に従って、ミッションログのイベントが時系列で一覧表示されます。 ミッションログは7つの列で構成されています。

- **TIME**. ミッション中に発生したイベントの時刻（時：分：秒）
- **INITIATOR**. イベントを起こしたユニット名 (例：武器発射、墜落、着陸など）
- **COUNTRY**. イベントユニットの所属国
- **TARGET**. イベントユニットが攻撃をした場合、ターゲットの名前がここに表示されます。
- **COUNTRY**. ターゲットの所属国
- **EVENT**. 起こったイベント内容（射撃、ヒット、破壊など）
- **DETAILS**. イベントユニットがターゲットの攻撃に使用した武器名など

デブリーフィング画面は、誰が、いつ、何をしたのか、どんな武器を使っていたのかを判断するのに便利なツールです。　多くの場合、シングルプレイやマルチプレイのミッションでは、誰が攻撃してきたのかを判断するために使用することができます。

ログフィルターセクションの下には5つのボタンがあります。

CLOSE. デブリーフィング画面を終了します。

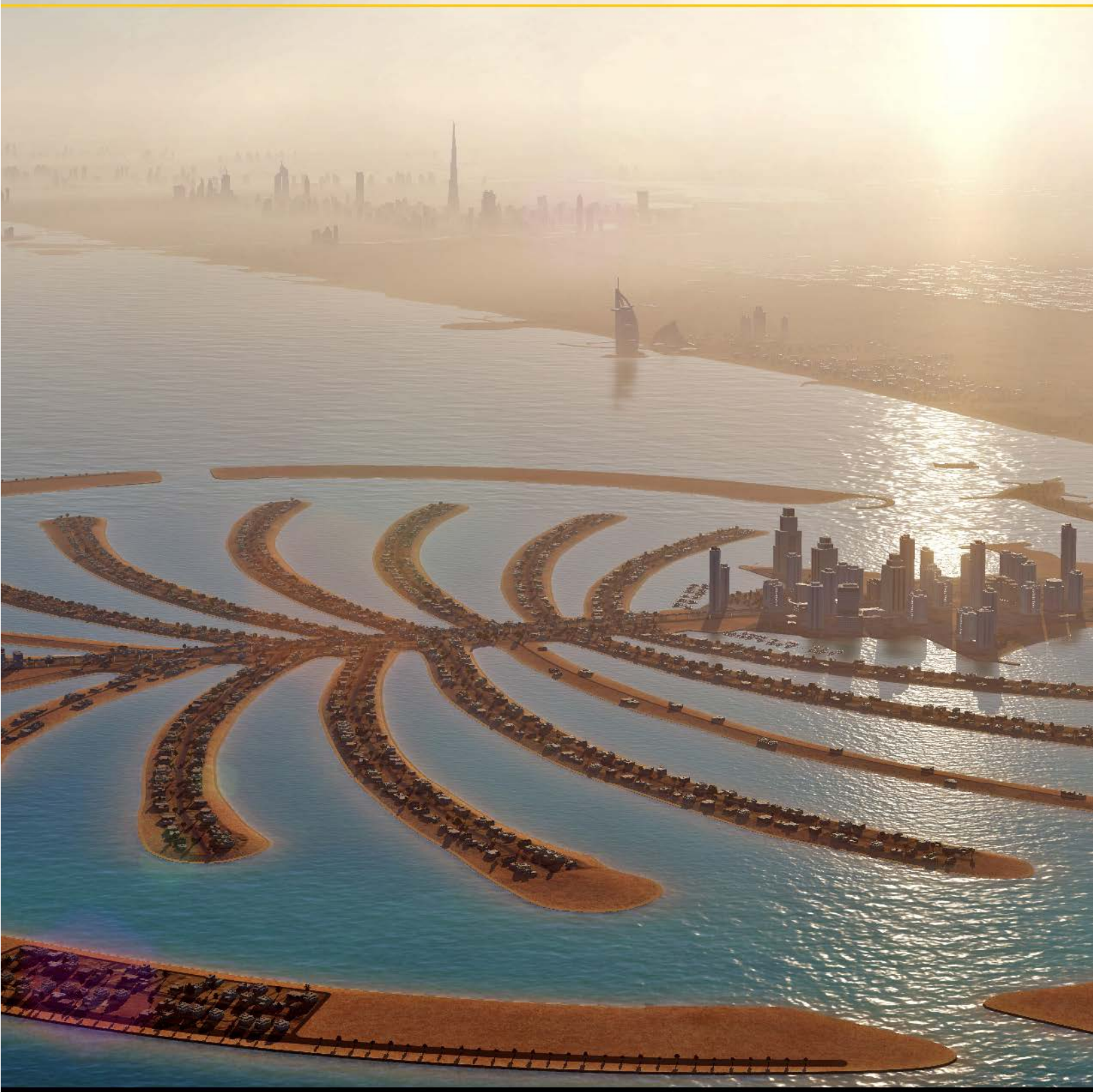
WATCH TRACK. 終了したばかりのミッションの録画（トラック）を見ることができます。
SAVE TRACK. ファイル保存画面が表示され、ミッション中に自動的に記録されたトラックファイルを保存できます。FILE 欄にファイル名を入力しSAVE TRACKで保存します。すべてのミッションでトラックファイルが記録されていることに注意してください。 同じミッションを再フライトすると、先に保存しておかない限り、トラックファイルは上書きされます。ゲームトラックは拡張子が "*.trk "で、デフォルトではC:\Users\<User Name>に保存されています。

SAVE DEBRIEFING. デブリーフィングの内容を保存します。SAVE DEBRIFING Fileウィンドウが表示され、保存先を選択してファイル名を入力し保存します。ゲームのデブリーフィングファイルの拡張子は「*.log」で、デフォルトでは「C:Users\<User Name>Saved Games\DCS\Logs」に保存されます。

FLY AGAIN（再飛行）. このボタンを押すと、ミッションを再飛行することができます。現在のトラックとデブリーフィングは保存しない限り「FLY AGAIN」を押すか、ブリーフィングを終了した時点で失われます。

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CREATE FAST MISSION

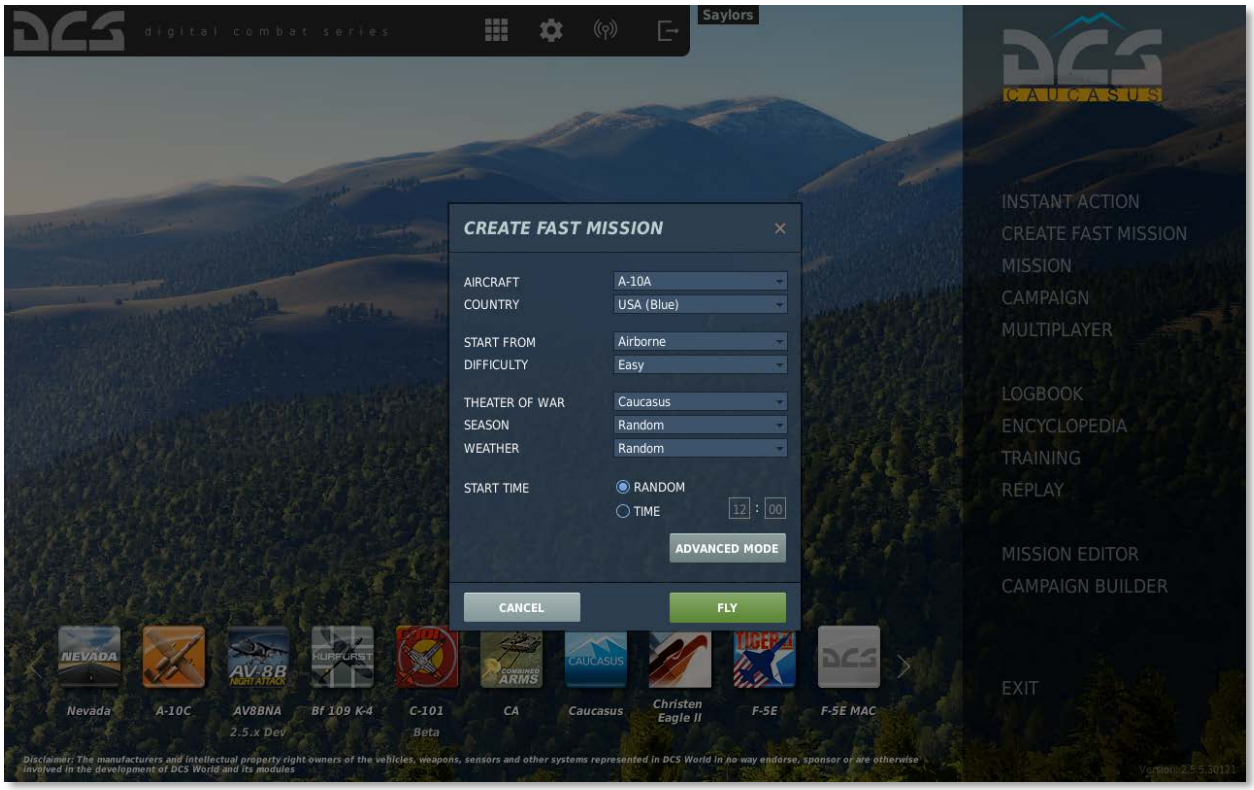
CREATE FAST MISSION （ファストミッション作成）

Create Fast Mission メニューでは、飛行するミッションシナリオを素早く設定することができます。



Create Fast Missionには2つのモードがあります。「シンプル」と「アドバンス」です。メインメニューから最初に開いたときは、シンプルモードが選択されています。

Simple Mode（シンプルモード）



Create Fast Mission (シンプルモード)のメニューには以下の設定があります。

AIRCRAFT. 操縦する機体の選択

COUNTRY. プレイヤー機の国籍

START FROM. どこから始めるかを指定します。

- **Airborne**. 空中スタート
- **From Runway**. 戦場に近い空港の滑走路で離陸可能状態でスタート
- **From Ramp**. 戦場に近い空港の駐機場でコールドスタート
- **Random**. 上記3つの中からランダムでスタート

DIFFICULTY. 難易度を指定します。

- **Easy**. 低い(low)スキルの敵が少数現れます。
- **Medium**. 平均（average)スキルの敵が平均数現れます。
- **Hard**. 高(high)スキルの敵が沢山現れます。

THEATER OF WAR. 戦場（シアター）を選択します。

- **Caucasus**. コーカサス地方でミッションを行います。 (Russia, Georgia, Abkhazia, South Osetia).

SEASON. 季節を指定します。

- **Summer**. 夏
- **Winter**. 冬
- **Fall**. 秋
- **Spring**. 春
- **Random**. ランダムに季節を決定します。

WEATHER. 天候を指定します。

START TIME. 開始時刻を指定します。

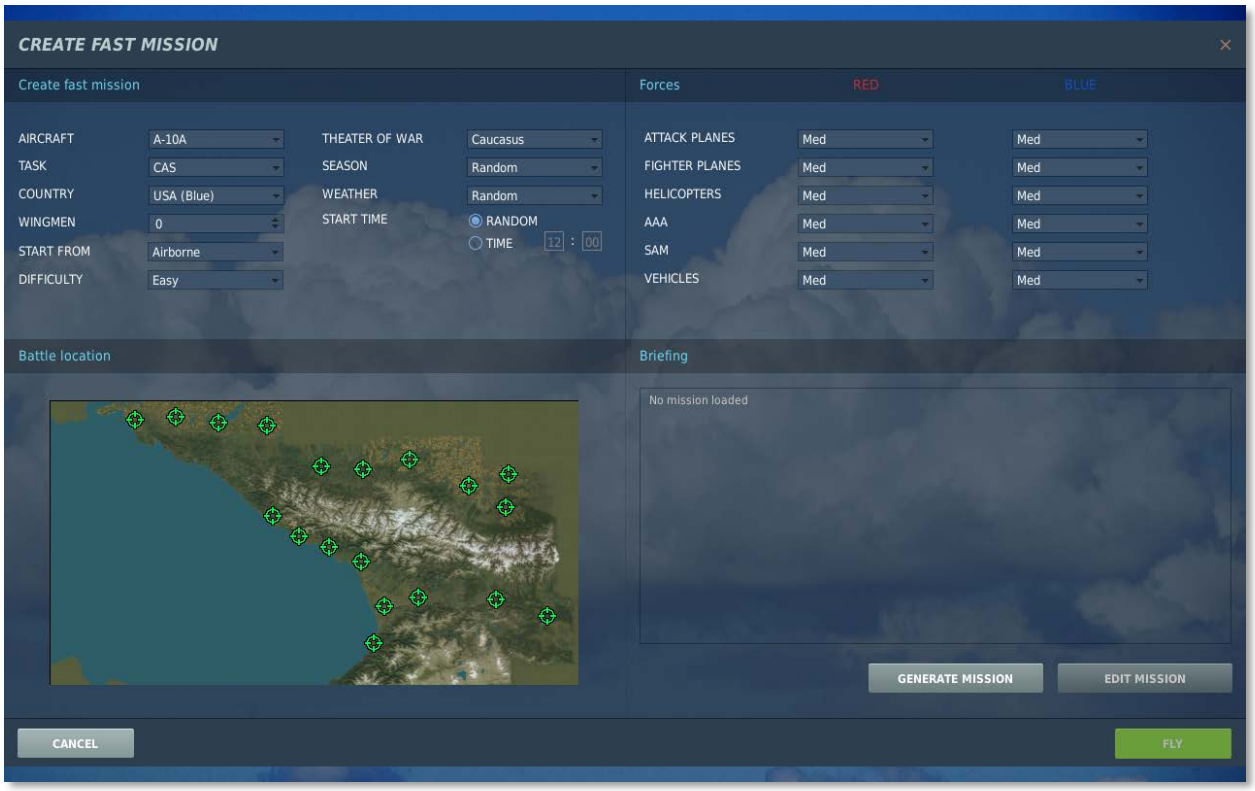
- Random. ランダムに決定します
- Time. 時間を24時間形式（HH:MM）で指定します。

ADVANCED MODE. Fast Create Mission (アドバンスモード) メニューに切り替えます。

FLY. ミッションを作成し、ブリーフィング画面を開きます。

CANCEL. メインメニューに戻ります。

Advanced Mode（アドバンスモード）



Fast Create Mission (アドバンスモード) のメニューには以下ものもがあります。

Create Fast Mission

ここでは、上記のシンプルモードで述べた全ての設定を行います。

Forces （兵力）

このセクションでは、ブルーとレッド陣営の戦力バランスを設定することができます。
これは6つの兵種項目があり、それぞれ5つの兵力レベルから1つを選択します。

- **No**. この種主の兵力を持ちません
- **Min**. 最小レベル
- **Med**. 中程度
- **Max**. 高レベル
- **Random**. Min, Mid or Max内でランダムに決定

各兵種の数は、兵種と設定した兵力レベルに応じて決定されます。

- **ATTACK PLANES**. Su-25やA-10などの攻撃機の数を設定します。
- **FIGHTER PLANES**. Su-27、Mig-29、F-15などの戦闘機の数を設定します。
- **HELICOPTERS**. Ka-50、Mi-24、AH-64などのヘリコプターの数を設定します。
- **AAA**. ZU-23、ZSU-23、バルカンなどの対空砲兵の数を設定します。
- **SAM**. M6、Hawk、Stinger、Igla、Osa、Buk、Strela-1などのSAMユニットの数を設定します。
- **VEHICLES**. 戦車、BMP、BTR、トラックなどの車両数を設定します。

赤軍と青軍の戦闘での戦術的な動きは、このセクションの設定に依存します。

Battle Location （戦闘地域）

このパネルでは、戦闘地域を選択することができます。エリアを指定するには、マウスを使ってクリックするだけです。

Briefing

ミッションが生成されると、ブリーフィングウィンドウに、ミッションのブリーフィングが表示されます。

ブリーフィングテキストエリアの下には、2 つのボタンがあります。

- **GENERATE MISSION**. ミッションを生成
- **EDIT MISSION**. 生成したミッションを微調整できるフルミッションエディタを開きます。

画面のしたには2つのボタンがあります。

- **CANCEL**. メインメニューに戻ります。
- **FLY**. ミッションを開始します。

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LOGBOOK

LOGBOOK （ログブック）

ログブックからは、好きなだけパイロットのプロファイルを作成することができます。 パイロットを作成してミッションを飛行することで、パイロットはミッションの統計情報や勲章を積み重ねていきます。

注意：*パイロットがフライト統計と勲章を記録するためには、ミッションでプレイヤー機が設定されていなければなりません。*
注意：*作成されたパイロットには必ず国を割り当ててください。プレイヤーミッションを作成する際、プレイヤーが飛行する国は使用するパイロットの国と一致していなければなりません。そうでない場合、ミッションの統計情報はログブックに記録されません。*

ログブックを見るには、メインメニューからLOGBOOKをクリックします。

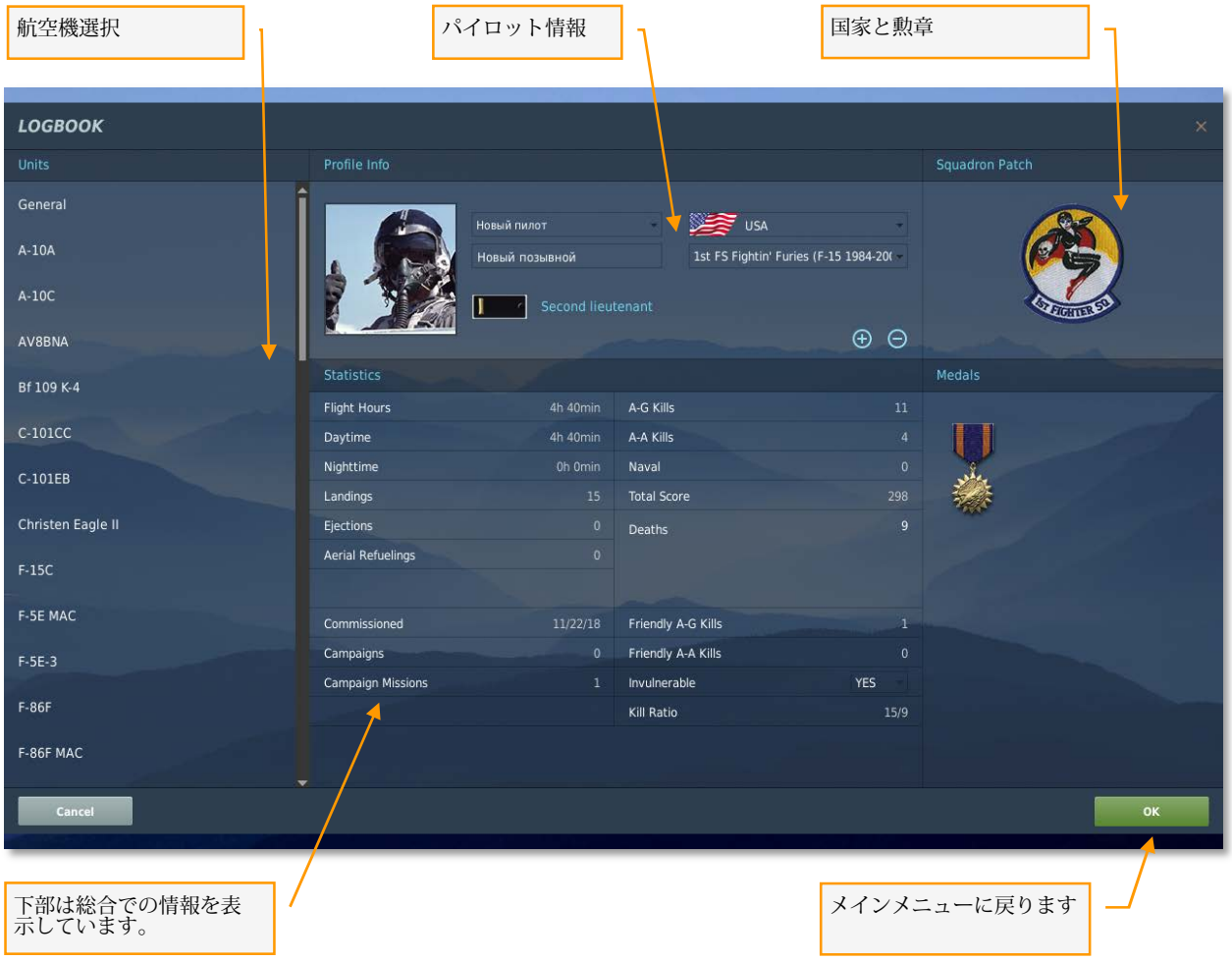


ログブックは３つの部分から構成されています。

1. Units（航空機）
2. Profile Info（パイロット情報）
3. Nation and Awards Page（国家と勲章）

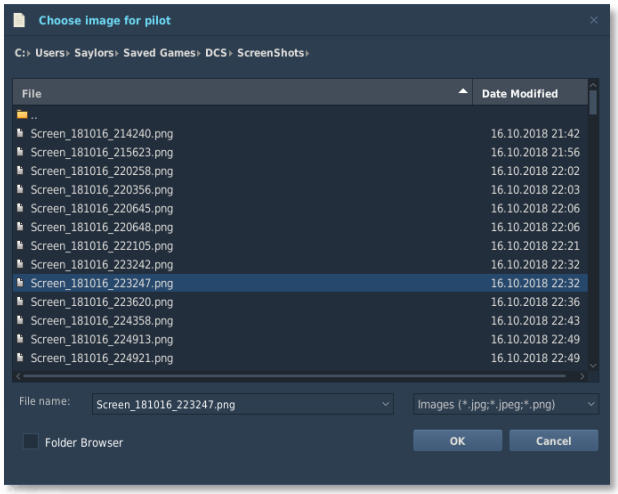
Pilot Page （パイロットページ）

パイロットのプロフィールはウィンドウの中央にあり、個人情報やパイロットの統計情報が表示されます。Generalでは全てを総合した情報が表示され、各機体を選択するとその機体での情報が個別に表示されます。上部には2つの丸いボタンがあります。(+)新規作成と(-)削除です。 新しいパイロットを作成するには、(+)新規ボタンを押してください。 既存のパイロットをログブックから削除するには、名前リストからパイロットを選択し、(-)削除ボタン(-)を押します。



パイロット情報

- **パイロット写真.** 左上には、128x128ピクセルのパイロット写真が表示されています。す。新しい写真を選択するには、画像上でマウスを左クリックして、ブラウザウィンドウを開き、 ご希望の写真（*.jpg; *.jpeg; *.png）を選択し、OKボタンを押してください。



- **NAME（名前）.** 新規パイロットを作成する際には、このフィールドにパイロットの名前を入力してください。 ドロップダウンメニューを使用して、以前に作成された他のパイロットを選択することもできます。
- **CALLSIGN（コールサイン）.** 新規パイロットを作成する際には、ここにパイロットのコールサインを入力してください。
- **RANK（階級）.** パイロットが経験を積むと自動的にランクが上がります。 ランクは、階級名と階級章で表示されます。 経験値は飛行時間と戦闘での実績によって得られます。

パイロットの経歴の統計を航空機ごとに分類表示します。

- **Flight Hours（飛行時間）.** パイロットがコックピットにいた時間の合計。
- **Daytime（日中）.** 日中のフライト時間の合計。
- **Nighttime（夜間）.** 夜間のフライト時間の合計。
- **Landings.** 着陸回数
- **Ejections.** パイロットが脱出した回数
- **Aerial Refuelings.** 空中給油の成功回数
- **A-G Kills.** 対地攻撃によるユニットの総撃破数。
- **A-A Kills.** 対空攻撃によるユニットの総撃墜数
- **Naval.** 艦船ユニットの総撃沈数
- **Total Score.** 任務遂行能力と飛行時間に基づくパイロットの総得点。
- **Deaths.** Invulnerableがチェックされている場合、パイロットが死亡するたびに、この値が増えます。

基本的なパイロットの記録の下には、全航空機、全ミッションで行ったパイロットの経歴がミッションキャリアとして記録されています。

- **Commissioned（キャリア開始）.** パイロットが作成された日時
- **Campaigns（キャンペーン）.** クリアしたキャンペーンの数（勝利と敗北の合計）
- **Campaign Missions.** 行ったキャンペーンミッションの総数
- **Friendly A-G Kills.** 破壊した友軍地上ユニットの総数
- **Friendly A-A Kills.** 撃墜した友軍航空機の総数

パイロットの情報

- **Invulnerable（不死身）.** ミッション中に死亡した場合の処理を決定します。YSE＝死亡しない、NO＝死亡
- **Kill Ratio（キルレシオ）.** この比率は敵ユニットの破壊数とパイロットの死亡数を比較したものです。(敵の破壊数/プレイヤーの死亡数).

Nation, Squadron, and Awards Page

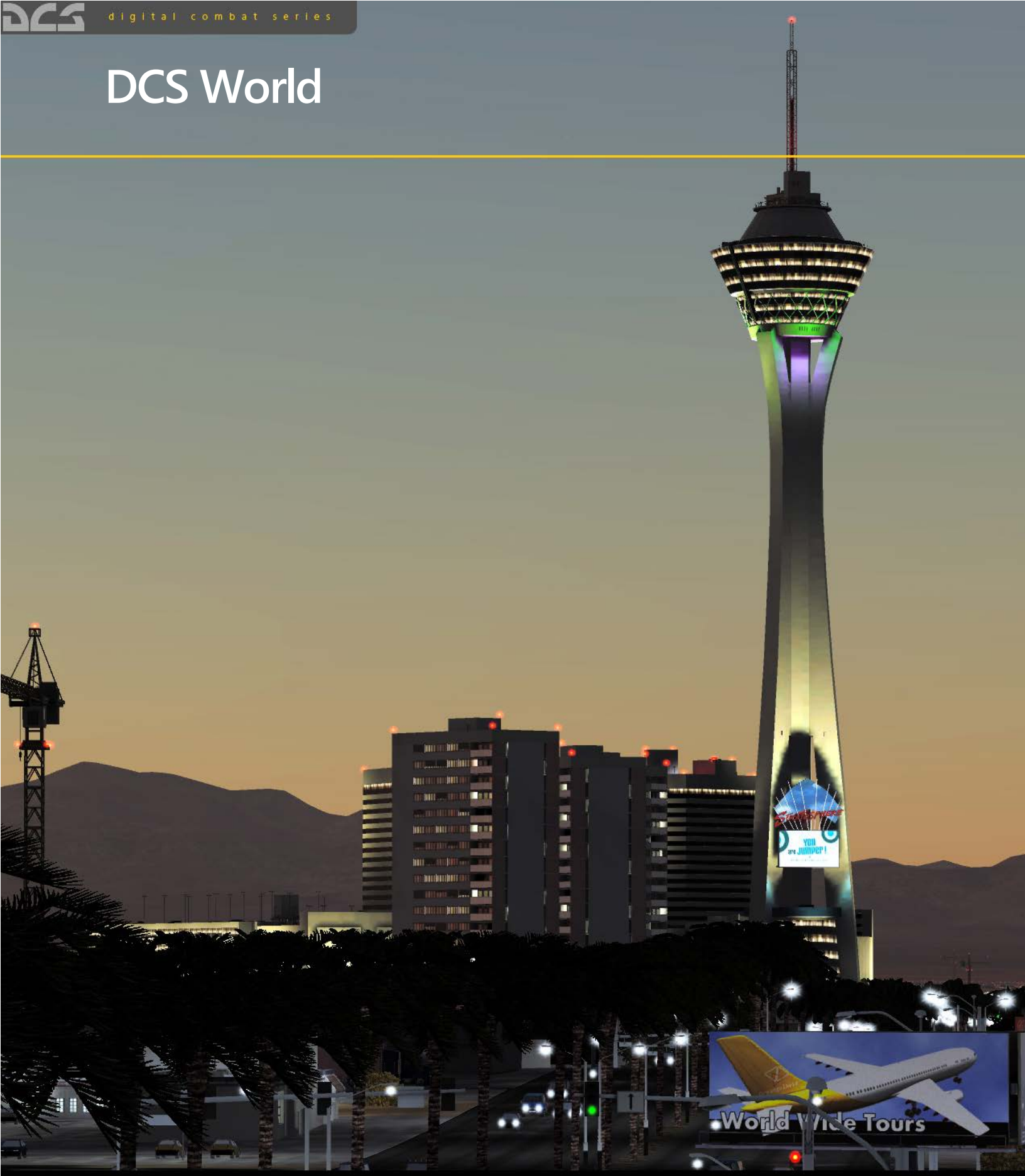
右側には、国家、スコードロン（squadron：中隊）、受賞した勲章が記載されています。

- **Nation.** ドロップダウンリストを使用して、パイロットの所属国を選択します。
- **Squadron.** ドロップダウンリストを使用して、パイロットの所属する軍の支部または中隊を選択します。 これらは選択した国によって異なります。
- **Awards.** パイロットが獲得したメダルが表示されます。

画面の一番下には、ログブックへの変更を保存せずにメインメニューに戻るCANCEL ボタンと変更を保存してメインメニューに戻るOKボタンがあります。
また、画面右上の黄色のXを押してログブックを終了することもできます。

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REPLAY

REPLAY(リプレイ)

ミッションをプレイするたびに、そのミッションのリプレイファイルが自動的に.trk（トラック）ファイルとして記録されます。 ミッションのデブリーフィング時に、リプレイをデフォルトとは別の名前で保存することができます。 ただし、これを行わずに同じミッションを再度プレイすると、元のリプレイは上書きされてしまいます。

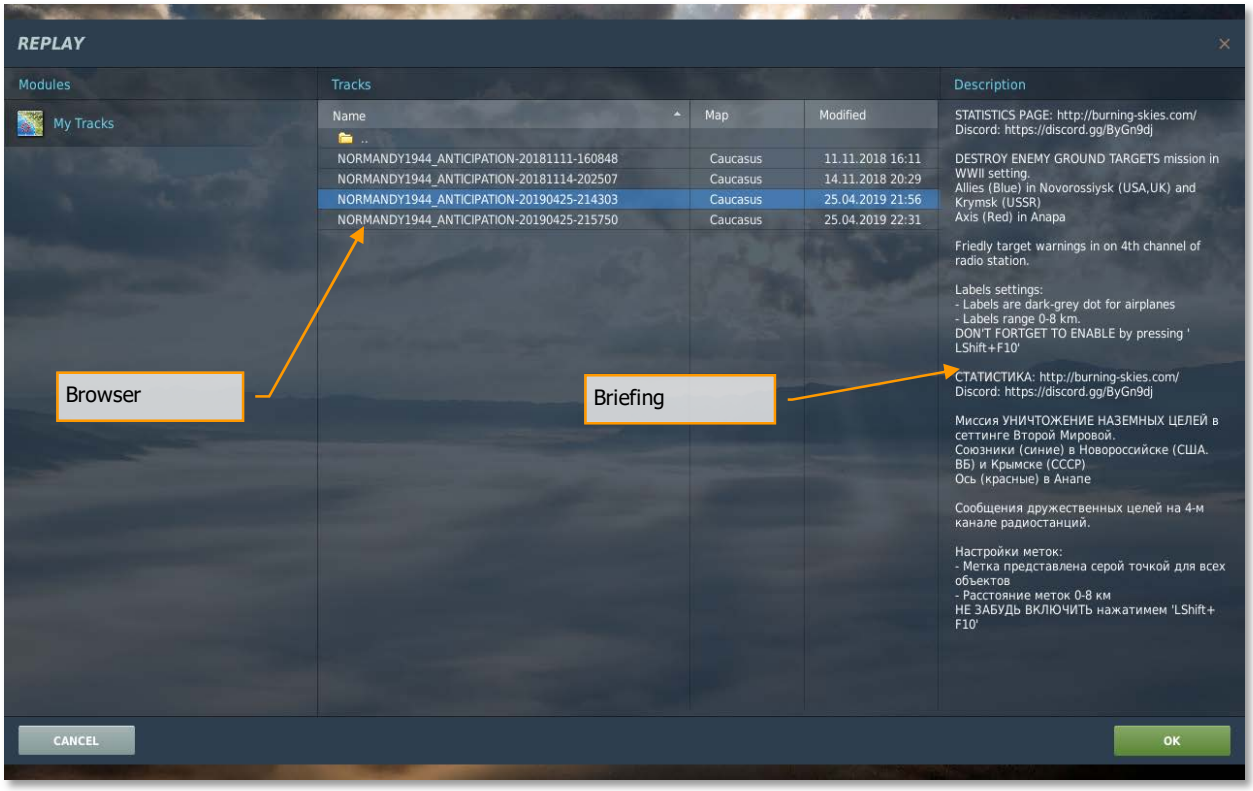
リプレイファイルを選択するには、メインメニューページからREPLAYを選択します。
MISSIONページやミッションエディタからはリプレイを見ることができません。



リプレイページを開くと、ミッションを開くページと非常に似たページが表示されます。 このページでは、ローカルディスクをブラウズし、保存されたリプレイファイルを選択して読み込むことができます（トラックファイルの拡張子は*.trkです）。
これを行うには、上部のドロップダウンリストで必要なドライブレターを選択し、トラックのあるフォルダに移動します。フォルダはダブルクリックで開き、上の行の「...」をダブルクリックすることで上の階層に戻ります。デフォルトでは、C:\Users\<ユーザー名>\Saved GamesDCS\Tracksフォルダが開きます。

ウィンドウの右側にはDESCRIPTIONがあり、トラックが作られたミッションの簡単な説明が書かれています。

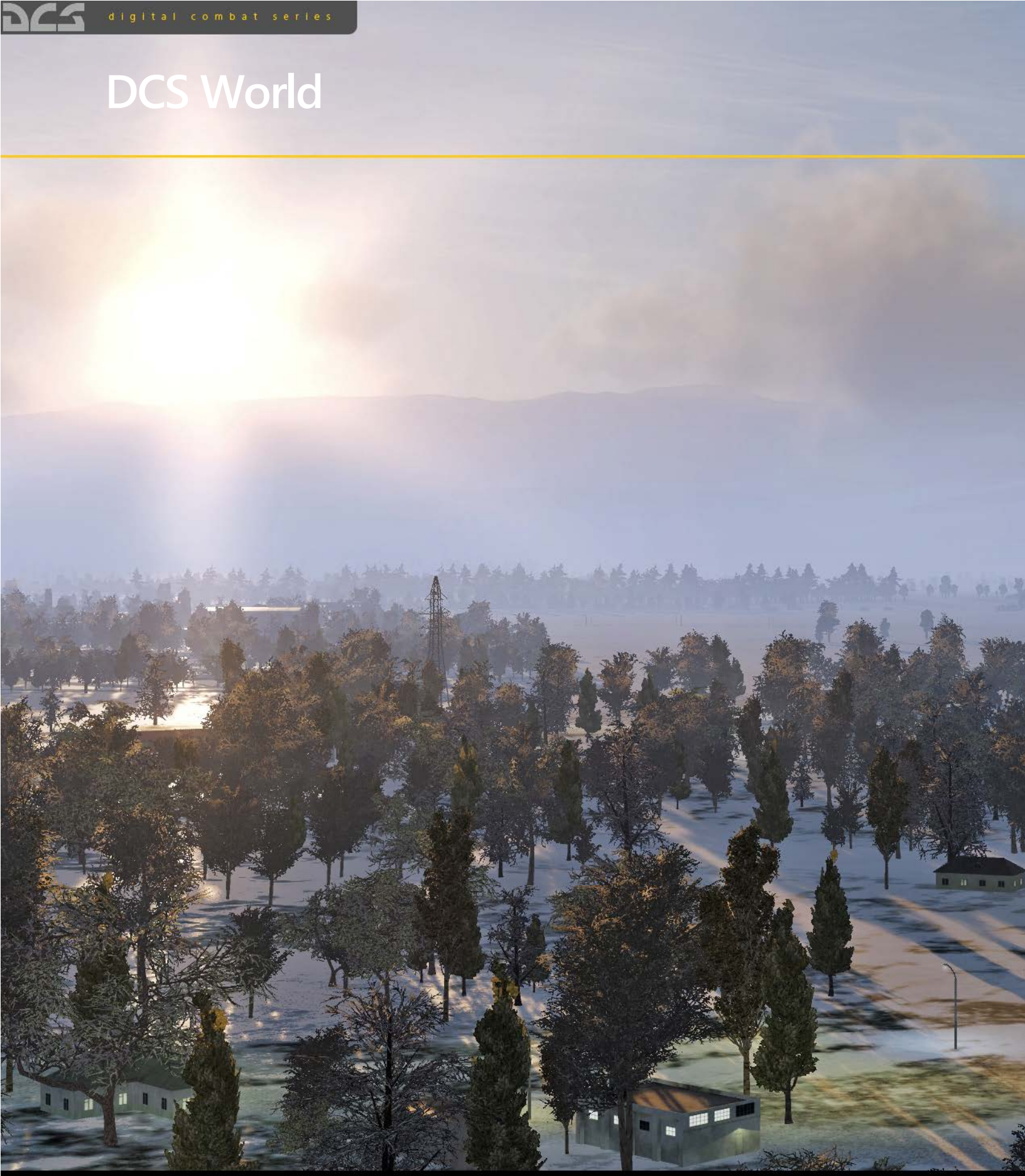
ウィンドウの下部には２つのボタンがあります。選択したトラックを再生するには OKを、メインウィンドウに戻るには CANCELを押します。



また、右上のXボタンを押すことで、ミッションをロードせずにウィンドウを終了することもできます。
ミッションを飛んだ後、Debriefingメニューからトラックを保存すると、デフォルトのディレクトリに保存されます。C:\Users\<ユーザー名>\Saved Games\Tracks.

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MISSION EDITOR

MISSION EDITOR

The Mission Editor (ME) of allows you to create stand-alone missions, campaign missions, training missions, and multiplayer missions. The ME consists of the following primary elements:

1. Interactive mapping system
2. Unit placement tools
3. Weather editor
4. File management system
5. Goal creation tool
6. Trigger system tool
7. Additional specialized panels

The mission editor is launched by pressing the MISSION EDITOR button in the main window.

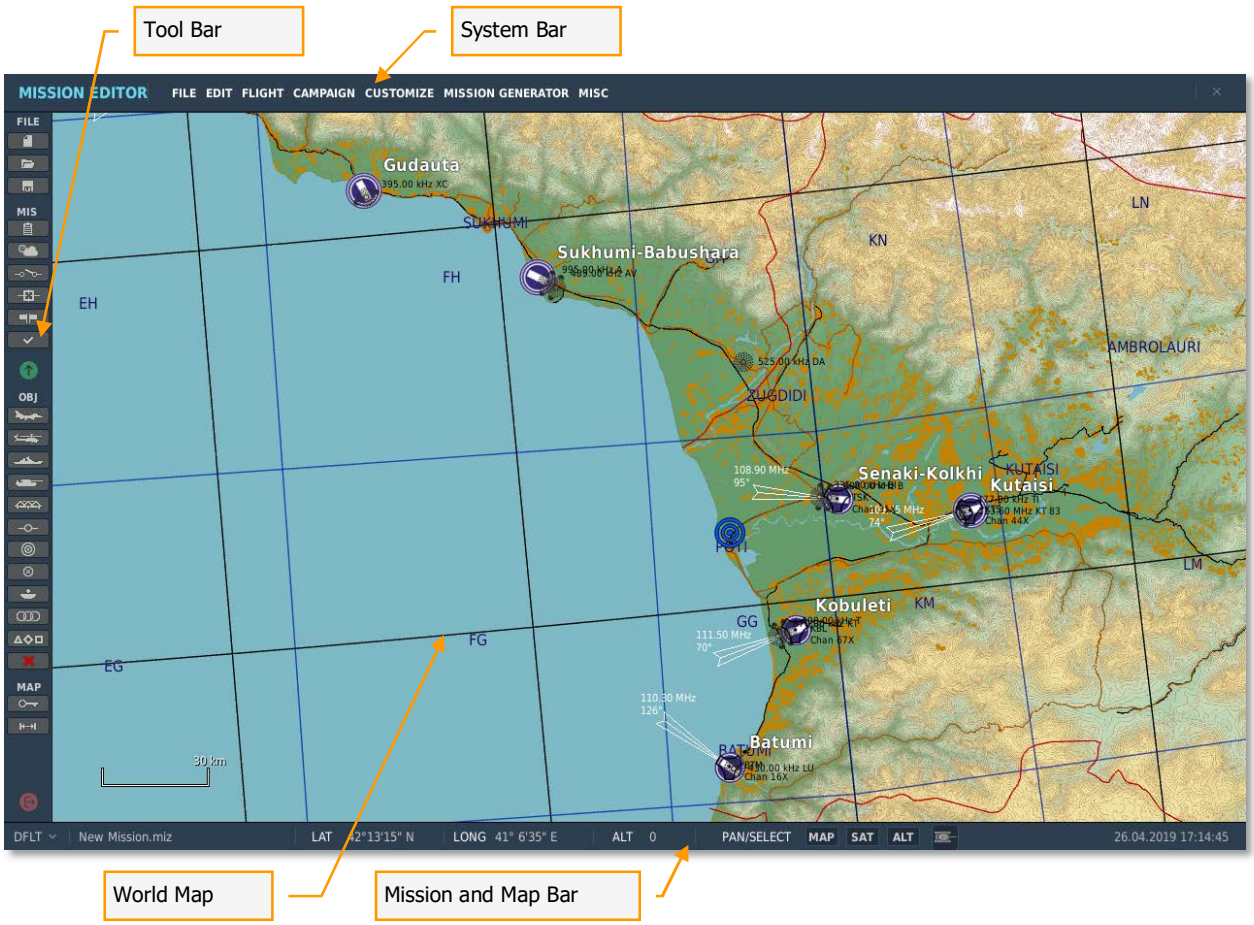


The Mission Editor Map and Navigation

The ME is divided into four primary areas:

- **World Map**. This area of the screen forms the majority of the screen and displays the topographic map, units, routes, and other mission elements.
- **Mission and Map Bar**. Along the bottom of the screen is the Mission and Map Bar and this provides you information on cursor location on the World Map as well as mission name and current time.
- **System Bar**. Along the top of the screen is the System Bar, and from here you control file management, access the campaign editor, the encyclopedia, credits, enable track recording, and duplicate several functions from the Tool Bar.
- **Tool Bar.** The Tool Bar, along the left side of the screen, provides you a large assortment of tools that allow you to create a mission and modify what is displayed on the World Map.

Each of these areas will be discussed in more detail below.



The World Map

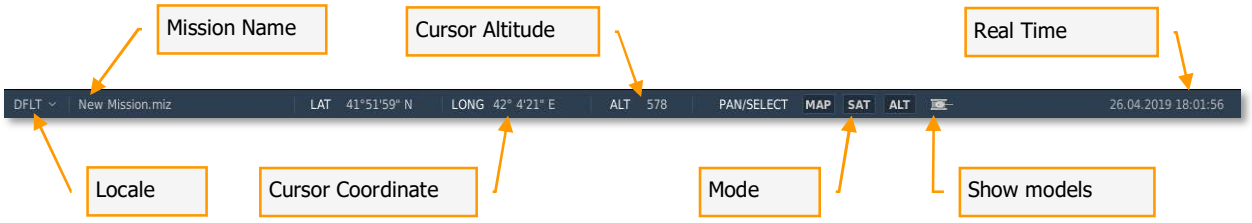
The World Map consists of the Black Sea region with detailed terrain along the eastern edge. While other areas of the map are represented, they are not as fully detailed as the eastern areas. Using the Options menu (described later), you may filter the data that is displayed on the map such as urban areas, rivers, roads, etc.

Navigating the Map

A right mouse button hold while moving the mouse will pan the view and rotating the mouse wheel controls the zoom level. Selecting an object or unit is done by clicking with the left mouse button. To zoom in on a specific location on the map, place the cursor over the desired location and rotate the mouse wheel forward.

The Mission and Map Bar

Located along the bottom of the screen is the Mission and Map Bar. This is an informative bar that tells you the name of the loaded mission, cursor position and altitude within the World Map, map mode, and the current time according to your Windows settings (not mission time).



Note that the coordinates and altitude are controlled according to the position of your mouse on the World Map. These numbers will change dynamically as your move your mouse and correspond to the real-life coordinates and altitude of the topography. This can be a useful tool when trying to determine the coordinates of an objective for a mission briefing. Cursor displayed altitude may be set to either feet or meters, depending on your Options/Gameplay/Units setting. The coordinate display can be toggled between Lat/Long, Lat/Long Decimal and MGRS, depending on your Options/Misc/Coordinate Display setting.

Note that "New Mission" will appear in the Mission Name field until you save a newly created mission.

The System Bar

Located along the top of the screen is the System Bar. The System Bar consists of several pull down menus. These are: FILE, EDIT, FLIGHT, CAMPAIGN, CUSTOMIZE, MISSION GENERATOR and MISC. To select one of these pull down menus, place your mouse over the text and left mouse click.

The System Bar pull down menus provide the following functions:

FILE

The File functions provide you basic file management of missions. These include:

1. **NEW**.

   Selecting New will allow you to exit the current mission loaded and start a new one from scratch. Before proceeding though, a consent message box is provided to you so that you do not accidentally lose a mission you are working on. After creating a new mission, you will be displayed the THEATER OF WAR and COALITIONS window.



   This window allows you to define a map and what countries (factions) will be assigned to one of the two sides. Unassigned countries will not take part in the mission. In the left side window all unassigned factions are listed; in the top right window the RED side factions are listed; and in the lower right window all BLUE side factions are listed. To move a faction, left click on the desired faction and then press either the left or right

arrow button to move it between the unassigned window and a side window or press the up and down arrows to move the faction to the other side directly.

At the bottom of the window are three buttons. The CLEAN COALITIONS button allows you to reset the current faction distribution to the default list. The SAVE button allows you to save the current faction distribution and will be set as the default. The OK button will use the current distribution for the mission but will not save it as the default.

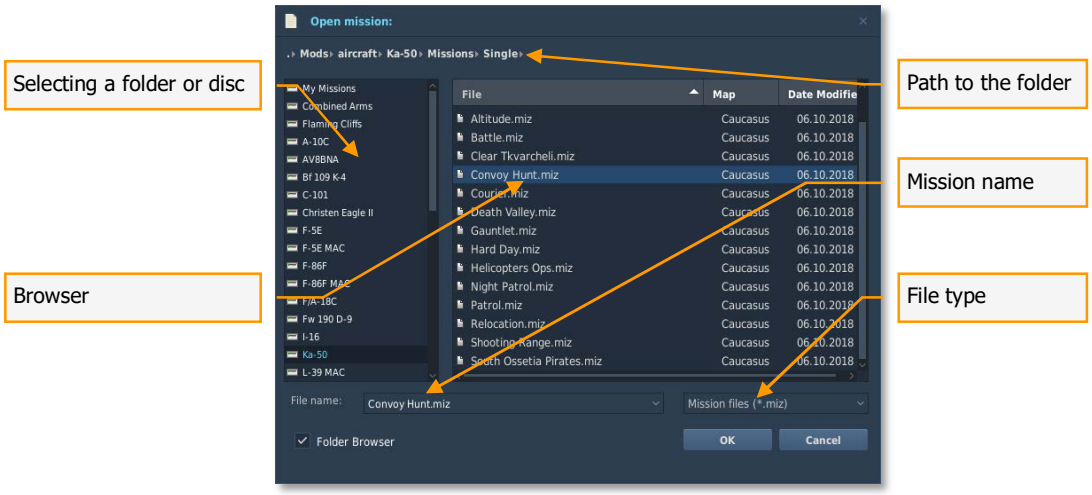
In the COALITION PRESETS field, you can select presets that can be changed at will:

- **Custom** – add countries by coalitions manually.
- **Modern** – based on modern realities.
- **WWII** – based on the layout of forces for the period of WWII.

***Note**: When creating missions, it is up to you to decide who will be allied with whom. You are free to create realistic alliances or make very fanciful ones. We leave it to you to decide.*

2. **OPEN**.

Selecting Open will display the Open Mission window. This window allows you to browse your dedicated mod folders or local disk(s) and select and load saved missions.



Using standard Windows file browser functionality, you may select the DRIVE you wish to search using the DRIVE field in the top right of the window and then select the desired mission from the folder/file listing. Once you select a mission file, the path to that file is displayed in the PATH field and the name of the mission file is displayed in the FILE field. Note that mission files are assigned a *.miz extension.

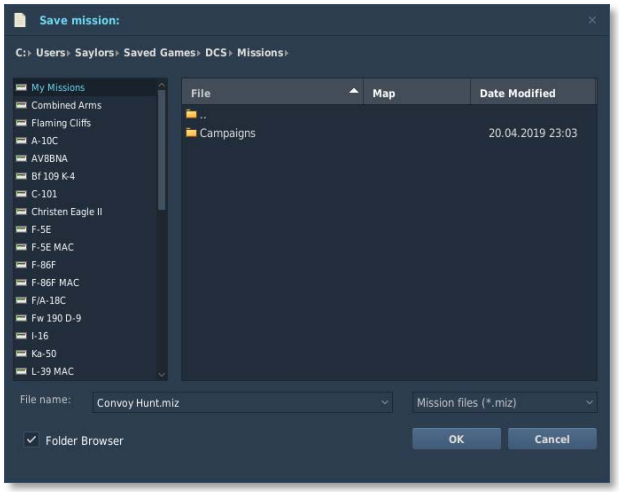
Once you have selected the desired mission file, press the OK button at the bottom of the window to load it. You may also exit the window without loading a mission or track file by pressing the CANCEL button at the bottom of the window or the X button in the top right corner.

3. **SAVE**.

   The SAVE button saves the mission file in the current directory while keeping the current filename. This will not allow you to change the filename or its directory.

4. **SAVE AS**.

   The Save As function operates much like the Open Mission window but it allows you save a mission file (*.miz or *.trk) and give it a unique name. The primary difference is that you may overwrite the name in the FILE NAME field as you wish and save the file under that name. To do so, erase the current name and type in the name you wish. You may then use the browser functions to determine where you wish to save the file. Once complete, you must press the OK button to complete the save.



5. **EXIT**. Pressing Exit will close the ME and return you to the Main Menu.

EDIT

The EDIT pull down provides you a second means of accessing the most important mission construction tools: ADD AIRPLANE, ADD HELICOPTER, ADD SHIP, ADD VEHICLE, ADD STATIC, ADD TEMPLATE, REMOVE, LOAD STATIC TEMPLATE and SAVE STATIC TEMPLATE. These tools are duplicated on the Tool Bar and we will discuss them there in detail.

FLIGHT

The Flight menu allows you start a loaded mission to play or start the mission to setup the player's aircraft avionics in PREPARE MISSION mode. This menu also allows you to convert a recorded (track) mission into an AVI video file and access the REPLAY menu for track playback.

1. **FLY MISSION**. Selecting this option will close the ME, load the mission into the simulation, and then start the simulation. The loading time of a mission will vary according to the number of units in the mission, the amount of scripting, and how much RAM your computer has.

2. **PREPARE MISSION**. The PREPARE MISSION mode is used for integrating navigation and environment data into the saved mission file. When near the end of creating a mission, you can start the mission in the PREPARE MISSION mode (it will look much like usual FLY MISSION start) and configure various avionics in the cockpit. Your avionics settings can then be saved and will start in the saved states whenever other players open the mission.

   The following avionics settings can be saved using the PREPARE MISSION function:

Ka-50

- Satellites – Mission name\Scripts\World\GPS_GNSS.lua,
- Additional ABRIS data – Mission name\ABRIS\Database\ ADDITIONAL.lua,
- Navigation database – Mission name\ABRIS\Database\NAVIGATION.lua,
- ABRIS routes – Mission name\ABRIS\Database\ROUTES.lua.

To save custom settings for the Satellites, edit the GPS_GNSS.lua file prior to running the Mission Prepare function. The GPS_GNSS.lua file can be found in \Scripts\World\GPS_GNSS.lua.

A-10C

- CDU. All setting except a mission editor flight plan that can't be changed.
- IFFCC. The block of test menu.
- MFCD. Lower OSB buttons functionality.
- SADL. Network user ID, callname etc. (NET page)
- TAD. Profiles.
- TGP. Laser codes, integration, calibration method, gain settings, TAAF altitude.
- DSMS. Mission Control Page settings. Weapon profiles.

Once your avionics are setup as desired for the mission, exit the mission and press the SAVE button in the ME once again to save all of the prepared data in the mission file.

***NOTE for mission makers**:*

If you use prepare mission mode, please indicate in the briefing that the player should not use the mission planner. If he'll change the route or weapons for the mission, the avionics scripts saved in the mission file using prepare mission will conflict with the new settings.

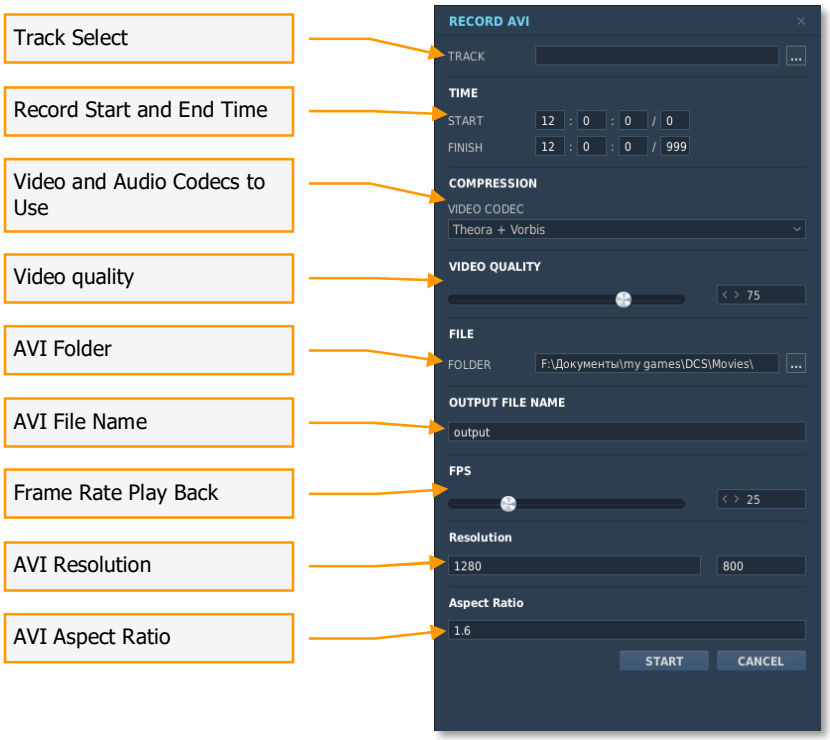
If you want to give the player the freedom of action to the use the mission planner, do not use mission prepare mode.

3. **RECORD AVI**.

   This option allows you to convert a Track file (which can only be viewed in the game) to an AVI video file that can be viewed by anyone using video playback software.

Once a Track file has been created, you may load the file into this tool and then set the video quality of the AVI video output. A few notes:

- Because the AVI tools render the Track file frame by frame and NOT in real-time, you can create an AVI with higher or lower frame rates than when the track was originally recorded.
- The higher the video quality and longer the recording, the longer it will take to render the AVI. Setting the FPS slider too high will also lengthen the time it takes. Please note that rendering long, high-quality AVIs can be a very time consuming process.
- Once the AVI is created, you may then play it back outside of the game or convert and compress it with other codecs.
- Track selection. Pressing the button labelled "…" brings up the track selection panel. Select the desired track and click OK.
- Using the START and FINISH fields, you can create an AVI out of just one part of a larger Track file.



- Video and Audio Codecs to Use. From the dropdown menu that appears, select a preset for the video and audio codecs to be used in rendering, as well as the quality (in percentage values).
- AVI Folder. Set the file save path.

- AVI File Name. Sets the file name.
- FPS. Use this slider select the desired number of frames per second for rendering video.
- Resolution. Use these two fields to set the horizontal and vertical resolution of the output video. The size of the video file is highly dependent on these parameters.
- Aspect Ratio. The aspect ratio of the output video is automatically calculated in this field.

To begin rendering click START. To stop rendering click CANCEL.

4. **REPLAY**. This command brings up the window of viewing previously recorded tracks.

CAMPAIGN BUILDER

The Campaign Builder is what we term a Staged Campaign System (SCS). We will describe the functionality of the Campaign Builder in detail in the Campaign Builder chapter.

CUSTOMIZE

The Customize menu allows you to configure a number of map display options and set gameplay options for the mission.

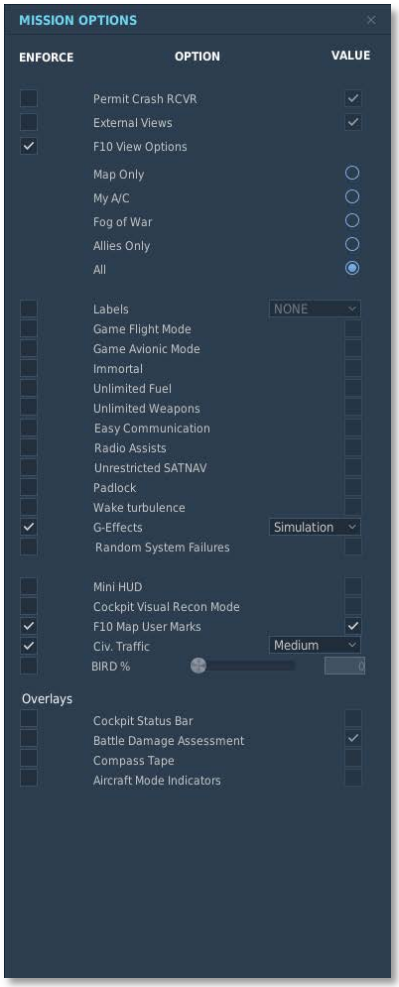
1. **MISSION OPTIONS**.

   The Mission Options menu can be used to lock certain gameplay options for the particular mission loaded. When the "USE THESE OPTIONS FOR ALL MISSIONS" check-box is unchecked in the Options\Gameplay menu, the locked mission options will override the settings in the main Options. However, when the "USE THESE OPTIONS FOR ALL MISSIONS" check-box is checked, the settings in the main Options screen will take priority.

   Along the left side of the window are the ENFORCE buttons. Checking an ENFORCE button will force the settings of this option on the player of the mission.

   Along the right side of the window are the VALUE buttons. VALUE buttons operate as on/off variables. When checked, the options will be forced ON. When unchecked, the option will be forced OFF.

   From the left side of the Mission Options window, you may select the following mission options to enforce:



- **Permit Crash RCVR**. If your aircraft is destroyed during a mission or you eject, you may respawn in an undamaged aircraft.
- **External Views**. When Enabled, external views are allowed.
- **F10 View Options**. When enabled, you may select one of the mutually exclusive radio buttons:
- **Map Only**. Map only view. No aircraft, vehicles or ships are displayed.
- **My A/C**. Only your aircraft.
- **Fog of War**. Fog of War hides all enemy activity in uncharted sections of the map.
- **Allies Only**. Only your aircraft and allied units.
- **All**. All units from both sides are displayed.

- **Labels**. When enabled, you can display easy-to-spot labels about friendly and enemy units. Close units will display the unit type name and the range from you.

Units at medium range will only show the range to unit. Units at long range will only be marked with a tic-mark.

- **Game Flight Mode**. Easy flight makes the aircraft much easier to fly when enabled and more analogous to an arcade game.
- **Game Avionic Mode**. Game Avionics Mode helps with flight orientation by displaying 3D markers in space.
- **Immortal**. When enabled your aircraft cannot be destroyed or even damaged.
- **Unlimited Fuel**. When enabled, fuel will not deplete from your aircraft. Note that you will always have 100% fuel when this is selected.
- **Unlimited Weapons**. When enabled, expended weapons will automatically be replenished.
- **Easy Communication**. Enables the simplified communication system for player aircraft.
- **Radio Assists**. When enabled, you are provided voice cues regarding approaching threats, when you fall within valid weapon launch parameters, and launch warnings.
- **Unrestricted SATNAV**. Satellite navigation available for all capable aircraft, regardless of country and date/time.
- **Padlock**. When enabled, you press the padlock enable key to keep your eyes on the vehicle or ground point in the center of your view.
- **Tool Tips**. When enabled and when you hover your mouse over a control in the cockpit (button, switch, dial, or lever), you will be shown a brief tool tip of the function of the control.
- **Wake turbulence**. Enables calculations of disturbance in the atmosphere that forms behind an aircraft as it passes through the air. Gives new feelings when air-refueling, flying in formations or sitting on a bandit`s six.
- **G-Effect**. When enabled, you may select the level of G-effect. Possible settings include: NONE, GAME, and SIMULATION.
- **Random System Failures**. Enables random system failures for the player unit/aircraft.

- **Mini HUD**. If flying Flaming Cliffs 3 aircraft, this options allows you to enable/disable the Mini HUD that provides basic HUD symbols to always be displayed regardless of where the view is directed.
- **Cockpit Visual Recon Mode**. Enables a possibility to mark any ground object on terrain and reveal it's location on the maps of allies. Use RMB display a cross and LMB select object.
- **F10 Map User Marks**. Gives opportunity to place and edit names of user marks in mission editor map.

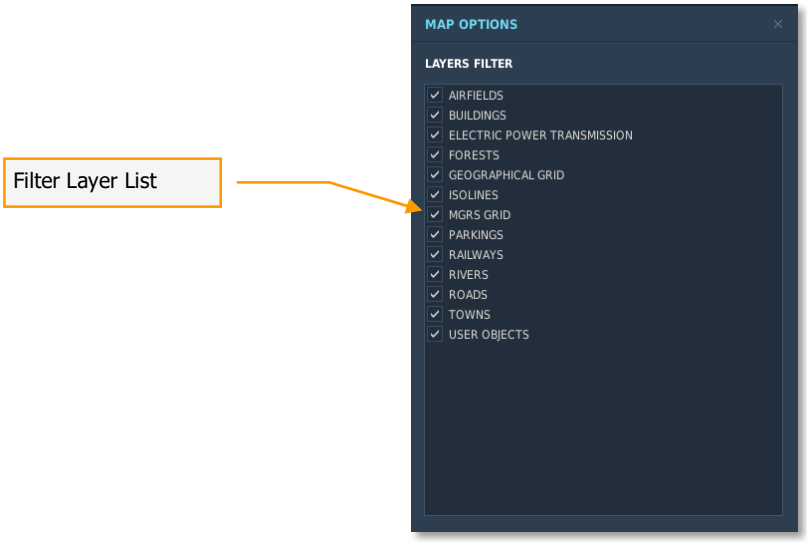
- **Civ Traffic**. When enabled, the world can be populated by moving cars, trucks and trains throughout the detailed world area. If enabled, possible settings include OFF, LOW, MEDIUM, and HIGH.
- **BIRD %**. This adjust the probability of a bird strikes when flying at altitude below 200 meters. 100% is the realistic bird strike probability setting. 0% - no birds. 1000% - the bird strike probability multiplier 10 times.

Overlays

- **Cockpit Status Bar**. Enables/disables a status bar that displays the current flight parameters and coordinates.
- **Battle Damage Assessment**. Shows/hides the damage done to an enemy unit in percentage values when a player scores a hit. Information about damage dealt and target destruction is displayed along the left border of the screen in real time. Destroyed targets are reported in red, while damaged targets are reported in yellow along with a percentage value of damage dealt.
- **Compass Tape**. Displays/hides a compass tape at the top of the screen.
- **Aircraft Mode Indicators**. Enables/Disables an overlay on the left side of the screen that displays information on the aircraft's systems - flightpath, combat modes, selected weaponry, and countermeasure systems (flare/chaff).

2. MAP OPTIONS

The Map Options selection from Customization allows you to filter the information that is displayed on the World Map as layers.



The majority of this window consists of the map filter window. Each item on the list has a check box, that when checked, displays the map data on the ME World Map. Filter items include:

- **AIRFIELDS**. Airport icons that orient in the correct runway direction.
- **BUILDINGS**. Individual buildings that are visible at low scales. This is OFF by default.
- **ELECTRIC POWER TRANSMISSION**. High tension power line towers and cables.
- **FORESTS**. Large stands of trees. Note the trees in and around urban areas are not indicated.
- **GEOGRAPHICAL GRID.** Latitude and longitude lines.
- **ISOLINES**. Continuous lines around terrain relief to indicate elevation change.
- **MGRS GRID**. Display MGRS coordinate grid over the map.
- **PARKINGS**. Aircraft parking at airfields.
- **RAILWAYS**. Display the railway system.
- **RIVERS**. Small rivers and streams.
- **ROADS**. Small and major road network.
- **TOWNS**. Urban areas are indicated as orange shaded areas and when at small map scales, individual buildings are indicated.
- **USER OBJECTS**. Any objects placed on the map by the user.

Airport Icons

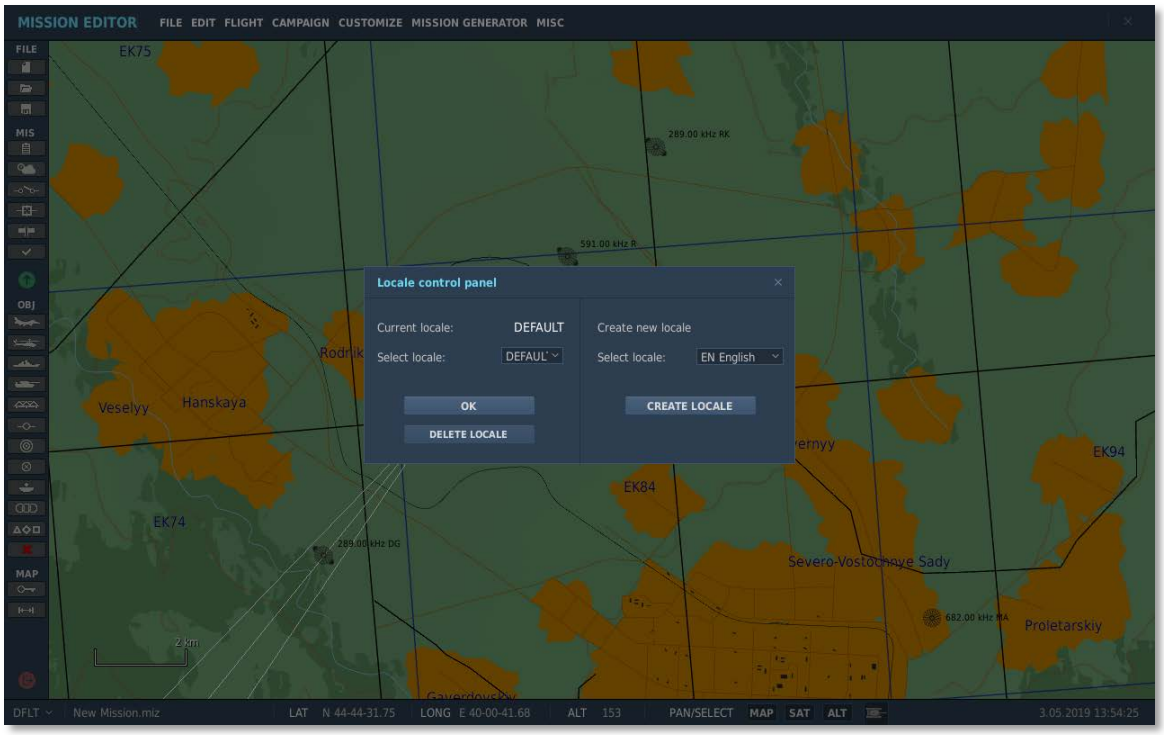
	Helipad. The temporary place of deployment of Army aviation.
	Field airdrome (not present in game)
	General aviation airfield (not present in game)
	Third class airdrome. 1200…1700 meters runway (not present in game)
	Second class airdrome. 1800…2400 meter runway.
	First class airdrome. 2500…3000 meter runway.

To close the window, click the X button in the top right corner of the window.

3. **LOCALE PANEL**.

   The panel allows you manage mission localization. If other localization versions are required, the mission designer can select a new localization from the drop-down list of languages found on the right side of the panel. Clicking the create button will add this

language to the left side of the panel, where it will be available for selection and installation in the mission. The selected language can be changed in the lower left corner of the mission editor window.



4. **SET POSITION**.

   Set the geographical coordinates of the current theater of operations in this panel.

MISSION GENERATOR

The Mission Generator selection opens the Mission Generator menu. The Mission Generator is a powerful tool, which can be used to create battle scenarios throughout the entire map. The Mission Generator creates missions for the Create Fast Mission application. Using the Mission Generator menu of the mission editor, you can edit or create new possible mission generation options for Create Fast Mission. The Mission Generator pull-down menu includes the following four items: GENERATE, NODES, TEMPLATES, and SAVE.

1. **GENERATE**… . Opens the Fast Create Mission (Advanced Mode) menu to generate a mission.

2. **NODES**. Opens the Mission Nodes panel and displays all default node templates on the map. This menu can be used to create new nodes or edit/remove default ones. Nodes define where blue and red forces can be spawned on the map.
3. **TEMPLATES**. Opens the Templates Editor panel. This menu can be used to create new templates or edit/remove default ones. Templates are used to edit the unit types, numbers, and formations used when populating nodes.
4. **SAVE**. Clicking SAVE will save the settings of all nodes and templates.

MISC

The Misc customization consists of two items: the Encyclopedia which is described further in this manual, and the game credits.

Resource Manager

The Resource Manager (RM) is Mission Editor a system that allows you to set a determined quantity of aircraft, fuel and equipment at each airbase warehouse and independent warehouse. Each warehouse (airbase or independent) can resupply other warehouses with defined speed, periodicity and size.

The RM is a very useful tool for single player and multiplayer missions in which the mission designer can limit the available aircraft, fuel and weapons to the player(s). Thereby adding a level of strategic thinking of how to best use limited resources.

Warehouse in Airport

Each airbase consumes resources from its own unique resource stockpile called a warehouse; these resources can be replenished from other airbases and warehouses.

By default, airbases are neutral and have infinite resources, meaning that each airbase can produce an infinite number of all types of aircraft, fuel and weapons.

The airbase settings panel can be opened in the mission editor by left-clicking on the zone of the airbase indicated by a circle.



- **COALITION** – airbase coalition affiliation: Neutral, Blue, Red.
- **NAME** – airbase name.
- **TOWER FREQUENCY** – the radio frequency used by the airbase's flight controller.

When you left mouse click on any airbase, you have two RM related items:

FULL INFO. Clicking this button will display the RM window for that airbase/warehouse.

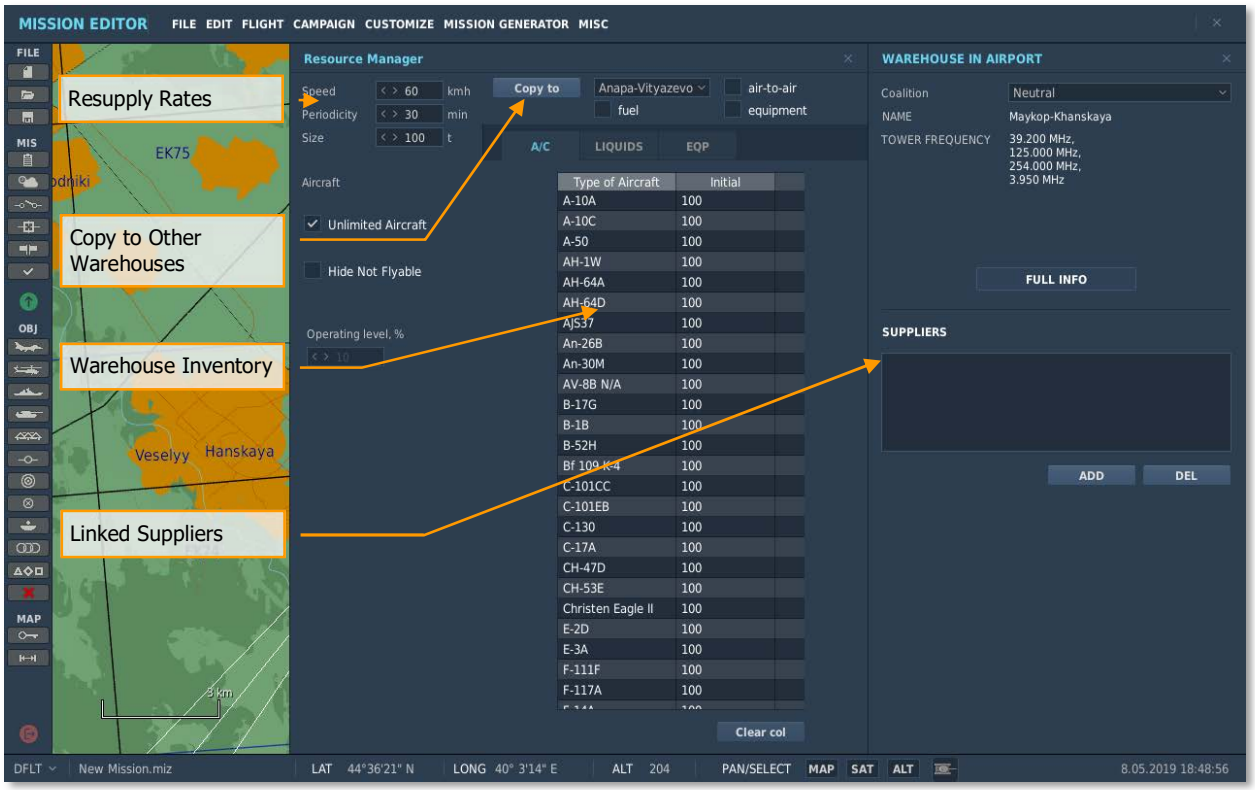
SUPPLIERS. Each warehouse can be linked to other warehouses, thus creating supply chains. This window displays all warehouses that the selected warehouse is supplying.

ADD, DEL – adds or deletes warehouses associated with airbases.

Airbases will only work with aircraft belonging to their own coalition. When planning a mission, the mission creator must first set the coalition allegiance for all airbases involved in the mission.

Resource Manager, Full Info

After clicking the Full Info button, the full display window for managing resources from the selected warehouse is displayed. Let's take a look at some of the options:



Supply Rates

In the top left of the window are three items that allow you to set how fast the warehouse can supply other linked warehouses. These include:

- **Speed** – This value determines the road speed by which virtual supply convoys travel from the selected airbase / warehouse to the warehouse being supplied.
- **Periodicity** – This value, in minutes, determines how often a supply convoy will be sent to the selected airbase / warehouse. This happens only after a supply request has been made. If the linked warehouse is the requester, that warehouse will be supplying with the set periodicity. If linked warehouse is not the requester, it does not supply.
- **Size** – This value, in tons, determines the amount of supplies that will be delivered from the supplier warehouses with each delivery.
- **Copy to** – copies the selected resource manager configuration for the categories marked by checkboxes (AIRCRAFT, FUEL, EQUIPMENT.) to another air base selected from the drop-down list.

- **Operating level, %** – When the resource is reduced to the set level, the nearest supplier will initiate a supply delivery. When the resource is reduced to the set level, the warehouse will send the request to upper-linked warehouse.

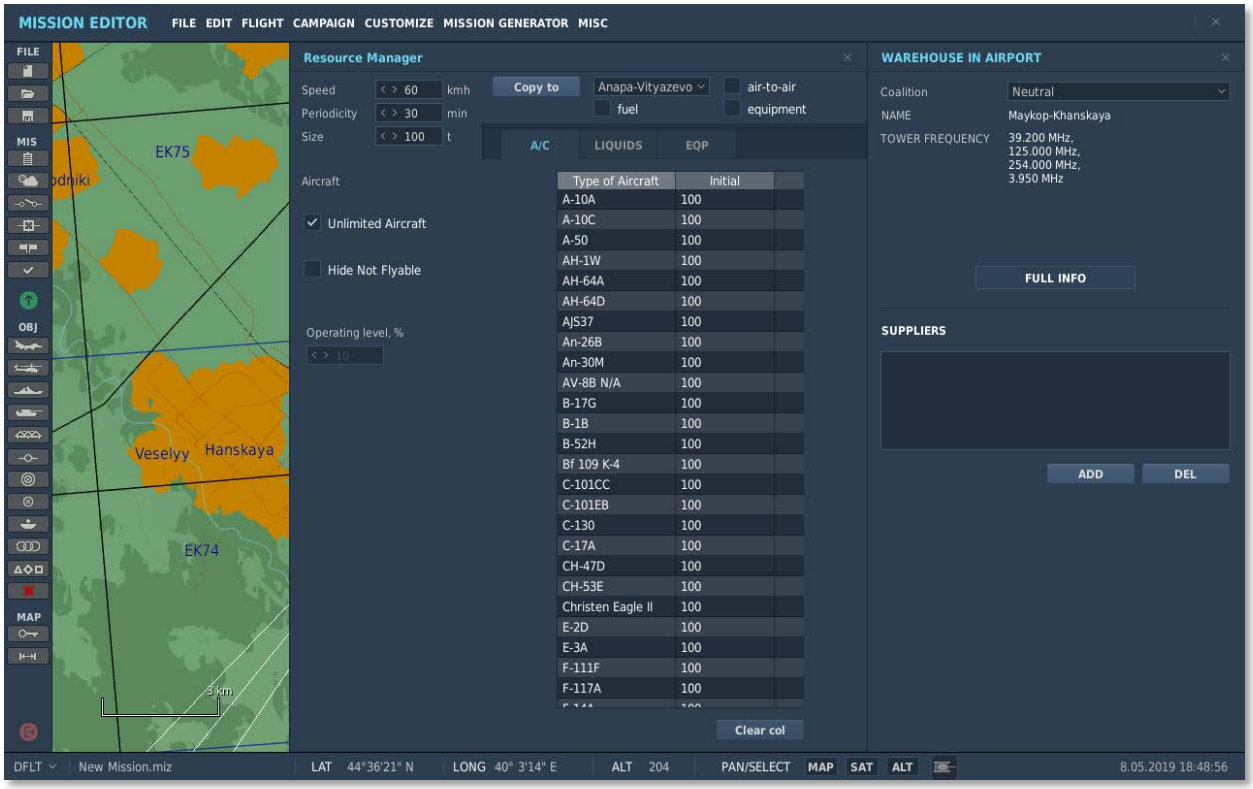
Warehouse Inventory

In the center of the display is a large, vertical column that displays the inventory of the selected warehouse. On the left side of the column is the type of inventory and to the right of each item is the number remaining. Above the column are three tabs: A/C, LIQUIDS and EQP.

To set all values to zero for a category, you can press the Clear Col button at the bottom of the column.

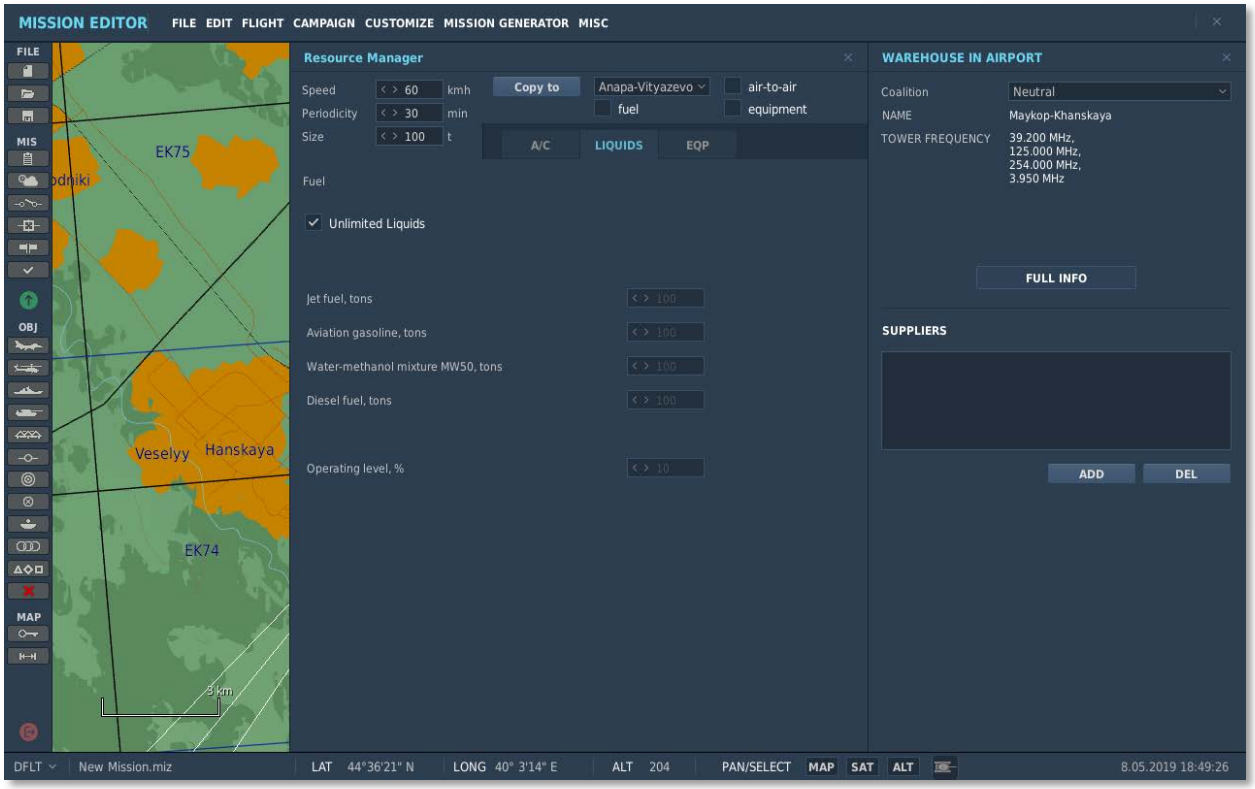
A/C Tab

The A/C tab lists the type and initial number of aircraft assigned to the warehouse. By left mouse clicking on an initial number, you can alter the value. If you wish to always have the airbase at maximum strength (100), check the **Unlimited Aircraft** check box to the left of the column. If you wish to only list aircraft that are player-flyable, check the **Hide Not Flyable** check box.



LIQUIDS Tab

This tab allows you to set the initial amount of fuel assigned to the warehouse. From the Initial Fuel Amount, tons box, you can increase or decrease the amount of fuel. If you wish for the fuel level to always be at 100 tons, check the Unlimited Fuel check box.



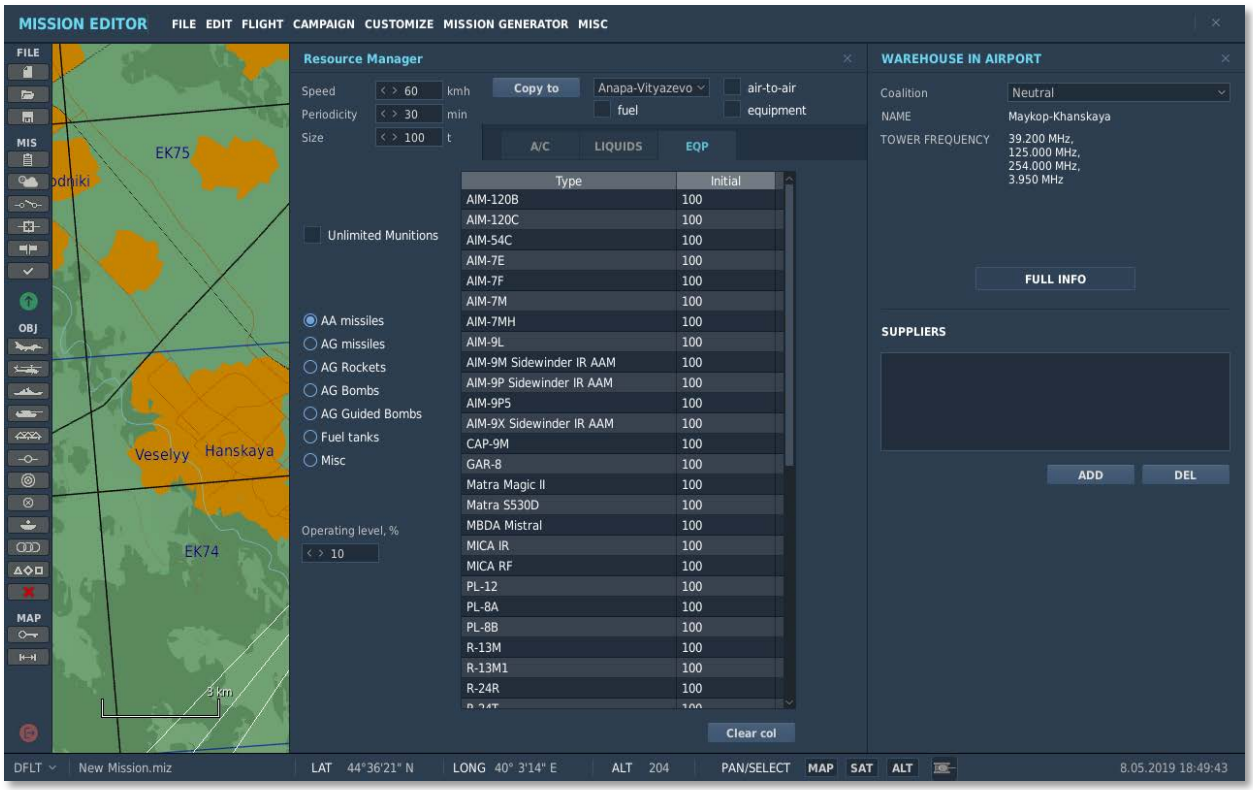
EQP (Equipment) Tab

The equipment tab lists all possible stores that can be loaded on aircraft. This includes weapons, pods, and fuel tanks. Like the aircraft selection, the name of the store is listed on the left side of the column and the initial number of the store is listed to the right. You can click on the initial number to adjust it, if the Unlimited Munitions check box is not checked.

Due to the wide variety and number of stores, seven store category buttons are listed to the left of the column. Use these filter buttons to display the type of store you wish to view / adjust.

1. AA missiles. Air-to-Air missiles
2. AG missiles. Air-to-Ground missiles
3. AG Rockets. Unguided Air-to-Ground rockets
4. AG Bombs. Unguided bombs
5. AG Guided Bombs. Guided bombs (such as laser and GPS)

6. Fuel tanks. External fuel tanks
7. Misc. Other stores such as ECM pods, smoke pods, gun pods, targeting pods, etc.



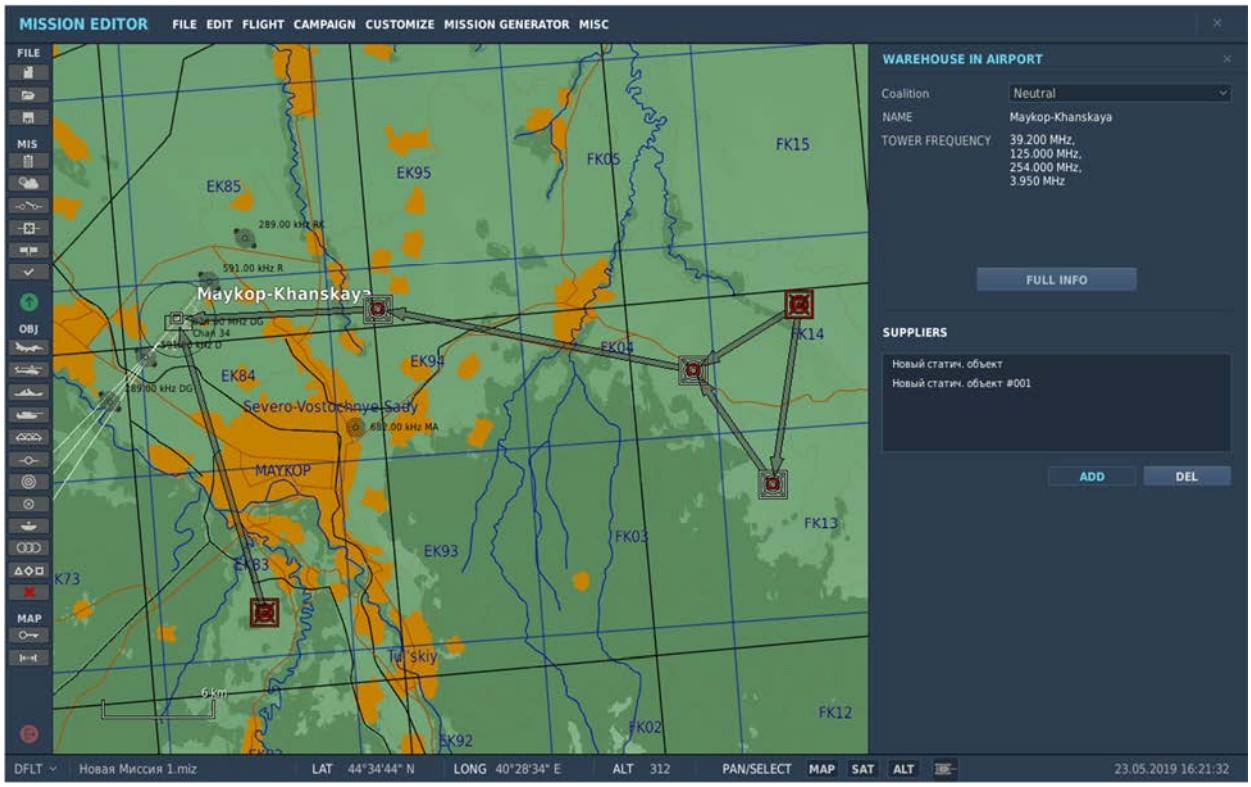
Copy to Other Warehouses

In the top center of the page is the Copy to option that allows you to copy the settings of the warehouse you are editing to another airbase warehouse. This is done by first selecting what resources you wish to copy (A/C, Fuel and Eqp) by selecting the check boxes and then selecting the airbase you wish to copy the settings to by using the drop-down airbase list.

Adding a Warehouse

In addition to the warehouse that is co-located with each airbase, you can also place independent warehouses. To do so, select warehouse from the Add or Modify Static Object menu. Once you place it, you can set its inventory values as you would an airbase warehouse.

***Note**: When connecting warehouses, they must be assigned to the same side.*

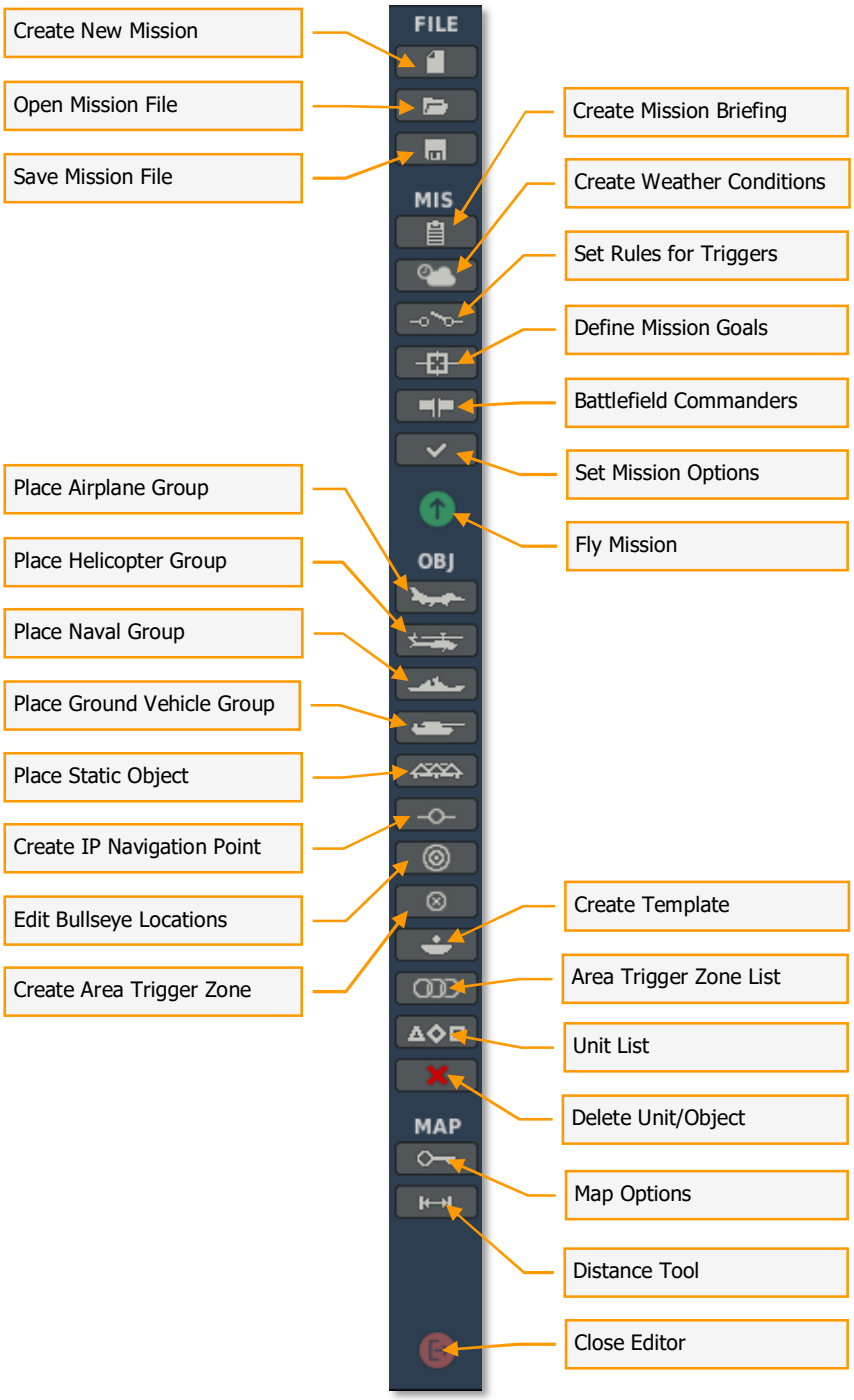


You can consecutive link many warehouses on one chain. Also, multiple suppliers and multiple consumers can be connected to a single warehouse.

If a current warehouse has a multiple suppliers, it is sending the request to the closest supplier at first. If the closest supplier can't provide the full request list, then the current warehouse will send the request to a more distant one, etc.

The Tool Bar

When creating a mission, the Tool Bar will probably be the most important tool you will be using. This bar provides quick access to common actions like unit placement, creating triggers, setting trigger zones, setting goals, and file management.



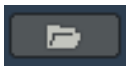
The below sections will review each of these Tool Bar elements in detail.

Create New Mission



Selecting Create New Mission will allow you to exit the current mission loaded and start a new one from scratch as mentioned in the File part of The System Bar chapter.

Open Mission File



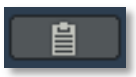
Selecting Open will display the Open Mission window. This window allows you to browse your local disk(s) and select and load saved missions. See description in the File part of The System Bar chapter.

Save Mission File



The Save As function. See description in the File part of The System Bar chapter.

Create Mission Briefing



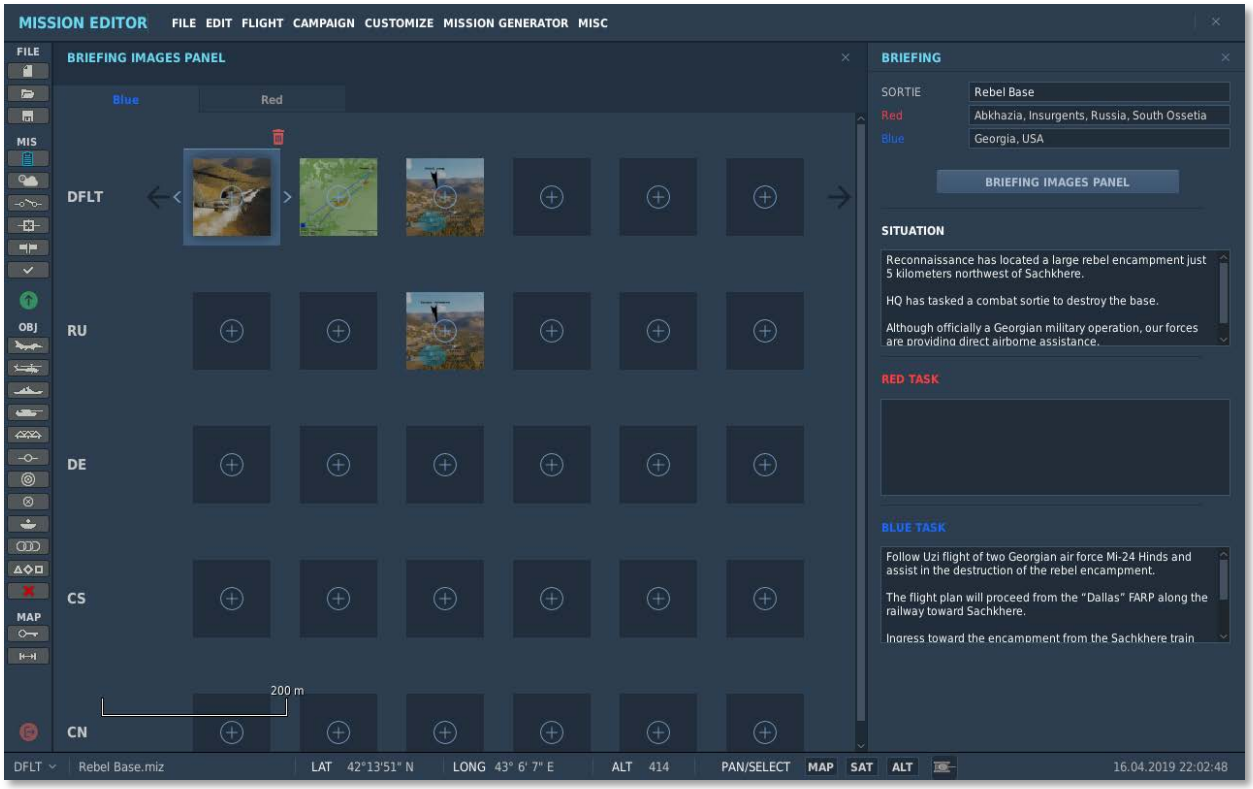
Upon selecting the Create Briefing button, the BRIEFING window will be displayed on the right side of the screen. From this window you can create the general mission briefing, individual briefings for Red and Blue sides, set briefing images for Red and Blue sides.

- At the top of the window is the **SORTIE**. Here you may enter a name/title for the mission. When you open the mission briefing to play it, this text will appear in the Title field.
- Next is the **RED** coalition field and this will automatically list the factions assigned to the red coalition.
- Below that is the **BLUE** coalition field and this will automatically list the factions assigned to the blue coalition.

- Below the data fields is the **BRIEFING IMAGES PANEL**, which consists of two boxes that allow you to assign unique briefing images for Red and Blue sides. To add a briefing image, click on the (+). Upon doing so, a file/folder browser will be displayed (much like the open and save dialogs), and from here you may select an image file. From the TYPE drop down you can select the image format (*.jpg; *.jpeg; *.png). The image should be 512x512 pixels minimum, recommended 1024x1024 pixels and the content can be of your choosing. For example, you may wish to include an image of the flight route, target area, specific target, or a unit type. Note that when in the simulation you can press the Print Screen button to capture an image that will be deposited in your Screen Shots folder. From there you may use an image editor to crop it, resize it, and annotate it. Once you select the image, a small version of it will be displayed in the box on the Briefing tool window. To remove a briefing image, click the red basket icon in the right corner above the cell.

  Graphic files are saved in the mission file.

  For different language locales, you can add the corresponding graphic files.



You can change a group's starting time using the group properties panels (described in detail further). Also note, the calendar starts with June 1st as day 0. For example, to set the mission date to July 1, set the day value to 30. To set the mission date to November 1, set the day value to 153. The default starting time for all missions is 12:00:00/0 (12:00 pm, June 1st).

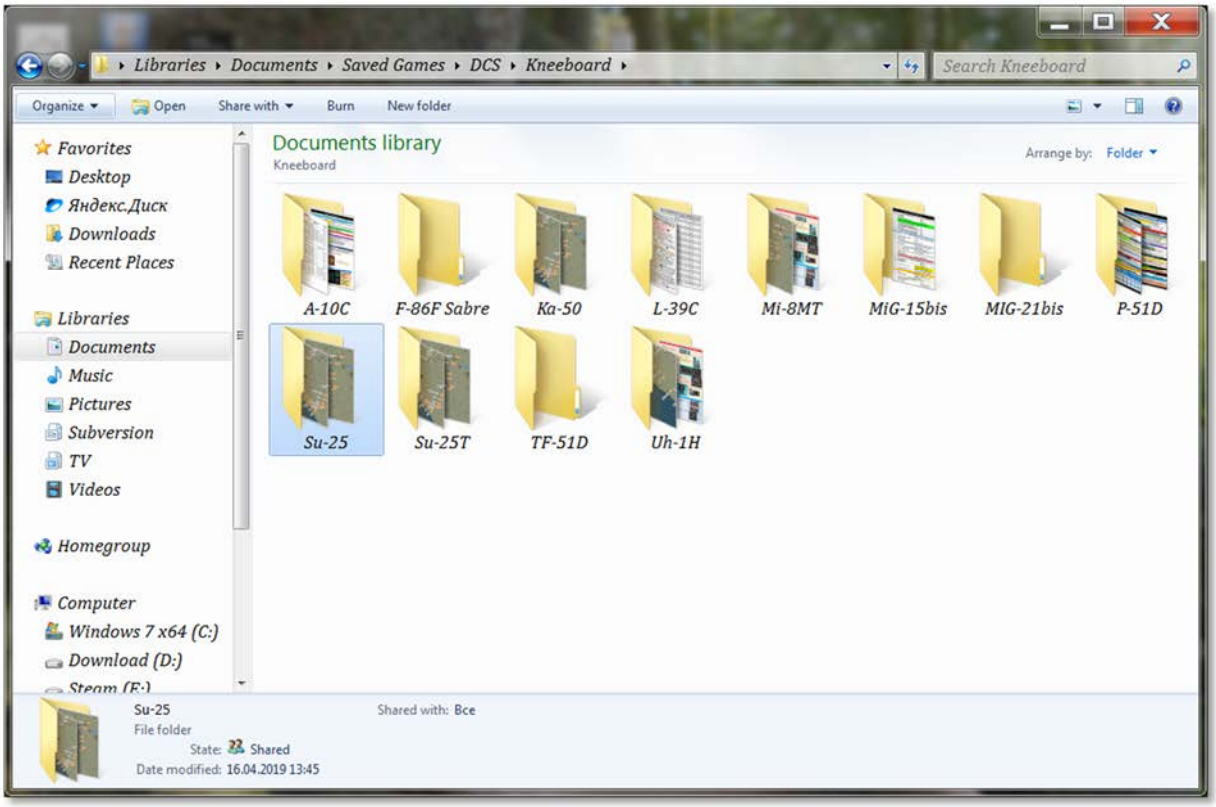
Changing the starting time and date will affect the position and motion of the sun, moon and stars to correspond to the time of day and year set.

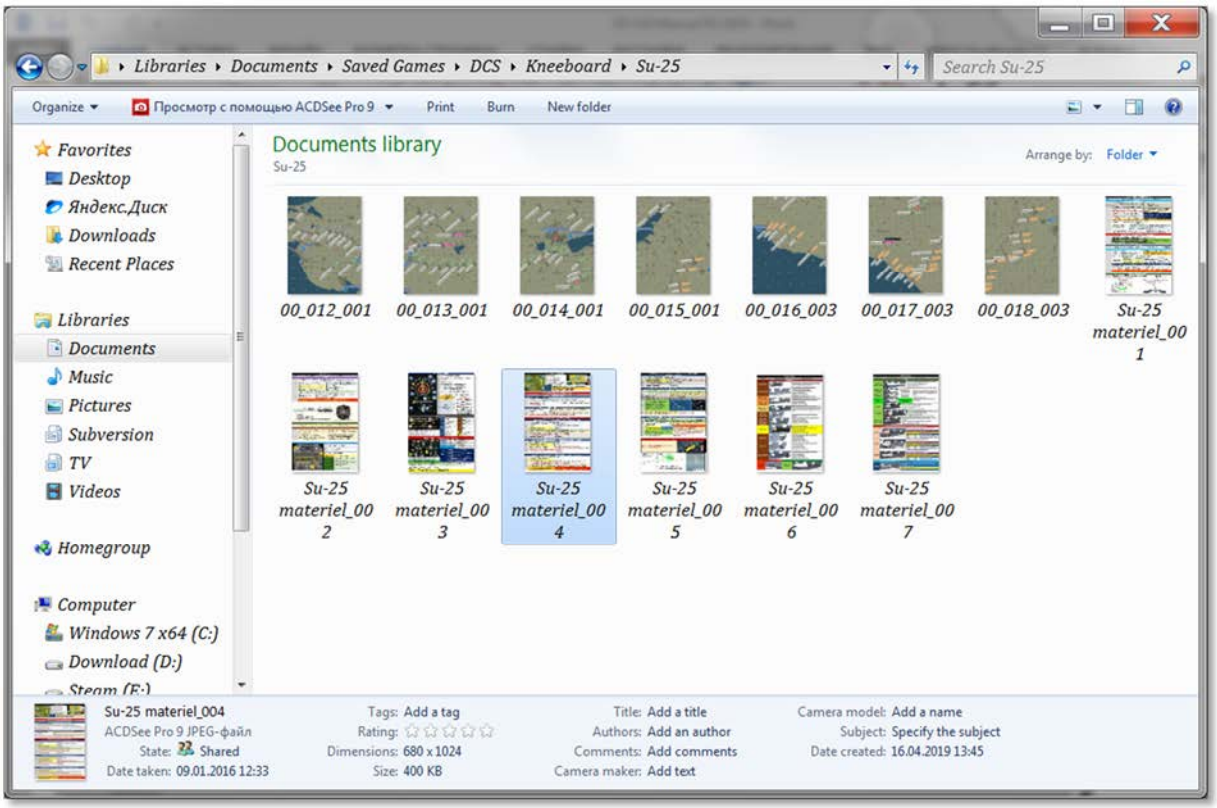
***Note**: If you wish to adjust the darkness of night missions, you can adjust the mission start date to a date when the moon is less visible.*

- The three text boxes that comprise the lower portion of the window allow you to enter a general mission briefing and individual briefings for Red and Blue sides (handy for multiplayer missions). Enter or copy [CTRL+C → CTRL+V] text in the **SITUATION** box and this text will appear in the SITUATION field when the mission briefing is displayed.
- Enter or copy [CTRL+C → CTRL+V] text in the **RED TASK** and **BLUE TASK** boxes to have unique text for each side that will be displayed in the OBJECTIVE field of each mission briefing.

Kneeboard

In addition to the standard airfield layouts, you can put graphic files with mission-relevant content, such as briefings, start / stop procedures, take-off / landing, etc., on the [RShift + K] kneeboard. Custom cards will be displayed at the end of the standard list. To access them immediately, cycle backwards through kneeboard cards. To add custom cards to the kneeboard, the mission creator must first create a folder named "Kneeboard" in the directory C:\Users\<User Name>\Saved Games\DCS\Kneeboard. Then, in this folder, create individual folders for each required module. The folder names must match the ones found in C:\Program Files\Eagle Dynamics\DCS World\Mods\aircraft. Afterwards, place the required image files (*.jpg; *.png) into the individual module folders. Filenames must not contain any non-latin script letters and must be 1024x680 or 1024x768 in size. While in the cockpit, bring up the kneeboard by pressing [RShift + K]. Press [or] to move back or forward a page.


Libraries ▸ Documents ▸ Saved Games ▸ DCS ▸ Kneeboard ▸
Search Kneeboard
Organize
Open
Share with
Burn
New folder
Favorites
Desktop
Яндекс.Диск
Downloads
Recent Places
Libraries
Documents
Music
Pictures
Subversion
TV
Videos
Homegroup
Computer
Windows 7 x64 (C:)
Download (D:)
Documents library
Kneeboard
Arrange by: Folder
A-10C
F-86F Sabre
Ka-50
L-39C
Mi-8MT
MiG-15bis
MIG-21bis
P-51D
Su-25
Su-25T
TF-51D
Uh-1H
Su-25
File folder
State: Shared
Date modified: 16.04.2019 13:45
Shared with: Все


Libraries ▸ Documents ▸ Saved Games ▸ DCS ▸ Kneeboard ▸ Su-25
Search Su-25
Organize
Просмотр с помощью ACDSee Pro 9
Print
Burn
New folder
Favorites
Desktop
Яндекс.Диск
Downloads
Recent Places
Libraries
Documents
Music
Pictures
Subversion
TV
Videos
Homegroup
Computer
Windows 7 x64 (C:)
Download (D:)
Documents library
Su-25
Arrange by: Folder
00_012_001
00_013_001
00_014_001
00_015_001
00_016_003
00_017_003
00_018_003
Su-25 materiel_001
Su-25 materiel_002
Su-25 materiel_003
Su-25 materiel_004
Su-25 materiel_005
Su-25 materiel_006
Su-25 materiel_007
Su-25 materiel_004
ACDSee Pro 9 JPEG-файл
State: Shared
Date taken: 09.01.2016 12:33
Tags: Add a tag
Rating:
Dimensions: 680 x 1024
Size: 400 KB
Title: Add a title
Authors: Add an author
Comments: Add comments
Camera maker: Add text
Camera model: Add a name
Subject: Specify the subject
Date created: 16.04.2019 13:45


Прицел АСП-17
1 Указатель крена
2 Прицельная марка
3 Шкала дальности
4 Диапазон разрешенных дальностей пуска
5 Зеленая лампа - лазерный дальномер включен
6 Оранжевая лампа - пуск разрешен
7 Красная лампа - выход из атаки
Применение оружия "Воздух-Воздух"
Шаг
Органы управления
Состояние
Применение ракет ближнего боя (Фиб)
1 Вкл. режим "Воздух-воздух"
2 Маневром самолета добиться положения цели вблизи прицельной марки
При наступлении захвата прицельная марка переохватывает на цель
Загорается оранжевая лампа "пуск разрешен" и появляется звуковой сигнал
3 Произвести пуск ракеты
Применение ВПУ и СППУ против воздушных целей
1 Вкл. режим "Воздух-воздух" (Выбрать Пушку или СППУ)
На отражателе прицела (АСП-17) появится прицельная марка
При необходимости вывести на прицел неподвижную сетку
2 Маневром самолета выбрать упреждение и произвести стрельбу
Применение оружия "Воздух-Земля"
Применение свободнопадающих бомб (с пикирования Hбомб>2500м)
1 Вкл. в режим применения оружия по земле
2 Выбрать Бомбы, контролируя тип по индикации на СУВ
3 Закрылки Установить в маневренное положение
4 Прицел Отключить
5 Лазерный дальномер Включить
В пологом пикировании поддерживать скорость 500-600 км/ч
6 Когда прицельная марка начнет двигаться вверх от нижней границы ИЛС, маневром самолета добиться совмещения прицельной мерки с целью
7 По загоранию оранжевой лампы произвести сброс бомб
8 Лазерный дальномер Выключить
Применение свободнопадающих бомб (с горизонтального полета)
1 Вкл. в режим применения оружия по земле
2 Выбрать Бомбы, контролируя тип по индикации на СУВ
3 Лазерный дальномер Включить
4 Маневром самолета наложить прицельную марку на цель
Зажать (Удерживать) боевую кнопку
Удерживать указатель крена на вертикальной линии
5 Дождаться автоматический сброс боеприпасов
6 Лазерный дальномер Выключить
Применение ракет класса "воздух-поверхность" Х-25МЛ, Х-29Л, С-25Л
1 Вкл. в режим применения оружия по земле
2 Выбрать управляемые ракеты
3 Лазерный дальномер Включить
4 Перемещением прицельной марки наложить марку на цель и выполнить Захват цели
5 По загоранию оранжевой лампы произвести пуск ракеты
6 Во время полета ракеты при необходимости корректировать положение прицельной марки на цели, при этом ракета будет отслеживать перемещение точки прицеливания
7 После поражения цели Выключить Лазерный дальномер
Для повторного включения ЛД, не дождавшись цикла охлаждения нажать 6 раз кнопку ЛД

Create Weather Conditions



The weather tool panel provides two options for setting the weather for the mission: Static conditions and Dynamic conditions.

1. Static conditions - standard static weather formed by user settings.
2. Dynamic conditions - weather formed by differences in the atmospheric pressure and dynamically changing during the mission.

Static conditions

The weather panel is divided into six functional sections: Season, Clouds and Atmosphere, Wind, Turbulence, Fog, Dust Smoke and Weather Templates.

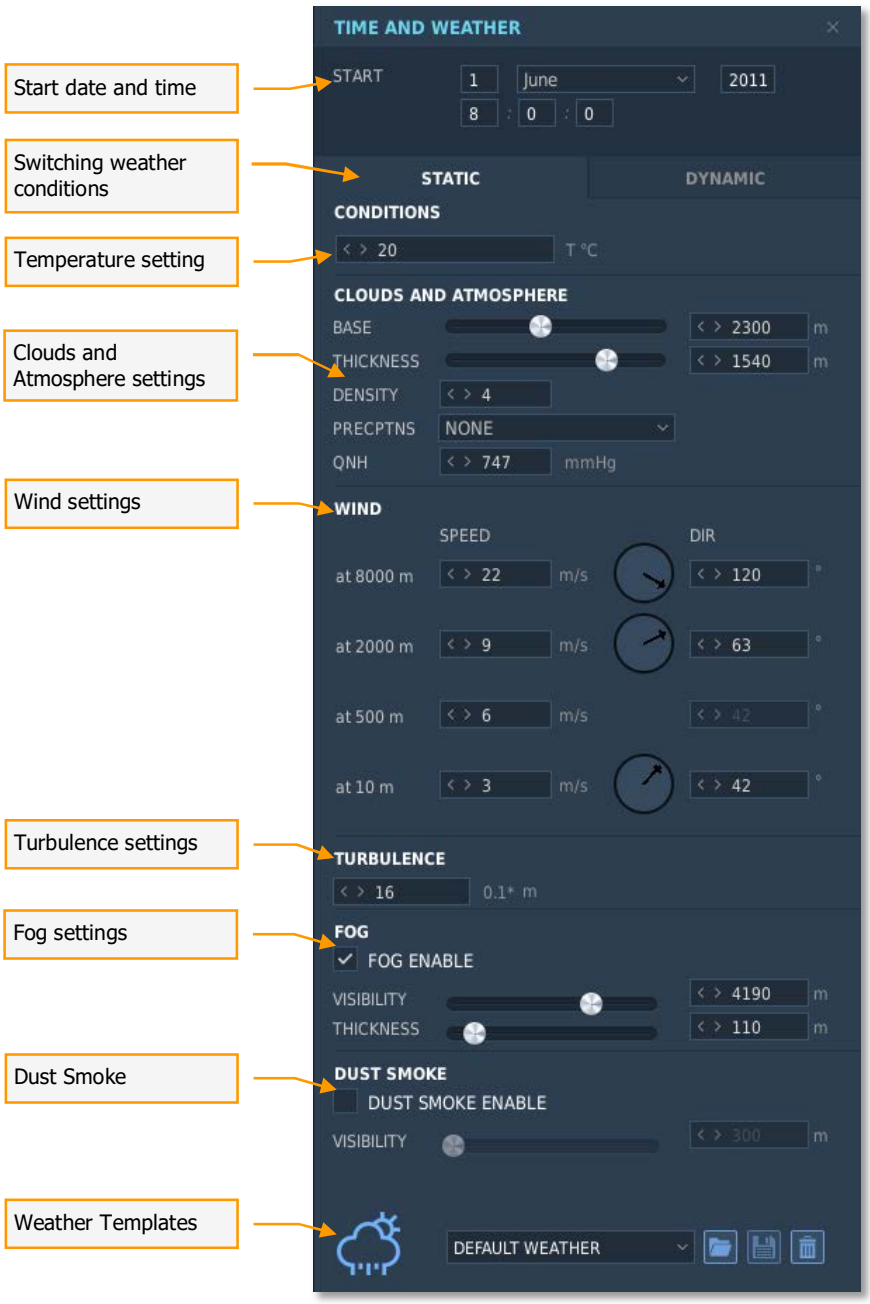
START. The top of the window consist of the Start section and allows you to set the season the mission will take place in, and the sea level air temperature (in Celsius). The left side drop down menu allows you to select from the four seasons: Summer, Winter, Spring, and Fall. The primary effect of changing the season will be to change how the terrain looks in the simulation. Additionally, many ground vehicles will have matching camouflage for the set season.

The right-side field allows you to set the sea level temperature of the mission. The value can be changed by either using the left and right arrows or manually typing the value into the field. Note that air temperature will affect aircraft performance.

CLOUDS AND ATMOSPHERE. This section allows you to define the cloud environment of the mission. Note that the cloud environment for the mission will be static and does not change during the course of the mission. From top to bottom, the cloud settings consist of the following controls:

- **BASE**. Defines the altitude above sea level that will form the bottom of the cloud layer. This can be set between 300 and 5,000 meters and is indicated in the right-side field. Use the left and right arrows to change the value or drag the slider bar in the left field.
- **THICKNESS**. Defines cloud layer thickness from the base level. For example, if the Base were set at 2,000 meters and the Thickness was set to 1,000 meters, you would have clouds between 2,000 and 3,000 meters above sea level. Note that thickness only applies to a solid cloud deck (Density 9 and 10) and not scattered clouds (Density 1 through 8). The cloud layer thickness can be adjusted using the left and right arrows or the slider bar. The selected thickness is displayed in the right-side field in meters.
- **DENSITY**. The Density defines the population density of clouds in the mission. Note that cloud cover is uniform throughout the simulation world. Density is according to a 0 to 10 scale. 0 defines no clouds; 1 through 8 defines scattered clouds with increasing levels of density; and 9 and 10 define a solid cloud layer deck (overcast). Use the left and right arrows to select the desired density setting.

- **PRECPTNS** (Precipitations). The precipitation drop down menu allows you to set what type, if any, of precipitation will fall during the mission. Choices include: None, Rain, Thunderstorm, Snow, and Snowstorm. The selections will vary according to the selected season, temperature and cloud density.
- **QNH.** Also referred to as "Q Code", this value sets the barometric air pressure in the mission. This is in reference to mmHg (millimeters of mercury) with Russian altimeters.



WIND. Rather than a single wind direction and speed, the wind function allows you to set four distinct wind altitude bands: 10 – 500 – 2,000 – 8,000 meters. These are indicated by the three lines within the Wind section of the Weather Tool. To set the speed in meters per-second (m/s), use the left and right arrows. Note that wind will be as a constant force without gusting. To the right of the wind speed setting is the wind direction setting. This can be set in two ways. The first is to click on the radial dial to denote the direction you want the wind to blow in (North is indicated as the top of the dial). The second option is to input the value using the left and right arrow buttons. The set direction that wind will blow in is indicated in the right field in degrees. For example, when set to 0 degrees (due north), the wind in the mission will blow from the south to the north.

TURBULENCE. You may set air turbulence at 0.1 m/s increments at ground level. Turbulence will then decrease as altitude increases.

FOG. The fog setting controls:

- **FOG ENABLE**. Set fog on or off.
- **VISIBILITY.** Visibility in meters of objects obscured by fog.
- **THICKNESS**. Use the scroll bar or left and right arrows to set how thick the fog layer will be. A 0 (zero) setting denotes sea level. For example, setting a value of 50 would place a uniform fog layer in the simulation world from 0 to 50 meters from sea level.

DUST SMOKE. Dust smoke setting parameters:

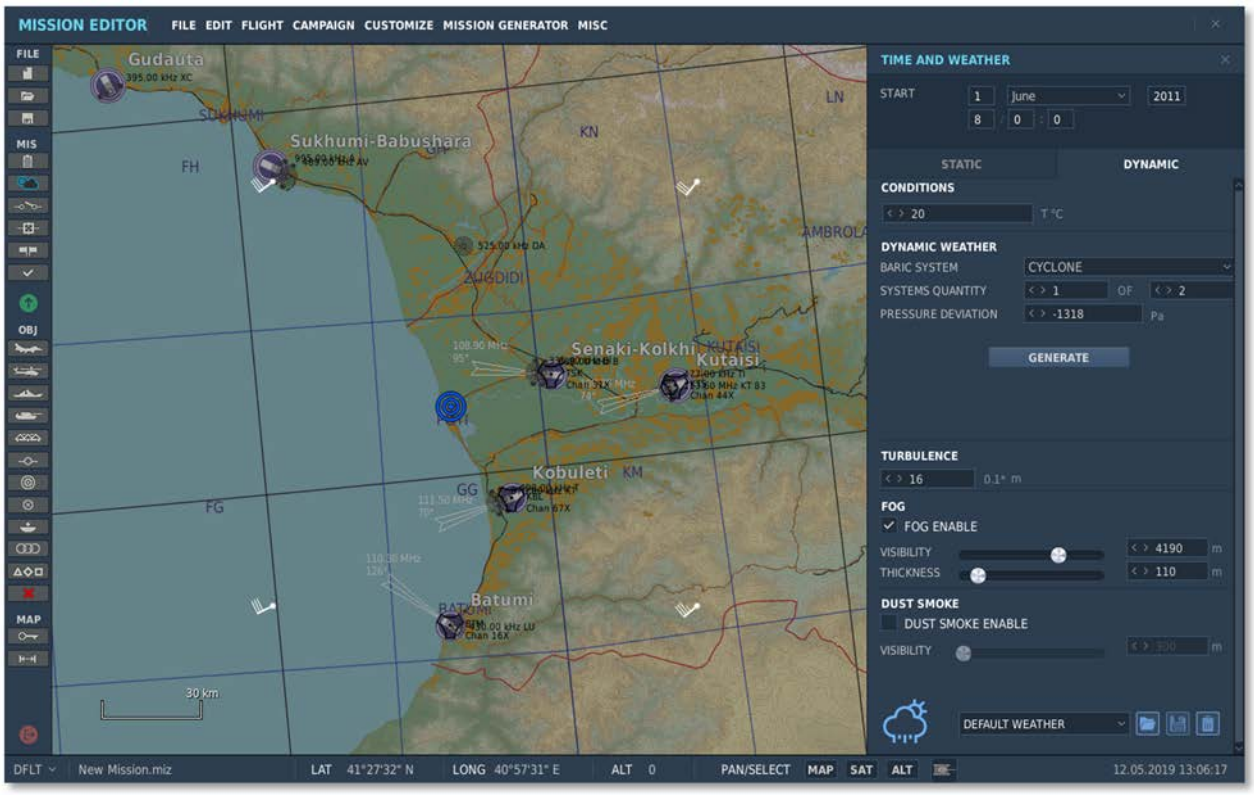
- **DUST SMOKE ENABLE**. Checkbox that enables or disables the dust smoke effect.
- **VISIBILITY**. Sets the visibility range (in meters) of objects during a dust storm.

WEATHER TEMPLATES. Rather than require you to create weather conditions each time you create a mission, the editor provides you the means to load saved weather templates and even modify them. These can be useful tools to save time when creating missions for a campaign. The Weather Template section has the following elements:

- **Template List**. On the left side of the template section is a drop-down list of saved weather templates. To select a template, you must left mouse click on it. Once selected, you will then need to click the LOAD button to load the template into the Weather settings tool.
- **LOAD selected template button**. Once you have selected a weather template from the template list, press this button to LOAD  the template into the Weather settings tool.
- **SAVE button**. If after loading a template you change its settings, you may then click the SAVE button  to save those changes to the template.
- **REMOVE button**. To remove a template from the Template List, select the template and then click the REMOVE button  .

Dynamic conditions

The Dynamic Conditions panel is divided into five functional sections: Start, Dynamic Weather, Turbulence, Fog, and Weather Templates.



Using the dynamic weather panel, you can create a weather system (areas of different atmospheric pressures), which will result in dynamic wind and cloud generation in the mission.

START. The top of the window consist of the Start section and allows you to set the season the mission will take place in, and the sea level air temperature (in Celsius). The left side drop down menu allows you to select from the four seasons: Summer, Winter, Spring, and Fall. The primary effect of changing the season will be to change how the terrain looks in the simulation. Additionally, many ground vehicles will have matching camouflage for the set season.

The right-side field allows you to set the sea level temperature of the mission. The value can be changed by either using the left and right arrows or manually typing the value into the field. Note that air temperature will affect aircraft performance.

DYNAMIC WEATHER. Settings of the dynamic weather system parameters.

- **BARIC SYSTEM**. Selection between cyclone (low pressure systems), anticyclone (high pressure systems) or none.
- **SYSTEMS QUANTITY**. Quantity of baric systems on the map.

- **PRESSURE DEVIATION**. Pressure deviation in the center of the selected pressure system from ISA International Standard Atmosphere, in Pascals.

GENERATE. Generate a randomized weather system using the set parameters.

Clouds and winds are generated on the entire map dynamically in the mission according to the type, locations and pressure deviations of the pressure systems. Wind conditions evolve during the mission according to the changes in the pressures and locations of the pressure systems.

TURBULENCE. You may set air turbulence at 0.1 m/s increments at ground level. Turbulence will then decrease as altitude increases.

FOG. The fog setting controls:

- **FOG ENABLE**. Set fog on or off.
- **VISIBILITY.** Visibility in meters of objects obscured by fog.
- **THICKNESS**. Use the scroll bar or left and right arrows to set how thick the fog layer will be. A 0 (zero) setting denotes sea level. For example, setting a value of 50 would place a uniform fog layer in the simulation world from 0 to 50 meters from sea level.

DUST SMOKE. Dust smoke setting parameters:

- **DUST SMOKE ENABLE**. Checkbox that enables or disables the dust smoke effect.
- **VISIBILITY**. Sets the visibility range (in meters) of objects during a dust storm.

WEATHER TEMPLATES. Rather than require you to create weather conditions each time you create a mission, the editor provides you the means to load saved weather templates and even modify them. These can be useful tools to save time when creating missions for a campaign. The Weather Template section has the following elements:

- **Template List**. On the left side of the template section is a drop-down list of saved weather templates. To select a template, you must left mouse click on it. Once selected, you will then need to click the LOAD button to load the template into the Weather settings tool.
- **LOAD selected template button**. Once you have selected a weather template from the template list, press this button to LOAD  the template into the Weather settings tool.
- **SAVE button**. If after loading a template you change its settings, you may then click the SAVE button  to save those changes to the template.
- **REMOVE button**. To remove a template from the Template List, select the template and then click the REMOVE button  .

You can also cancel the window by pressing the yellow X in the top right corner of the window.

Set Rules for Triggers



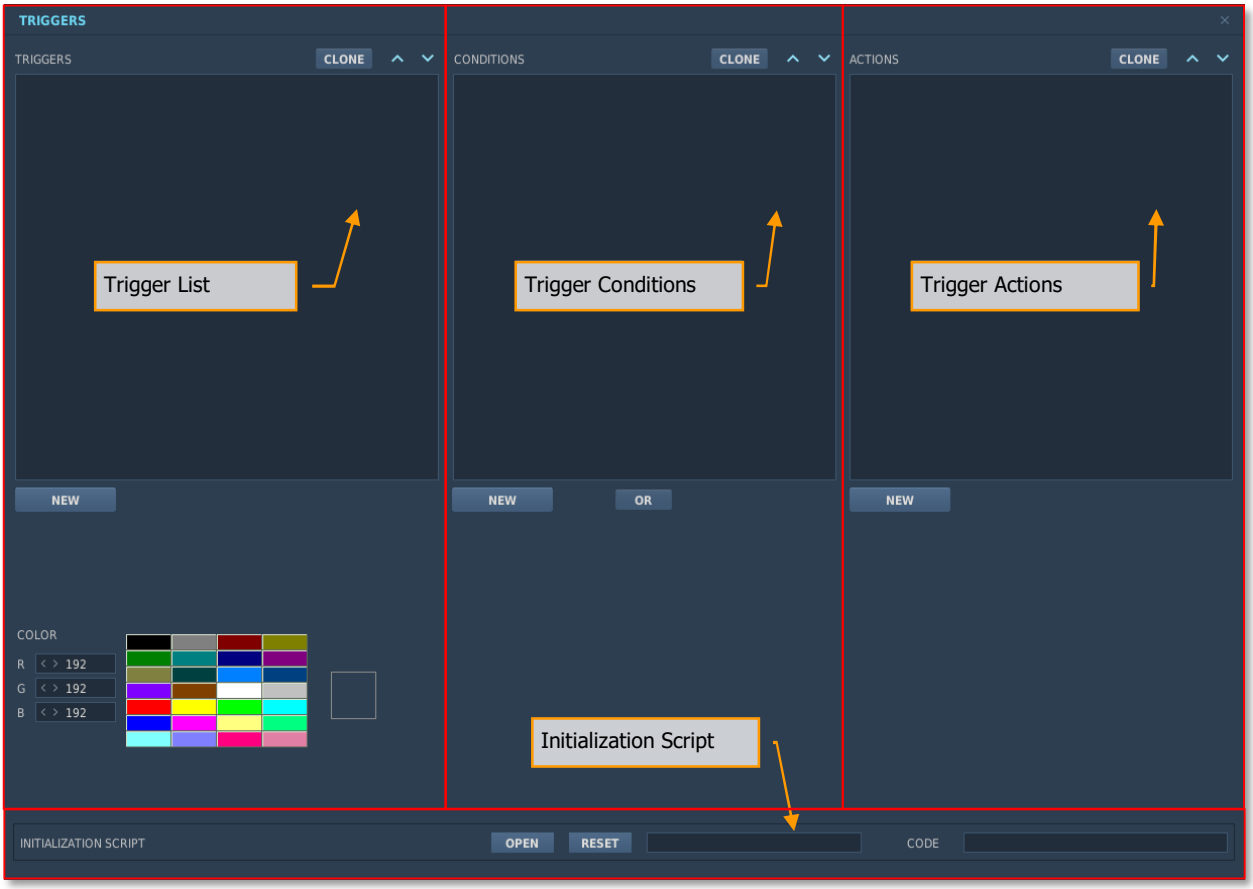
An important part of making a good mission is the ability to script actions during the mission that lead to a more immersive experience for the player. By setting actions that respond to the player and other AI units in an intelligent manner, the simulated battlefield can have a much more interactive and interesting environment. Such actions could be the activation of units, text, and voice messages, or setting a flag state. The conditions that are used to trigger such actions can range from units going in and out of defined areas of the map, units being destroyed or damaged, specified times, flag states, and even random states. Using these tools and an active imagination, you can create your own engaging missions that no automatic mission generator can match. The missions and campaigns use the same scripting system.

The trigger system is not an "event" type system, but rather a "condition" type system; this means that a trigger does not occur when something occurs, but rather when a condition becomes TRUE.

Setting a trigger condition is a three step process:

1. Create a NEW trigger
2. Create the condition(s) for the trigger
3. Create the action(s) that will result from the trigger when the condition(s) are true.

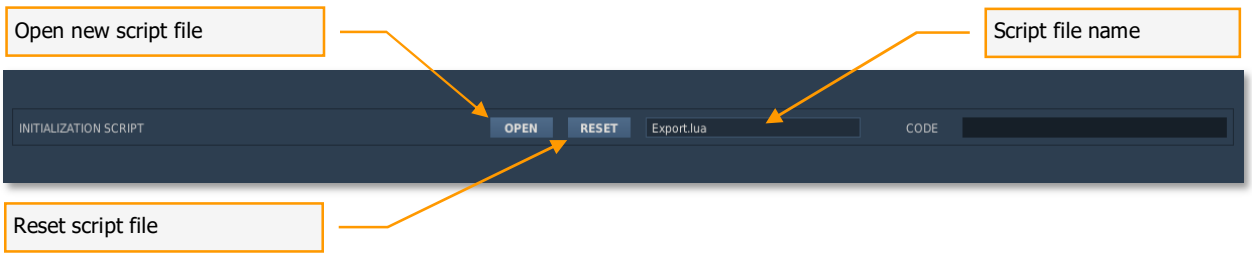
Upon selecting the set Triggers button, the TRIGGERS window will appear. There are four primary elements to creating a mission trigger and each has its own pane in the window: Trigger List, Trigger Conditions, Trigger Actions and Initialization Script. For convenience, individual triggers may be assigned a color from the color palette provided below.



The CREATE button found under each list is used for the creation of new elements.

Initialization Script

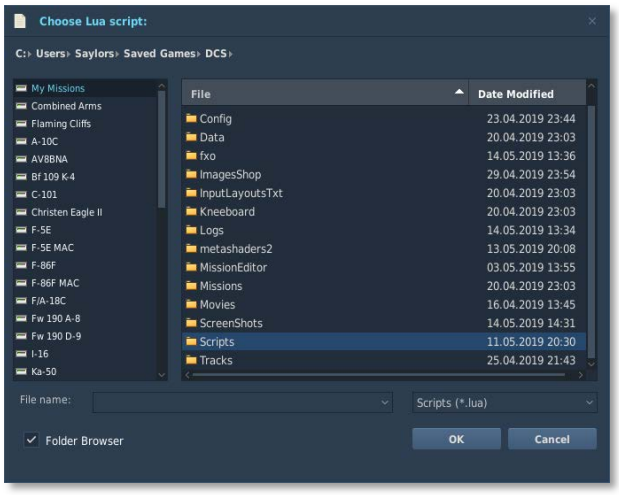
The INITIALIZATION SCRIPT column is located on the lower block (of the mission editor.) This is used to load and reset the LUA script file in the mission to set the conditions for random spawning of unit groups. In the CONDITION field in the aircraft / helicopter / sea / ground group panel, the expression is written to the LUA using SSE tools.



OPEN. Opens a selection panel for LUA script files.

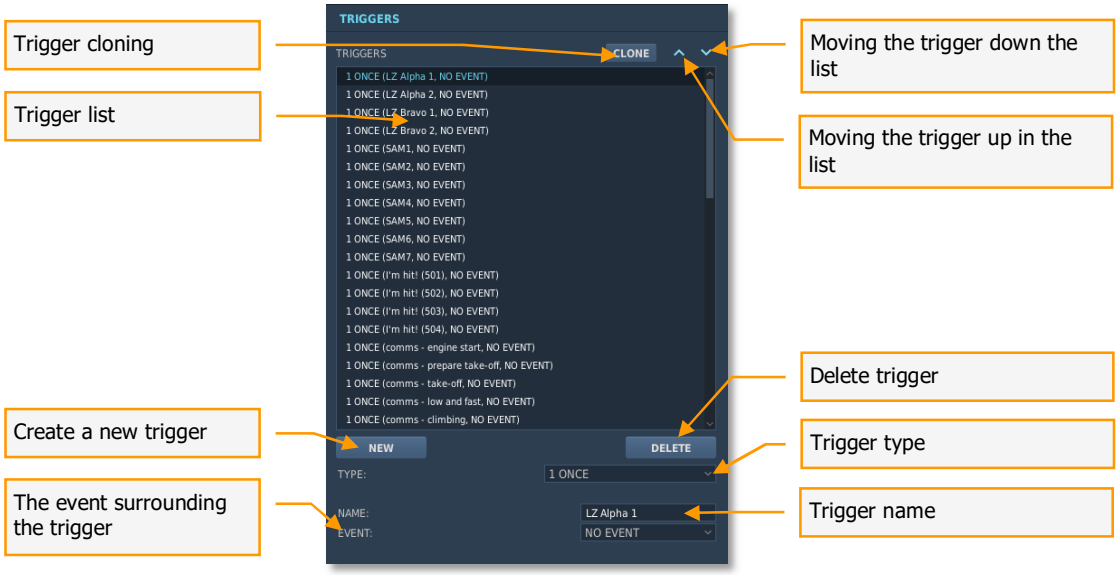
RESET. Resets the installed script file.

The LUA script selection panel is shown below:



Trigger List

This left-most pane is used to create new triggers and list existing ones. In the triggers listing pane, each trigger will be listed as a Type of trigger and then the name of the trigger in parenthesis. For example: "ONCE (Set Area 01)".



At the bottom and top of the trigger list panel there are four controls that allow you to control triggers:

NEW button. Press the NEW button to create a new trigger. When you create a new trigger, you will need to set the TYPE and give it a unique name that briefly describes its function. Be careful in naming your triggers so that there is not mass-confusion when you have a long list of triggers!

DELETE button. Press the DELETE button to remove a trigger from the list. You will first need to click on the trigger you wish to remove before pressing the DELETE button.

CLONE button. Used to create an identical copy of the selected trigger in the list.

Up and **Down** arrows. Used to move the selected trigger up or down in the list.

TYPE drop down field. Using this drop down, you can set when and how the trigger event will be cued. For example, you may want the trigger to only happen once, or you may want it to happen each time defined conditions are met. You have four options:

1. **ONCE**. The trigger will be executed only once after the condition is evaluated as true. Once set as true, the condition is removed from memory. For example, if you want a

message to show only the first time an aircraft enters an area, you would use the ONCE option.

Trigger Behavior:

Condition: Evaluated Continuously, until it is evaluated as "True". Then removed from the list of evaluations.
Action: Activated Once, when condition is "true".

Example:

You want to spawn a AAA backup-group when another Armor-group is dead.

- Add a "ONCE" Trigger.
- Add a "Group Dead" condition and select the Armored-group that must be dead.
- Add a "Activate Group" and select the AAA-group you want to activate.

The group being activated will be activated once when the first group is dead.
Hint: Remember to set the AAA-group to 23-hours of delay, so it stays inactive until activated.

2. **REPETITIVE ACTION**. The Repetitive Action trigger checks the trigger's conditions each second. Each second the conditions return true, the actions are performed; each second the conditions are false, no action is not performed.

   Trigger-Behavior:

   Condition: Evaluated Continuously.

   Action: Repeated every time as long as the conditions stays "True".

   Example:

   You want to set up a (short) alarm-sound after a group of units have entered a Zone.

   - Add a "REPETITIVE ACTION" trigger.
   - Add a "Unit in Zone" condition, select the appropriate unit and zone to enter.
   - Add a "Sound" action, select the <nnn-sound> in the sound browser.

   The sound will now replay every 1 second after the unit has entered the Zone.

3. **SWITCHED CONDITION**. The trigger will perform the set action(s) every time the trigger's condition is checked and evaluated as true and its previous state was false. For example, such a trigger can be used to show a text message whenever a particular unit enters a trigger zone. The first time the unit enters the zone, the trigger action will be activated, because the previous check of the trigger conditions was evaluated as false (unit was not in zone). The trigger will not activate any longer as long as the unit is inside the zone, because the previous check of the trigger conditions will be evaluated as true (unit remains inside zone). However, if the unit exists the zone and then re-enters it again, the conditions will have once again switched from false to true and the trigger action will be activated.

Trigger Behavior:

Condition: Evaluated Continuously.

Action: Repeatable, Once per condition-set changes from "False" -> "True". Actions will be performed again on the next "Switched state-change".

Example:

You want to perform a action each time a unit enters a zone. This differs from them "being in a zone".

- Add a "SWITCHED CONDITION" trigger.
- Add a "Unit in Zone" condition, select a unit and zone to check.
- Add a "Message" of "A unit is trespassing our Zone!"

Now each time the unit enters the Zone you will get the message Once. The unit needs to leave and reenter the Zone again to show the message again.

4. **MISSION START**. The trigger will only be checked at mission start. For example, if you wanted to set multiple units to a random activation, you would use this option to evaluate this condition and determine which units are included in the mission according to the set percentage.

   Trigger Behavior:

   Condition: Evaluated Once, at mission startup.

   Action: Performed Once, at mission startup.

   Example:

   You want to activate a random group at startup of the mission.

   - Add a "MISSION START" Trigger.
   - Add a "RANDOM" condition, set it to 10%
   - Add a "Activate group" action, select the group you want to give 10% chance.

NAME field. Manually enter the name of the trigger in this field.

EVENT field. Setting an event for a trigger limits the trigger's conditions to being evaluated only when the set events occur. With no event set, trigger conditions are evaluated once per second throughout the duration of the mission, unless already expired and removed from memory. In a trigger-heavy mission, this can cost considerable CPU cycles and affect gameplay performance. Setting triggers to events can reduce the CPU workload and also provide some unique trigger mechanisms. The following events are available:

- NO EVENT. There's no event.
- ON DESTROY. Trigger's conditions are evaluated whenever any unit is destroyed.
- ON SHOT. Trigger's conditions are evaluated whenever any weapon is fired.
- ON CRASH. Trigger's conditions are evaluated whenever the player crashes.

- ON EJECT. Trigger's conditions are evaluated whenever the player ejects.
- ON REFUEL. Trigger's conditions are evaluated whenever the player begins to air refuel.
- ON PILOT DEAD. Trigger's conditions are evaluated whenever the player is killed.
- ON BASE CAPTURED. Trigger's conditions are evaluated whenever an airbase or FARP is captured.
- ON TAKE CONTROL. Trigger's conditions are evaluated whenever the player takes control of an aircraft.
- ON REFUEL STOP. Trigger's conditions are evaluated whenever the player stops to air refuel.
- ON FAILURE. Trigger's conditions are evaluated whenever the player's aircraft suffers a system failure.
- ON MISSION START. Trigger conditions are evaluated whenever a mission starts.
- ON MISSION ENDS. Trigger conditions are evaluated whenever the mission ends.

Trigger Conditions

The central unit of the trigger panel contains a list of trigger conditions.



Once you have created a trigger, you will then need to set the conditions that govern when the trigger will be set to true or false. To do so, you first need to click on the trigger you want to set conditions for by clicking on it from the trigger list. Upon doing so, press the NEW button below the CONDITIONS pane.

NEW button. Press the NEW button to create a new condition for the selected trigger. Please note that you can create multiple conditions for a single trigger! When you create a new condition, you will need to set the TYPE and define additional data that varies according to the TYPE. Each new condition you create will be listed in the condition pane list.

When multiple trigger conditions are set, they operate with an "AND" operator. In other words, all of the conditions must be true for the trigger to be activated. To separate conditions with an "OR" operator, press the **OR** button in the conditions pane. In this case, the condition(s) either above or below the "OR" operator will activate the trigger.

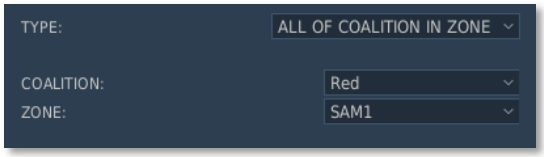
DELETE button. Press the DELETE button to remove a condition from the list. You will first need to click on the condition you wish to remove before pressing the DELETE button.

CLONE button. Used to create an identical copy of the selected condition in the conditions list.

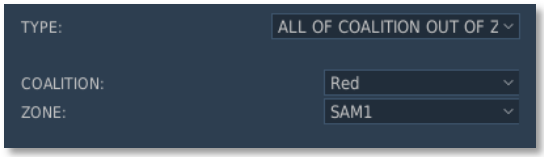
Up and **Down** arrows. Used to move the selected condition up or down in the condition list.

Below is a list of available rules in the drop-down list of TYPEs:

- **ALL OF COALITION IN ZONE**. When creating a Trigger Zone, you can set a trigger to true when all of the set coalition units (aircrafts, vehicles, ships) are inside a defined trigger zone. Upon selecting this condition, two new fields will be displayed. The COALITION drop down menu allows you to select the coalition that when inside the selected trigger zone will set the condition to true. The ZONE drop down will list all the trigger zones you have created according to the names you created for them.



- **ALL OF COALITION OUT OF ZONE**. This condition works the same as the ALL OF COALITION IN ZONE, but the condition will be set to true when the all coalition's units are outside the defined zone.



- **ALL OF GROUP IN ZONE**. When creating a Trigger Zone, you can set a trigger to true when all units of the set group are inside a defined trigger zone. The GROUP drop down menu allows you to select the group that when inside the selected trigger zone will set the condition to true. The ZONE drop down will list all the trigger zones you have created according to the names you created for them.



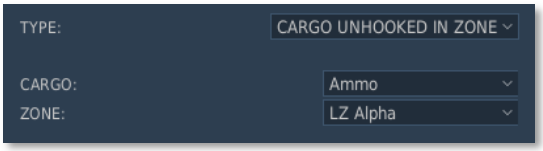
- **ALL OF GROUP OUT OF ZONE**. This condition works the same as the ALL OF GROUP IN ZONE, but the condition will be set to true when the all group's units are outside the defined zone.



- **BOMB IN ZONE**. Checks to see if a bomb has been dropped and is currently in a trigger zone. When choosing this condition, the mission creator must also specify the type and quantity of the bomb to be checked, as well as the trigger zone.



- **CARGO UNHOOKED IN ZONE**. Checks to see if a cargo object has been unhooked within a trigger zone. The cargo object and trigger zone must be specified when choosing this condition.



- **COALITION HAS AIRDROME**. If the set coalition has captured set airfield, the action(s) will be activated.



All airfields with no aircraft assigned for take-off or landing and with no ground units of either coalition within a 2,000 m. radius are neutral. Any ground unit present within a 2,000 m. radius of an airfield will capture the airfield for the coalition of the ground unit. If ground units of both coalitions are present within a 2,000 m. radius of an airfield, the airfield is contested and not assigned to either coalition. In cases when armed ground units of one coalition are present within a 2,000 m. radius of an airfield at the same time as unarmed ground units of the other coalition, the coalition of the armed units will capture the airfield.

- **COALITION HAS HELIPAD**. If the set coalition has captured set FARP, the action(s) will be activated.



Note, FARP capture rules are identical to airfield capture rules (see above).

- **DEAD ZONE**. Trigger rule for the condition of the specified zone. When choosing this condition, you must specify the name of the trigger zone.



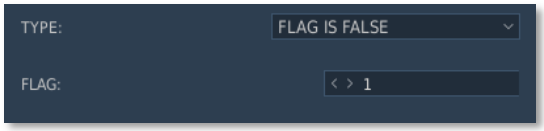
- **FLAG EQUALS**. This condition checks whether the value of a set flag is equal to the set value.



- **FLAG EQUALS FLAG**. This condition checks whether the value of one set flag is equal to the value of another set flag.



- **FLAG IS FALSE**. When selected, a FLAG field is displayed below. By either using the left and right arrows or typing in the field, set the flag number that when set to false will set the condition to false.



- **FLAG IS LESS.** When selected, FLAG and VALUE fields are displayed below. By either using the left and right arrows or typing in the field, set the flag number to check the value for and the flag value, below which the trigger condition will be true.



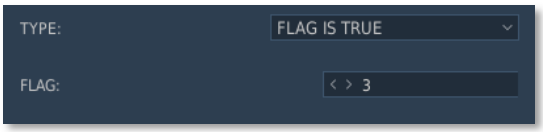
- **FLAG IS LESS THAN FLAG**. This condition checks whether one set flag has a smaller value than another set flag.



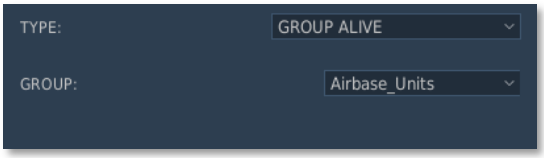
- **FLAG IS MORE.** When selected, FLAG and VALUE fields are displayed below. By either using the left and right arrows or typing in the field, set the flag number to check the value for and the flag value, above which the trigger condition will be true.



- **FLAG IS TRUE**. When selected, a FLAG field is displayed below. By either using the left and right arrows or typing in the field, set the flag number that when set to true will set the condition to true.



- **GROUP ALIVE**. When selected, a GROUP drop down menu will be displayed below. From this list, select the group that will activate the trigger as long as at least one unit of the group is alive. Groups are listed according to their "NAME" from the unit placement windows.

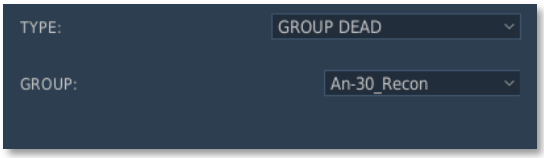


- **GROUP ALIVE LESS THAN**. When selected, a GROUP drop down list and "%" field will be displayed below. From the GROUP list, select the group that will activate the trigger when the number of live units in the group corresponds to the per cent value set in the "%" field.



For example, if the group has 10 units and you entered 40%, the trigger will activate when 3 units remain alive.

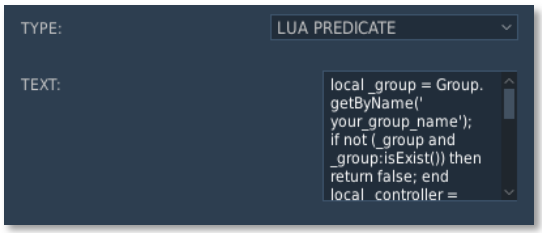
- **GROUP DEAD**. When selected, a GROUP drop down menu will be displayed below. From this list, select the group that will activate the trigger when all units of the group are destroyed (dead). Groups are listed according to their "NAME" from the unit placement windows.



- **LUA PREDICATE**. When selected, you can write the code in the LUA script syntax.

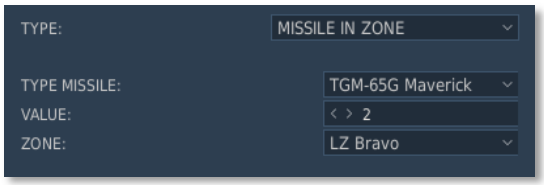
Example code:

```
local _group = Group.getByName('your_group_name');
if not (_group and _group:isExist()) then return false; end
local _controller = _group:getController();
local _targets = _controller:getDetectedTargets();
for _index, _target in ipairs(_targets) do
        if _target.visible then return true; end -- I see the enemy!!!
end
return false; -- I don't see the enemy...
```

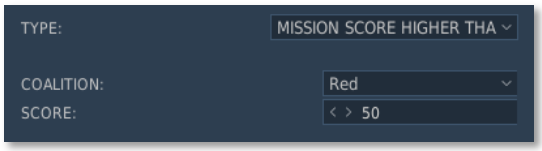


- If you type this text into the condition, the trigger will be triggered each time the specified group sees an adversary.

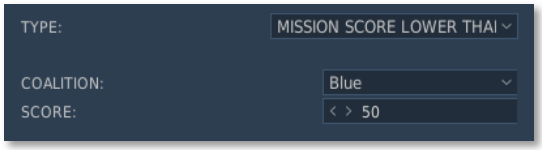
- **MISSILE IN ZONE**. Checks to see if a missile has been launched and is currently in a trigger zone. When choosing this condition, the mission creator must also specify the type and quantity of the missile to be checked, as well as the trigger zone.



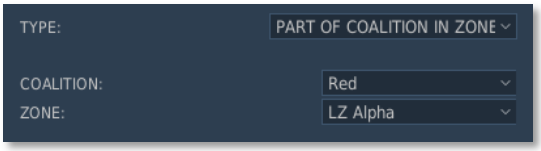
- **MISSION SCORE HIGHER THAN**. If the set coalition has a score higher than the set amount, the action(s) will be activated.



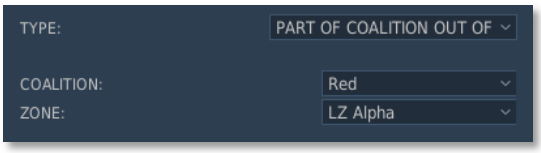
- **MISSION SCORE LOWER THAN.** If the set coalition has a score lower than the set amount, the action(s) will be activated.



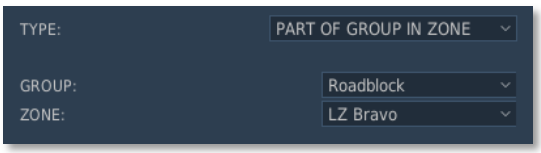
- **PART OF COALITION IN ZONE**. The trigger will be set to true if any of coalition's unit will appear inside the selected trigger zone. The COALITION drop down menu allows you to select the coalition. The ZONE drop down will list all the trigger zones you have created according to the names you created for them.



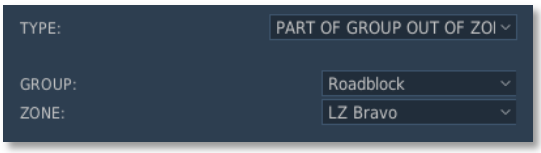
- **PART OF COALITION OUT OF ZONE**. This condition works the same as the PART OF COALITION IN ZONE, but the condition will be set to true when the any of coalition's unit are outside the defined zone.



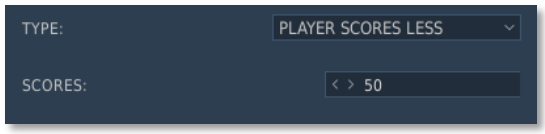
- **PART OF GROUP IN ZONE**. The trigger will be set to true if any of group's unit will appear inside the selected trigger zone. The GROUP drop down menu allows you to select the group. The ZONE drop down will list all the trigger zones you have created according to the names you created for them.



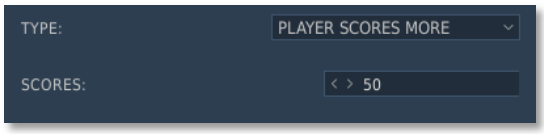
- **PART OF GROUP OUT OF ZONE**. This condition works the same as the PART OF GROUP IN ZONE, but the condition will be set to true when the any of group's unit are outside the defined zone.



- **PLAYER SCORES LESS**. When selected, a SCORES field is displayed below. By either using the left and right arrows or typing in the field, enter the score, below which the condition will become true.



- **PLAYER SCORES MORE**. When selected, a SCORES field is displayed below. By either using the left and right arrows or typing in the field, enter the score, above which the condition will become true.



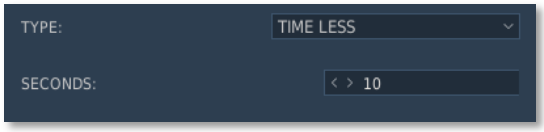
- **PLAYER'S UNIT ARGUMENT IN RANGE**. Trigger rule for using the argument player 3D model's animation within the specified range. When choosing this condition, enter the argument number and its minimum / maximum response values in the additional fields.



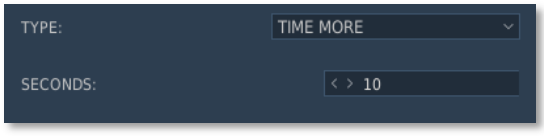
- **RANDOM**. The Random condition allows you to set a percentage chance of action(s) assigned to the trigger from happening. Use the left and right arrows or type directly into the field to adjust the value. Values can range from 0 (no chance) to 100 (certain) as a percentage. The most common action you will use this condition with is Activate Group. This will allow you to set a random appearance of units in a mission.



- **TIME LESS**. When selected, a SECONDS field is displayed below. By either using the left and right arrows or typing in the field, enter the time in seconds when the condition will become false. For example, if you set a value of 120, the condition will become false two minutes into the mission.



- **TIME MORE**. When selected, a SECONDS field is displayed below. By either using the left and right arrows or typing in the field, enter the time in seconds when the condition will become true. For example, if you set a value of 120, the condition will become true two minutes into the mission.



- **TIME SINCE FLAG**. When selected, FLAG and SECONDS fields are displayed below. By either using the left and right arrows or typing in the field, set the flag number that when set to true will set the condition to true in the FLAG field. In the SECONDS field, enter the delay in seconds that the condition will be set to true after the flag is set to true.



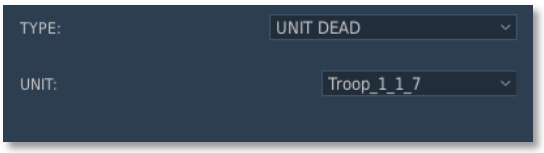
- **UNIT ALIVE**. When selected, a UNIT drop down menu will be displayed below. From this list, select the unit that will activate the trigger as long as the unit is alive. Units are listed according to their "UNIT NAME" from the unit placement windows.



- **UNIT DAMAGED**. When selected, a UNIT drop down menu will be displayed below. From this list, select the unit that will activate the trigger when the unit is damaged. Units are listed according to their "UNIT NAME" from the unit placement windows.



- **UNIT DEAD**. When selected, a UNIT drop down menu will be displayed below. From this list, select the unit that will activate the trigger when the unit is destroyed (dead). Units are listed according to their "UNIT NAME" from the unit placement windows.

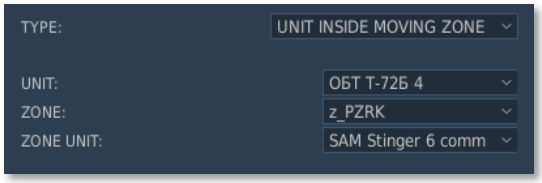


- **UNIT HITS**. Checks to see if the player unit has achieved successful hits on an object. The target object and number of hits required must be specified.

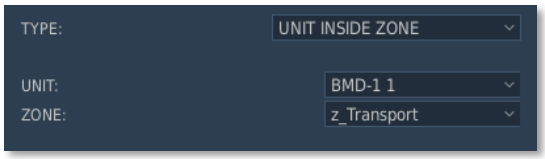


- **UNIT INSIDE MOVING ZONE**. This trigger allows you to attach a trigger area zone to a unit and the zone will move along with the assigned unit. Note, the trigger zone can be placed anywhere on the map, but will in fact be always attached to the center of the

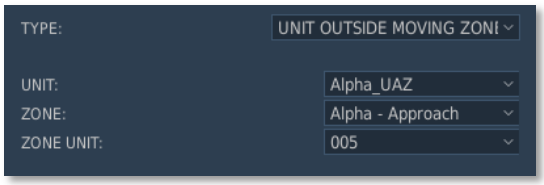
assigned unit in-game. When selected, you should first select the unit that the area trigger will be attached to by selecting it from the ZONE UNIT drop down list. Next, select the area trigger zone that will be attached to the unit from the ZONE drop down list. Lastly, you need to set the unit that will set the trigger to true when it enters the zone unit's zone. Do this by selecting a unit from the UNIT drop down list.



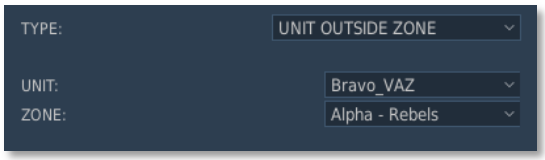
- **UNIT INSIDE ZONE**. When creating a Trigger Zone (discussed later in this document), you can set a trigger to true when a defined unit enters a defined trigger zone. Upon selecting this condition, two new fields will be displayed. The UNIT drop down menu allows you to select the unit that when inside the selected trigger zone will set the condition to true. The ZONE drop down will list all the trigger zones you have created according to the names you created for them.



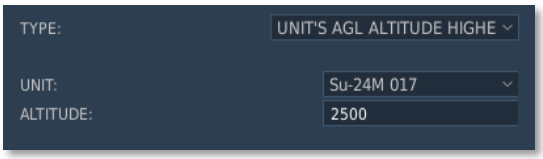
- **UNIT OUTSIDE MOVING ZONE**. This trigger allows you to attach a trigger area zone to a unit and the zone will move along with the assigned unit. Note, the trigger zone can be placed anywhere on the map, but will in fact be always attached to the center of the assigned unit in-game. When selected, you should first select the unit that the area trigger will be attached to by selecting it from the ZONE UNIT drop down list. Next, select the area trigger zone that will be attached to the unit from the ZONE drop down list. Lastly, you need to set the unit that will set the trigger to true when it is outside the zone unit's zone. Do this by selecting a unit from the UNIT drop down list.



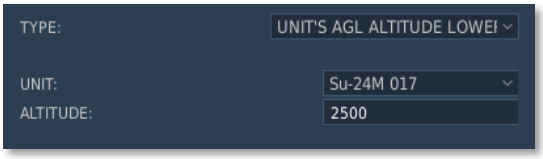
- **UNIT OUTSIDE ZONE**. This condition works the same as the UNIT INSIDE ZONE, but the condition will be set to true when the unit is outside the defined zone.



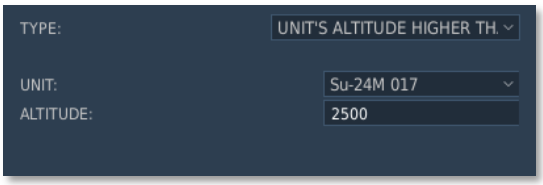
- **UNIT'S AGL ALTITUDE HIGHER THAN**. Checks to see if a unit (aircraft or ground unit) has achieved an altitude higher than a certain value. The checked unit and target altitude must be specified.



- **UNIT'S AGL ALTITUDE LOWER THAN**. Checks to see if a unit (aircraft or ground unit) has achieved an altitude lower than a certain value. The checked unit and target altitude must be specified.



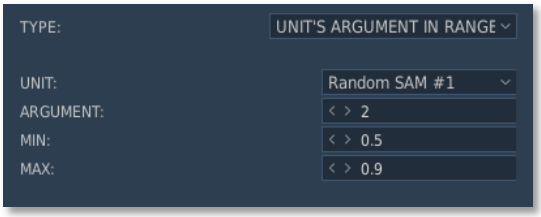
- **UNIT'S ALTITUDE HIGHER THAN**. This condition allows you to set a unit and altitude in meters (MSL). If the set unit flies above the set altitude, the trigger will be set to true.



- **UNIT'S ALTITUDE LOWER THAN**. This condition allows you to set a unit and altitude in meters. If the set unit flies below the set altitude, the trigger will be set to true.



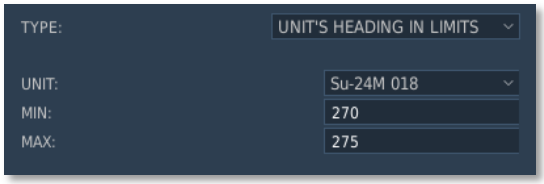
- **UNIT'S ARGUMENT IN RANGE**. Trigger rule for using the argument of the unit 3D model's animation within the specified range. For example, you can determine that the cargo doors on a helicopter are open by changing the animation argument of the cargo doors. When choosing this condition, you must enter the object, the argument number and its minimum / maximum trigger values.



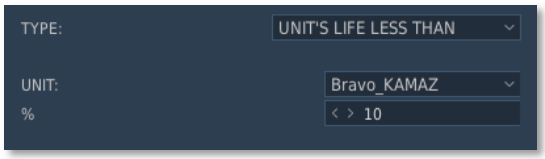
- **UNIT'S BANK IN LIMITS**. When selected, a UNIT drop down list, MIN and MAX fields will be displayed below. From UNIT list, select the unit that will activate the trigger when the bank angle is within the MIN and MAX values set in the corresponding fields. Note, negative angles denote left bank and positive angles denote right bank.



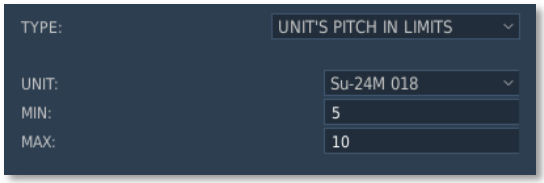
- **UNIT'S HEADING IN LIMITS**. When selected, a UNIT drop down list, MIN and MAX fields will be displayed below. From UNIT list, select the unit that will activate the trigger when the unit's heading is within the MIN and MAX values set in the corresponding fields.



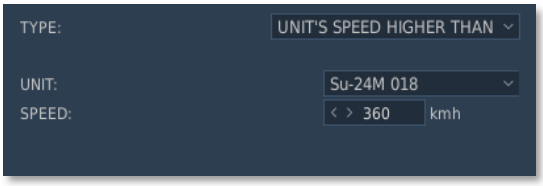
- **UNIT'S LIFE LESS THAN**. When selected, a UNIT drop down list and "%" of life will be displayed below. From the UNIT list, select the unit that will activate the trigger when the per cent of "life" of the unit corresponds to the value set in the "%" field.



- **UNIT'S PITCH IN LIMITS**. When selected, a UNIT drop down list, MIN and MAX fields will be displayed below. From UNIT list, select the unit that will activate the trigger when the pitch angle is within the MIN and MAX values set in the corresponding fields. Note, negative angles denote negative pitch (nose down) and positive angles denote positive pitch (nose up).



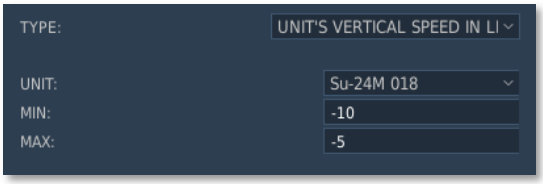
- **UNIT'S SPEED HIGHER THAN**. This condition allows you to set a unit and speed in indicated meters per second. If the set unit flies faster than set speed, the trigger will be set to true.



- **UNIT'S SPEED LOWER THAN**. This condition allows you to set a unit and speed in indicated meters per second. If the set unit flies under than set speed, the trigger will be set to true.



- **UNIT'S VERTICAL SPEED IN LIMITS**. When selected, a UNIT drop down list, MIN and MAX fields will be displayed below. From UNIT list, select the unit that will activate the trigger when the vertical velocity in meters per second is within the MIN and MAX values set in the corresponding fields.



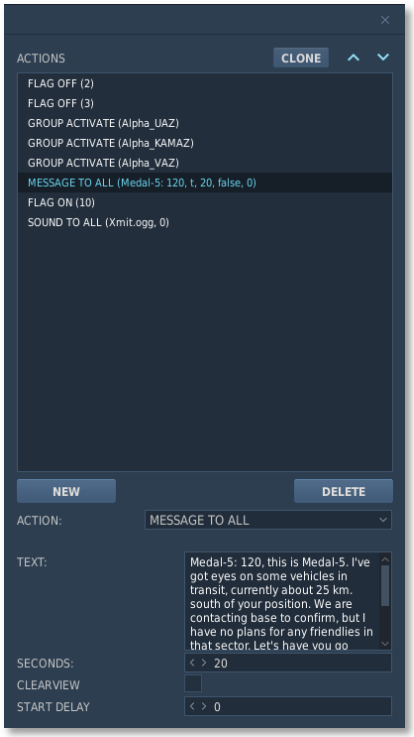
All conditions with the "X" prefix are designed for use in creating cockpit triggers and intended for training missions. See description next.

Trigger Actions

Once you have defined the condition(s) that will determine when a trigger will be true or false, you now define the actions that will result.

To create one or more actions for a trigger, click the NEW button at the bottom of the pane.

With a new action created, new functions will be displayed at the bottom on the actions pane:



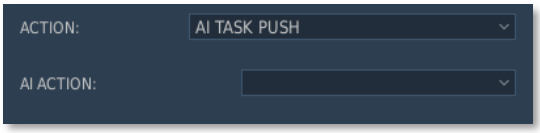
NEW button. Press the NEW button to create a new action for the selected trigger. Please note that you can create multiple actions for a single trigger! When you create a new action, you will need to set the ACTION and define additional data that varies according to the action. Each new action you create will be listed in the actions pane list.

DELETE button. Press the DELETE button to remove an action from the list. You will first need to click on the action you wish to remove before pressing the DELETE button.

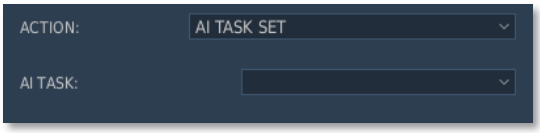
ACTION drop down field. Using this drop down, you can set an action for the trigger.

Actions include:

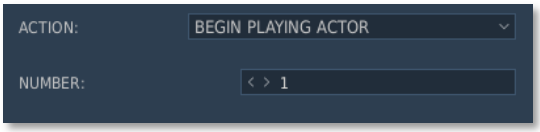
- **AI TASK PUSH**. Performs the action set in the TRIGGER ACTIONS panel while planning tasks for an AI unit group. When creating this condition, the mission creator must set the task to be fulfilled by the AI unit group.



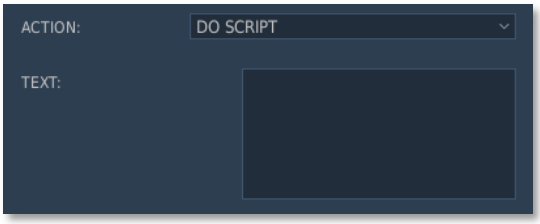
- **AI TASK SET**. An AI action set in the TRIGGERED ACTIONS panel of the AI Group Properties Advanced Actions menu will be executed. The desired AI action identified by its name should be selected from drop down list.



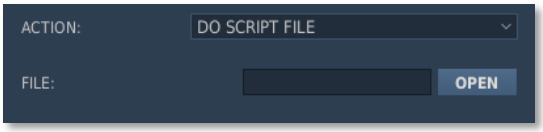
- **BEGIN PLAYING ACTOR**. This action generates an action from the player. When selecting this action, a player number must be set.



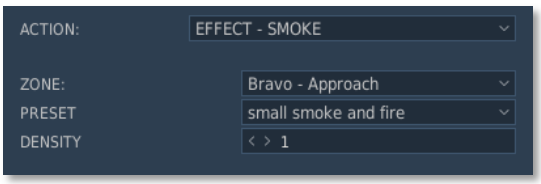
- **DO SCRIPT**. Runs a LUA script as entered in the TEXT filed.



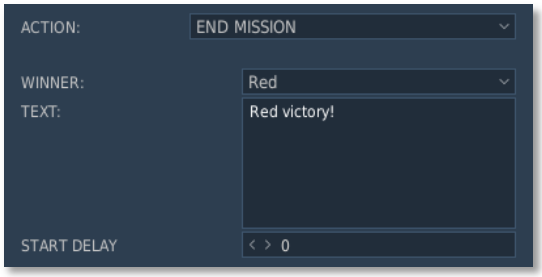
- **DO SCRIPT FILE**. Runs a LUA script file that is linked to in the TEXT field.



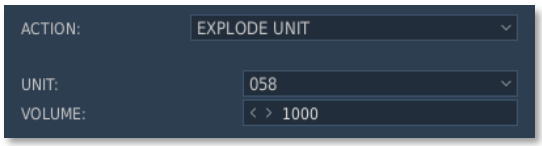
- **EFFECT – SMOKE**. Generates a smoke effect within the trigger zone. A trigger zone, smoke effect preset, and density value must be set when selecting this action.



- **END MISSION**. If you wish to have the mission end according one or more specified conditions being set to true, you can use this action. This can be a handy action to end a mission after all mission objectives have been met. In the text box, you can also enter text that will appear when one side has won the mission.



- **EXPLODE UNIT**. From this action, select a unit that you wish to explode as a result of this action. You can use the volume setting to determine the size of the explosion.



- **EXPLOSION**. To have an explosion appear in the mission based on a resulting action, you can use this action. The three values you need to set include the name of the zone that the explosion will occur in, the altitude the explosion will occur, and the size (volume) of the explosion.



- **FLAG DECREASE**. This action allows you to set a defined flag to decrease in value. Use the FLAG field left and right arrows to set the flag number or manually enter it in the field. Use the VALUE field left and right arrow keys to set the value by which to decrease or manually enter it in the field.



Example. Flag 5 is indicated (with a current value of 10) and given value 4. When the action is triggered, the current value of 10 will be reduced by the given value, 4, and as a result, the resulting value of the flag will be 6.

The flag value may have negative numbers. If, as a result of subtraction, the flag value becomes equal to 0, the flag will be removed.

- **FLAG INCREASE**. This action allows you to set a defined flag to increase in value. Use the FLAG field left and right arrows to set the flag number or manually enter it in the field. Use the VALUE field left and right arrow keys to set the value by which to increase or manually enter it in the field.



Example. Flag 5 is indicated (with current value 1) and given value 10. When the action is triggered, the given value of 10 will be added to the current value of 1, and as a result, the flag value will then be 11.

- **FLAG OFF**. This action allows you to clear a defined flag number to false. Use the FLAG field left and right arrows to set the flag number or manually enter it in the field. The value of the removed flag is 0.



- **FLAG ON**. This action allows you to set a defined flag number to be true. Use the FLAG field left and right arrows to set the flag number or manually enter it in the field.

  In the current implementation, flags have two attributes: the flag name (natural number) and the flag value (natural number). When a flag is created, the name is set in the field labelled FLAG, and the default assigned value is 1.



Example. Flag 5 has a value of 1.

You can increase or decrease the value of the flag with the corresponding actions.

- **FLAG SET RANDOM VALUE**. This action allows you to set a defined flag to a random value. Use the FLAG field left and right arrows to set the flag number or manually enter it in the field. Use the VALUE LIM MIN and VALUE LIM MAX fields left and right arrow keys to set the range of possible values for the flag or manually enter them in the fields.



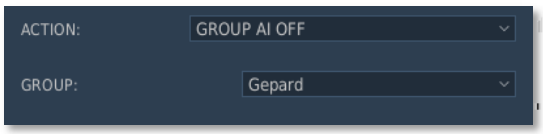
Example. Flag 1 is specified with the minimum value of 10 and maximum value of 20. When the action is triggered, flag 1 will be assigned a random value from 10 to 20, for example 12.

- **GROUP ACTIVATE**. By default, units appear automatically in a mission once the mission starts. However, using the Activate group action, you can set unit groups to

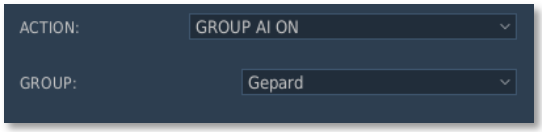
only appear in the mission according to the set conditions. Common rules to govern the appearance of an Activated group are TIME MORE, FLAG IS TRUE, UNIT IN ZONE and RAMDOM. IMPORTANT: FOR A GROUP TO BE ACTIVATED VIA A CONDITION YOU MUST SET THE GROUP'S START TIME BEYOND ANY ANTICIPATED PLAY TIME IN THE MISSION, SUCH AS AT LEAST 1 DAY AFTER THE MISSION STARTS! For example, if the mission starting time is 12:00:00/1 (default mission start time), you can set the group to start at 12:00:11/2.



- **GROUP AI OFF**. Disable the artificial intelligence of all units in the selected group. The group will stop all movement, sensor operations and will not engage in combat.



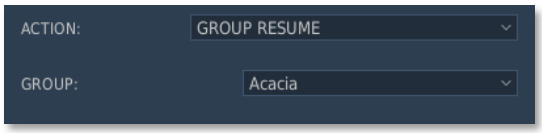
- **GROUP AI ON**. Activate the artificial intelligence of all units in the selected group. Only possible after use of the GROUP AI OFF action on the group.



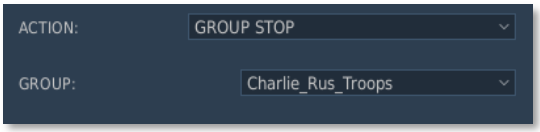
- **GROUP DEACTIVATE**. To have a group removed from a mission while in progress, you can use this action tied to a condition. When enabled, the specified group will disappear from the mission.



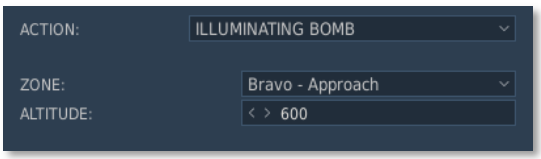
- **GROUP RESUME**. This action orders to group to resume movement along its planned route.



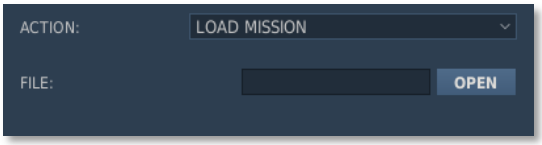
- **GROUP STOP**. This action sends the ground unit group an order to halt movement. A target group must be specified when selecting this action.



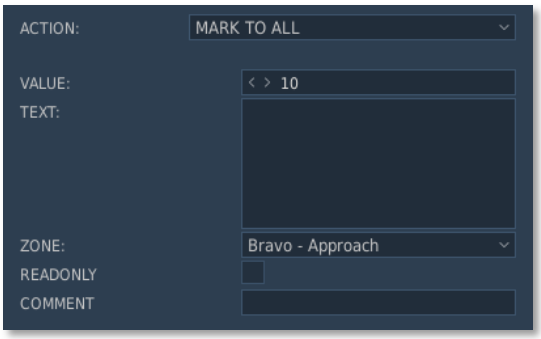
- **ILLUMINATING BOMB**. For night missions, you can use this action to have an illumination bomb appear in a specified zone at a specified altitude. You can use this to simulate artillery fired illumination rounds. The flash bomb provides illumination for a period of 2 minutes, during which it descends to the ground by parachute for a distance of approximately 1000 meters.



- **LOAD MISSION**. This is a very useful action that allows you to load a new mission in multiplayer, as a result of the rule conditions. As such, you could have different missions loaded depending on what is happening in the mission. In this way, you can use this action to create multiplayer campaigns. Working in the multiplayer only.



- **MARK TO ALL**. This action generates visual markers on the F10 map for all players in the session. When selecting this option, a value, text message, zone, read-only value, and comment must be set.



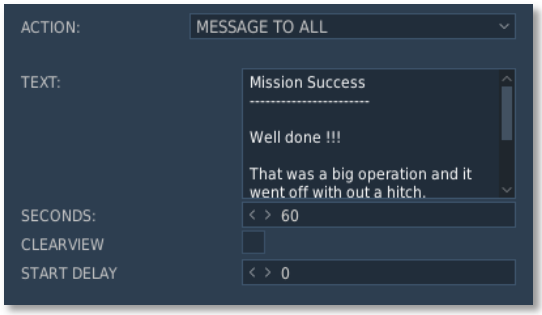
- **MARK TO COALITION**. This action generates visual markers on the F10 map for all players of the selected coalition. When selecting this option, a value, text message, zone, coalition, read-only value, and comment must be set.



- **MARK TO GROUP**. This action generates visual markers on the F10 map for all players of the selected unit group. When selecting this option, a value, text message, zone, group, read-only value, and comment must be set.



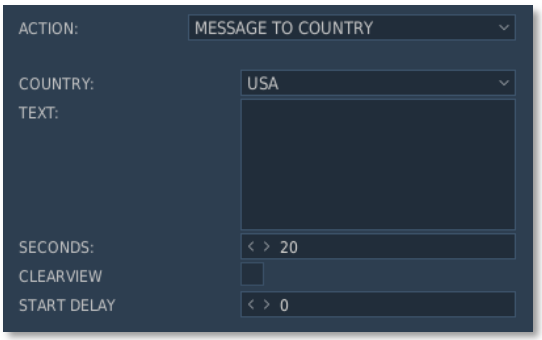
- **MESSAGE TO ALL**. To have a trigger display a text message on the screen, you will use this action. Upon selecting this action, a TEXT entry box will appear. Type the message you wish to be displayed here. Below the text box you can set how long you wish the message to be displayed in seconds.



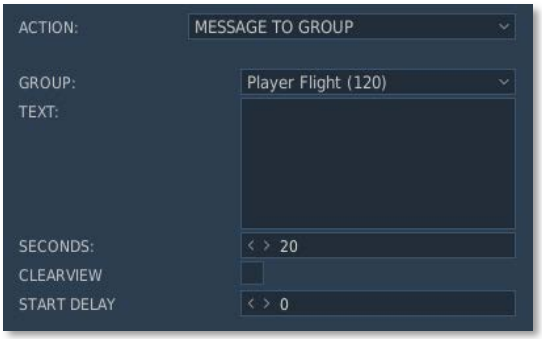
- **MESSAGE TO COALITION**. This action allows a text message to just be sent to the defined coalition. As with the Message action, you can enter text and message duration.



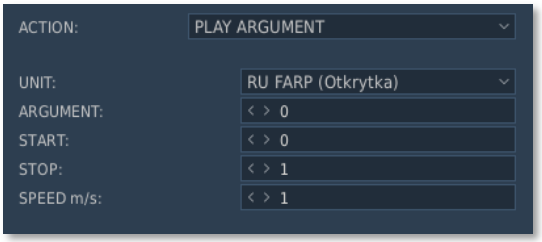
- **MESSAGE TO COUNTRY**. This action allows a text message to just be sent to the defined country. As with the Message action, you can enter text and message duration.



- **MESSAGE TO GROUP**. Display a custom text message only to a set unit group (useful for multiplayer).



- **PLAY ARGUMENT**. Turn on a graphics animation argument (such as turning of a radar antenna) of the selected static object.



- **RADIO ITEM ADD**. This action allows you to add a custom radio message to the player's radio communications menu in-game. Enter the desired radio message in the RADIO TEXT field. In the FLAG field, set the flag number that will be set to ON when the player selects this radio message.



Note, this trigger does not work in multiplayer.

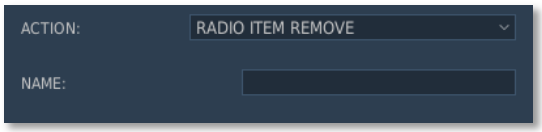
- **RADIO ITEM ADD FOR COALITION**. This action adds a radio item (command) in the F10 OTHER radio menu for players of the chosen coalition. When creating this action, a coalition and item name must be set, as well as the flag the value of which will be raised when the command is issued.



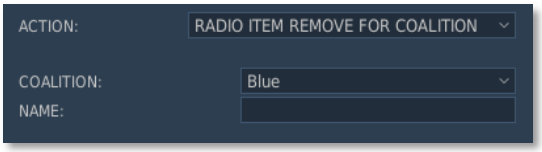
- **RADIO ITEM ADD FOR GROUP**. This action adds a radio item (command) in the F10 OTHER radio menu for players of the chosen unit group. When creating this action, a unit group and item name must be set, as well as the flag the value of which will be raised when the command is issued.



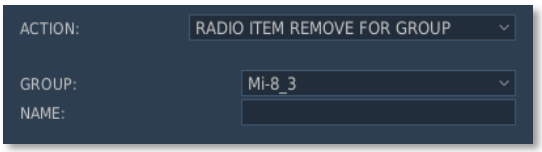
- **RADIO ITEM REMOVE**. Delete a custom radio message from the player's radio communications menu created earlier using the RADIO ITEM ADD action.



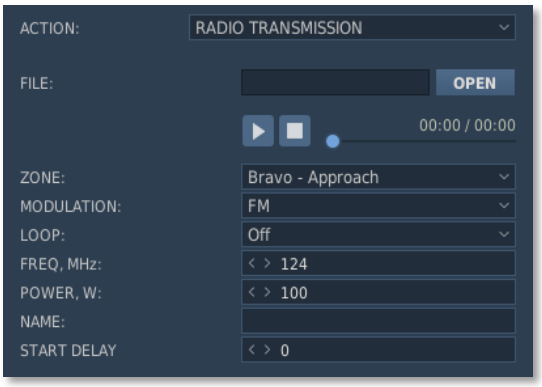
- **RADIO ITEM REMOVE FOR COALITION**. This action removes a radio item from the radio menu for the selected coalition. When creating this action, a coalition must be selected, and radio item to be removed must be specified.



- **RADIO ITEM REMOVE FOR GROUP**. This action removes a radio item from the radio menu for the selected unit group. When creating this action, a unit group must be selected, and radio item to be removed must be specified.



- **RADIO TRANSMISSION.** Allows the user to define a sound file to play and the transmitter's location, emission power, frequency and modulation.



ZONE: From the drop-down menu, select the zone in which the transmission will be broadcast.

MODULATION: From the drop-down menu, select a modulation value for the FM or AM radio station.

LOOP: Select the playback mode of the file: single (OFF) or multiple (ON).

FREQ, MHz: Specify the frequency of the radio station. To listen in on this frequency, it must be included in the operating range of the player aircraft's radio station.

POWER, W: specify the power of the radio station. The more powerful the radio station, the further it can be heard. The typical power of military radio stations is tens of watts.

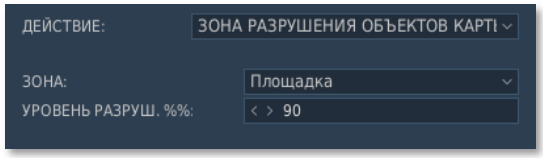
NAME: sets the name of the radio transmitter.

START DELAY: sets the delay in seconds before turning on the radio.

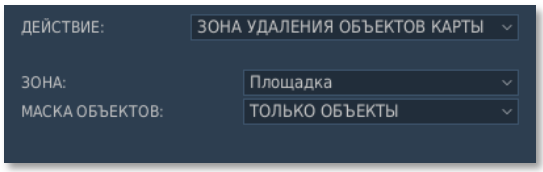
- **REMOVE MARK**. Removes a marker from the F10 map. A value must be specified when selecting this action.



- **SCENERY DESTRUCTION ZONE**. This action reproduces the destruction of zone objects, for example, after an artillery barrage. When selecting this action, it is necessary to select a target zone and the level of destruction (in %) of the objects within this zone.



- **SCENERY REMOVE OBJECTS ZONE**. This action despawns objects from the map, for example, in order to lessen the system load caused by rendering these objects after an artillery barrage or aerial bombardment. When selecting this action, the following values must be set: target zone (area in which objects will be despawned) and target object type (ALL OBJECTS, TREES ONLY, STRUCTURES ONLY).



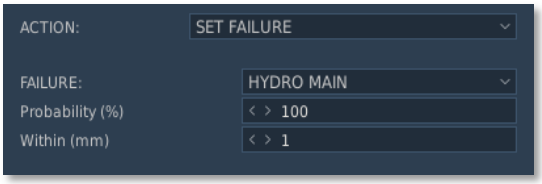
- **SET ACTIVE HELPER GATE TO POINT**. This action generates arcade-style flight gates to the selected object's waypoint. These are green squares through which the player needs to fly, typically present in training and certain quick missions. When choosing this action, you must specify the target object (unit) and the waypoint number.



- **SET BRIEFING**. This action This action displays a briefing message in the game window for the selected coalition players (RED or BLUE). When creating this action, you must select a coalition in the appropriate fields, enter the briefing text to be displayed and if necessary, select one image file to be displayed. (*.jpg; *.jpeg; *.png).



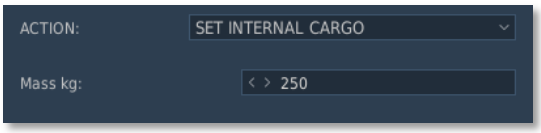
- **SET FAILURE**. This action is used to setup system failures for the player's aircraft. Select the system to fail in the FAILURE drop down menu. Set the probability of failure in per cent in the PROBABILITY % field. Set the desired time span after the trigger conditions are true for the failure to occur in the WITHIN field.



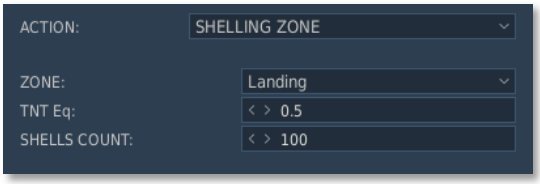
- **SET FLAG VALUE**. This action sets the selected flag to a specified value. The flag number and the target value must be specified when selecting this action.



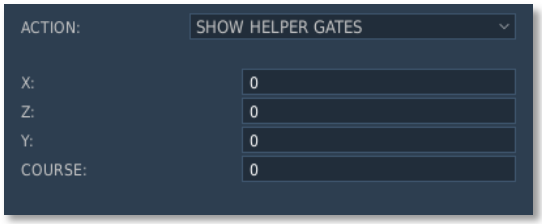
- **SET INTERNAL CARGO**. This action adds the mass of the virtual cargo to the mass of the helicopter simulating the loading of the cargo compartment. When creating this action, you must specify the mass of the cargo in kilograms.



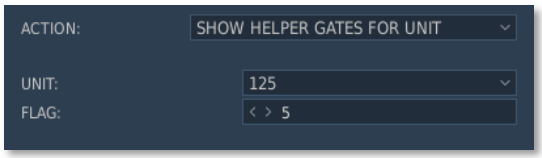
- **SHELLING ZONE**. This action creates an artillery bombardment event inside a specified zone. When creating this action, the following must be specified in the additional fields: bombardment zone, amount of TNT released during the explosion of one shell, and the number of shells that will strike the target zone.



- **SHOW HELPER GATES**. This action generates arcade-style flight gates - green squares through which the pilot must fly. These elements are in some training missions and quick mission presets. When creating this action, you must specify the location coordinates and the course direction of the gates.



- **SHOW HELPER GATES FOR UNIT**. This action generates arcade-style flight gates for the selected unit. When choosing this action, you must select the unit (object) and the flag number.



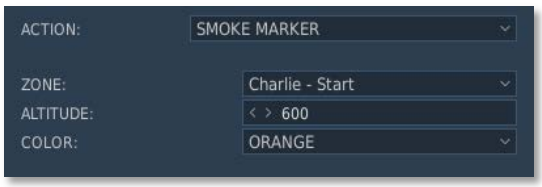
- **SIGNAL FLARE**. Use this action to have signal flares (flare gun) fired from a specified zone. You can also select the color of the flare. This can be a useful action to allow friendly units to mark their location to avoid friendly fire.



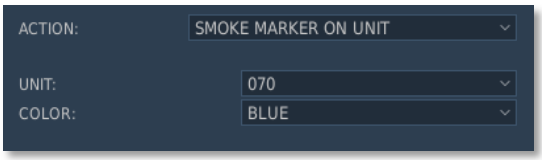
- **SIGNAL FLARE ON UNIT**. This works the same as the previous action, but the signal flare is fired from the location of a specified unit. This should be used when the unit is moving.



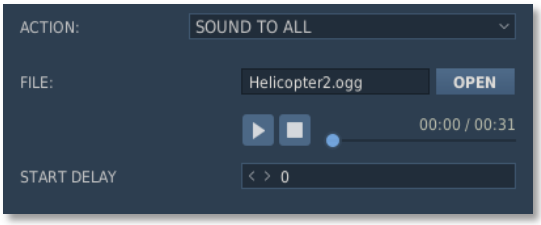
- **SMOKE MARKER**. To have a white phosphorous (WP) marker appear in the mission, you can use this action. When activated, a white WP marker will appear in the specified zone and at the specified altitude. This can be a useful tool when creating Forward Air Controller (FAC) missions.



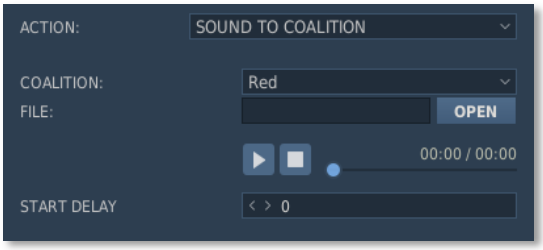
- **SMOKE MARKER ON UNIT**. This acts the same as the previous action, but instead of occurring within a specified zone, it will appear on a specified unit.



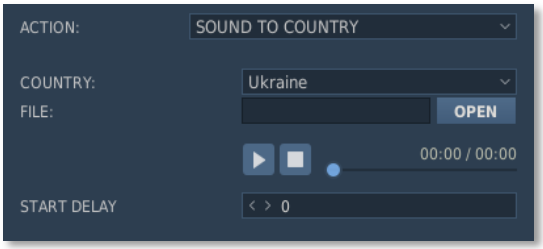
- **SOUND TO ALL**. The Sound action enables you to add an audio file to play as a trigger function. This is often used in conjunction with a Message action to add voice messages with subtitles to the mission. Once selected, a FILE field will be displayed along with an OPEN button. Press the open button and then use the file/folder browser to select the desired file ("*.ogg" or "*.wav" formats). The name of the audio file will then be displayed in the FILE field.



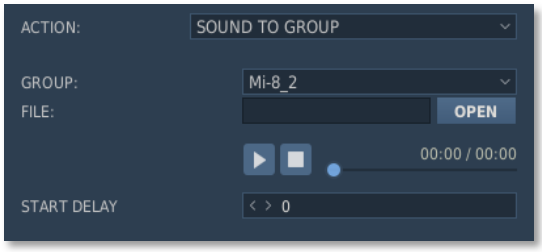
- **SOUND TO COALITION**. This action allows an audio file to just be play to the defined coalition. Once selected, a FILE field will be displayed along with an OPEN button. Press the open button and then use the file/folder browser to select the desired file ("*.ogg" or "*.wav" formats). The name of the audio file will then be displayed in the FILE field.



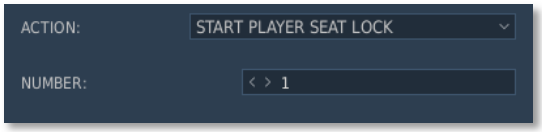
- **SOUND TO COUNTRY**. This action allows an audio file to just be play to the defined country.



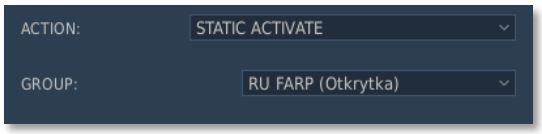
- **SOUND TO GROUP**. Play a custom sound ("*.ogg" or "*.wav" formats) only to a set group (useful for multiplayer).



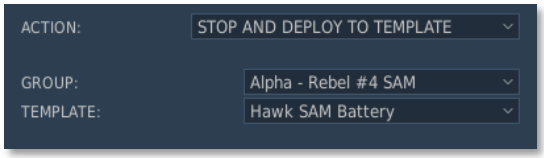
- **START PLAYER SEAT LOCK**. Used for multi-seat aircraft, this action forces the player to the spot corresponding to the specified number and binds them to it. When creating this action, you must the pilot's cockpit seat number.



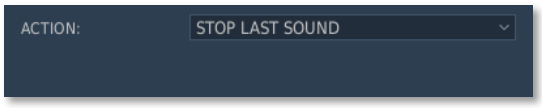
- **STATIC ACTIVATE**. This action generates the delayed activation of static objects that are present in the mission file but will not spawn in the mission until the moment of activation. When creating this action, you must specify the group of static objects.



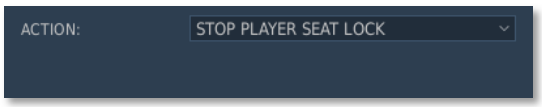
- **STOP AND DEPLOY TO TEMPLATE**. Using this action, an AI group can be made to stop movement along its route and take up a position as defined by a formation template (for example, an artillery battle position). To create a custom template, see the Templates panel in the Tool Bar of the mission editor.



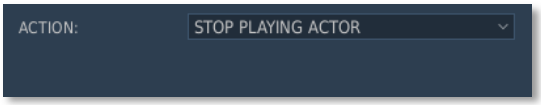
- **STOP LAST SOUND**. This action forcibly stops the audio played by the previous trigger.



- **STOP PLAYER SEAT LOCK**. This action removes the bind previously set by the START PLAYER SEAT LOCK command, giving the chosen player the opportunity to transfer to another seat in a multi-seat aircraft.



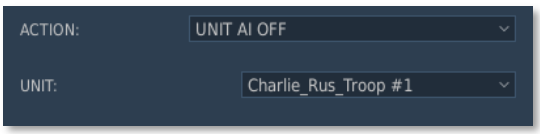
- **STOP PLAYING ACTOR**. The action automatically ends on behalf of the player.



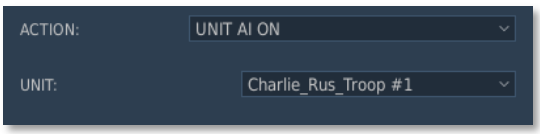
- **STOP RADIO TRANSMISSION**. This action turns off the virtual radio station playing sound files to any unit (air, ground, sea). When creating this action, you must specify the name of the radio transmitter.



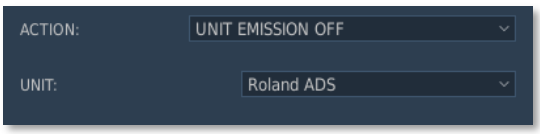
- **UNIT AI OFF**. Disable the artificial intelligence of the selected unit. The unit will stop all movement, sensor operations and will not engage in combat.



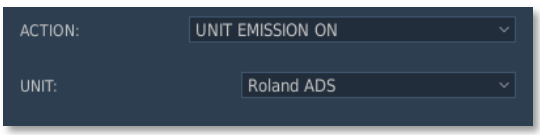
- **UNIT AI ON**. Activate the artificial intelligence of the selected unit. Only possible after use of the UNIT AI OFF action on the unit.



- **UNIT EMISSION OFF**. This action switches on radar emission from the specified unit or aircraft, for example, when the unit is approaching an enemy unit or entering a combat zone. When choosing this action, you must select a unit.



- **UNIT EMISSION ON**. This action switches off radar emission from the specified unit or aircraft, for example, when the unit is distancing from enemy units or exiting the combat zone. When choosing this action, you must select a unit.



All actions with the "X" prefix are designed for use in creating cockpit triggers and intended for training missions. See description next.

Cockpit Triggers Guide

The cockpit triggers utilized in ED Simulators allow a mission designer to create a wide variety of triggers that evaluate the state of cockpit switches and instruments. This data can be used to provide feedback and instruction to the player using text and voice prompts, as well as 3D graphical highlights of cockpit elements.

Although cockpit triggers are considered a developer's tool, most of their functionality is open to the player. There are some examples, however, which require the use of the in-game console, which is disabled in public builds.

Cockpit triggers are utilized similarly for all DCS flyable aircraft. A number of aircraft script files (.LUA) are used to obtain data used by the triggers. Each DCS aircraft includes a collection of script files in its own dedicated folder, found in X:\DCS\Mods\aircrafts\Aircraft Name\Cockpit\Scripts. The specific .LUA files used by cockpit triggers will be referenced below. To ease the workload of a mission designer working with cockpit triggers, a cockpit arguments guide has been prepared for the A-10C cockpit and can be found here: A-10C Cockpit arguments.

.LUA files should only be opened with a proper code editor, such as Notepad++. When working with the scripts, the Search function [Ctrl +F] can be very helpful in locating the desired sections of code.

CONDITIONS

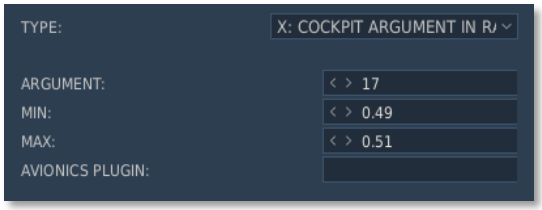
The conditions and actions of the triggers use the command numbers, which can be found in the corresponding Scripts folder of each aircraft along the path DCS\Mods\aircraft\Aircraft Name\Cockpit\Scripts:

\Mods\aircraft\Aircraft Name\Cockpit\Scripts\command_defs.lua – command numbers;

\Mods\aircraft\Aircraft Name\Cockpit\Scripts\clickabledata.lua – button command numbers;

\Mods\aircraft\Aircraft Name\Cockpit\Scripts\devices.lua – device ID numbers in the cockpit.

X: **COCKPIT ARGUMENT IN RANGE**. The position of every cockpit control and instrument is defined by a custom argument ID and valid value range. This trigger checks for a set cockpit argument to be within a set value range. This is most often used to evaluate switch and button states with discrete argument values (such as '0' for *off* and '1' for *on* or in decimal values between 0 and 1 for multi-position switches). However, it can also be used to evaluate the position of needle instruments, such as airspeed or vertical velocity, etc.

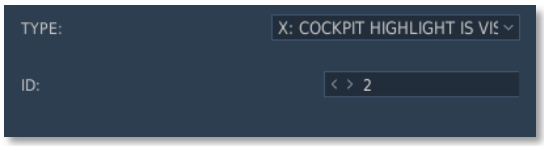


- **ARGUMENT** – sets the ID number of the cockpit argument to check. Clickable cockpit functions (switches, buttons, knobs, etc.) are described in *DCS\Mods\aircrafts\Aircraft Name\Cockpit\Scripts\clickabledata.lua*. Non-clickable cockpit arguments, such as instruments are described in *DCS\Mods\aircrafts\Aircraft Name\Cockpit\Scripts\mainpanel_init.lua*. The variable "*arg"* describes the argument ID. The variable "*arg_value"* describes the increment of each movement of the argument. The variable "*arg_limit"* describes the limits of the argument's range of movement.
- **MIN** – sets the minimum value of the set argument, which will satisfy the condition to execute the trigger. Valid value range is -1 to 1.
- **MAX** – sets the maximum value of the set argument, which will satisfy the condition to execute the trigger. Valid value range is -1 to 1.

Note, to successfully evaluate an argument, you may need to enter a MIN and MAX range that is inclusive of the desired value. For example, when evaluating an argument to equal to "0.5", enter "0.49" for MIN and "0.51" for MAX.

X: **COCKPIT HIGHLIGHT IS VISIBLE.** Checks for a set cockpit highlight to be displayed. Each cockpit highlight is assigned an ID by the mission designer. This trigger evaluates whether a specific highlight is displayed in the cockpit.

- **ID** – sets the ID value of the cockpit highlight to check (as set by X: COCKPIT HIGHLIGHT ELEMENT/INDICATION/POINT trigger actions).



X: **COCKPIT INDICATION TEXT IS EQUAL TO** (requires console) – checks the contents of a set cockpit digital indication string. Digital indicators in the cockpit are individually identified by an ID number in the .LUA files. Each indicator displays a collection of display elements. This trigger evaluates whether a specific display element of a set indicator matches the desired string of characters. This is a very useful method of evaluating whether the correct avionics parameters are set-up on the digital displays.

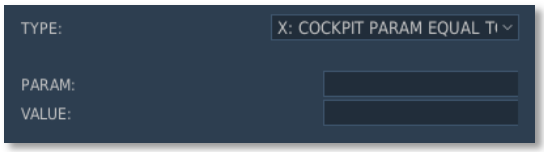
- **INDICATOR ID** – sets the ID value of the cockpit indicator containing the desired element. Indicator ID's are described in *DCS|Mods|aircrafts|Aircraft Name|Cockpit|Scripts|device_init.lua*. To quickly locate the indicator ID's in the .LUA file, perform a search function for "indicators". Indicator ID's begin with '0' at the top of the table and are increased by 1 with each subsequent indicator in the table. For example, the HUD ID for A-10C is '5'.

- **ELEM NAME** (element name) – sets the name of the display element containing the desired string. To obtain an element name, run a mission with the desired element string displayed in the cockpit, open the game console and execute the "*console.out(list_indication(n))"* command (where '*n*' is the desired indicator's ID). The game console can be accessed by pressing the "`" command in-game. The *console.out(list_indication(n))* console command will list all currently displayed elements with their associated strings for the set indicator. Note, the element name entry must match the element name exactly, including capitals and spaces (spaces may be present as initial characters).
- **VALUE** – sets the character string, which will satisfy the condition to execute the trigger. Note, the string entry must match the element string exactly, including capitals and spaces (spaces may be present as initial characters).



EXAMPLE: To setup a trigger in A-10C that checks for the TAD map range to be set to 40 miles, enter "1" for indicator ID to set this to the left MFCD, enter "TAD_scale" as the element name and enter "40" as the desired string.

X: **COCKPIT PARAM EQUAL TO** (requires console) – checks for the desired setting of a set cockpit parameter. This trigger evaluates a number of specially-designed cockpit parameter trackers for the A-10C (such as TGP zoom level, steerpoint name, etc.).



- **PARAM** (parameter) – sets the name of the parameter to check. To obtain a parameter name, run a mission with the desired parameter active, open the game console and execute the "*console.out(list_cockpit_params())*" command. The game console can be accessed by pressing the "`" command in-game. The *console.out(list_cockpit_params())* console command will display the current states of all cockpit parameters.
- **VALUE** – sets the parameter setting, which will satisfy the condition to execute the trigger. Note, depending on the parameter, the value may be a numerical or alphabetical string.

X: **COCKPIT PARAM IN RANGE** (requires console) – checks for the set cockpit parameter to be within the set value range. This trigger evaluates a number of specially-designed cockpit parameter trackers (such as TGP zoom level, steerpoint name, etc.).

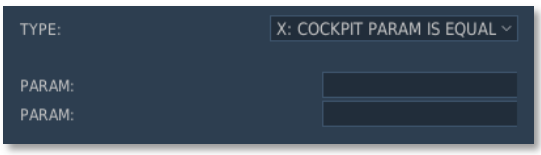
- **PARAM** (parameter) – sets the name of the parameter to check. To obtain a parameter name, run a mission with the desired parameter in effect, open the game console and execute the "*console.out(list_cockpit_params())*" command. The game console can be

accessed by pressing the "`" command in-game. The *console.out(list_cockpit_params())* console command will display the current states of all cockpit parameters.

- **MIN** – sets the minimum value of the set parameter, which will satisfy the condition to execute the trigger. Valid value range is -1^9 to 1^9.
- **MAX** – sets the maximum value of the set parameter, which will satisfy the condition to execute the trigger. Valid value range is -1^9 to 1^9.



X: **COCKPIT PARAM IS EQUAL TO ANOTHER** – checks for the set parameter value to be identical to another parameter's value. This trigger can be used in conjunction with the X: COCKPIT PARAM SAVE AS action in order to compare parameter values at different points in the mission.



- **PARAM** (parameter) – sets the name of the first parameters to evaluate. To obtain a parameter name, run a mission with the desired parameter in effect, open the game console and execute the "*console.out(list_cockpit_params())*" command. The game console can be accessed by pressing the "`" command in-game. The *console.out(list_cockpit_params())* console command will display the current states of all cockpit parameters.
- **PARAM** (parameter) – sets the name of the second parameter to evaluate.

Note, one of the two parameter variables may be a custom parameter name saved using the X: COCKPIT PARAM SAVE AS action.

ACTIONS

The conditions and actions of the triggers use the command numbers, which can be found in the corresponding Scripts folder of each aircraft along the path DCS\Mods\aircraft\Aircraft Name\Cockpit\Scripts:

\Mods\aircraft\Aircraft Name\Cockpit\Scripts\command_defs.lua – command numbers;

\Mods\aircraft\Aircraft Name\Cockpit\Scripts\clickabledata.lua – button command numbers;

\Mods\aircraft\Aircraft Name\Cockpit\Scripts\devices.lua – device ID numbers in the cockpit.

X: **COCKPIT HIGHLIGHT ELEMENT** – enables a cockpit highlight of the specified cockpit control (switch, button, knob, etc.).

- **ID** – sets the ID of the highlight. The ID can be referenced by other triggers to evaluate the status of or remove the highlight.
- **ELEM NAME** (element name) – sets the name of the cockpit control to highlight. Clickable cockpit functions (switches, buttons, knobs, etc.) are described in *Scripts*|*Aircrafts*|*Aircraft Name*|*Cockpit*|*clickabledata.lua*.
- **SIZE OF BOX** – sets the span of the highlight in meters. A setting of '0' will automatically conform the highlight to the 3D shape of the cockpit control.



X: **COCKPIT HIGHLIGHT INDICATION** (required console) – enables a cockpit highlight of the specified digital indication element (such as HUD display elements).

- **ID** – sets the ID of the highlight. The ID can be referenced by other triggers to evaluate the status of or remove the highlight.
- **INDICATOR ID** – sets the ID value of the cockpit indicator containing the desired indication. Indicator ID's are described in *DCS*|*Mods*|*aircrafts*|*Aircraft Name*|*Cockpit*|*Scripts*|*device_init.lua*. To quickly locate the indicator ID's in the lua file, perform a search function for "indicators". Indicator ID's begin with '0' at the top of the table and are increased by 1 with each subsequent indicator in the table. For example, the HUD ID for A-10C is '5'.
- **ELEM NAME** (element name) – sets the name of the display element to highlight. To obtain an element name, run a mission with the desired element string displayed in the cockpit, open the game console and execute the "*console.out(list_indication(n))*" command (where '*n*' is the desired indicator's ID). The game console can be accessed by pressing the "`" command in-game. The *console.out(list_indication(n))* console command will list all current element strings of the set indicator.
- **SIZE OF BOX** – sets the span of the highlight in meters.



X: **COCKPIT HIGHLIGHT POINT** – enables a cockpit highlight centered at the specified 3D cockpit coordinates. This action allows you to specify the 3D dimensions of the highlight, which

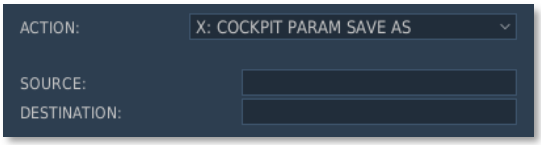
can be useful for creating "flat" 2D highlights of panel instruments in either horizontal or vertical planes.

- **ID** – sets the ID of the highlight. The ID can be referenced by other triggers to evaluate the status of or remove the highlight.
- **X (FWD/AFT)** – sets the highlight position on the X (forward/aft) axis of the cockpit in meters.
- **Y (UP/DN)** – sets the highlight position on the Y (up/down) axis of the cockpit in meters.
- **Z (RGT/LFT)** – sets the highlight position on the Z (right/left) axis of the cockpit in meters.
- **BOX DIM, X** – sets the span of the highlight's X axis in meters.
- **BOX DIM, Y** – sets the span of the highlight's Y axis in meters.
- **BOX DIM, Z** – sets the span of the highlight's Z axis in meters.



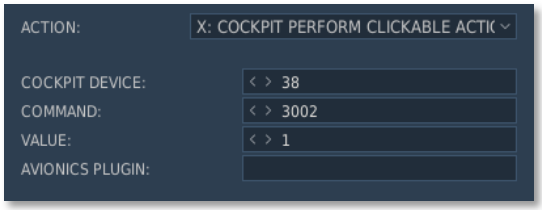
X: **COCKPIT PARAM SAVE AS** (requires console) – save the current value of the specified cockpit parameter as a new parameter. This trigger allows you to save the state of a parameter for later evaluation during the mission.

- **SOURCE** – sets the name of the parameter for which the value will be saved. To obtain a parameter name, run a mission with the desired parameter in effect, open the game console and execute the "*console.out(list_cockpit_params())*" command. The game console can be accessed by pressing the "`" command in-game. The *console.out(list_cockpit_params())* console command will display the current states of all cockpit parameters.
- **DESTINATION** – sets the name of the new parameter, under which the value of the saved parameter will be stored while the mission is running.



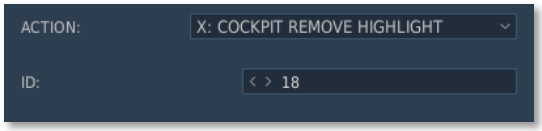
X: **COCKPIT PERFORM CLICKABLE ACTION** – sets the specified cockpit control to the specified position. This trigger can be used to force-set a required cockpit control (for example to engage autopilot mode).

- **COCKPIT DEVICE** – sets the ID value of the cockpit device manipulated by the required control. Device ID's are described in *DCS|Mods|aircrafts|Aircraft Name|Cockpit|Scripts|devices.lua*. Device ID's begin with '1' at the top of the table and are increased by 1 with each subsequent device in the table. For example, the ID for *autopilot* is '38'.
- **COMMAND** – sets the command value for the required control, starting with a base value of 3000. Command values are described in *DCS|Mods|aircrafts|Aircraft Name|Cockpit|Scripts|clickabledata.lua*, using the argument "*device_commands.Button_n"* (where '*n*' is the command value of interest). To quickly locate a desired command in the .LUA file, perform a search for relevant terms (such as "autopilot" or "MFCD"). For example, the command to engage the autopilot in A-10C is scripted as "*device_commands.Button_2"*. As such, it would be entered in the Command field as "3002".
- **VALUE** – sets the argument value (control position in the cockpit). Control arguments are described in *DCS|Mods|aircrafts|Aircraft Name|Cockpit|Scripts|clickabledata.lua.* The variable "*arg"* describes the argument ID (not necessary for this trigger). The variable "*arg_value"* describes the increment of each movement of the argument. The variable "*arg_limit"* describes the limits of the argument's range of movement.

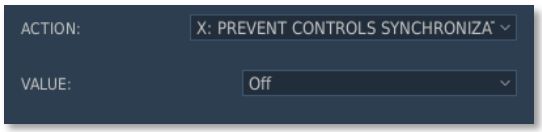


X: **COCKPIT REMOVE HIGHLIGHT** – removes the specified cockpit highlight.

- **ID** – sets the ID of the cockpit highlight to remove.



X: **PREVENT CONTROLS SYNCHRONIZATION** – This trigger can be used at the start of the mission to disable synchronization of cockpit controls with the actual controls on the player's input hardware. When enabled in the player's *Options > Miscellaneous* tab, control synchronization may negatively affect missions with cockpit triggers by automatically setting cockpit controls to positions other than default.



X: **SET COMMAND** – inputs the specified game command.

- **COMMAND** – sets the ID value of the command to input. The command ID can be obtained in the *OPTIONS>CONTROLS* screen by hovering the mouse over the command description (left side of the screen). This functionality can be enabled in *Mission Editor/Mission Editor.lua* by changing the *"OPTIONS_ADD_COMMAND_CODES_TO_TOOLTIP"* line from *"false"* to *"true"*.



X: **SET COMMAND WITH VALUE** – inputs the specified game command with the specified value.

- **COMMAND -** sets the value of the command to input. The command value can be obtained in the *Game Options>Controls* screen by hovering the mouse over the command description (left side of the screen). This functionality can be enabled in *Mission Editor/Mission Editor.lua* by changing the *"OPTIONS_ADD_COMMAND_CODES_TO_TOOLTIP"* line from *"false"* to *"true"*.
- **VALUE** – sets the value with which to enter the command.

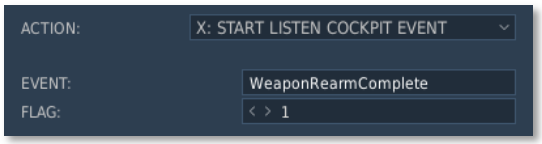


X: **START LISTEN COCKPIT EVENT** – start listening for the specified cockpit event to set the specified flag to *on*. Cockpit events are limited to the Ka-50.

- **EVENT** – sets the event name to listen to. The following events are currently scripted in the engine.
  - "setup_HMS"
  - "setup_NVG"
  - "DisableTurboGear"
  - "GroundPowerOff"
  - "EnableTurboGear"
  - "GroundPowerOn"
  - "OnNewNetHelicopter"
  - "initChaffFlarePayload"
  - "switch_datalink"
  - "OnNewNetPlane"
  - "WeaponRearmFirstStep"
  - "WeaponRearmComplete"

- "initChaffFlarePayload"
- "repair"
- "LinkNOPtoNet"
- "UnlinkNOPfromNet"

- **FLAG** – sets the flag number to set to *on* when the specified event occurs.



X: **START LISTEN COMMAND** – start listening for the specified input command to set the specified flag to *on*.

- **COMMAND** – sets the value of the command to listen to. The command value can be obtained in the *Game Options>Controls* screen by hovering the mouse over the command description (left side of the screen). This functionality can be enabled in *Mission Editor/Mission Editor.lua* by changing the *"OPTIONS_ADD_COMMAND_CODES_TO_TOOLTIP"* line from *"false"* to *"true"*.
- **FLAG** - sets the flag number to set to *on* when the specified event occurs.
- **HIT COUNT** – sets the number of times the command must be entered before the specified flag is set to *on*.
- **VAL LIM MIN** – sets the minimum value of the command input.
- **VAL LIM MAX** – sets the maximum value of the command input.
- **COCKPIT DEVICE** – cockpit system ID number.

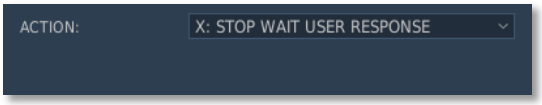


X: **START WAIT USER RESPONSE** – start listening for input of the SPACE BAR key to set the specified flag to *on*. This can be a convenient way to progress a training mission based on the player's signal. Using this trigger will prevent an input command normally mapped to the Space Bar key (usually fire gun).

- **FLAG** – sets the flag number to set to *on* when the SPACE BAR key is pressed while the action is in effect.



X: **STOP WAIT USER RESPONSE** – stop listening for input of the SPACE BAR key. This trigger disables the START WAIT USER RESPONSE action.



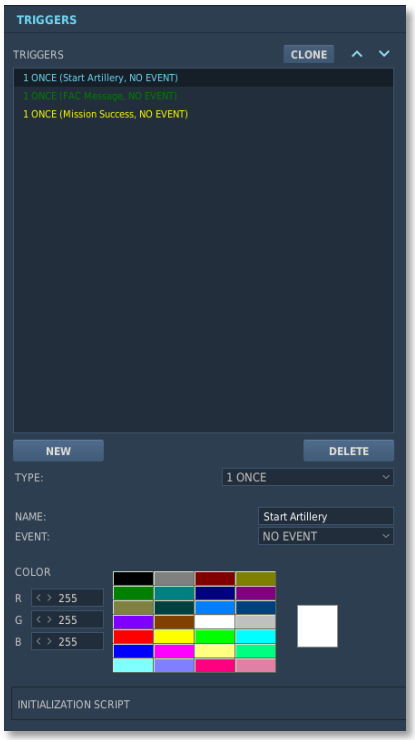
This action can be useful when creating mission scenarios where the player is to command AI groups to perform various actions, such as advance forward or hold position, by binding the desired AI actions to the flag number of the corresponding custom radio message.

Trigger Practical Exercise

Now that we have reviewed the mechanics of the Trigger system, we will create a basic mission using trigger scripting. In this mission, we will do the following:

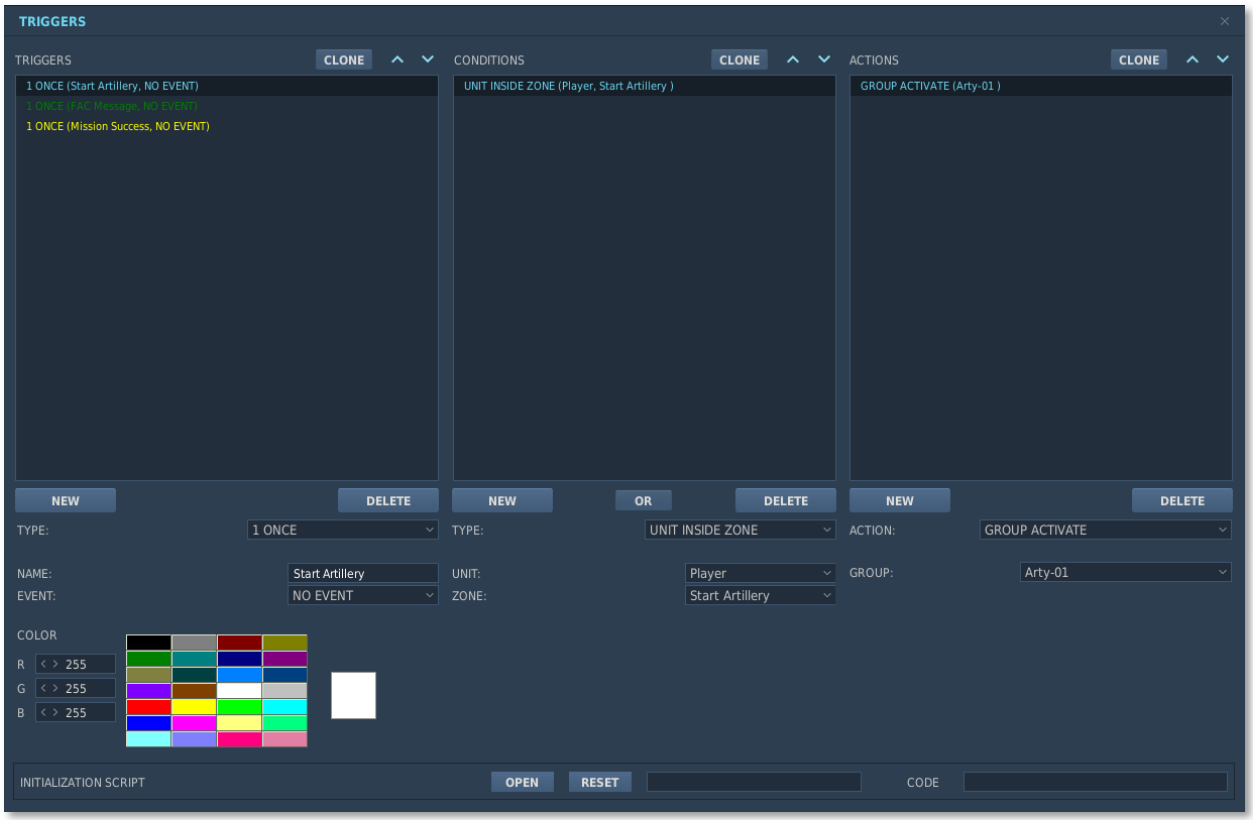
1. Player will fly to battle area, and when nearing this area, friendly artillery will bombard an enemy position.
2. As the player flies closer, they will receive a Forward Air Controller message about tasked targeting.
3. When the player destroys enemy armor in the target area, friendly ground forces will advance.
4. When friendly forces reach their objective, they will radio the player of a successful mission.

The first thing we will do is set up the Zone triggers. So, I will create three different trigger zones, and I will call these Start Artillery, FAC Message, and Mission Success. I will set the TYPE as ONCE for each of them.



I will now set up the conditions and actions for each of these triggers.

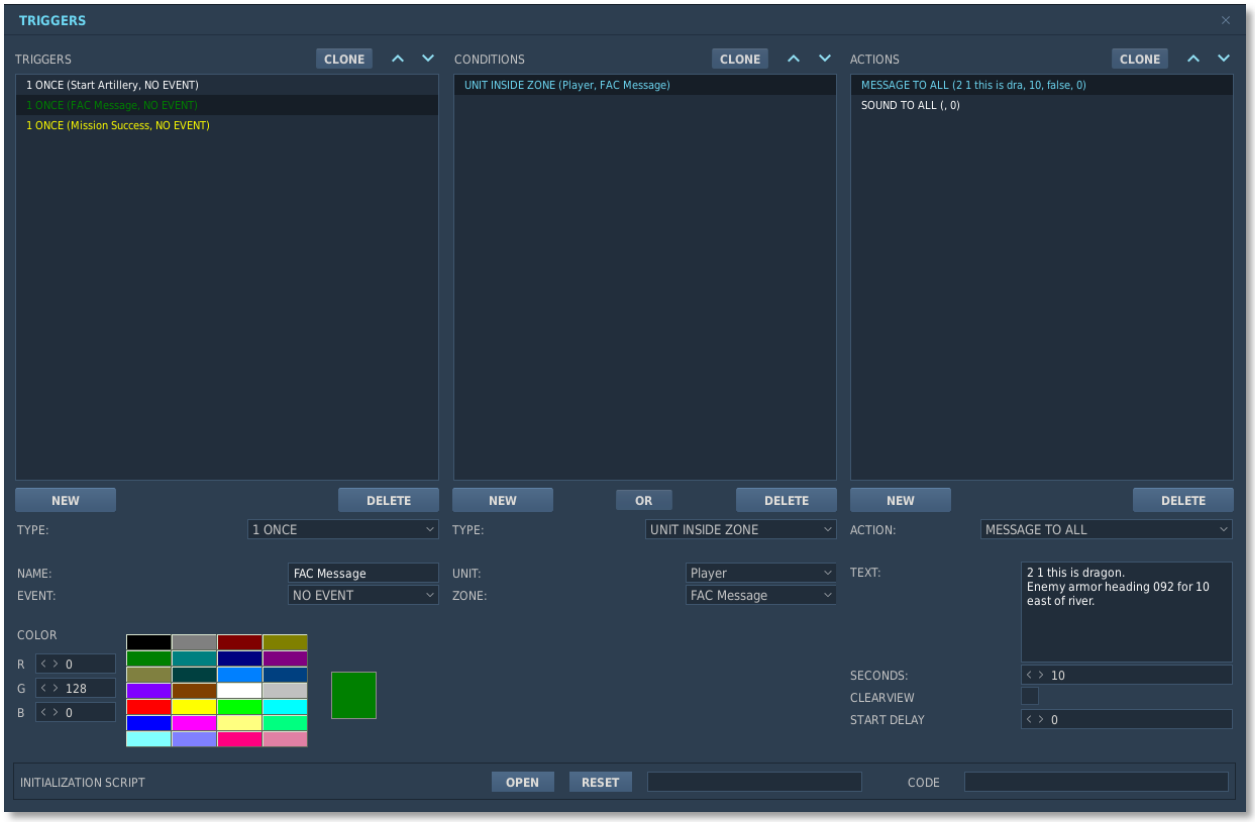
For the Start Artillery trigger, I will create a NEW condition and set the TYPE to UNIT INSIDE ZONE. In the UNIT field I will select the player's aircraft (Player) and in the ZONE I will select Start Artillery. The Start Artillery zone is one I created with the trigger zone creation tool that we'll discuss later.



To set the action, I will press NEW and then select GROUP ACTIVATE as the ACTION. With that selected, I will set the Arty-01 as the Group. Because this is to be an activated group, I set its START TIME in the Group Properties to a time past the anticipated mission start time.

The next trigger zone we will script is the FAC Message. To do so I will select the FAC Message from the triggers list and then click the NEW button in the Conditions pane. Again, I will select UNIT IN ZONE as the TYPE and the UNIT as PLAYER. However, for this trigger I will set the ZONE as FAC Message.

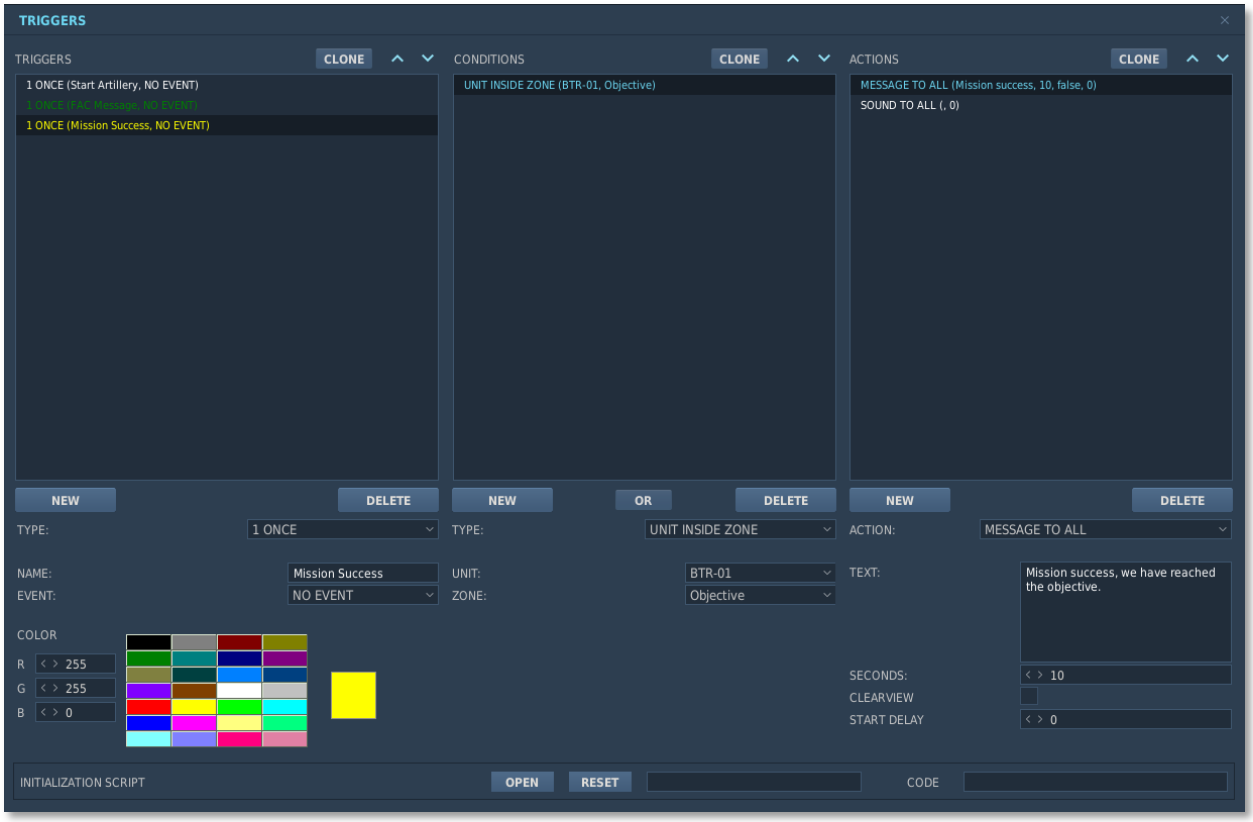
For the action, I want both a text message and audio message to play. So, I will first press the NEW button in the Actions pane and check that the MESSAGE TO ALL action is selected. I will then enter a text message in the TEXT box.



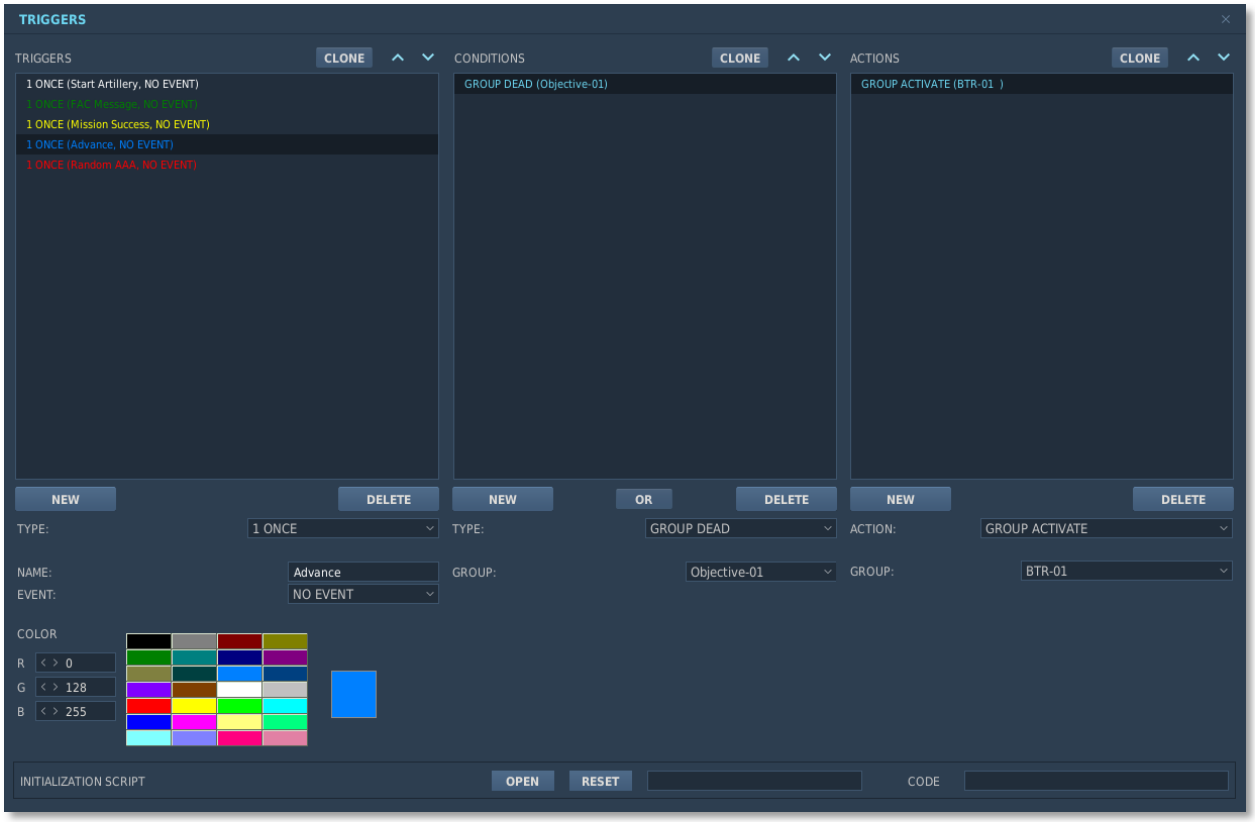
I now need to set the second (audio) action for this trigger by pressing the NEW button again but this time selecting the SOUND TO ALL option as my ACTION. I will then press the OPEN button and use the browser to select the audio file I wish to play.

The third trigger zone condition action we will create will be for the Mission Success message. This will trigger when the friendly ground forces reach the objective. The objective will be defined as one of the trigger zones I had created.

I will create a NEW condition and set the TYPE to UNIT IN ZONE. In the UNIT field, I will select one of the friendly armor units (BTR-01) and in the ZONE field, I will select Objective.

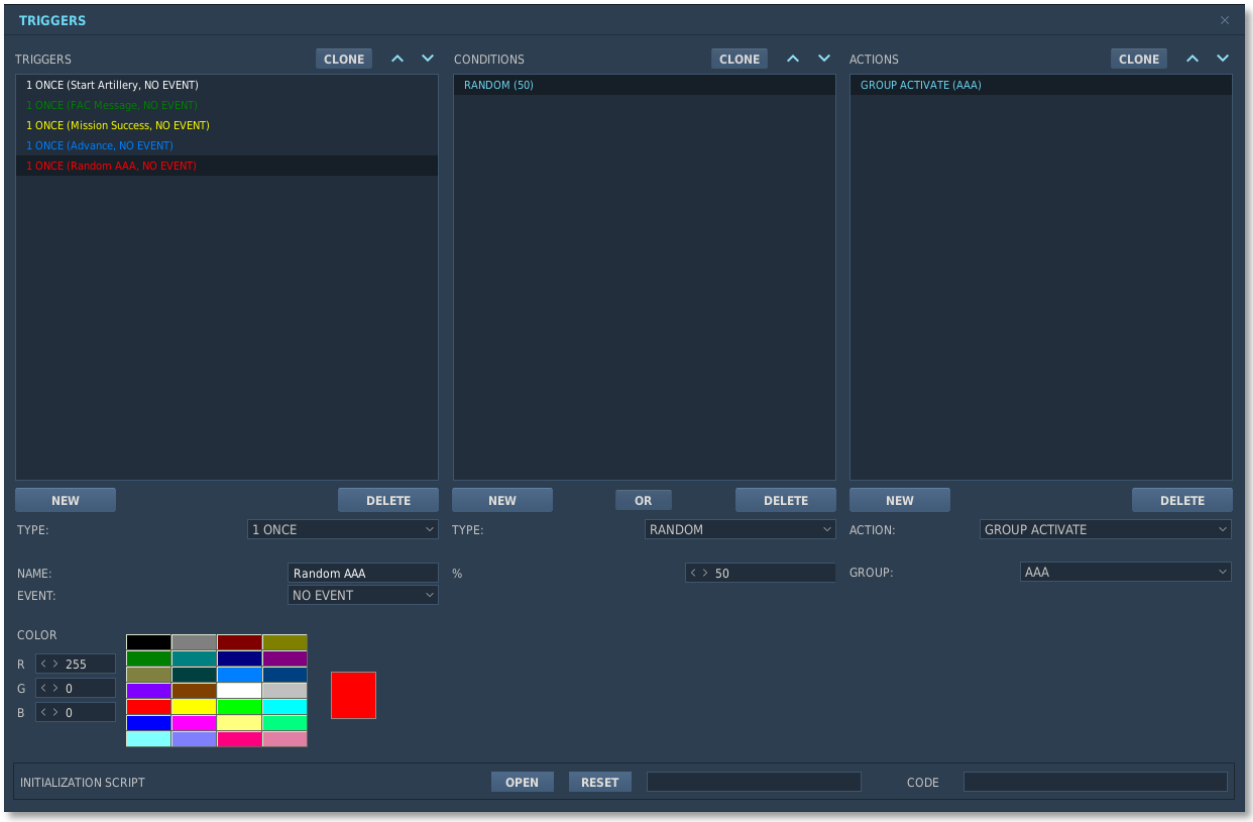


For the action, I want both a text message and audio message to play. So, I will first press the NEW button in the Actions pane and check that the MESSAGE TO ALL action is selected. I will then enter a text message in the TEXT box. I now need to set the second (audio) action for this trigger by pressing the NEW button again but this time selecting the SOUND TO ALL option as my ACTION. I will then press the OPEN button and use the browser to select the audio file I wish to play.



The next trigger we will create is one that allows the friendly ground units to advance from their starting location once the blocking enemy force is destroyed. To do so, we will first create a new trigger as a ONCE TYPE and call it Advance. Now we will create a NEW condition and set the TYPE to GROUP DEAD. From there, we will set the UNIT to the enemy armor units that are blocking our path (Objective-01). For the Action, we will create a new one by clicking the NEW button and then setting the ACTION to GROUP ACTIVATE. From the GROUP drop down list I will select our friendly ground force platoon that will activate and advance when the blocking force is destroyed. Remember that I needed to set the START TIME value of the friendly ground forces (BTR-01) to a time past the mission play time.

The final trigger we will place in this example is an enemy air defense that is assigned a random chance of appearing in the mission. We will create a new ONCE trigger and call it Random AAA. Next, we will create a NEW condition and set the TYPE to RANDOM. In the % field, we will choose 50 (as in 50%).



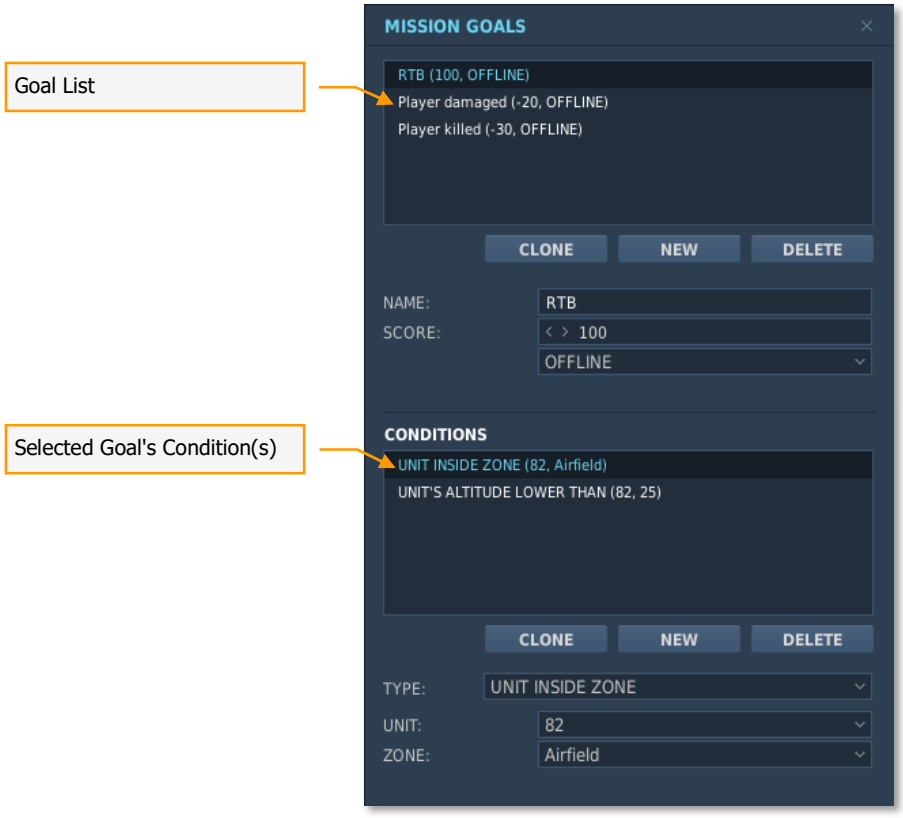
To set the action, we will create a NEW action and set the ACTION to GROUP ACTIVATE. From the GROUP drop down we will select the enemy air defense vehicle I placed in the mission called AAA. In the unit placement window, there is also the option to have activated unit visible or invisible before it is active in the world.

Define Mission Goals



When evaluating a mission for success, draw, or failure, the simulation uses point totals that are assigned by the mission builder. If total points at the end of a mission are 49 or less, the mission is a failure; if total points equal 50, the mission is a draw. If total points are 51 or higher, the mission is deemed a success. This point total is also used to define which stage and mission may be chosen next in a campaign.

The same set of conditions used in the Trigger system is also used to define mission goals. In the Goal window, the list of all created goals is displayed in the top pane. Once a goal is selected by clicking on it, the conditions of the goal are listed in the bottom pane.



To create a New goal, press the NEW button under the MISSION GOALS pane. Upon doing so, functions are displayed beneath the pane:

- **NEW** button. Press this button to create/add a new Goal to the Goal List.
- **DELETE** button. To remove a Goal from the list, click on it from the list and then click on the DELETE button.

- **CLONE** button. Press this button to clone the selected Goal into the Goal list.
- **NAME**. Type in the name you wish to assign to the Goal. This name, along with its score and assignment, will be displayed in the Goal List.
- **SCORE**. Each goal may be assigned a point score by either using the left and right buttons or manually entering it into the field.

The bottom drop down field allows you to assign who the points will be awarded to. There are three options:

- **OFFLINE**. This setting is used for single player missions; points are only in regards to the Player.
- **RED**. A Goal assigned to RED will result in the assigned points awarded to the RED side.
- **BLUE**. A Goal assigned to BLUE will result in the assigned points awarded to the BLUE side.

Once a Goal has been created, you must click the NEW button under the CONDITIONS pane to determine the conditions by which the Goal is based. The creation of Goal conditions is the same as that used to create Trigger conditions.

NEW button. Press the NEW button to create a new Goal condition. When you create a new Goal you will need to set the TYPE. You may create multiple conditions for a single Goal.

DELETE button. Press the DELETE button to remove a condition from the Goal. You will first need to click on the condition you wish to remove before pressing the DELETE button.

CLONE button. Press the CLONE button to clone the selected Goal condition.

TYPE drop down field. Using this drop down, you can set the condition by which the Goal is accomplished.

Enter or select additional parameters depending on the type of condition.

The mission is considered completed when all its specified conditions are met. If even one of these conditions is not yet satisfied, the mission will not be completed.

Battlefield Commanders



The Battlefield Commanders button opens the BATTLEFIELD COMMANDERS panel of the Mission Editor. This panel allows you to set the various roles for single and multiplayer missions.



PILOT CAN CONTROL VEHICLES. This checkbox enables/disables the ability of a player assigned to an aircraft to assign movement and targeting commands to AI ground units via the F10 map view during the mission. If you are creating a single player mission, you will probably want to enable this.

MISSION/MULTIPLAYER ROLES (Red, Blue). This table sets the number of slots open in a multiplayer mission for each side:

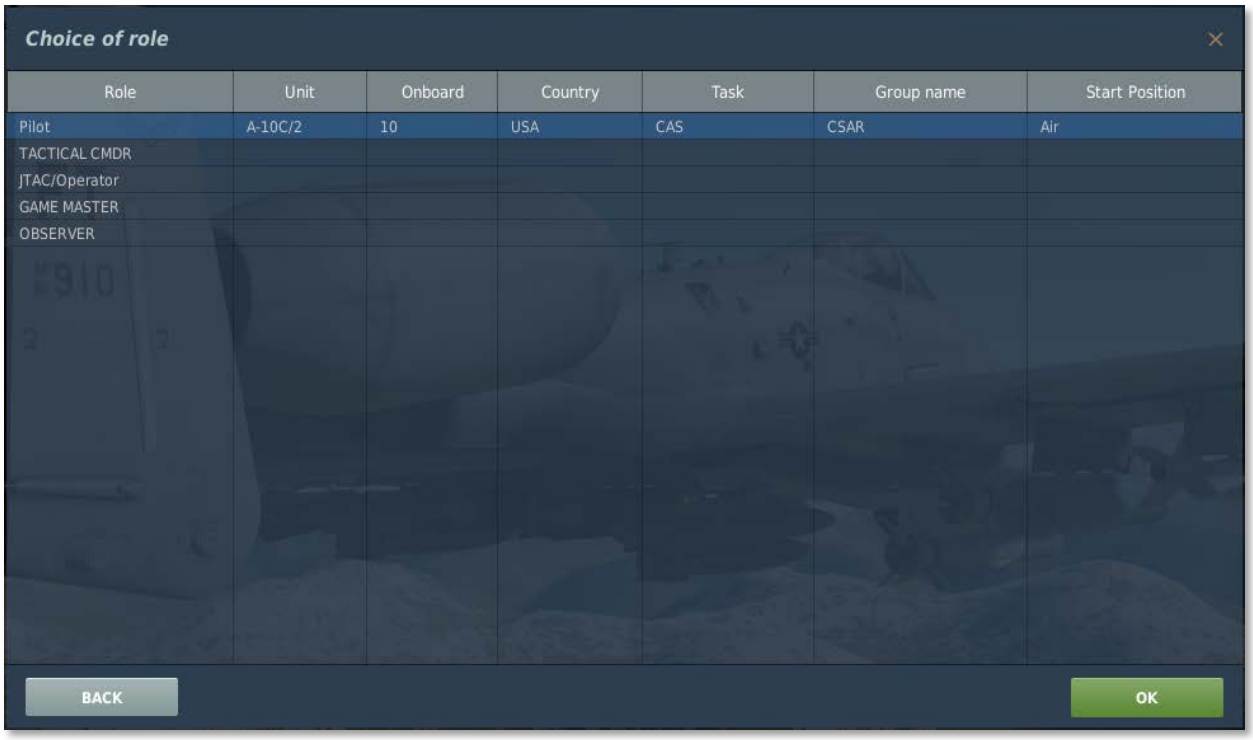
- **GAME MASTER**. This allows the player to observe and play as JTAC and control ground units for both sides. The Game Master can view all units. As a Game Master, you can dynamically alter the ground war during the mission.

> ***Note 1**: Although there are both Red and Blue side Game Master roles, both sides act the exact same and both control units from both sides.*
>
> ***Note 2**: For Head-to-Head multiplayer matches, it is best not to enable any Game Master roles.*

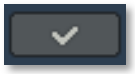
- **TACTICAL CMDR (COMMANDER)**. This allows players to strategically control ground units from the F10 Command Map. This includes indirect fire systems like artillery and multiple rocket launcher systems. This also allows first person vehicle control, including JTAC capability.
- **JTAC/ Operator**. This allows first person vehicle control, including JTAC capability, but does not have strategic control over ground units.
- **OBSERVER**. Players cannot play as JTAC or control ground units, but it has unlimited camera access to observe the action.

Once a mission has been started, each player will choose which command role to play from the Choose Role window. Once you select the desired role, press the Start button.

Aircraft that are set to human control will also be listed.



Set Mission Options



The Mission Options panel has been described in detail previously in the System Bar chapter of the manual.

Fly Mission



Press the green flight button to exit the Mission Editor and open the mission Briefing menu in preparation for the flight.

Task Planning for Unit Groups

Fundamental Concepts

Mission building involves creating unit groups, planning of group routes and combat tasks, and developing a dynamic battlefield atmosphere using triggered events, mission weather and time of day settings.

When designing a mission, you can choose one of two basic approaches to creating battle scenarios:

1. Simple: Place and route groups with minimal adjustments of default route and task settings (actions to determine behavior will be created automatically).

2. Advanced: Use the Advanced Actions Panel of the Group Properties Menu for precise manual control over group actions.

The first method requires minimum use of the interface to create simple missions with the idea that AI groups will act generally as envisioned by the designer, but using automatic behavior actions programmed into the AI based on factors such as presence of enemy units, contact range, etc. In this approach, the AI will generally move along its route and engage enemy units as soon as they are within engagement range, which can vary based on the unit type and task (in case of aircraft groups).

However, the automated AI actions created by default may not always produce the desired results. In particular, if the mission calls for complex AI actions, for example attacking only certain types of ground units in different target areas or restrictions on the use of particular weapons, the desired behavior may be achieved more effectively by using the Advanced Actions Panel to create and configure group actions in detail. Although more time consuming and complex, this allows for much greater flexibility and control in scripting AI behavior in the mission.

Actions refer to the various behaviors a mission designer can script for the AI to perform in the mission. Before we describe all of the advanced actions in detail, you should understand the following terms: *Group Artificial Intelligence, Task, Enroute Task, Command* and *Option.*

Tasks and Enroute Tasks are the primary combat actions and include a number of different targeting and engagement options for air, ground and naval groups. For example, a task to attack an enemy group or to act as a Forward Air Controller (FAC) in providing targeting commands to other friendly units. All Tasks and Enroute Tasks are actions performed over a period of time. Commands are instantaneous actions, such as setting a radio frequency for the group. Options are rules or limitations imposed on the group in executing an action.

All actions include a number of conditions the mission designer can set to control the starting and ending of the action. These include setting a mission time for starting and/or ending the action, action duration time, flag states (on/off), action activation probability, and custom scripting in LUA code language.

Group Artifical Intelligence (AI)

Group Artificial Intelligence is a virtual object (represented by the lead unit of a group) that controls game units or a group of units. The AI executes the actions set for the group in the mission editor. In doing so, the object follows behavior rules programmed into the simulation by the developers, however its behavior can also be adjusted by the mission designer using the tasking options available in the Advanced Actions Panel. Note, actions are set for entire groups only and not for individual units within a group.

When creating a group route, the mission designer can add or edit the actions for the group to perform at each waypoint (WP) of the route. Actions of a given waypoint are activated once the group leader reaches the waypoint. The actions are performed sequentially based on their order and/or priority setting. Priority is set as a whole number, starting with 0 as the highest priority value. Actions can also be set in the Trigger Menu of the mission editor to be activated independently of the route waypoints.

Perform Task

Task actions involve combat engagement, targeting and maneuvering. By setting a Perform Task, the mission designer commands the AI to perform a specific combat function at a specific point in the mission. For example: orbit on station, engage an enemy group, provide targeting to other groups as a FAC. Tasks have the highest execution priority for the AI and are generally used to set the primary group action for a waypoint. The AI will perform multiple Tasks one at a time according to their order or priority setting.

Task execution can be ended either by the AI automatically in-game or according to the action stop conditions set by the mission designer. For example, the AI will cease to attack a designated group of enemy vehicles when all of the vehicles in the group are destroyed or the AI group has exhausted its supply of usable weapons. Alternatively, the AI can be forced to cease the attack by setting an action stop condition, such as an action duration time limit.

Start Enroute Task

Like Perform Tasks, Start Enroute Tasks are combat actions that involve targeting or engagement actions. However, Start Enroute Tasks are designed to be active for the length of the group's route and can be executed multiple times and in multiple locations as long as the conditions for executing the Start Enroute Task are in effect and the AI is able to execute the action. As such, Start Enroute Tasks assume periodic interruptions of route navigation to perform the action, such as attacking "pop-up" enemy units detected along the route.

Whereas Tasks involve definitely known and pre-set targets, Start Enroute Tasks allow for some uncertainty in target type or location. The key phrase to keep in mind when using Start Enroute Tasks is "*Search For...*". This means the AI will actively search for the target and perform an attack only when the target is detected.

Start Enroute Tasks can only be ended by action stop conditions set by the mission designer. Otherwise, an Start Enroute Task will be active for the duration of the AI's "life" in the mission.

Numerous Start Enroute Tasks can be active simultaneously, however only one Start Enroute Task can be executed at a time. If conditions for multiple Start Enroute Tasks are true (for

example, enemy groups designated in two different Start Enroute Tasks are detected simultaneously), one Start Enroute Task will be selected by the AI to perform based on its order or priority setting.

Note, any Task will always have a higher priority over any concurrent Start Enroute Task (unless set otherwise manually by the mission designer). As such, an Start Enroute Task can be interrupted by a Task and resumed after its completion.

Perform Command

A Command is an instantaneous group action that is executed immediately upon activation. For example, a *Set Frequency* command action will change the group's radio communications frequency. Like Tasks and Enroute Tasks, the conditions for executing Commands can be customized by the mission designer and include mission time, flags state, command activation probability, and custom scripting in LUA code language.

Set Option

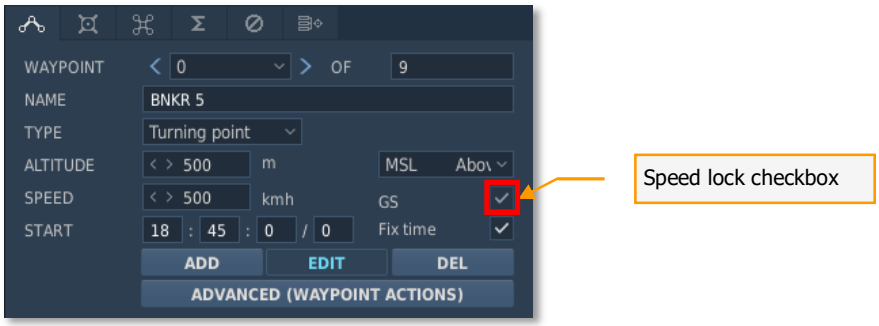
Options set rules for the group for the duration of the mission, unless started and ended at specific waypoints. Options are set using the format: *<variable>=<value>*, such as *Formation=Trail.* Using such options, the mission designer can control the AI's use of sensors and weapons when performing actions. For example, the *<Radar Use>* option setting to *<Never>* can be used to script a radar-silent intercept.

All actions can be tied to route waypoints or set independently of the route using the AI Task action of the Trigger Menu of the mission editor. In the latter case, the action will be executed when the rules of the trigger are in effect, as opposed to passing a waypoint.

A more thorough overview of all AI actions will be provided below.

Panel Flags

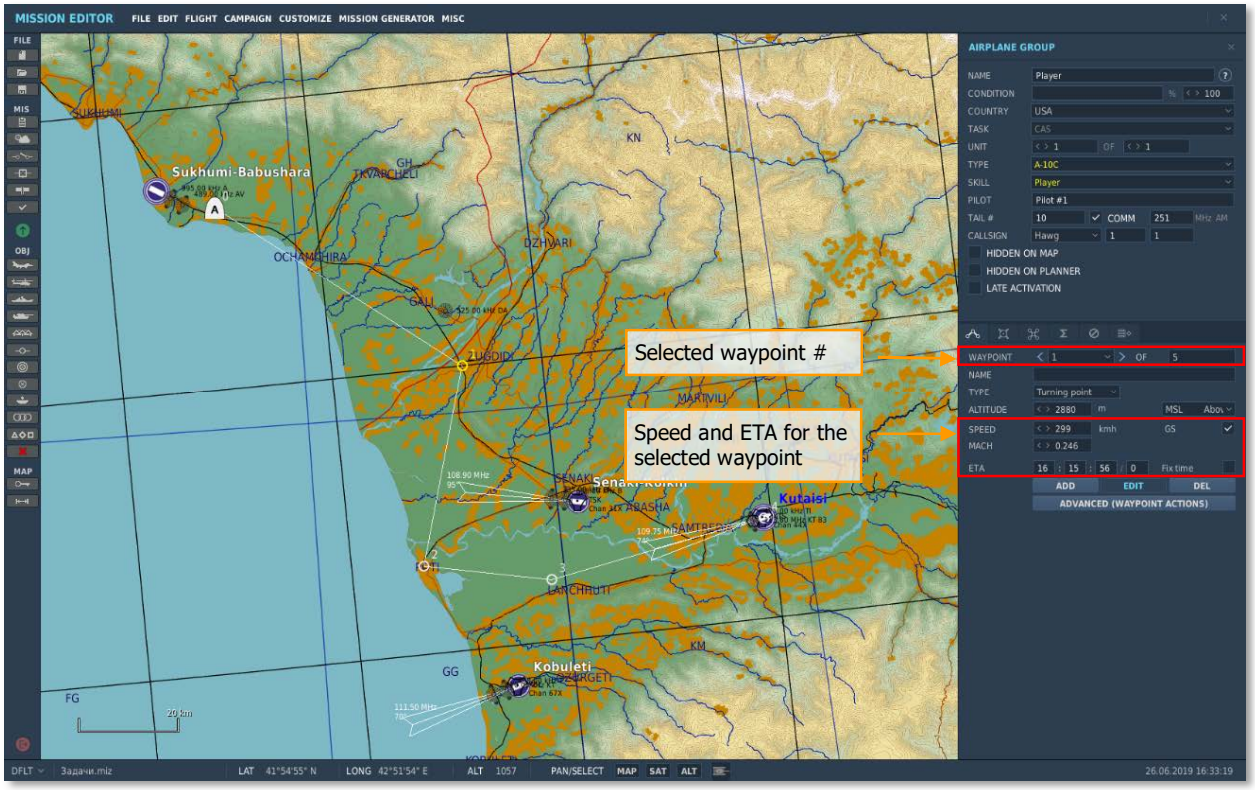
"Flag" variables in the form of a checkbox are used extensively to enable/disable various options in the panels of the Group Properties Menu. Do not confuse this with the FLAG ON/FLAG OFF trigger rule and action in the Trigger Menu of the mission editor. You can think of panel flags as "locks".



In conclusion, some of the main steps in creating a mission include: creating unit groups, placing the group routes on the map, setting up group main tasks (for aircraft groups), configuring the group actions using the Advanced Actions Panel (Tasks, Enroute Tasks, Commands, Options), creating triggers to script certain group actions in addition to those set with route waypoints.

Group Route Planning

When placing group routes, one of the basic functions to set will be the desired speed ("SPEED") and estimated time of arrival ("ETA") for each waypoint.



Depending on the plans of the mission designer, one, both or neither of these values can be "locked" at each waypoint using the panel flags. When either value is locked by the user, the mission editor automatically calculates the other value based on the known range between consecutive waypoints.

Locking speed and ETA values requires a logical consistency between waypoint lock settings as not every combination of speed and ETA lock is always possible. The route waypoint locks must be arranged such that the AI always retains a reference time value somewhere along the route. Using an example of a relatively simple route consisting of 4 waypoints, the following 6 combinations of lock settings are valid.

The Red Coalition has the following three designations for WP:

1. *SP – Starting Point,*
2. *WP … – Waypoint Route Number,*
3. *DP – Destination Point.*

The Blue Coalition has all its WP set to zero by default.

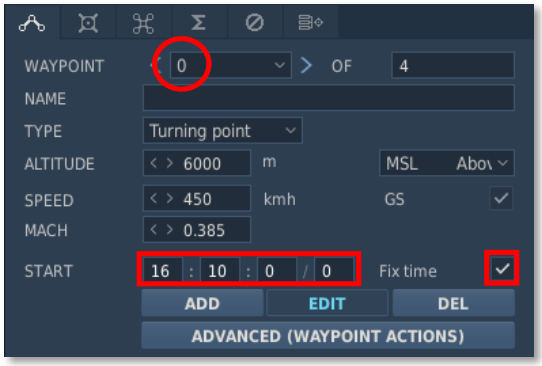
Example 1

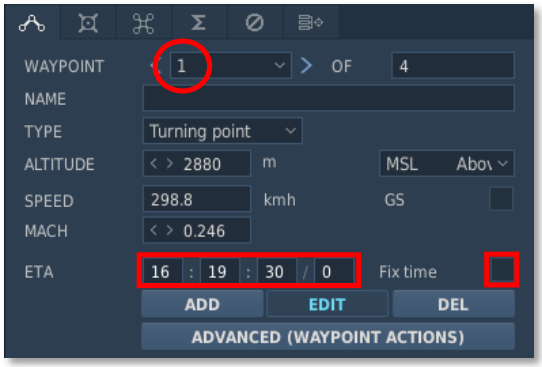
User locks:

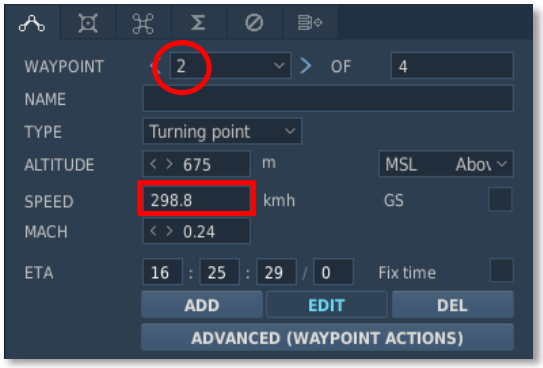
- Group start time (T_0), i.e. ETA for WP0;
- ETA for the final waypoint (T_3);

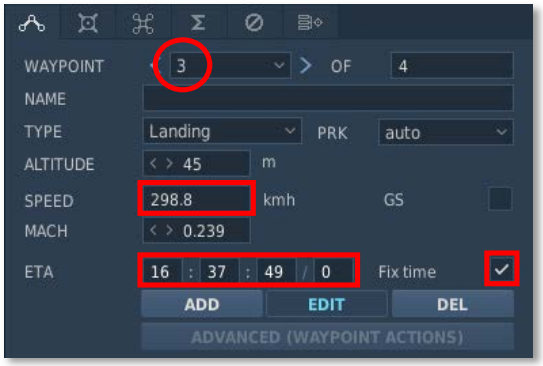
In this case, the user doesn't control the speed and ETAs for waypoints 1 and 2 or the speed for WP3. These values are calculated by the mission editor automatically. The mission editor calculates the average speed for the route to reach WP3 at the designated ETA. Note that the desired speed refers to the speed *to* the waypoint, i.e. the speed the AI group will assume once it passes the preceding waypoint.

The AI group will attempt to adjust its speed for WP1, WP2 and WP3 as necessary to arrive at WP3 at the locked ETA.









If the user unlocks the ETA for either WP0 or WP3, the route will no longer be valid, because the mission editor will not have a time reference for calculating the desired speeds for WP1 and WP2. When a route is invalid, the speed and ETA checkboxes become framed in red to indicate an error and a corresponding message appears when the user attempts to close the Group Properties Menu or save the mission.

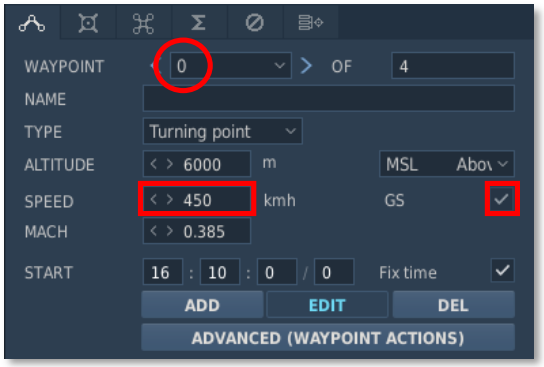
In order to unlock both the speed and ETA of a waypoint, at least one preceding and subsequent waypoint must have a locked ETA!

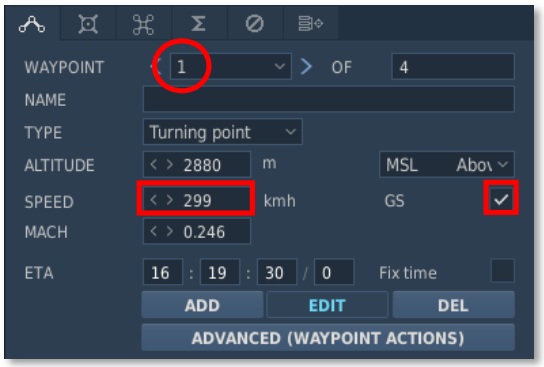
Example 2

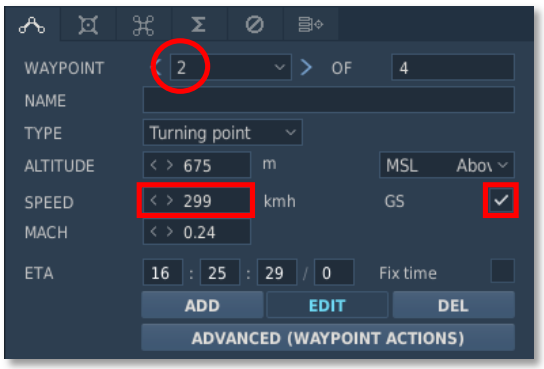
User locks*:*

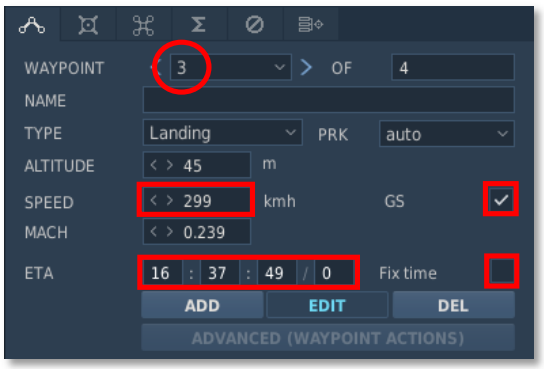
- Group start time (T_0), i.e. ETA for WP0;
- Speed ($V_0 - V_3$) for each waypoint;

In this case, the user doesn't control the ETAs for any waypoints after the initial waypoint, but wants to ensure the group maintains a desired speed along the route. The mission editor automatically calculates the ETAs for WP1, WP2, WP3 based on the known range and locked desired speed of the waypoints:


WAYPOINT 0 OF 4
NAME
TYPE Turning point
ALTITUDE 6000 m MSL
SPEED 450 kmh GS
MACH 0.385
START 16 : 10 : 0 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)


WAYPOINT 1 OF 4
NAME
TYPE Turning point
ALTITUDE 2880 m MSL
SPEED 299 kmh GS
MACH 0.246
ETA 16 : 19 : 30 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)


WAYPOINT 2 OF 4
NAME
TYPE Turning point
ALTITUDE 675 m MSL
SPEED 299 kmh GS
MACH 0.24
ETA 16 : 25 : 29 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)

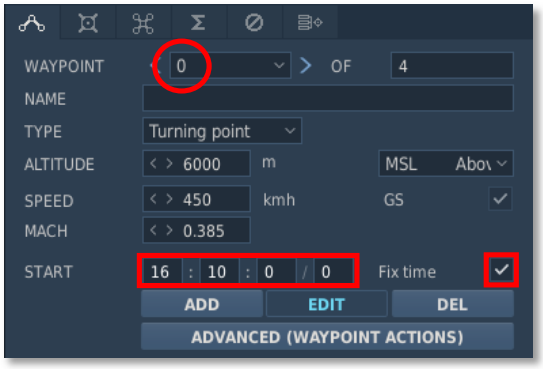

WAYPOINT 3 OF 4
NAME
TYPE Landing PRK auto
ALTITUDE 45 m
SPEED 299 kmh GS
MACH 0.239
ETA 16 : 37 : 49 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)

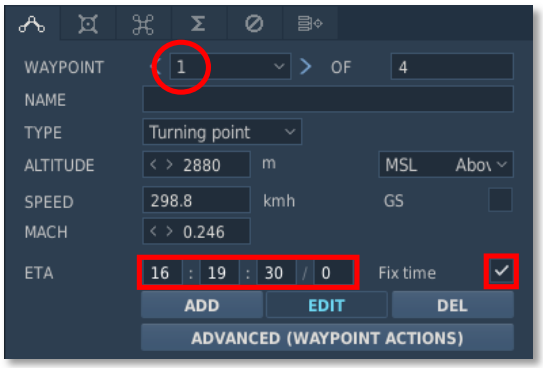
Example 3

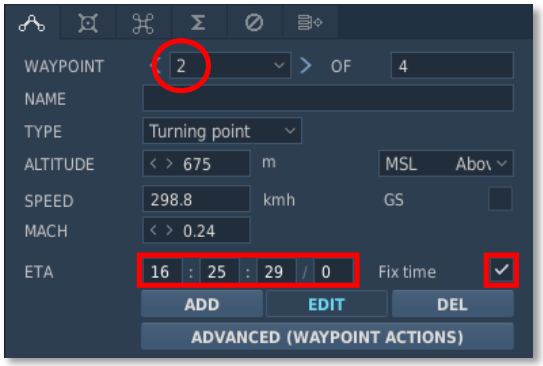
User locks:

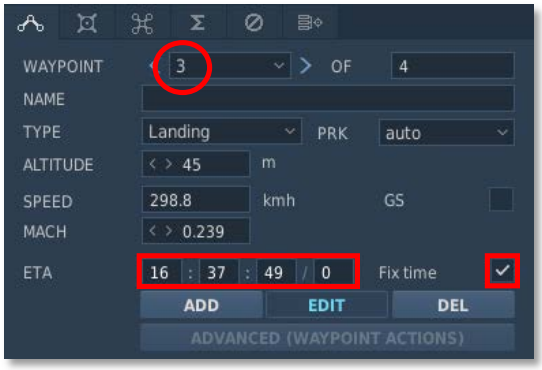
- Group start time (T_0), i.e. ETA for WP0;
- ETA for all other waypoints (T_1, T_2, T_3);

In this case, the user wants to ensure the waypoints are reached at the designated ETAs. The AI group will attempt to maintain speed for each waypoint such that it arrives at each waypoint at the designated ETA.
The mission editor calculates the desired speed for each waypoint based on the known range and ETA between consecutive waypoints:







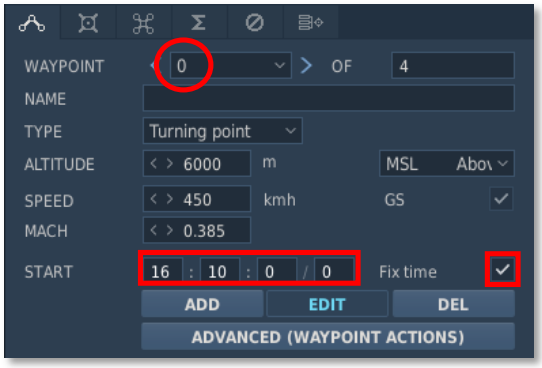


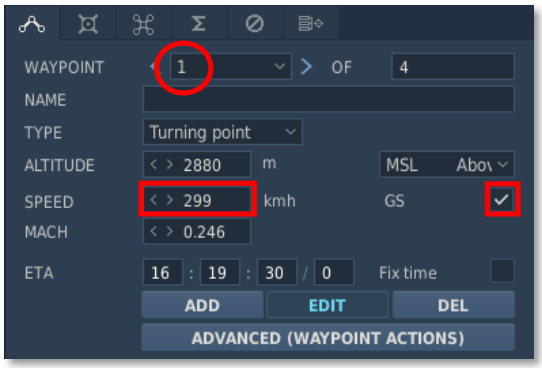
Example 4

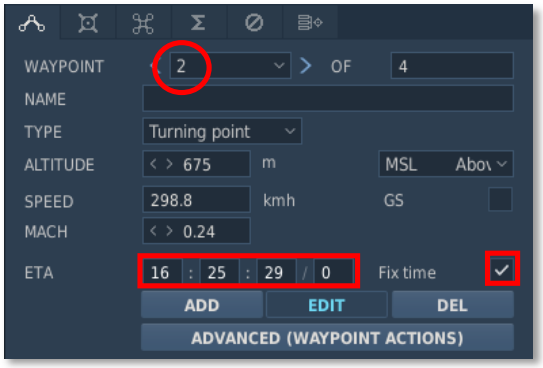
User locks:

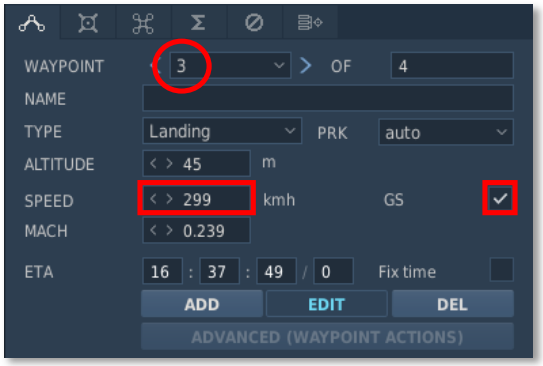
- Group Start time (T_0), i.e. ETA for WP0;
- Desired speed for WP1 and WP3 (V_1, V_3);
- ETA for WP2 (T_2);

In this case, the user controls the ETA for WP2 and the travel speed for WP1 and WP3.
The mission editor automatically calculates the ETAs for WP1 and WP3, as well as the speed for WP2:









Example 5

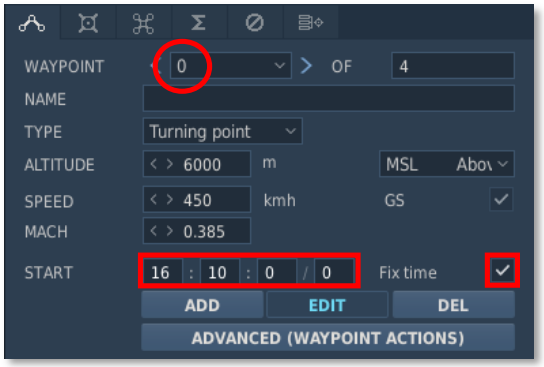
User locks:

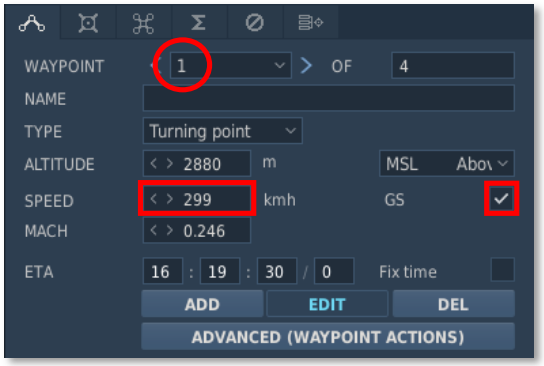
- Group start time (T_0), i.e. ETA for WP0;
- ETA for the final waypoint (T_3);
- Desired speed for WP1 and WP3 (V_1, V_3);

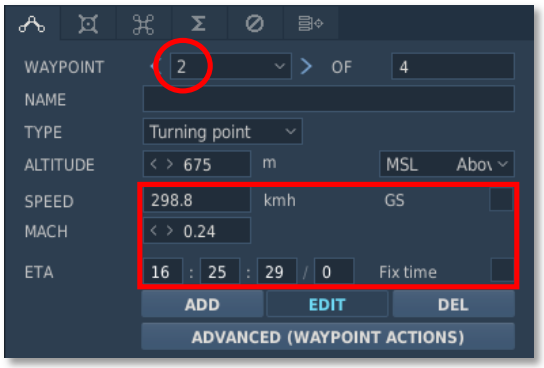
In this case, the user controls the ETA for WP3 and also the travel speed for WP1 and WP3. Such a route is valid, because there is at least one waypoint (in this case WP2) between the starting and ending waypoints, which has both the speed and ETA unlocked, allowing the mission editor to automatically calculate both values as needed to meet the desired settings for WP0, WP1 and WP3.

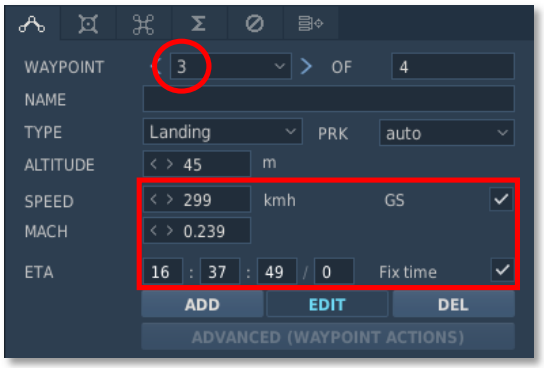
The AI group will attempt to adjust its speed for WP2 such that it will arrive at WP3 at the designated ETA and desired speed.

The mission editor automatically calculates the ETA for WP1 and WP2, as well as the desired speed for WP2:


WAYPOINT 0 OF 4
NAME
TYPE Turning point
ALTITUDE 6000 m MSL Abov
SPEED 450 kmh GS
MACH 0.385
START 16 : 10 : 0 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)


WAYPOINT 1 OF 4
NAME
TYPE Turning point
ALTITUDE 2880 m MSL Abov
SPEED 299 kmh GS
MACH 0.246
ETA 16 : 19 : 30 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)


WAYPOINT 2 OF 4
NAME
TYPE Turning point
ALTITUDE 675 m MSL Abov
SPEED 298.8 kmh GS
MACH 0.24
ETA 16 : 25 : 29 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)

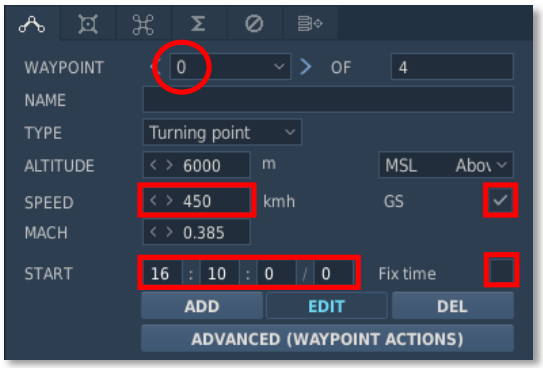

WAYPOINT 3 OF 4
NAME
TYPE Landing PRK auto
ALTITUDE 45 m
SPEED 299 kmh GS
MACH 0.239
ETA 16 : 37 : 49 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)

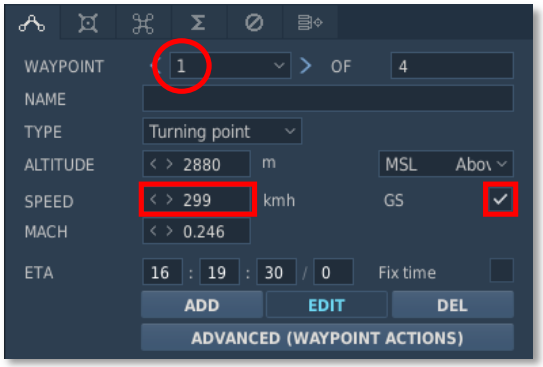
Example 6

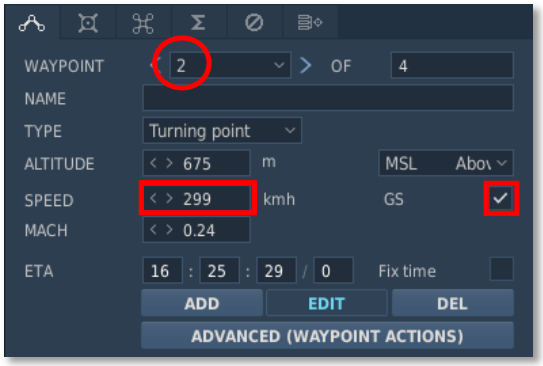
User locks:

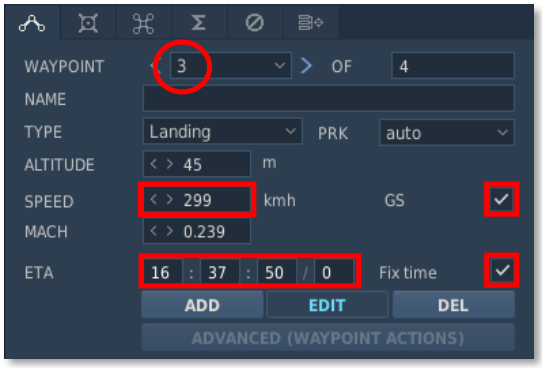
- ETA for the final waypoint (T_3);
- Desired speed for WP1 – WP3 (V_1 - V_3);

In this case, the mission editor automatically calculates the starting time (ETA for WP0):

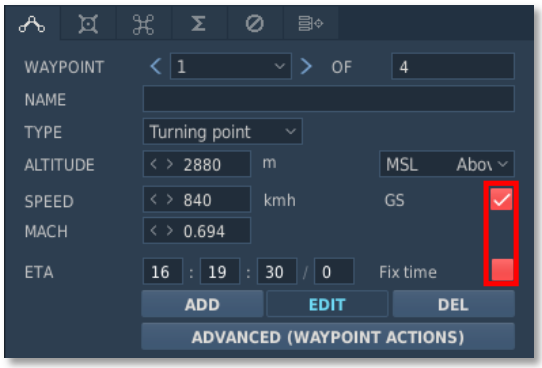




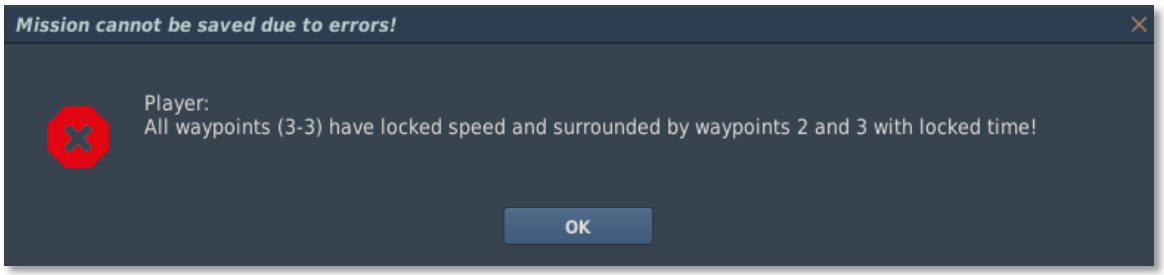




If the SPEED or ETA value is outside of the valid range for the particular unit (such as desired speed below the minimum flying speed), the SPEED and ETA checkboxes become framed in red to indicate an invalid value warning and a corresponding message appears when the user attempts to close the Group Properties Menu or save the mission. To correct such an error, enter a valid value in the locked field.



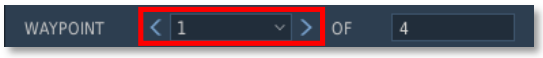
If the user attempts to set an invalid route, for example lock the speed for all waypoints and at the same time lock the ETA for the starting and final waypoints, the SPEED and ETA checkboxes become framed in red to indicate a flag error warning and a corresponding message appears when the user attempts to close the group properties panel or save the mission:



To correct such an error, set waypoint locks to a valid route combination.

Hints

- To avoid repetitive error messages when closing the group properties panel, use the waypoint edit panel to cycle between waypoints rather than selecting waypoints on the map:



- To correct invalid SPEED or ETA value warnings (red font), enter a valid value.
- To correct invalid flag warnings (red checkboxes), set the SPEED and ETA locks to a valid route combination.
- Every route must have at least one waypoint with a locked ETA! This can be the initial waypoint to set the group starting time or any subsequent waypoint.
- The Mission Editor does not prevent you from setting any combination of speed and ETA locks, however it informs you when an invalid combination is set with either an invalid value warning or invalid flag warning. A warning window will pop-up if attempt to save the group properties or the mission whenever an invalid route is present.

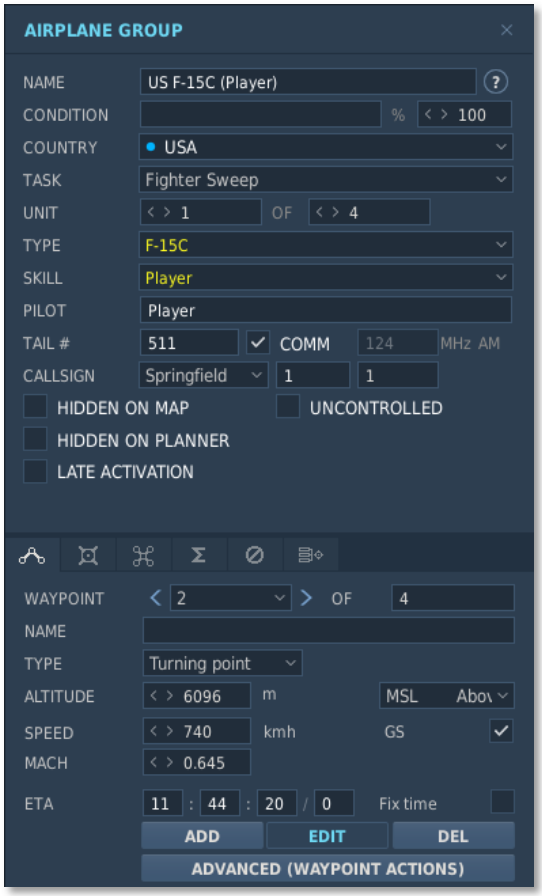
Place Airplane and Helicopter Group



At the top of the Object (OBJ) Tool Bar are the Place Airplane and Place Helicopter buttons. You will use these buttons to place aircraft groups in the mission, set their routes, and the actions they will take. When placing an aircraft group, it is important to remember that the groups can consist of one to four units (aircraft). Because the placement functions of fixed wing and rotary wing (helicopters) are very similar, we will discuss them jointly.

Note, the mission editor supports group copy/paste functionality. To copy a group, select it and press [CTRL+C]. To paste the group with identical properties, position the mouse over the desired point on the map and press [CTRL+V].

The aircraft Group Properties Mode consist of numerous functions, which are described below. We will discuss them from top to bottom.



NAME. In the Name field, you may enter in a unique name for the aircraft group. If you do not enter one, a default name will be generated. The Name you create will be used when assigning some types of Triggers such as a Group Dead condition. Always be careful not to assign more than one group with the same Name.

CONDITION. The condition for the random spawning of a group in a mission. For this, LUA expressions used in the Simulator Scripting Engine Documentation are entered into the field. First, the initialization script should be loaded in the trigger panel. The script is executed before the mission is loaded and its result is used to set the conditions for the random group spawn.

COUNTRY. The Country drop down list will display all the countries that have been assigned to either the RED or BLUE sides when the mission was initially created by use of the CREATE NEW MISSION button. The Country selected will filter the TYPE of aircraft available.

TASK. The Task drop down menu allows you to select the group task. This setting will be in effect for the group for the duration of the mission. Note, the group task serves as a filter of available and automatically generated actions in the Advanced Actions Panel as well as the default payloads available.



- **Nothing**. This mission task assumes a non-combat flight along a route. Such an aircraft does not take part in any active actions against enemy forces and just follows its route. Under the threat of an enemy attack the aircraft will try to evade it.
- **AFAC**. The Airborne Forward Air Controller (AFAC) mission task will set the assigned aircraft to mark targets with smoke rockets or illumination flares. For night missions, this can be a useful task to assign an airplane to support a player flying a Close Air Support (CAS) mission.

- **Anti-ship Strike**. A mission task to attack enemy marine vessels using anti-ship missiles.
- **AWACS**. (Airborne Warning and Control System). The AWACS aircraft flies according to a planned straight or circular route using looped waypoints, and it alerts allied aircraft, SAM sites, and ships when it detects enemy aircraft. Certain SAM systems can receive targeting data directly from the AWACS even when their own acquisition radar systems have been destroyed. Note that AWACS detection can be limited by range, very low target altitudes, and terrain masking.
- **CAP** (Combat Air Patrol). The CAP mission task implies flying a large race-track pattern using looped waypoints around a defined route to defend an area from enemy aircraft incursion. This type of task does not involve spotting and destroying enemy ground targets or a significant deviation from the planned route to intercept aircraft. Be aware that a high-altitude CAP will make life for low level interdiction easier for your aircraft. A combination of a high/low CAP sandwich is the most balanced deployment. The crucial factor while patrolling will be the fuel load limiting the distance and duration of the CAP. Unless manually disabled by the mission designer in the Advanced Actions Panel, all AI aircraft will stop patrolling and return to base in a straight route as soon as their fuel falls to the guaranteed minimum required for the return flight (Bingo fuel state).
- **CAS** (Close Air Support). CAS involves actively searching for enemy ground targets on the battlefield and destroying them in support of friendly ground units under fire. This task is most suitable for dedicated attack aircraft and helicopters.
- **Escort**. This mission task is allocated to fighters and attack helicopters and involves escorting allied aircraft (transport aircraft, bombers, or attack aircraft) and defending them along the route from possible attacks of enemy aircraft and air defense systems. In doing so, the escorts should not engage in fights with the enemy if they do not pose a threat or are significantly off the course line.
- **Fighter Sweep**. The fighter sweep mission task is a combat task that involves penetrating enemy air space to attack enemy fighters or other types of aircraft. The main objective of a fighter sweep is winning air superiority and to ensure unimpeded use of the air space by friendly aircraft. Since the aircraft taking part in a fighter sweep may find themselves at a considerable distance from their airfields and take part in prolonged dogfights, their fuel load will be a crucial factor.
- **Ground Attack**. This mission task is used to attack enemy ground targets using a variety of air to surface weapons. This type of mission usually involves unguided bombs weighing from 500 to 20000 lbs. and unguided rockets. Unless otherwise set in the advanced tasking options, the AI will generally prioritize use of longer-range weapons, such as guided air-to-surface missiles, over shorter-range weapons, such as unguided rockets and cannon.
- **Intercept**. This is a defensive mission task whereby the aircraft must carry out an active search of incoming enemy aircraft and/or receive targeting data from ground based or airborne radar. This type of combat task is reserved for large scale defense and active patrolling and you should not use it while defending a small area or a local installation.

While chasing the enemy, the interceptor may deviate far from its planned route and the area that is to be defended will be left undefended.

- **Pinpoint Strike**. The Pinpoint strike mission task involves detecting and attacking surface targets using precision-guided weapons.
- **Reconnaissance**. The aircraft will fly directly over the assigned Reconnaissance Task waypoint to best acquire intelligence.
- **Refueling**. This mission task is reserved for aerial tankers. An aircraft assigned this mission will refuel any thirsty allied aircraft during its flight.
- **Runway Attack**. This is a specialized form of ground attack that will allow the aircraft to automatically align the axis of its attack along the length of the targeted runway. This is most useful when deploying runway denial weapons. To do so, set the Targeting area over the airfield to attack and select Airfields from the Target Categories.
- **SEAD** (Suppression of Enemy Air Defenses). Similar to the CAS Task, this mission task allows you to use attack aircraft in a more specialized role to attack enemy air defense sites with anti-radar missiles or other weapon types.
- **Transport**. An aircraft assigned to a Transport mission task does not take part in any active actions against enemy forces and just follows its route. Under the threat of an enemy attack the aircraft will try to evade it.

UNIT. The Unit selection is composed of two fields and they allow you to select how many aircraft units will be part of the flight group (1 to 4). The right field is used to set the total number of airplanes in the group; to do so, use the left and right arrow buttons. The left field is used to select an airplane within the group; to do so, use the left and right arrow keys.

TYPE. Depending on the Country and Task selections, a list of appropriate aircraft is listed in this pull-down list.

SKILL. The Skill level allows you to select how competent the aircraft pilot is when flown by the artificial intelligence. This setting can control such factors on how much G the pilot can pull, at what range it will attack, and how accurate its weapon delivery skills are. There are five Skill options for AI aircraft:

- **Average** – an AI pilot with average training.
- **Good** – an AI pilot with good training.
- **High** – an AI pilot with high training.
- **Excellent** – an AI pilot with excellent training.
- **Random** – This option will randomly select from one of the four options above.

For non-AI groups, you have two selections:

- **Client** – If you wish the aircraft to be flown by a human in a multiplayer game, set it as Client. Do not set aircraft to Player in a multiplayer game.
- **Player** – Set the aircraft to Player when you want it to be flown by a human player in a single-player mission/campaign.

Note that a unit's skill level will affect its detection ranges, reaction times and targeting errors.

PILOT. Enter a unique name for each unit within the aircraft Group. If you do not, a default name will be created automatically. This name is important because it will be used to set Trigger Conditions. For example, if you create a condition based on the destruction of a unit, the unit will be selected from a list using its PILOT name.

TAIL #. Enter in this field either a three- or two-digit number you wish to have displayed on the aircraft.

COMM. Frequency for radio communications in MHz. The AI group will conduct radio communications at the specified frequency. Example: 124 MHz AM means a frequency of 124 MHz in the AM (amplitude modulation) band.

CALLSIGN. Type in the callsign of the aircraft you wish it to use. For Russian aircraft, a three-digit number is used. This callsign will be used in communications with the flight, AWACS, and ground controllers.

HIDDEN ON MAP. check box. After creating a mission, you may wish to hide certain units so that people playing your mission cannot see them. For example, you may want to hide mobile enemy units from being seen by the player on the briefing and F10 maps. Press this check box to hide the selected aircraft group from the ME World Map and F10 view. You can view all hidden groups using the Units List window (discussed later in this manual).

UNCONTROLLED. When setting an AI group to a "Takeoff from Ramp" (cold start) waypoint, the Uncontrolled check box can be used to have the AI group appear at their starting location on the ramp, but not begin the spool up sequence until initiated by an AI TASK trigger as set by the START Command in the Triggered Actions of the group. In this case, the group aircraft will appear in the mission in parked configuration and remain as such until triggered to "come alive".

HIDDEN ON PLANNER. This option hides aircraft from the mission map in the mission planner. This option can be useful if you need to hide an object from the player.

LATE ACTIVATION. This option spawns a group of aircraft in the mission after a certain time after reaching any condition or event.

Modal Buttons

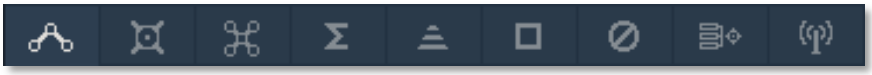
The following modal buttons are displayed on the lower half of the Group Properties Panel:

For AI groups:



- route,
- payload,
- triggered actions,
- route summary.

For player-controlled aircraft:



- route,
- payload,
- triggered actions,
- route summary,
- INU fix point,
- navigation target points,
- failures,
- waypoint properties,
- radio presets.

Route Mode



When in Route mode, the data in the lower half of the Group Properties Menu is in regards to waypoint management. Waypoints are arbitrary points on the map (Lat, Long, and altitude) that can be chained together to create a flight route. During the course of a mission, the group will fly from one waypoint to the next along the route line and at each waypoint you can assign unique characteristics.

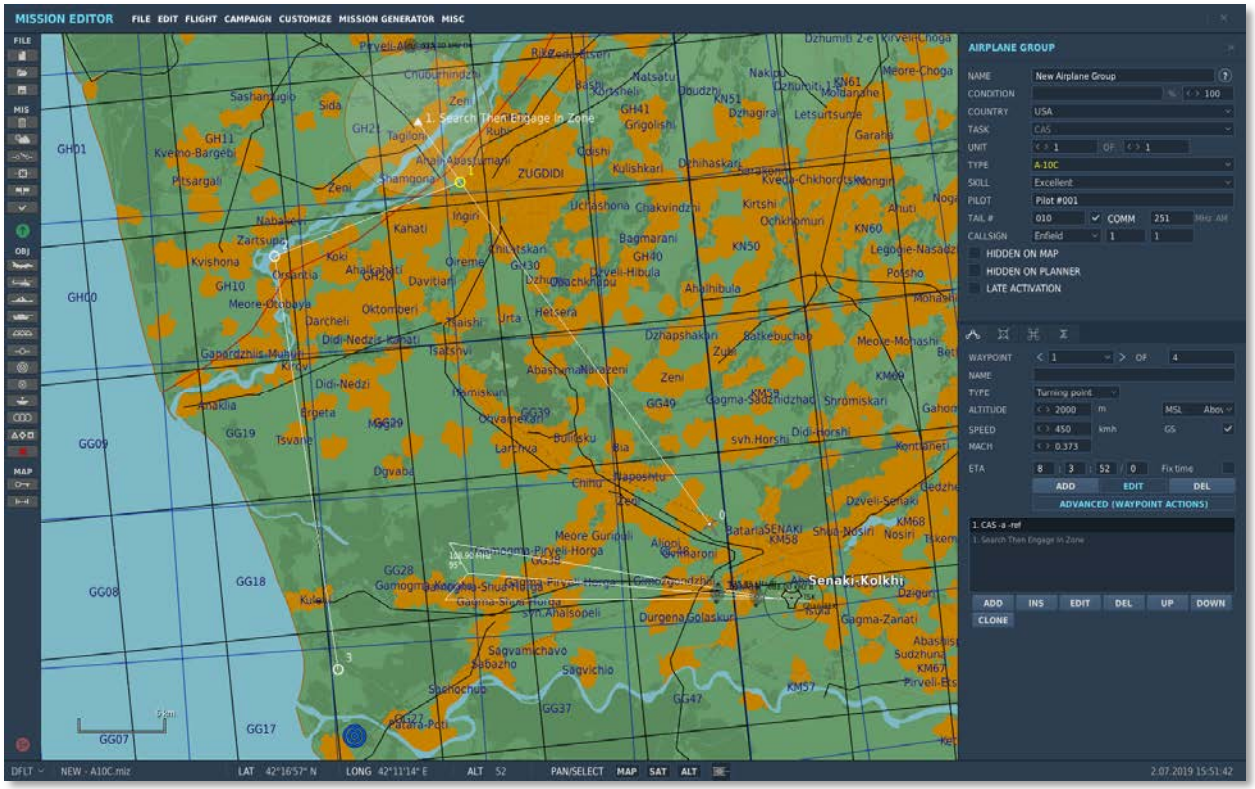
To place a new aircraft group, you will need to be in the Route mode and have an aircraft selected. You may then left click on the map to place the groups starting point, which will also be waypoint 1. When you place a group, its waypoint marker (circle with its waypoint number next to it) and route line will be colored appropriately:

- White - Selected unit.
- Red - A Red side unit that is not selected.
- Blue - A Blue side unit that is not selected.

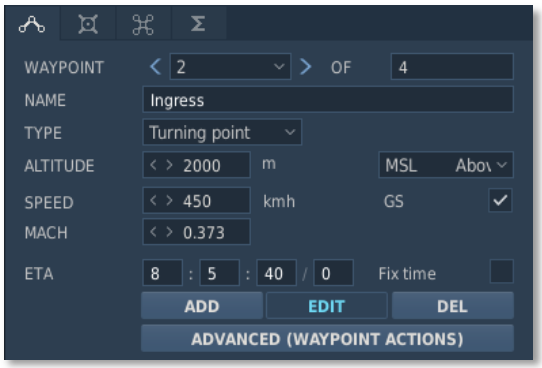
At the starting waypoint, there will be a unit icon instead of a standard waypoint circle. Different icons denote different types of units:

Aircraft Editor Icons (Russian/Western)

Russia	NATO	Unit Type
		Utility helicopter
		Attack helicopter
		Reconnaissance helicopter
		Anti-ship/Anti-submarine helicopter
		Fighter aircraft
		Attack aircraft
		Reconnaissance aircraft
		Bomber aircraft
		Transport aircraft
		AWACS (Airborne Early Warning and Control)
		Anti-ship/Anti-submarine aircraft
		Tanker
		UAV (Unmanned Aerial Vehicle)



For each waypoint of the route, the following options can be set in the waypoint control panel [as described above.](#)

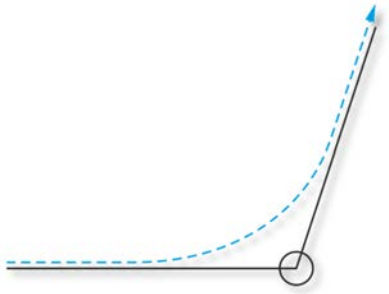


WAYPOINT. The WAYPOINT fields allow you to cycle between waypoints you have created, and by left clicking on the World Map you will add a waypoint. The left field displays the currently selected waypoint and you can cycle it by pressing the left and right arrow buttons. The field on the right displays the total number of waypoints in the route. The circle and waypoint number of the selected waypoint will be colored yellow on the map.

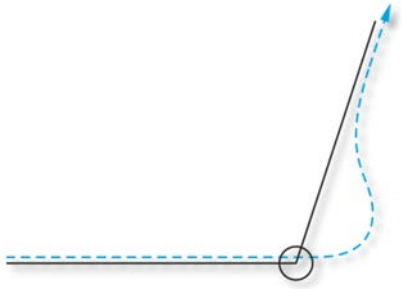
NAME. For each waypoint you can assign a unique name. Type the name of the waypoint in this field and this name will then appear next to the waypoint on the map.

TYPE. Each waypoint can be assigned a type of action that the aircraft will perform when it is at that waypoint. These include:

- **Turning point.** This is the most common and basic waypoint and will result in the aircraft performing a lead turn ahead of the actual waypoint location to complete the turn on course for the following waypoint.



- **Fly over point.** Much like a Turing Point, a Fly Over Point will sequence the aircraft from the preceding waypoint to the next, but only after passing directly over the waypoint location. This will result in a course correction required to come on course for the following waypoint.



- **Takeoff from runway.** This type is only available when waypoint 1 is selected and it allows the aircraft to start on the runway threshold with all systems up and running. When set to Takeoff, the waypoint will snap to the nearest airfield or FARP.
- **Takeoff from ramp.** This type is only available when waypoint 1 is selected and it allows the aircraft to start on the parking apron (ramp) with all systems shut down. When set to Takeoff, the waypoint will snap to the nearest airfield or FARP.
- **Takeoff from parking hot.** This type is only available when waypoint 1 is selected and it allows the aircraft to start on the parking apron (ramp) with all systems up and running. When set to Takeoff, the waypoint will snap to the nearest parking apron (ramp).
- **LandingReFuAr**. This type is available when selecting any waypoint other than 1. When set to LandingReFuAr, the waypoint will snap to the nearest airfield or FARP.
- **Landing.** This type is only available when the last waypoint in the route is selected; the waypoint will automatically snap to the nearest airfield or FARP.
- **Takeoff from ground.** This type is only available when waypoint 1 is selected and it allows the aircraft to start from the ground with all systems shut down. In this case, it takes some time to prepare for takeoff.
- **Takeoff from ground hot.** This type is only available when waypoint 1 is selected and it allows the aircraft to start from the ground with all systems up and running.

ALTITUDE. The altitude setting will set the altitude in meters that the aircraft will be at when it reaches the waypoint. To set this, you can either use the left and right arrow buttons or type it into the field.

- **Altitude type**. Altitude can be set in meters above Mean Sea Level (MSL) - altitude measured relative to sea level (constant) or in meters Above Ground Level (AGL) - altitude measured directly over the ground below. Because terrain elevations vary throughout the theater, setting an AGL altitude will result in the aircraft following the terrain contours along the route leg for the waypoint. On the other hand, setting an MSL altitude will result in level flight at the set altitude, unless obstructed by terrain (such as when attempting to maintain a low-level MSL altitude in mountainous terrain).

SPEED. Indicates the desired ground speed for the waypoint as set by the user when the speed is locked or as automatically calculated by the mission editor when the speed is unlocked. Note, at high altitudes the ground speed is significantly different (higher) than indicated airspeed, which is displayed in the aircraft cockpit.

GS (GROUND SPEED LOCK CHECKBOX). Selects between manual and automatic entry of the desired speed for the waypoint. Not applicable to a starting waypoint.

If the speed is locked, the group will attempt to maintain the desired speed on the route leg from the preceding waypoint to the current one.

MACH. The Mach number shows the ratio of the speed of the aircraft relative to the speed of sound. Depends on the height, temperature and pressure specified in the editor. It is shown only in the aircraft group panel.

START/ETA (Estimated Time of Arrival). Indicates the desired arrival time for the waypoint (or group start time in the case of the initial waypoint). This is set by the user if the ETA is locked or calculated automatically by the mission editor if the ETA is unlocked.

Fix Time (ETA LOCK CHECKBOX). Selects between manual and automatic entry of the desired ETA for the waypoint. The ETA is locked for the initial waypoint (group start time) by default.

When locked, the ETA field can be used to set a delayed group activation in the mission by setting an ETA for the initial waypoint at a time past the mission starting time. For example, if the mission is set to start at 12:00:00/1 and the user wishes to have the group appear in the mission 15 minutes after the mission starts, he can lock the ETA for waypoint-1 at 12:15:00/1. Furthermore, if the user wishes for the group to appear in the mission based on a triggered event of unknown time, he can set the ETA for the starting time past any likely mission time, such as 100 days later (12:00:00/100). In such a scenario, the group will not appear until triggered by the scripted event or the starting ETA is reached.

If the ETA for the waypoint results in calculated speeds that are too slow or too fast for the aircraft type, the font will become red to indicate an invalid entry and a corresponding error message will appear when the user attempts to close the properties panel or save the mission. See above for more information.

Waypoint Mode Buttons

These three buttons allow you to add, edit and delete route waypoints.

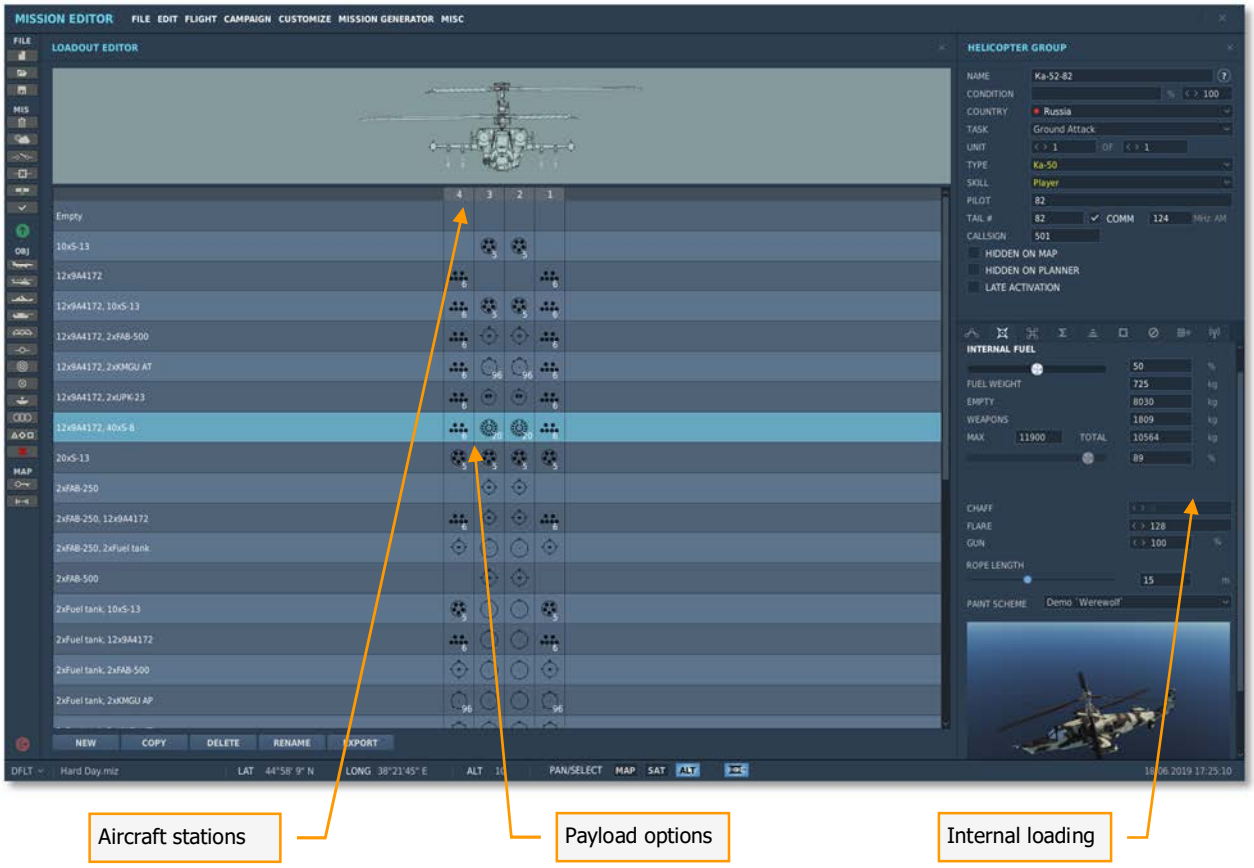


- **ADD** – In ADD mode, each press of the left mouse button over the map adds a new waypoint. This mode is selected on by default when a Group Properties Mode is open.
- **EDIT** – In EDIT mode, existing waypoints can be selected on the map for editing in the waypoint properties panel.
- **DEL** – Pressing this buttons deletes the selected waypoint.

Payload Mode



The Payload screen allows you to set the aircraft's external stores (weapons, fuel tanks, and pods), internal fuel, quantities of chaff, flares and gun rounds, and the paint scheme.



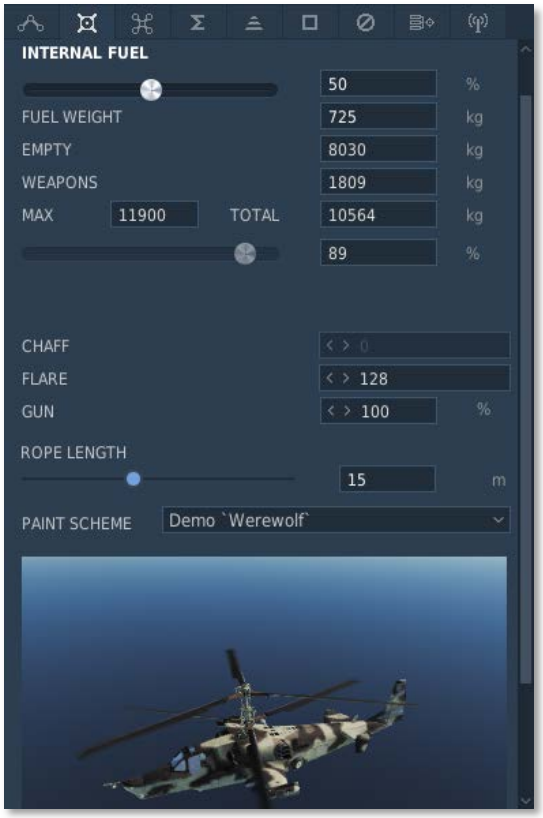
At the top of the Payload page is a drawing of the aircraft (head-on) that illustrates the station number assigned to each station. Stations can load weapons, fuel tanks, and pods according to the aircraft and station.

Below the illustration is the Loading Chart and this lists each loading payload package as a separate line. To the left of each line is the name of the payload package, and each column to the right represents a different station and what is loaded on it. To select a payload package, click on it and it will be highlighted. If you right click your mouse on a station box in the chart, a pop-up window lists the possible types of payloads that can be hung on that station and by then moving your mouse to the right you can select the specific payload to load on that station. At the bottom of the pop-up window is the Remove option that allows you to clear the station.

Below the Loading Chart is a group of five buttons that allow you to manage the chart. These include:

- **NEW**. In addition to the existing payload packages, you may press the NEW button at the bottom of the Loading Chart to create your own. Upon doing so, a pop-up window will prompt you to create a name for the new package. After you create a name and press the OK button to accept it, a new line on the loading chart with the name you entered will be displayed. You may then load each station using the right-mouse click function on each station.
- **COPY**. You may also create new payload packages by making a copy of an existing one and then modifying it. To do so, select the package you want to use as a base and then press the COPY button. When you do so, you will be prompted to create a new name for the package. Once you do so, a new payload package line will be created with the new name but using the package assignment of the base-payload you selected.
- **DELETE**. To remove a payload package from the chart, select the package you want to remove and then press the DEL (delete) button.
- **RENAME**. If you are not satisfied with the name of a payload package, you may select the package and then press the Rename button. You will then be prompted to enter a new name for the package. Once you press OK, the package will be renamed.
- **EXPORT**. When you create an aircraft group and set its TASK, a group of payload packages will be automatically associated with that task. The Export button allows you to associate a payload package with another Task. For example, if you have an all-ATGM package in the CAS Task that you also want to use as a Ground Attack Task package, you could select the package in CAS and then Export it to Ground Attack. The next time you create a Ground Attack Tasked group, the package would then appear as one of the payload options.

To the right of the Payload screen is a set of controls that allow you to adjust the internal loading of the aircraft and the paint it will wear.



- **INTERNAL FUEL** (slider). The INTERNAL FUEL slider can be dragged left/right to set the amount of fuel loaded on the aircraft in percent of maximum.
- **FUEL WEIGHT.** Loaded internal fuel as a kilogram (kg) value.
- **EMPTY.** Empty weight of the aircraft without fuel and payload package in kilograms.
- **WEAPONS.** Total weight of all loaded stores on stations in kilograms.
- **MAX.** This field displays the maximum, safe total weight of the aircraft in kilograms.
- **TOTAL.** Total weight of aircraft to include empty weight, fuel weight, and payload package weight in kilograms.

The non-interactive slider bar below the MAX and TOTAL fields provides a visual indication of the maximum percent of loading on the aircraft as a percentage.

Non-fuel internal loading of the aircraft is listed next:

- **CHAFF**. Number of chaff bundles loaded. This can be decreased/increased using the "<" - ">" arrow buttons.
- **FLARE**. Number of flares loaded. This can be decreased/increased using the "<" - ">" arrow buttons.

Note, the total number of chaff and flares is limited by the number of countermeasure cartridges on the aircraft, so increasing one will decrease the other.

- **GUN**. Percent of maximum number of cannon rounds that can be loaded.
- **AMMO TYPE.** Selection of the ammunition type loaded for the gun.
- **ROPE LENGTH**. Adjuster of the length of the rope of the cargo transported on the external suspension of the helicopter.
- **PAINT SCHEME.** Selection of the aircraft paint scheme ("skin"). The number of choices depends on the aircraft.

Triggered Actions Mode

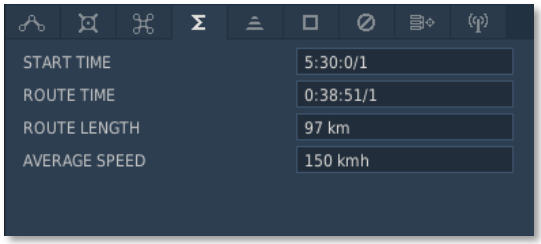


The Triggered Actions button opens the Triggered Actions Panel. This panel can be used to add or edit group actions analogous to the Advanced Actions Panel, but activated based on trigger rules as scripted in the Triggers Menu using the AI TASK trigger action.

Route Summary Mode



The Summary panel provides information on the entire flight route of the selected air group.



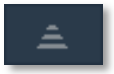
START TIME. The mission start time for the group in Hour:Minute:Second/Day.

ROUTE TIME. How long it will take the aircraft to fly the route, assuming no departures from the planned route. This is indicated in Hour:Minute:Second/Day format.

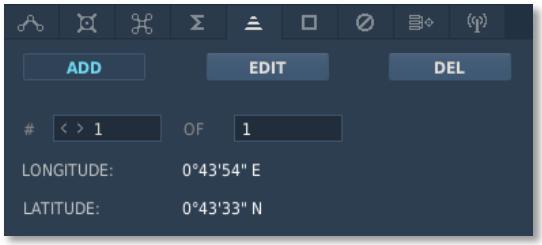
ROUTE LENGTH. Total distance of the route in meters.

AVERAGE SPEED. The average indicated airspeed of the route by totaling the assigned airspeed of each route leg and then dividing by the number of legs.

INU Fix Point Mode



The INU Fix Point button opens the INU FIX POINT panel. This panel is used to place reference points with known earth coordinates for the Inertial Navigation Unit (INU) of the player aircraft. These points can be used to update the INU when flying over or designating them using the targeting system. Usually visible landmarks are used as update locations, such as bridges or other easily identified landmarks.



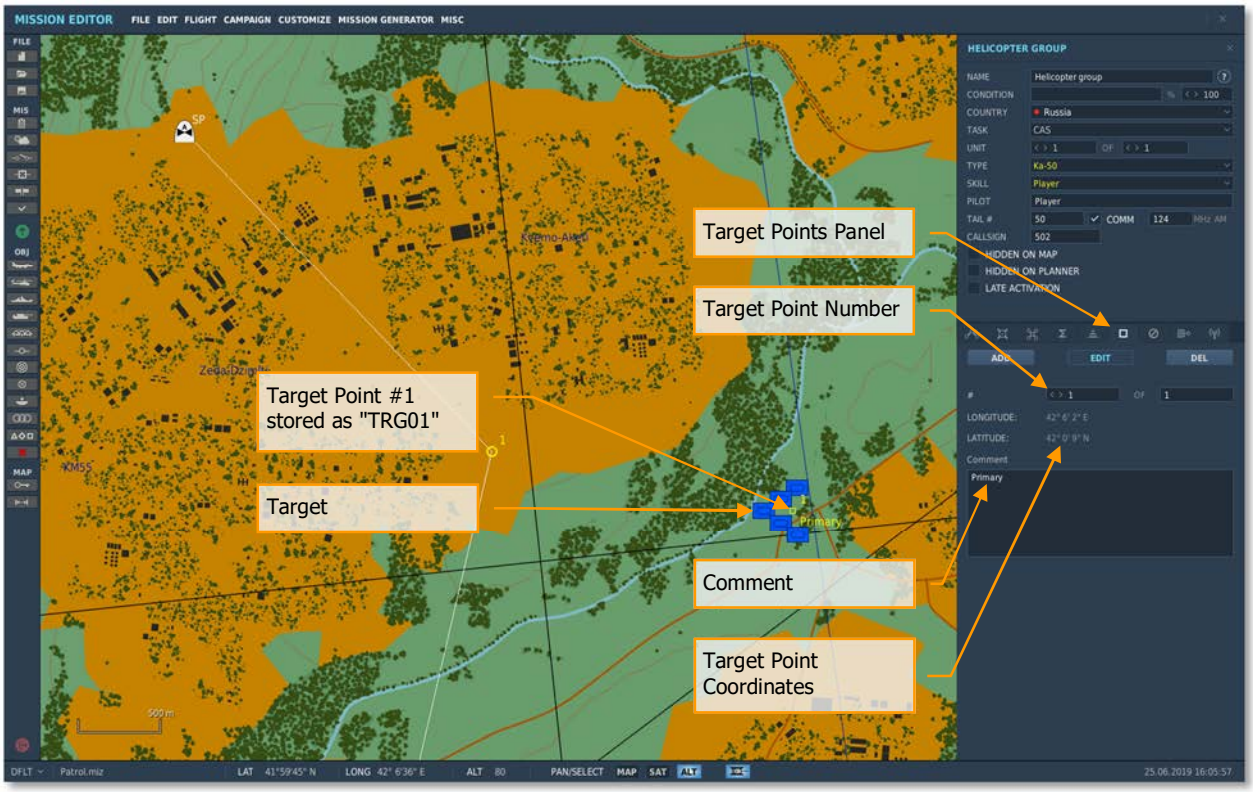
The coordinates of the point will appear in the INU FIX POINT panel. Fix Points can be placed directly on the map using the mouse in ADD mode or moved over the map in EDIT mode. Pressing the DEL button will delete the fix point.

Target Point Mode



The Target Point mode allows you to set target points for the navigation system.

Target Points can be used by the mission designer to designate coordinate points on the ground as targets. These points can be selected by the pilot in flight using the "OT" button on the navigation system and are indicated on the PVI-800 and ABRIS displays.



ADD. Add a new Target Point.

EDIT. In EDIT mode, the Target Point can be moved over the map by clicking and dragging the point.

DEL. Delete the selected Target Point.

The Target Point panel indicates the latitude/longitude coordinates of the selected Target Point.

The Comment field allows you to add a custom comment for the selected Target Point.

Note, the navigation system of the Ka-50 can store up to 10 Target Points.

In flight, Target Points 0 - 9 can be selected on the PVI control panel.



A detailed description of operations using the operational points panel is given in the pilot's manual of the aircraft equipped with the operational points function.

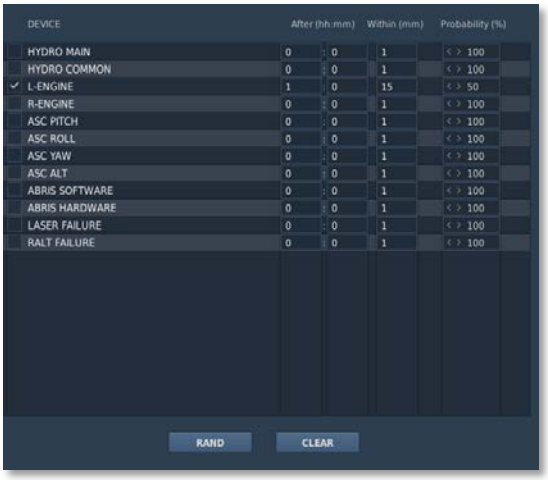
Failures Mode



The FAILURES Panel allows you to set the aircraft's failures. The list of failures depends of aircraft type.

Your aircraft might be damaged in combat as the result of a missile or shell hit. To be ready for such a situation and to be able to fly the aircraft when onboard systems have failed, this tool allows you to simulate such failures. Select the Failures button and specify in the failures panel the device you want to fail. Specify the exact failure time for the device or a Time Between Failure (TBF) if you wish to simulate failures without an exact failure time, but will occur within that specified time frame. For example: if you specify 30 in the Within field for the left engine, this means that your left engine will fail within the next 30 minutes. If you were to specify 0:15 in the After field, the failure will occur with set probability within the next 30 minutes after 15 minutes after takeoff.

Additionally, you can adjust the probability of failures by adjusting the Probability (%) field using the arrow buttons. The Probability ranges between 0% (no chance) to 100% (certain failure). Note that failure(s) only apply to the Player aircraft.



Along the left side of the window is the DEVICE list that can be used to set an in-mission failure.

To the right of each device line is a set of fields that allow you to set a time frame from mission start that the failure will occur and the probability that the failure will occur.

- **AFTER** (hh:mm). Input the hour and minutes from mission start that the failure will occur.
- **WITHIN** (mm). Use the third field to determine the time that the failure may happen.
- The **PROBABILITY** (%) field allows you to enter the probability as a percentage (0 – 100) that the failure will occur in the mission.

The steps needed to set aircraft failures include:

1. Check the checkbox of the desired failure.
2. Set the time at or after which the failure is to take place (hh:mm).
3. Set the time within which the failure is to take place (mm).
4. Set the probability for the failure to occur (%).

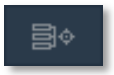
For example, to set a left engine failure an hour into the flight within a 15-minute time span and a 50% probability, check the L-ENGINE Device box, enter 01:00 in the After field, enter 15 in the Within field, and enter 50 in the Probability field.

At the bottom of the window are two additional controls:

- **RAND**. Pressing the random button will randomly select devices to fail and set random time windows and probabilities.
- **CLEAR**. The clear button will disable all failures and set all time windows and probabilities to zero.

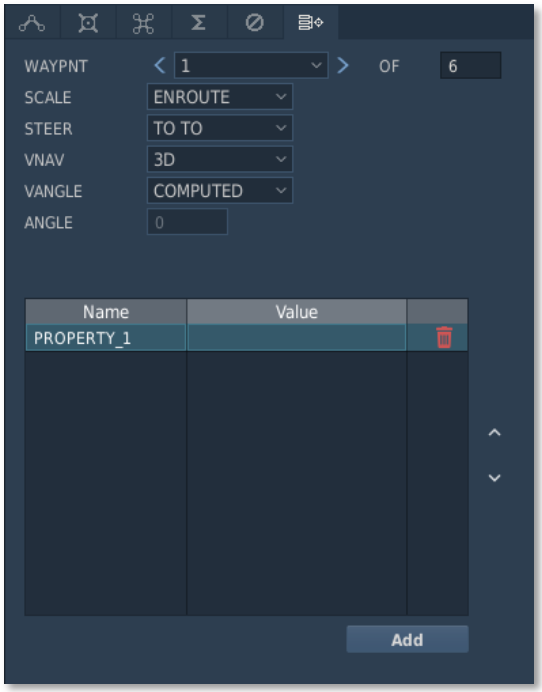
Note, the mission editor supports group copy/paste functionality. To copy a group, select it and press [CTRL+C]. To paste the group with identical properties, position the mouse over the desired point on the map and press [CTRL+V].

Waypoint Properties Mode



A detailed description of the operations for using the waypoint parameters panel is given in the manual for the NS 430 navigator and the pilot's manual of the aircraft equipped with the waypoint function.

Example for A-10C. Attributes define the steering modes selected for the desired steerpoint. These steering modes are; scale, steer, vertical navigation (2D/3D), vertical angle (when in 3D mode only), and selected vertical angle (when in 3D mode only).



Radio Presets Mode



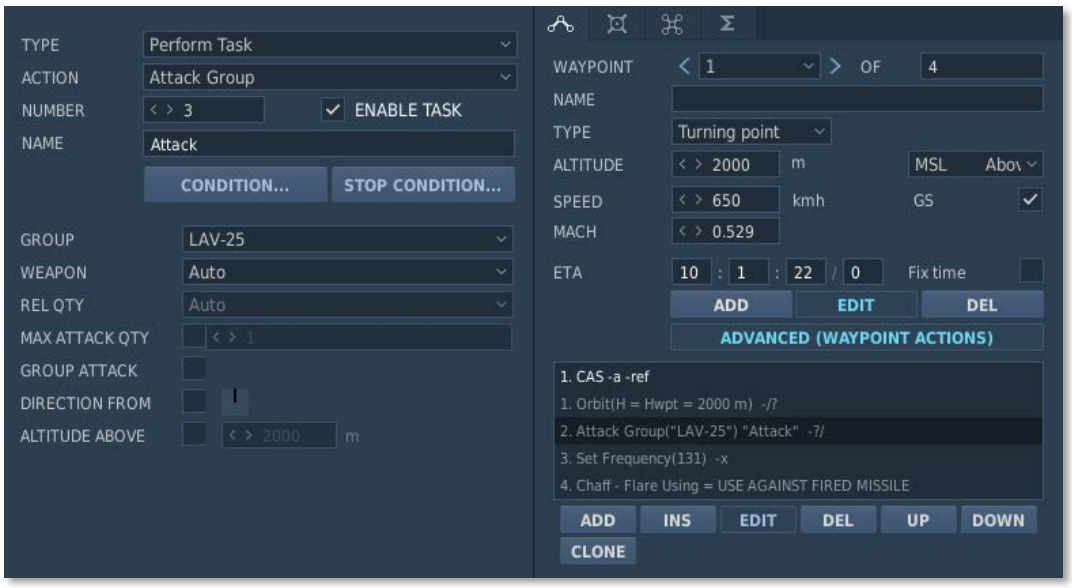
The panel is used to preset frequencies on the channels of the radio stations on this type of aircraft. User-set frequencies are saved to the mission file.



Advanced Actions Mode (aircraft groups)



Pressing the Advanced tab open the Advanced Actions Panel, which lists the actions associated with the selected waypoint:



This panel is used to set (add/edit/delete) group actions (Perform Task, Start Enroute Task, Perform Command, Set Option), which are tasked to the AI at this waypoint by the user or generated automatically by the Mission Editor.

Panel Interface Controls

The following buttons are located at the bottom of the Advanced Actions Mode:

- **ADD** – Adds a new action to the list. When pressed, a default No Task action is created, which can be edited in the Action Settings pop-up panel to the left.
- **INS** – Inserts a new action above the currently selected one.
- **EDIT** – Opens the Action Settings pop-up panel to display and edit the settings of the action. The same functionality is duplicated when clicking on any route waypoint on the map.
- **DEL** – Deletes the selected action.
- **UP** – Moves the selected action up the list.
- **DOWN** – Moves the selected action down the list.
- **CLONE** – Clones the selected action down the list.

Actions List

The Actions list includes all Perform Tasks, Start Enroute Tasks, Perform Commands, and Set Options associated with the currently selected waypoint (to be more precise, associated with arrival at the currently selected waypoint by lead AI unit of the group). The actions are executed sequentially or as prioritized once the group leader reaches the waypoint.

As discussed previously, Perform Tasks and Start Enroute Tasks are executed as processes, i.e. requite time to complete. Perform Commands are executed instantaneously. Options set rules and limitations for the group.

Only one Task can be performed at a time. Some additional priority rules apply when triggered actions are present in addition to standard waypoint actions, which we will discuss below.

Multiple Start Enroute Tasks can be active at the same time, but only one can be executed at a time. All Start Enroute Tasks are stopped if a Task needs to be performed. If an Start Enroute Tasks is stopped to perform a Task, the Start Enroute Task is reactivated upon completion of the Task.

Commands and Options are executed instantaneously according to the order in the actions list or priority setting.

All actions listed in the actions list are displayed using a standard format of Action Number, Action Type with parameter(s), Action Name, Action Attribute:

<number>.<type>(<parameter #1>, <parameter #2>, … <parameter #n>)"<name>" -<attribute #1> -<attribute #2> …-<attribute #n>

The Action Name is an optional custom name for the action for easier identification in the actions list. This field can be left blank (empty).

The Action Attributes are used to reference one or more of the following possible attributes:

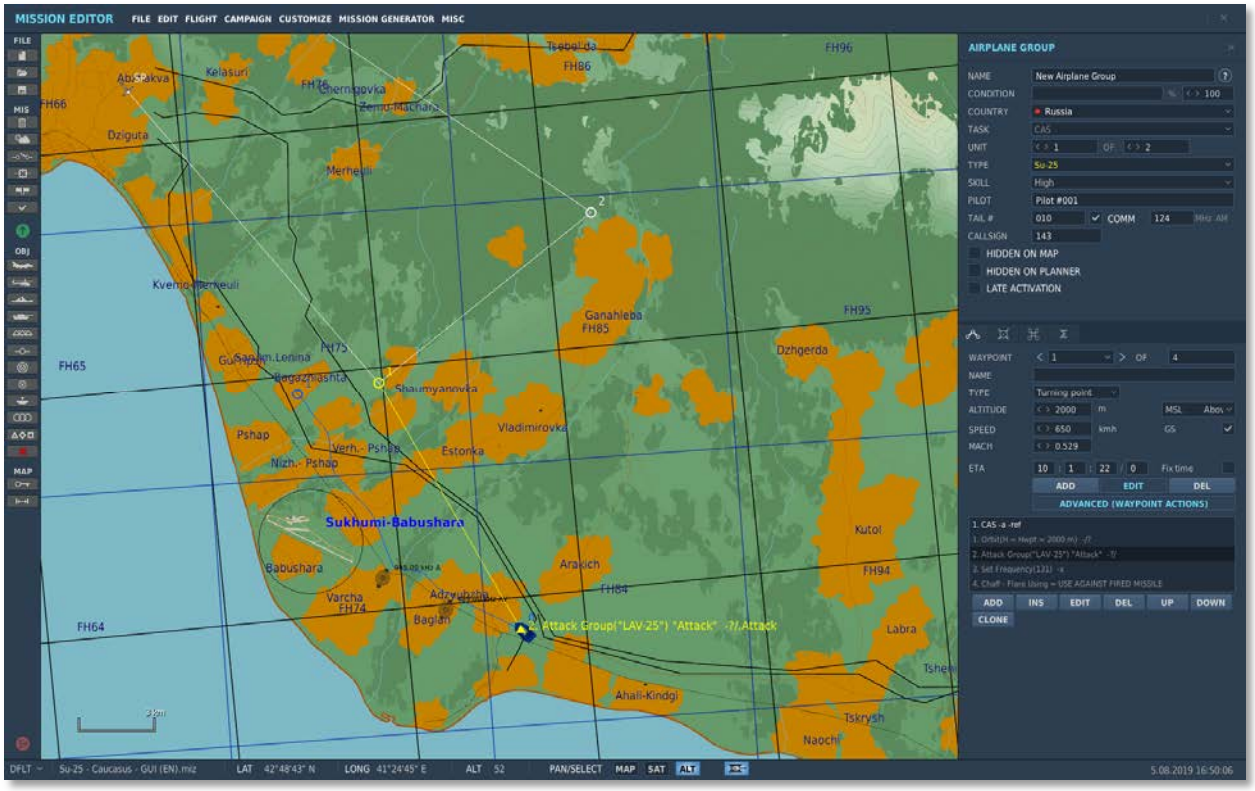
- "-**x**" – disabled action;
- "-**a**" – automatically generated action;
- "-**!**" – invalid action. This attribute indicates an incorrect action, usually resulting from the action being incompatible with the group's mission task as set in the Group Properties Panel. This may be due to the mission designer changing the group's mission task after creating the action;
- "-**?/**" – action includes a start condition;
- "-**/?**" – action includes a stop condition;
- "-**?/?**" – action includes both a start and stop condition;
- "-**ref**" – action was started at a previous waypoint.

Additionally, black, grey and red font colors are used to denote:

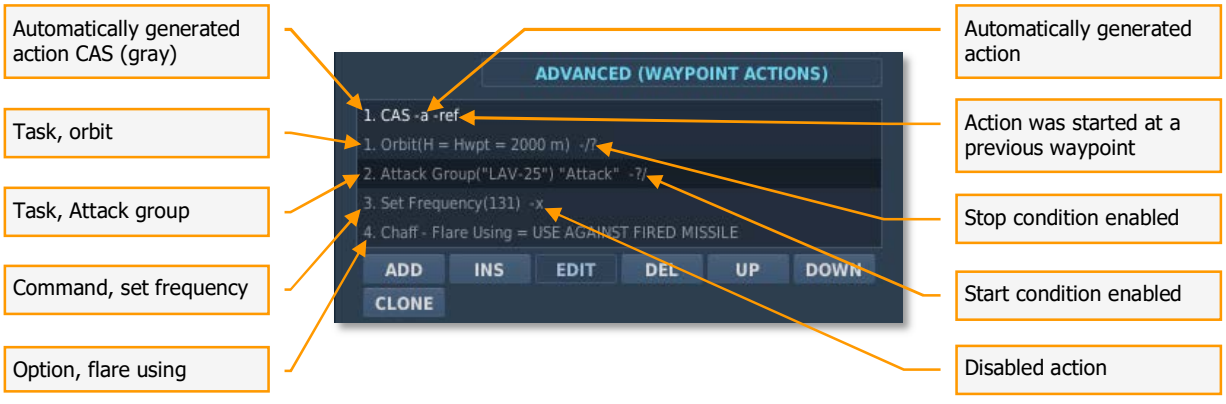
grey – automatically generated action;

white – valid action;

red – invalid action.

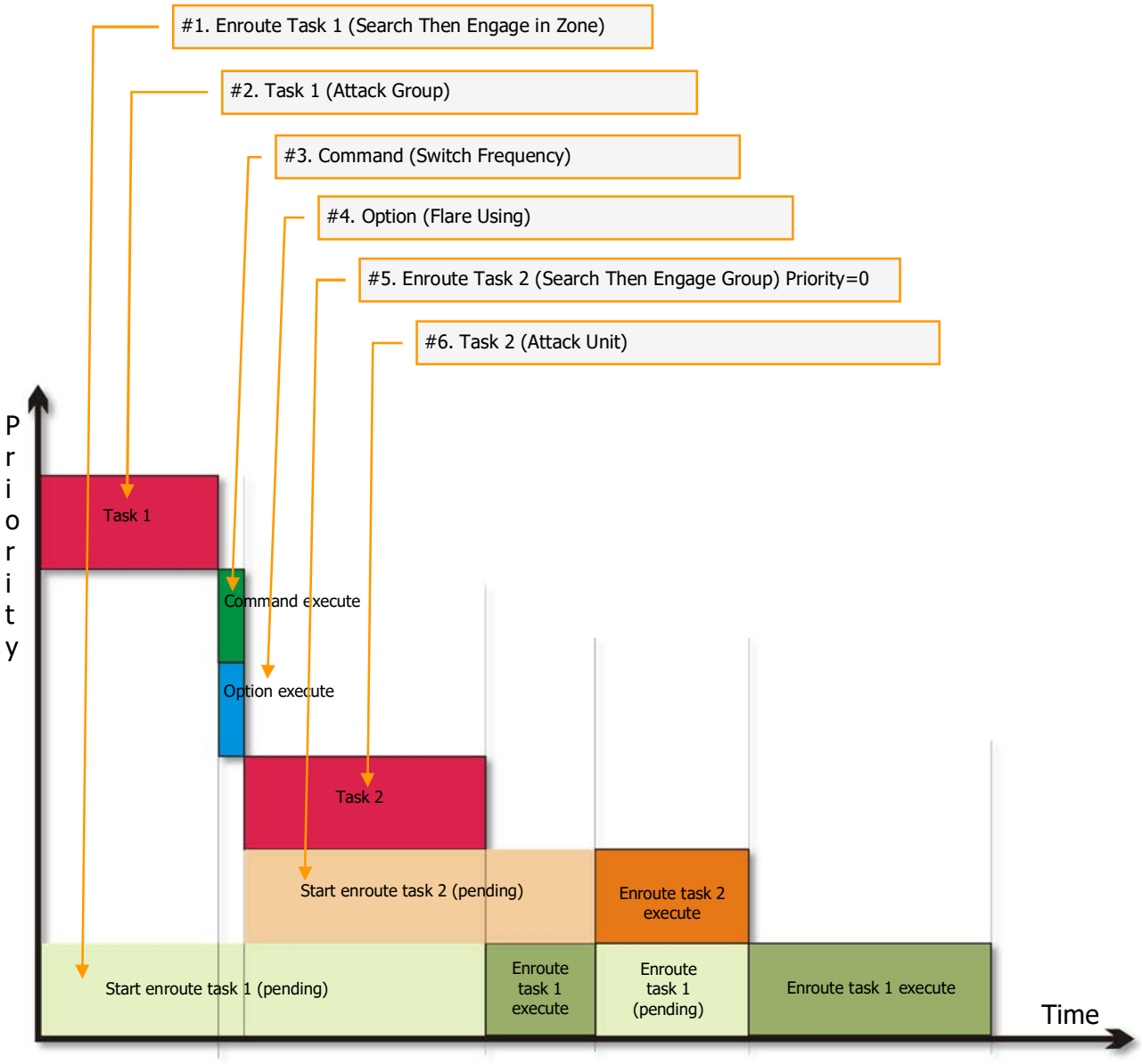


Example of actions for an Su-25 attack group at WP1.





Let's examine a sample scenario of 6 actions to better understand the execution sequencing based on the action type, list order, and priority setting:



At mission start, the AI has two concurrent actions: action #1 (Enroute Task 1 <Search Then Engage in Zone>) and action #2 (Task 1 <Attack Group>). Unless set otherwise by the mission designer, any Task has a higher priority over any Enroute Task, so the AI begins to execute action #2, while storing action #1 in memory. Once action #2 is completed (or is no longer executable), the AI proceeds to execute action #3 (Command <Set Frequency>), action #4 (Option <Flare Using>), action #6 (Task <Attack Unit>), while also starting #5 (Enroute Task 2

<Search Then Engage Group>) and storing it in memory. Although both action #1 and action #5 (enroute tasks) are now running in the background, they cannot be executed while action #6 (a task) is underway. Once action #6 is completed (or is no longer executable), the AI can return its attention to actions #1 and #5. The conditions for action #1 become true first, so the AI begin to execute this action. However, action #5 has the higher priority setting of <0>, so when the conditions for both action #1 and action #5 are in effect, action #1 is stopped and action #5 is executed. Once action #5 is completed, the AI can return to executing action #1.

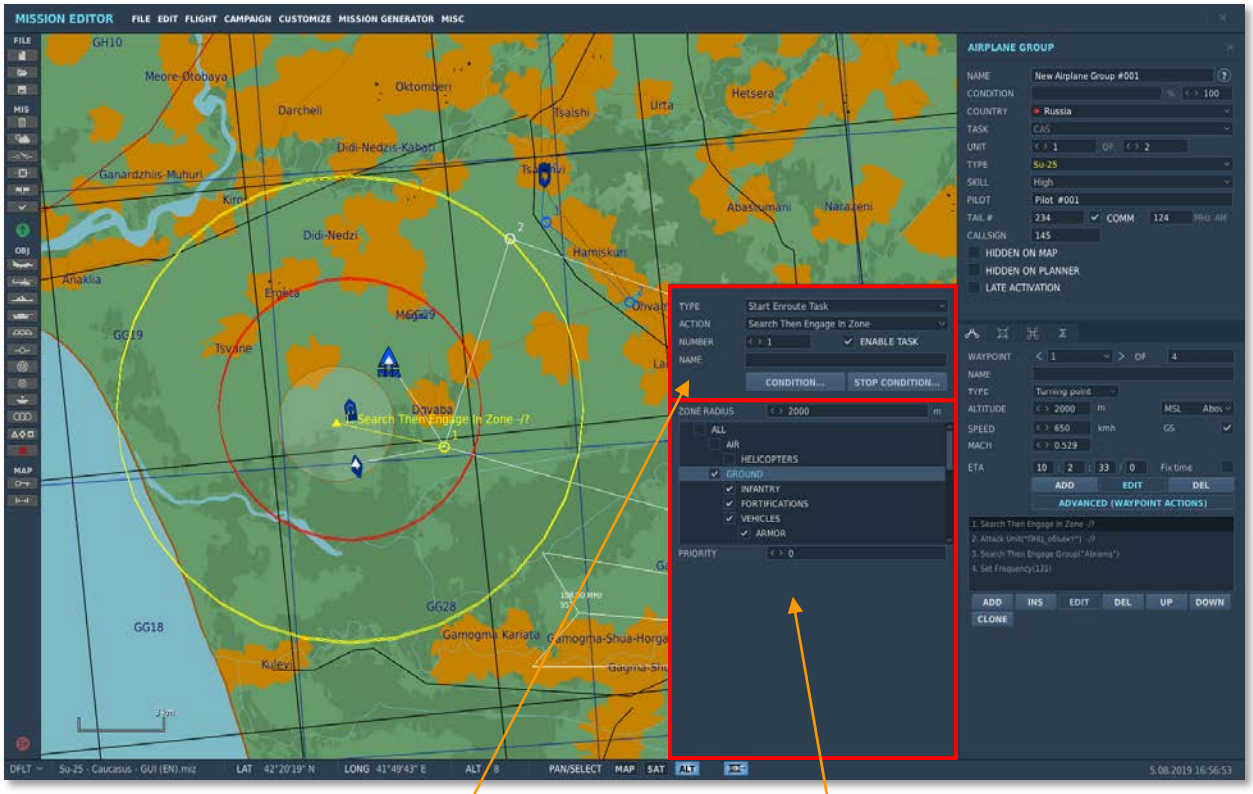
Triggered Tasks

Tasks activated by a trigger will have a higher execution priority over waypoint Tasks. For example, if a triggered Task is activated while a waypoint Task is underway, the waypoint Task will be stopped and stacked in memory while the AI switches to the triggered Task. Once the triggered Task is completed (or skipped due to not being possible), the previously running waypoint Task will be resumed.

Triggered Enroute Tasks

Enroute Tasks activated by a trigger will be added to the list of currently active Enroute Tasks. As with waypoint Enroute Tasks, triggered Enroute Tasks remain active until stopped by a stop condition.

Action Properties Panel

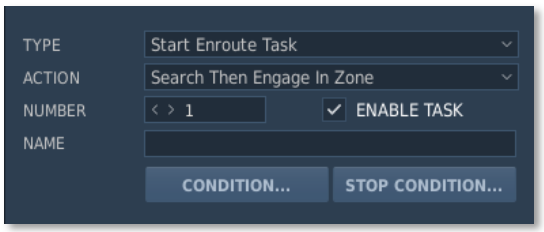


The Action Properties Panel is used to configure the properties of the action selected in the actions list. The panel can be opened by pressing the ADD, INS or EDIT buttons in the action list, or by double-clicking an action. By default, pressing the ADD button adds a "No Task" action and opens a black Action Properties Panel. In the Action Properties Panel, the actions in the actions list can be cycled using the NUMBER </> buttons in the properties panel.

Upper Action Properties Panel

The Upper Action Properties Panel is used to set the main action properties, including the action type, as well as access the start and stop condition panels of the action:

TYPE – action type drop-down menu.



- Perform Task – set a Task action;
- Start Enroute Task – start an Enroute Task action;
- Perform Command – set a Command action;
- Set Option – set an Option action.

ACTION – action drop down menu to select the specific action of the action type chosen in the TYPE menu (specific actions discussed below).

NUMBER – indicates the selected action's sequence number in the actions list panel.

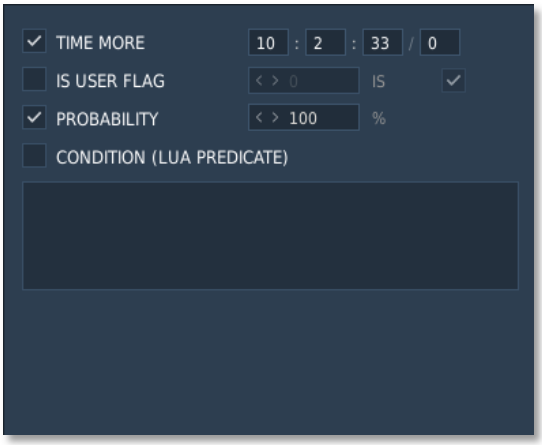


ENABLE TASK (check-box) – removing the check disables the selected action and adds an "-x" attribute in the actions list panel. This function allows you to test AI behavior with and without the selected action without having to delete and re-create it in the actions list.

NAME (input string) – input in this field is optional and can be used to enter a custom action ID to ease navigation and identification of the action for the mission designer or other potential users.



CONDITION – this button opens and closes the Start Conditions Panel. Start conditions can be used to set rules for the selected action to begin and can include: mission time, logical variable (flag state), probability, and custom scripting in LUA code language.

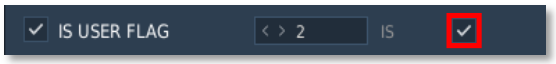


All of the start condition operate using a "OR" rule. When multiple conditions are set, any one being true will activate the action.

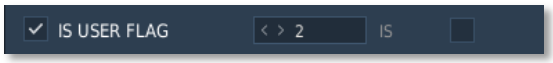
TIME MORE – sets the in-game mission time after which this condition is in effect.

IS USER FLAG – sets a flag number and status (on/true or off/false) as a start condition. The flag number can be entered manually or cycled using the left/right arrow keys. The flag state is set using the checkbox to the right of the flag number. If the box is checked, the flag needs to be on/true for the condition to be in effect. If the box is unchecked, the flag needs to be

off/false for the condition to be in effect. Flags can be set on and off using the trigger functions in the Trigger Menu.



If flag 2 is ON/TRUE



If flag 2 is OFF/FALSE

PROBABILITY – sets the probability in per cent of the action being started.

CONDITION (LUA PREDICATE) – this window can be used to enter a custom expression for the action using Lua code language. The code entered will be generated as a function:

function <generated_name>()

return <lua code>

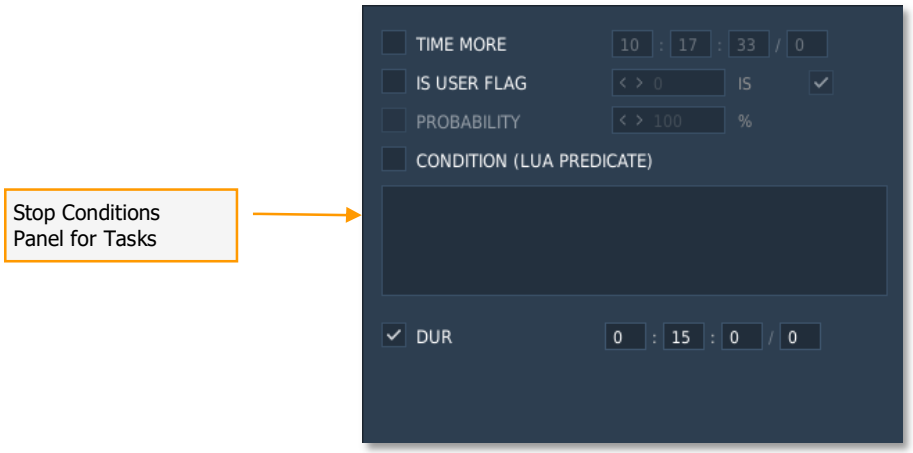
end,

This function will be run periodically by the simulation and used as a start condition for the action.

STOP CONDITION – this button opens and closes the Stop Conditions Panel. Stop conditions can be used to set rules for the selected action to end and can include: mission time, logical variable (flag state), custom scripting in Lua code language, duration in time since the action was started, and a route waypoint (for Enroute Tasks only).

While Tasks can be stopped either by a stop condition or by the AI in-game when the task is completed (or skipped), Enroute Tasks are designed to be in effect for the duration of the route, so can only be stopped using a stop condition.

The Stop Conditions Panel varies slightly between Tasks and Enroute Tasks:





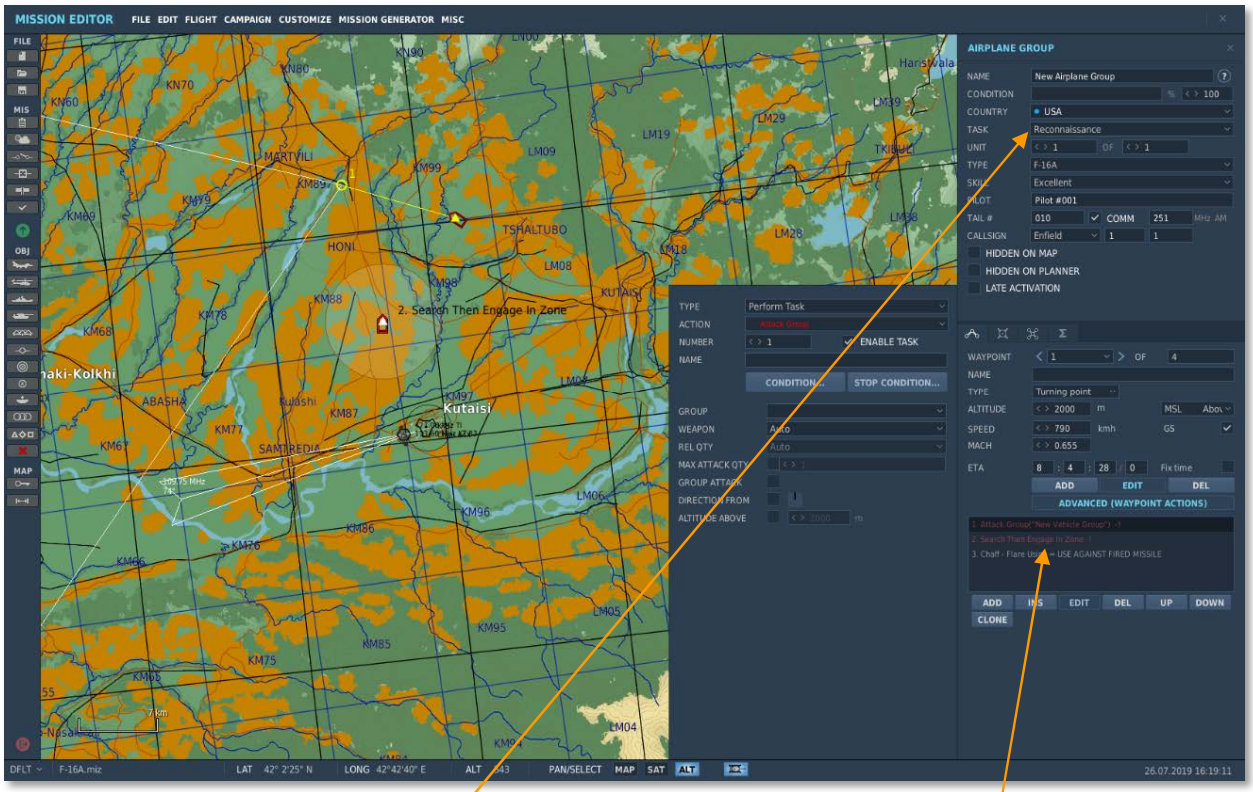
The TIME MORE, IS USER FLAG, PROBABILITY and CONDITION (LUA PREDICATE) functions are identical to the Start Conditions Panel.

The **DURATION** function sets the maximum duration in time for the action to be active. The action will be stopped once the set duration time is exceeded. This can be a useful condition for Orbit tasks to limit the AI's time on station.
The **LAST WPT** functions sets the waypoint at which the action will be stopped.

Configuring Advanced Actions

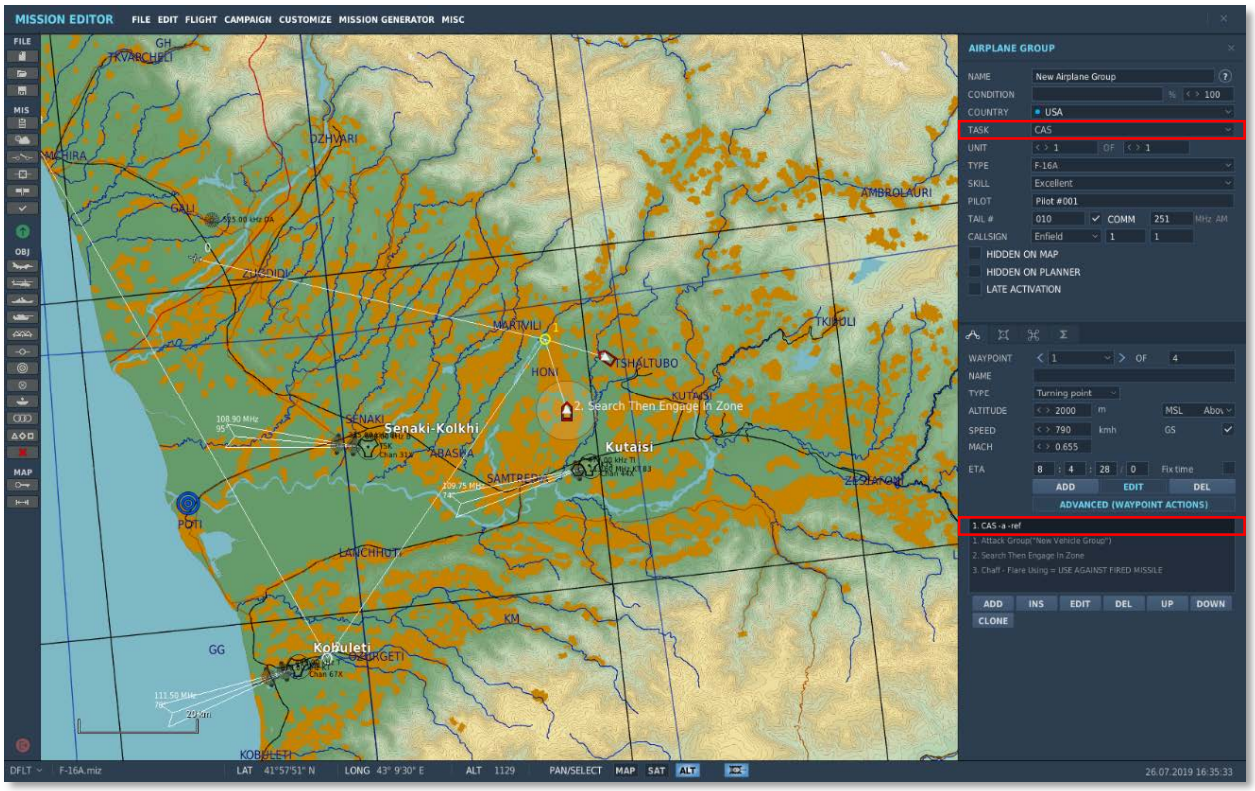
Once an action is selected in the ACTION drop down menu of the Upper Action Properties Panel, the action properties appear in the Lower Action Properties Panel and can be customized by the mission designer.

The specific actions available for selection depend on the type of group being edited and the group task (CAS, Intercept, etc.). If the group task is changed after initial group creation, it may conflict with the actions already set in the actions list. In such cases, the invalid action(s) will be presented in red font and a "-!" attribute will be added to indicate an invalid action. An error window will also pop up if the user attempts to close the group Properties Panel or save the mission.



Automatic Actions

An air group's task setting (CAS, Intercept, etc.) does not affect its behavior in-game, but only serves to filter the available actions in the Advanced Actions Panel.
To allow users to build missions without involving themselves with the Advanced Actions Panel, an initial action is automatically generated for the air group at the initial waypoint. This action appears as a repeater of the group task, however in this case it *does* determine the groups behavior in-game. Any automatic action includes the "-a" attribute in the actions list. This action is automatically repeated as a reference with a "-ref" attribute at all other route waypoints.



If the group task is changed after creation, the automatic actions are automatically changed accordingly. The group task actions can also be added manually using the Advanced Actions Panel.

The following group tasks generate automatic actions of the same type: "AWACS", "Refuleing", "CAS", "CAP", "Fighter Sweep", "SEAD" and "Anti-ship". In all cases other then "AWACS" and "Refueling", the automatic actions are in fact Enroute Tasks to search for then engage targets appropriate to the task. Because these actions are automatic, the properties of these actions are pre-determined and cannot be edited.

Perform Task

As mentioned above, the Perform Tasks available for a group depend on the group type and the group task.

The following Perform Tasks may be available, depending on the group task:

- No Task
- Aerobatics
- Attack Group
- Attack Map Object
- Attack Unit
- Bombing
- Bombing Runway
- Cargo Transportation
- Embarking
- Disembarking
- Escort
- Follow
- Ground Escort
- FAC – Assign Group
- Orbit
- Land
- Refueling
- WW2: Big Formation
- WW2: Carpet Bombing

No Task

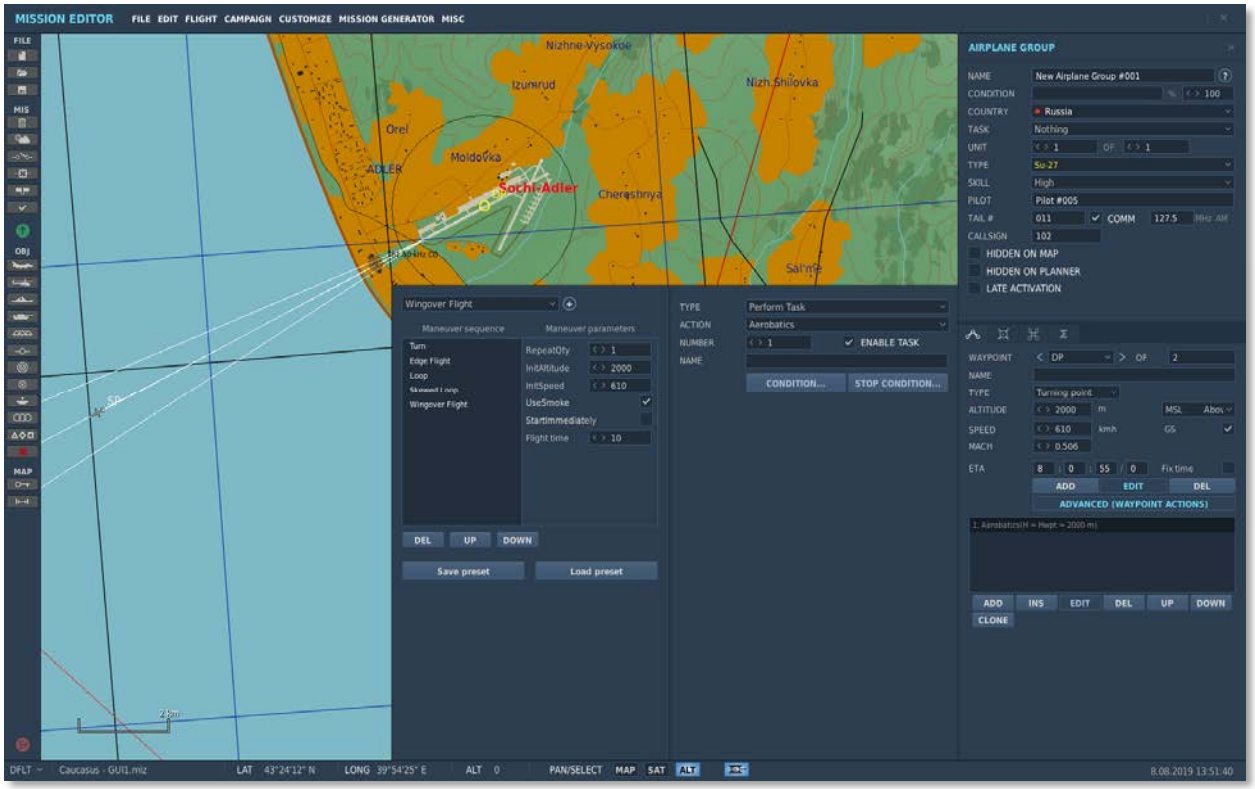
Available for all group types and all mission tasks. A No Task action does not set any action for the group to perform.

Aerobatics

The action is designed to perform maneuvers and aerobatic figures. Applicable to:

Group type(s): fixed-wing groups;

Group Task(s): "Nothing", "CAP", "CAS", "Fighter Sweep", "Ground Attack", "Intercept", "Reconnaissance", "Transport".



During aerobatic task selection, a panel opens, showing a list where you can select the necessary maneuver and add (+) them to the Maneuver sequence column. The buttons below: DEL, UP, DOWN – are used to set the required sequence of maneuvers. The SAVE PRESET and LOAD PRESET buttons save and load preset series of aerobatic figures and maneuvers, respectively.

Maneuver parameters:

RepeatQty. Number of repetitions.

InitAltitude. Initial altitude for maneuver execution.

InitSpeed. Initial speed for maneuver execution.

UseSmoke. Enables smoke.

StartImmediately. Executes the maneuver immediately after mission start.

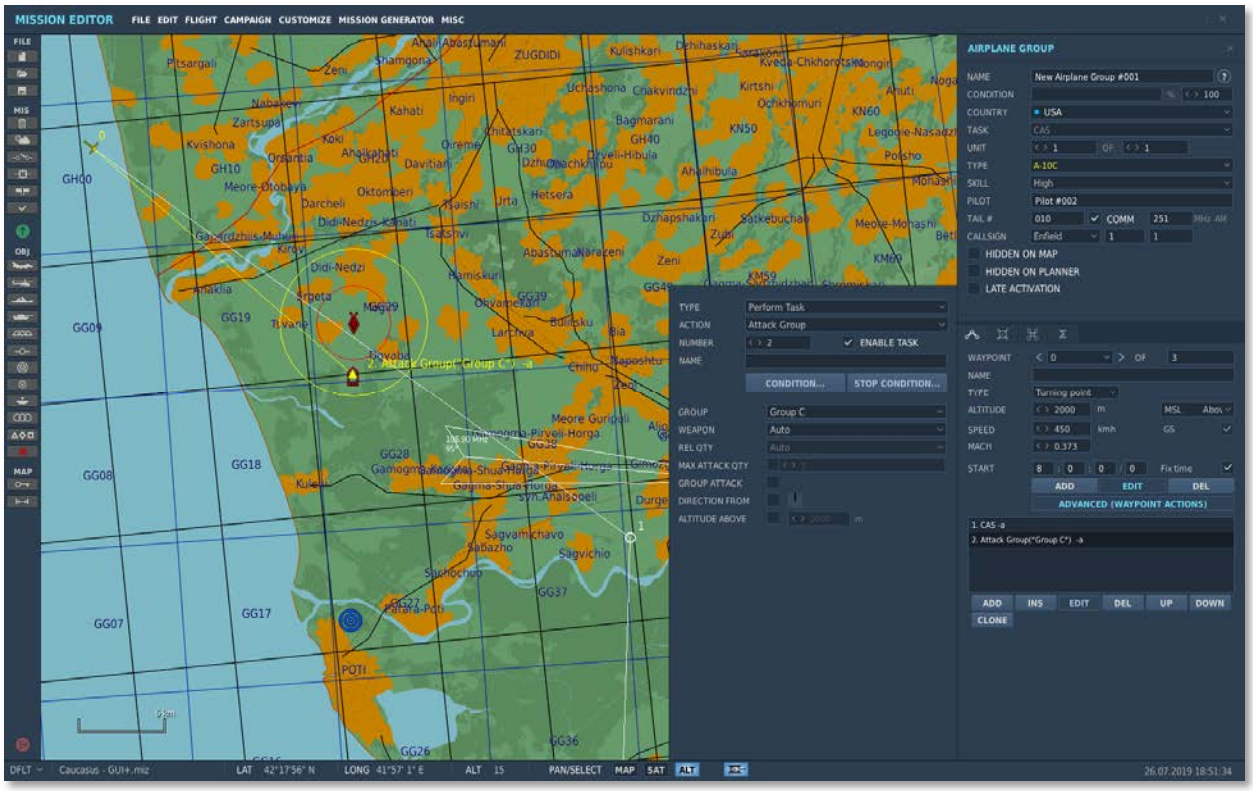
Flight time. Sets the time for maneuver execution.

Attack Group

Attack a specific enemy air, ground or naval group as designated by the mission designer. Applicable to:

Group type(s): fixed-wing and helicopter groups;

Group Task(s): "SEAD", "CAS", "AFAC" (can only designate ground groups as targets), "Anti-ship" (can only designate naval groups as targets), "Intercept" (can only designate air groups as targets).



***Note**: The Attack Group Task assumes full situational awareness for the attacking group. The target's location will always be known and the group will always attempt an attack if conditions permit.*

The target group can be designated by selecting it from the GROUP drop-down menu in the Lower Action Properties Panel or by left-clicking over the target group on the map.

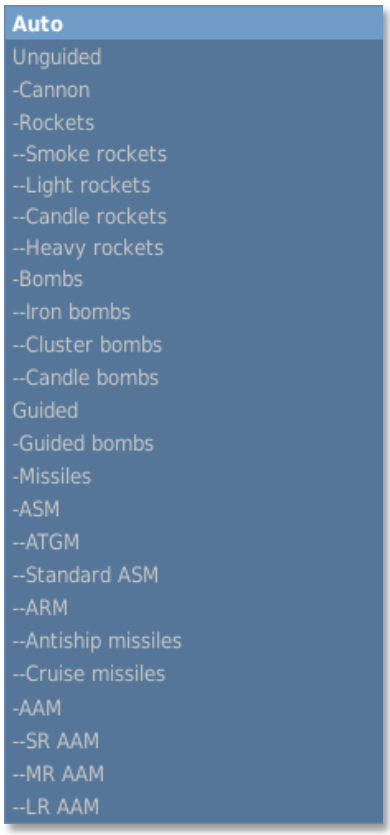
When designated as the target, a red triangle appears over the target group and a dashed line connects the target triangle to the waypoint of the action.

The Lower Action Properties Panel also allows you to set the specific weapon type for the attacking group to employ against this target or, alternatively, leave this choice up to the AI using the default "Auto" setting. If a specific weapon type is set, but the attack is not possible using this weapon type (for example, if the weapon type is missing from the payload or has been exhausted), the attack will not be performed.

Target Panel:

GROUP. Drop down menu to select the target group from all applicable enemy groups available in the mission.

WEAPON. Drop down menu to select the weapon types cleared to use for this action. Depending on the task to be performed for the AC AI, the list of weapons in the menu may differ. Below is a complete list of available weapons for AI in the game and its description. Each menu item assigns a specific category/type of weapon to use.



- **Auto**. Automatic selection of weapons by the AI.
- **Unguided**. Unguided bombs, rockets, cannon.
  - **Cannon**. Cannon only.
  - **Rockets**. Unguided rockets only.
    - **Smoke rockets**. Smoke rockets only.
    - **Light rockets**. Low-caliber unguided rockets.
    - **Candle rockets**. Illumination rockets only.
    - **Heavy rockets**. High-caliber unguided rockets.
  - **Bombs**. Unguided bombs only.
    - **Iron bombs**. General-purpose free-fall bombs.
    - **Cluster bombs**. Cluster bombs only.

        - **Candle bombs**. Illumination bombs only.
- **Guided**. Guided munitions (bombs, missiles).
    - **Guided bombs**. Guided bombs only.
    - **Missiles**. The use of all air-to-surface and air-to-air guided missiles aboard an aircraft.
    - **ASM**. Application of tactical guided missiles of air-surface class of all types on board the aircraft. The surface can be water and land.
        - **ATGM**. Anti-tank guided missiles only.
        - **Standard ASM**. Guided missiles on ground targets only.
        - **ARM**. Anti-radiation missile only.
        - **Antiship missiles**. Antiship missiles only.
        - **Cruise missiles**. Cruise missiles only.
    - **AAM**. Use of air-to-air guided missiles of all types on board aircraft.
        - **SR AAM**. Guided short-range missiles only.
        - **MR AAM**. Guided medium-range missiles only.
        - **LR AAM**. Guided long-range missiles only.

Note, by limiting the AI on the types of weapons cleared for use in the action, you may prevent the AI from successfully executing it (for example, when the weapon type has been exhausted).

REL QTY. The amount of payload cleared to expend in executing this action.

MAX ATTACK QTY. The maximum number of attack runs cleared for this action.

GROUP ATTACK. When checked, the entire flight group will participate in the attack of the target. By default, each aircraft in a flight group is assigned one target.

DIRECTION FROM. Allows you to set the course (azimuth) of the aircraft group for reaching the target for an attack.

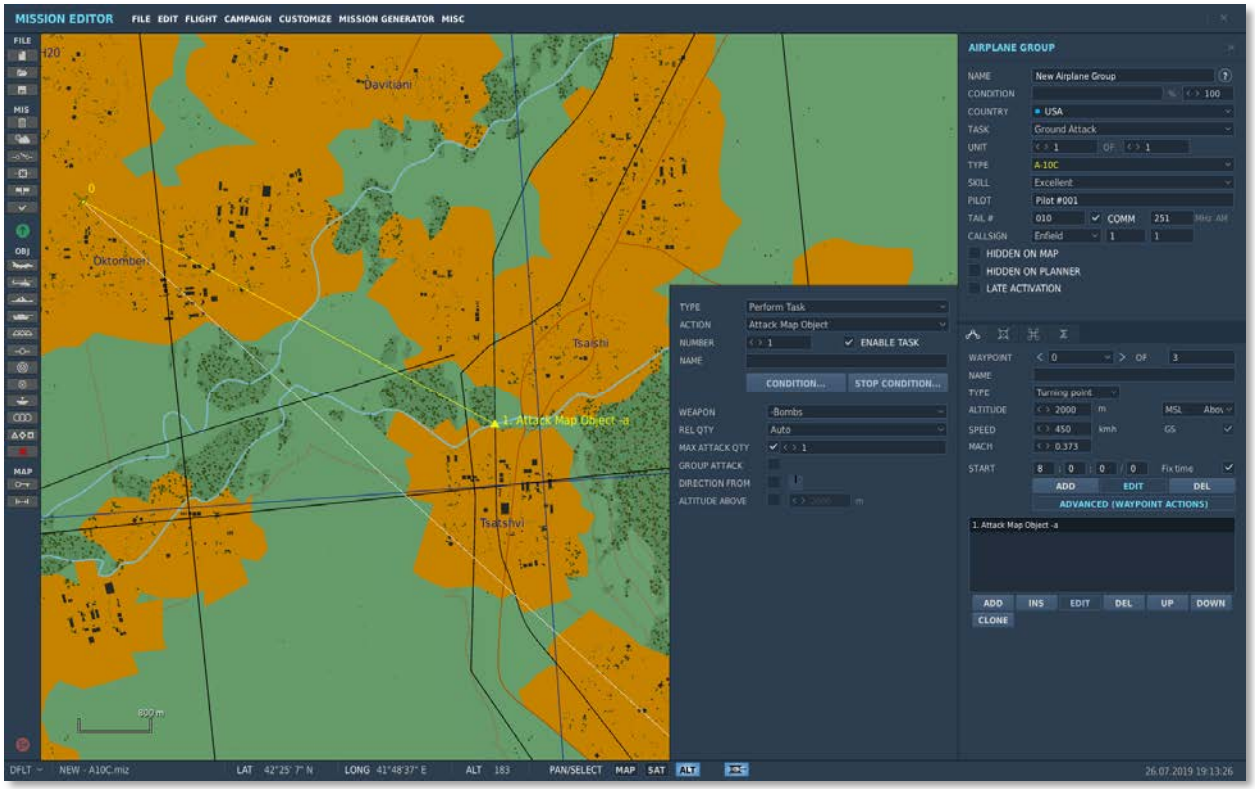
ALTITUDE ABOVE. Sets the altitude of the aircraft group on approach to target.

Attack Map Object

Attack a specific structure or object on the map, such as a bridge, bunker, etc., based on a map coordinate position. Applicable to:

Group Type(s): fixed-wing and helicopter groups;

Group Task(s): "Pinpoint Strike", "Ground Attack", "Runway Attack".



Because world structures are not individually selectable on the map, this Task designates a target point on the map with a red triangle and dashed line connecting the target point to the Task waypoint. You can drag the target point using the mouse to position it over the desired target on the map. Once the Task is initiated, the AI will attempt to find the closest structure or object to the target point and attack it using the weapons set in the task properties.

Target Panel:

WEAPON. Drop down menu to select the weapon types cleared to use for this action.

REL QTY. The amount of payload cleared to expend in executing this action.

MAX ATTACK QTY. The maximum number of attack runs cleared for this action.

GROUP ATTACK. When checked, the entire flight group will participate in the attack of the target. By default, each aircraft in a flight group is assigned one target.

DIRECTION FROM. Allows you to set the course (azimuth) of the aircraft group for reaching the target for an attack.

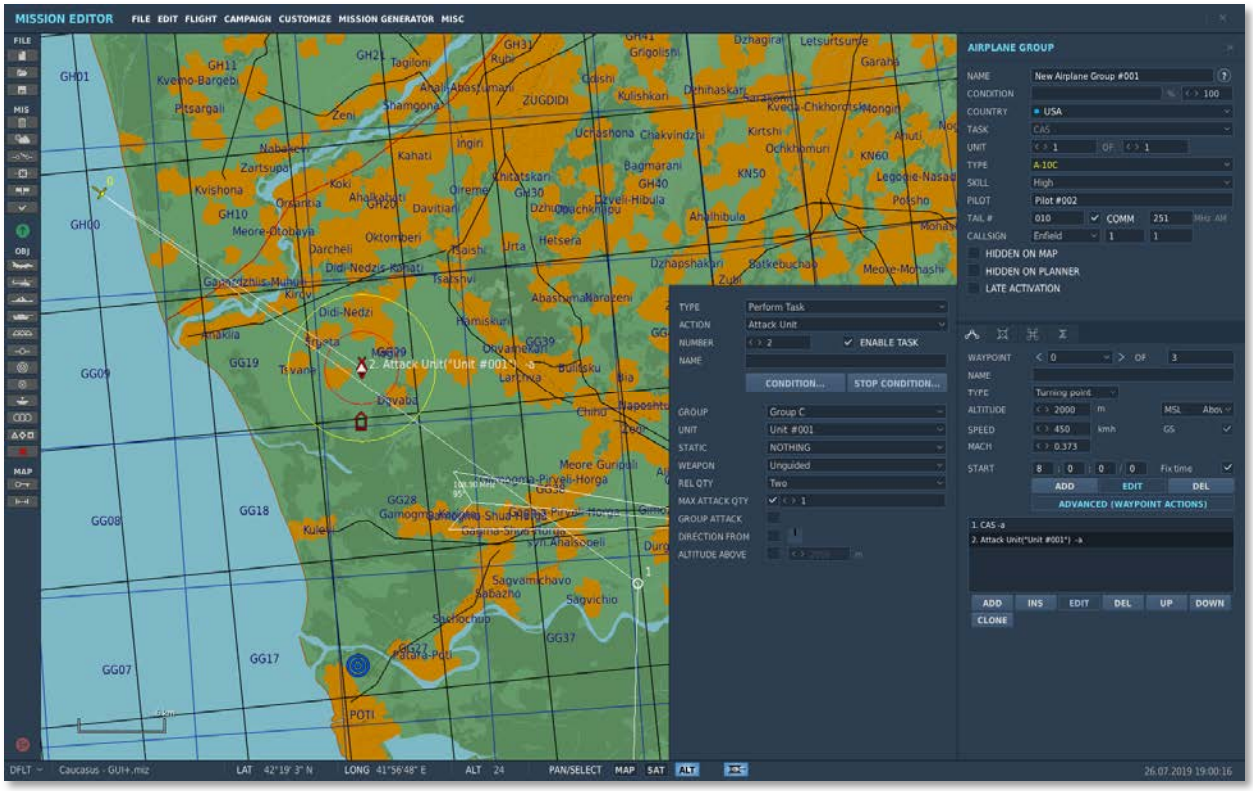
ALTITUDE ABOVE. Sets the altitude of the aircraft group on approach to target.

Attack Unit

Similar to the Attack Group Task, but allows the mission designer to designate a specific unit within the target group to attack. Applicable to:

Group Type(s): fixed-wing and helicopter groups;

Group Task(s): "Pinpoint Strike", "Ground Attack", "Runway Attack".



In addition to selecting a specific group unit as a target, the Attack Unit Task can be used to designate a static object as a target using the STATIC drop-down menu or by left-clicking a static object on the map. This task also allows the mission designer to set the amount of munitions to expend and number of attacks runs.

Target Panel:

GROUP. Drop down menu to select the target group from all applicable enemy groups available in the mission.

UNIT. Drop down menu to select the specific unit of the target group to attack.

STATIC. Drop down menu to select a static object as the target (bunker, FARP, etc.)

WEAPON. Drop down menu to select the weapon types cleared to use for this action.

REL QTY. The amount of payload cleared to expend in executing this action.

MAX ATTACK QTY. The maximum number of attack runs cleared for this action.

GROUP ATTACK. When checked, the entire flight group will participate in the attack of the target. By default, each aircraft in a flight group is assigned one target.

DIRECTION FROM. Allows you to set the course (azimuth) of the aircraft group for reaching the target for an attack.

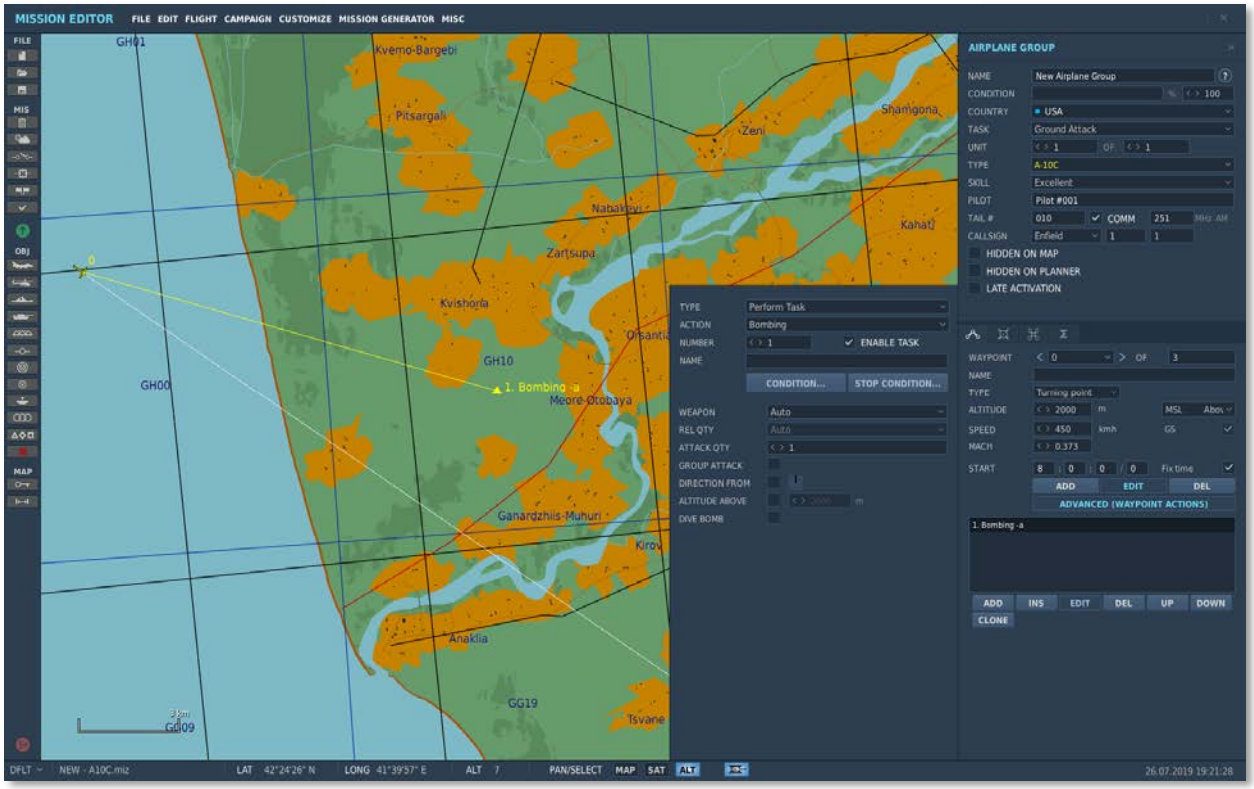
ALTITUDE ABOVE. Sets the altitude of the aircraft group on approach to target.

Bombing

Deliver air-to-ground ordnance onto the map coordinate as designated by the mission designer. Applicable to:

Group Type(s): fixed-wing and helicopter groups;

Group Task(s): "Pinpoint Strike", "Ground Attack", "Runway Attack".



Unlike the Attack Map Object Task, the Attack Coordinate Task will set the AI to attack the specific point on the map and not the closest map object in the vicinity. In other words, if you place the target point over an empty piece of dirt, the AI will attempt to deliver weapons onto this particular point.

As with other attack Tasks, you can configure the Task properties, including the weapons to be employed.

Target Panel:

WEAPON. Drop down menu to select the weapon types cleared to use for this action.

REL QTY. The amount of payload cleared to expend in executing this action.

MAX ATTACK QTY. The maximum number of attack runs cleared for this action.

GROUP ATTACK. When checked, the entire flight group will participate in the attack of the target. By default, each aircraft in a flight group is assigned one target.

DIRECTION FROM. Allows you to set the course (azimuth) of the aircraft group for reaching the target for an attack.

ALTITUDE ABOVE. Sets the altitude of the aircraft group on approach to target.

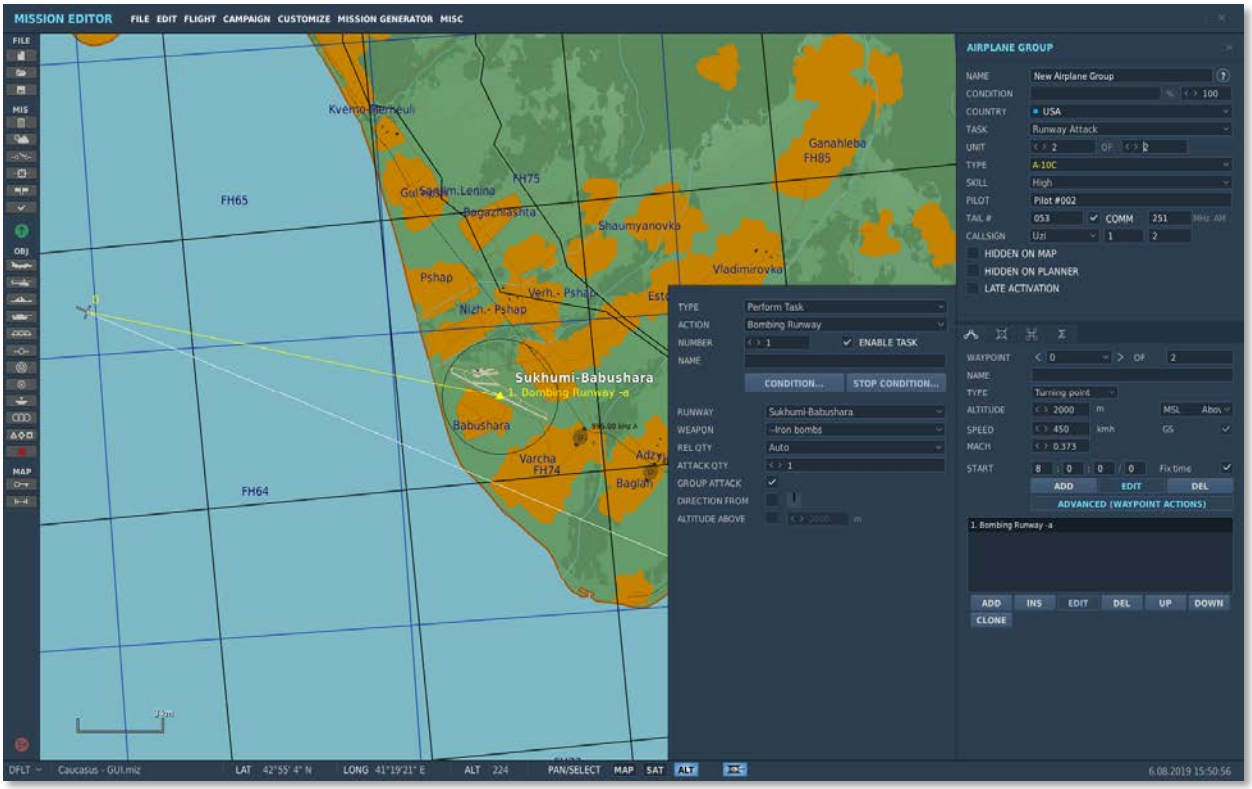
DIVE BOMB. May be used against targets against which it is impossible to perform low-altitude attacks due to the following factors: terrain, target type, lack of necessary ammunition.

Bombing Runway

Attack enemy airbase runways. Applicable to:

Group Type(s): fixed-wing groups;

Group Task(s): "Runway Attack".



The target airbase can be selected from the RUNWAY drop-down menu or by selecting the target airbase on the map using the mouse.

Target Panel:

RUNWAY. Drop down menu to select the target airbase.

WEAPON. Drop down menu to select the weapon types cleared to use for this action.

REL QTY. The amount of payload cleared to expend in executing this action.

MAX ATTACK QTY. The maximum number of attack runs cleared for this action.

GROUP ATTACK. When checked, the entire flight group will participate in the attack of the target. By default, each aircraft in a flight group is assigned one target.

DIRECTION FROM. Allows you to set the course (azimuth) of the aircraft group for reaching the target for an attack.

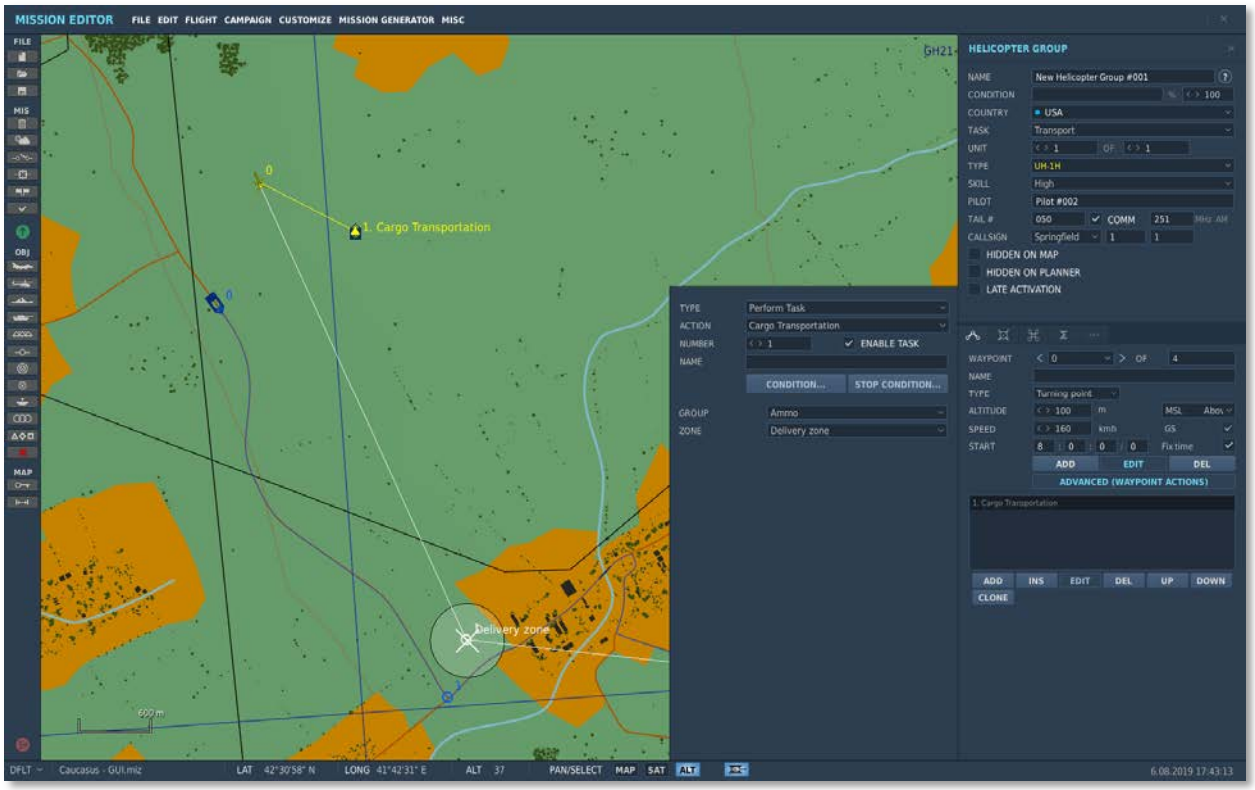
ALTITUDE ABOVE. Sets the altitude of the aircraft group on approach to target.

Cargo Transportation

The action is used to organize the transportation of cargo by a helicopter group to the specified zone. Applicable to:

Group Type(s): helicopter groups;

Group Task(s): "Anti-ship Strike", "CAS", "Transport", "Escort".



GROUP. Selection of cargo to be transported by the helicopter group.

ZONE. Delivery trigger zone.

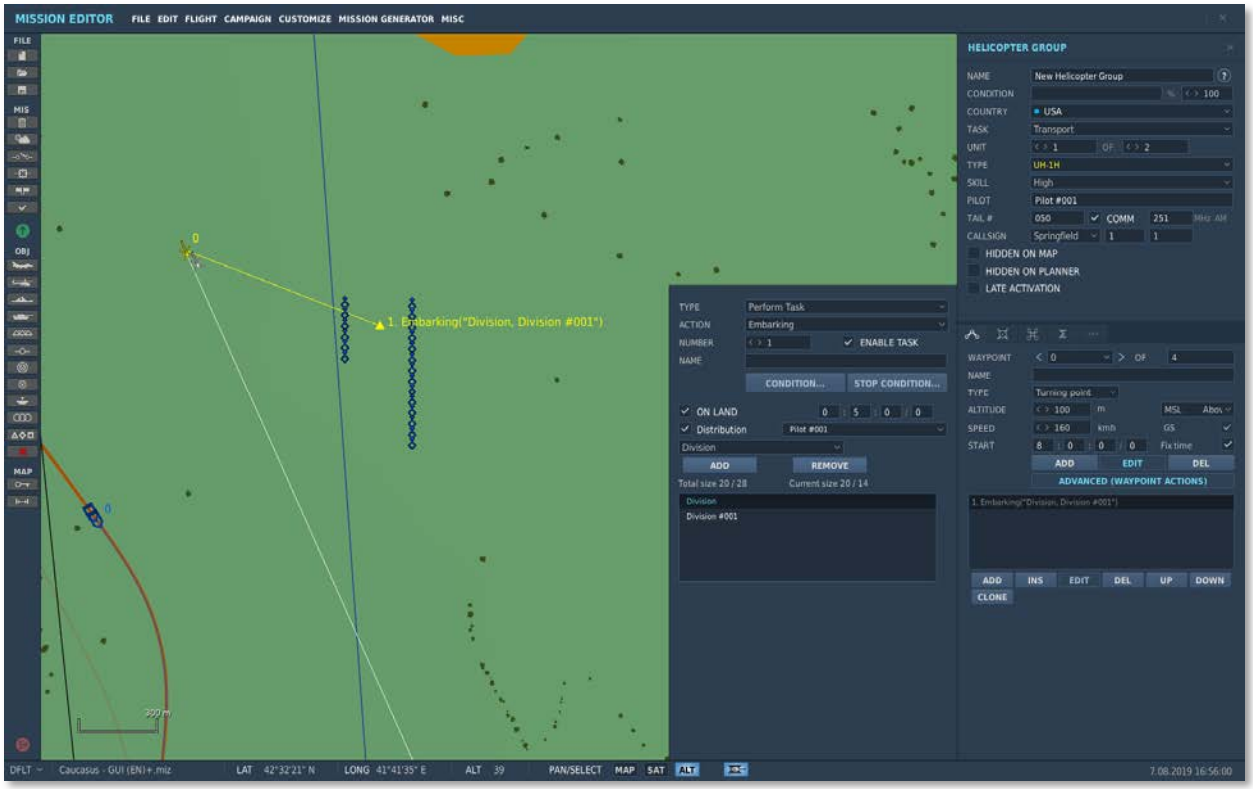
When this action is selected, the triangular marker will be automatically set to the selected cargo group. The helicopter will then fly up to the cargo, carry out its pick-up and then deliver it to the specified delivery area.

Embarking

The action is for organizing the embarkation of assault personnel. Applicable to:

Group Type(s): helicopter groups;

Group Task(s): "Ground Attack", "CAS", "Transport".



ON LAND. This flag activates the helicopter group's set waiting time on the ground for personnel embarkation, in the format (hours: minutes: seconds / day). The helicopter group will remain grounded for the entirety of the specified duration even when personnel embarkation has already been completed successfully. If the flag is unchecked, the helicopters will take off immediately after finishing embarkation.

Distribution. Selects the helicopter pilot for the embarkation task.

List of divisions. List of infantry squads for embarkation. In order to be able to add a ground group to the list of squads, each infantry squad must be set to "Perform Task – Embark to transport", and must have the flag – "TRANSPORTABLE" activated in the vehicle group panel.

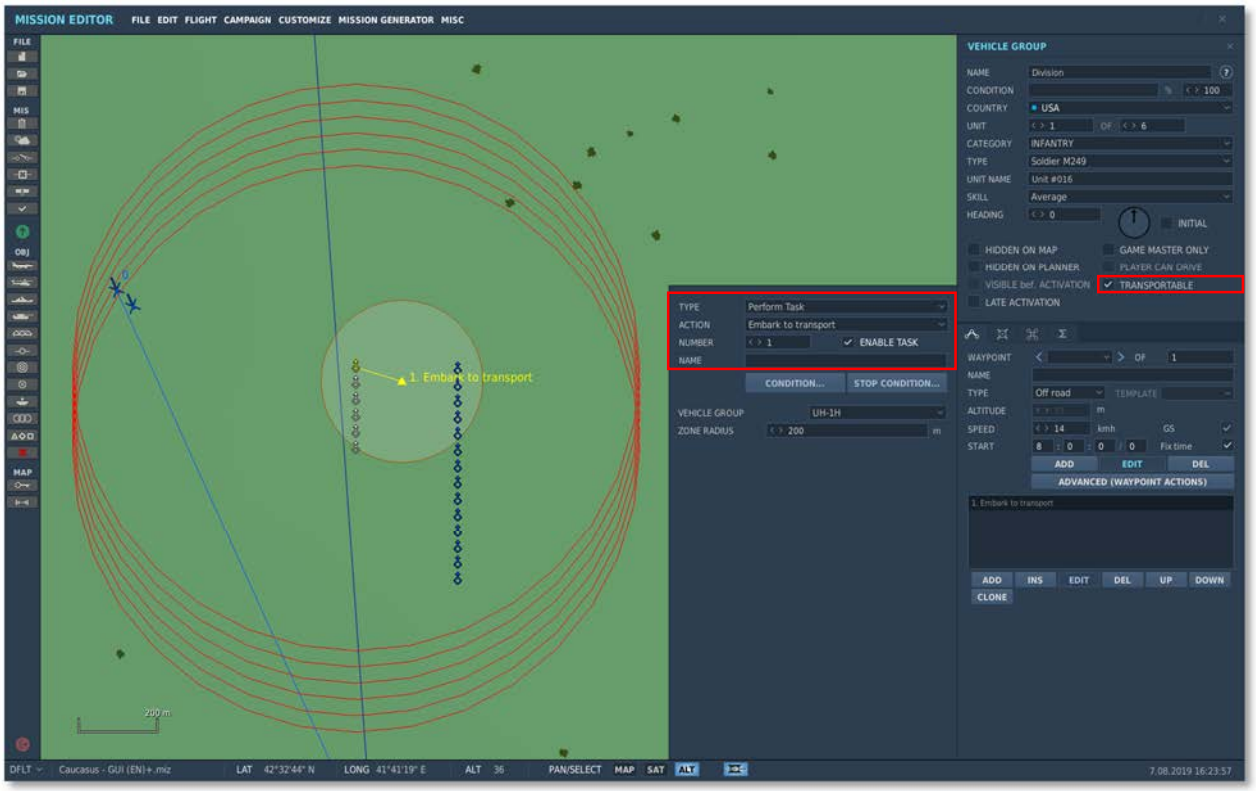
ADD. The button adds the infantry unit (selected from the list above) into the embarkation list below.

REMOVE. Removes the selected infantry unit from the embarkation list.

Total size .../... . The total number of infantrymen in the squads and the total possible number of loading places in the helicopter group.

Current size .../... . The number of infantrymen in the added (selected) infantry unit, and the possible number of loading places in one helicopter from the group.

When you select this action, a triangular marker will automatically appear near the waypoint, connected to it by a line. This marker must be dragged with the mouse to the map point chosen for the personnel embarkation.



VEHICLE GROUP. Selects the vehicle group for the embarkation task.

ZONE RADIUS. The radius of the embarkation area of the infantry squads, in meters.

After the helicopter group lands at the indicated point (in the radius of the zone), the landing will begin its loading:





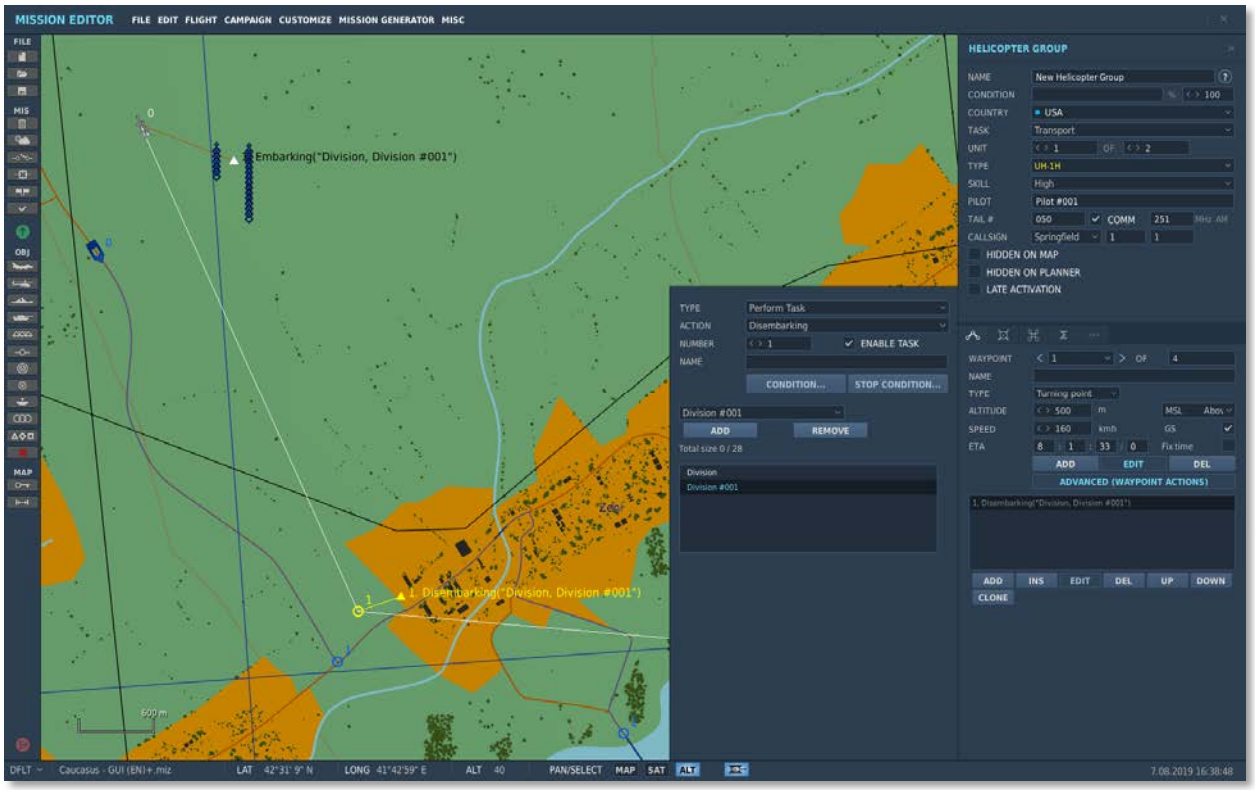
There are currently no in-game models for infantrymen in-transit (located inside the helicopter cabin).

Disembarking

This action is for organizing the disembarkation of assault personnel. Applicable to:

Group Type(s): helicopter groups;

Group Task(s): "Ground Attack", "CAS", "Transport".



List of divisions. Shows a list of infantry squads available for unloading. In order to be able to add an infantry group this list, the infantry group must have the main task "Embarking".

ADD. The button adds the infantry unit (selected from the list above) into the disembarkation list below.

REMOVE. Removes the selected infantry unit from the disembarkation list.

When you select this action, a triangular marker will automatically appear near the waypoint, connected to it by a line. This marker must be dragged with the mouse to the map point chosen for the personnel disembarkation.

The helicopter group will drop off the infantry groups in the specified zone, after which it will take off and continue to follow its route and assigned tasks.

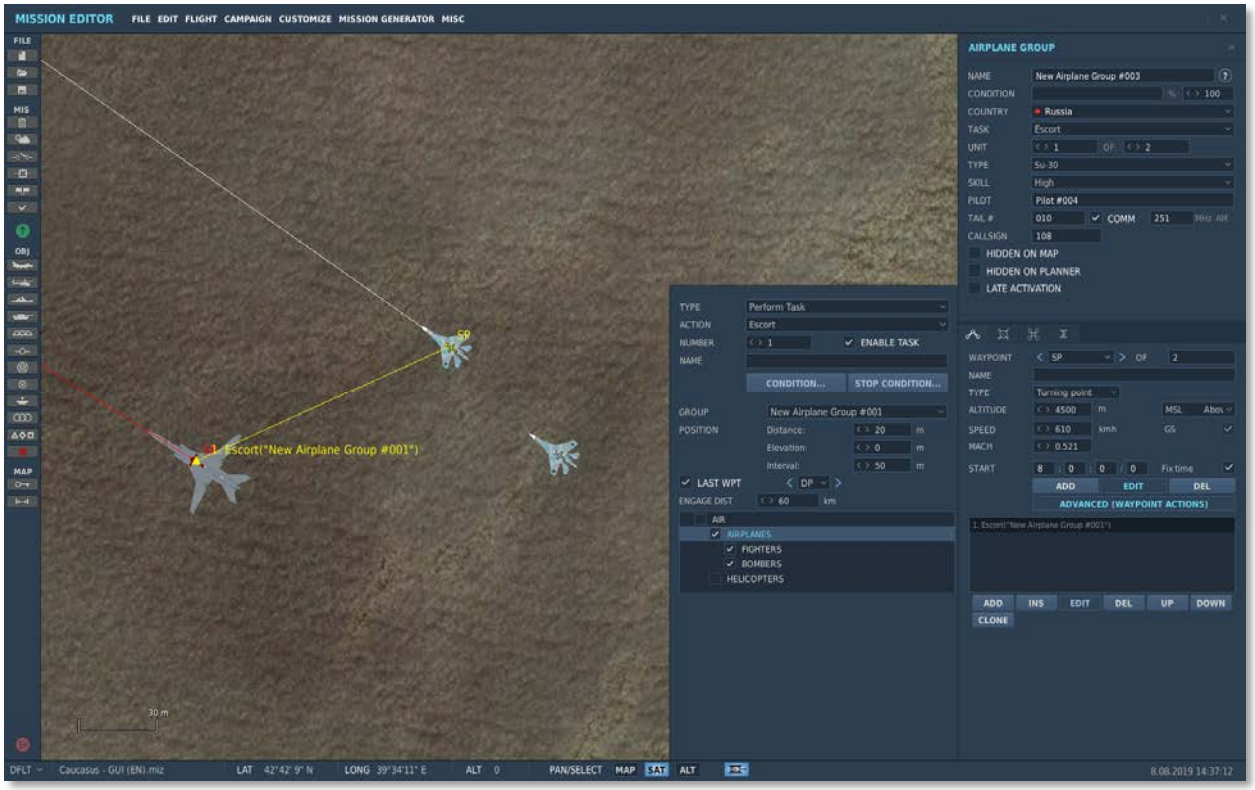


Escort

Air group to escort other air group. Escort group will engage threats to escorted air group. Applicable to:

Group Type(s): fixed-wing groups.

Group Task(s): "Escort", "SEAD",



The target air group can be selected from the GROUP drop-down menu or by selecting the air group to escort on the map using the mouse.

Target Panel:

GROUP. Drop down menu to select the air group.

- **Distance** - Formation forward and back location of group to group being followed.
- **Elevation** - Formation altitude difference of group to group being followed.
- **Interval** - Formation side-to-side location of group to group being followed.

LAST WPT. Waypoint at which Follow action ends.

ENGAGE DIST. Distance between threat and escorted flight that will cause the escort flight to engage the threat.

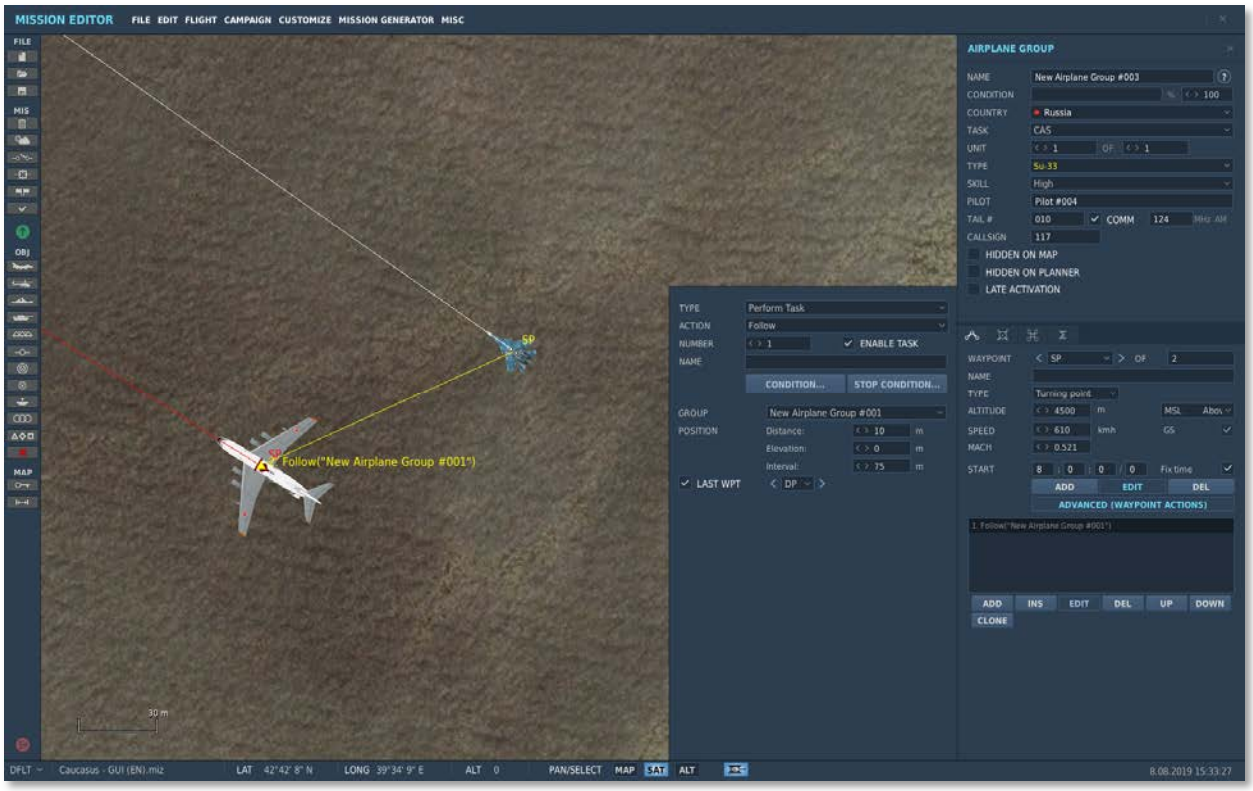
Threat Type List. Check types of threats that escort flight is cleared to engage as part of escort task.

Follow

Air group to follow other air group. Applicable to:

Group Type(s): fixed-wing and helicopter groups;

Group Task(s): All.



The target air group can be selected from the GROUP drop-down menu or by selecting the air group to follow on the map using the mouse.

Target Panel:

GROUP. Drop down menu to select the air group.

- **Distance** - Formation forward and back location of group to group being followed.
- **Elevation** - Formation altitude difference of group to group being followed.
- **Interval** - Formation side-to-side location of group to group being followed.

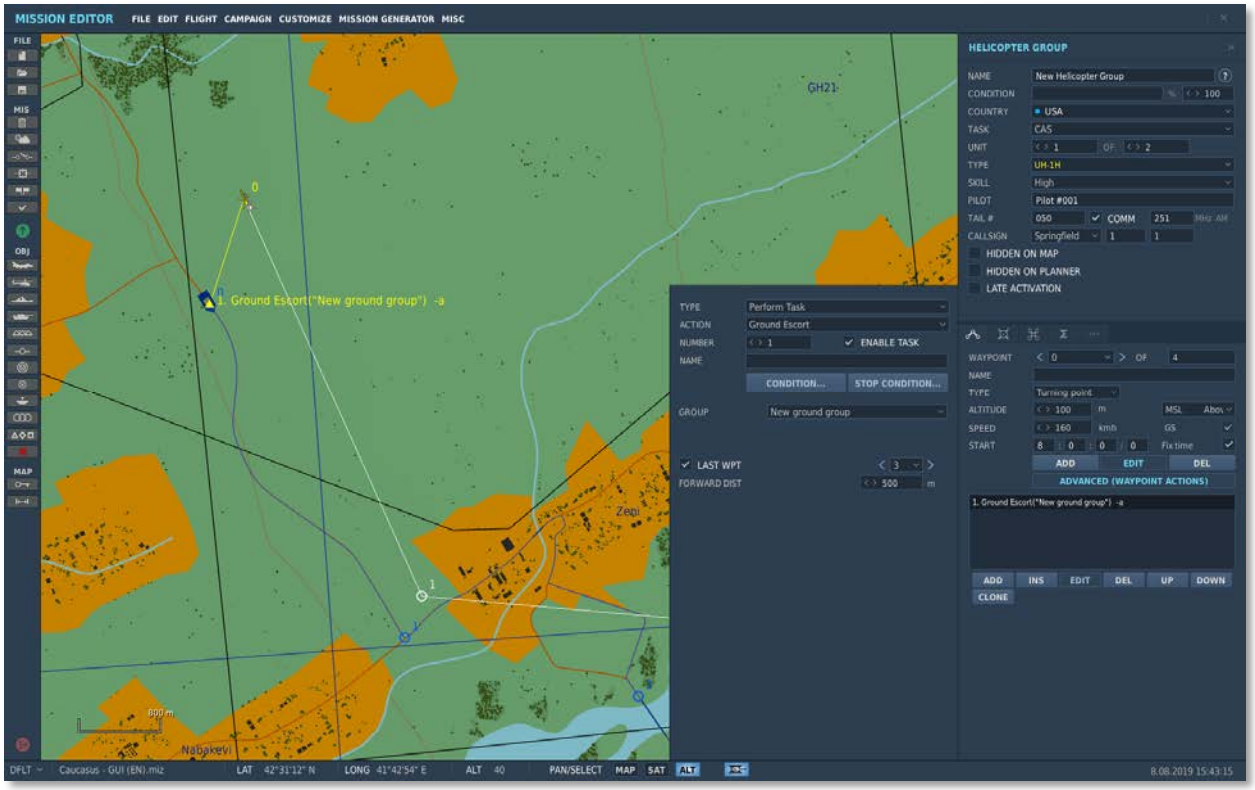
LAST WPT. Waypoint at which Follow action ends.

Ground Escort

This action is for organizing the aerial escort (performed by a helicopter group) of a ground unit group along the ground unit group's travel route. Applicable to:

Group Type(s): helicopter groups;

Group Task(s): "CAS", "Ground Attack", "Escort".



Target Panel:

GROUP. Selects the ground unit group to be escorted by the helicopter group.

LAST WPT. Waypoint at which Follow action ends.

FORWARD DISTANCE. Forward distance to the ground unit group's target area, in meters.

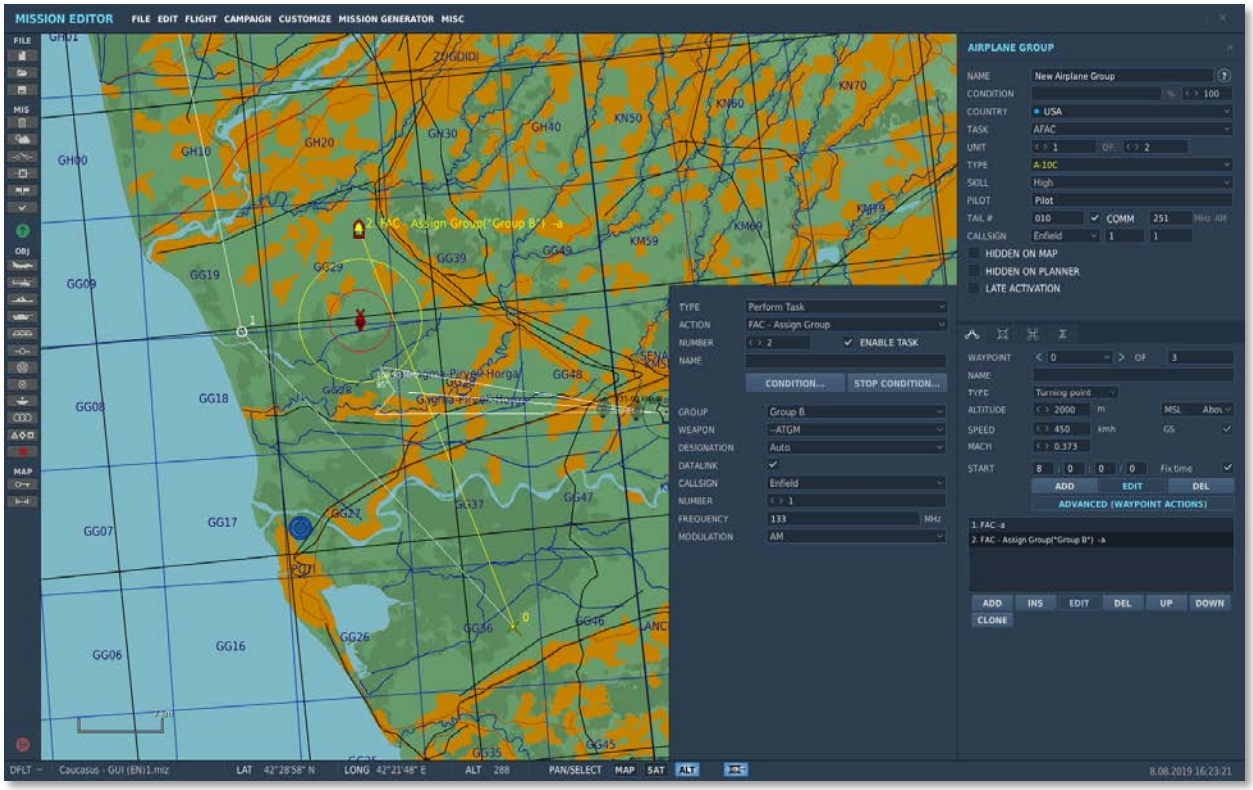
The target ground group can be selected from the GROUP drop-down menu or by selecting the ground group to follow on the map using the mouse.

FAC – Assign Group

This task creates an Airborne Forward Air Controller (AFAC) to direct friendly aircraft onto a target group designated by the mission designer. Applicable to:

Group Type(s): fixed wing and helicopter groups in BLUE coalition only;

Group Task(s): "AFAC".



Targeting actions will begin when friendly units arrive in the area and establish communication with the AFAC.

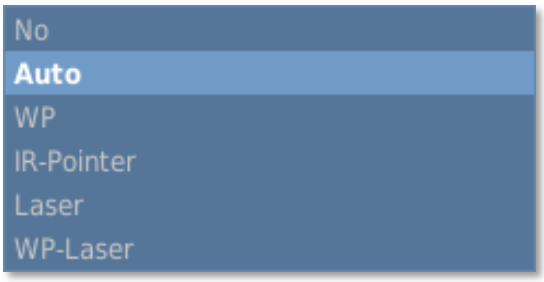
***Note**: the FAC: Assign Group Task assumes full situational awareness for the AFAC. The target's location will always be known and the AFAC will always attempt prosecute the attack if conditions permit, even when the target has not been independently detected by the AFAC.*

Target Panel:

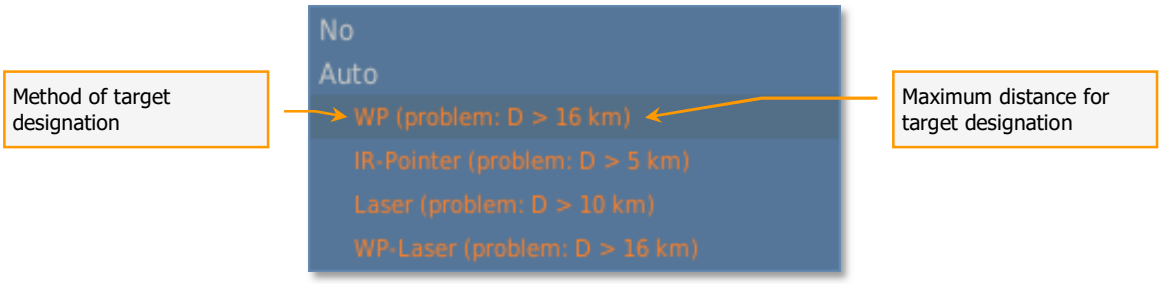
GROUP. Drop down menu to select the target group from all applicable enemy groups available in the mission. The FAC will only provide targeting against the group designated as the target in the mission editor and will not target other groups.

WEAPON. Drop down menu to select the weapon types cleared to use for this action. The FAC-Assign Group Task allows you to set the weapon type(s) for the FAC to request from the attacking groups.

DESIGNATION. Selects the method of target designation (Auto, smoke marker ("Willy Pete", WP), IR-Pointer, Laser, WP-Laser. Target designation is carried out on the player's radio request at a fixed radio range.



If the FAC unit is located from the enemy at a distance greater than the operational range of the target designator equipment, then the maximum ranges for each of the available equipment will be displayed in the menu.



In this case, it is necessary to adjust the distance of the Forward Air Controller to the enemy such that it is within the specified maximum range.

DATALINK. Automatic data transmission system for the exchange of target information between ground units and aircraft.

CALLSIGN. Callsign of the Forward Air Controller.

NUMBER. Selection field for the FAC's number.

FREQUENCY. Set the FAC group's radio communications frequency in MHz.

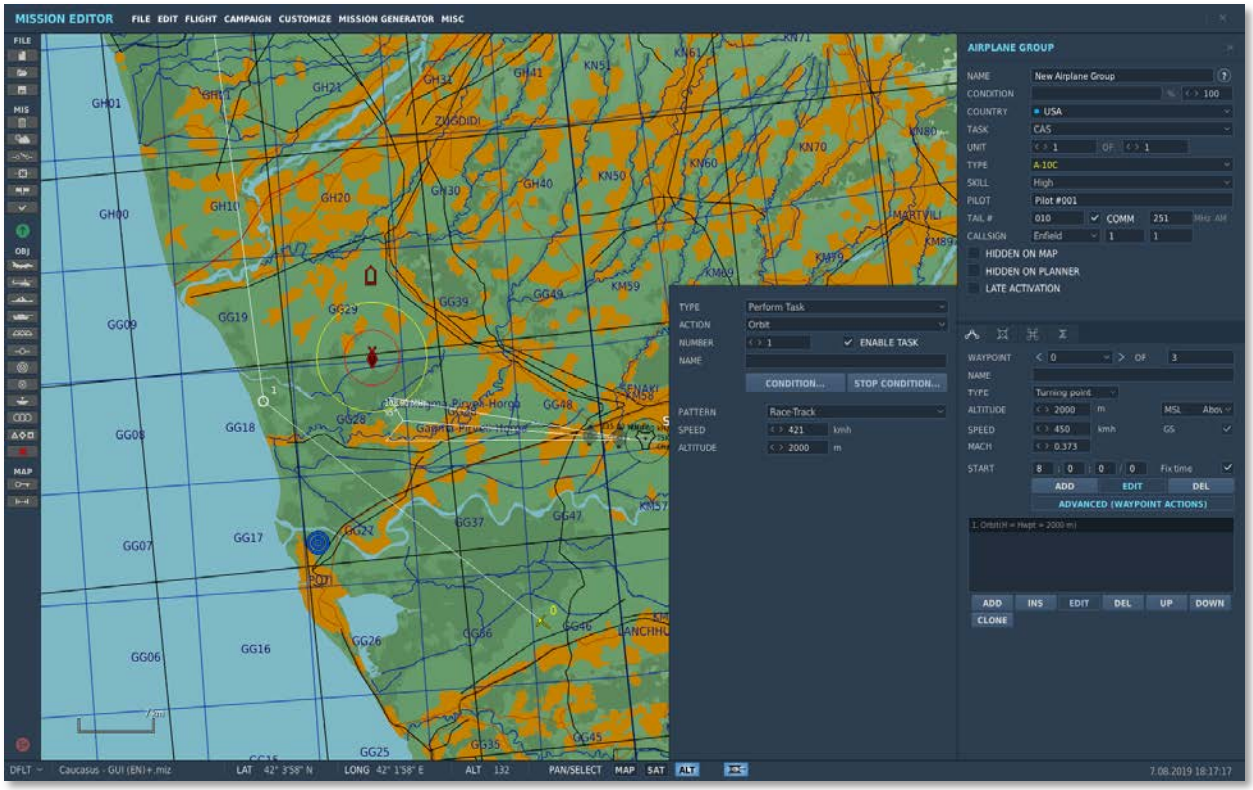
MODULATION. Set the FAC group's radio communications AM/FM modulation.

Orbit

Set the air group to orbit "on station" while awaiting for further conditions to proceed with the mission. Applicable to:

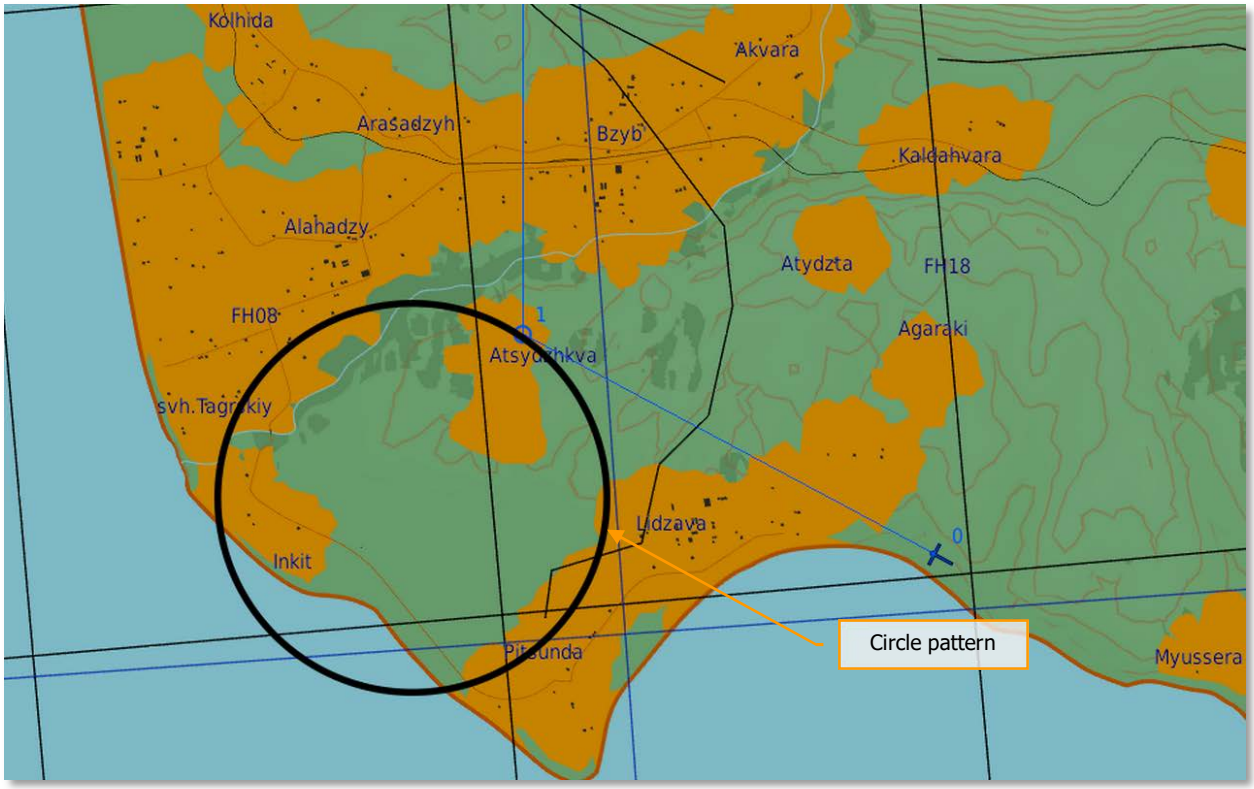
Group Type(s): fixed-wing and helicopter groups;

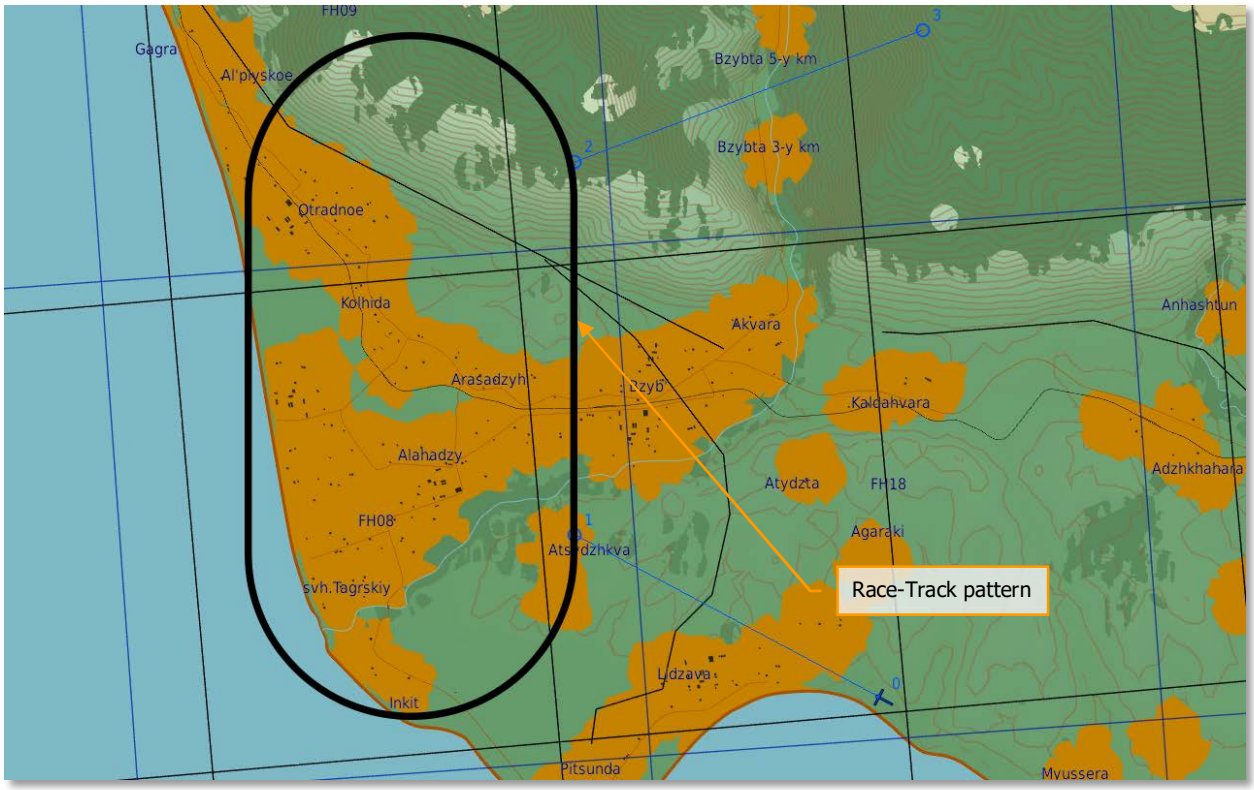
Group Task(s): all mission tasks.



The PATTERN drop-down menu allows the mission designer to select between a race-track pattern (loop between the current and subsequent waypoints) or a circle pattern. When circle pattern is selected, the orbit will be centered on the orbit waypoint.

The SPEED and ALTITUDE settings set the airspeed in kilometers per hour and altitude in meters for the pattern.


Kolhida
Akvara
Arasadzyh
Bzyb
Kaldahvara
Alahadzy
Atydzta
FH18
FH08
Agaraki
Atsydzhkva
svh.Tagrskiy
Ldzava
Inkit
Pitsunda
Myussera
Circle pattern


FH09
Gagra
Al'pyskoe
Bzybta 5-y km
Bzybta 3-y km
Otradnoe
Kolhida
Akvara
Anhashtun
Arasadzyh
Bzyb
Kaldahvara
Alahadzy
Atydzta
FH18
Adzhkhahara
FH08
Agaraki
Atsydzhkva
svh.Tagrskiy
Race-Track pattern
Ldzava
Inkit
Pitsunda
Myussera

An Orbit Task will be stopped by the AI automatically upon reaching Bingo fuel state, when the AI only has enough fuel to return to base (RTB).

The Task can also be stopped using the Stop Conditions Panel by setting one or more stop conditions for the task.

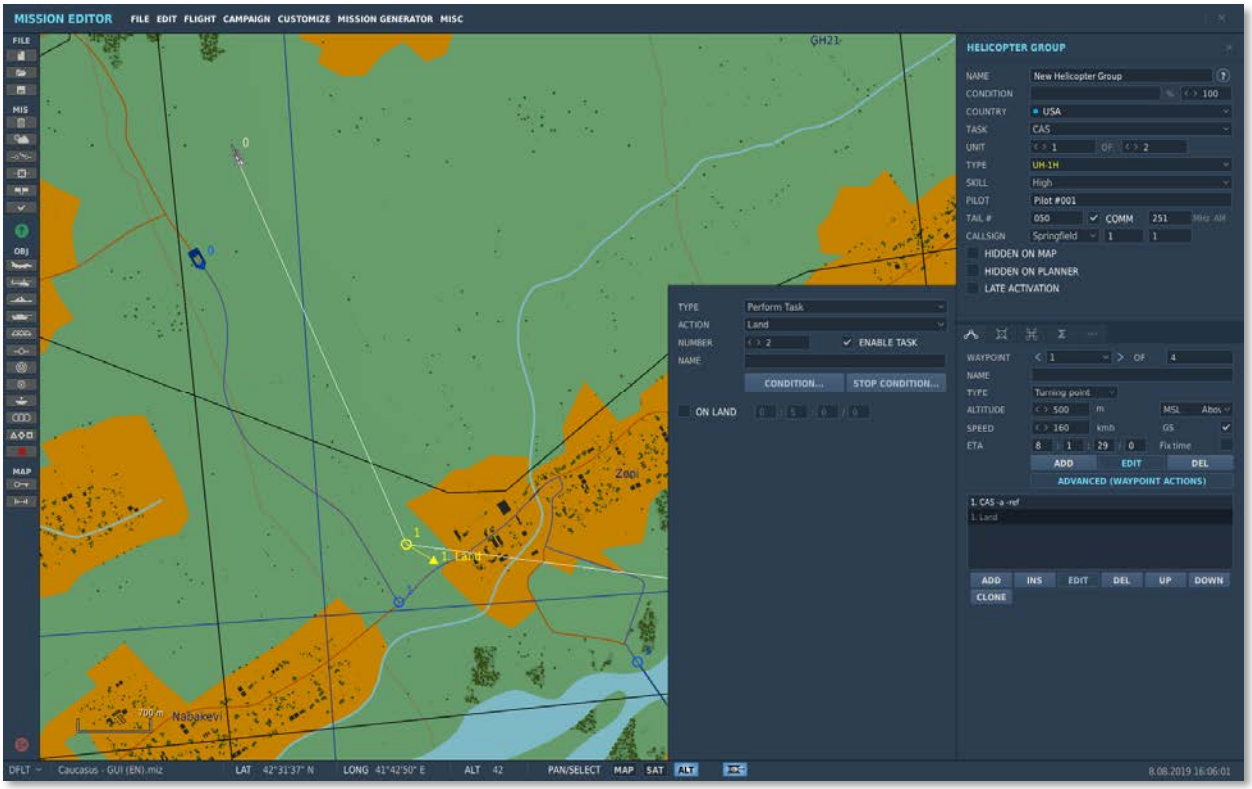
The Orbit Task allows the mission designer to create scenarios for the AI group to hold off on proceeding with the route until certain conditions are met. These can include a radio command from the player scripted through the Triggers Menu, the destruction of an enemy unit, such as a SAM threat, or the presence of particular units inside a designated zone. Any actions anticipated for the group after entering an orbit station should be added in the actions list below the Orbit Task.

Land

The action is for organizing of the en-route stopovers of the helicopter group between the waypoints of its travel route. Applicable to:

Group Type(s): helicopter groups;

Group Task(s): all mission tasks.



ON LAND. This flag activates the helicopter group's set waiting time on the ground after landing, in the format (hours: minutes: seconds / day).

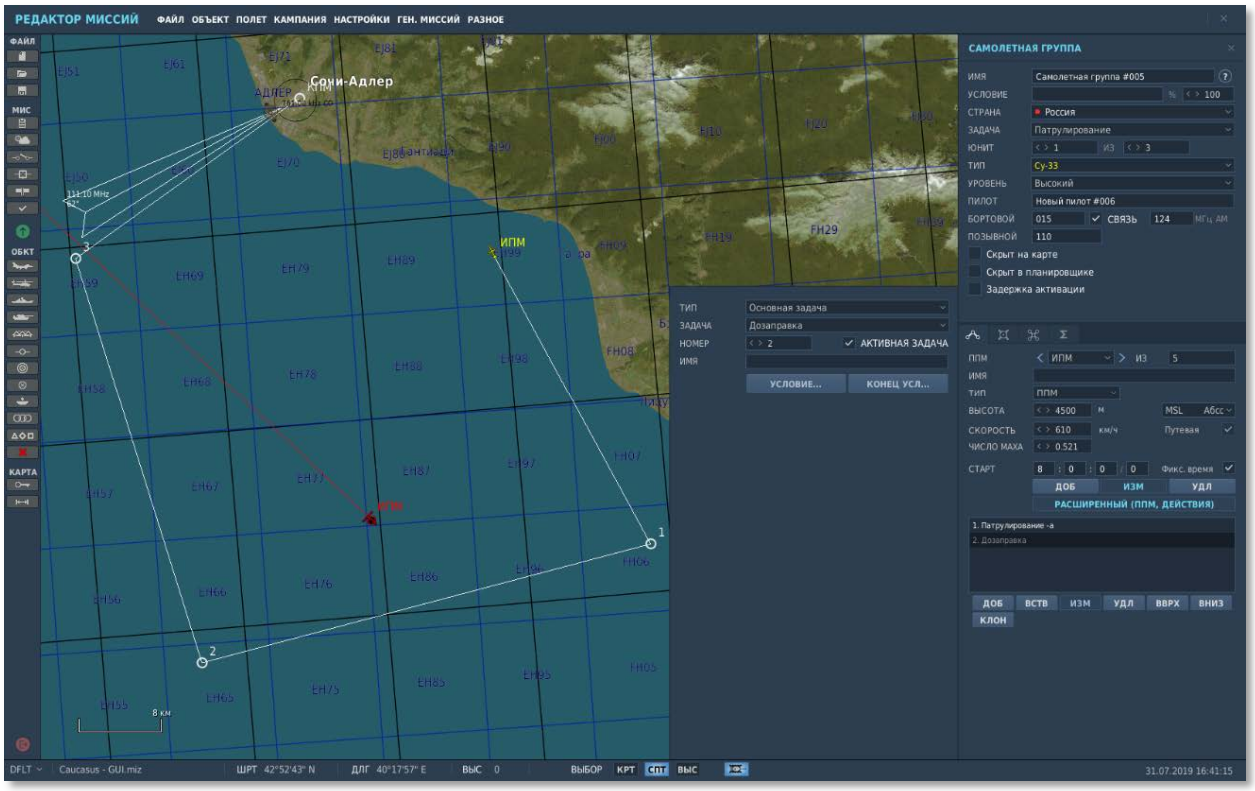
The target area can be placed on the map using the left mouse button. A line connects the center of the target area (triangle marker) with the action waypoint.

Refueling

Set the air group to refueling for forced following to nearest tanker and refueling. Applicable to:

Group Type(s): fixed-wing groups;

Group Task(s): all mission tasks.



During the course of an aircraft group's TASKS (roles), for example, escort, patrol, attacking air / ground targets, etc., you can add refueling as one of the main tasks. When the aircraft group reaches a critical fuel level, the group will set course for the nearest friendly tanker, perform its approach, and refuel from the UPAZ suspended refueling units.

Standard options for airborne fuel transfer for the Il-78M tanker aircraft are: fuselage-mounted refueling apparatus for refueling heavy bomber class aircraft, and the paired underwing refueling units for refueling fighter-class aircraft. Options may vary for tankers of NATO countries.





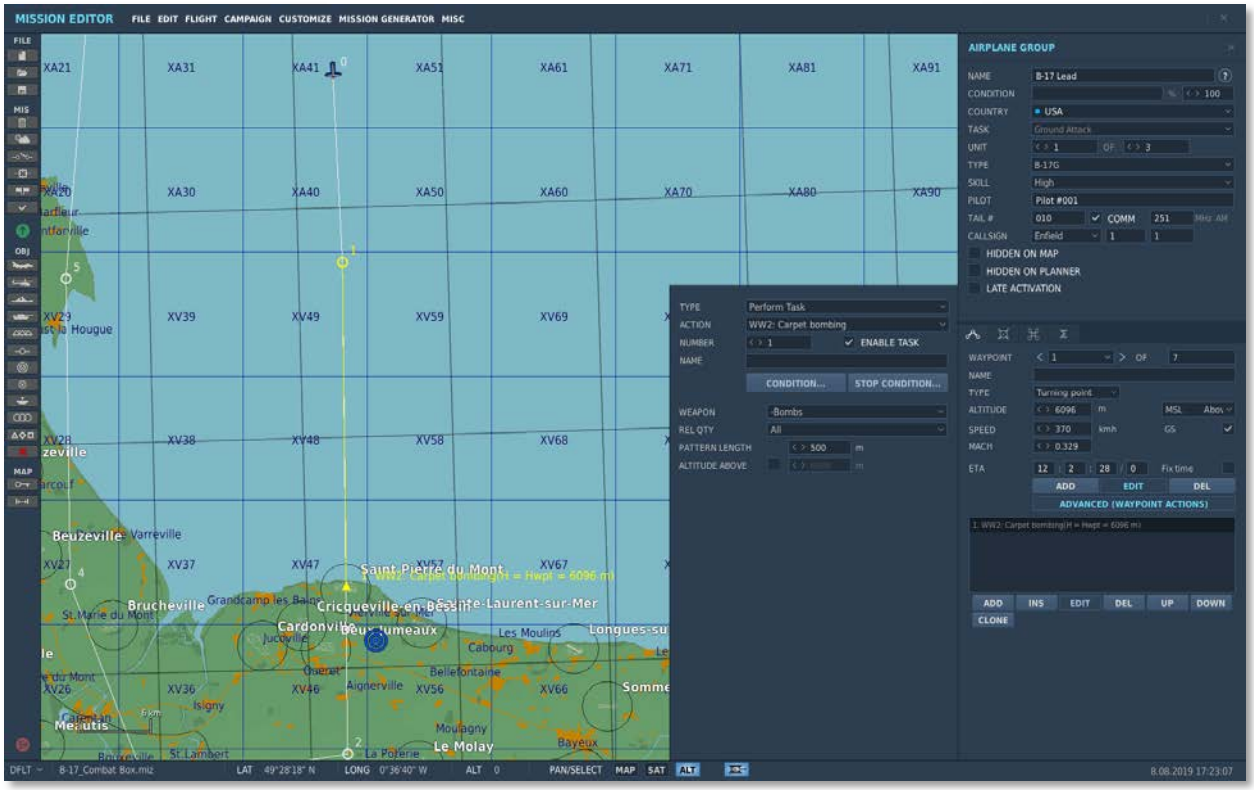
After refueling the group will resume assigned route.

WW2: Carpet Bombing

The action is for the organization of carpet bombardment of area-type targets, for missions during the WW2 period. Applicable to:

Group Type(s): fixed-wing groups, primarily for WW2 module bomber aircraft;

Group Task(s): "Ground Attack".



WEAPON. Drop down menu to select the weapon types cleared to use for this action.

REL QTY. The amount of payload cleared to expend in executing this action.

PATTERN LENGTH. Length of the aircraft group's formation.

ALTITUDE ABOVE. Altitude of the aircraft group on approach to target.

The target area can be placed on the map using the left mouse button. A target line connects the center of the target area (triangle marker) with the action waypoint (industrial areas, cities, or enemy ground unit groups). Keep in mind that this action for bombers should be set from the waypoint that comes after the starting point of the travel route. In this example, these waypoints are Waypoint 1 and Waypoint 0 respectively. For the starting point of the route, Waypoint 0, the main tasks to be set are for the construction and type of the battle formation.

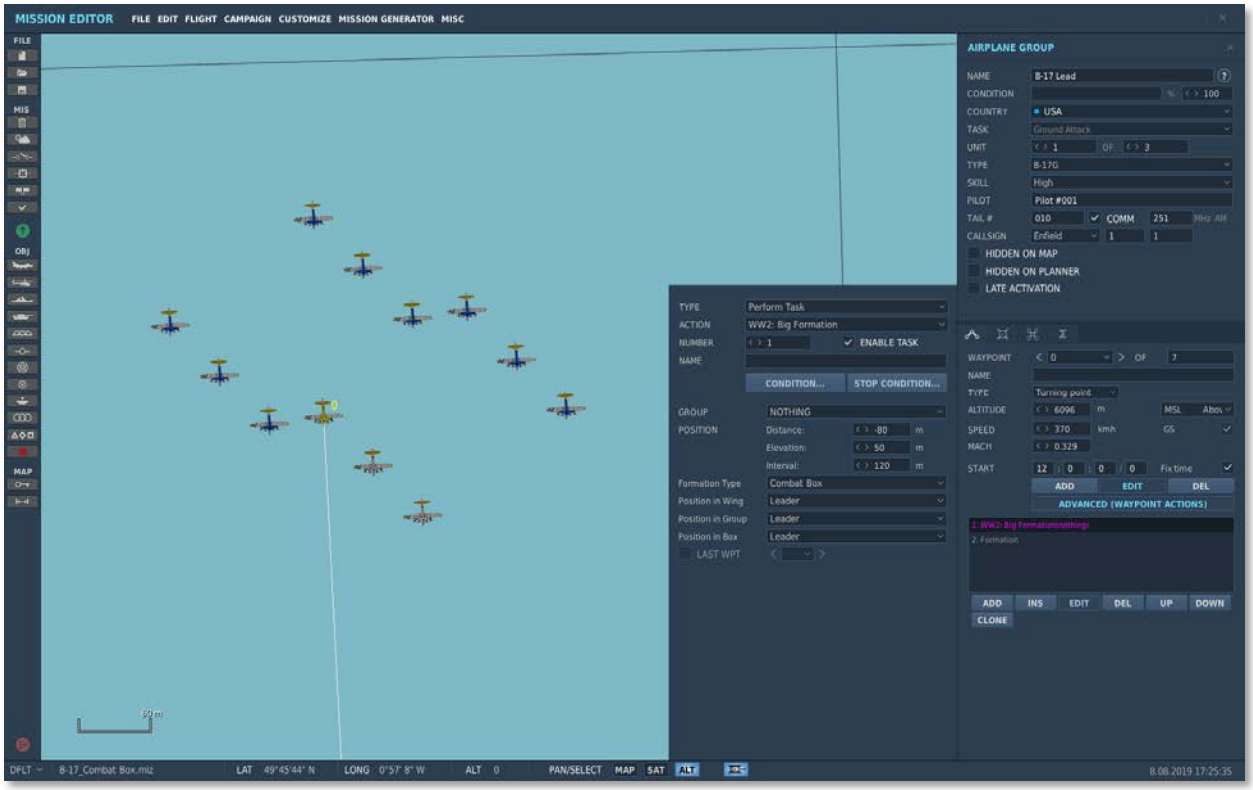
As with other attack Tasks, you can configure the Task properties, including the weapons to be employed. If the specified weapons are unavailable at the time of the planned attack, the aircraft will not execute any attack.

WW2: Big Formation

The action is for the organization of carpet bombardment of area-type targets, for missions during the WW2 period. Applicable to:

Group Type(s): fixed-wing groups, primarily for WW2 module bomber aircraft;

Group Task(s): "Ground Attack".



GROUP. Selects the lead aircraft from the list. The group's flight formation will be constructed with respect to this lead aircraft.

POSITION. Selects the position of the aircraft group highlighted on the map with respect to the lead aircraft group.

- **Distance** – Formation forward and back location of group to group being followed.
- **Elevation** – Formation altitude difference of group to group being followed.
- **Interval** – Formation side-to-side location of group to group being followed.

Below are the options for the battle formation and the location of the group relative to the leader in the formation.

- **Formation Type** – battle formation used by USAF bombers throughout the WWII ("Combat Box" and "Javelin Down" wedge).
- **Position in Wing** – position of the aircraft group within the wing (a relatively large formation of aircraft) with respect to the lead group, to a max of 500 meters.

- **Position in Group** – position of the group in the middle formation, to a max of 250 meters.
- **Position in Box** – position of the group in its battle formation, to a max of 80 meters.

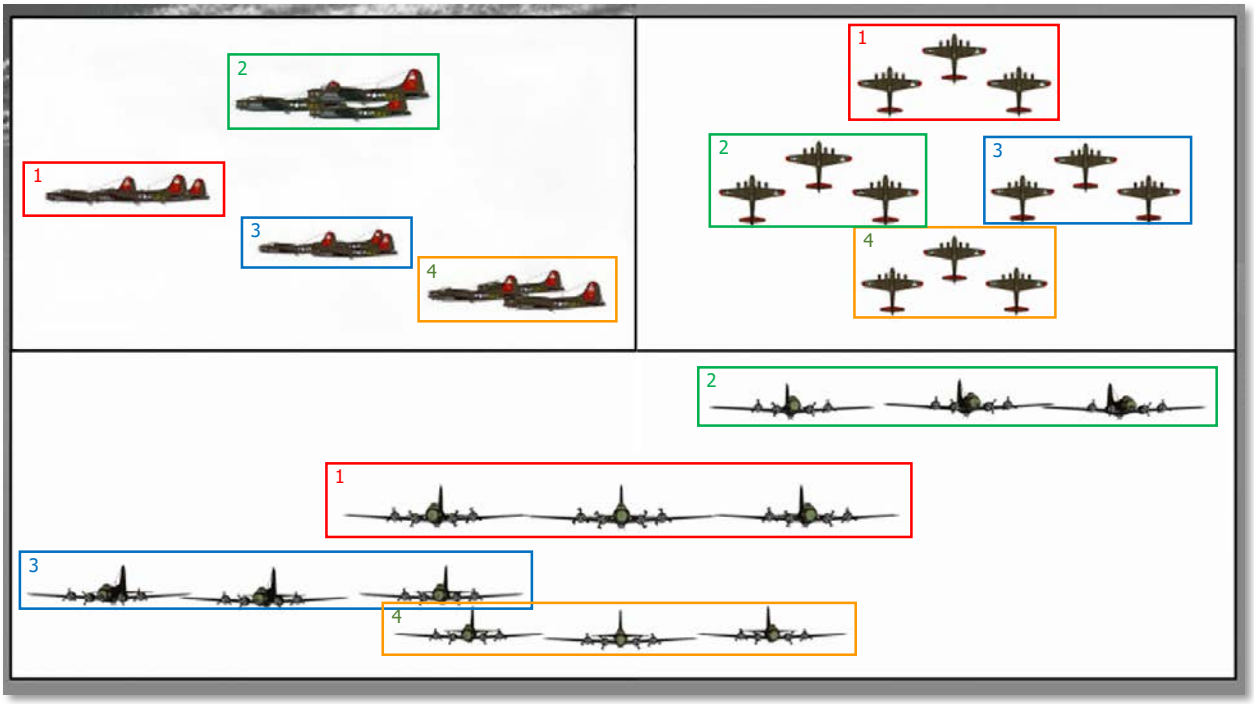
When you select a position in the formation, the distance/excess/interval are all automatically assigned. They can also be adjusted manually only after selecting all items. Keep in mind that the AI bomber groups will perceive short distances as critical and will maneuver to avoid a collision.

LAST WPT. Saving (execution) of this type of formation to the specified route point. Used for trailing groups following the route of the lead group.

Below are some examples of historically-accurate formations:

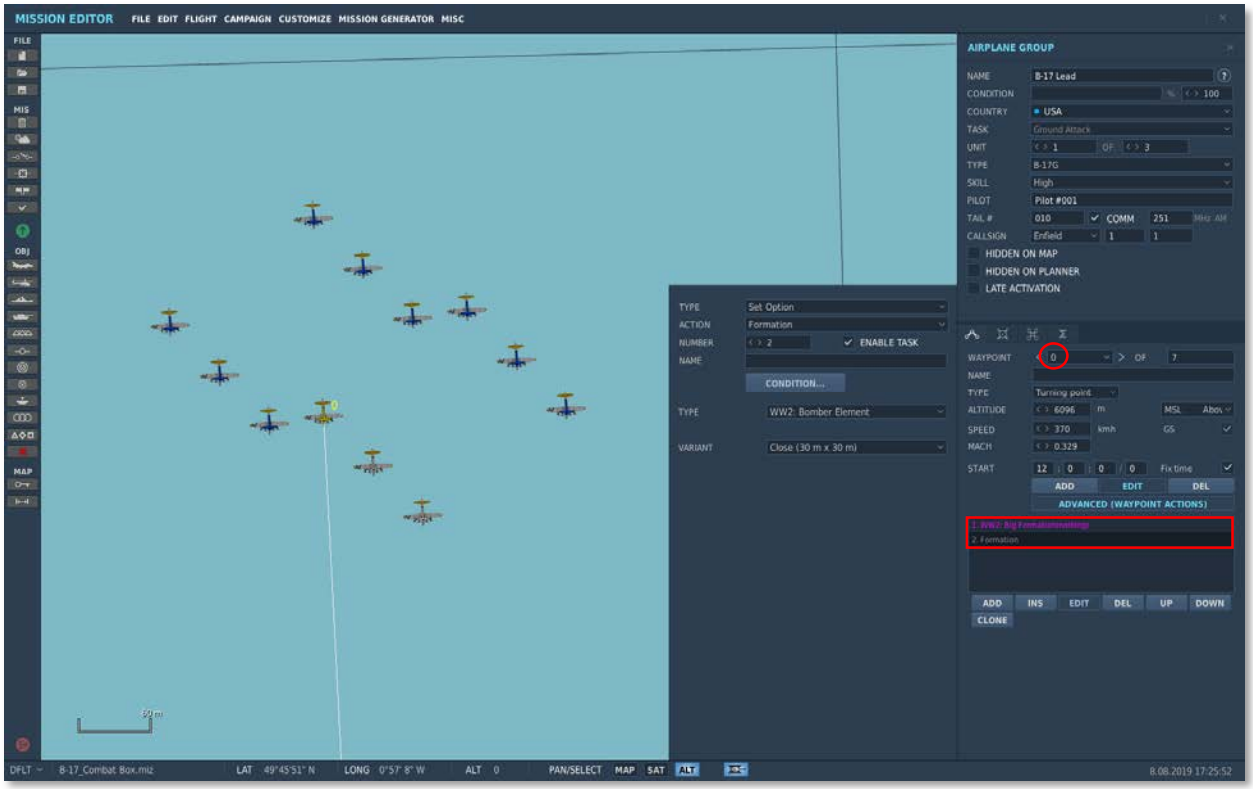
Example 1: **"Combat Box"**.

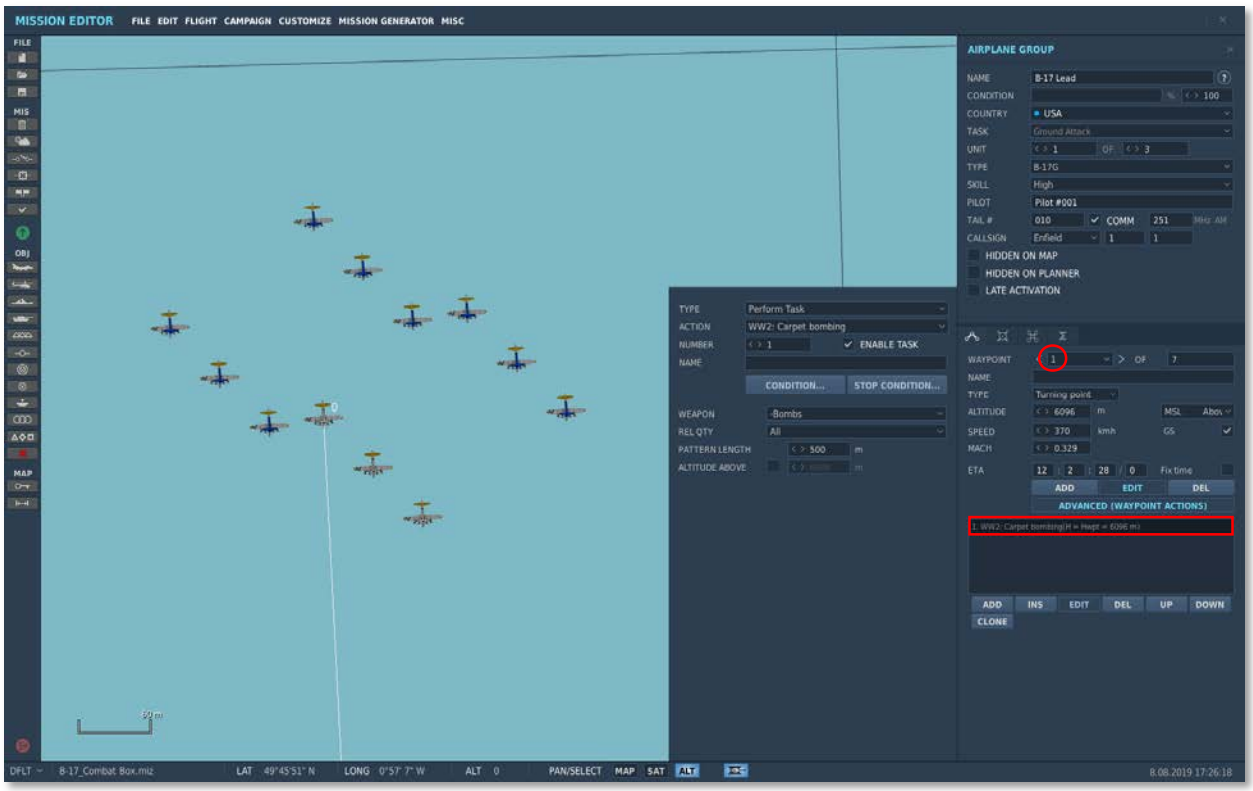
Below is a diagram of the "Combat Box" formation employed by USAF bombers during WW2.



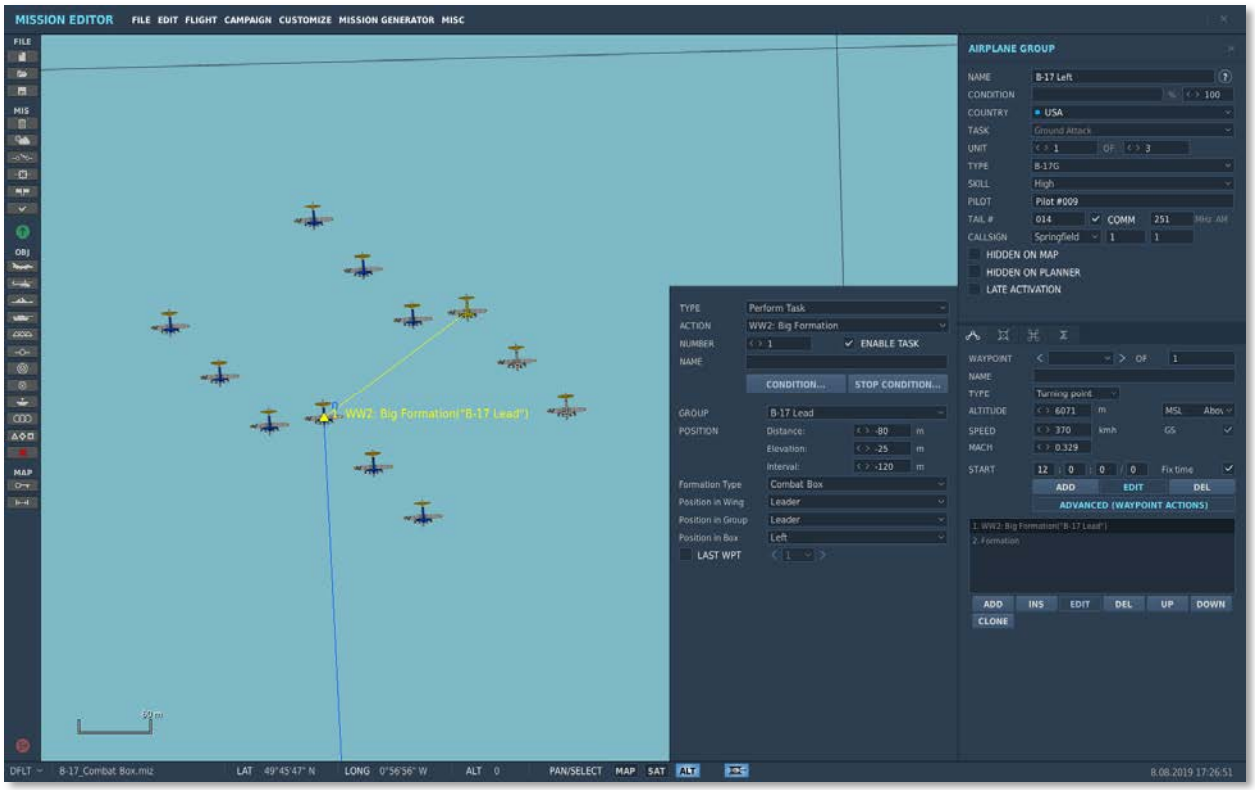
The diagram shows four wings (three aircraft in each) located at different heights relative to the lead wing. These wings together close the formation, in our case, a combat box. This formation made it possible for the wings to maintain a tight structure and conduct a circular fire defense covering the entire aircraft group.

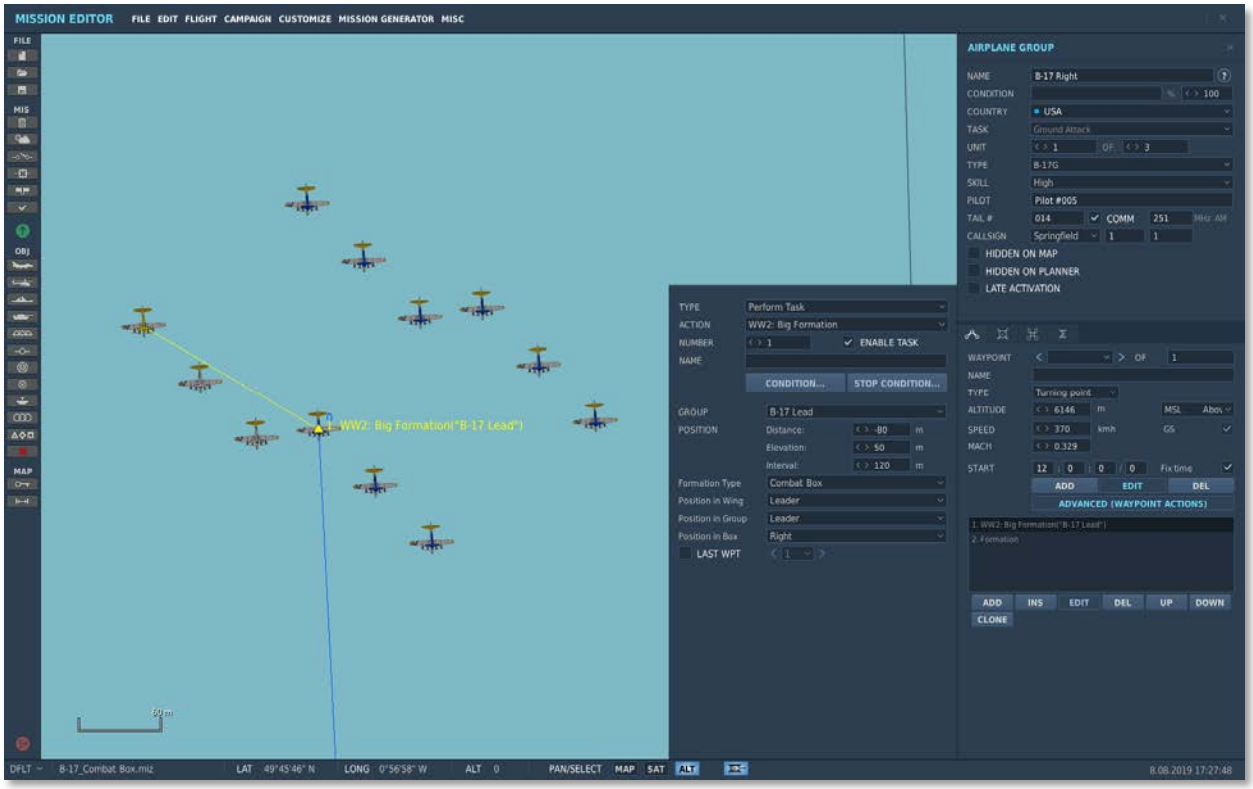
The first lead wing in the formation is assigned the WPT route from 0 to 7. Under the actions for WPT 0 the main task is then set to "WW2: Big Formation" (formation type "Combat Box", with Leader in all positions). The second action for WPT 0 is set to the behavior option "VARIANT" (type "WW2: Bomber Element", any formation). At WPT 1, we then set the main task to "WW2: Carpet Bombing". The remaining wings will then orient themselves according to the tasks assigned to the lead wing for WPTs 0 and 1.

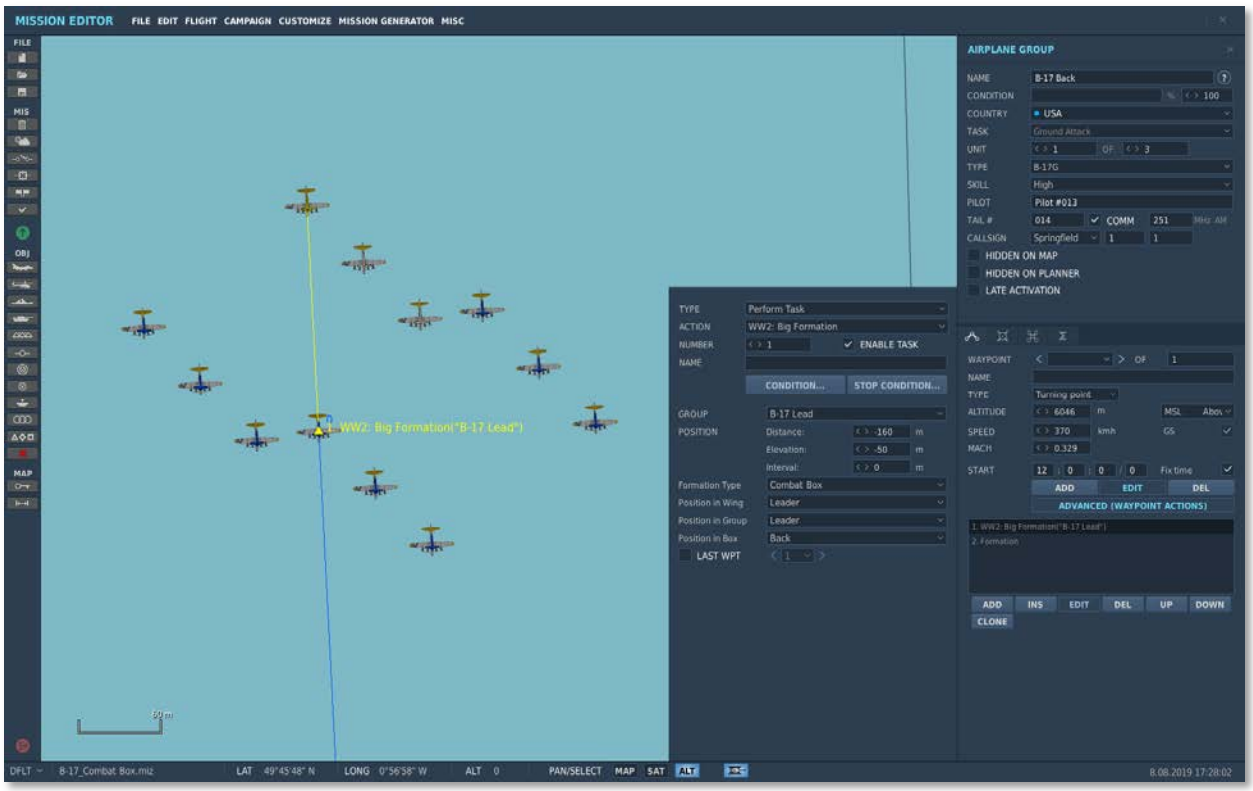

MISSION EDITOR
FILE EDIT FLIGHT CAMPAIGN CUSTOMIZE MISSION GENERATOR MISC
AIRPLANE GROUP
NAME B-17 Lead
CONDITION
COUNTRY USA
TASK Ground Attack
UNIT
TYPE B-17G
SKILL High
PILOT Pilot #001
TAIL # 010
COMM 251
CALLSIGN Enfield
HIDDEN ON MAP
HIDDEN ON PLANNER
LATE ACTIVATION
TYPE Set Option
ACTION Formation
NUMBER 2
ENABLE TASK
NAME
CONDITION...
TYPE WW2: Bomber Element
VARIANT Close (30 m x 30 m)
WAYPOINT 0 OF 7
NAME
TYPE Turning point
ALTITUDE 6096 m MSL
SPEED 370 kmh GS
MACH 0.329
START 12 0 0 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)
1 WW2: Big Formation
2 Formation
ADD INS EDIT DEL UP DOWN
CLONE
DFLT 8-17_Combat Box.miz
LAT 49°45'51" N
LONG 0°57' 8" W
ALT 0
PAN/SELECT MAP SAT ALT
8.08.2019 17:25:52


MISSION EDITOR
FILE EDIT FLIGHT CAMPAIGN CUSTOMIZE MISSION GENERATOR MISC
AIRPLANE GROUP
NAME B-17 Lead
CONDITION
COUNTRY USA
TASK Ground Attack
UNIT
TYPE B-17G
SKILL High
PILOT Pilot #001
TAIL # 010
COMM 251
CALLSIGN Enfield
HIDDEN ON MAP
HIDDEN ON PLANNER
LATE ACTIVATION
TYPE Perform Task
ACTION WW2: Carpet bombing
NUMBER 1
ENABLE TASK
NAME
CONDITION... STOP CONDITION...
WEAPON -Bombs
REL QTY All
PATTERN LENGTH 500 m
ALTITUDE ABOVE m
WAYPOINT 1 OF 7
NAME
TYPE Turning point
ALTITUDE 6096 m MSL
SPEED 370 kmh GS
MACH 0.329
ETA 12 2 28 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)
1 WW2: Carpet bombing(H = Height = 6096 m)
ADD INS EDIT DEL UP DOWN
CLONE
DFLT 8-17_Combat Box.miz
LAT 49°45'51" N
LONG 0°57' 7" W
ALT 0
PAN/SELECT MAP SAT ALT
8.08.2019 17:26:18

For each of the three trailing wings added in the example, we repeat the extended actions for WPT 0 as we did for the lead wing - this is necessary to build a large formation. In the GROUP field of the trailing wings, indicate the leading wing (B-17 Lead), with their position within the box set to Left / Right / Rear. To copy a group, select it and press [CTRL+C]. To paste the group with identical properties, position the mouse over the desired point on the map and press [CTRL+V].




MISSION EDITOR
FILE EDIT FLIGHT CAMPAIGN CUSTOMIZE MISSION GENERATOR MISC
AIRPLANE GROUP
NAME B-17 Right
COUNTRY USA
TASK Ground Attack
TYPE B-17G
SKILL High
PILOT Pilot #005
HIDDEN ON MAP
HIDDEN ON PLANNER
LATE ACTIVATION
TYPE Perform Task
ACTION WW2: Big Formation
ENABLE TASK
CONDITION... STOP CONDITION...
GROUP B-17 Lead
POSITION
Formation Type Combat Box
Position in Wing Leader
Position in Group Leader
Position in Box Right
LAST WPT
WAYPOINT
ALTITUDE 6146
SPEED 370
MACH 0.329
START
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)
ADD INS EDIT DEL UP DOWN
CLONE
WW2: Big Formation("B-17 Lead")
DFLT B-17_Combat Box.miz
LAT 49°45'46" N
LONG 0°56'58" W
ALT 0
PAN/SELECT MAP SAT ALT
8.08.2019 17:27:48


MISSION EDITOR
FILE EDIT FLIGHT CAMPAIGN CUSTOMIZE MISSION GENERATOR MISC
AIRPLANE GROUP
NAME B-17 Back
COUNTRY USA
TASK Ground Attack
TYPE B-17G
SKILL High
PILOT Pilot #013
HIDDEN ON MAP
HIDDEN ON PLANNER
LATE ACTIVATION
TYPE Perform Task
ACTION WW2: Big Formation
ENABLE TASK
CONDITION... STOP CONDITION...
GROUP B-17 Lead
POSITION
Formation Type Combat Box
Position in Wing Leader
Position in Group Leader
Position in Box Back
LAST WPT
WAYPOINT
ALTITUDE 6046
SPEED 370
MACH 0.329
START
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)
ADD INS EDIT DEL UP DOWN
CLONE
WW2: Big Formation("B-17 Lead")
DFLT B-17_Combat Box.miz
LAT 49°45'48" N
LONG 0°56'58" W
ALT 0
PAN/SELECT MAP SAT ALT
8.08.2019 17:28:02

This results in the "Combat Box" formation:



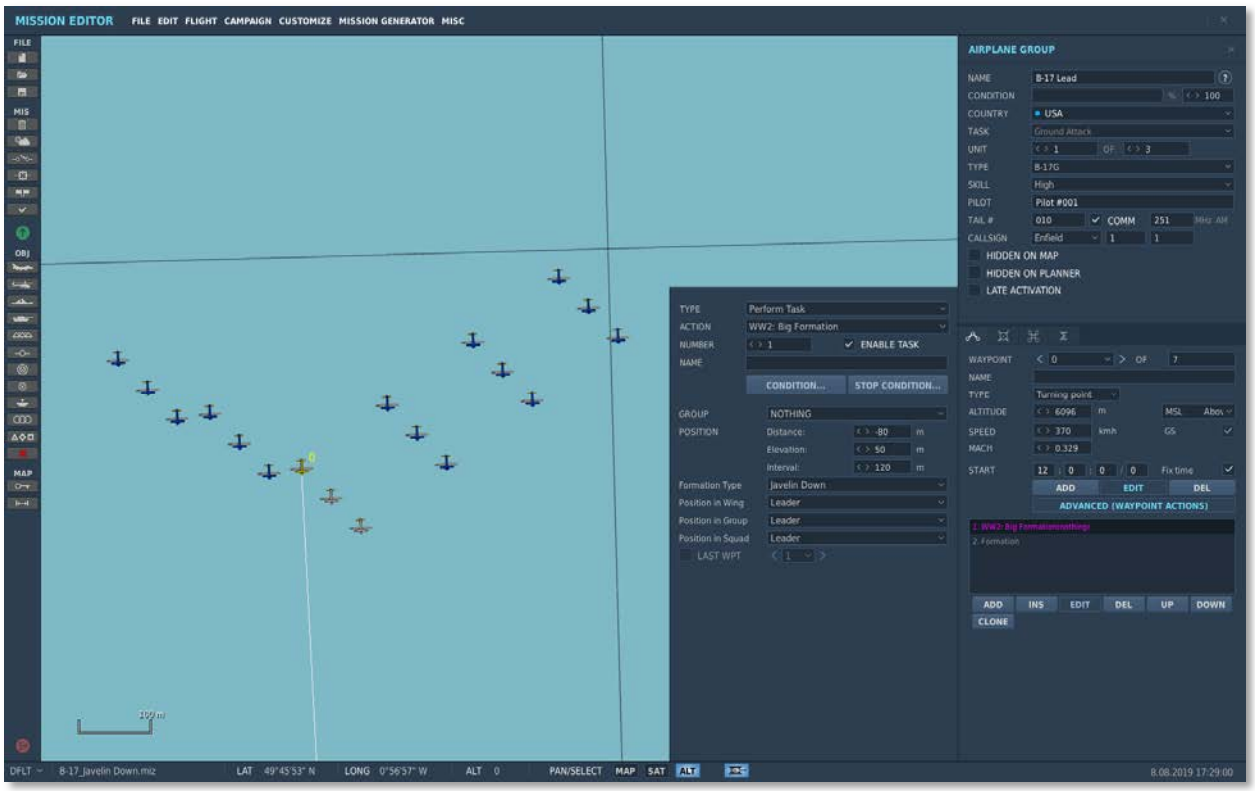


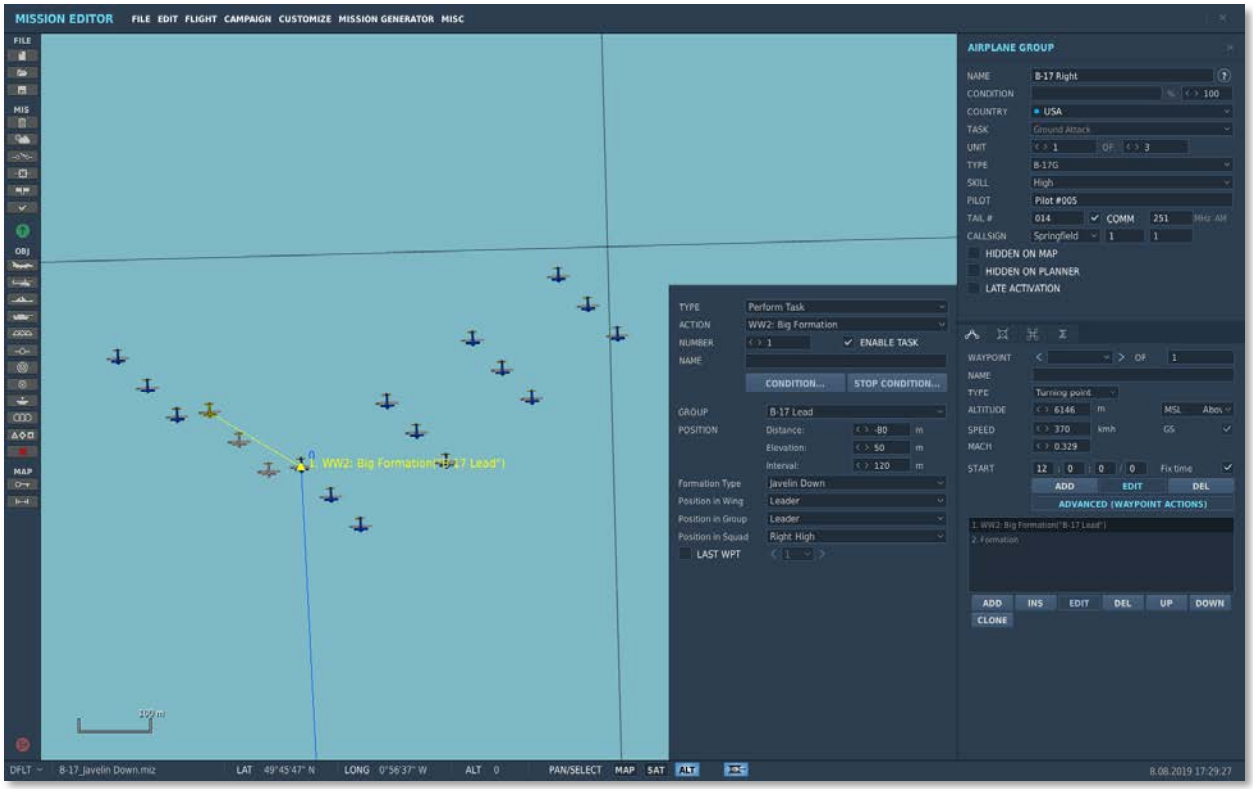
Bombing by cue of the lead wing:

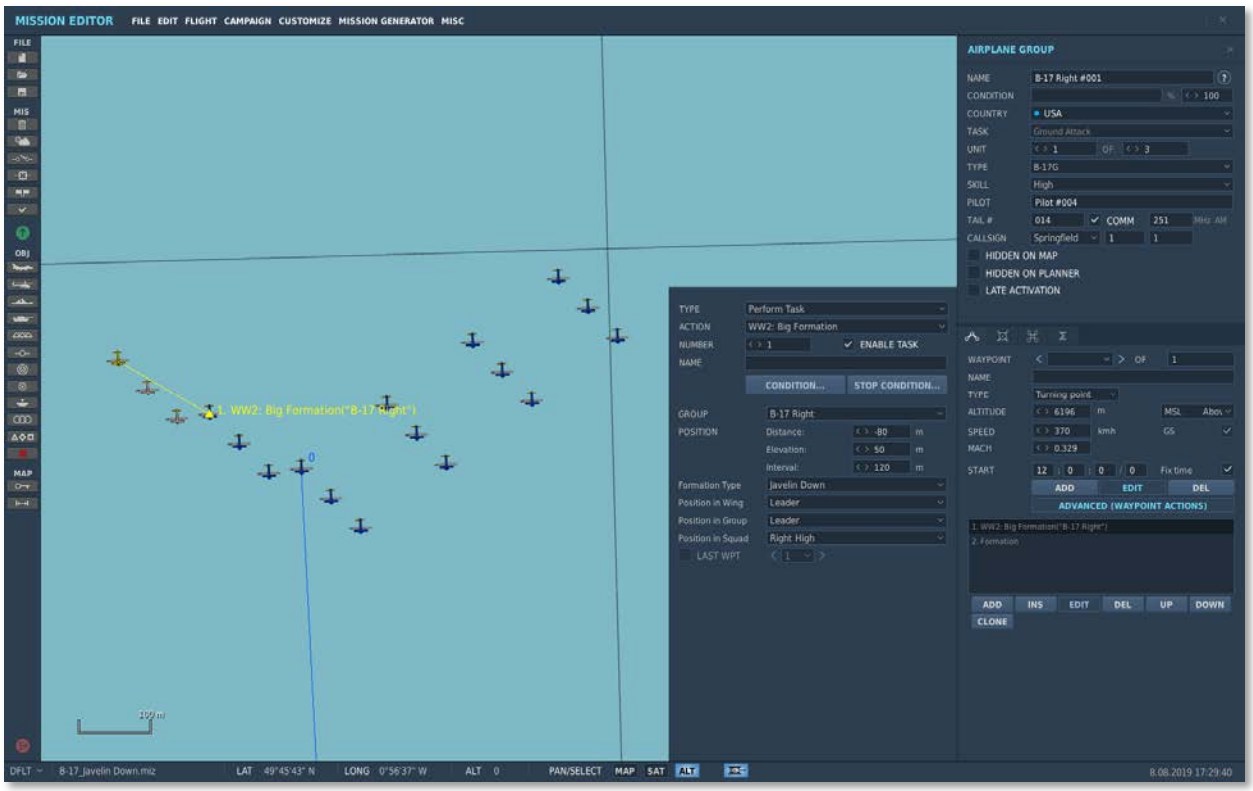


Example 2: "Javelin Down".

The "Javelin Down" formation it is formed in the same way as the Combat Box formation by assigning extended actions to the lead wing for WPTs 0 and 1, and WPT 0 for the trailing wings. Since the construction is a wedge, the closest wing (looking towards the lead wing) must be selected in the GROUP field for all the remaining trailing wings. To copy a group, select it and press [CTRL+C]. To paste the group with identical properties, position the mouse over the desired point on the map and press [CTRL+V].




MISSION EDITOR
FILE EDIT FLIGHT CAMPAIGN CUSTOMIZE MISSION GENERATOR MISC
AIRPLANE GROUP
NAME B-17 Right
CONDITION
COUNTRY USA
TASK Ground Attack
UNIT
TYPE B-17G
SKILL High
PILOT Pilot #005
TAIL # 014
COMM 251
CALLSIGN Springfield
HIDDEN ON MAP
HIDDEN ON PLANNER
LATE ACTIVATION
TYPE Perform Task
ACTION WW2: Big Formation
NUMBER
ENABLE TASK
NAME
CONDITION...
STOP CONDITION...
GROUP B-17 Lead
POSITION
Distance: -80 m
Elevation: 50 m
Interval: 120 m
Formation Type Javelin Down
Position in Wing Leader
Position in Group Leader
Position in Squad Right High
LAST WPT
WAYPOINT OF 1
NAME
TYPE Turning point
ALTITUDE 6146 m MSL
SPEED 370 kmh GS
MACH 0.329
START 12 : 0 : 0 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)
1. WW2: Big Formation("B-17 Lead")
2. Formation
ADD INS EDIT DEL UP DOWN
CLONE
1. WW2: Big Formation("B-17 Lead")
DFLT 8-17_Javelin Down.miz
LAT 49°45'47" N
LONG 0°56'37" W
ALT 0
PAN/SELECT MAP SAT ALT
8.08.2019 17:29:27


MISSION EDITOR
FILE EDIT FLIGHT CAMPAIGN CUSTOMIZE MISSION GENERATOR MISC
AIRPLANE GROUP
NAME B-17 Right #001
CONDITION
COUNTRY USA
TASK Ground Attack
UNIT
TYPE B-17G
SKILL High
PILOT Pilot #004
TAIL # 014
COMM 251
CALLSIGN Springfield
HIDDEN ON MAP
HIDDEN ON PLANNER
LATE ACTIVATION
TYPE Perform Task
ACTION WW2: Big Formation
NUMBER
ENABLE TASK
NAME
CONDITION...
STOP CONDITION...
GROUP B-17 Right
POSITION
Distance: -80 m
Elevation: 50 m
Interval: 120 m
Formation Type Javelin Down
Position in Wing Leader
Position in Group Leader
Position in Squad Right High
LAST WPT
WAYPOINT OF 1
NAME
TYPE Turning point
ALTITUDE 6196 m MSL
SPEED 370 kmh GS
MACH 0.329
START 12 : 0 : 0 / 0 Fix time
ADD EDIT DEL
ADVANCED (WAYPOINT ACTIONS)
1. WW2: Big Formation("B-17 Right")
2. Formation
ADD INS EDIT DEL UP DOWN
CLONE
1. WW2: Big Formation("B-17 Right")
DFLT 8-17_Javelin Down.miz
LAT 49°45'43" N
LONG 0°56'37" W
ALT 0
PAN/SELECT MAP SAT ALT
8.08.2019 17:29:40

The left side of the formation is set in the same manner as with the right. This produces the "Javelin Down" formation:







Start Enroute Task

As mentioned previously, the Start Enroute Tasks available for a group depend on the group type (fixed wing or helicopter group) and the group task. Remember, unlike Tasks, Start Enroute Tasks involve target groups the AI will have to independently detect prior to being able to conduct an attack. The AI's ability to detect enemy groups depends on a number of factors, including the sensors carried by the aircraft, AI skill setting, weather, etc.

Aircraft sensors:

- Air-to-air radar
- Air-to-surface radar
- Multimode radar
- Radar Warning Receiver (RWR)
- Daytime electro-optical systems
- Night-time electro-optical system
- Visual detection (subject to cockpit visibility restrictions)

Attacking AI settings:

- Skill setting
- Airspeed
- Cockpit visibility restrictions

Environmental and target factors:

- Illumination level
- Background terrain (forest, field, road, open water)
- Direct visibility limitations due to atmospheric conditions such as fog, rain and snow
- Cloud interference
- Target unit size
- Target group size
- Target speed
- Firing activity around the target

When creating scenarios using Start Enroute Tasks, all of the above conditions should be taken into account. For example, sending a daytime CAS aircraft to attack a ground group at night or obscured in dense fog conditions may result in the failure of the aircraft to detect the target. In addition, consider the distance of the target zone from the AI's flight route and make sure it is within the AI's detection range.

The following Start Enroute Tasks may be available, depending on the group type and task:

- No Enroute Task

- Search Then Engage
- Search Then Engage in Zone
- Search Then Engage Group
- Search Then Engage Unit
- Refueling
- AWACS
- FAC
- FAC - Engage Group

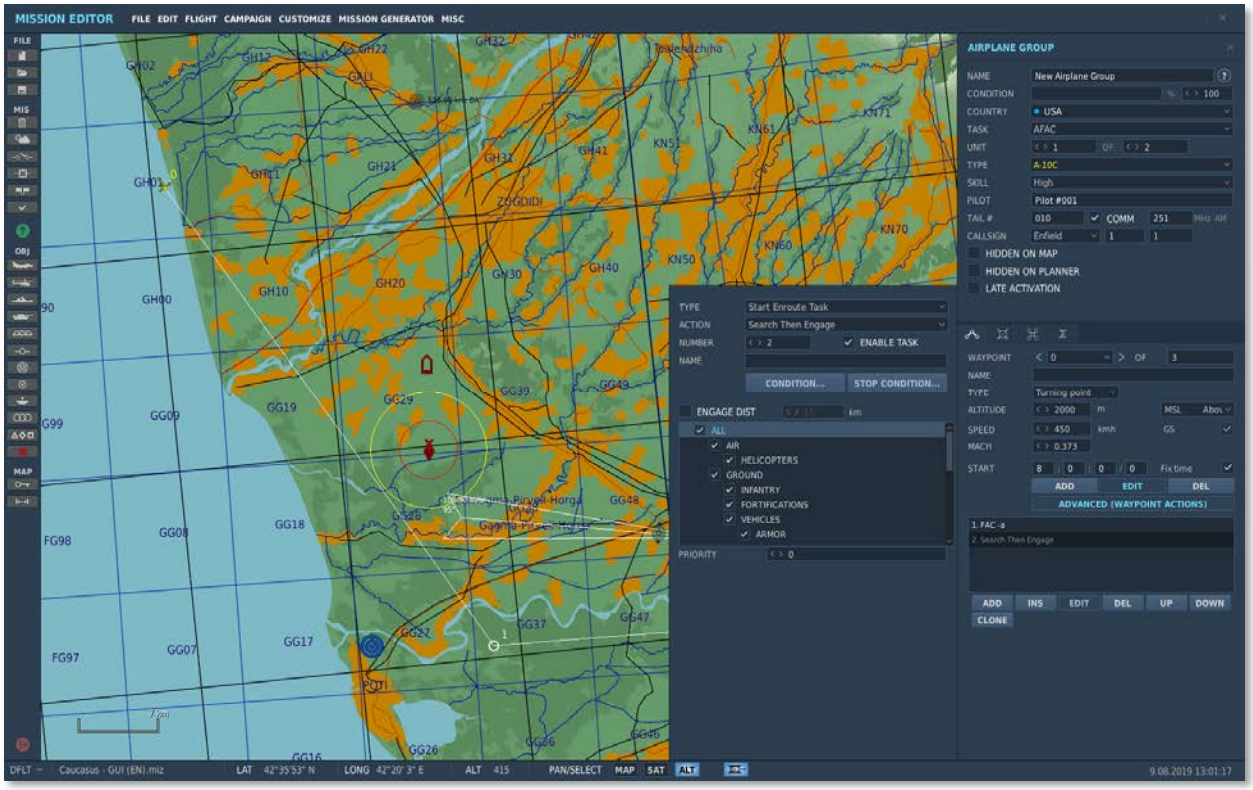
No Enroute Task

Available for all group types and all mission tasks. A No Enroute Task action does not set any action for the group to perform.

Search Then Engage

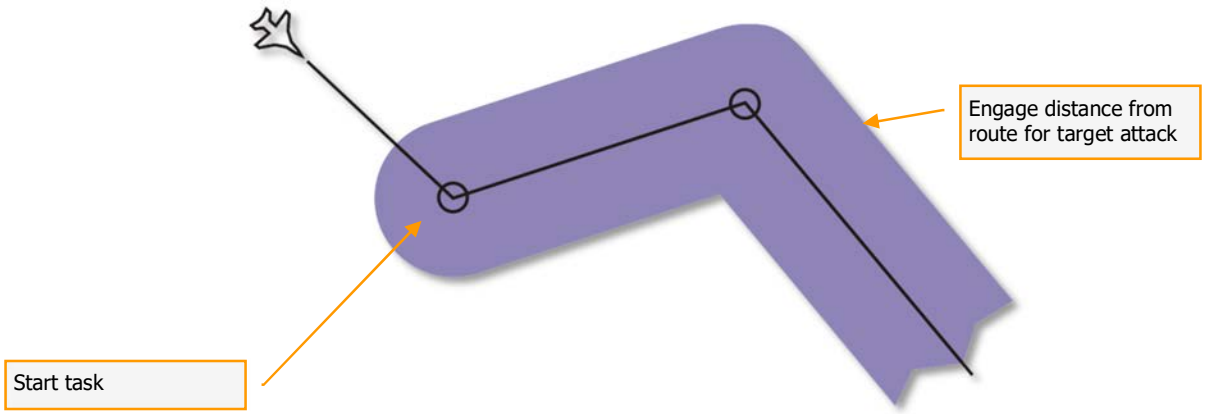
As long as the action is in effect, the AI will actively search for the target types set in the task properties and engage as soon as possible without conflicting with other tasks. Applicable to:

Group Type(s): fixed wing and helicopter groups.



Target Panel:

ENGAGE DISTANCE. This checkbox allows you to lock a specific range from the route leg for valid targets of the action. Targets beyond this range will not be attacked.



TYPES OF TARGETS. The list of available targets depends on the group task and the targets to be engaged are selected using check-boxes in the Lower Action Properties Menu:

1. SEAD.
    - Air Defense
        - AAA (Anti-Aircraft Artillery)
        - SAM (Surface-to-Air Missile systems)
            - SR SAM (Short-Range SAM systems)
            - MR SAM (Medium-Range SAM systems)
            - LR SAM (Long-Range SAM systems)
2. Anti-Ship.
    - Naval
        - Ships
            - Armed Ships
                - Heavy Armed Ships
                    - Aircraft Carrier
                    - Cruisers
                    - Destroyers
                    - Fregates
                    - Corvettes
                - Light Armed Ships
                - Unarmed Ships

- Submarines

3. CAS and AFAC
   - All
     - Air
       - Helicopters
     - Ground
       - Infantry
       - Fortifications
       - Vehicles
         - Armor
           - Tanks
           - IVF (Infantry Fighting Vehicles)
           - APC (Armored Personnel Carriers)
         - Artillery
         - Unarmed
       - Air Defense
         - AAA (Anti-Aircraft Artillery)
         - SAM (Surface-to-Air Missile systems)
           - SR SAM (Short-Range SAM systems)
           - MR SAM (Medium-Range SAM systems)
           - LR SAM (Long-Range SAM systems)
     - Naval
       - Light Armed Ships
4. Fighter Sweep and Intercept.
   - Airplanes
     - Fighters
     - Bombers

PRIORITY. This number is used to set the action priority in relation to other actions of the waypoint. The priority number represents the order this action will take relative to other actions, starting with '0' as the highest priority.

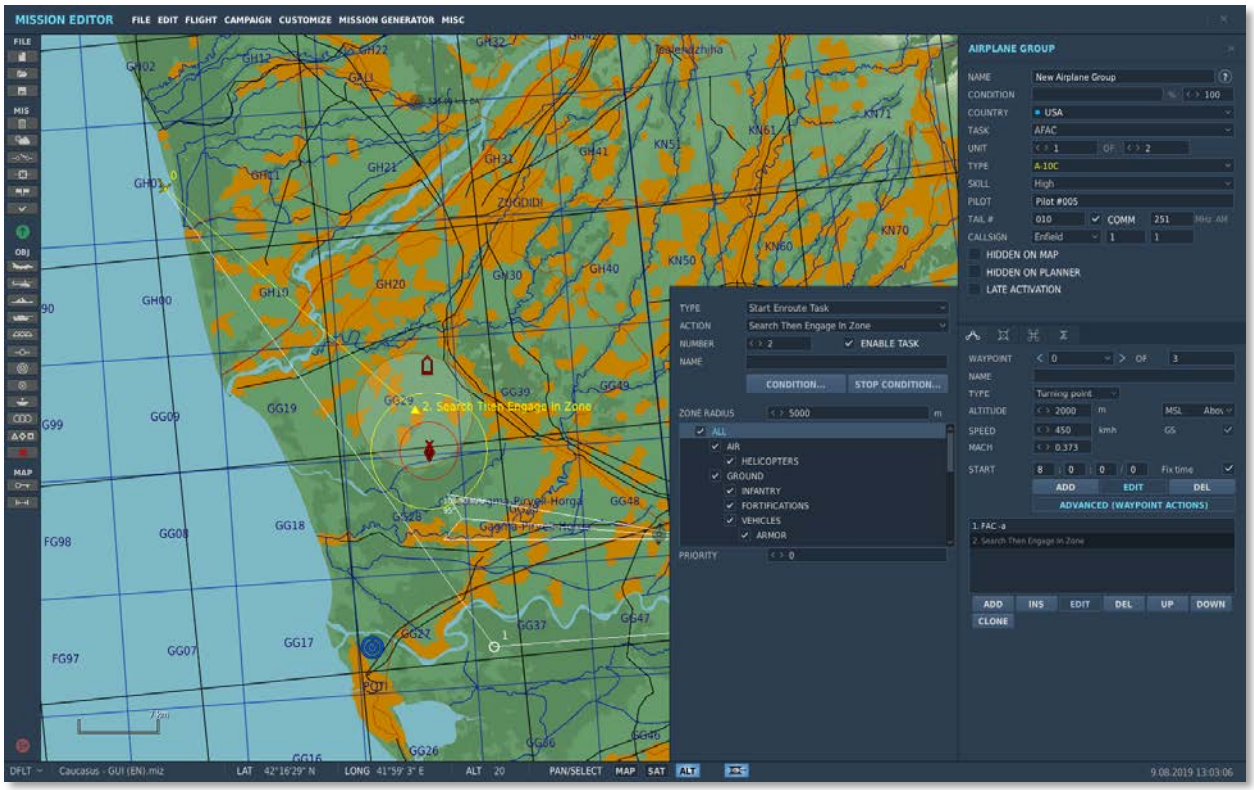
The Search Then Engage Enroute Task can be used to create scenarios where the air group maintains a holding pattern and leaves it to attack discovered enemy units within the limitations set by the task properties. To do this, create the Search Then Engage Enroute Task in the actions list after the Orbit Task.

Search Then Engage in Zone

Same as the Search Then Engage Enroute Task, but with an added restriction of a user-defined target area. The target area is a circular zone the user can place anywhere on the map and adjust the radius. Applicable to:

Group Type(s): aircraft and helicopter groups;

Group Task(s): all group tasks, except "Fighter Sweep" and "Intercept".



The target area can be placed on the map using the left mouse button. A target line connects the center of the target area (triangle marker) with the action waypoint.

Target Panel:

ZONE RADIUS. This field is used to enter the radius of the target area in meters.

TYPES OF TARGETS. The list of available targets depends on the group task and the targets to be engaged are selected using check-boxes in the Lower Action Properties Menu.

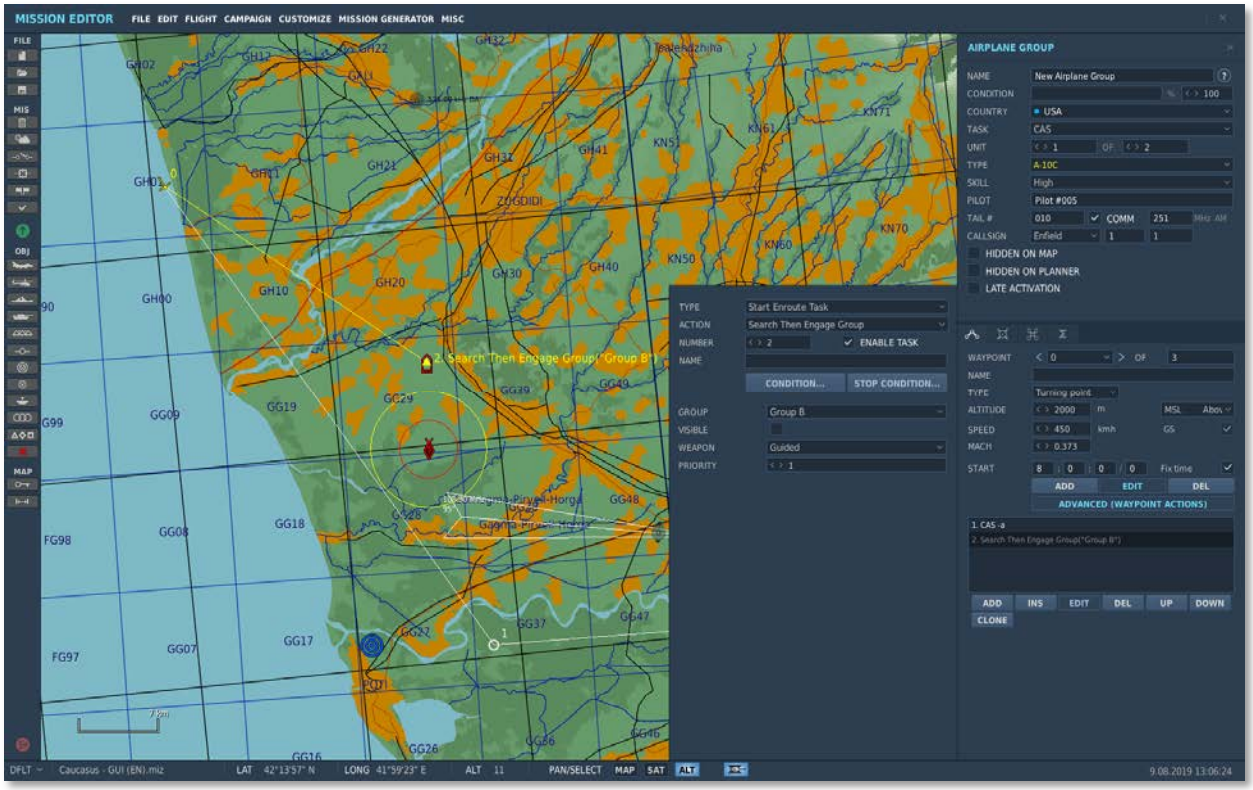
PRIORITY. This number is used to set the action priority in relation to other actions of the waypoint. The priority number represents the order this action will take relative to other actions, starting with '0' as the highest priority.

Like the Search Then Engage Enroute Task, the Search Then Engage in Zone Enroute Task can be used to create scenarios where the air group maintains a holding pattern and leaves it to attack discovered enemy units within the target area and the limitations set by the task

properties. To do this, create the Search Then Engage in Zone Enroute Task in the actions list after the Orbit Task.

Search Then Engage Group

Analogous to the Attack Group Task, but requires the attacking group to independently detect the target group prior to conducting an attack.



Target Panel:

GROUP. Drop down menu to select the target group from all applicable enemy groups available in the mission.

VISIBLE. The VISIBLE option allows the unit to know or not know the location of target position at the start of the mission.

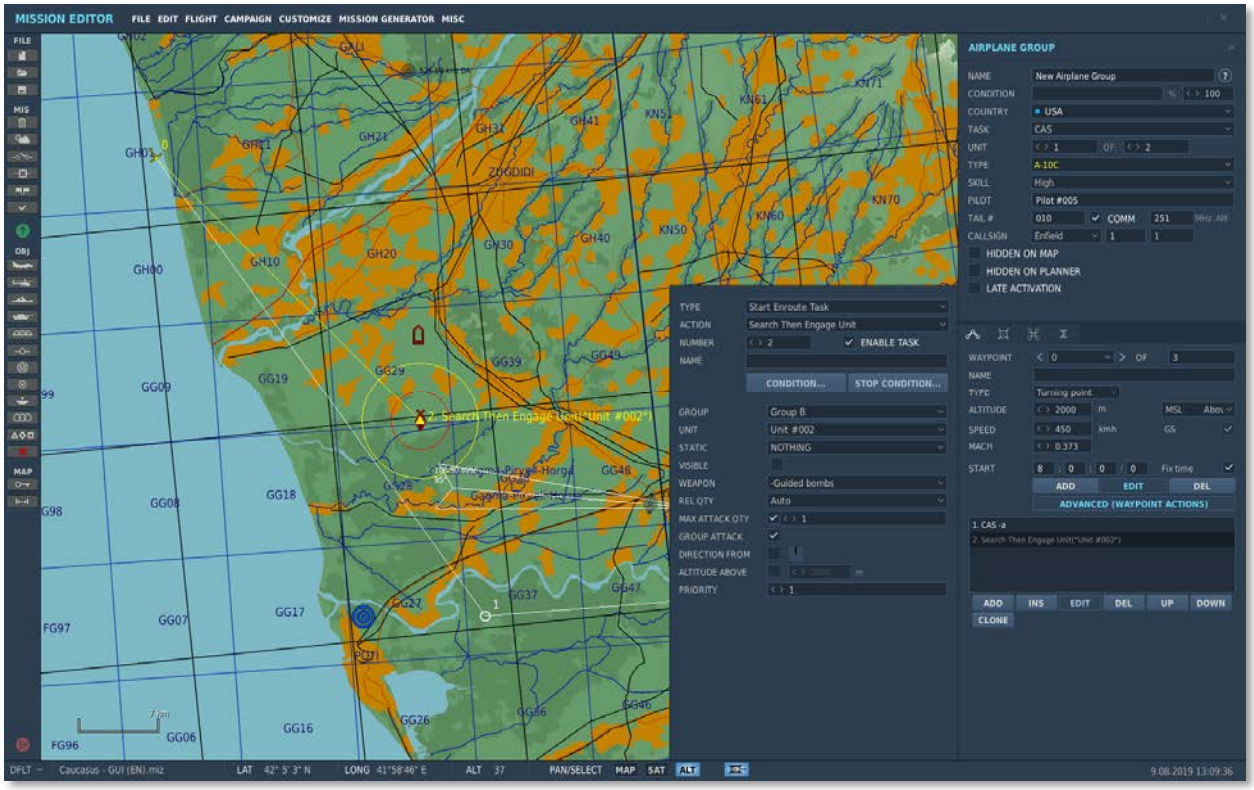
WEAPON. Drop down menu to select the weapon types cleared to use for this action.

PRIORITY. This number is used to set the action priority in relation to other actions of the waypoint. The priority number represents the order this action will take relative to other actions, starting with '0' as the highest priority.

The Search Then Engage Group Task can be used to create scenarios where the air group maintains a holding pattern and leaves it to attack the designated group whenever it is detected. To do this, create the Search Then Engage Group Enroute Task in the actions list after the Orbit Task.

Search Then Engage Unit

Analogous to the Attack Unit Task, but requires the attacking group to independently detect the target unit prior to conducting an attack.



Target Panel:

GROUP. Drop down menu to select the target group from all applicable enemy groups available in the mission.

UNIT. Drop down menu to select the specific unit of the target group to attack.

STATIC. Drop down menu to select a static object as the target (bunker, FARP, etc.)

VISIBLE. The VISIBLE option allows the unit to know or not know the location of target position at the start of the mission.

WEAPON. Drop down menu to select the weapon types cleared to use for this action.

REL QTY. The amount of payload cleared to expend in executing this action.

MAX ATTACK QTY. The maximum number of attack runs cleared for this action.

GROUP ATTACK. When checked, the entire flight group will participate in the attack of the target. By default, each aircraft in a flight group is assigned one target.

DIRECTION FROM. Allows you to set the course (azimuth) of the aircraft group for reaching the target for an attack.

ALTITUDE ABOVE. Sets the altitude of the aircraft group on approach to target.

PRIORITY. This number is used to set the action priority in relation to other actions of the waypoint. The priority number represents the order this action will take relative to other actions, starting with '0' as the highest priority.

The Search Then Engage Unit Task can be used to create scenarios where the air group maintains a holding pattern and leaves it to attack the designated enemy unit whenever it is detected. To do this, create the Search Then Engage Group Enroute Task in the actions list after the Orbit Task.

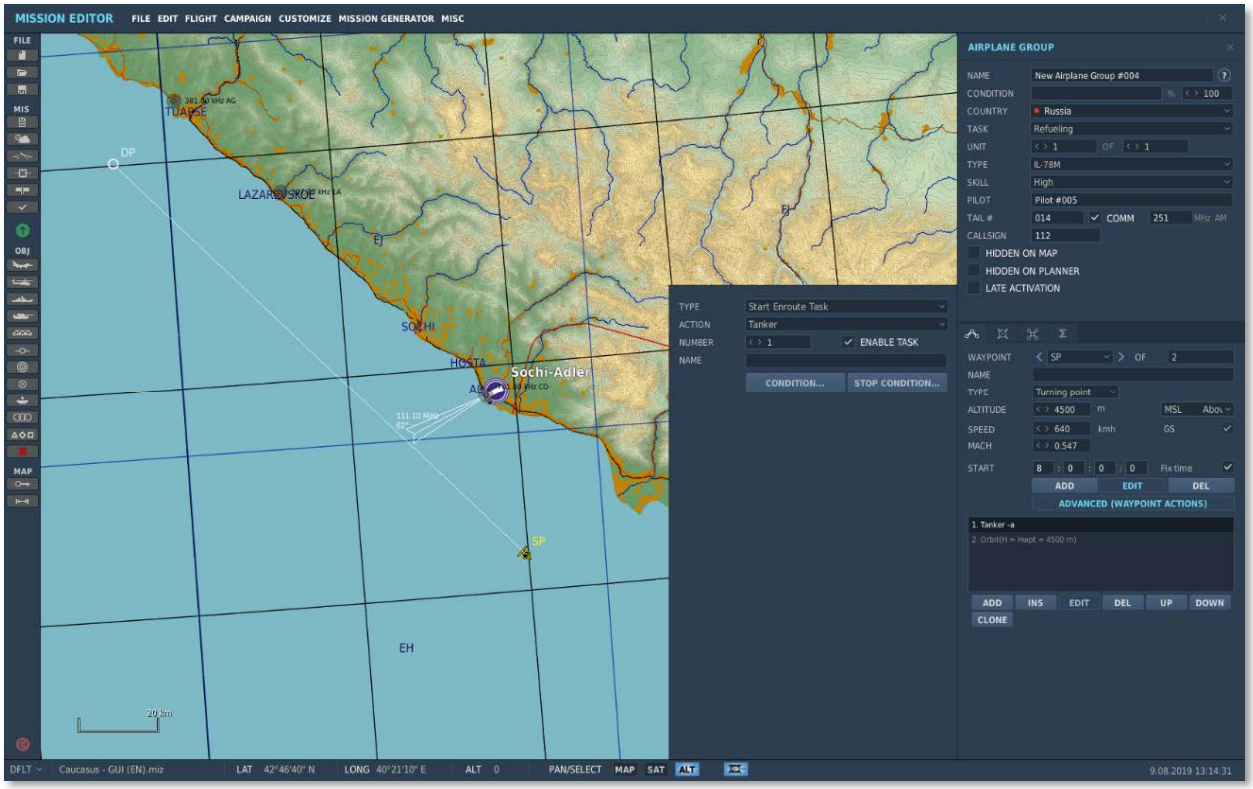
Refueling

The aircraft will perform aerial refueling. Applicable to:

Group Type(s): airborne tanker aircraft, including IL-78 and KC-135;

Group Task(s): Refueling.

When the Refueling Mission Task is created, a Refueling Enroute Task is created automatically as an automatic action at the initial waypoint. This action cannot be deleted or edited.



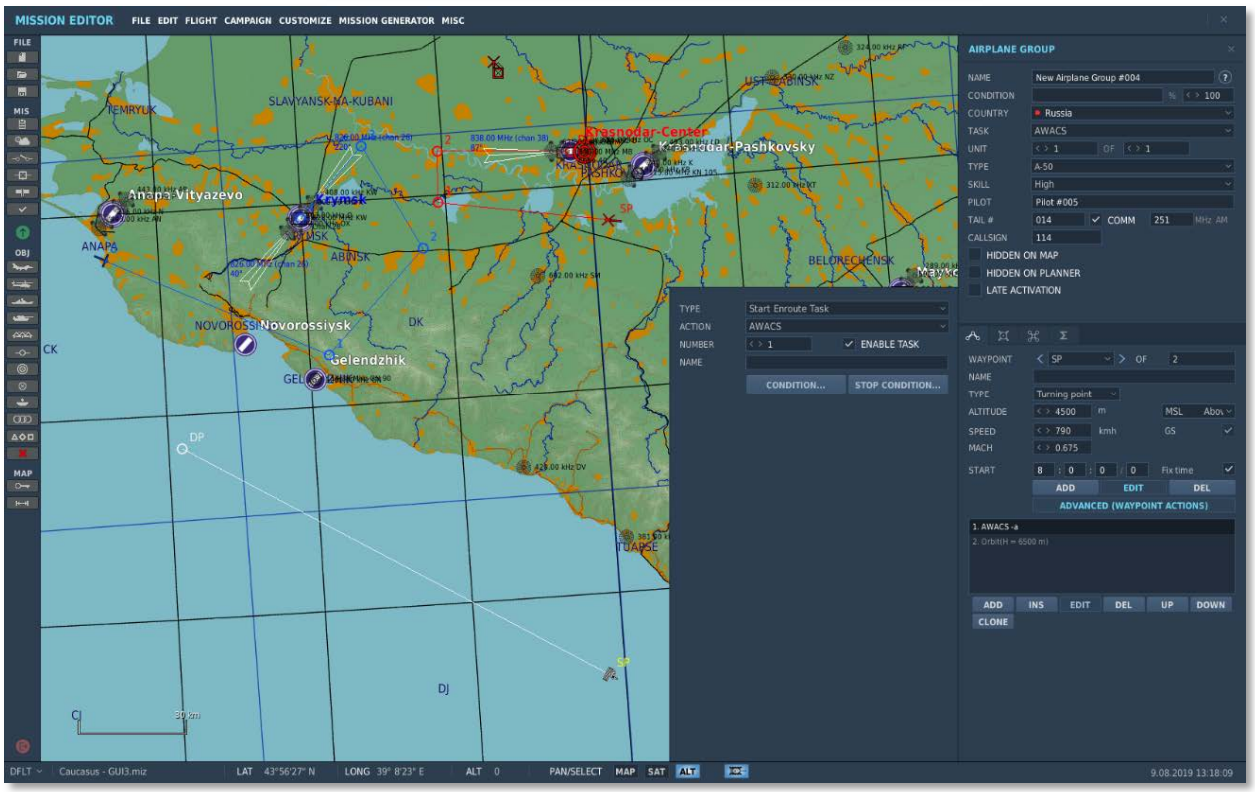
AWACS

The aircraft will perform Airborne Warning and Control for friendly units. Applicable to:

Group Type(s): dedicated AWACS aircraft, including A-50, E-2B, E-3A;

Group Task(s): AWACS.

AWACS detects aerial targets and transmits data to friendly aircraft via radio and data links. This information may be used by the player. When the AWACS Mission Task is created, an AWACS Enroute Task is created automatically as an automatic action at the initial waypoint. This action cannot be deleted or edited.



FAC

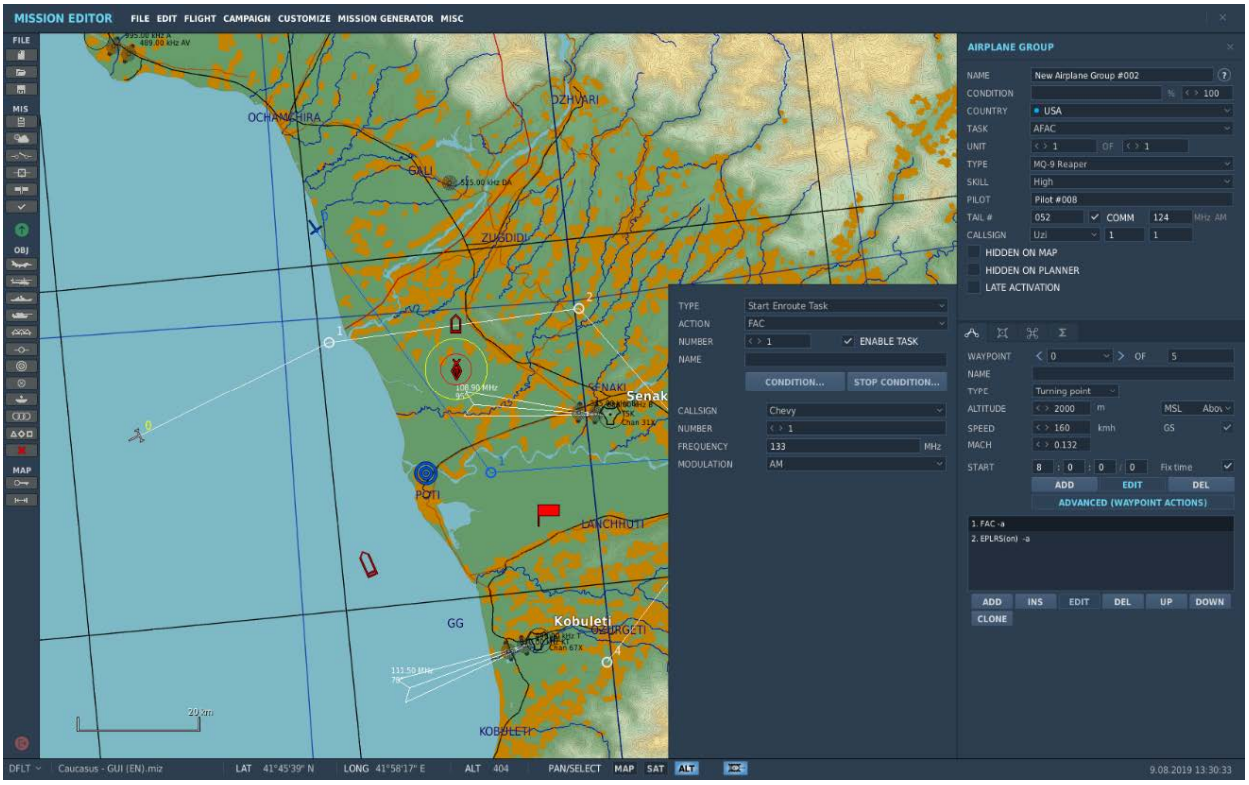
The aircraft will act as an Airborne Forward Air Controller (AFAC) to provide targeting orders for friendly units in the area. Applicable to:

Group Type(s): fixed wing and helicopter groups in BLUE coalition only;

Mission Task(s): AFAC.

Under a FAC Enroute Task, the FAC will automatically prioritize targets for the player or attacking AI groups and issue targeting orders throughout the attack phase.

***Note**: The FAC's target selection may not correspond to the targets intended for the player or attacking AI by the mission designer, because no specific target group is selected.*



Target Panel:

CALLSIGN. FAC's callsign

NUMBER. FAC's number.

FREQUENCY. Set the FAC group's radio communications frequency in MHz.

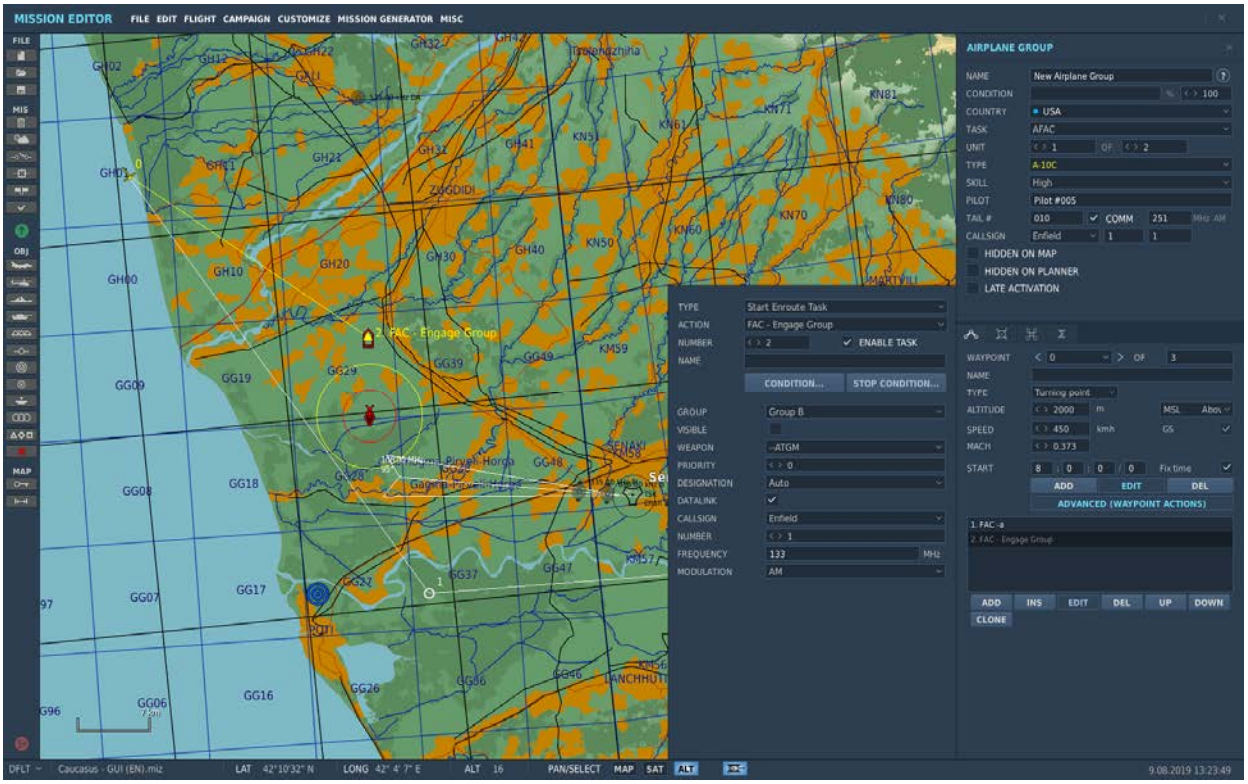
MODULATION. Set the FAC group's radio communications AM/FM modulation.

FAC – Engage Group

Analogous to the FAC Enroute Task, except in this case a specific enemy group is selected by the mission designer for the AFAC to target. Applicable to:

Group Type(s): fixed-wing and helicopter groups in BLUE coalition only;

Mission Task(s): AFAC.



Target Panel:

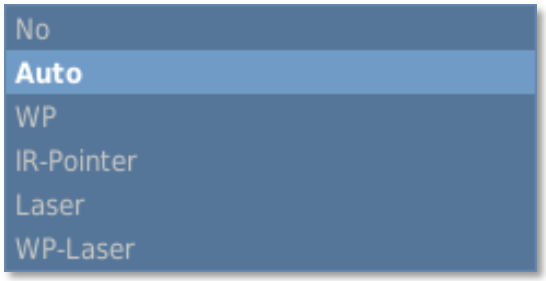
GROUP. Drop down menu to select the target group from all applicable enemy groups available in the mission.

VISIBLE. The VISIBLE option allows the unit to know or not know the location of target position at the start of the mission.

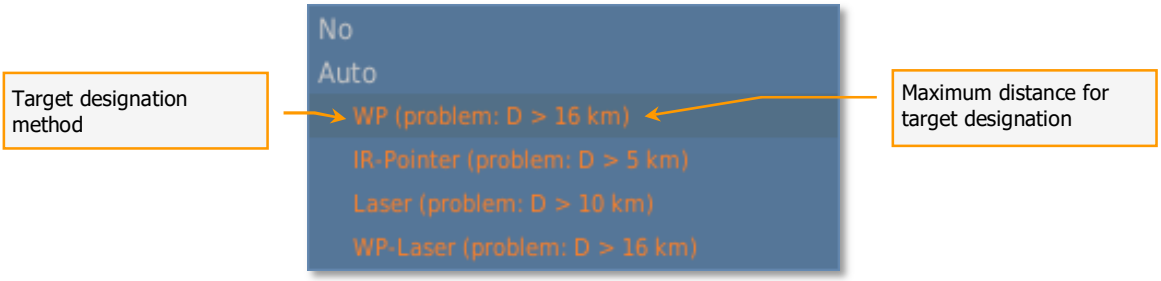
WEAPON. Drop down menu to select the weapon types cleared to use for this action. In the case of an AFAC, the WEAPON selection refers to weapons the AFAC will request to be employed against the target.

PRIORITY. This number is used to set the action priority in relation to other actions of the waypoint. The priority number represents the order this action will take relative to other actions, starting with '0' as the highest priority.

DESIGNATION. Selects the method of target designation (Auto, smoke marker ("Willy Pete", WP), IR-Pointer, Laser, WP-Laser. Target designation is carried out on the player's radio request at a fixed radio range.



If the FAC unit is located from the enemy at a distance greater than the operational range of the target designator equipment, then the maximum ranges for each of the available equipment will be displayed in the menu.



In this case, it is necessary to adjust the distance of the forward observer to the enemy such that it is within the specified maximum range.

DATALINK. Automatic data transmission system for the exchange of target information between ground units and aircraft.

CALLSIGN. Callsign of the FAC.

NUMBER. Selection field for the FAC's number.

FREQUENCY. Set the FAC group's radio communications frequency in MHz.

MODULATION. Set the FAC group's radio communications AM/FM modulation.

Remember, because this is an Enroute Task, the AFAC will have to independently detect the target group prior to being able to issue targeting commands against it. As described above, target detection will depend on a number of dynamic variables in-game, including possible terrain obstructions or weather interference. If the mission designer has not assigned any groups, the AFAC will not issue any targeting commands.

Perform Command

The following commands can be set as actions upon reaching a waypoint:

No Action

No command is set. Default selection.

Run Script

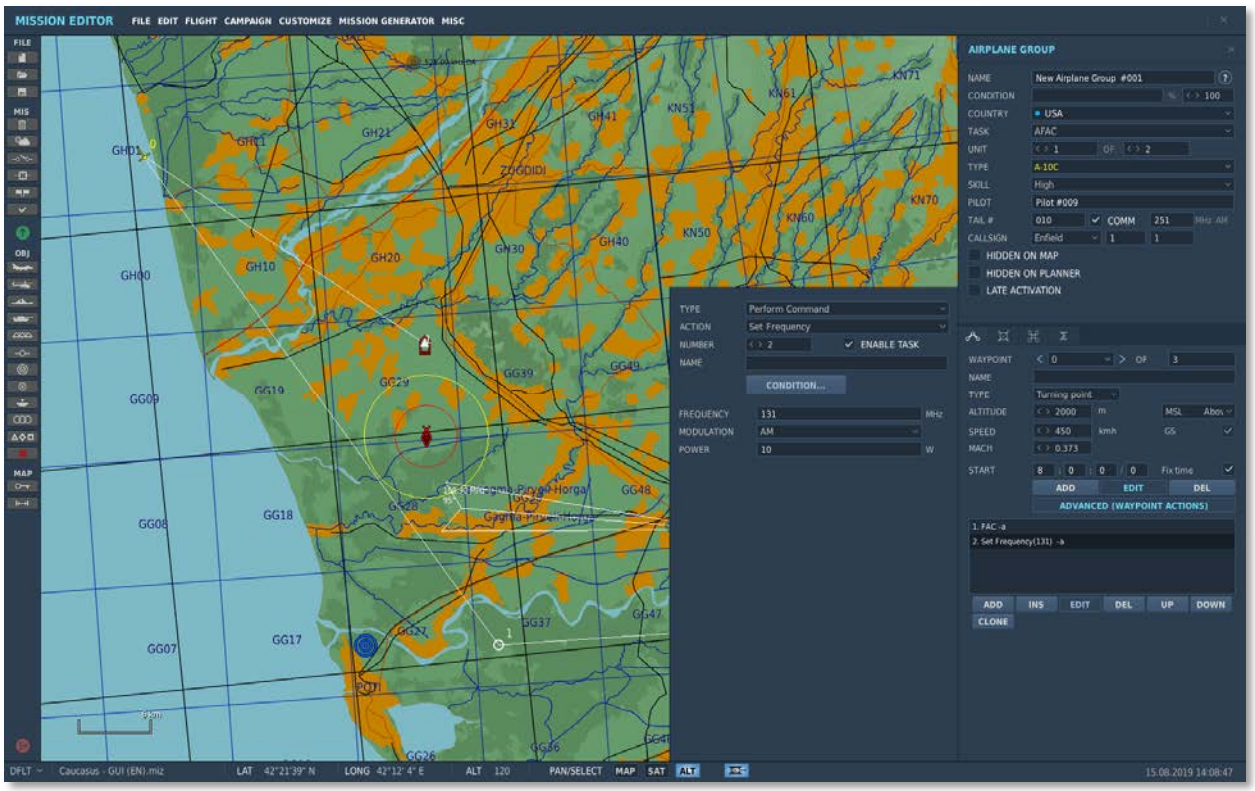
Run the Lua-script entered in the scratchpad.

Script File

Selects the LUA-script file.

Set Frequency

Set the group's radio communications frequency in MHz and AM/FM modulation.



FREQUENCY. Set the group's radio communications frequency in MHz.

MODULATION. Set the group's radio communications AM/FM modulation.

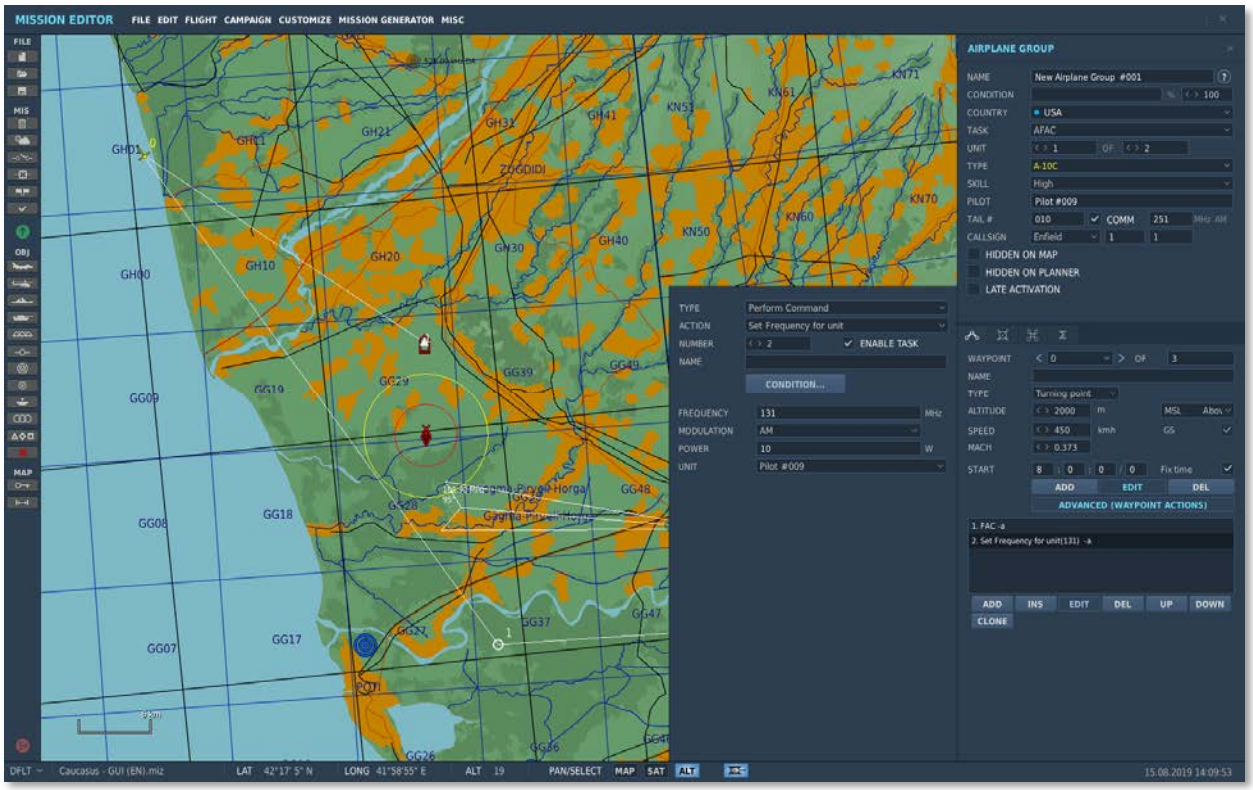
- **AM** - Amplitude modulation.

- **FM** - Frequency modulation.

POWER. Transmitter power. This affects communication range.

Set Frequency for unit

Command to switch to the specified radio frequency for the unit.



FREQUENCY. Set the group's radio communications frequency in MHz.

MODULATION. Set the group's radio communications AM/FM modulation.

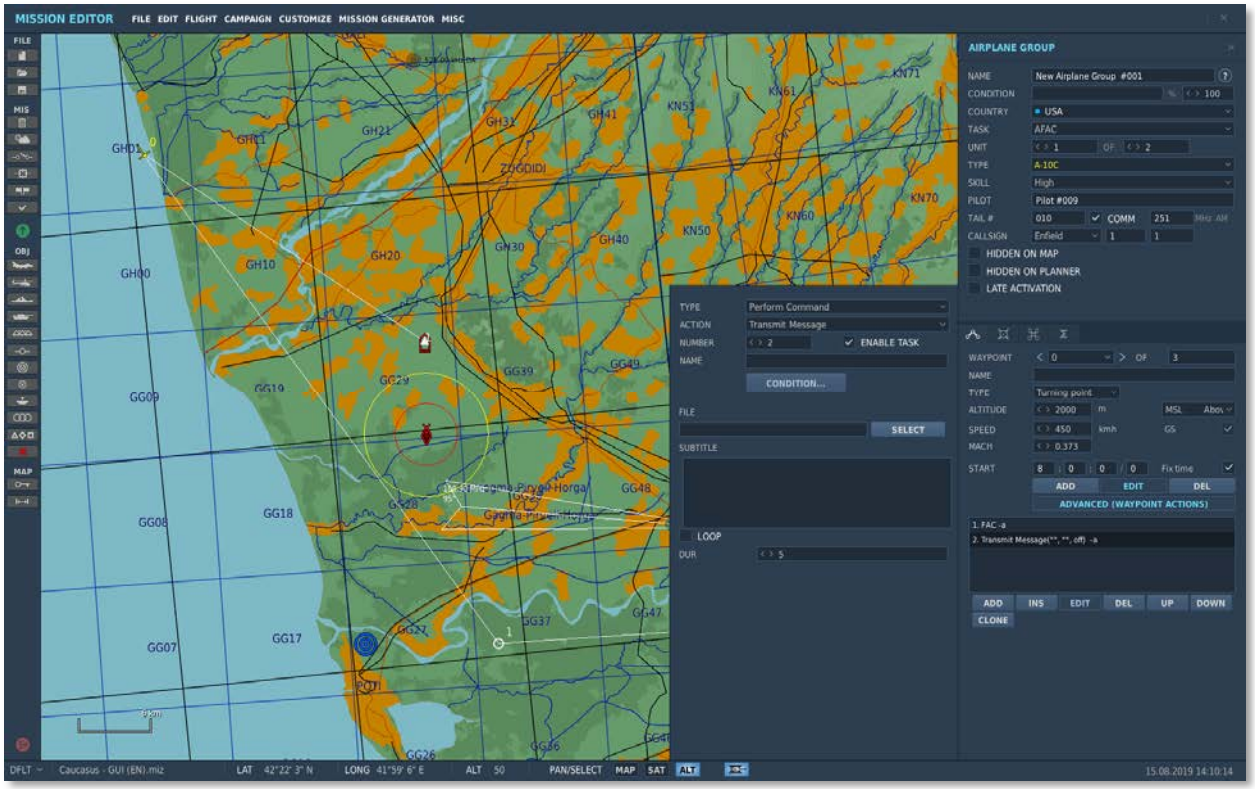
- **AM** - Amplitude modulation.
- **FM** - Frequency modulation.

POWER. Transmitter power. This affects communication range.

UNIT. Selects the unit to which the frequency is assigned.

Transmit Message

Command to send a message from the lead group.



FILE. Selects the audio file to be played during the radio transmission (*.ogg;*.wav).

SUBTITLE. Field for inputting the text message (subtitles) to be displayed during the radio transmission.

LOOP. Settings for repeating the message.

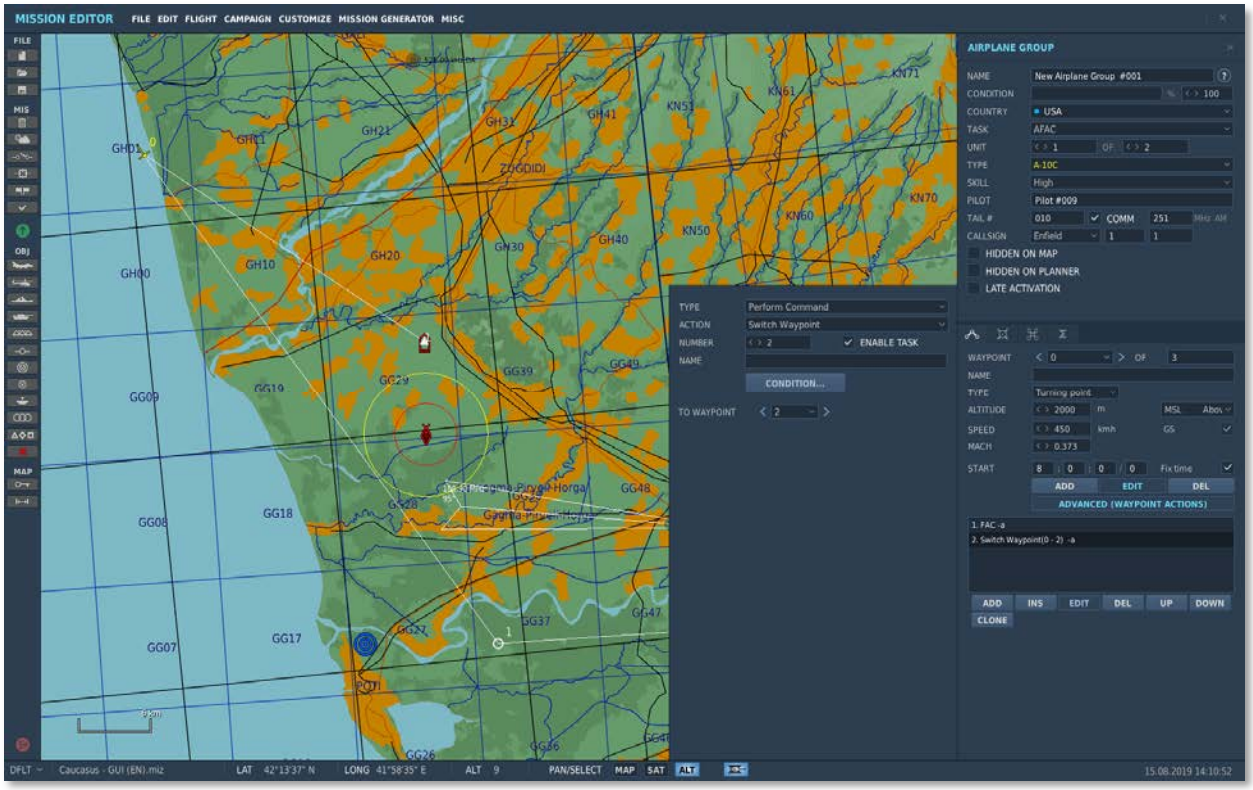
DURATION. Duration of the message, in seconds.

Stop Transmission

Stops the transmission from the lead group. This command has no available options.

Switch Waypoint

When the conditions for this action are in effect, the group will navigate from the current waypoint directly to the one set by this action. For example, the group can be made to fly from waypoint 3 directly to waypoint 6. When this command is executed, all other actions associated with the waypoint are stopped.



By default, this is set to the next waypoint in the route.

All main tasks, commands, and behavior options, missed waypoints, will not be fulfilled.

This command can also be used to create a looping route by switching to an earlier waypoint. The looping command can be stopped using a stop condition, such as a mission time or a flag state.

Invisible

Set the group to be invisible to all enemy AI units. Enabled and disabled using the ENABLE checkbox.

This command may be useful when creating ground-based FACs in close proximity to enemy positions as a means of simulated a well camouflaged position and preventing the FAC from being destroyed.

Immortal

Set the group to be immortal. Enemy AI units will be able to detect the group, but their weapons will have no effect.

This command may be useful when creating prolonged engagement scenarios, in which persistence of fire is a desired element.

EPLRS

Set the unit's position datalink on/off.

Smoke On_Off

This command activates the a smoke generator, if available, located on the aircraft's wing mounts. It has a single option – the VALUE checkbox, which allows or disables the execution of the command.

Set Option

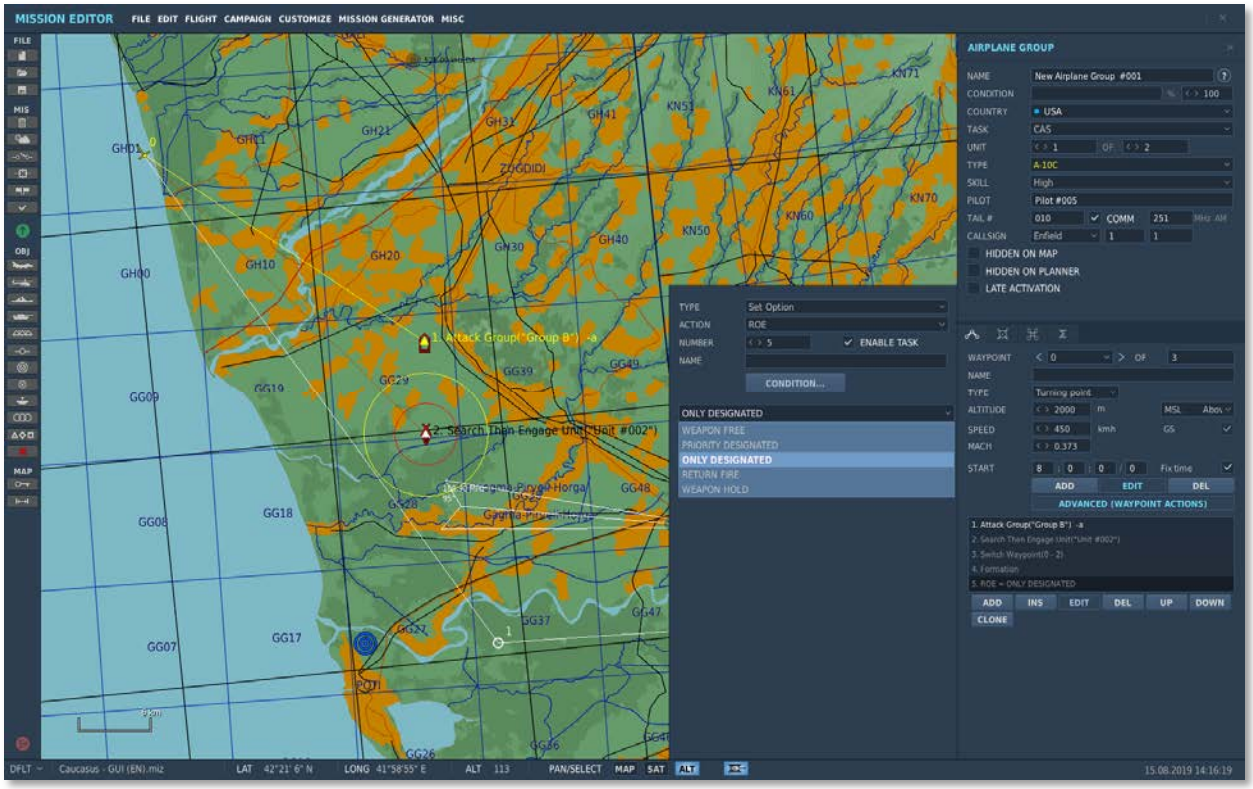
The following Options can be set as actions upon reaching a waypoint:

No Option

No option is set. Default selection.

ROE (Rules of Engagement)

Set fire rules to determine the group's combat behavior when encountering enemy units.

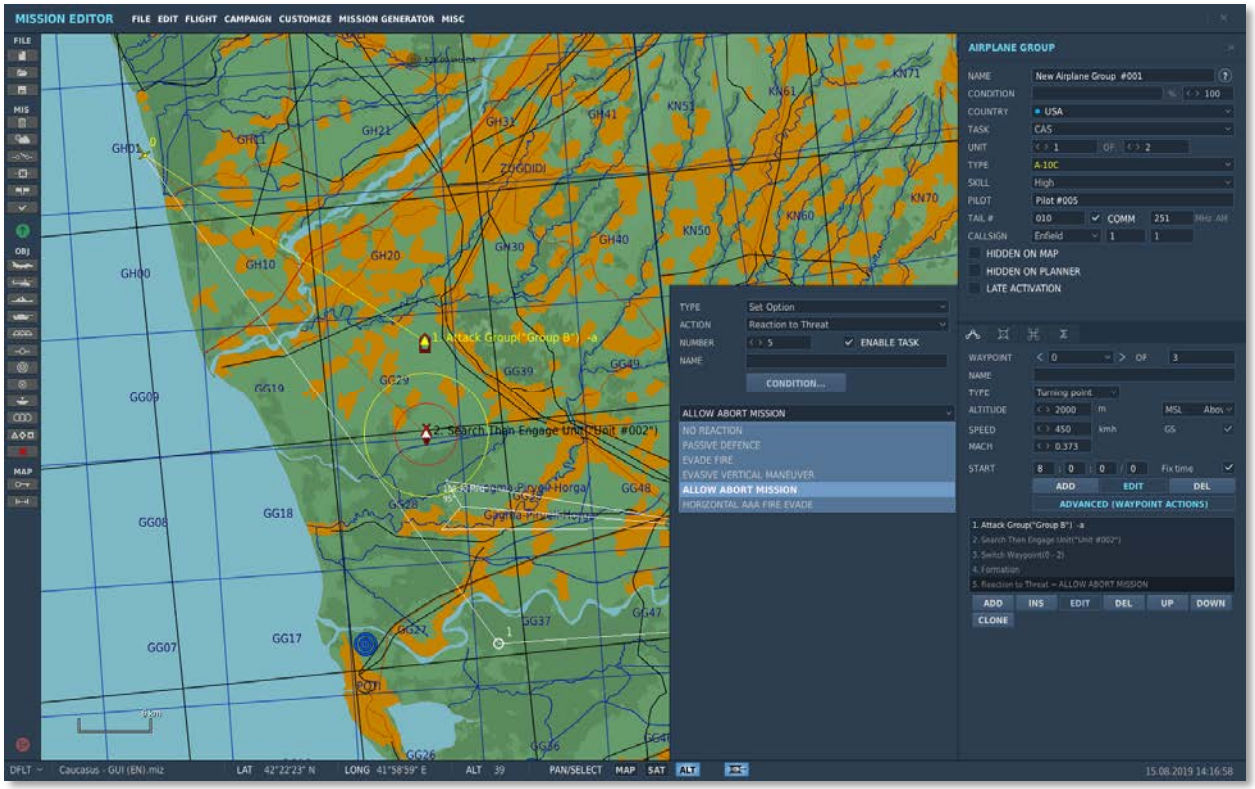


- **Weapon free** – engage any enemy group in contact. Target prioritization is performed automatically by the group based on the situation.
- **Priority designated** – engage any enemy groups in contact. However, the enemy group designated as the target for the action will be the primary target and engaged first. Once the primary target is destroyed, further target prioritization is performed automatically by the group based on the situation.
- **Only designated** – engage only the enemy group designated as the target for the action.
- **Return fire** – return fire only; do not engage first.
- **Weapon hold** – do not fire under any circumstances.

The default setting for all groups is *Only designated.*

Reaction to Threat

Set defensive rules to determine the group's combat behavior when encountering enemy units.



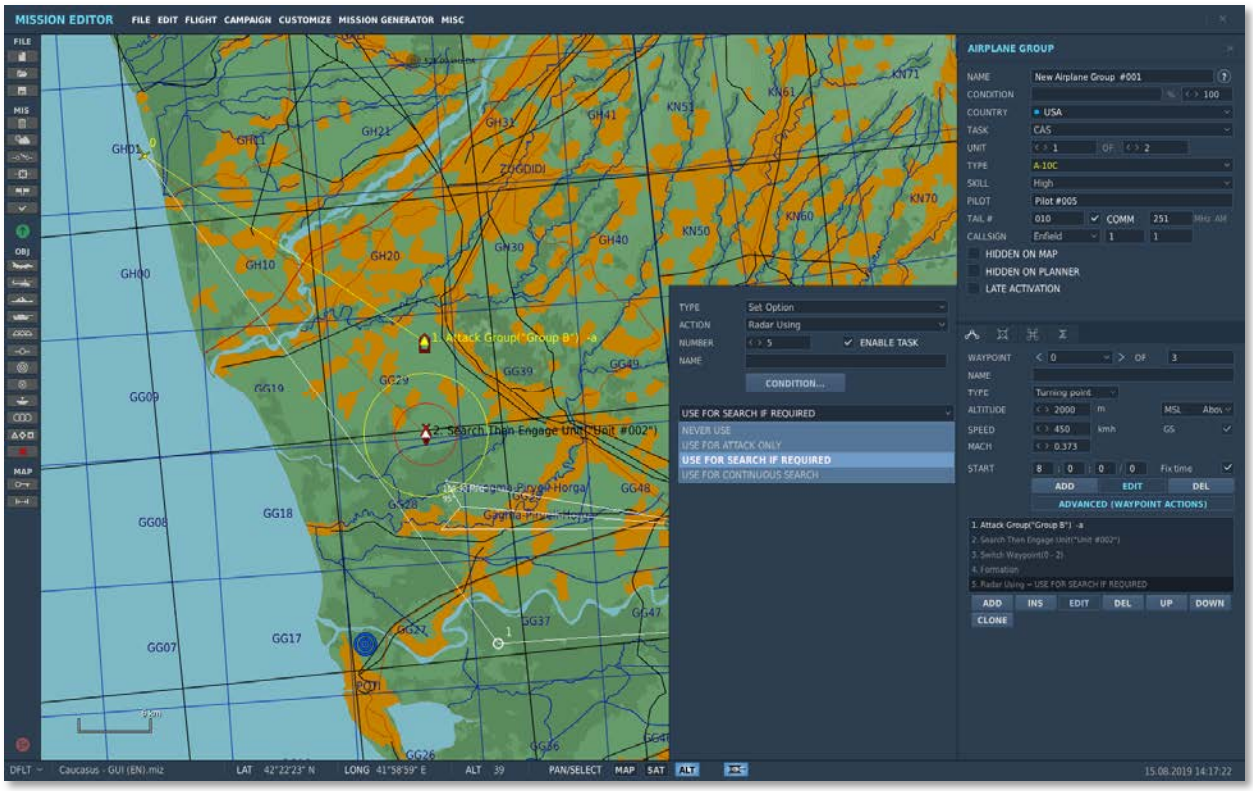
- **No reaction** – no defensive actions to counter threats.
- **Passive defense** – utilize passive and active defense measures, but no defensive maneuvering against incoming threats.
- **Evade fire** – utilize passive and active defense measures as well as defensive maneuvering against incoming threats.
- **Evasive vertical maneuver** – the group will be able to increase or decrease altitude to avoid a known threat zone. The group will also utilize active and passive defense measures.
- **Allow Abort Mission** – allow group to cease mission if engaged.
- **Horizontal AAA fire evade** – The group performs evasive maneuvers against enemy anti-aircraft fire without changing altitude.

The Reaction on Threat Options are only in effect in defense. When performing an attacking action, the ROE option is in effect. In other words, the ROE option has a higher priority over the Reaction on Threat option.

The default setting for all groups is *Allow Abort Mission*.

Radar Using

Set the rules for the use of radar by the group.



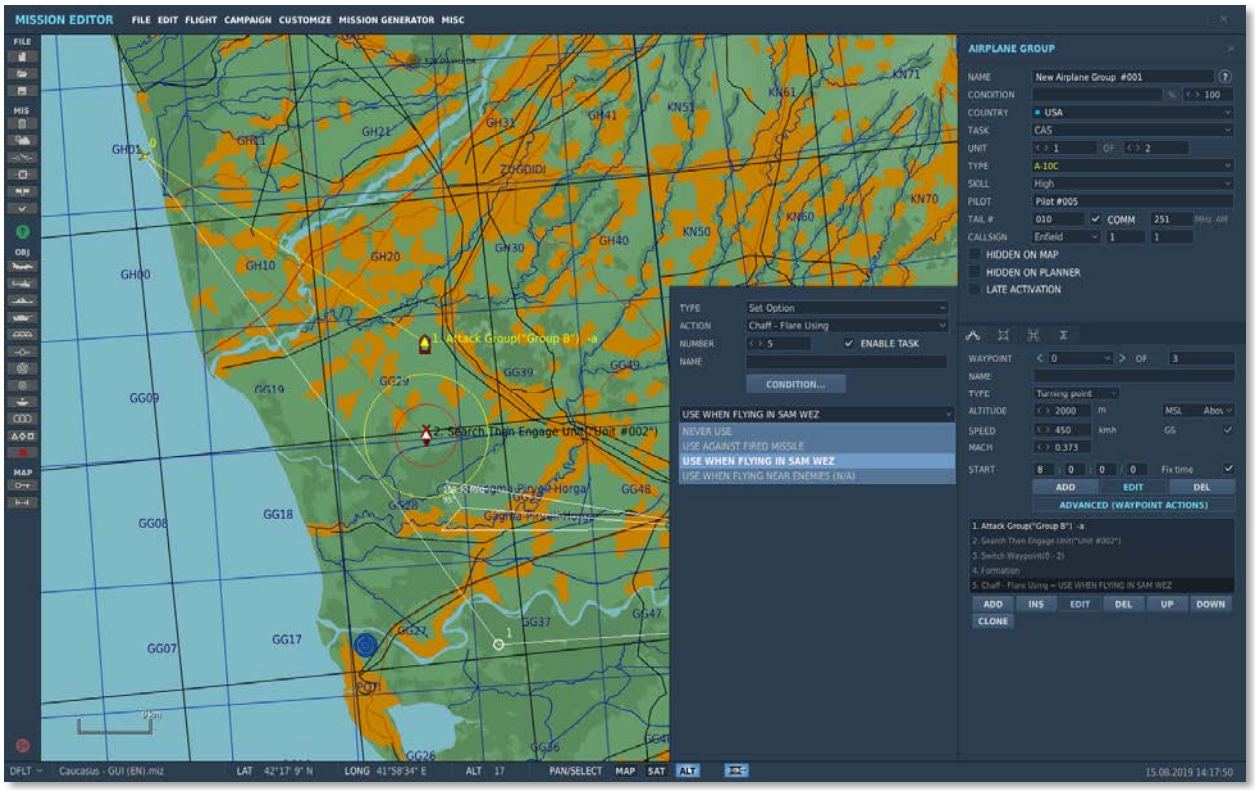
- **Never use** – do not use radar to perform target search or engagement (missile guidance).
- **Use for attack only** – do not use radar to perform target search, but use it if necessary to engage the target (missile guidance). Target search can be performed by other sensors if available.
- **Use for search if required** – allows the group to use radar for target search if necessary (when detection by other means is not successful).
- **Use for continuous search** – maintain radar on continuously.

The default setting is *Use for search if required*.

Keep in mind that radar illumination is a telltale signature and can ruin the element of surprise in combat operations, and as such pilots should ideally minimize radar uptime in covert operations.

Chaff – Flare Using

Set the rules for the release of chaff and/or flares by the group.

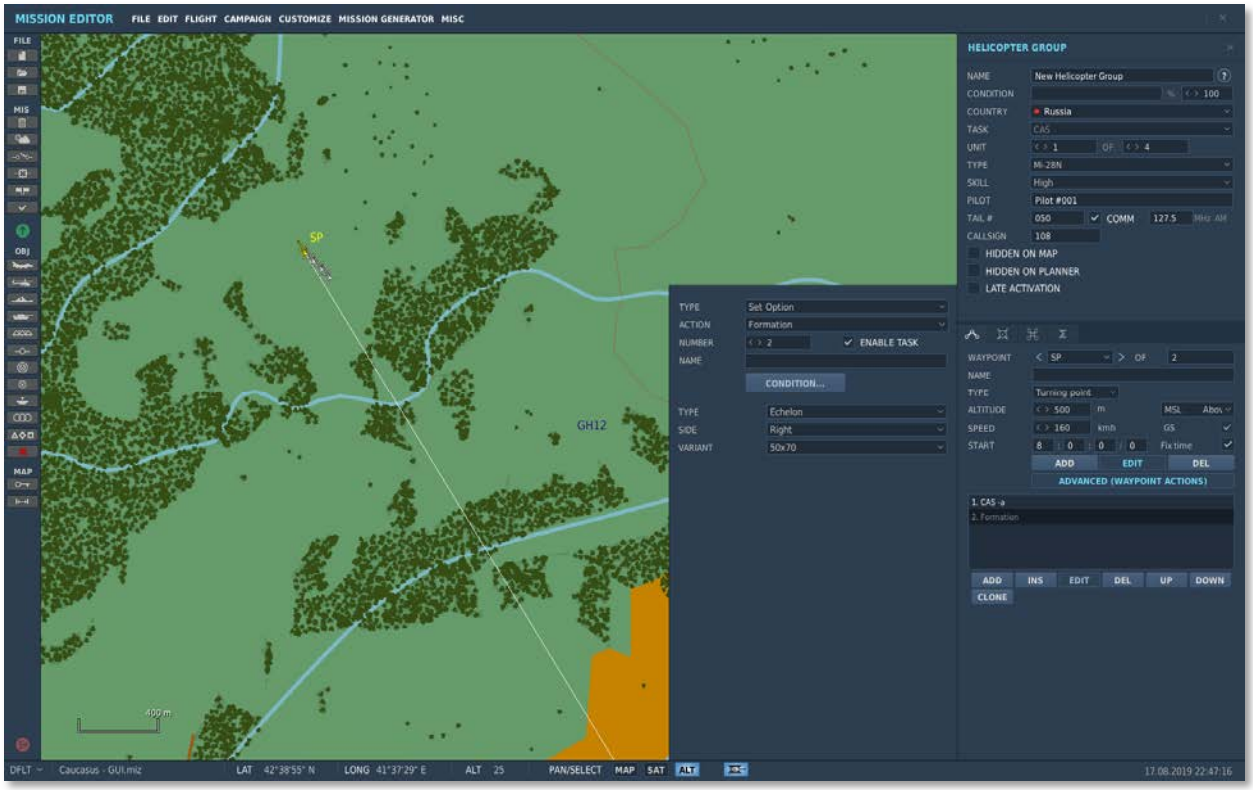


- **Never use** – do not release chaff and/or flares when engaged.
- **Use against fired missiles** – release chaff and/or flares only when incoming missiles have been detected.
- **Use when flying in SAM WEZ** – release chaff and/or flares as a preventative measure when inside a known enemy SAM threat zone or within weapons range of a known enemy aircraft armed with IR-homing missile.
- **Use when flying near enemies (n/a)** – The aircraft group will use countermeasures/decoys over enemy territory.

The default setting is *Use when flying in SAM WEZ*.

Formation

Set the formation for the group to assume upon reaching the waypoint. The following formations are available: Line Abreast, Trail, Wedge, Echelon Right, Echelon Left, Finger Four, Spread Four, WW2: Bomber Element, WW2: Bomber Element Height Separation and WW2: Fighter Vic.



TYPE. Formation type. 10 options are available for aircraft combat formations:

- **Line Abreast**.
- **Trail**.
- **Wedge**.
- **Echelon Right**, the default setting.
- **Echelon Left**.
- **Finger Four**.
- **Spread Four**.
- **WW2**: **Bomber Element**.
- **WW2**: **Bomber Element Height Separation**.
- **WW2**: **Fighter Vic**.

For helicopter combat formations, 4 options are available:

- **Wedge**, the default setting.
- **Front**.
- **Echelon**.
- **Column**.

SIDE. Choice of port/starboard side for asymmetrical formations. Used only for helicopter formations when choosing line abreast/echelon formation.

VARIANT. Choice of formation variant when selecting formation type, differing in size (indicated in meters).



For a more detailed explanation of the formations, including illustration, consult the Flight Manual.

The default setting is Echelon Right.

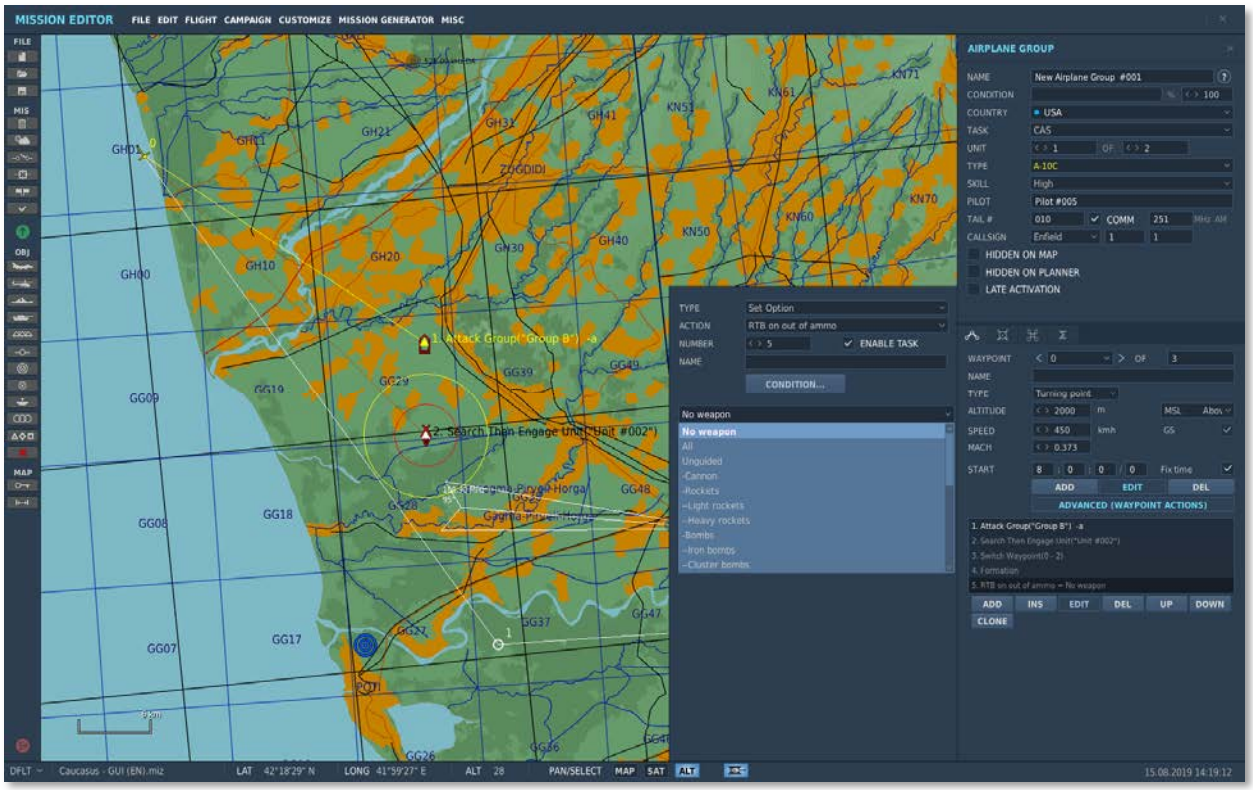
RTB on Bingo Fuel

This is a flag setting to allow/restrict the group to return to base upon reaching Bingo Fuel state (when there is only enough fuel left to return to base.)

The default setting is ON (allow).

RTB on out of ammo

This option allows for the group to return to base when all ammunition has been spent.



The default setting is *No weapon*.

Silence

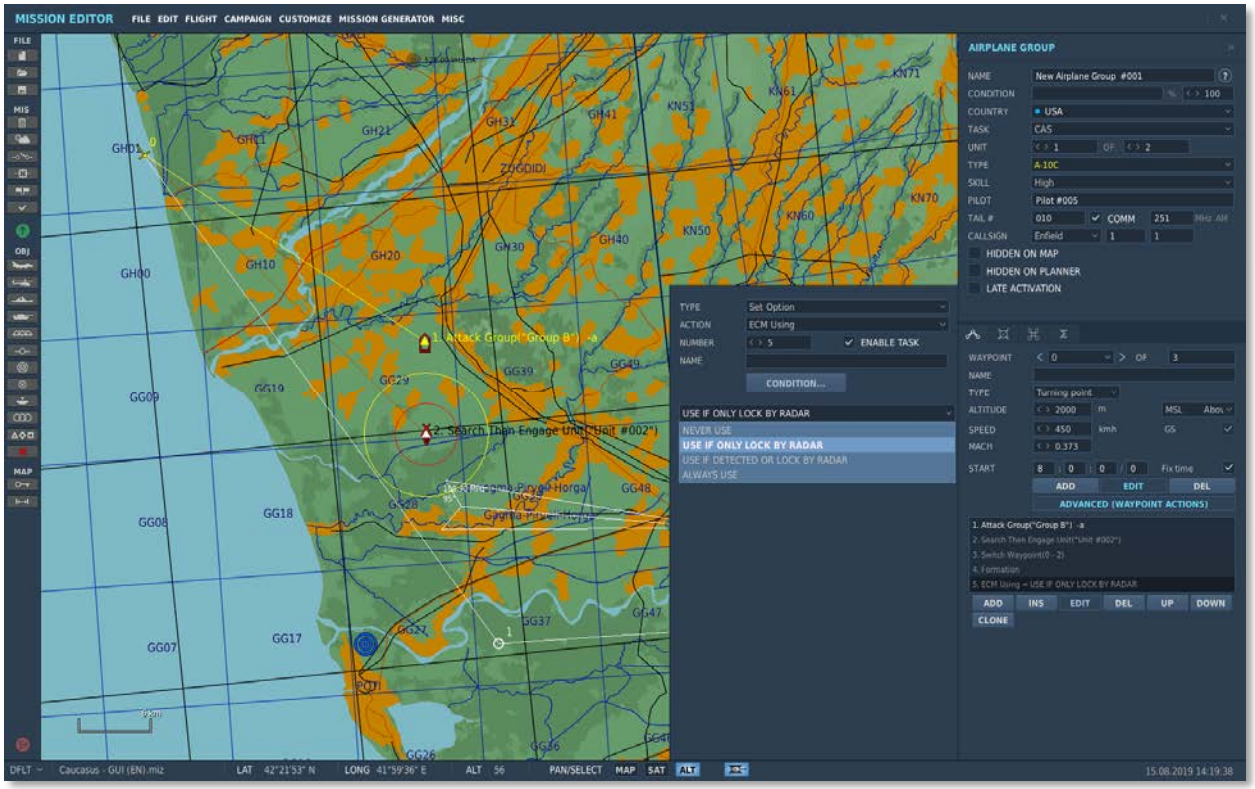
This is a flag setting to force the group to maintain radio silence upon reaching the waypoint.

The default setting is OFF (radio communications are allowed).

If no Options are set in the Actions List, the default settings for all options are assumed.

ECM Using

The "ECM Using" option determines the procedure for using the aircraft's onboard electronic warfare suite during the course of a combat mission.



The default setting is *Use if only lock by radar*.

Restrict Air-to-Air Attack

This option forbids units from attacking airborne targets. Tick the checkbox labelled VALUE to forbid this action.

The default setting is OFF.

Restrict Jettison

This option restricts units from jettisoning external stores. Tick the checkbox labelled VALUE to forbid this action.

The default setting is OFF.

Restrict Afterburner

This option restricts units from using afterburners, reducing IR visibility and improving survivability against heat-seeking missiles. Tick the checkbox labelled VALUE to forbid this action.

The default setting is OFF.

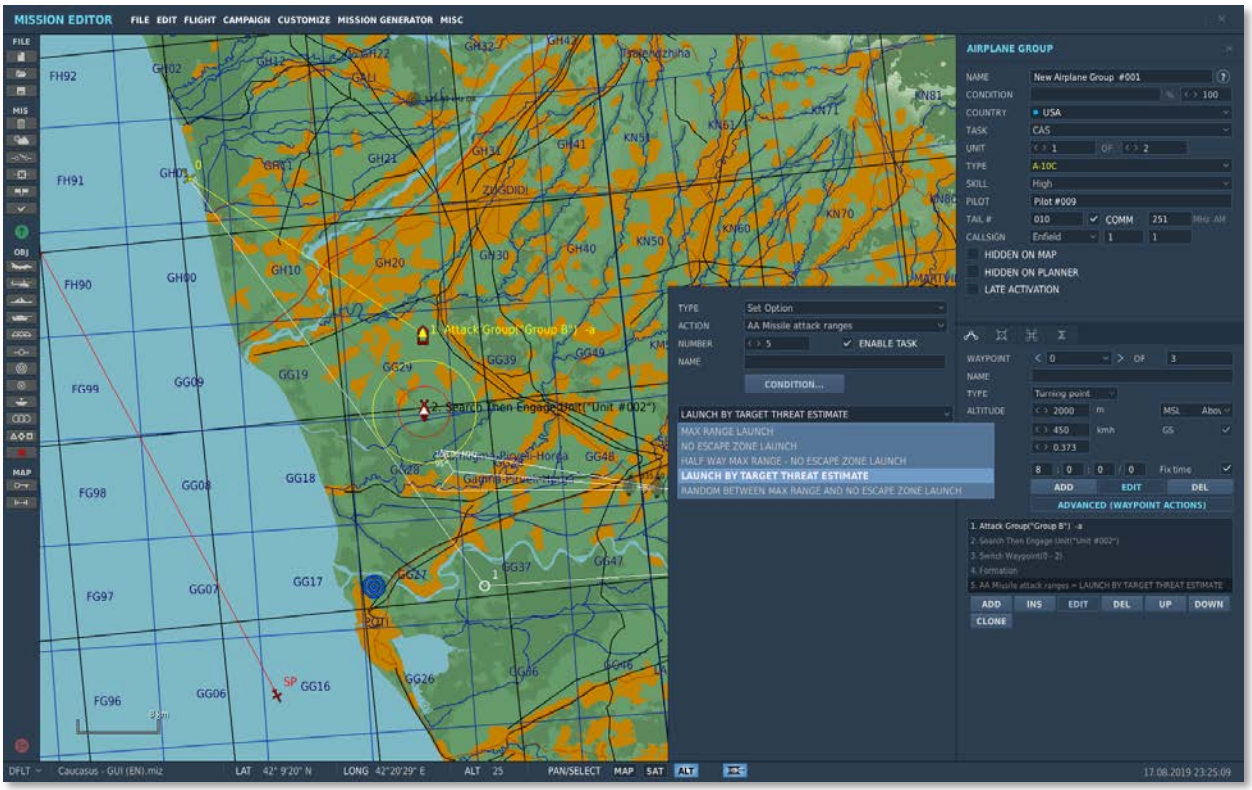
Restrict Air-to-Ground Attack

This option forbids units from attacking ground targets. Tick the checkbox labelled VALUE to forbid this action.

The default setting is OFF.

AA Missile attack ranges

This option determines the distance at which air-to-air missiles may be fired.



- **Max range launch** – Missile launch is permitted at the maximum possible range.
- **No escape zone launch** – Missile launch is permitted only when the target has entered the missile's no escape zone.
- **Half way max range - No escape zone launch** – Missile launch is permitted at a distance between the missile's maximum possible range and it's no escape zone.
- **Launch by target threat estimate** – the default setting. Missile launch according to the threat level of the target: the closest hostile target that is closing in on the unit, with an active radar; either locked on or actively attacking.
- **Random between max range and no escape zone launch** – Missile launch is permitted at a random distance between the missile's maximum possible range and it's no escape zone.

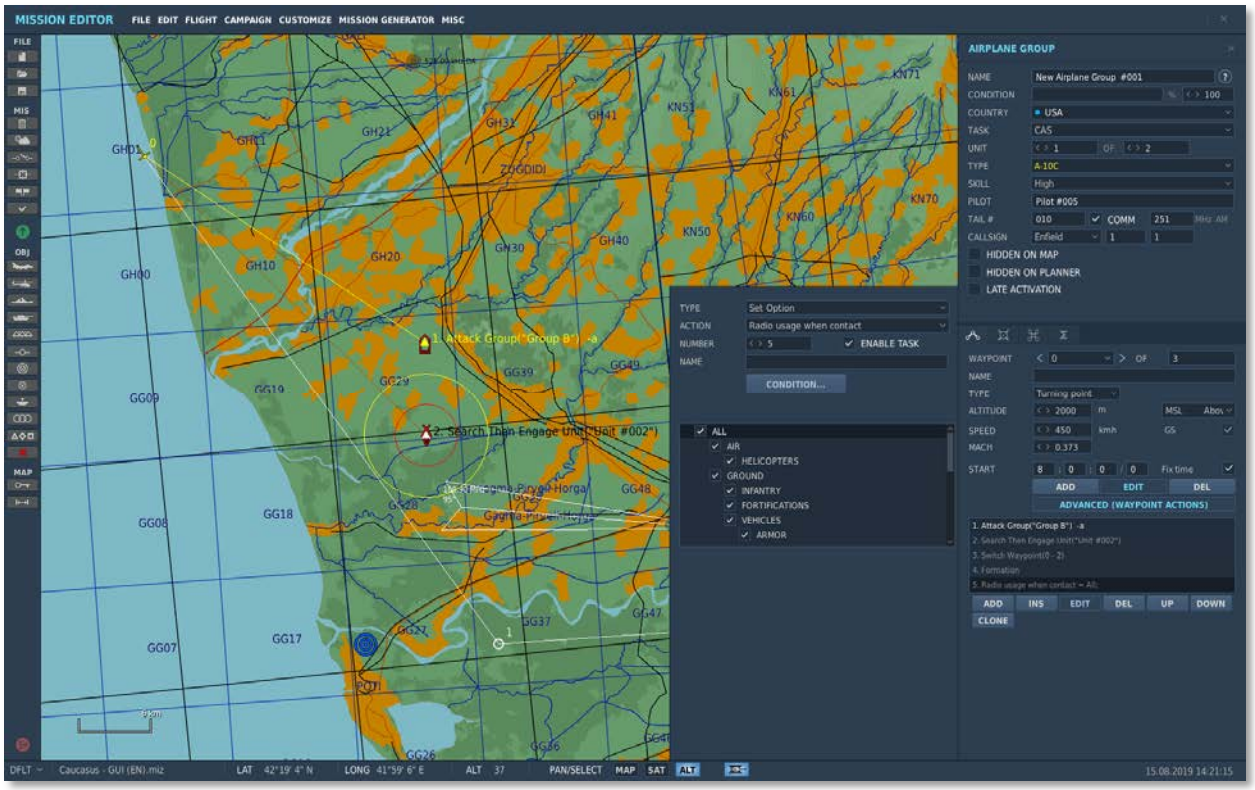
No Report Waypoint Pass

This option forbids the aircraft group from reporting on their passage through a waypoint. Tick the checkbox labelled VALUE to forbid this action.

The default setting is OFF.

Radio usage when contact

The option determines the type of targets for radio communications when the group is in contact with the enemy.



The default setting is ALL targets.

Radio usage when engage

The option determines the type of targets for radio communications when the group is actively engaged in combat with the enemy. The target panel is similar to that of the previous option.

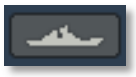
The default setting is ALL targets.

Radio usage when kill target

The option determines the type of targets for radio communications when the group has damaged or destroyed an enemy target. The target panel is similar to that of the previous option.

The default setting is ALL targets.

Place Ship Group



You will use this window to place ship groups in a mission, set their routes, and determine the actions they will take. Unlike aircraft groups, you can place up to 99 units within a single group (not recommended though). When placing ships, you will be automatically restricted to placing them on large bodies of water.

Note, the mission editor supports group copy/paste functionality. To copy a group, select it and press [CTRL+C]. To paste the group with identical properties, position the mouse over the desired point on the map and press [CTRL+V].

The Place ship windows consist of numerous functions, which are described below. We will discuss them from top to bottom.



NAME. In the Name field, type in a unique name for the naval group. If you do not enter one, a default one will be generated. This Name you create will be used when assigning some types of Triggers such as Activate Unit. Always be careful not to assign more than one group with the same Name.

CONDITION. The condition for the random spawning of a group in a mission. For this, LUA expressions used in the Simulator Scripting Engine Documentation are entered into the field. First, the initialization script should be loaded in the trigger panel. The script is executed before the mission is loaded and its result is used to set the conditions for the random group spawn.

COUNTRY. The Country drop down list displays all the countries that have been assigned to either the RED or the BLUE sides when the mission was initially created by use of the CREATE NEW MISSION button. The Country selected will filter the TYPE of ships available.

UNIT. The Unit selection is composed of two fields and they allow you to select how many ships will be part of the naval group (1 to 99). The right field is used to set the total number of ships in the group; to do so, use the left and right arrow buttons. The left field is used to select a ship within the group; to do so, use the left and right arrow keys.

TYPE. Depending on the Country, a list of appropriate ships is listed in this pull-down list.

UNIT NAME. Enter a unique name for each unit within the naval Group. If you do not, a default name will be created automatically. This name is important because it will be used to set Trigger Conditions based on units. For example, if you create a condition based on the destruction of a unit, the unit will be selected from a list using its UNIT NAME.

SKILL. The Skill level allows you to select how competent the aircraft pilot is when flown by the artificial intelligence. This setting can control such factors on how much G the pilot can pull, at what range it will attack, and how accurate its weapon delivery skills are. There are five Skill options for AI aircraft:

- Average
- Good
- High
- Excellent
- Random. This option will randomly select from one of the four options above.

Note that a unit's skill level will affect its detection ranges, reaction times and targeting errors.

HEADING. The course dial can be used to set a unit's standing course when no route has been created to determine the direction of travel.

HIDDEN ON MAP check box. After creating a mission, you may wish to hide certain units so that people playing your mission cannot see them. Check this box to hide the selected naval group from the ME World Map. You can view all hidden groups using the Units List window (discussed later in this document).

HIDDEN ON PLANNER. This option hides aircraft from the mission map in the mission planner. This option can be useful if you need to hide an object from the player.

VISIBLE bef. ACTIVATION. If you choose to have a unit appear according to an Activate Group/Unit trigger, you can choose to have that unit visible in the world or invisible in the world until the unit is activated. If you wish to have the unit visible in the world but inactive until activation, check this box.

GAME MASTER ONLY. This option indicates that the selected group is available only to the game master – the owner of the game session.

LATE ACTIVATION. This option spawns a group of aircraft in the mission after a certain time after reaching any condition or event.

FREQUENCY. Frequency for radio communications in MHz. The AI group will conduct radio communications using the indicated frequency. Example: 127.5 MHz AM means 127.5 MHz in the AM (amplitude modulation) range.

Modal buttons

Five modal buttons control what is displayed on the lower half of the Naval Group Properties placement window. These include: ROUTE, AMMO, TRIGGERED ACTIONS, SUMMARY and SUPPLIERS.



Route Mode



When in Route mode, the data in the lower half of the ship placement window is in regards to waypoint management. A waypoint is an arbitrary point on the map (Lat, Long) that can be chained together to create a route. During the course of a mission, the group will steer from one waypoint to the next along the route line, and at each waypoint you can assign unique actions.

To place a new naval group, you will need to be in the Route mode with a ship Type selected. You may then left click on the map to place the group's starting point, which will be waypoint 1. When you place a group, its waypoint marker (circle with its waypoint number next to it) and route line will be colored appropriately:

- White - Selected unit.
- Red - A Red side unit that is not selected.
- Blue - A Blue side unit that is not selected.

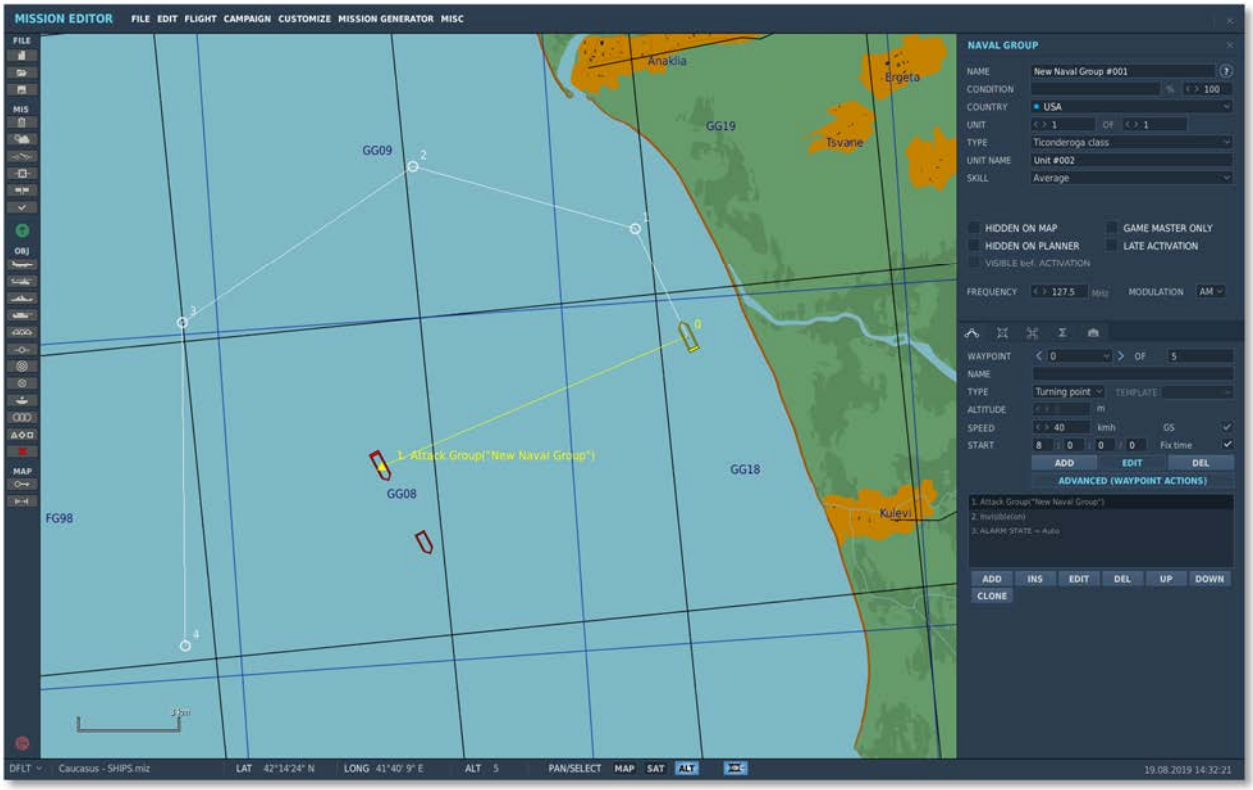
At waypoint one there will be a unit icon instead of a standard waypoint circle. Different icons denote different types of units:

Naval Editor Icons (Russian/Western)

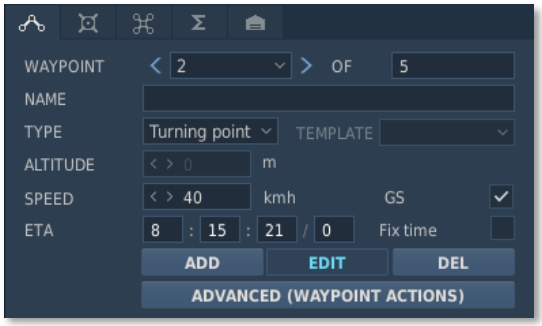
Russian	NATO	Unit Type
		Submarine

		Frigate
		Aircraft carrier
		Heavy cruiser
		Cruiser
		Medium ship/frigate
		Cargo (commercial) ship

To shift the entire group's position on the map, click and drag the group's lead unit to the desired location. To adjust the position of other units within the group, click and drag the desired unit relative to the leader's position.



Below the Modal buttons are the Route controls. For each waypoint of the route, the following options can be set in the waypoint control panel as described above. From top to bottom:



WAYPOINT. The WAYPOINT fields allow you to cycle between waypoints you have created. The left field displays the currently selected waypoint and you can cycle it by pressing the left and right arrow buttons. The field on the right displays the total number of waypoints in the route. The circle and waypoint number of the selected waypoint is colored yellow on the map.

NAME. For each waypoint you can assign a unique name. Enter the name of the waypoint in this field and this name will then appear next to the waypoint on the map.

TYPE. Each waypoint can be assigned a type of action that the group will perform when it reaches the waypoint. These include:

- Turning Point. This is the most common and basic waypoint and the group will simply pass through such a waypoint and proceed to the next.

ALTITUDE. Not adjustable for naval groups.

SPEED. The speed setting will set the speed in kilometers per hour (km/h) that the ship group will have achieved when it reaches the waypoint. To set this, you can either use the left and right arrow buttons or enter it into the field. The group's maximum speed will not exceed the maximum speed of the slowest ship in the group.

GS (GROUND SPEED LOCK CHECKBOX). Selects between manual and automatic entry of the desired speed for the waypoint. Not applicable to a starting waypoint.

If the speed is locked, the group will attempt to maintain the desired speed on the route leg from the preceding waypoint to the current one. See above "Group Route Planning".

START/ETA. (Estimated Time of Arrival). Indicates the desired arrival time for the waypoint (or group start time in the case of the initial waypoint). This is set by the user if the ETA is locked or calculated automatically by the Mission Editor if the ETA is unlocked.

Fix Time (ETA LOCK CHECKBOX). Selects between manual and automatic entry of the desired ETA for the waypoint. The ETA is locked for the initial waypoint (group start time) by default.

When locked, the ETA field can be used to set a delayed group activation in the mission by setting an ETA for the initial waypoint at a time past the mission starting time. For example, if the mission is set to start at 12:00:00/1 and the user wishes to have the group appear in the mission 15 minutes after the mission starts, he can lock the ETA for waypoint-1 at 12:15:00/1. Furthermore, if the user wishes for the group to appear in the mission based on a triggered

event of unknown time, he can set the ETA for the starting time past any likely mission time, such as 100 days later (12:00:00/100). In such a scenario, the group will not appear until triggered by the scripted event or the starting ETA is reached.

If the ETA for the waypoint results in calculated speeds that are too slow or too fast for the aircraft type, the font will become red to indicate an invalid entry and a corresponding error message will appear when the user attempts to close the properties panel or save the mission. See above for more information.

Waypoint Mode Buttons

These three buttons allow you to add, edit and delete route waypoints.

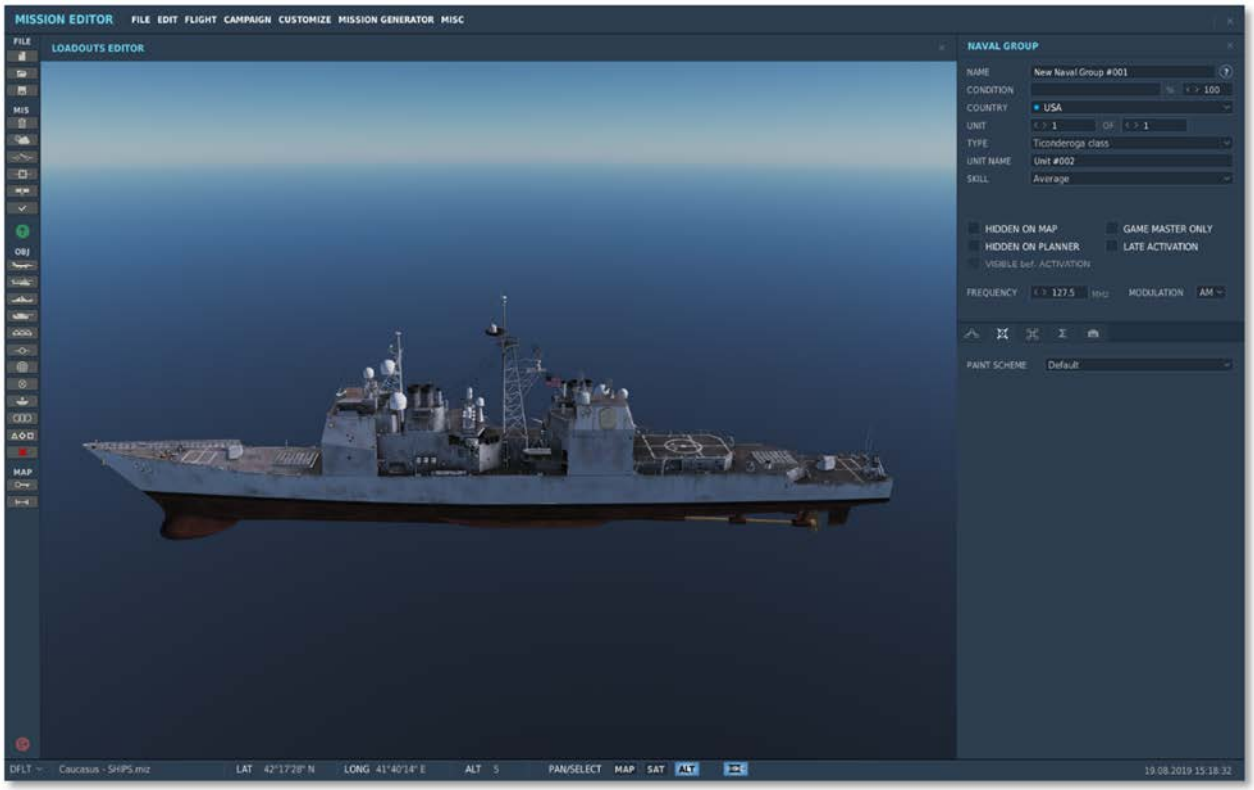


- **ADD** – In ADD mode, each press of the left mouse button over the map adds a new waypoint. This mode is selected on by default when a group properties panel is open.
- **EDIT**– In EDIT mode, existing waypoints can be selected on the map for editing in the waypoint properties panel.
- **DEL** – Pressing this button deletes the selected waypoint.

Ammo Mode



The LOADOUTS EDITOR panel is opened by pressing on the button labelled AMMO and allows the user to select the color scheme of the chosen unit.



Triggered Actions Mode



The Triggered Actions button opens the Triggered Actions Panel. This panel can be used to add or edit group actions analogous to the Advanced Actions Panel, but activated based on trigger rules as scripted in the Triggers Menu using the AI TASK trigger action.

Route Summary Mode



When in Summary mode, you are provided basic mission data in the lower half of the window. This data includes:



START TIME. The mission start time for the group in Hour:Minute:Second/Day.

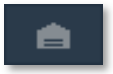
ROUTE TIME. Estimated en-route time for the chosen route when travelling at the given speed. This is indicated in Hour:Minute:Second/Day format.

ROUTE LENGTH. Total distance of the route in kilometers.

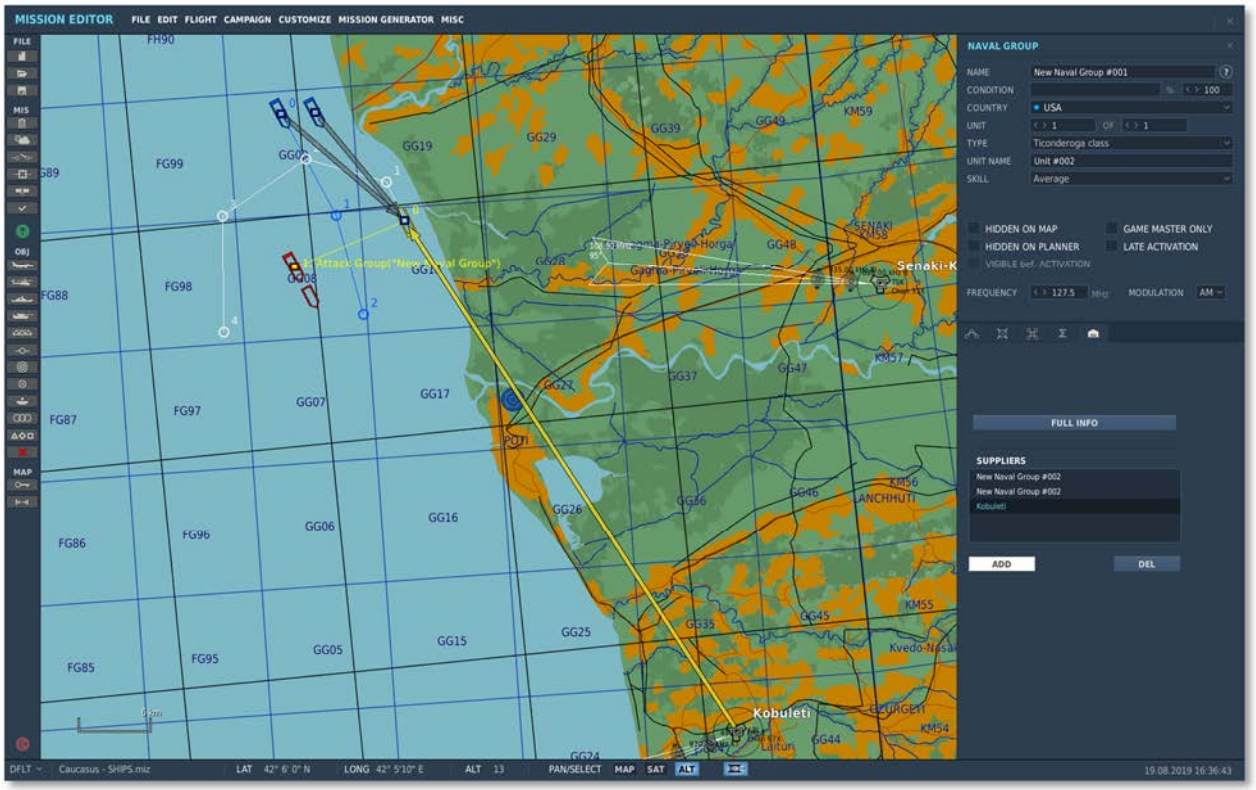
AVERAGE SPEED. The average indicated speed of the route by totaling the assigned speed of each route leg and then dividing by the number of legs.

Note, the mission editor supports group copy/paste functionality. To copy a group, select it and press [CTRL+C]. To paste the group with identical properties, position the mouse over the desired point on the map and press [CTRL+V].

Suppliers



The SUPPLIERS button brings up an information panel with the ammunition suppliers for the chosen ship group.



FULL INFO. Opens the Resource Manager panel.

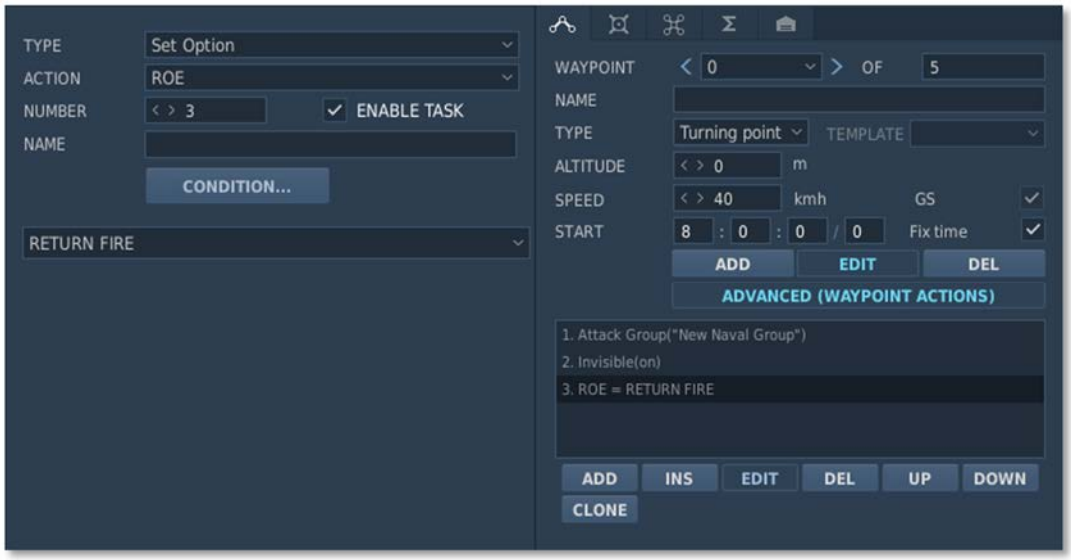
ADD. Adds a warehouse. When clicked, it marks all ship groups and airfields with checkboxes, that when ticked designates these groups as suppliers. This will create arrowed lines that go from these supplier units to the original ship group.

DEL. Removes the selected suppliers from the panel list.

Advanced Actions Mode (naval groups)



Pressing the Advanced tab opens the Advanced Actions Panel, which lists the actions associated with the selected waypoint:



This panel is used to set (add/edit/delete) group actions (Perform Task, Start Enroute Task, Perform Command, Set Option), which are tasked to the AI at this waypoint by the user or generated automatically by the Mission Editor.

Panel Interface Controls

The following buttons are located at the bottom of the Advanced Actions Menu:

- **ADD** – Adds a new action to the list. When pressed, a default No Task action is created, which can be edited in the Action Settings pop-up panel to the left.
- **INS** – Inserts a new action above the currently selected one.
- **EDIT** – Opens the Action Settings pop-up panel to display and edit the settings of the action. The same functionality is duplicated when clicking on any route waypoint on the map.
- **DEL** – Deletes the selected action.
- **UP** – Moves the selected action up the list.
- **DOWN** – Moves the selected action down the list.
- **CLONE** – Clones the selected action down the list.

Action Properties Panel for Naval Groups

The Action Properties Panel for Naval Groups is analogous to the panel for aircraft groups. In other words, this panel is used to set Perform Tasks, Start Enroute Tasks, Perform Commands and Set Options for the group to perform, as well as the starting and ending conditions of the actions. However, the specific types of actions available for naval groups differs from aircraft groups.

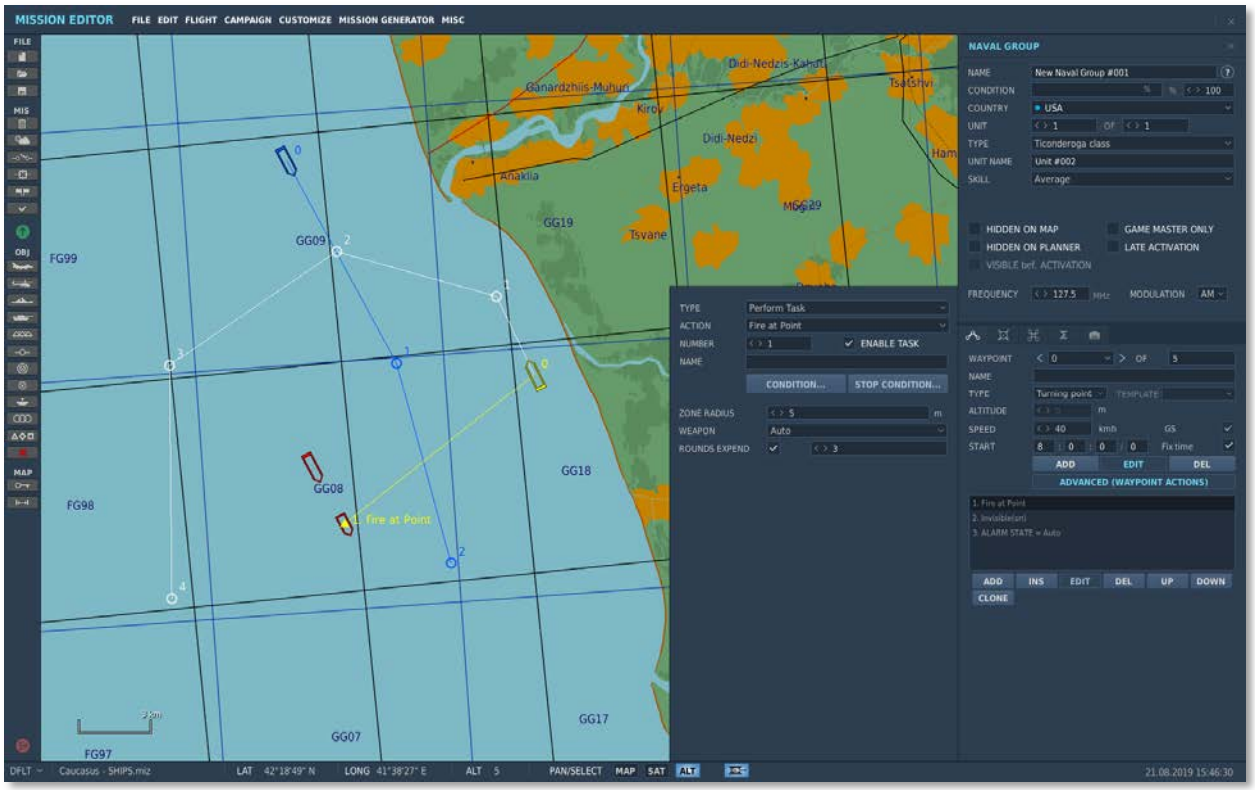
Perform Task

No Task

No action is assigned to the group to perform.

Fire at Point

Fire against a coordinate point on the map as designated by the mission designer.



To perform this action, move the triangular marker that appears near the group when this action is selected, to a desired point on the map.

Target Panel:

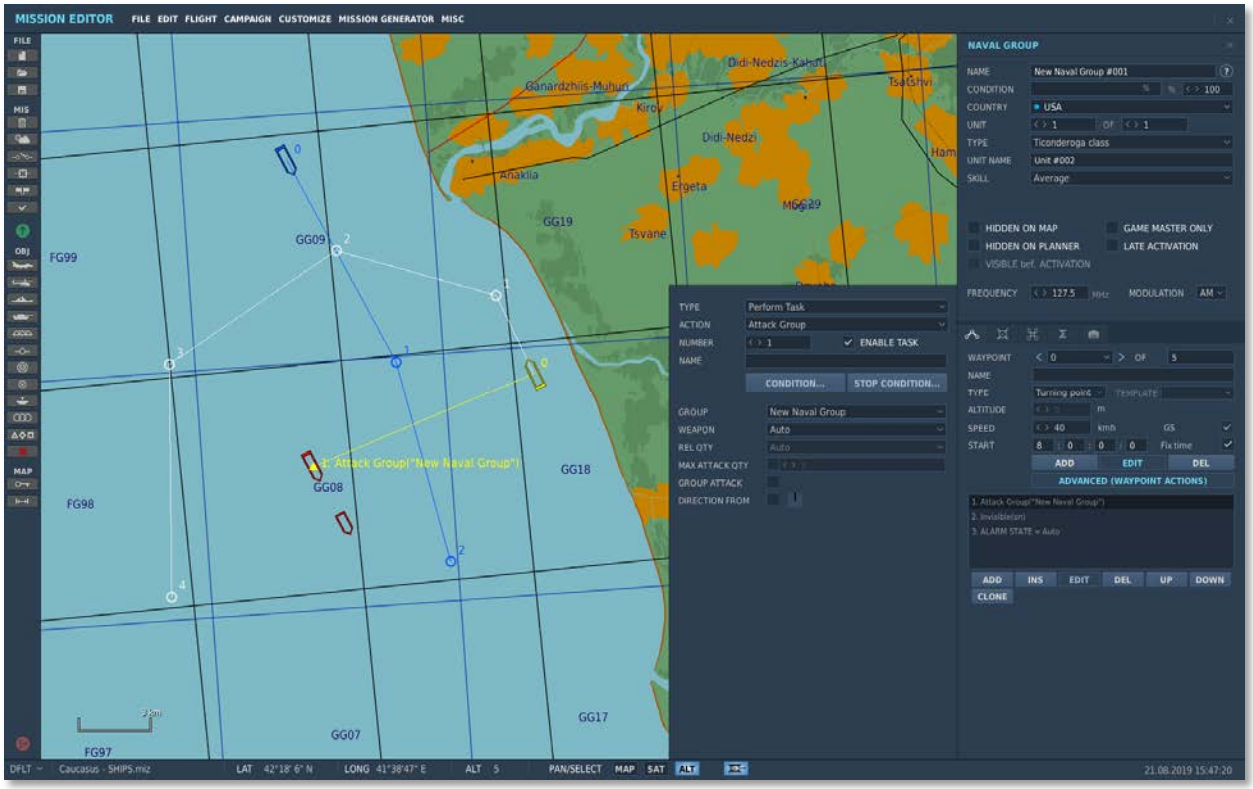
ZONE RADIUS. The value sets the radius of the target area. When firing artillery, the ship will spread rounds within the entire target area. When firing cruise missiles, the ship will target the center coordinate of the target area.

WEAPON. Drop down menu to select the weapon types cleared to use for this action. This menu is dependent on the type of ship and its available types of weaponry.

ROUNDS EXPEND. Setting this flag allows you to set the amount of ammunition consumed.

Attack Group

This action is intended to organize an attack of a specified group of ships.



GROUP. Selecting the target air/ground/ship group from the list.

WEAPON. Drop down menu to select the weapon types cleared to use for this action. This menu is dependent on the type of ship and its available types of weaponry.

REL QTY. The amount of payload cleared to expend in executing this action.

MAX ATTACK QTY. The maximum number of attack runs cleared for this action. The ship's group stops an attack if the target is defeated, even if the specified number of attacks has not yet been performed. If the target is not hit within the specified number of attacks, the ship's group also stops the attack.

GROUP ATTACK. When checked, the entire naval group will participate in the attack of the target. By default, each ship in a naval group is assigned one target.

DIRECTION FROM. Allows you to set a course (azimuth) for the selected ship group for its approach to the target.



The target marked for destruction is the enemy group that is marked with a triangular marker. A line will also be drawn from the waypoint to the target.

Start Enroute Task

Naval groups cannot be assigned any Start Enroute Task actions. Therefore, the only available selection is No Enroute Task.

Perform Command

The following commands can be set as actions upon reaching a waypoint:

No Action

No command is set.

Run Script

Run the Lua script entered in the scratchpad.

Script File

Selects the LUA script file.

Invisible

Set the group to be invisible to all enemy AI units. Enabled and disabled using the ENABLE checkbox.

Immortal

Set the group to be immortal. Enemy AI units will be able to detect the group, but their weapons will have no effect.

Set Option

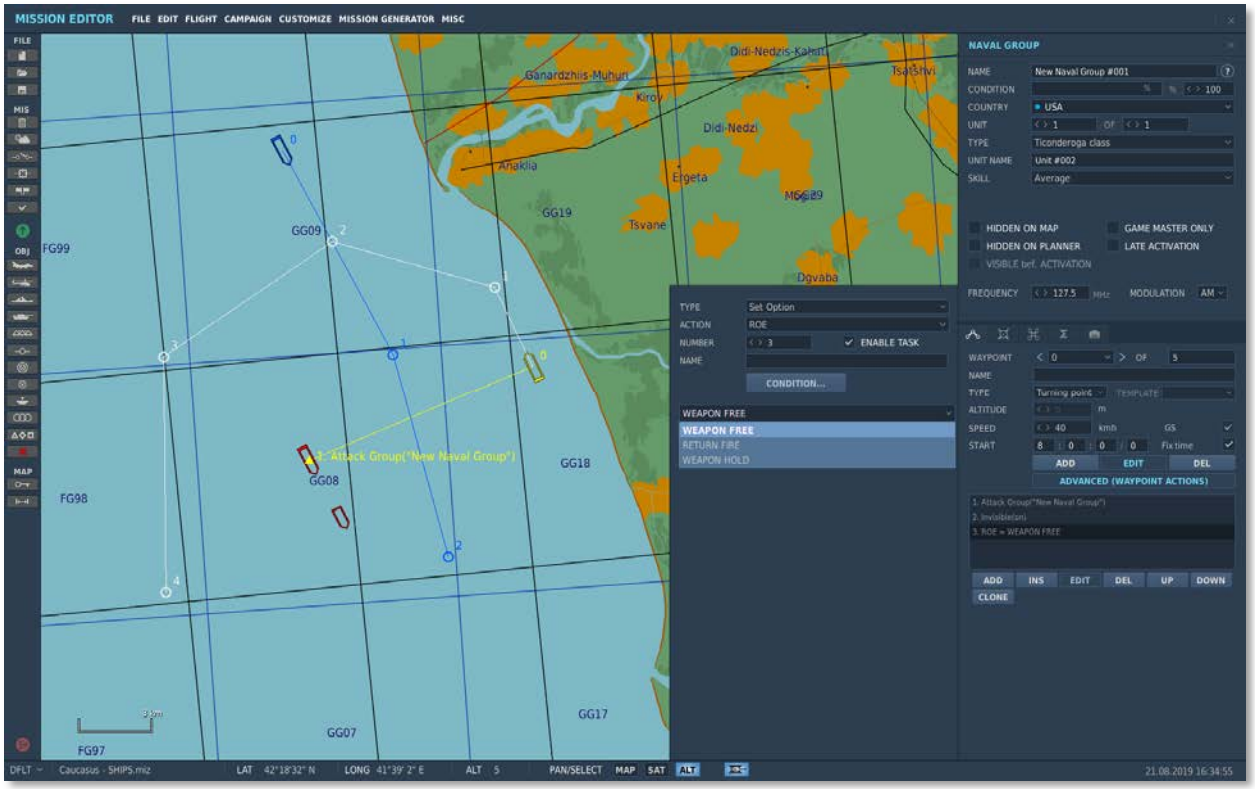
The following Options can be set as actions upon reaching a waypoint:

No Option

No option is set. Default selection.

ROE (Rules of Engagement)

Set fire rules to determine the group's combat behavior when encountering enemy units.

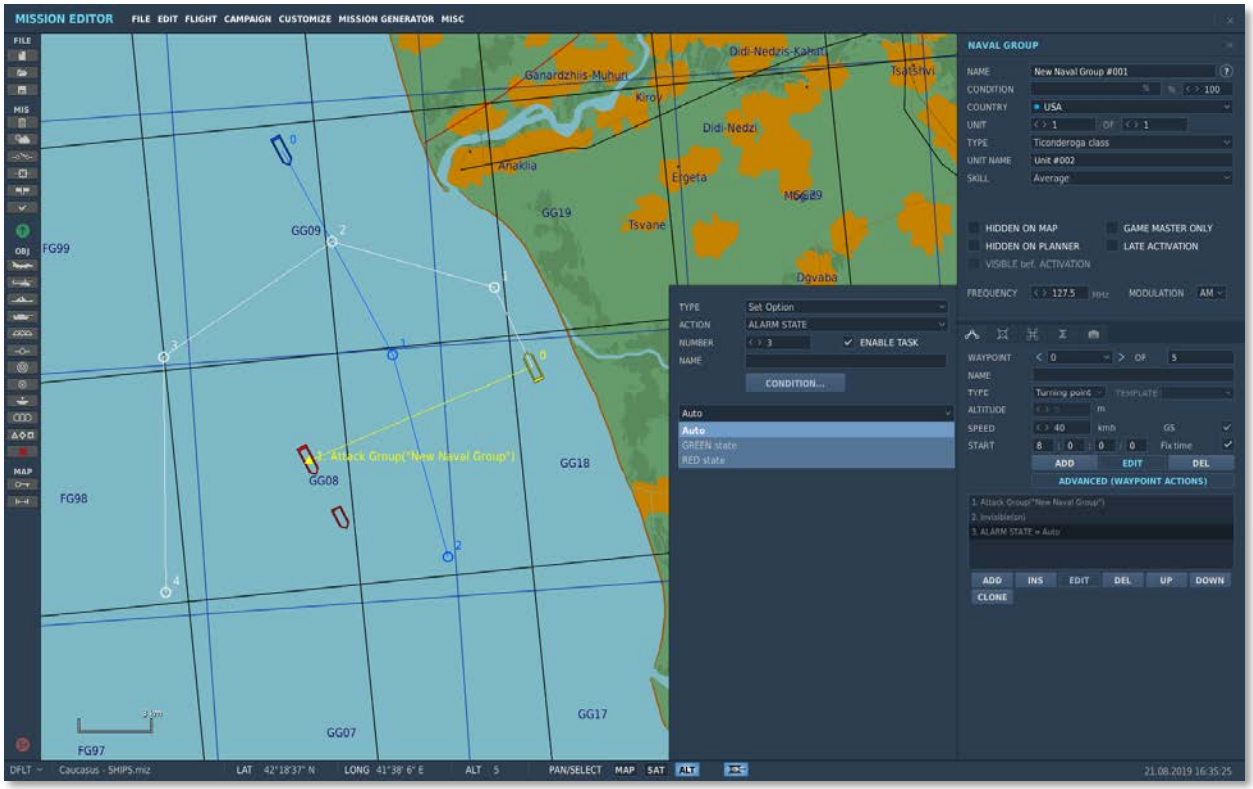


- **WEAPON FREE** – engage any enemy group in contact. Target prioritization is performed automatically by the group based on the situation. The default setting.
- **RETURN FIRE** – return fire only; do not engage first.
- **WEAPONS HOLD** – do not fire under any circumstances.

As mentioned above, all behavior options work regardless of the active task. If the behavior options are not added to the action list, then the default behavior options apply.

Alarm State

This option determines the readiness of the group for combat and has three definitions.



- **Auto** – Auto mode. The default option. The group switches from travel mode to combat alert when an enemy is detected. In the absence of an enemy, the group automatically returns to travel mode.
- **GREEN state** – Group is not ready for combat. Sensors and weapons are stowed.
- **RED state** – The group is combat ready; weapons and sensors are ready to fire and ships are actively searching for targets.

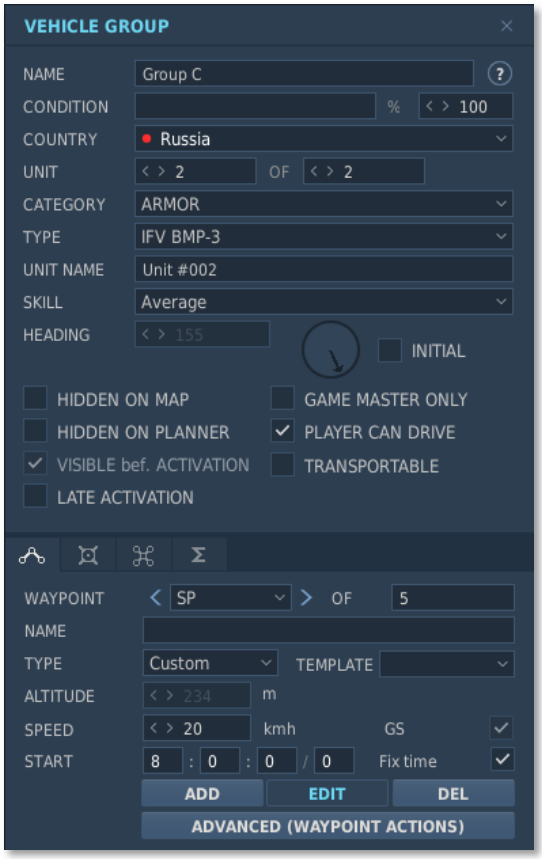
Place Ground Unit



You will use this button to place ground unit groups in the mission, set their routes, and create their actions. When placing a ground group, it is important to remember that the groups can consist of one to 99 vehicles. When placing groups, only the group lead vehicle will be visible on the map when zoomed out. As you zoom in though, the individual units of the group will appear on the map. This is important for accurate unit placement.

Note, the mission editor supports group copy/paste functionality. To copy a group, select it and press [CTRL+C]. To paste the group with identical properties, position the mouse over the desired point on the map and press [CTRL+V].

The ground group placement windows consist of numerous functions which are described below. We will discuss them from top to bottom.



NAME. In the Name field, you may enter a unique name for the vehicle group. If you do not enter one, a default one will be generated. The Name you create will be used when assigning some types of Triggers such as Activate Unit. Always be careful not to assign more than one group with the same Name.

CONDITION. The condition for the random spawning of a group in a mission. For this, LUA expressions used in the Simulator Scripting Engine Documentation are entered into the field. First, the initialization script should be loaded in the trigger panel. The script is executed before the mission is loaded and its result is used to set the conditions for the random group spawn.

COUNTRY. The Country drop down list displays all the countries that have been assigned to either the RED or BLUE sides when the mission was initially created by use of the CREATE NEW MISSION button. The Country selected will filter the TYPE of ground units available.

UNIT. The Unit selection is composed of two fields and they allow you to select how many ground units will be part of the group (1 to 99). The right field is used to set the total number of vehicles in the group; to do so, use the left and right arrow buttons. The left field is used to select a vehicle unit within the group; to do so, use the left and right arrow keys.

***Note**: For Surface to Air Missile (SAM) batteries like SA-10, SA-11, etc., all the needed units of the SAM battery must all be part of the same Vehicle Group.*

***Note**: If you place a ground unit to act as a JTAC, use a unit that is armed and has sensors. The armed HMMWV is a good choice.*

CATEGORY. Drop-down list for selecting the unit category:

- Air Defence.
- Armor.
- Artillery.
- Fortification.
- Infantry.
- Missiles.
- Train.
- Unarmed

TYPE. Depending on the Country selection, a list of appropriate ground units will populate this pull-down list.

UNIT NAME. Enter a unique name for each unit within the Group. If you do not, a default name will be created automatically. This name is important because it will be used to set some Trigger Conditions. For example, if you create a condition based on the destruction of a unit, the unit will be selected from a list using its UNIT NAME.

SKILL. The Skill level allows you to select how competent the unit is controlled by the artificial intelligence. This setting can control such factors as what range it will attack from, and how accurate its weapon delivery skills are. There are five Skill options for AI ground units:

- **Average**
- **Good**
- **High**
- **Excellent**

- **Random**. This option will randomly select from one of the four options above.

***Note**: A unit's skill level will affect its detection ranges, reaction times and targeting errors.*

***Note**: Regardless of skill level, AI aircraft will have great difficulty spotting infantry units unless the infantry unit is moving or firing weapons.*

HEADING. The course dial can be used to set a unit's standing course when no route has been created to determine the direction of travel.

HIDDEN ON MAP check box. After creating a mission, you may wish to hide certain units so that people playing your mission cannot see them. Check this box to hide the selected ground group from the ME World Map. You can view all hidden groups using the Units List window (discussed later in this document).

HIDDEN ON PLANNER. This option hides ground unit from the mission map in the mission planner. This option can be useful if you need to hide an object from the player

VISIBLE before ACTIVATION. If you choose to have a unit appear according to an Activate Group/Unit trigger, you can choose to have that unit visible in the world or invisible in the world until the unit is activated. If you wish to have the unit visible in the world but inactive until activation, check this box.

LATE ACTIVATION. This option spawns a group of units in the mission after a certain time after reaching any condition or event.

GAME MASTER ONLY. This option indicates that the selected group is available only to the game master – the owner of the game session.

PLAYER CAN DRIVE. This option enables the player to drive the selected units by using [RAlt+J].

TRANSPORTABLE. This option allows the selected infantry, transport, armored, and cargo units for transport by helicopter.

Modal Buttons

Four modal buttons control what is displayed on the lower half of the Vehicle Group Properties placement window. These include: ROUTE, AMMO, TRIGGERED ACTIONS, and SUMMARY.



Route Mode



When in Route mode, the data in the lower half of the vehicle placement window is in regards to waypoint management. Waypoints are arbitrary points on the map (Lat / Long) that can be

chained together to create a route. During the course of a mission, the group will drive from one waypoint to the next along the route line and at each waypoint you can assign unique actions (like a formation).

To place a new vehicle group, you will need to be in the Route mode and have a vehicle group selected. You may then left click on the map to place the group's starting point, which will also be waypoint 1. When you place a group, its waypoint marker (circle with its waypoint number next to it) and route line will be colored appropriately:

- White - Selected unit.
- Red - A Red side unit that is not selected.
- Blue - A Blue side unit that is not selected.

At waypoint one will be a unit icon instead of a standard waypoint circle. Different icons denote different types of units:

Ground Unit ME Icons (Russian/Western)

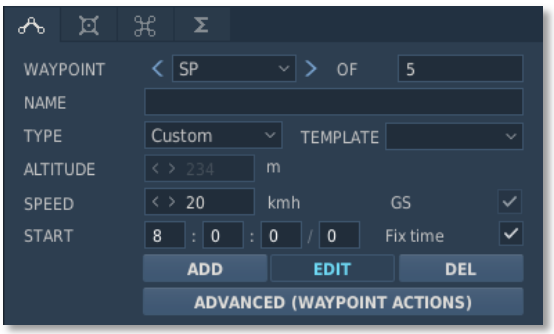
Russian	NATO	Unit Type
		Tank
		Infantry Fighting Vehicle (IFV)
		Combat reconnaissance vehicle
		Armored Personnel Carrier (APC)
		Transport
		Civilian vehicle
		Engineers
		Self-Propelled Gun (SPG)
		Multiple Launch Rocket System (MLRS)
		MLRS Medium
		MLRS Heavy
		Anti-tank Guided Missile (ATGM) tracked
		Anti-tank Guided Missile (ATGM) wheeled

		Forward Air Controller (FAC) vehicle
		Infantry
		Man-Portable Air Defense (MANPAD) SAM system
		Stinger SAM
		Anti-Aircraft Artillery (AAA)
		Self-propelled AAA
		Radar-guided AAA
		Wheeled AAA
		Tracked AAA
		SAM short range, wheeled
		SAM short range, tracked
		SAM short range, wheeled, radar-guided
		SAM short range, tracked, radar-guided
		SAM medium range, tracked
		SAM medium range, wheeled
		SAM medium range, tracked, radar-guided
		SAM long range, tracked, radar-guided
		SAM long range, wheeled
		Avenger SAM system
		M6 Linebacker SAM system
		Chaparral SAM system

		M-163 Vulcan AAA system
		Mobile radar
		Radio-navigation beacon
		Stationary radar
		Air controller
		Air defense direction center
		Battle command center
		Forward Air Controller (FAC)
		Command center
		Bunker
		Checkpoint
		Storage
		Fuel depot
		Structure (building)

To shift the entire group's position on the map, click and drag the group's lead unit to the desired location. To adjust the position of other units within the group, click and drag the desired unit relative to the leader's position.

Below the Modal buttons are the Route controls. From top to bottom:

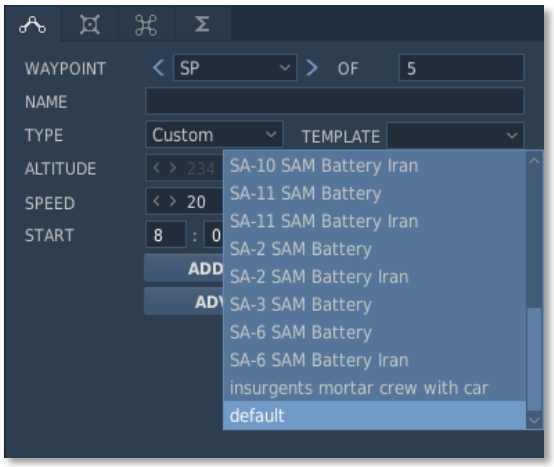


WAYPOINT. The WAYPOINT fields allow you to cycle between waypoints you have created. The left field displays the currently selected waypoint and you can cycle it by pressing the left and right arrow buttons. The field on the right displays the total number of waypoints in the route. The circle and waypoint number of the selected waypoint is colored yellow on the map.

NAME. For each waypoint you can assign a unique name. Enter the name of the waypoint in this field and this name will then appear next to the waypoint on the map.

TYPE. Each waypoint can be assigned a type of action that the group will perform when it reaches the waypoint. These include:

- **Offroad**. The vehicle group will drive to the next waypoint one in front of the other in a line.
- **On road**. When set to On Road, the waypoint will snap to the nearest road. When the next waypoint is also set to On Road, the editor will automatically plot a path along the road network between the two points. When placing an On Road waypoint, avoid doing so at a road intersection. If placing multiple groups on the same road, try to keep them at least 500 meters apart to avoid route conflicts.
- **Line Abreast**. The group vehicles will assume a line abreast formation.
- **Cone**. The group vehicles will assume a cone formation.
- **Vee**. The group vehicles will assume a vee formation.
- **Diamond**. The group vehicles will assume a diamond formation.
- **Echelon Left**. The group vehicles will assume an echelon left formation.
- **Echelon Right**. The group vehicles will assume an echelon right formation.
- **Custom**. Select a custom formation for the group using the template drop-down menu. Group templates can be created using the Templates Menu of the mission editor.



ALTITUDE. Not adjustable for vehicle groups. This field will indicate the altitude of the waypoint.

SPEED. The speed setting sets the speed in kilometers per hour (km/h) that the vehicle will drive to reach its next waypoint. To set this, you can either use the left and right arrow buttons or type it into the field.

GS (GROUND SPEED LOCK CHECKBOX). Selects between manual and automatic entry of the desired speed for the waypoint. Not applicable to a starting waypoint.

If the speed is locked, the group will attempt to maintain the desired speed on the route leg from the preceding waypoint to the current one.

START/ETA. (Estimated Time of Arrival). Indicates the desired arrival time for the waypoint (or group start time in the case of the initial waypoint). This is set by the user if the ETA is locked or calculated automatically by the Mission Editor if the ETA is unlocked.

Fix Time (ETA LOCK CHECKBOX). Selects between manual and automatic entry of the desired ETA for the waypoint. The ETA is locked for the initial waypoint (group start time) by default.

When locked, the ETA field can be used to set a delayed group activation in the mission by setting an ETA for the initial waypoint at a time past the mission starting time. For example, if the mission is set to start at 12:00:00/1 and the user wishes to have the group appear in the mission 15 minutes after the mission starts, he can lock the ETA for waypoint-1 at 12:15:00/1. Furthermore, if the user wishes for the group to appear in the mission based on a triggered event of unknown time, he can set the ETA for the starting time past any likely mission time, such as 100 days later (12:00:00/100). In such a scenario, the group will not appear until triggered by the scripted event or the starting ETA is reached.

If the ETA for the waypoint results in calculated speeds that are too slow or too fast for the aircraft type, the font will become red to indicate an invalid entry and a corresponding error message will appear when the user attempts to close the properties panel or save the mission. See above for more information.

Waypoint Mode Buttons

These three buttons allow you to add, edit and delete route waypoints.



- **ADD** – In ADD mode, each press of the left mouse button over the map adds a new waypoint. This mode is selected on by default when a Group Properties panel is open.
- **EDIT** – In EDIT mode, existing waypoints can be selected on the map for editing in the waypoint properties panel.
- **DEL** – Pressing this button deletes the selected waypoint.

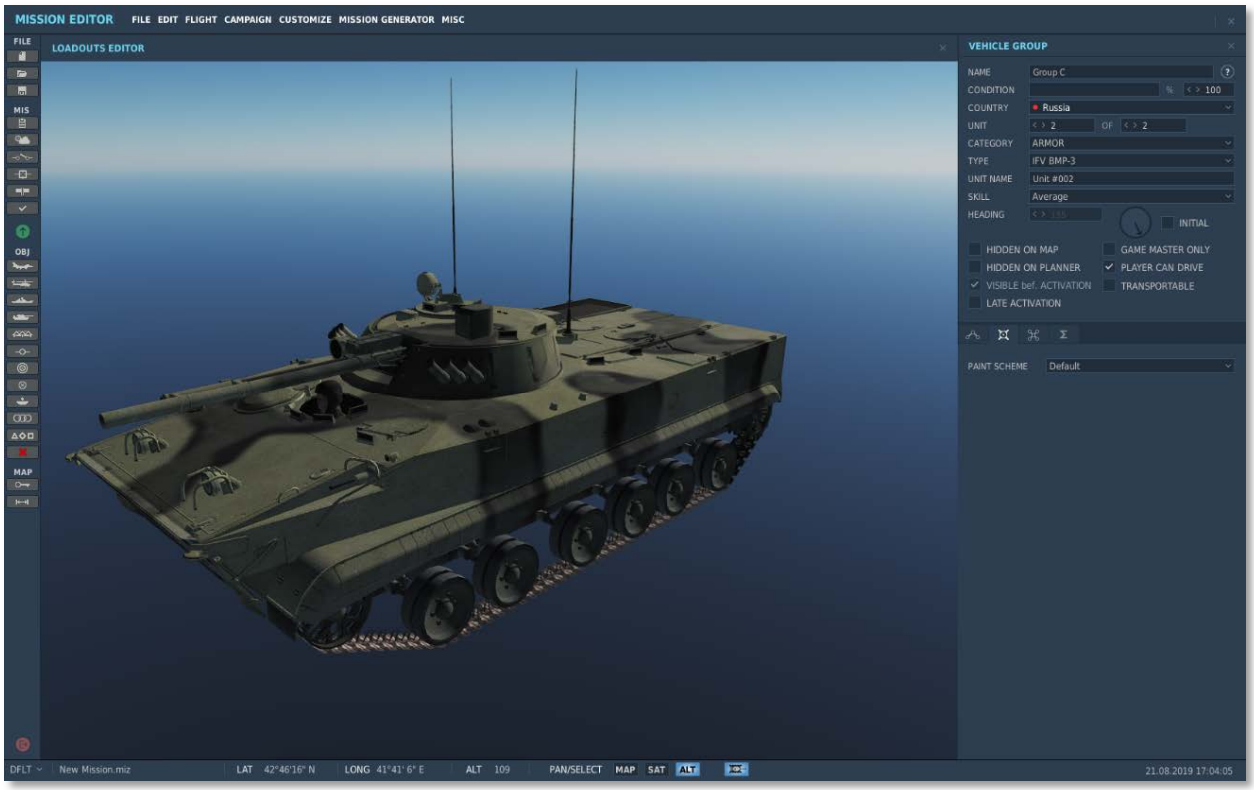
When planning engagements, consider some of the following: vehicle weapon fire is less accurate when on the move. Each vehicle has separate armor values depending on facing (front, sides, rear and top). As such, you generally want to keep units facing towards the enemy where their armor is the thickest. You may also want to have as many barrels pointing at the enemy as

possible, so formations like Line Abreast or Echelon Left/Right may be most effective. It's also helpful to do a careful study of the terrain when planning the mission. Consider the line of sight between friendly and enemy units to make sure vehicles can be detected (or not) in accordance with your plans.

Ammo Mode



The LOADOUTS EDITOR panel is opened by pressing on the button labelled AMMO and allows the user to select the color scheme of the chosen unit.



Triggered Actions Mode

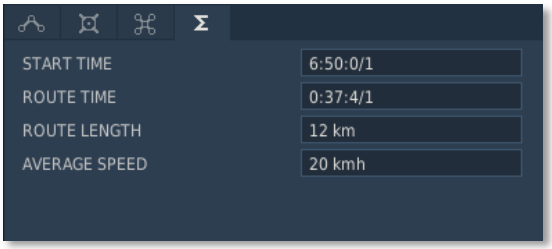


The Triggered Actions button opens the Triggered Actions Panel. This panel can be used to add or edit group actions analogous to the Advanced Actions Panel, but activated based on trigger rules as scripted in the Triggers Menu using the AI TASK trigger action.

Route Summary Mode



When in Summary mode, you are provided basic mission data in the lower half of the window. This data includes:



START TIME. The mission start time for the group in Hour:Minute:Second/Day.

ROUTE TIME. Calculated travel time while maintaining a given speed. This is indicated in Hour:Minute:Second/Day format.

ROUTE LENGTH. Total distance of the route in kilometers.

AVERAGE SPEED. Average speed while travelling the given route.

Notes regarding airbase and FARP ground vehicle support equipment

At Forward Arming and Refueling Points (FARP)s, ground equipment resources are tracked to determine the level of support that can be provided to the player. This can include electrical power, radio communications, fuel, and weapons for both AI helicopters and the player. For more details on contacting ground personnel, see the Radio Communications section in the Flight Manual.

The ground equipment units must be placed within a 150 meters radius from the center of the FARP. The units required vary between eastern and western forces:

Table of objects/units at the airfield/FARP

Service	Required objects (Eastern forces)	Required objects (Western forces)	Conditions
Airfield, Air Control	Tower	Tower	Not destroyed
Airfield, Ground Crew, rearming, refueling, electrical power, aircraft repair	Not required	Not required	Repairs take 3 minutes after rotors (supporting propellers) stop
FARP, Air Control	SKP-11, FARP Command Post	M1025 HMMWV, FARP Command Post	150 meters radius from the center of the FARP
FARP, Ground Crew, rearming	GAZ-3308, GAZ-66, KAMAZ-43101, KrAZ-6322, Ural-375 KUNG, Ural-375, Ural-4320-09-31, Ural-4320T, FARP Ammo Storage	M818, FARP Ammo Storage	150 meters radius from the center of the FARP
FARP, Ground Crew, refueling	ATMZ-5, ATZ-10, FARP Fuel Depot	M978 HEMTT, FARP Fuel Depot	150 meters radius from the center of the FARP
FARP, Ground Crew, electrical power	GPU APA-5D, GPU APA-80	M818	150 meters radius from the center of the FARP

FARP, night lighting system	SKP-11	M1025 HMMWV	150 meters radius from the center of the FARP
FARP, aircraft repair	UAZ-469, Ural-4320-09-31, Ural-4320T, ZiL-131 KUNG, KAMAZ-43101, FARP Tent	M818, FARP Tent	150 meters radius from the center of the FARP. Repairs take 3 minutes after rotors (supporting propellers) stop

The vehicles must be within a radius of 150 meters from the center of the FARP to provide appropriate service.

For example, if you want to provide on FARP radio communication with the CP (command post) and electrical power, you must supply at least two vehicles – SKP-11 and one of the APAs for the red side or the HMMVW M1025 and M818 for the blue side.

If any of the above units are absent or destroyed, the assigned resource will be unavailable. In case a FARP is attacked and all units are destroyed, you may wish to set up a trigger to move new units within 150 meters of the destroyed base to provide support functions.

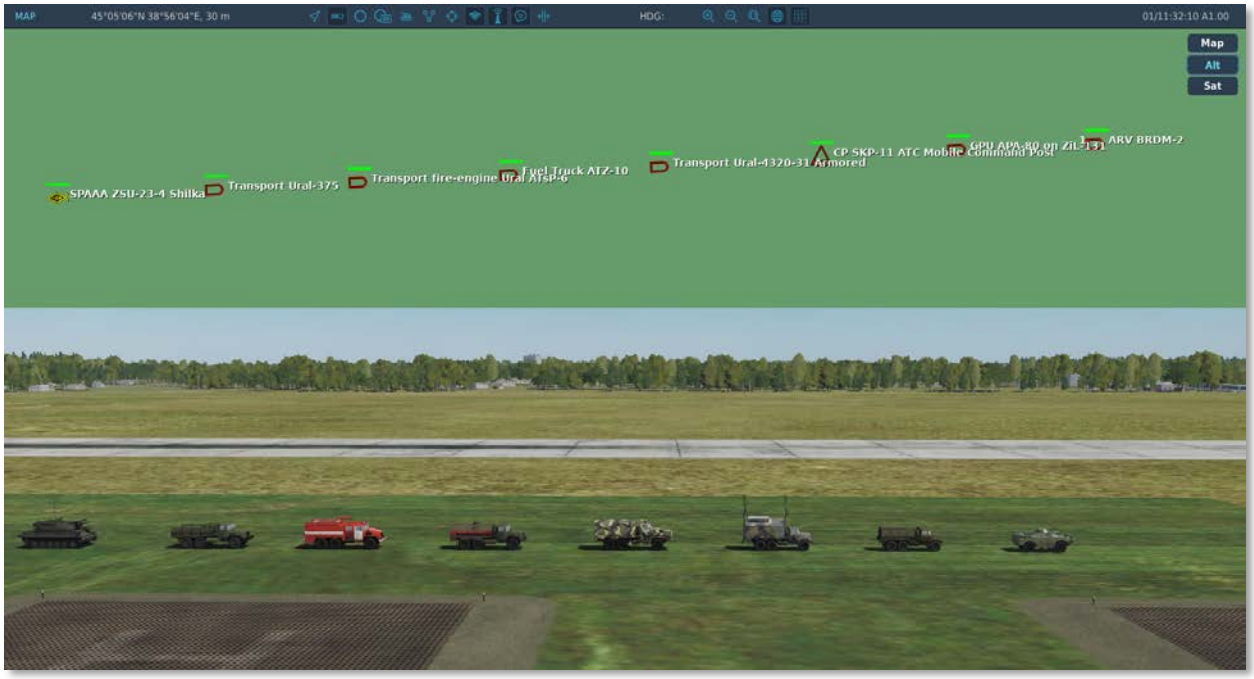
Service vehicles shall be located within a radius of 150 meters from the center of the FARP. The size of the FARP shall be approximately 200x200 meters.

The rearm, refuel, and electric power at the airfields do not depend on vehicles. Again though, the radio communications to request these depend on the tower (the tower must not be destroyed). If the tower is destroyed, the communications can be restored with either the SKP-11 for the red side or the HMMVW M1025 for the blue side. To prepare for such an event, you can set the mission up so that if the tower is destroyed a mobile command vehicle will be dispatched to the airbase.

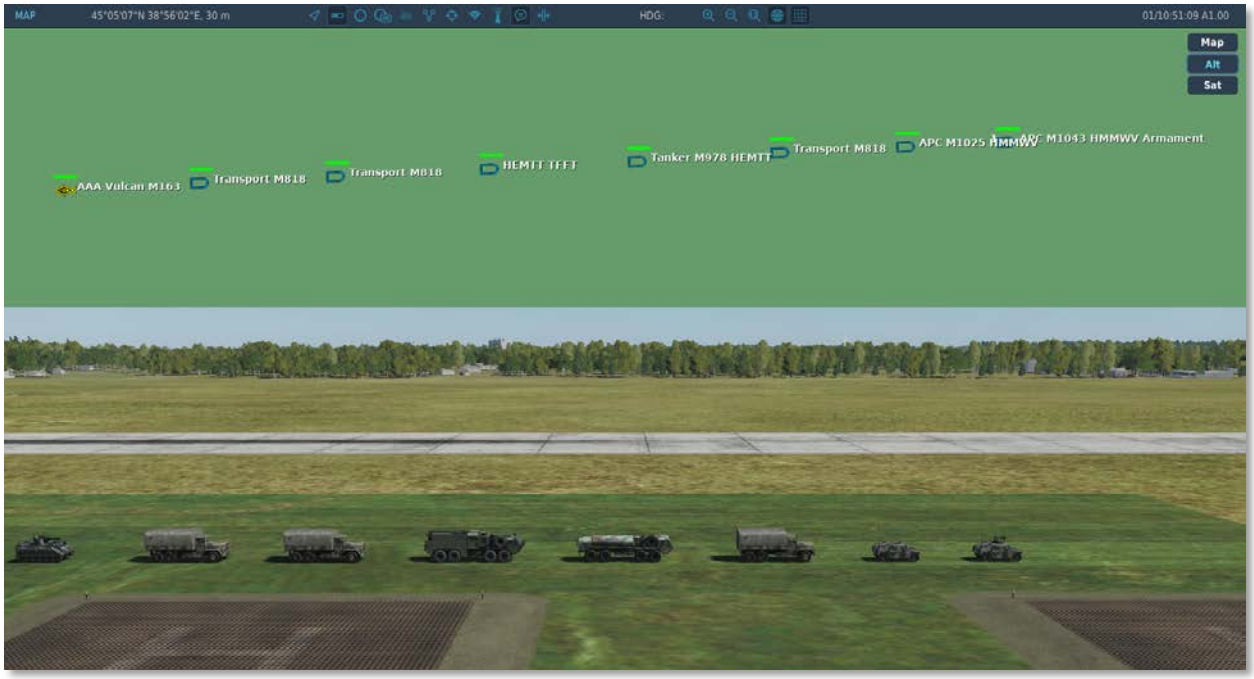
Repair of the player's helicopter is carried out in the airfield or FARP within 3 minutes after the rotors (supporting propellers) stop.

Below is one of the many possible options for symmetrical service convoys for the red and blue coalitions for the FARP. The units in the columns perform all the functions as shown in the table above, plus column security is added.

Service Column (Red Coalition):



Service column (blue coalition):



Ammo supply for ground units

Upon exhausting all their ammunition, ground units, such as air defense systems, missile defense systems, self-propelled guns, infantry fighting vehicles and tanks, may all have their ammunition stores be replenished using supply vehicles. Supply vehicles are counted as unlimited warehouses. They can be included in the composition of both a static group of units (SAM positions) and mobile unit groups (self-propelled gun convoy or armored vehicles). A group can have several supply vehicles, and if a coalition of a group has different types of these machines, then the creator of the mission can diversify the composition of the group for added realism. One supply vehicle replenishes the ammunition of all units located within a radius of 200 meters from it. If the unit is outside this 200 meter radius, its ammunition will not be replenished. The creator of the mission in the editor should take this into account when forming diverse positions of air defense groups and convoys of armored vehicles, especially if they are large in length and do not fit into a radius of 200 meters. Ammunition replenishment time depends on the type of unit.

Supply vehicles for ground forces

Supply vehicle (warehouse)	Country	The ammo supply condition
GAZ-3308	Russia	Within a radius of 200 meters of the supply vehicle
GAZ-66	Russia	Within a radius of 200 meters of the supply vehicle
KAMAZ-43101	Russia	Within a radius of 200 meters of the supply vehicle
KrAZ-6322	Ukraine	Within a radius of 200 meters of the supply vehicle
M818	USA	Within a radius of 200 meters of the supply vehicle
Ural-375 KUNG	Russia	Within a radius of 200 meters of the supply vehicle
Ural-375	Russia	Within a radius of 200 meters of the supply vehicle
Ural-4320-09-31	Russia	Within a radius of 200 meters of the supply vehicle
Ural-4320T	Russia	Within a radius of 200 meters of the supply vehicle

The following are examples of the location of the supply vehicle in mobile vehicle formations and fixed unit groups (examples are not a limitation on the mission creator's intent):

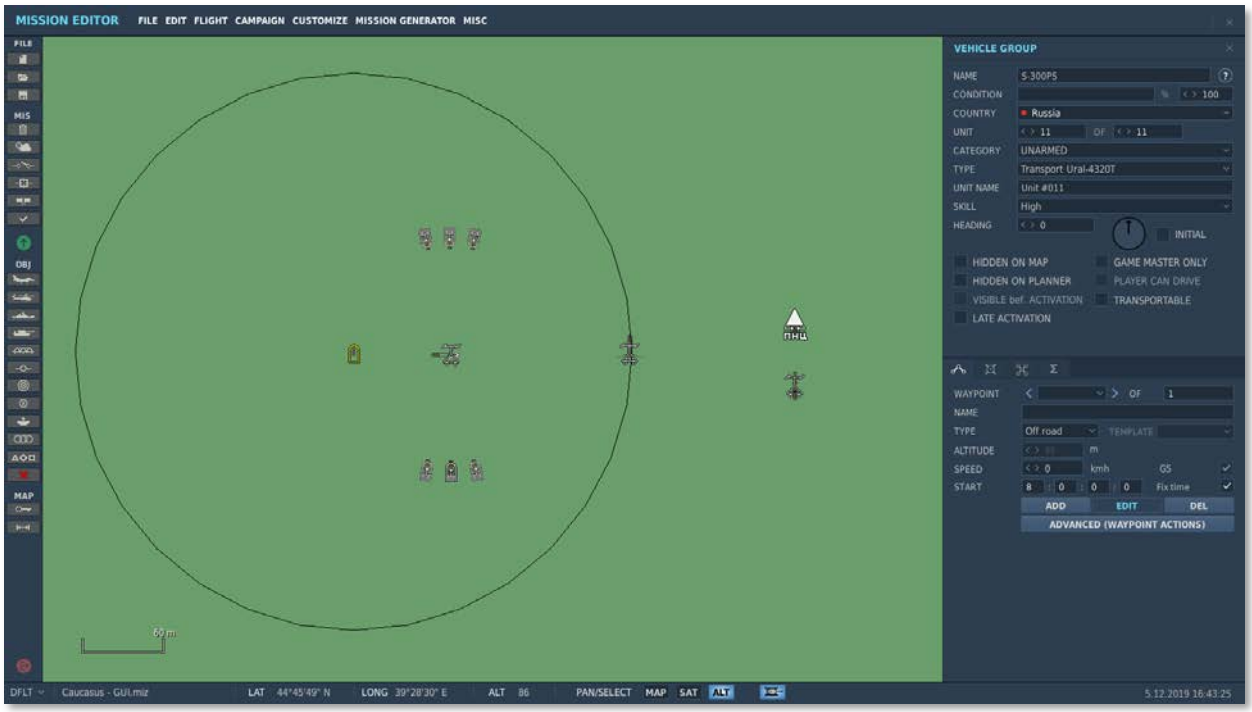
Example 1. Supply vehicle in a column of self-propelled guns (one for 4 self-propelled guns).



Example 2. Supply vehicle at a Patriot SAM site (one for each launcher).



Example 3. Supply vehicle at a S-300PS SAM site (one for all launchers).



Replenishment time consists of the time required to move the launchers or ammunition to a given unit, replace them (replenish them), and connect them, such as on the chassis of Patriot, S-300PS, Buk, Hawk SAMs and other units according to their type of weapon.

Supply ammo for SAM

SAM	Armament	Supply ammo time from a supply vehicle (warehouse), sec.
SA-19 Tunguska 2S6M	Cannons 2x30мм	2400
	Missiles	960
SA-6 Kub LN 2P25	Missiles	1800
SA-3 S-125 LN 5P73	Missiles	600
SA-10 S-300PS LN 5P85C(D)	Missiles	7200
SA-8 Osa 9A33	Missiles	1800
SA-9 Strela-1 9P31	Missiles	1200

SA-13 Strela-10M3 9A35M3	Missiles	960
SA-11 Buk LN 9A310M1	Missiles	780
SA-15 Tor 9A331	Missiles	1080
Patriot LN M901	Missiles	3600
SPAAA Gepard-1A2	Cannons 2x35мм	1800
Hawk LN M192	Missiles	420
SA-18 Igla MANPADS	Missiles	120
Linebacker M6	Missiles	200
	Cannon 25мм	880
Chaparral M48	Missiles	360
Vulkan M163	Cannon 20мм	1200
Avenger M1097	Missiles	400
Roland ADS	Missiles	300
Stinger MANPADS	Missiles	120
SPAAA ZSU-23-4 Shilka	Cannons 4x23мм	20
ZU-23 Emplacement	Cannons 2x23мм	60
ZU-23 Closed	Cannons 2x23мм	60
ZU-23 on Ural-375	Cannons 2x23мм	60

Supply ammo for MLRS

MLRS	Armament	Supply ammo time from a supply vehicle (warehouse), sec.
9K51 Grad BM-21	Solid propellant rockets 40x122мм	420
9K57 Uragan BM-27	Solid propellant rockets 16x220мм	840
9A52 Smerch	Solid propellant rockets 12x300мм	2160
M270 MLRS	Solid propellant rockets 12x227мм	560

Supply ammo for Howitzers

Howitzers	Armament	Supply ammo time from a supply vehicle (warehouse), sec.
2B11 Mortar	Mortar tube 120мм	30
2S1 Gvozdika	Gun 122мм	940
2S3 Akatsia	Gun 152мм	1186
2S9 Nona-S	Gun 120мм	500
2S19 Msta-S	Gun 152мм	1080
	Machine gun 12,7мм	110
M109 Paladin	Gun 155мм	936
SpGH vz.77 Dana	Gun 152мм	1488

Supply ammo for IFV/APC

IFV/APC	Armament	Supply ammo time from a supply vehicle (warehouse), sec.
AAV7	Grenade launcher 40мм	210
	Machine gun 12,7мм	90
BMD-1	Gun 73мм	600
	ATGM "Malutka"	80
	Machine guns 3x7,62мм	60
BMP-1	Gun 73мм	600
	ATGM "Malutka"	80
	Machine gun 7,62мм	60
BMP-2	Cannon 30мм	1800
	ATGM "Konkurs"	120
	Machine gun 7,62мм	600
BMP-3	Cannon 30мм	1800
	Gun 100мм	710
	ATGM "Kastet"	120
	Machine guns 3x7,62мм	600
MT-LB	Machine gun 7,62мм	150
BRDM-2	Machine gun 14,5мм	230

	Machine gun 7,62мм	140
BTR-80	Machine gun 14,5мм	230
	Machine gun 7,62мм	140
BTR-RD	ATGM "Konkurs"	125
	Machine guns 2x7,62мм	60
Cobra	Machine gun 12,7мм	100
LAV-25	Cannon 25мм	1200
	Machine guns 2x7,62мм	480
M2A2 Bradley	Cannon 25мм	240
	ATGM "TOW"	140
	Machine gun 7,62мм	480
M113	Machine gun 12,7мм	75
M1043 HMMWV	Machine gun 12,7мм	75
M1045 HMMWV	ATGM "TOW"	180
M1126 Stryker ICV	Machine gun 12,7мм	125
M1134 Stryker ATGM	ATGM "TOW"	280
	Machine gun 7,62мм	-
Marder	Cannon 20мм	2000

	Machine gun 7,62мм	400
MCV-80 Warrior	Cannon 20мм	380
	Machine gun 7,62мм	75
TPz 1 Fuchs	Machine gun 7,62мм	120
ZBD-04A	Cannon 30мм	1800
	Gun 100мм	710
	ATGM "Kastet"	120
	Machine gun 7,62мм	600

Supply ammo for Tanks

Tank	**Armament**	**Supply ammo time from a supply vehicle (warehouse), sec.**
Challenger 2	Gun 120мм	450(AP), 440(HE)
	Machine gun 7,62мм	-
Leclerc	Gun 120мм	530(AP), 230(HE)
	Machine gun 7,62мм	-
Leopard 1A3	Gun 105мм	515(AP), 290(HE)
	Machine gun 7,62мм	40
Leopard 2A5	Gun 120мм	400(AP), 275(HE)
	Machine gun 7,62мм	126

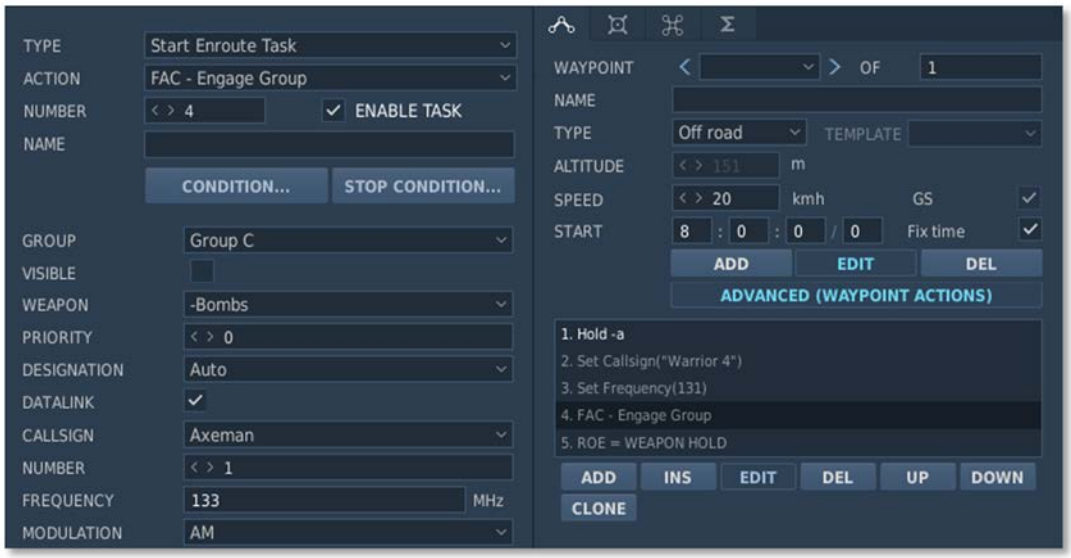
M1A2 Abrams	Gun 120мм	400(AP), 260(HE)
	Machine gun 12,7мм	90
	Machine gun 7,62мм	580
M1128 Stryker MGS	Gun 105мм	260(AP), 140(HE)
	Machine gun 12,7мм	95
	Machine gun 7,62мм	500
Merkava Mk.4	Gun 120мм	465(AP), 300(HE)
	Machine gun 2x7,62мм	545
T-55	Gun 100мм	495(AP), 184(HE)
	Machine gun 12,7мм	110
	Machine gun 7,62мм	-
T-72B	Gun 125мм	416(AP), 308(HE)
	ATGM "Refleks"	130
	Machine gun 12,7мм	110
	Machine gun 7,62мм	-
T-80U	Gun 125мм	406(AP), 316(HE)
	ATGM "Refleks"	120
	Machine gun 12,7мм	150

	Machine gun 7,62мм	-
T-90A	Gun 125мм	408(AP), 310(HE)
	ATGM "Refleks"	130
	Machine gun 12,7мм	50
	Machine gun 7,62мм	-

Advanced Actions Mode (ground groups)



Pressing the Advanced tab open the Advanced Actions Panel, which lists the actions associated with the selected waypoint:



This panel is used to set (add/edit/delete) group actions (Perform Task, Start Enroute Task, Perform Command, Set Option), which are tasked to the AI at this waypoint by the user or generated automatically by the Mission Editor.

Panel Interface Controls

The following buttons are located at the bottom of the Advanced Actions Mode:

- **ADD** – Adds a new action to the list. When pressed, a default No Task action is created, which can be edited in the Action Settings pop-up panel to the left.
- **INS** – Inserts a new action above the currently selected one.
- **EDIT** – Opens the Action Settings pop-up panel to display and edit the settings of the action. The same functionality is duplicated when clicking on any route waypoint on the map.
- **DEL** – Deletes the selected action.
- **UP** – Moves the selected action up the list.
- **DOWN** – Moves the selected action down the list.
- **CLONE** – Clones the selected action down the list.

Action Properties Panel for Ground Groups

The Action Properties Panel for Ground Groups is analogous to the panel for aircraft groups. In other words, this panel is used to set Perform Tasks, Start Enroute Tasks, Perform Commands and Set Options for the group to perform, as well as the starting and ending conditions of the actions. However, the specific types of actions available for vehicle groups differs from aircraft groups.

Perform Task

No Task

No action is assigned to the group to perform.

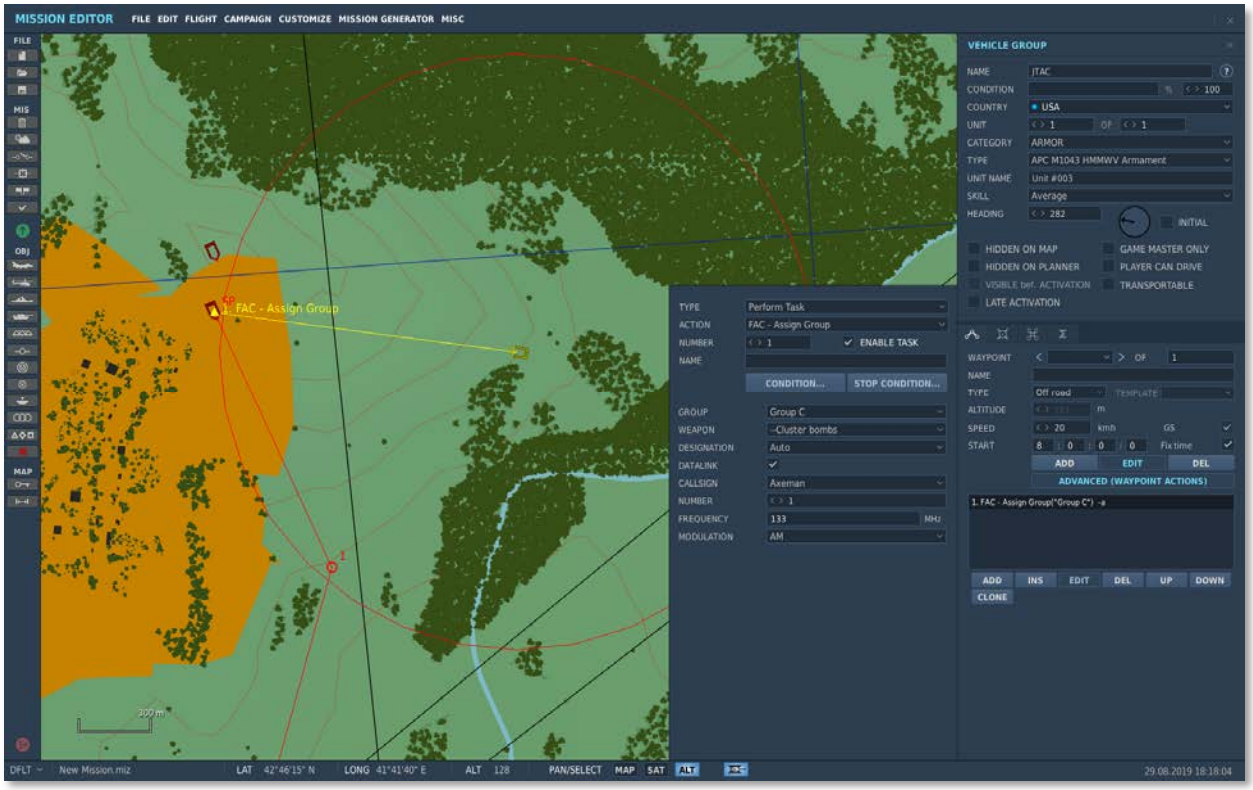
FAC – Assign Group

This task creates a Forward Air Controller (FAC) to direct friendly aircraft onto a target group designated by the mission designer. To achieve full FAC functionality, the FAC unit should be an armed vehicle with direct line of sight to the target and placed within a few kilometers range.

Targeting actions will begin when friendly units arrive in the area and establish communication with the FAC.

***Note**: The FAC: Assign Group Task assumes full situational awareness for the FAC. The target's location will always be known and the FAC will always attempt prosecute the attack if conditions permit.*

The FAC will only provide targeting against the group designated as the target in the mission editor and will not target other groups.

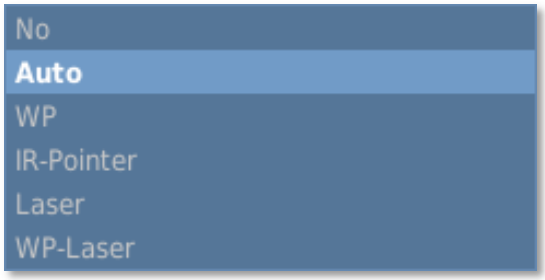


Target Panel:

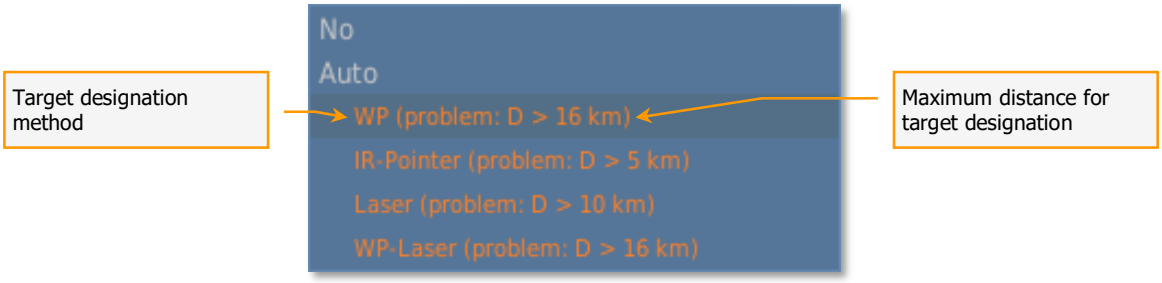
GROUP. Drop down menu to select the target group from all applicable enemy groups available in the mission.

WEAPON. Drop down menu to select the weapon types cleared to use for this action. The FAC-Assign Group Task allows you to set the weapon type(s) for the FAC to request from the attacking groups.

DESIGNATION. Selects the method of target designation (Auto, smoke marker ("Willy Pete", WP), IR-Pointer, Laser, WP-Laser. Target designation is carried out on the player's radio request at a fixed radio range.

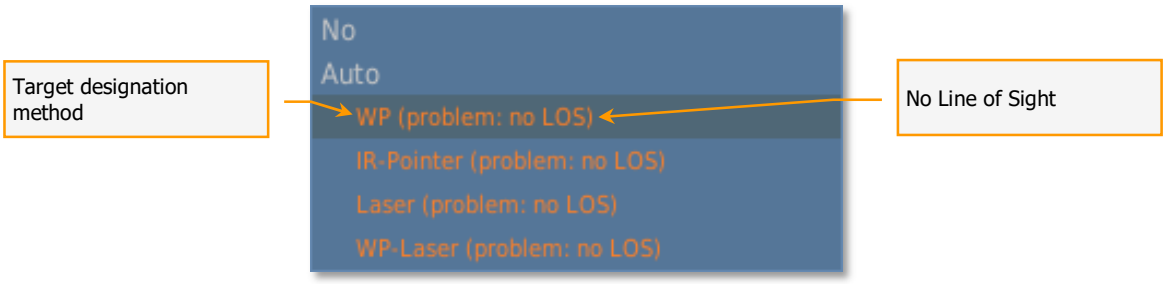


If the FAC unit is located from the enemy at a distance greater than the operational range of the target designator equipment, then the maximum ranges for each of the available equipment will be displayed in the menu.



In this case, it is necessary to adjust the distance of the FAC to the enemy such that it is within the specified maximum range.

If an enemy unit is obscured from the forward air observer's line of sight (for example, due to terrain), the message "problem: no LOS" will be displayed in brackets, signifying that target designation is currently impossible for the player.



For target designation the FAC must first maneuver until the target is in their line of sight.

DATALINK. Only ground units that have EPLRS (as shown in waypoint advanced options) can transmit target location data via datalink to suitably equipped aircraft.

CALLSIGN. Callsign of the FAC.

NUMBER. Selection field for the FAC's number.

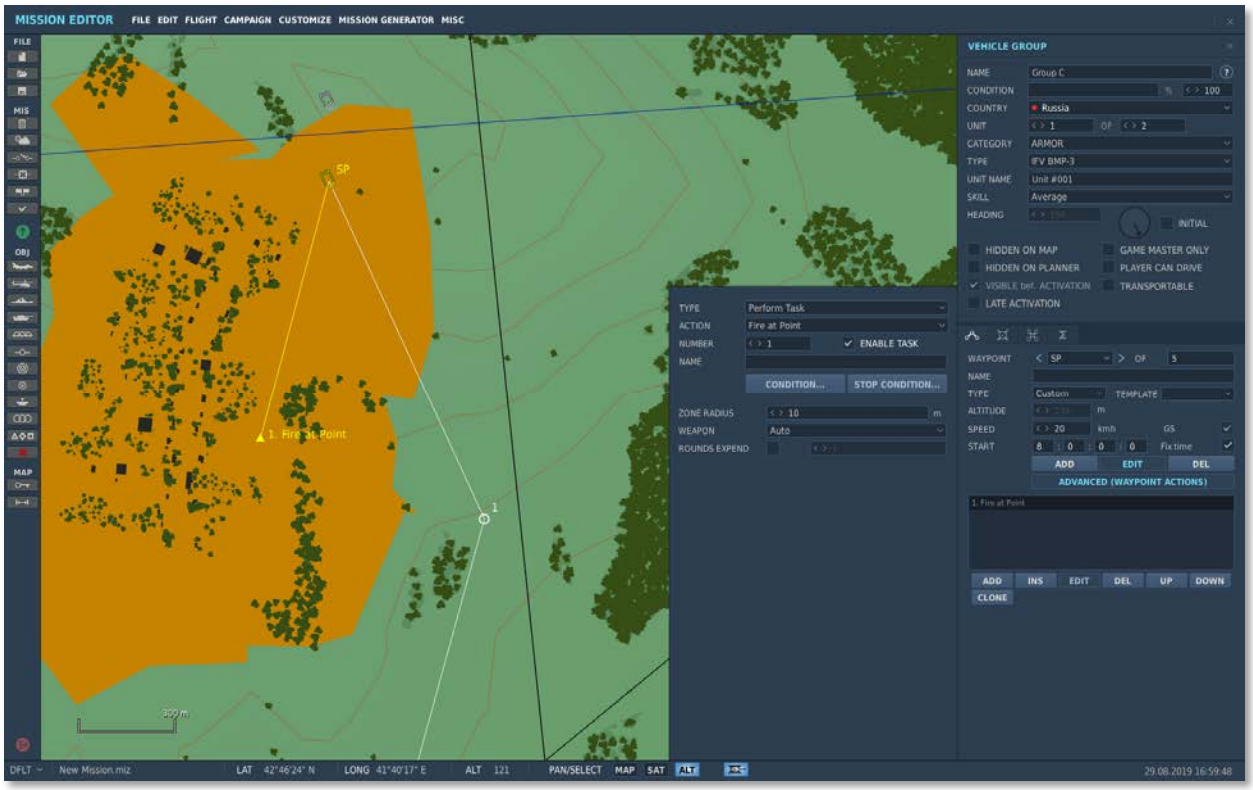
FREQUENCY. Set the FAC group's radio communications frequency in MHz.

MODULATION. Set the FAC group's radio communications AM/FM modulation.

Fire at Point

Fire against a coordinate point on the map as designated by the mission designer.

To perform this action, move the triangular marker that appears near the group when this action is selected, to a desired point on the map.



Target Panel:

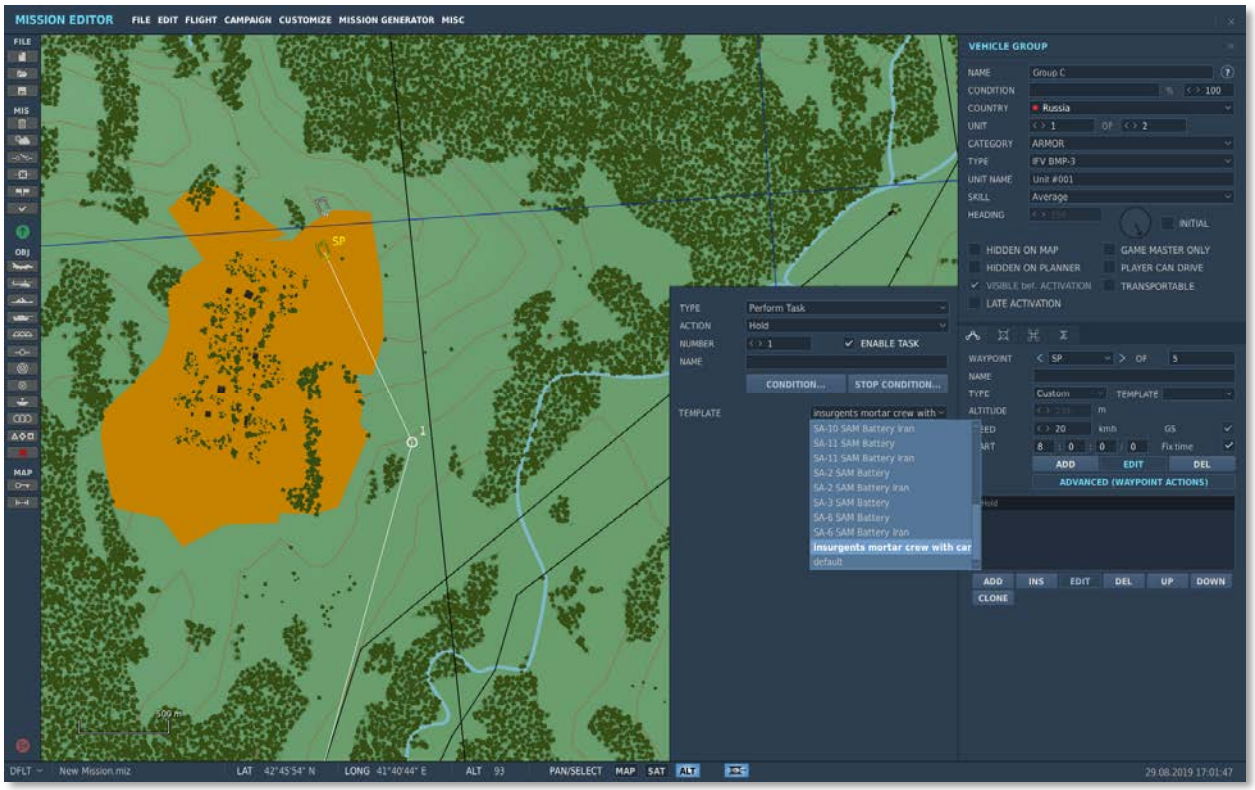
ZONE RADIUS. This parameter affects the distribution of fire: artillery systems will shell the entire area of the zone, whereas missiles will hit only the center of the zone.

WEAPON. Drop down menu to select the weapon type(s) cleared to use for this action. This menu is dependent on the type of armoured vehicles and its available types of weaponry.

ROUNDS EXPEND. Setting this flag allows you to set the amount of ammunition consumed.

Hold

Stop at the waypoint location and maintain position as long as the action is in effect.



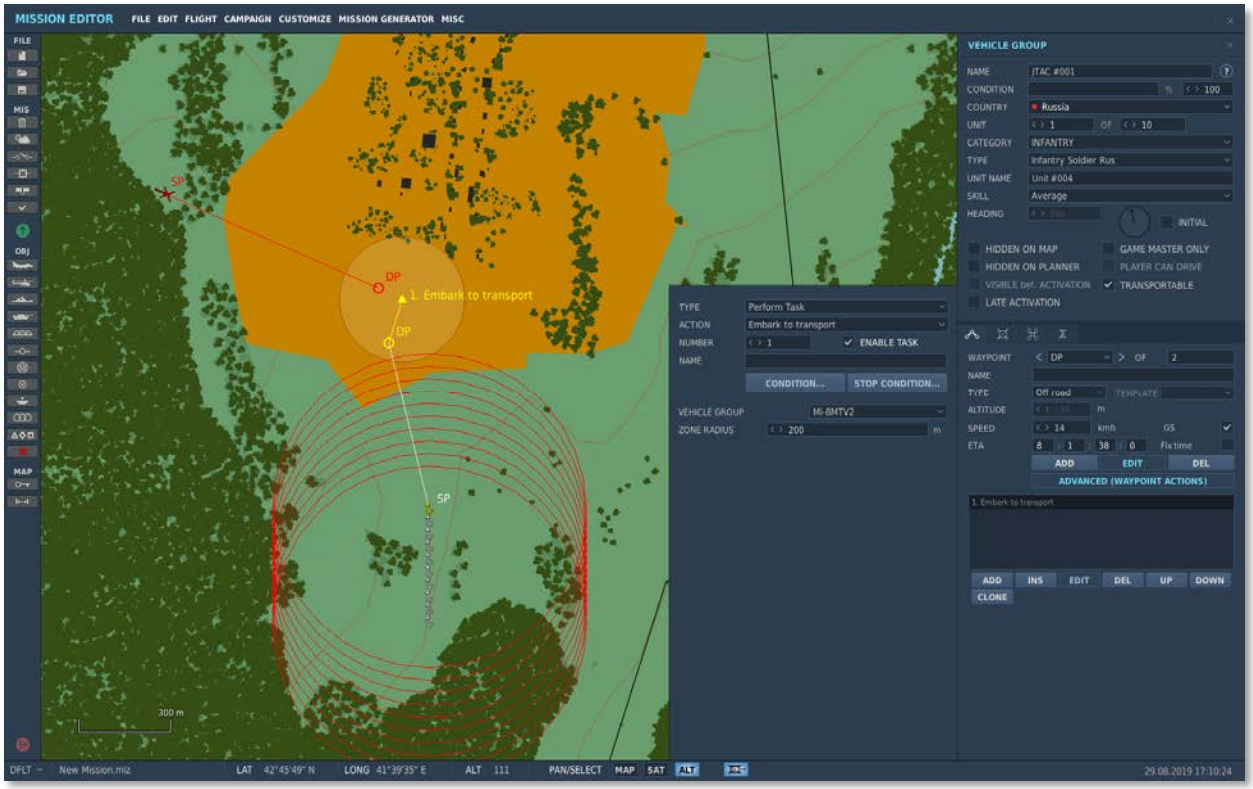
The Lower Action Properties Panel can be used to select a custom formation for the group to take while holding.

This function may be useful for various combat tasks, such as post patrols or artillery fire. For example, an artillery battery can be set to hold in a fire mission formation to conduct fire.

Embark to Transport (work in progress)

Facilitates ground unit embarkation to transport. Available for the following ground unit group categories: "INFANTRY".

Each infantry squad must have "Perform Task – Embark to transport" set, and the TRANSPORTABLE flag activated in the vehicle group panel.



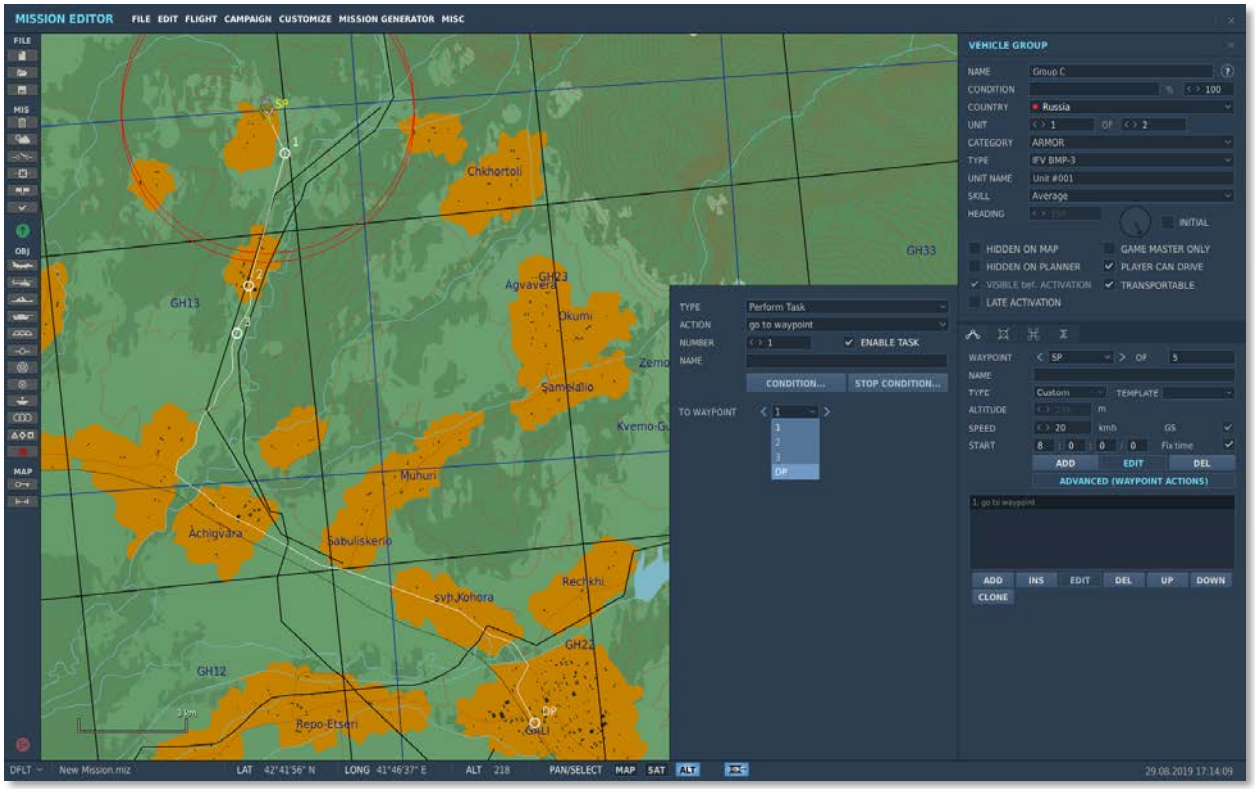
Target Panel:

VEHICLE GROUP. Selects the vehicle group for the embarkation task.

ZONE RADIUS. The radius of the embarkation area of the infantry squads, in meters.

Go to Waypoint

The group will follow the route until it reaches the specified waypoint.



Target Panel:

TO WAYPOINT. Specifies the destination waypoint.

Start Enroute Task

No Enroute Task

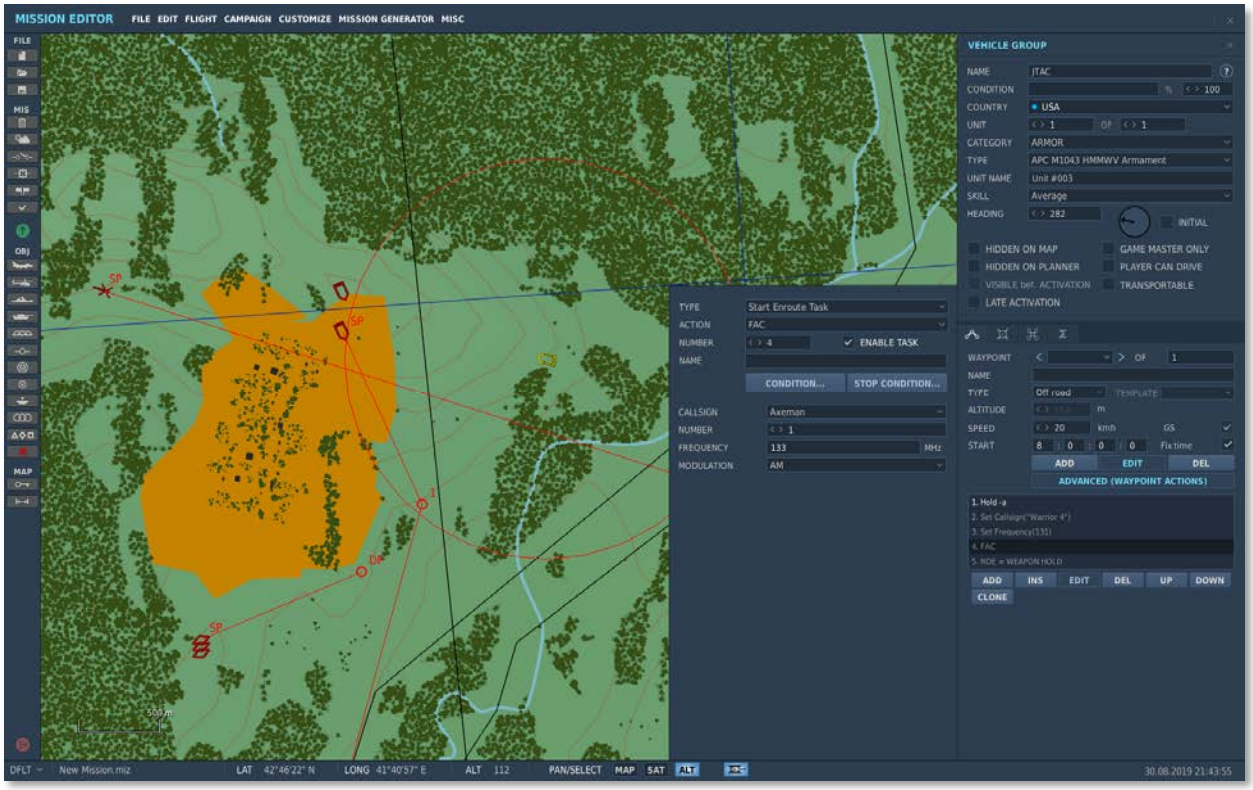
A No Enroute Task action does not set any action for the group to perform.

FAC

The group will act as a Forward Air Controller (FAC) to provide targeting orders for friendly units in the area.

Under a FAC Enroute Task, the FAC will automatically prioritize targets for the player or attacking AI groups and issue targeting orders throughout the attack phase.

***Note**: The FAC's target selection may not correspond to the targets intended for the player or attacking AI by the mission designer.*



Target Panel:

CALLSIGN. Callsign of the FAC.

NUMBER. Selection field for the FAC's number.

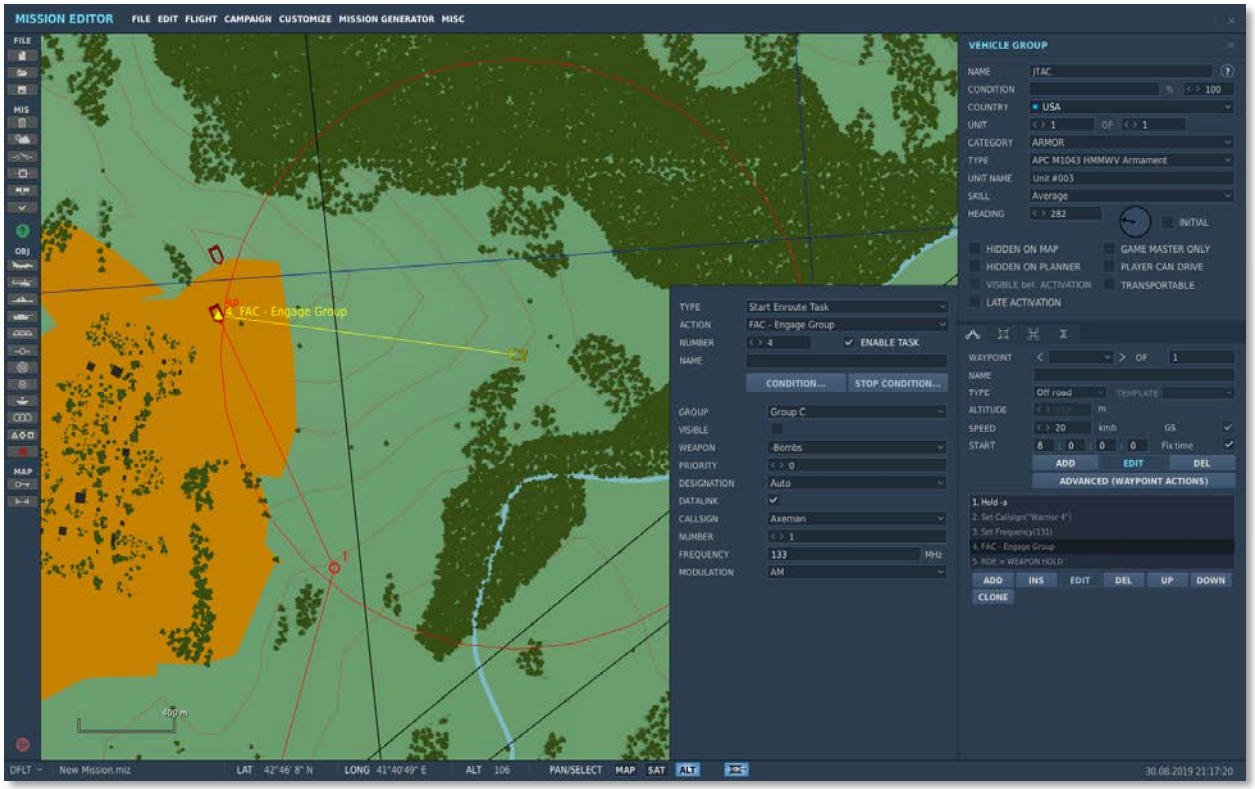
FREQUENCY. Set the FAC group's radio communications frequency in MHz.

MODULATION. Set the FAC group's radio communications AM/FM modulation.

FAC – Engage Group

Analogous to the FAC Enroute Task, except in this case a specific enemy group is selected by the mission designer for the FAC to target.

Remember, because this is an Enroute Task, the FAC will have to independently detect the target group prior to being able to issue targeting commands against it. As described above, target detection will depend on a number of dynamic variables in-game, including possible terrain obstructions or weather interference. If no group is designated by the mission designer, the FAC will not issue any targeting commands.



Target Panel:

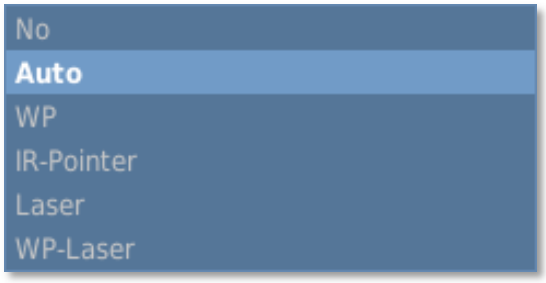
GROUP. Drop down menu to select the target group from all applicable enemy groups available in the mission.

VISIBLE. The VISIBLE option allows the unit to know or not know the location of target position at the start of the mission.

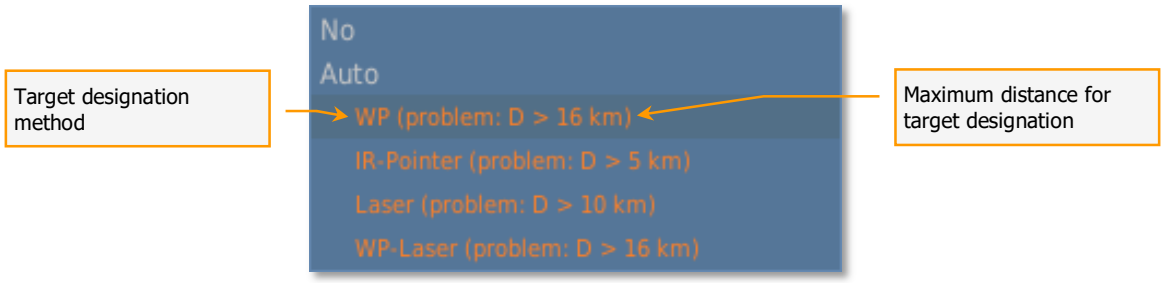
WEAPON. Drop down menu to select the weapon types cleared to use for this action. The FAC-Assign Group Task allows you to set the weapon type(s) for the FAC to request from the attacking groups.

PRIORITY. The PRIORITY number is used to set the action priority in relation to other actions of the waypoint. The priority number represents the order this action will take relative to other actions, starting with '0' as the highest priority.

DESIGNATION. Selects the method of target designation (Auto, smoke marker ("Willy Pete", WP), IR-Pointer, Laser, WP-Laser. Target designation is carried out on the player's radio request at a fixed radio range.

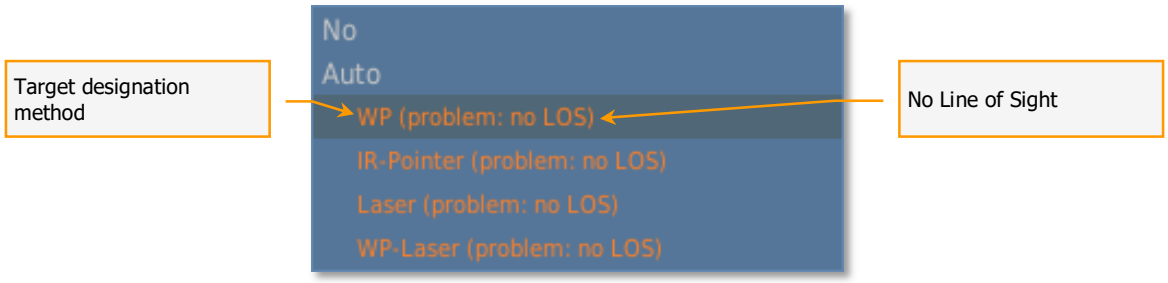


If the FAC unit is located from the enemy at a distance greater than the operational range of the target designator equipment, then the maximum ranges for each of the available equipment will be displayed in the menu.



In this case, it is necessary to adjust the distance of the FAC to the enemy such that it is within the specified maximum range.

If an enemy unit is obscured from the FAC's line of sight (for example, due to terrain), the message "problem: no LOS" will be displayed in brackets, signifying that target designation is currently impossible for the player.



For target designation the FAC must first maneuver until the target is in their line of sight.

Only armed units can designate a target with smoke and laser. (e.g. APC M1043 HMMWV Armament).

DATALINK. Only ground units that have EPLRS (as shown in waypoint advanced options) can transmit target location data via datalink to suitably equiped aircraft.

CALLSIGN. Callsign of the FAC.

NUMBER. Selection field for the FAC's number.

FREQUENCY. Set the FAC group's radio communications frequency in MHz.

MODULATION. Set the FAC group's radio communications AM/FM modulation.

Perform Command

The following commands can be set as actions upon reaching a waypoint:

No Action

No command is set.

Run Script

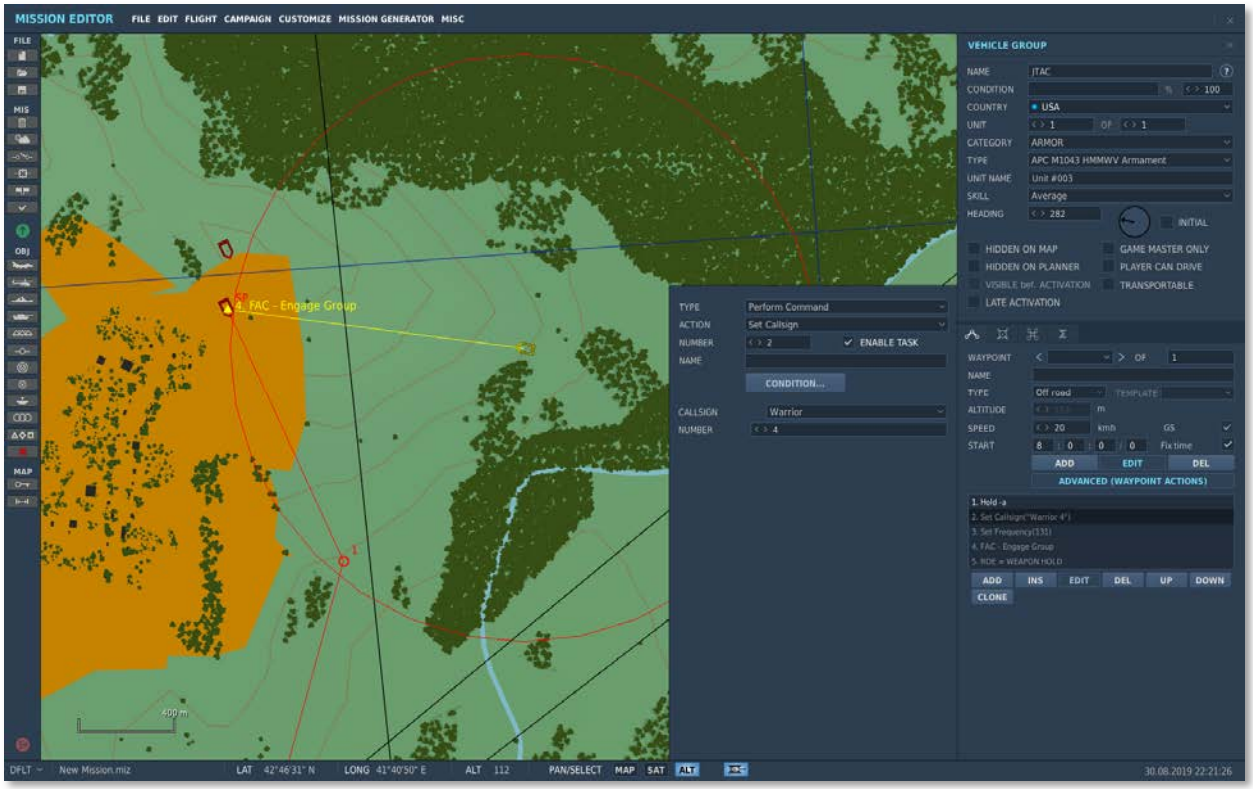
Run the Lua script entered in the scratchpad.

Script File

Selects the LUA script file.

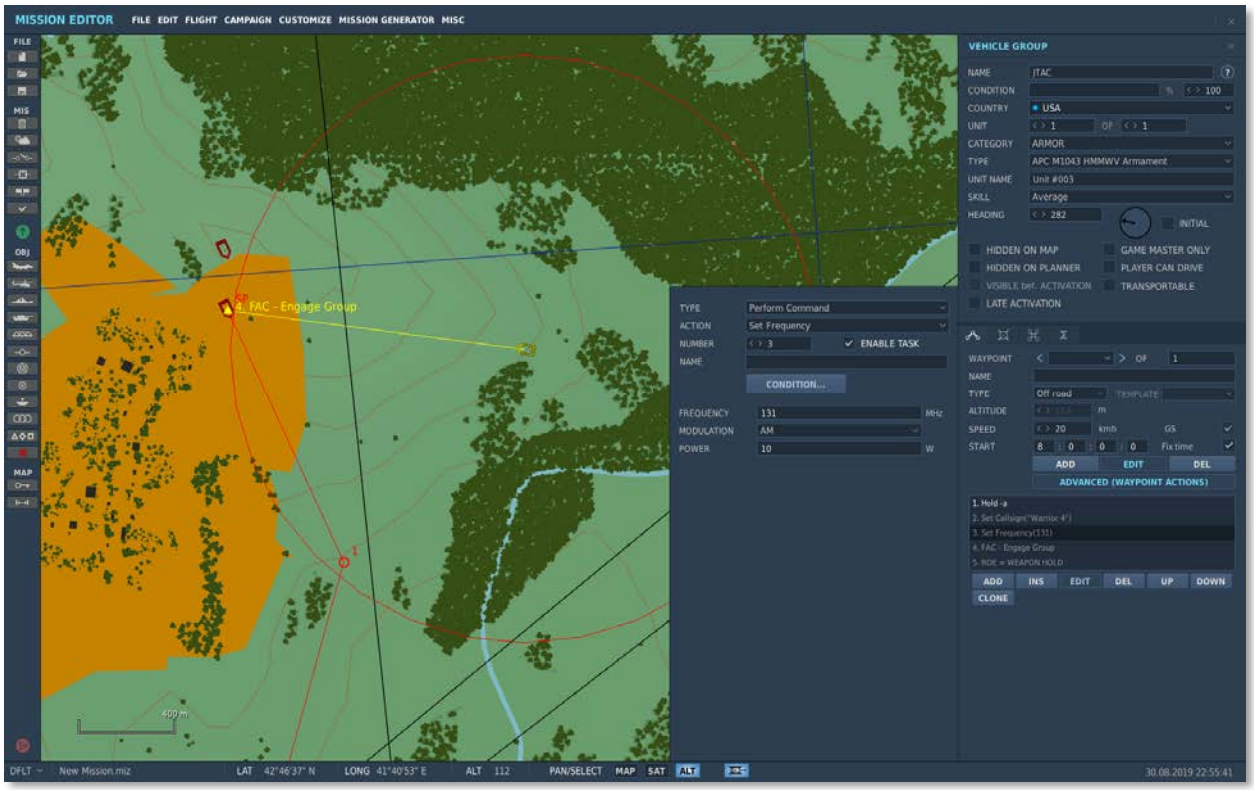
Set Callsign

Set the group's callsign. For Western units, the callsign consists of a primary designation and a numerical ID. For example, "Warrior 4".



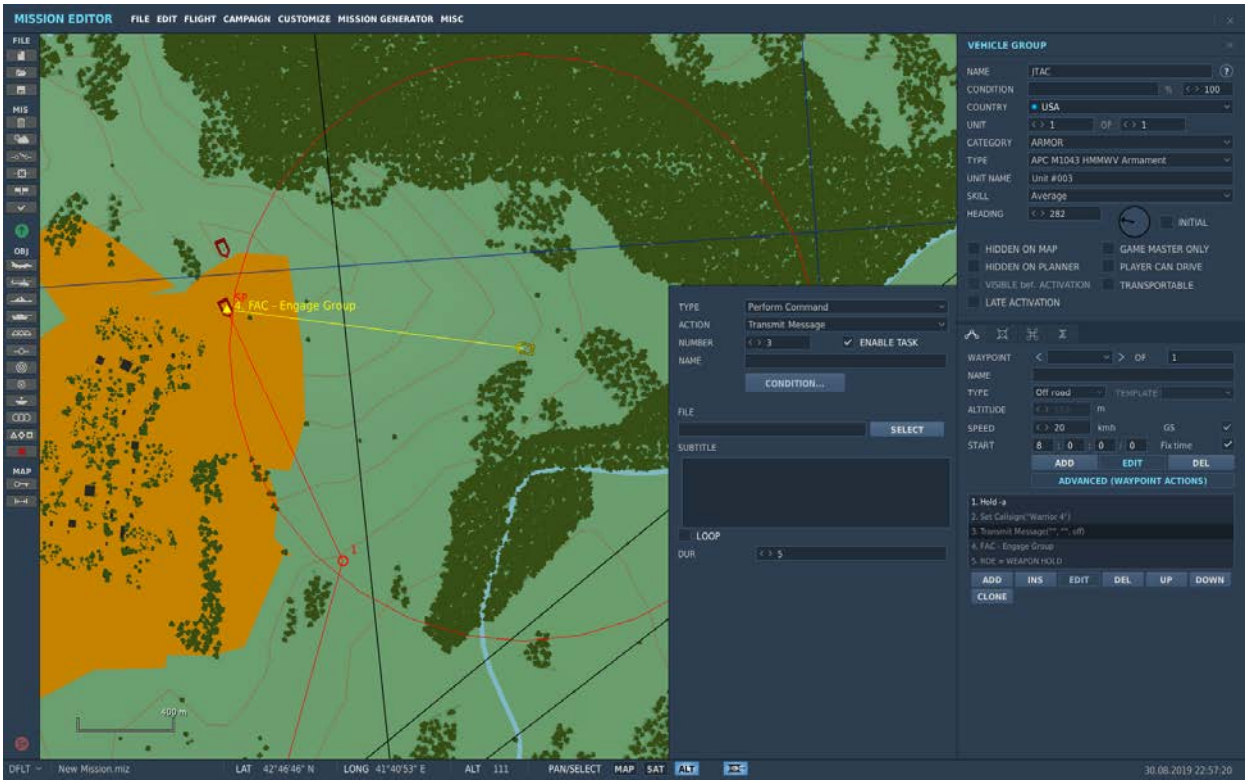
Set Frequency

Set the group's radio communications frequency in MHz and AM/FM modulation.



Transmit Message

Command to send a message from the lead group.



FILE. Selects the audio file to be played during the radio transmission (*.ogg;*.wav).

SUBTITLE. Field for inputting the text message (subtitles) to be displayed during the radio transmission.

LOOP. Settings for repeating the message.

DURATION. Duration of the message, in seconds.

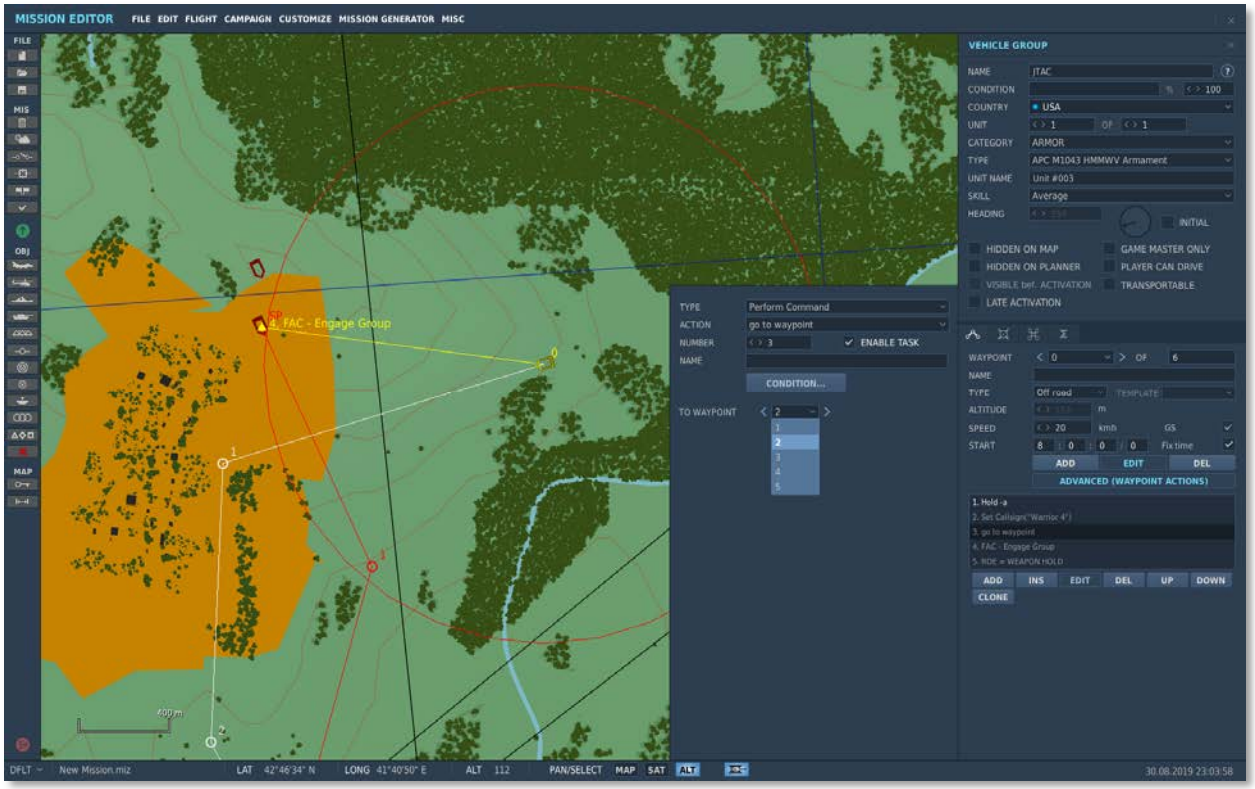
Stop Transmission

Stops the transmission from the lead group. This command has no available options.

Go to Waypoint

When the conditions for this action are in effect, the group will navigate from the current waypoint directly to the one set by this action.

For example, you can send a unit from WPT 0 immediately to WPT 2, or vice versa, from WPT 2 back to WPT 0.



By default, this is set to the next waypoint in the existing route.

All main tasks, commands, and behavior options, missed waypoints, will not be fulfilled.

This action can be used when organizing a circular route. For example, if you send a unit to WPT 0 after WPT 2, it will return along the route to WPT 2 and again go to WPT 0 while the command is active. You can limit the command with logical conditions by using CONDITION ..., for example, to set a time limit for executing the command.

Invisible

Set the group to be invisible to all enemy AI units. Enabled and disabled using the ENABLE checkbox.

This command may be useful when creating ground-based FACs in close proximity to enemy positions as a means of simulated a well camouflaged position and preventing the FAC from being destroyed.

Immortal

Set the group to be immortal. Enemy AI units will be able to detect the group, but their weapons will have no effect.

EPLRS

This command enables the usage of the EPLRS (Enhanced Position Location and Reporting System) system and the transmission of EPLRS data to units in the group. Has one option for enabling or disabling this function.

Set Option

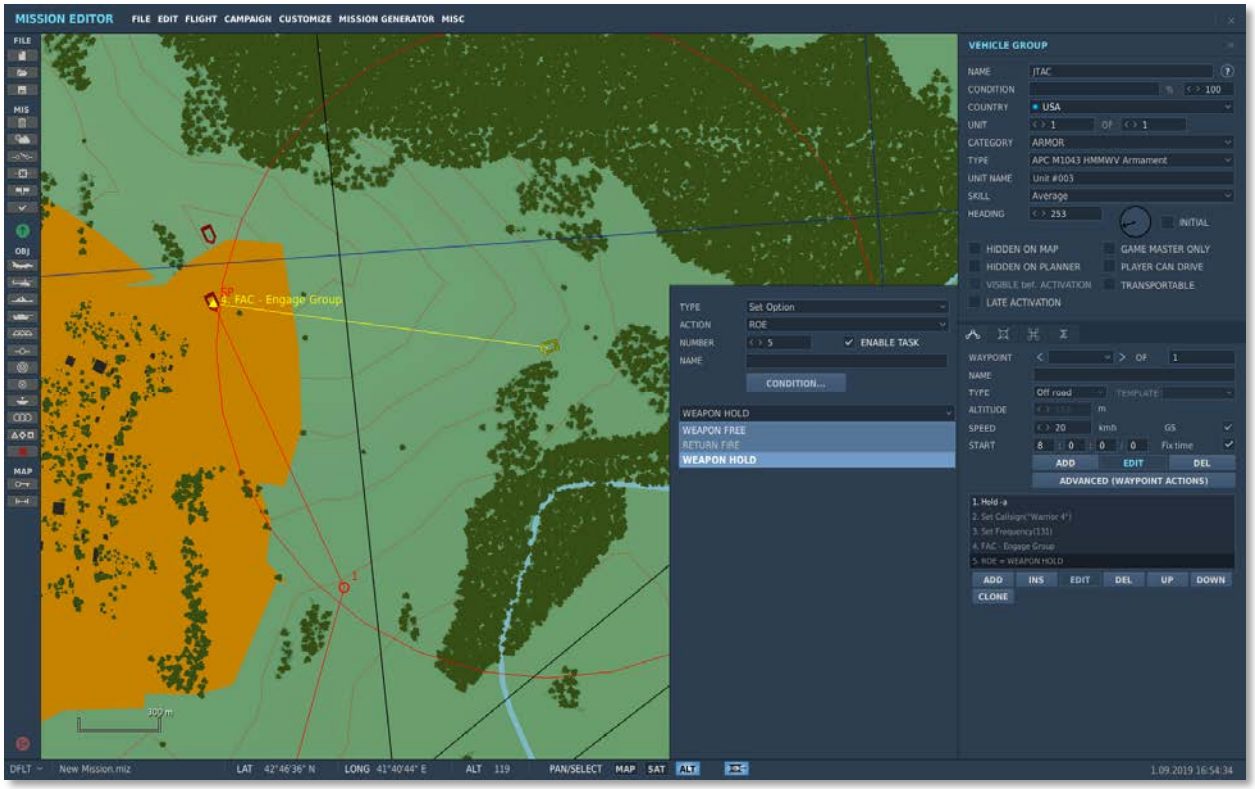
The following Options can be set as actions upon reaching a waypoint:

No Option

No option is set. Default selection.

ROE (Rules of Engagement)

Set fire rules to determine the group's combat behavior when encountering enemy units.



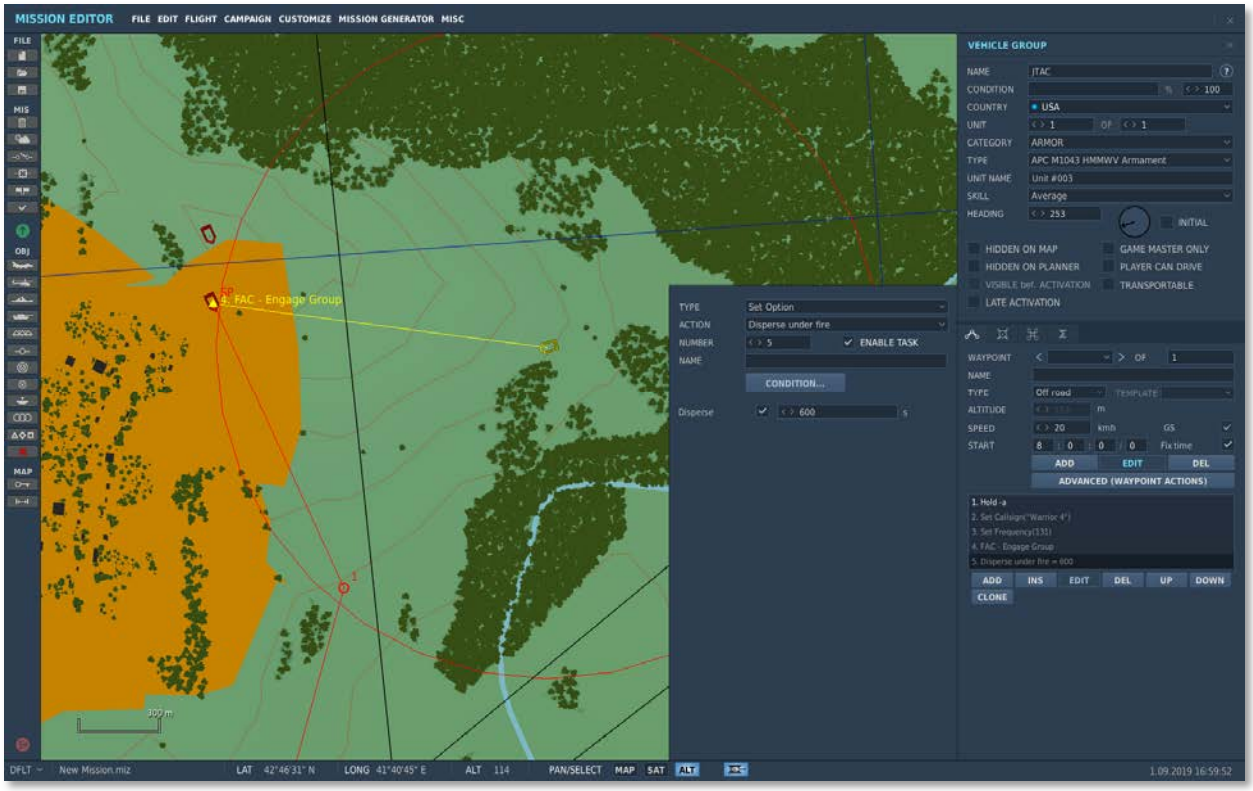
- **WEAPON FREE** – engage any enemy group in contact. Target prioritization is performed automatically by the group based on the situation.
- **RETURN FIRE** – return fire only; do not engage first.
- **WEAPONS HOLD** – do not fire under any circumstances.

The default setting is WEAPONS FREE.

Disperse Under Fire

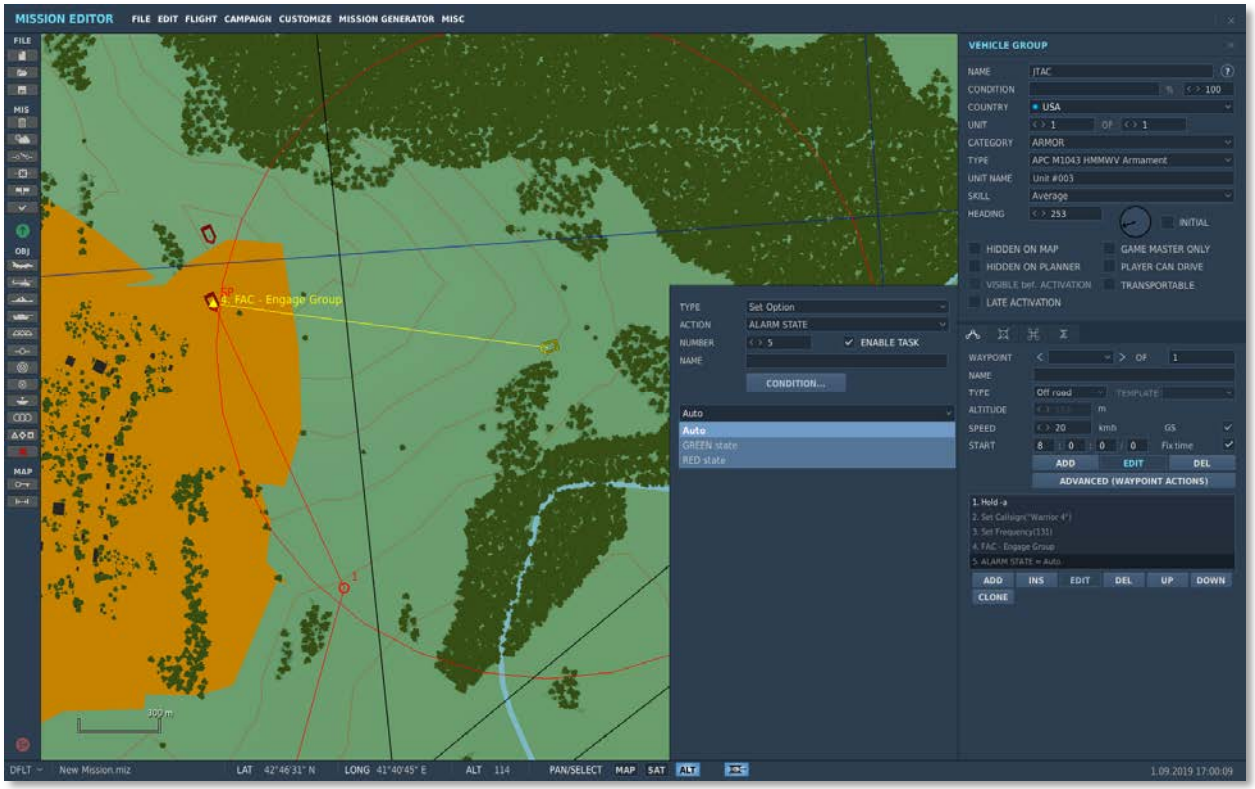
When enabled, this option will allow the group to disperse when under attack. The default value is Allow. In this case, the unit group will disperse when any one of its units is destroyed or damaged. The dispersal duration is indicated in seconds, after which the group will return to

formation and continue moving along the route. If this option is not enabled, the group will continue to move in formation while under fire.



Alarm State

Sets the combat readiness state of the group.



- **Auto** – default setting, where the group is not combat ready until the presence of enemy units is detected. Note, the AI may "sense" the presence of enemy units without actually detecting them with own sensors.
- **GREEN state** – group is not combat ready. It does not perform target search or engagement, maintains sensors off and stowed if possible.
- **RED state** – group is combat ready and actively searching for targets. Depending on the unit, it may or may not be able to move while in combat ready state.

Engage Air Weapons

This option allows the ground group to intercept and destroy aircraft-launched weapons on their approach to target. It has one checkbox which allows or forbids this action.

Place Static Object



In addition to placing active air, land, and sea units, you may also place static versions of these objects to populate a mission. Static objects share the same external model as their active-object counterparts, but they are immobile and do not use sensors or weapons. Unlike active units, you can only have one unit per group.

Note, the mission editor supports group copy/paste functionality. To copy a group, select it and press [CTRL+C]. To paste the group with identical properties, position the mouse over the desired point on the map and press [CTRL+V].

The Static Object window includes the following functions from top to bottom:



NAME. In the Name field, enter in a unique name for the static object. If you do not enter one, a default one will be generated. The Name you create will be used when assigning some types of Triggers such as Unit Dead. To avoid confusion, be careful not to assign more than one group with the same Name.

COUNTRY. The Country drop down list will display all the countries that have been assigned to either the RED or BLUE sides when the mission was initially created by use of the CREATE NEW MISSION button.

CATEGORY. Static Objects are divided into six general categories:

- **Airfield and deck equipment** enables the usage of airfield-based maintenance equipment for aircraft.
- **Animals** enables the spawning of a set of certain types of animals.
- **Cargos** that includes a range of different types of cargo for helicopters.
- **Effects** that includes a set of smoke effects.
- **Ground Vehicles** that include vehicles appropriate to the selected country.
- **Helicopters** that include those belonging to the selected country.
- **Heliports** that allow the placement of a Forward Arming and Refueling Point (FARP) for forward helicopter operations. When setting a FARP, you can choose the FARP callsign for radio communications from the CALLSIGN drop-down menu and set the radio frequency in MHz in the FREQUENCY field.
- **Planes** that include fixed-wing aircraft of the selected country.
- **Sea Shelf Objects** that includes drilling rigs and gas production platforms.
- **Ships** that belong to the selected country.
- **Structures** that includes pillboxes and many other types of buildings (military and civil).
- **Warehouses** that includes a set of warehouses and tanks.

TYPE. Depending on the Category selection, a list of appropriate units is listed in this pull-down list.

HEADING. You may use this control to set the heading of the unit in the simulation world. You can do this by either using the left and right arrow keys, typing in the heading value, or clicking on the heading dial.

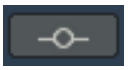
HIDDEN. If you wish to hide the static object on the World Map and make it invisible to players in the briefing map, check this box.

HIDDEN ON PLANNER. This option hides aircraft from the mission map in the mission planner. This option can be useful if you need to hide an object from the player.

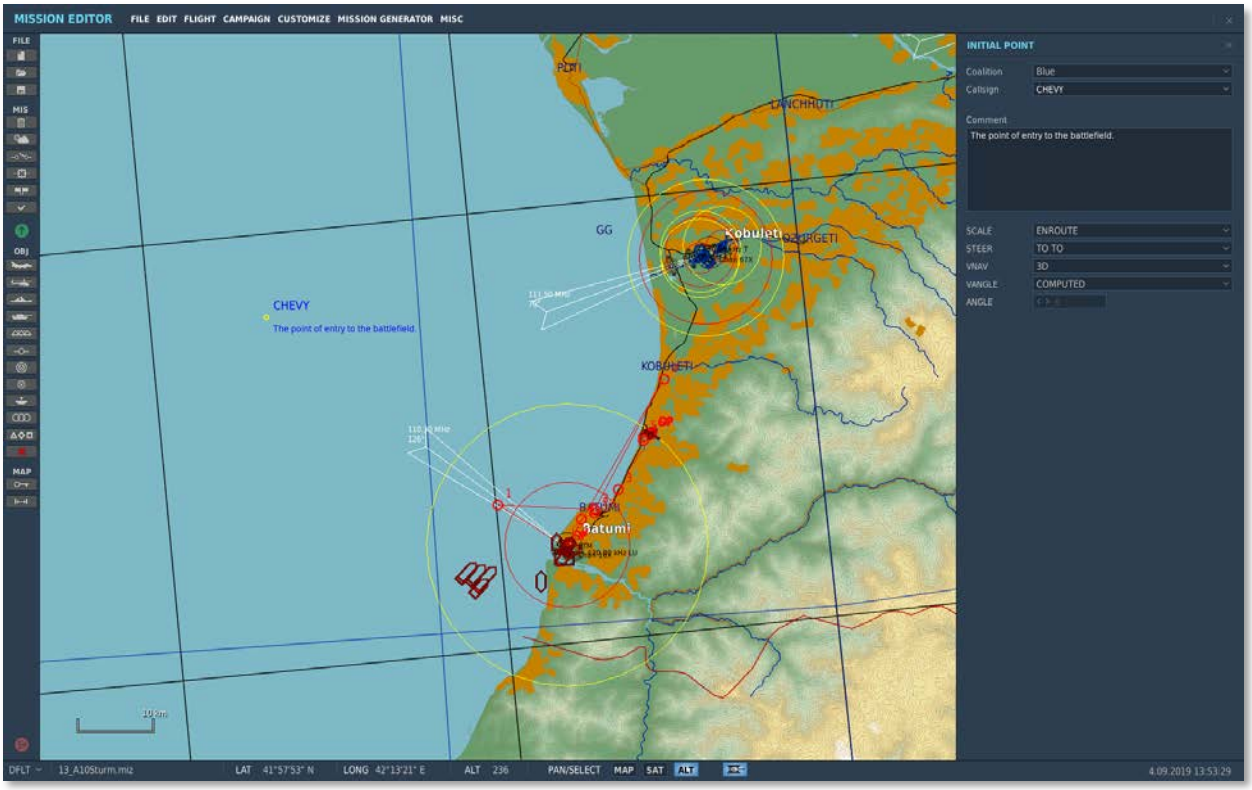
DEAD. In addition to the normal static object, you may also populate the world with the destroyed version of the object by checking the DEAD box.

PAINT SCHEME. Allows the user to select the paint scheme for static objects.

Place Initial Point (A-10C only)



The Initial Point (IP) button opens the Initial Point window, which is used to place an IP on the map. This point will be stored in the player's A-10C navigation system as an independent navigation point.



The IP window allows you to set the following options:

COALITION. Select coalition the navigation point will be available to.

CALLSIGN. Select the IP callsign (ID) of the point from the list of eight available choices.

COMMENT. Optional comments to be indicated on the Mission Editor map.
Below the comment field, you can set the IP properties as specific to the navigation system of the aircraft. These settings have no function in the A-10C.

SCALE. The scale mode of the waypoint.

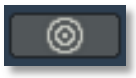
STEER. The steering mode of the waypoint.

VNAV. The vertical navigation mode of the waypoint.

VANGLE. When VNAV is set to 3D, the vertical angle computation mode.

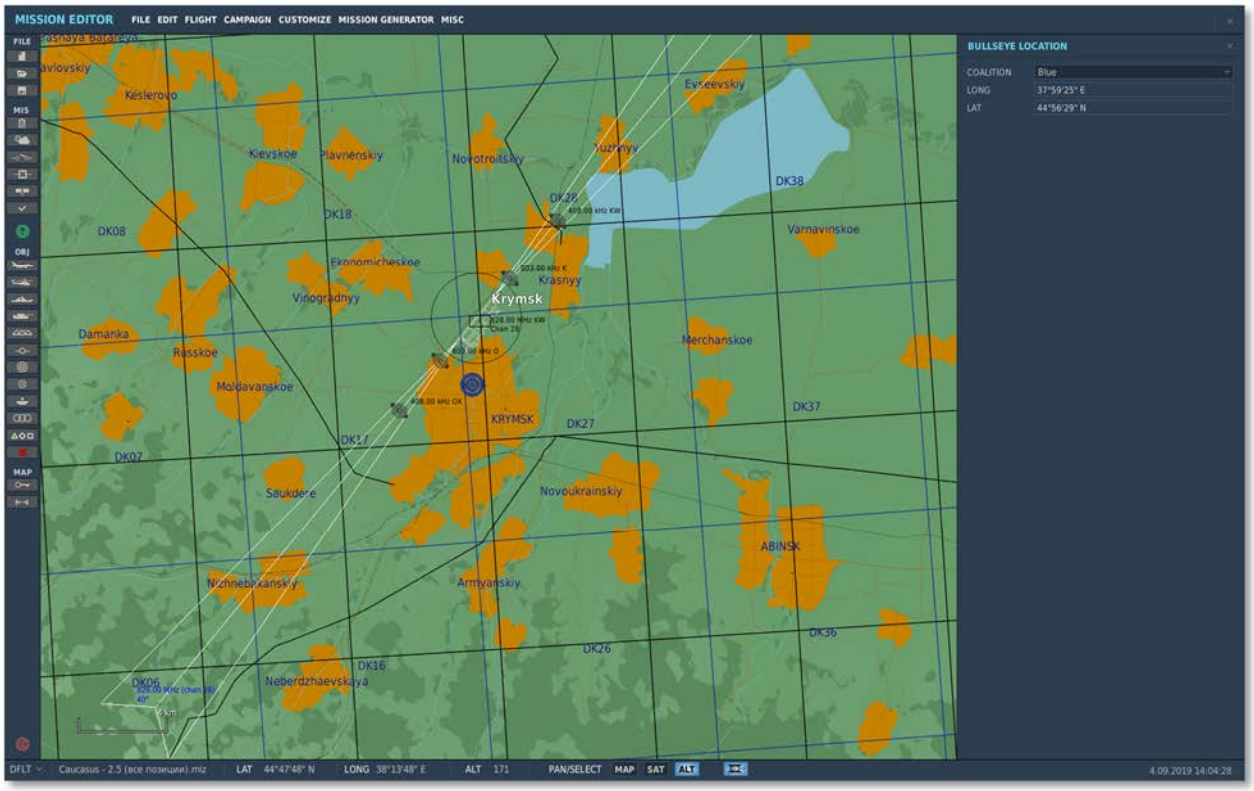
ANGLE. When VNAV is set to 3D and VANGLE is set to ENTERED, the vertical angle of the waypoint.

Place Bullseye

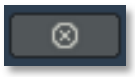


The Bullseye button opens the Bullseye window, which is used to place a Bullseye point on the map. Bullseye is a reference point used by friendly forces to communicate their position. Each coalition is provided a single Bullseye point. To move the position of the Bullseye on the map, simply click and drag it to the desired location.

The Bullseye window allows you to select the coalition of the Bullseye to edit and indicates the current world coordinates of the selected Bullseye.

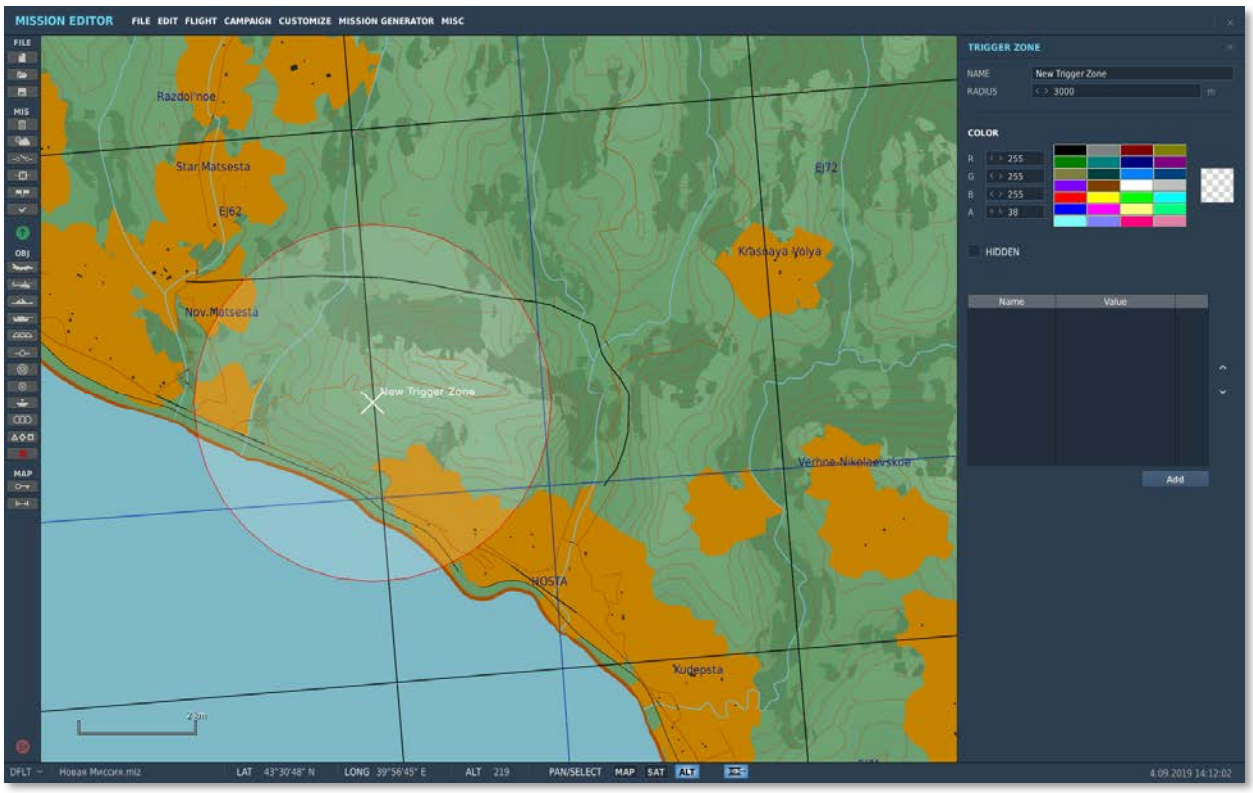


Create Area Trigger Zone



Area trigger zones are a powerful tool that allow you set triggered conditions based on a specified unit entering or exiting a defined area on the map. Trigger zones can be placed anywhere on the map, can be any size, and can vary in color. You can also create a trigger zone and assign that zone to a mobile unit in the mission.

Once you open the trigger zone tool, left click on the map where you want the zone to be centered (all trigger zones are circular). Upon doing so, the trigger zone will be displayed on the map with a default name.



On the right-side trigger zone window, you have the following options:

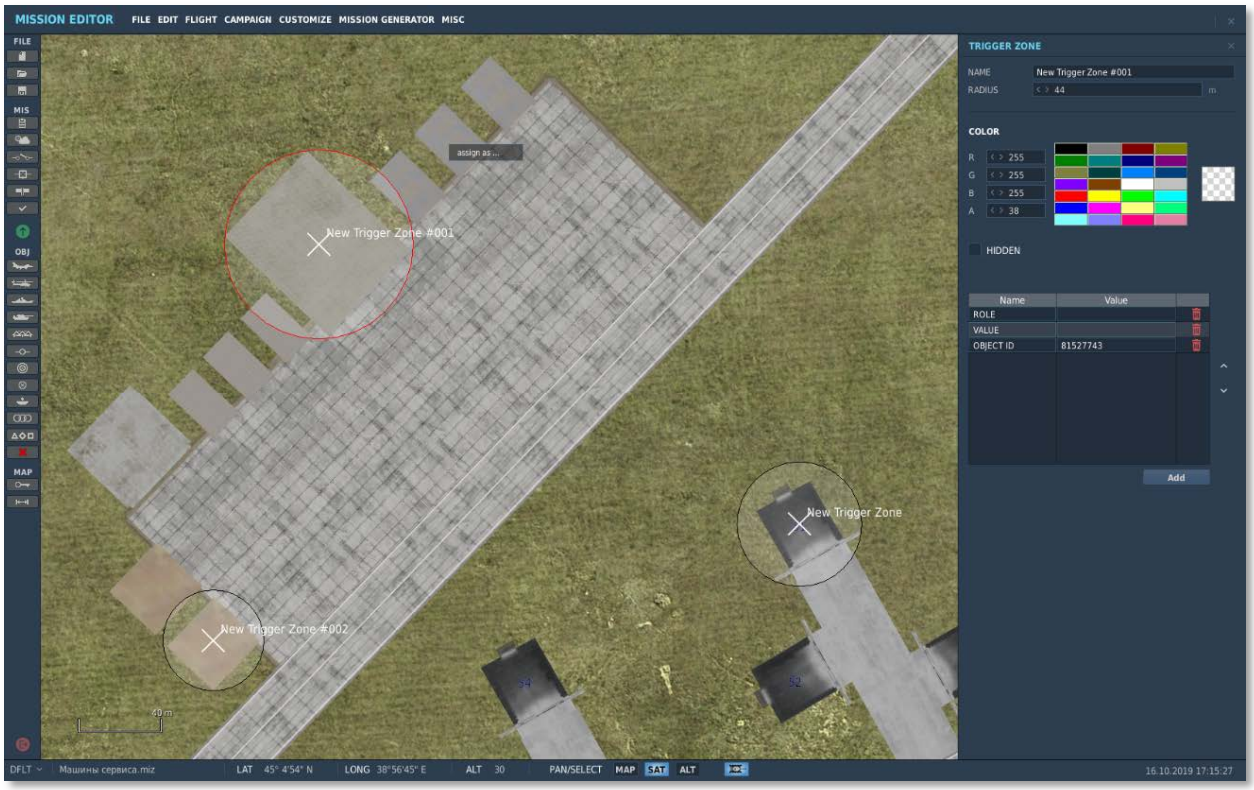
NAME. Enter in the name to give the trigger zone in this field. If no name is entered, a default one will be used. This is the name you will use when setting up a trigger condition.

RADIUS. The trigger zone may vary in size by adjusting this value. The value can be changed by either clicking on the left and right arrows or by entering the value in the field. The radius in meters is displayed in the field.

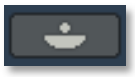
COLOR. To help distinguish trigger zones from each other, you may use the default colors or create a custom one. To select a default color, click on one of the 28 colors. To create a custom color, use the left and right arrows of the R (red), G (green), and B (blue) color values. The A (alpha) value adjusts the transparency. The selected/created color is displayed in the RGB values.

HIDDEN. When the mission is played by others, you will probably not want the trigger areas to be visible on the map. To hide them you must check this box.

It is also possible to add a trigger zone to the ground structures, such as an airfield hangar or a building in a settlement, for example. If you right-click on a building on the ground, select assign as ... and set up the trigger zone as desired:



Create Unit Template



Because you can create ground vehicle groups that consist of many different types of vehicles, it may be useful for you to save a vehicle group composition for later use. A good example might be an artillery battery that consists of artillery units, ammunition trucks, a command and control truck, security APCs, etc. You can place these all in a single vehicle group and then use the Template tool to save it and use it again later. Additionally, templates may be useful in conjunction with the STOP AND DEPLOY TO TEMPLATE action of the Trigger Menu.

To create a new template:

1. Create a new ground vehicle group with all of the group's required units.
2. Position the units as desired for the template. The figure shows one of the options for the SA-10 (S-300PS) battery position. Keep in mind that moving the lead unit of a group will move the entire group.
3. With the desired group selected, press the Create and Modify Templates button.
4. Select the appropriate country for the group from the COUNTRY drop-down menu.
5. Confirm that the ground vehicle group that you wish to make a Template of is listed in the SELETED GROUP field.
6. In the TEMPLATE NAME field, enter a name that you wish to save for the Template.
7. Press the SAVE TEMPLATE button.
8. Close the TEMPLATES panel by pressing the Create and Modify Templates button again or "X" in the upper right corner of the panel.

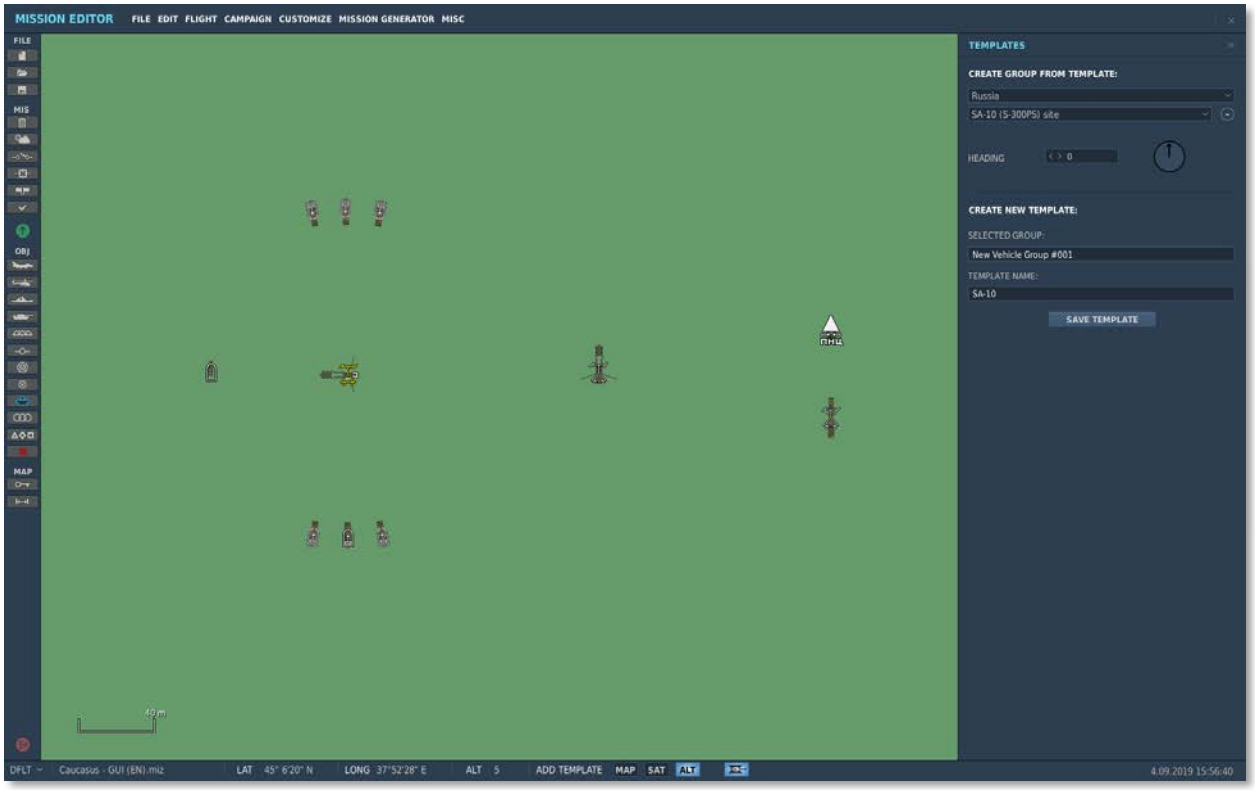
User templates are saved in the file

C:\Users\UserName\Saved Games\DCS\MissionEditor\templates.lua.

You have now created and saved a template.

You have controls at the top of the Template window to manage your templates.

- The top field allows you to select the country for which to save or open Templates.
- The field below the country field is a drop-down menu that lists all the Templates you have created for the selected country.
- Using the heading field, you can orient the template using the left and right arrows or by clicking on the dial. The set orientation is indicated in the HEADING field in degrees.



Below are step-by-step instructions for placing a ground vehicle group from the TEMPLATES panel:

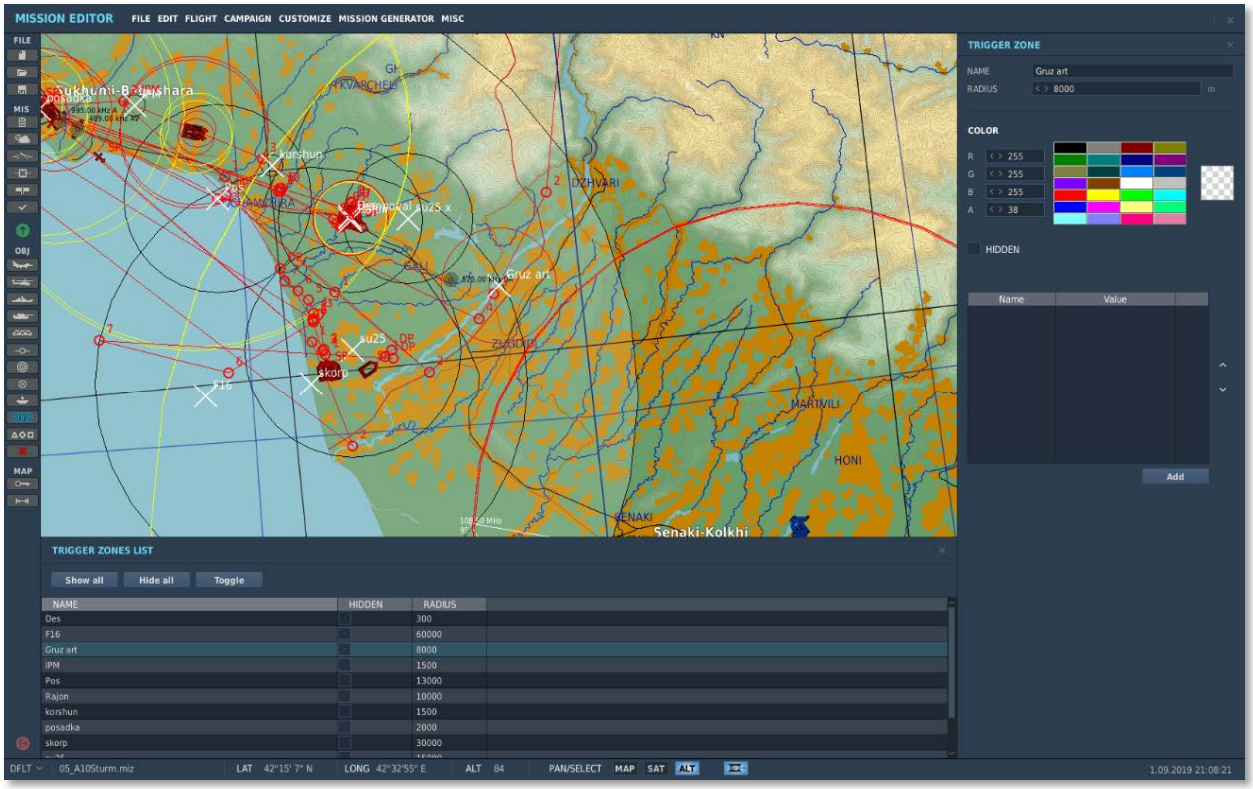
1. Open the TEMPLATES panel by clicking on the Create and Modify Templates button.
2. In the first drop-down list, select a country.
3. In the second drop-down list, select the template for the ground unit group.
4. Place the template on the map by left-clicking on the desired start position.
5. Set the desired start orientation of the ground unit group by using the HEADING dial.

After all the desired templates have been placed on the map, close the panel by clicking the Create and Modify Templates button again, or by clicking the "X" in the upper right corner of the panel.

Area Trigger Zone List



When creating area trigger zones, you have the option to hide them so that users playing your missions cannot see them. Pressing the area trigger zone list button will display the area zone trigger tool on the right side of the screen and list all the zones you have created for the mission along the bottom.



At the top of the zone list are three buttons that allow you to enable and disable the hide function for all the listed trigger zones.

Show All. Pressing this button will set all trigger zones to a visible state.

Hide All. Pressing this button will set all trigger zones to a Hide state.

Toggle Selection. The Hide state of the trigger zones will toggle.

In the trigger zone listing window there are three columns, and each row lists a different trigger zone. The columns include:

NAME. The name of the trigger zone as entered in the NAME field.

STATE. This can either be blank (indicating a visible state) or HIDDEN to indicate that the trigger zone is currently hidden.

RADIUS. The radius of the trigger zone in meters.

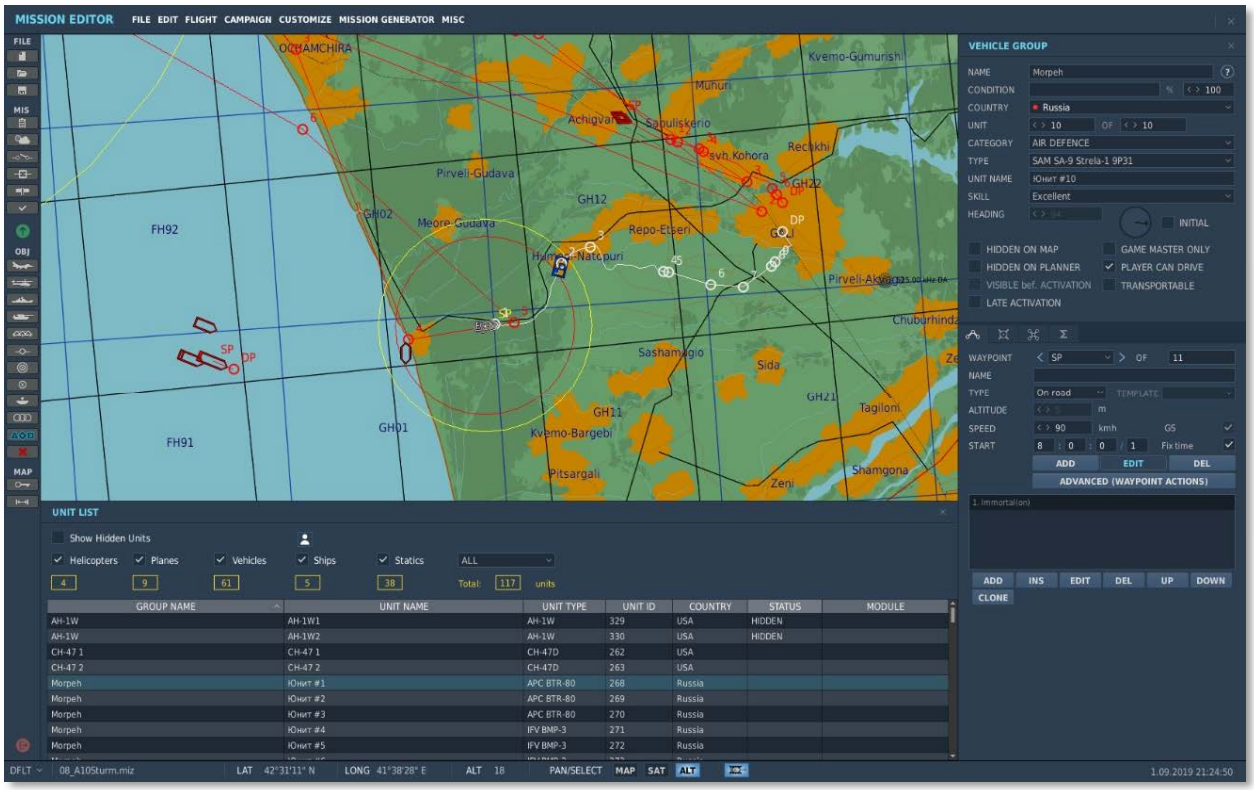
To close the widow, you may press the X button in the top right corner of the window.

Unit List



The Unit List displays the list of units (active and static) that you have placed in the mission, and allows you to quickly find a unit and view data on it.

At the top of the list are five check boxes that allow you to filter the types of units that are displayed on the list. These include Show Hidden Units, Helicopters, Planes, Vehicles, Ships, and Statics. To view one of the types in the list, check the box next to the category name.



Each row of the list represents a different unit group; you can use the scroll bar on the side for when the list becomes long.

 – This button centers the map on the player-controlled aircraft.

There are seven columns that provide data for each group. They are:

GROUP NAME. Name of the group as set when the group was created.

UNIT NAME. Name of the unit.

UNIT TYPE. Unit type.

UNIT ID. Unit Identifier.

COUNTRY. The name of the country that the group is assigned to.

STATUS. This can either be blank to indicate that it is not hidden or say HIDDEN to indicate that the group is currently in a Hide state.

MODULE. Name of the module required for the chosen unit (e.g. WWII).

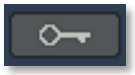
Note: To quickly Hide or Show a group, you may double-click on it.

Delete Unit/Object



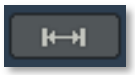
To remove a group entirely from the mission, including waypoints, select the group and then press the Delete button [DELETE].

Map Options



The Map Options panel has been described in detail previously in the System Bar chapter of the manual.

Distance Tool



You can use the Distance Tool to measure distance on the World Map. When enabled by pressing the Distance Tool button, left mouse button click, and hold and drag to measure a distance. The distance line is drawn in blue and the distance (in meters) and bearing (degrees) are in black. You must press the Distance Tool button again in order to disable it, and resume normal map functioning.

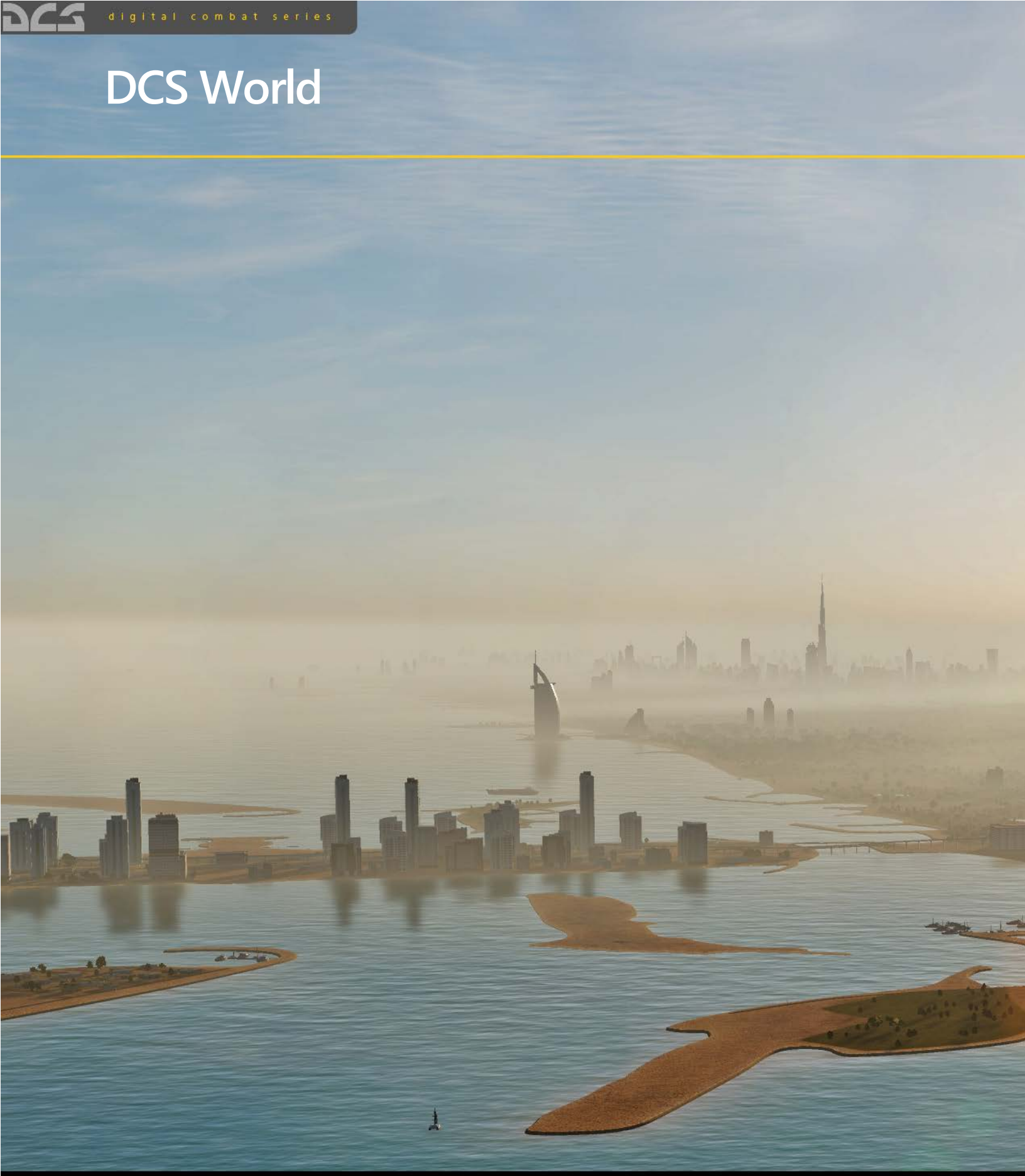
Exit Mission Editor



To exit the Mission Editor, press the Exit button.

digital combat series

DCS World



CAMPAIGN

CAMPAIGN

To select a new campaign to play or continue an existing one, select the CAMPAIGN button from the Main Menu screen.



The CAMPAIGN screen is divided into four primary areas that allow you to select modules, campaigns, view statistics, and view the general briefing of the campaign. The screen consists of the following elements:

MODULES. Select the module to be used for the current campaign.

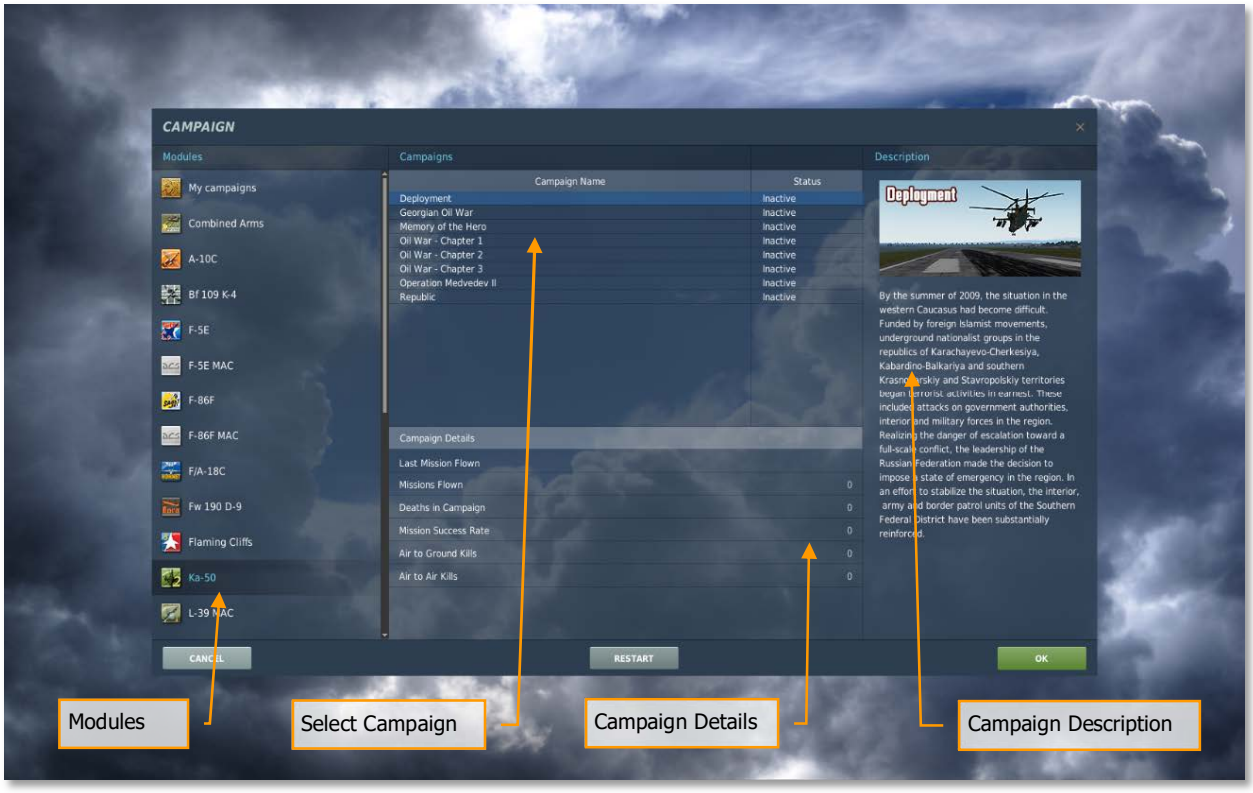
SELECT CAMPAIGN. Along the left side of the screen is a list of all new campaigns you may start, and saved campaigns in progress. The left column of the list displays the name of the campaign and the right column displays its status. To select a campaign, left click on it to highlight it. Once selected, progress statistics are displayed in the Campaign Details section where you can review the general briefing of the campaign.

Campaign Details. Once a campaign has been selected, details of the progress of the campaign are listed in this pane. Information includes:

- **Last Mission Flown**. Date and time of the last flown mission.
- **Missions Flown**. How many missions the player has flow in the campaign.
- **Deaths in Campaign**. How many times the player has died in the campaign.

- **Mission Success Rate**. Mission success rate as a percentage.
- **Air to Ground Kills**. Number of ground units destroyed by player.
- **Air to Air Kills**. Number of air units destroyed by player.

CAMPAIGN DESCRIPTION. When a campaign is created in the Campaign Builder, a general campaign briefing is written. Here you can view that briefing.



At the bottom of the screen are three additional buttons:

- **CANCEL**. Press the CANCEL button to return to the Main Menu screen. You may also exit the Campaign screen by pressing the yellow X in the top right corner of the screen.
- **RESTART**. Press this button to RESTART the campaign from its original start point and reset all Campaign Details.
- **OK**. Once you have selected a new or saved campaign to play, press the OK button to begin the next mission in the selected campaign.

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CAMPAIGN BUILDER

CAMPAIGN BUILDER

The Campaign Builder creates what we term a Staged Campaign System (SCS). A SCS is somewhere between a dynamic campaign system that automatically generates missions, and a linear, scripted campaign that plays the same way through every time. A SCS may consist of one or more (many more if you wish) stages and each stage can consist of one or more (many more if you wish) missions. Each of these missions is created in the ME as a single mission. As such, each mission may consist of numerous triggers and random settings. Using the Campaign Builder, you may create your campaigns by creating stages and populating them with the missions (.miz) you create.

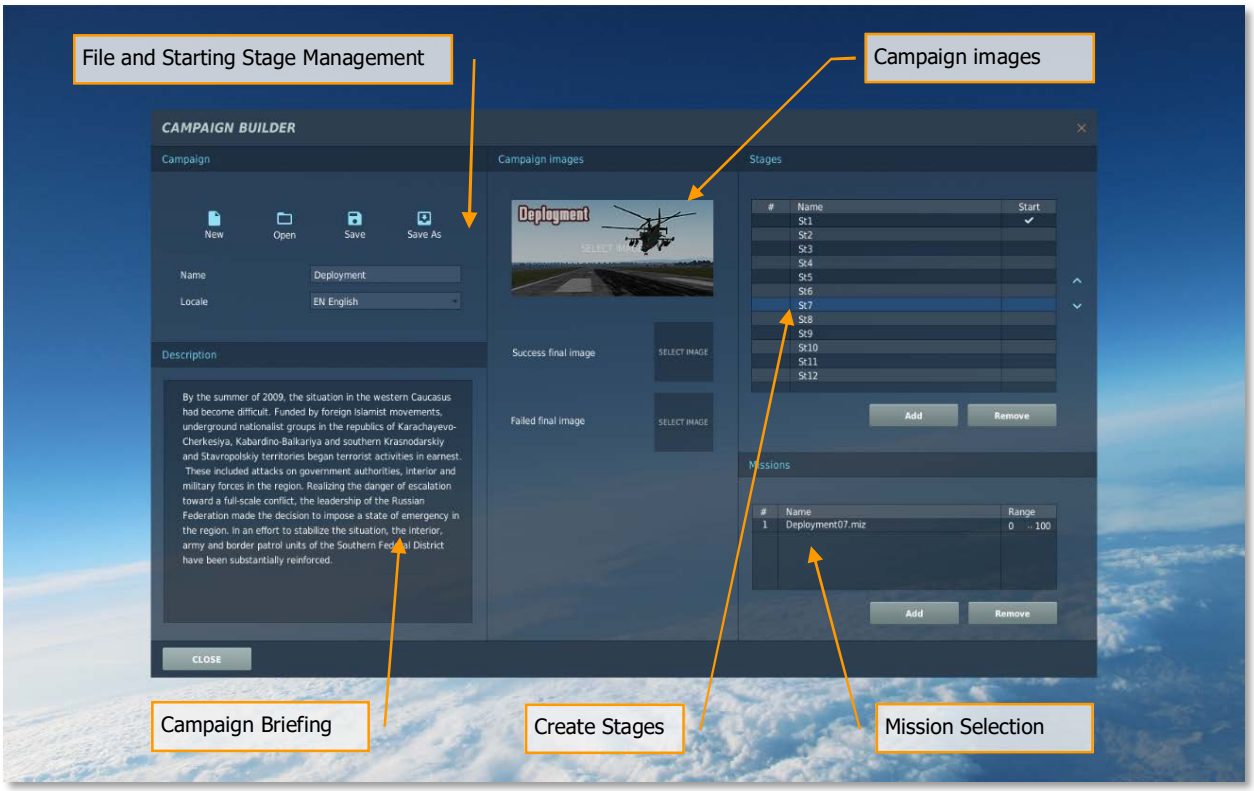
Upon selecting the CAMPAIGN BUILDER from the Main Menu, you will then be shown the campaign builder menu.



In the top left corner are functions to manage the files and set the stage that the campaign will start on. From here you may Open an existing campaign, Save the current campaign as is, create a New campaign, or Save the current campaign As a new file. Below the File and Starting Stage Management is a field that displays the briefing created for the mission. In the section Description, you can also delete the default text and enter your own.

At the top right section of the screen is the area where you define how many stages the campaign will consist of. At the bottom of the section are two buttons: Add and Remove. To add a stage to the campaign, press the add button. To remove a stage, click on the stage and

then press the Remove button. You may also use the Up and Down buttons to rearrange the order of the stages.



The window of the campaign editor has five logical sections:

FILE AND STARTING STAGE MANAGEMENT. This section contains the following options:

- **New**. Creates a new campaign.
- **Open**. Opens an existing campaign.
- **Save**. Saves all changes to the current campaign.
- **Save As**. Saves the currently-open campaign under a new filename.
- **Name**. Specify the campaign name in this field.
- **Locale**. Drop-down list for choosing the localization language for the campaign.

CAMPAIGN IMAGES. In this section you may choose the image files to be used for the campaign:

- **Campaign Image**. Clicking on the SELECT IMAGE field opens a standard file selection window in which you can select an image (*.jpg; *.jpeg; *.png) for the campaign. The image must be 273x130 pixels in size.
- **Success final image**. Clicking on the SELECT IMAGE field opens a standard file selection window in which you can select an image (*.jpg; *.jpeg; *.png) to be displayed upon campaign completion. The image must be 512x512 pixels in size.

- **Failed final image**. Clicking on the SELECT IMAGE field opens a standard file selection window in which you can select an image (*.jpg; *.jpeg; *.png) to be displayed upon failure to complete the campaign. The image must be 512x512 pixels in size.

CAMPAIGN BRIEFING. In this field you may enter a brief description of the campaign.

CREATE STAGES. List of campaign stages. This section contains 4 buttons for controlling the stages:

- **Up**. Shifts the selected stage one spot higher in the list.
- **Down**. Shifts the selected stage one spot lower in the list.
- **Add**. Adds a new stage.
- **Remove**. Removes the highlighted stage.

MISSION SELECTION. Displays a list of missions for the selected stage:

- **Name**. Mission name.
- **Range**. In this field you need to set the range of points according to which missions will be assigned to the player, depending on the previous results garnered.
- **Add**. Adds a mission to the stage.
- **Remove**. Removes a mission from the stage.

Below of the Stages section is the Missions section. It is from here that you will populate each stage with missions. To do so, first select the Stage you wish to populate and then press the Add button in the Missions section. Upon doing so you will be presented a folder/file selection dialog window to select the mission you want. Once you select it, press the OK button on the file browser. The mission will now be listed as part of the selected stage, and its mission name and value range will be displayed. Each mission can be assigned a value range which will determine mission selection within a phase.

When creating a mission, you may assign values to triggered events such as the destruction of a unit, a unit reaching a defined area, a time value, etc. At the end of a mission, these values are totaled and used to determine what the next stage will be and which mission from within that stage will be chosen.

If mission total values are 49 or lower, the player will fall back one stage. If the value is 50, they stay in the same stage. If the mission total value is 51 or above, the player will advance to the next stage. By populating a stage with multiple missions, each with different values, you can create a campaign that flows back and forth according to mission results.

If two missions are assigned the same value range in a stage, the mission will be randomly selected.

The assigned value range of a mission within a stage is listed in the Range column.

At the bottom of the section, next to the Add button, is the Remove button. Use this to remove a mission from a stage.

From the start stage field, you may set the starting stage for the campaign. You generally will not want to start the campaign on stage 1 in case the player loses the first mission and subsequently loses the campaign! The starting stage for the campaign will be marked with an asterisk (*) on the stage listing.

Some notes about campaign building:

- The more stages you put in a mission and the more missions you place in each stage, reduces the chances of mission repetition.
- When creating a stage, it can often save time by creating a template for the stage. This "template" mission would include general force layouts so that you could simply add to and/or modify the mission many times resulting in unique missions for a particular stage.
- Use random settings and triggers as much as you can. These can be used to create unpredictable force set ups including air defense units and variable AI skill settings.
- By placing front lines adjacent to each other over several stages, you can reproduce a front line that moves back and forth according to mission results.

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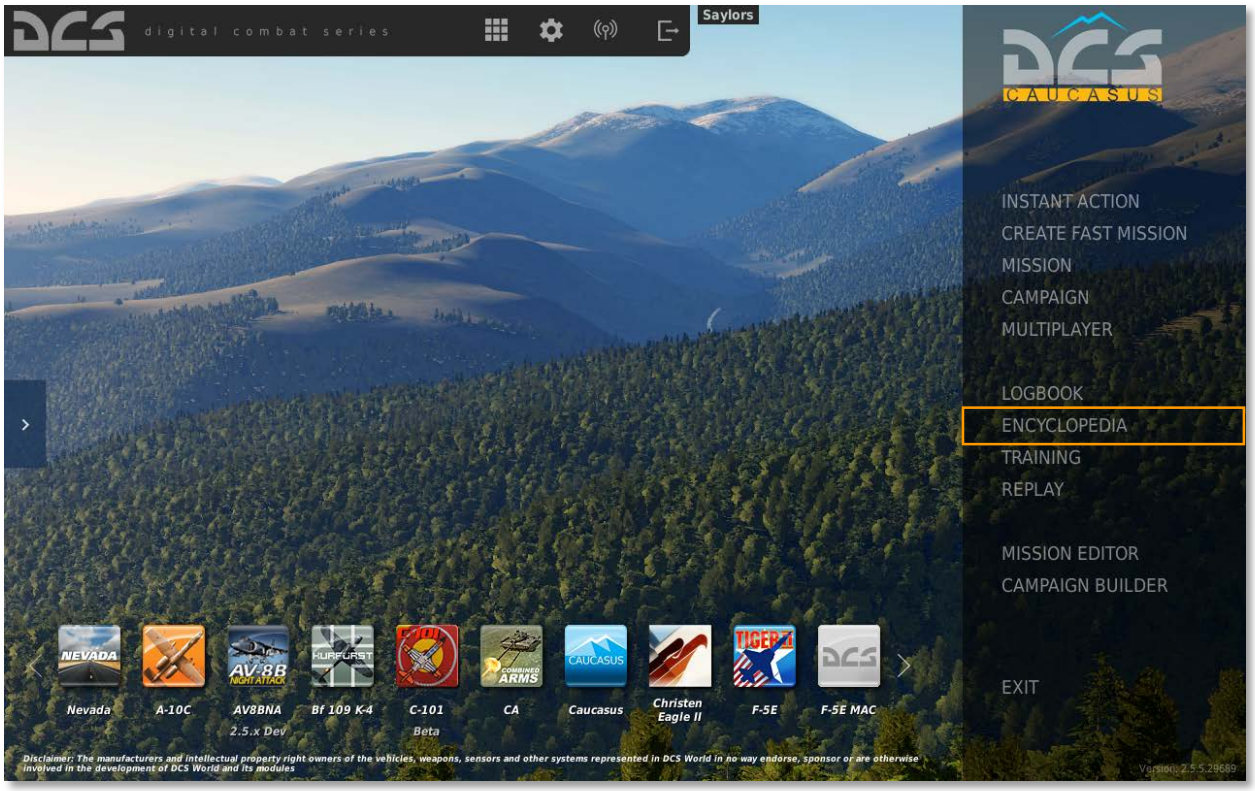


ENCYCLOPEDIA

ENCYCLOPEDIA

The ENCYCLOPEDIA is an invaluable tool for visual recognition of various units in the game, and is also an excellent source of technical data for each unit.

To select the Encyclopedia, click the ENCYCLOPEDIA button on the Main Menu page.



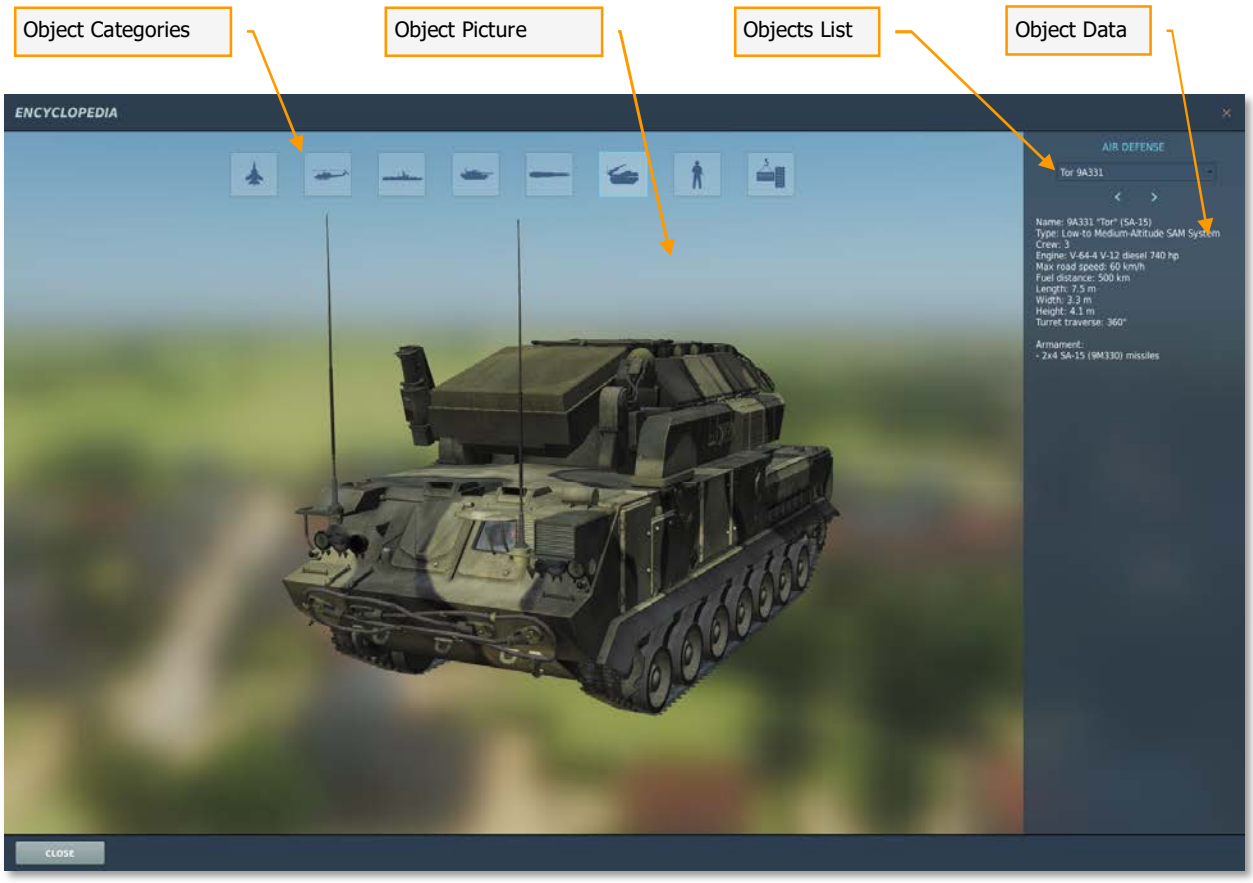
The Encyclopedia consists of two primary parts: the object picture on the left, and the object data window on the right.

Along the top of the screen are a series of eight tabs that correspond to eight categories of objects (unit types) in the game. These include:

1. **AIRPLANES**. All fixed-wing aircraft.
2. **HELICOPTERS**. All rotary-wing aircraft.
3. **SHIPS**. All naval vessels.
4. **VEHICLES**. All non-air defense-related ground vehicles.
5. **WEAPONS**. Air, land, and sea weapons that can be launched/dropped from a unit.
6. **AIR DEFENSE**. Air defense systems.
7. **PERSONNEL ASSETS**. Aircraft carrier deck personnel and infantry.

8. **MISCELLANIOUS**. Ground structures, helicopter cargo, etc. that are available to the user in the mission editor as static objects.

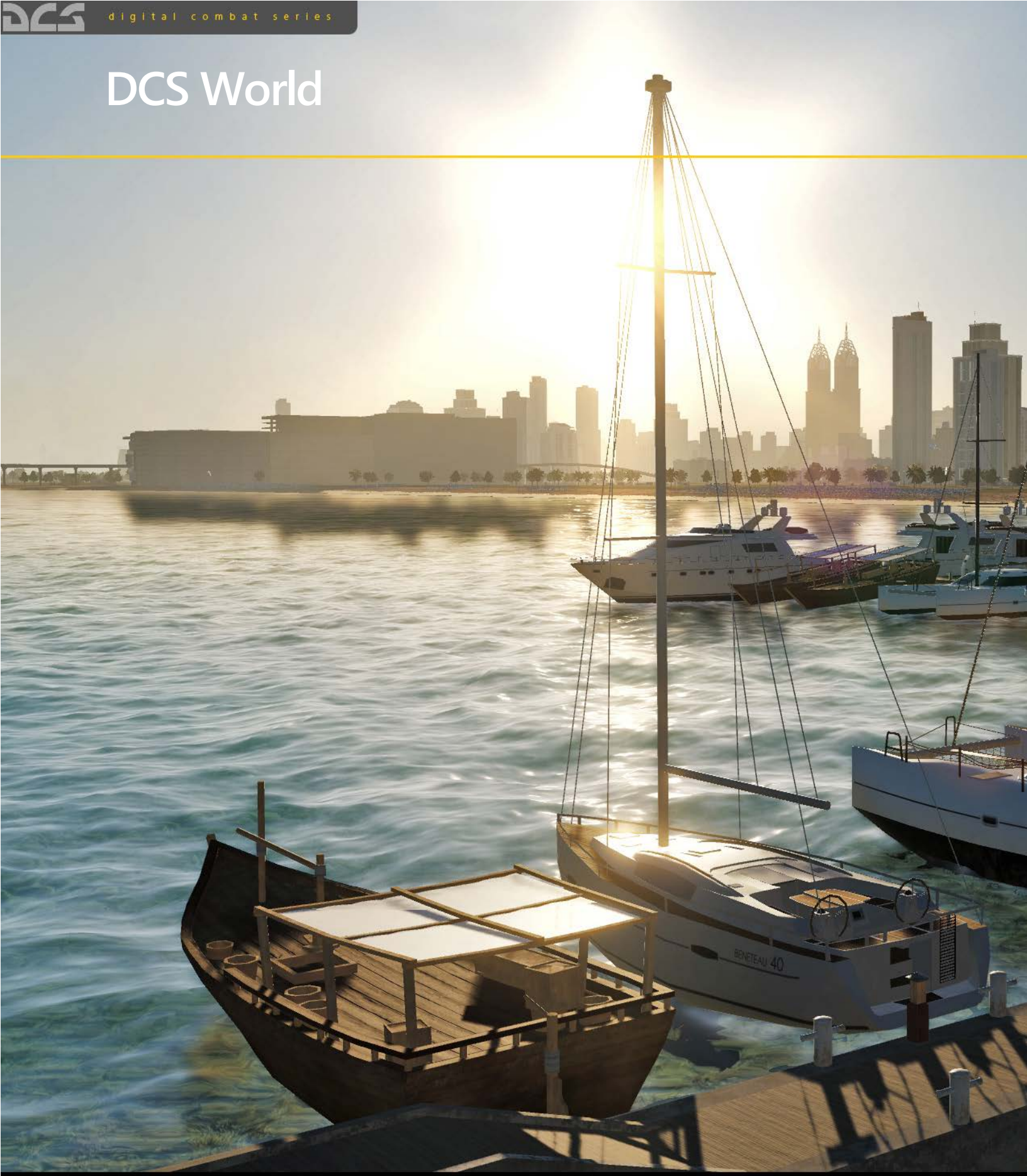
After selecting an Object Category, you may use the Object List drop down list on the right side of the screen to view all units in that category that are represented in the game. Once you have selected a unit, left mouse click on it to view a picture on the left side of the screen, and view its data on the right side.



To exit the Encyclopedia, press the CLOSE button at the bottom of the screen or press the yellow X in the top right corner.

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MULTIPLAYER

MULTIPLAYER

ATTENTION! AN INTERNET CONNECTION IS REQUIRED FOR MULTIPLAYER INCLUDING LAN PLAY

Multiplayer missions can be created by you in the Mission Editor. When creating a multiplayer mission though, it is important to remember that all player-controlled aircraft in the mission must be set to the SKILL of CLIENT and not PLAYER. Also, keep in mind that only one CLIENT aircraft can exist in an aircraft group. For example, to create a mission for four online pilots, you will need to create four aircraft groups with a single unit set to CLIENT in each.

To access the multiplayer menu, click mouse on MULTIPLAYER in the Main Menu.



In the Main Multiplayer Menu screen, you can create own server or join another server as a client.

In the upper part of the window is the SERVER LIST SETTINGS block, the CONNECT BY IP button and the player's call sign.

- **CONNECT BY IP** – To connect to a game over the Internet, you use the CONNECT BY IP option. To do so, press the CONNECT BY IP button. Upon doing so, the Connection to server window will be displayed.



This window consists of two fields:

IP/URL: Enter the IP number or URL of the server you wish to join.

PASSWORD: If the server is password protected, enter the password into this field. If the server is not password protected, you may leave this field blank.

After completing these fields, you can either press the CANCEL button to exit the window and not join the server, or press the OK button to join the server.



- **CHANGE NICKNAME** – The pencil icon in the upper right corner of the network game window opens the input / selection window of the player callsign. Enter the name you wish other multiplayer members to see you as here. This name will be visible in the menu screens and also as a unit label in the simulation. If no name is entered, a default one will be used. Callsigns may be a max of 25 characters long, and up to five different callsigns may be saved.



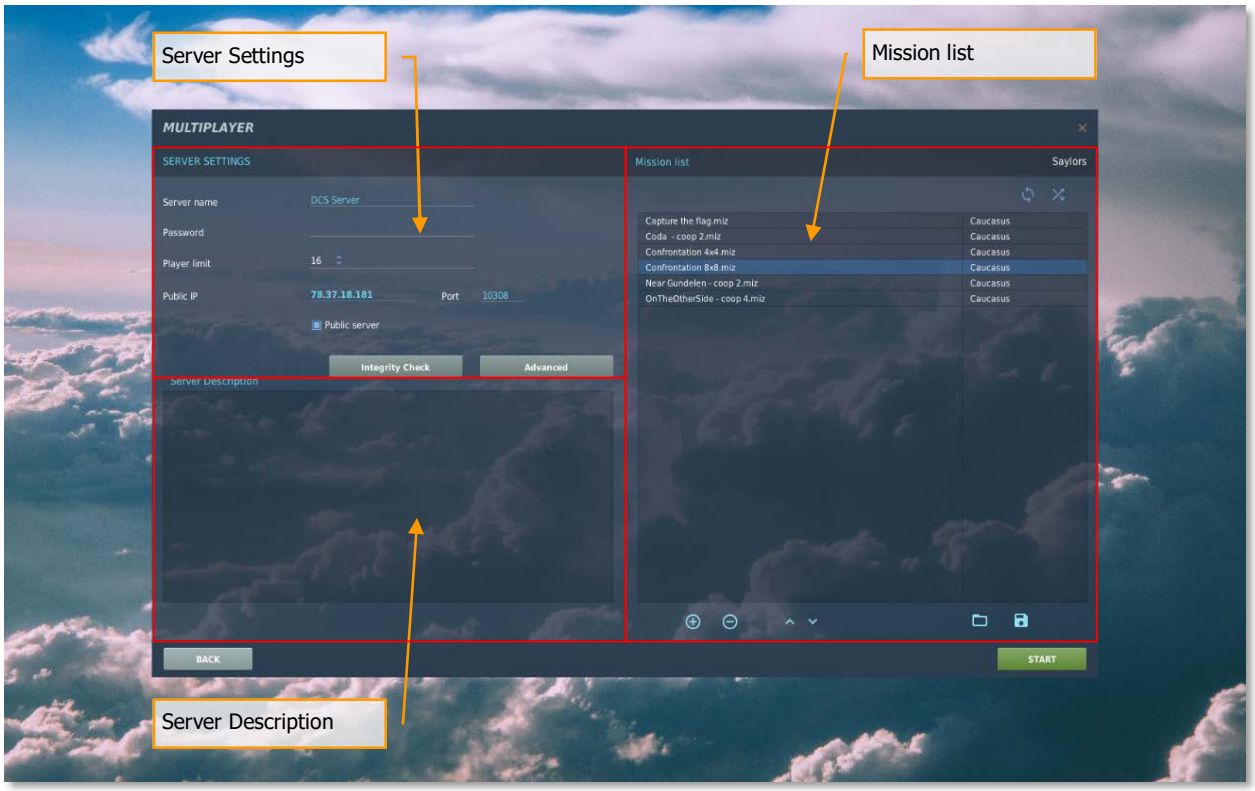
In the center of the window is a list of available servers. At the bottom are the buttons for the main actions.

- **EXIT**. The EXIT button will exit Multiplayer and return you to the main menu.
- **NEW SERVER**. The NEW SERVER window allows you to host a mission that other players (clients) can join.
- **JOIN**. After you have selected a server from the Server List, press the JOIN button to join the server.

New Server

For a mission to run as a multiplayer game, one user must be set up as a Server (host) that runs the mission that other players (clients) can join. This is done as either a mission listing on a server list or as a direct IP connection.

To start a mission Server, select the NEW SERVER button on the Main Multiplayer Menu screen.



The Server screen has three main areas: Server Settings, Server Description, and Mission list. After you have selected a mission and set the server options, you will press the START button at the bottom of the screen to start the server and allow clients to join it. It will also direct you to the MULTIPLAYER – Select role screen.

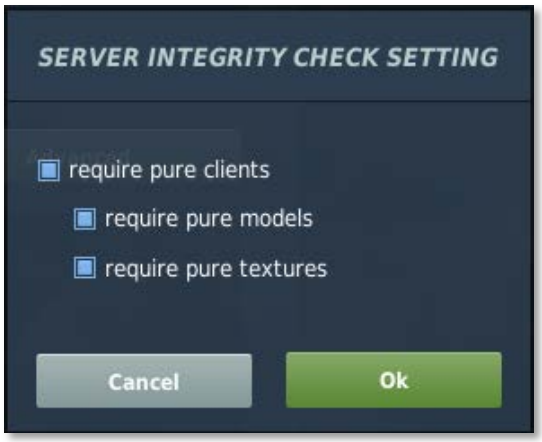
Server settings

The Server Options allow you to setup and view the basic parameters of the mission server. Some of the items can be modified and some are information only and cannot be modified. These are:

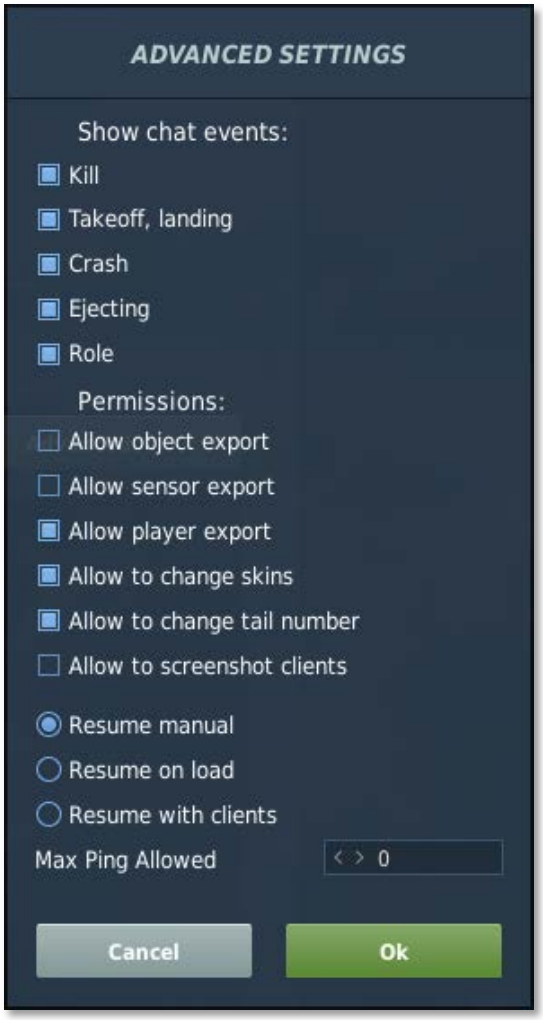
- **Server Name**. Enter the name of the server in this field. This name will then appear in a client's server list if on a LAN network.

- **Password**. To prevent unwanted guests from joining your server, you may enter a password here. Any client joining the server will be required to enter this password before joining.
- **Player limit**. Enter the maximum number of players that may join the server.
- **Public IP**. The IP of the server computer is displayed in this field. Note that if you are behind a router, the router IP may be listed and this will be an invalid IP for clients to directly connect to. Router IP numbers generally start with 192.
- **Port**. Each server can be assigned a unique port by entering it here. By default, the port number is 10308. However, in order to access the server behind a firewall, you may need to change this port number or open the default port number in the firewall.
- **Public server**. This indicates the type of server detected.

INTEGRITY CHECK. This button brings up the server integrity check settings panel:



ADVANCED. This button brings up a panel of advanced server settings:



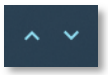
Server description

When a mission is created in the Mission Editor, the mission designer has the option to include a general mission briefing that will be displayed to both Red and Blue sides. This briefing will be displayed in the Server Description field.

Mission list

Using the standard Windows folder and file browser function, select the drive and folder that holds your multiplayer missions. In general, they will be stored in the Missions/Multiplayer folder. Left click on the desired mission and it will be highlighted (selected).

The list of network missions is located on the right side of the window. The user may manage the mission list using the following options:

-  – The LOOP button cycles through the missions in the server mission playlist.
-  – The SHUFFLE button randomizes the order of missions in the server mission playlist.
-  – The ADD MISSION button opens a browser window for selecting and adding missions into the server mission playlist.
-  – The REMOVE MISSION button removes the selected mission from the server mission playlist.
-  – The UP/DOWN arrows move the highlighted mission one step up or down the server playlist.
-  – The OPEN button opens a browser window for selecting and adding mission playlists (*.lst) into the server playlist.
-  – The SAVE AS button opens a browser window for saving mission playlists (*.lst) under a new filename into the chosen folder.

At the bottom of the window are the START and BACK buttons:

START. Pressing the START button will allow clients to join the server and direct you to the Select role screen.

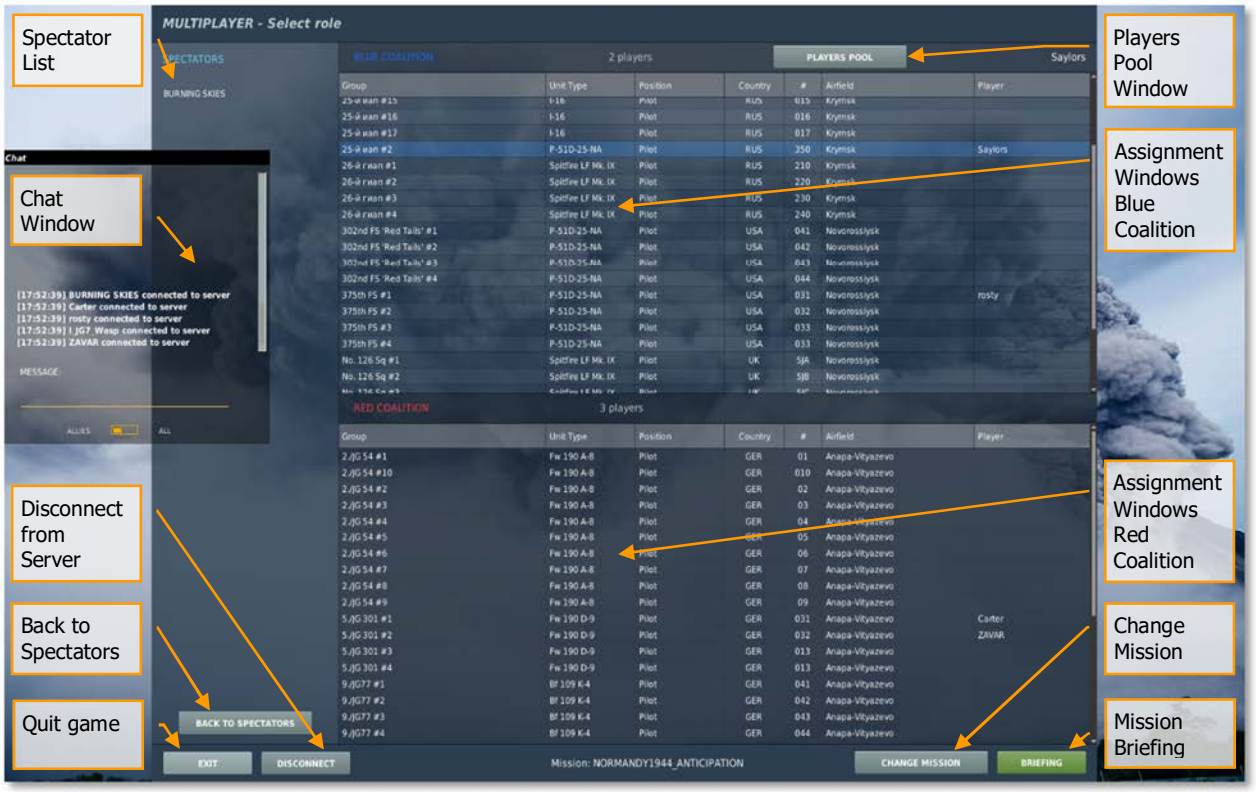
BACK. At the bottom of the screen is the BACK button which will direct you back to the Main Multiplayer Menu screen without saving changes.

You can also cancel the screen by pressing the yellow X in the top right corner of the screen.

Select role (Server)

The next step after creating the server is to select a coalition and an aircraft.

The SELECT ROLE window appears after clicking the START button.



SPECTATORS. When a player first enters the Join screen, they are not assigned a side or aircraft and are instead listed in the Spectator List. When in this list, they are effectively neutral, and may view all players on the server. However, once the player has joined an aircraft, they are automatically removed from the Spectator List.

CHAT. The chat window is used for server and player messages.

[Tab] - Chat open/close.
[LCtrl+Tab] - Chat channel to ALLIES.
[LShift+Tab] - Chat channel to ALL.

To enter text, left click in the message window until you see a flashing cursor. You may then type in your message. When complete, press the Enter button on your keyboard to send it.

You can also switch the message recipient by clicking on the slider.

The assigned chat keys cannot be changed. The window can be moved by clicking and dragging the window by the title bar (in this case, the chat should be in recording mode). The size of the chat window cannot be changed.

Remove the envelope icon from the screen using the key combination [LCtrl+LShift+Y].

BACK TO SPECTATORS. At the bottom of the list is the BACK TO SPECTATORS button; press this button to return your pilot to the Spectator List.

EXIT. Disconnect from the server and return to the Main menu.

DISCONNECT. Disconnect from the server and return to the Server list.

ASSIGNMENT WINDOWS. These two windows display all Red Coalition and Blue Coalition unit type that are assigned to the mission and are capable of being flown by players (set as Client in the mission editor). Both windows have seven columns that provide information about each unit type listed:

- **Group**. Name of unit group.
- **Unit Type**. Unit type.
- **Position**. Role/position selection in a multi-seat unit.
- **Country**. Country that unit is assigned to.
- **#**. Onboard number assigned to the unit.
- **Airfield**. Home airfield.
- **Player**. Player Name assigned to the unit.

PLAYERS POOL. Player list open / close button [`] (apostrophe). Pressing this button displays a semi-transparent table that lists all players on the server and basic information about each.



The following data is displayed in the players pool window:

- **Callsign**. Player callsign.

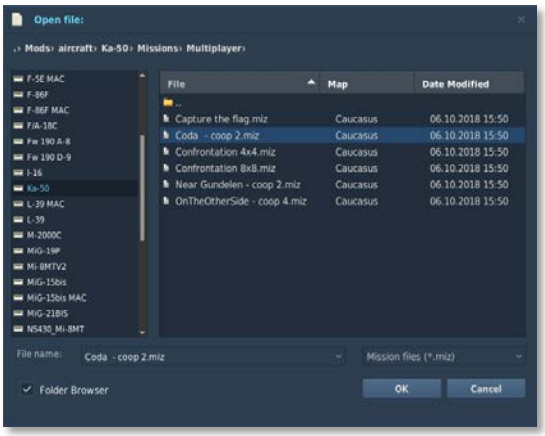
- **Ping**. The connection latency between you and the server. Lower numbers equate to better Ping numbers.
- **Unit**. Player aircraft/unit type.
- **Tac**. **No**. Player unit TAC number.
- **Score**. Total score the player has made on the mission.
- **A/C**. Number of downed aircraft.
- **Ground**. Number of ground units destroyed.
- **Ships**. Number of ships destroyed.
- **Loss**. Number of times the player has been killed.

At the top of the panel is the BAN LIST button:

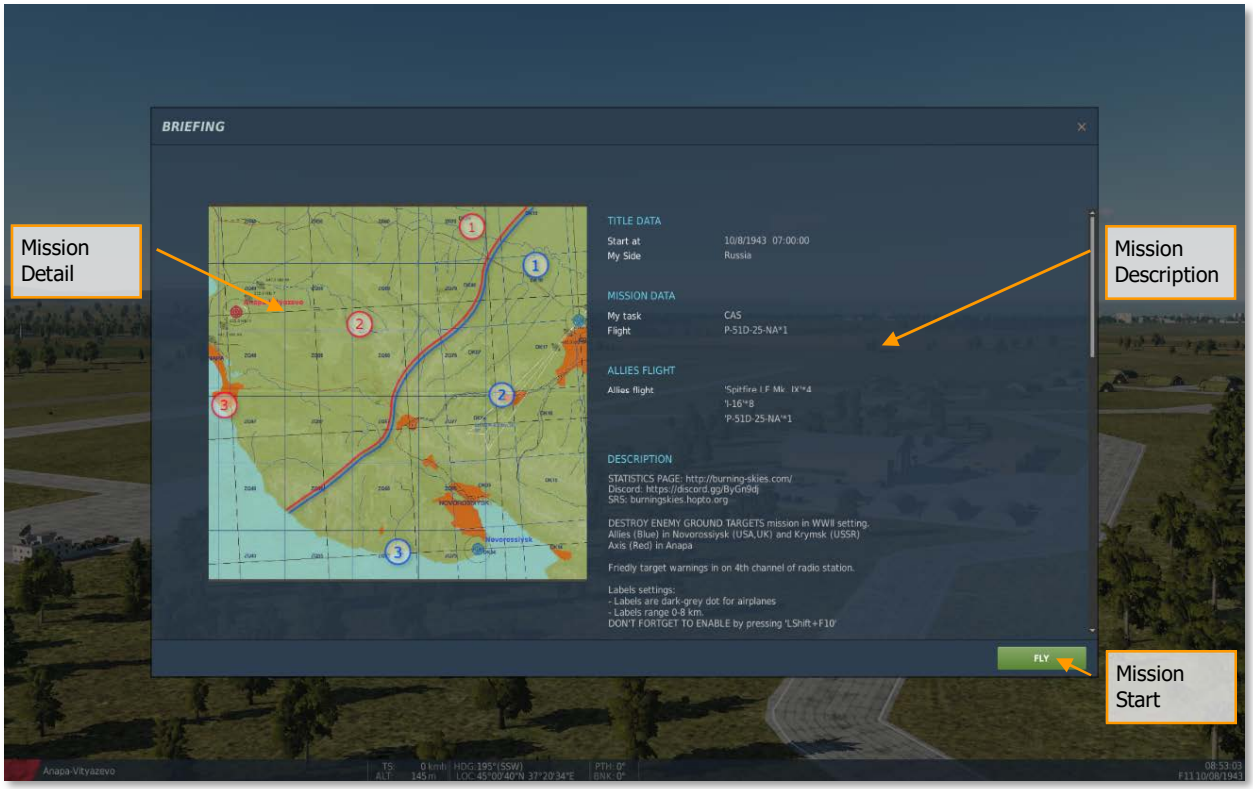
BAN LIST. The player hosting the server of the mission (Host) will also have access to the BAN LIST button. After selecting a player from the list, the Host may press the BAN LIST button to remove the player from the server.

To exit the Player Pool window, press the CLOSE button.

CHANGE MISSION. If you are the server host, you also have access to the CHANGE MISSION button on the bottom of the screen. Using the standard Windows folder and file browser function, select the drive and folder that holds your multiplayer missions. In general, they will be stored in the Missions/Multiplayer folder. Left click on the desired mission and it will be highlighted (selected). Press the OK button load the new mission onto the server.



BRIEFING. After selecting your coalition and aircraft, press the BRIEFING button to view the briefing for your coalition (side). This briefing consists of an image on the left page and a text briefing on the right page. When you are ready to enter the simulation, press the blue FLY button along the bottom of the screen.



- **ОПИСАНИЕ**. The mission description and data are displayed on the right side of the BRIEFING window.
- **ДЕТАЛИ**. The picture(s) with mission details for the selected coalition is displayed on the left side of the window.
- **FLY**. Starts the mission.

Client to server connection

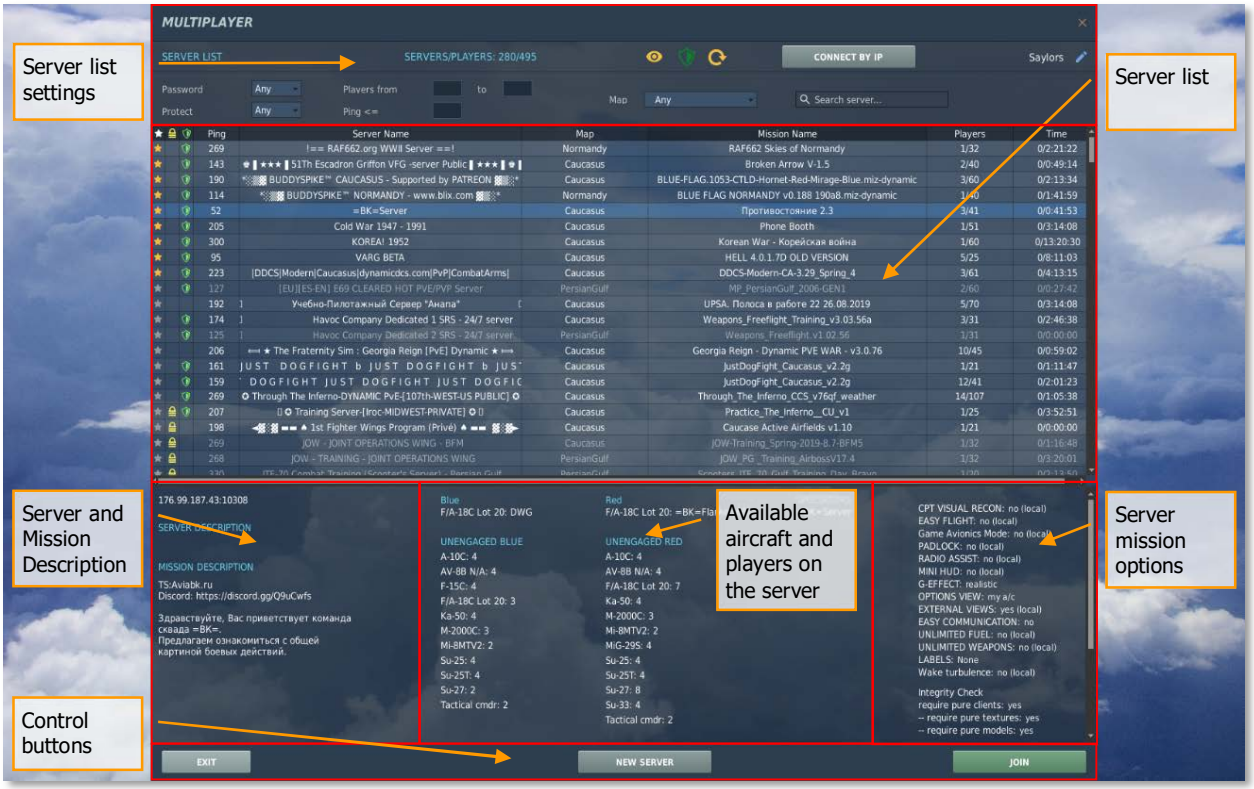
The MULTIPLAYER client window is designed to connect to existing servers in a multiplayer game. Choosing and connecting to a server is the first step in a network game.

After you have selected a server from the Server List, press the JOIN button at the bottom of the screen to join the server.

The SERVER LIST displays all available servers and takes up the majority of the screen. The SERVER LIST is designed to show all detected Servers running the game over the Internet or a LAN. Each row of the Server List represents a different server. Each column in the Server List provides different information about each Server.

Below is the appearance of the window with explanations of the control elements.

The MULTIPLAYER window consists of six logical blocks.



The SERVER LIST SETUP block consists of the following elements:

- **SERVERS/PLAYERS**: .../... – Shows the total number of currently available servers and players online.
- **Password** – Sorts the server list by password (Any/Yes/No).
- **Protect** – Sorts the list of servers by game file integrity check (Any/Yes/No).
- **Players from** ... **to** ... – Sorts the server list by the number of players on the server.

- **Ping <=** – Sorts the server list by ping.
- **Map** – Sort the list of servers by map/map module.
-  – The button allows you to toggle between displaying servers with only modules available to the player, or all servers regardless of modules.
-  – Integrity check passed. The green shield icon shows the integrity of the player's game files. If the player has any third-party modified files installed in the game, the shield icon will be red. In this case, connecting to servers where file integrity verification is required will be impossible.
-  – The Refresh button will refresh the data for the servers currently in the server list.
- **Search server** … – Searches for servers by server name.
- **CONNECT BY IP** – To connect to a game over the Internet, you use the CONNECT BY IP option. To do so, press the CONNECT BY IP button. Upon doing so, the Connection to server window will be displayed.

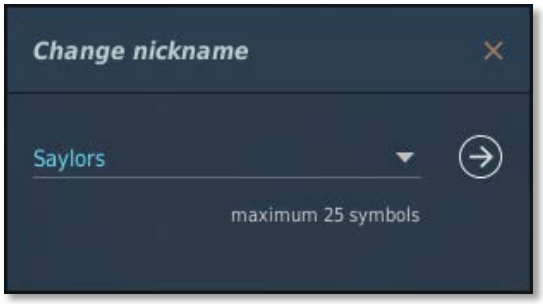


This window consists of two fields:

IP/URL: Enter the IP number or URL of the server you wish to join.

PASSWORD: If the server is password protected, enter the password into this field. If the server is not password protected, you may leave this field blank.

- **CHANGE NICKNAME** – The pencil icon in the upper right corner of the network game window opens the input / selection window of the player callsign. Enter the name you wish other multiplayer members to see you as here. This name will be visible in the menu screens and also as a unit label in the simulation. If no name is entered, a default one will be used. Callsigns may be a max of 25 characters long, and up to five different callsigns may be saved.



The SERVER LIST block displays all found game servers and consists of the following elements (which may be sorted by clicking):

-  – Favorite servers are marked by the star icon displayed in the first column.
-  – Padlock symbol. This column will be marked with a padlock symbol if the server is password protected.
-  – A shield icon is displayed in the third column if a game file integrity check is set on the server.
- **PING**. The connection latency between the player and the server. Lower numbers equate to a better Ping number.
- **SERVER NAME**. The name of the server is listed in this column.
- **MAP**. The map module on which the mission on the server is running.
- **MISSION NAME**. The title of the mission being run on the server.
- **PLAYERS**. Maximum number of players allowed on the server and current number on it.
- **TIME**. How long the mission has been running.

SERVER AND MISSION DESCRIPTION block:

- **SERVER DESCRIPTION**. Displays information about the server.
- **MISSION DESCRIPTION**. When a mission is created in the Mission Editor, the mission designer has the option to include a general mission briefing that will be displayed to both Red and Blue sides. This briefing will be displayed in the Mission Description field.

The AVAILABLE AIRCRAFT and PLAYERS ON THE SERVER block shows the number and type of available aircraft for each coalition, the callsigns of the players who are occupying the aircraft for each coalition, and the callsigns of the players currently spectating.

The forced difficulty settings for the mission on the server are shown in the SERVER MISSION OPTIONS block. Forced server settings are applied over the player's difficulty settings.

In the bottom part of the window are the control buttons:

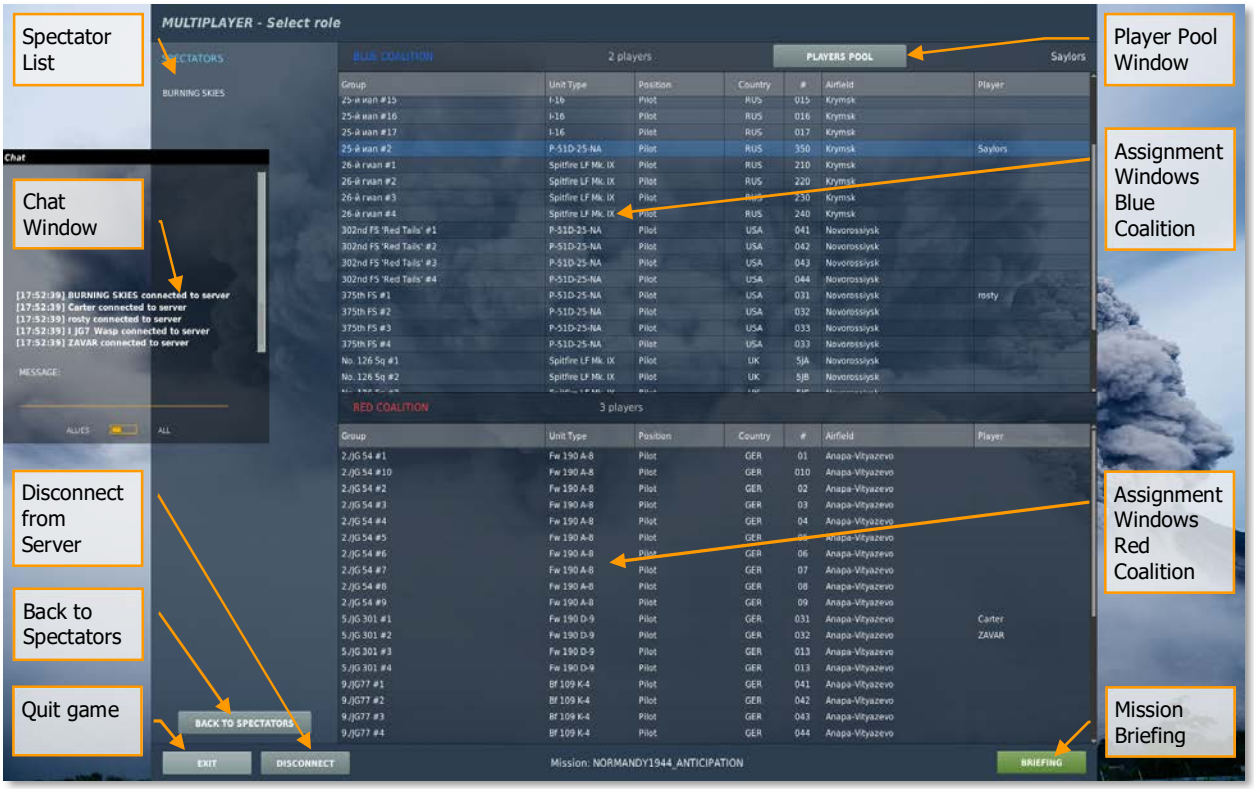
- **EXIT**. The EXIT button will exit Multiplayer and return you to the Main Menu.
- **NEW SERVER**. The NEW SERVER window allows you to host a mission that other players (clients) can join.
- **JOIN**. After you have selected a server from the Server List, press the JOIN button to join the server.

Select role (Client)

After you have joined a mission (Server List or Connect by IP), you are directed to the Join Window. From this window you may select a side, select your aircraft, chat with other mission players, view the briefing, and view the server pool to see who is on the server.

The next step after connecting to the server is coalition and role selection, as well as mission briefing.

The ROLE SELECTION window on the client side is similar to the ROLE SELECTION window on the server side, but the CHANGE MISSION button at the bottom is missing in the window.



Game Server Creation (Server)

Below are step-by-step instructions for creating a server for online play:

1. Open the MULTIPLAYER window.
2. Press the NEW SERVER button to bring up the server creation window.
3. In the SERVER SETTINGS section, enter the server name, select the type of game, specify the maximum number of players. If necessary, select the network interface (if several network cards are installed on the computer) and the port.
4. In the MISSION LIST section, select the desired missions for the game and configure their rotation.
5. Press the START button.
6. In the SELECT ROLE window, select the unit for the game or remain in the list of spectators (recommended).
7. Press the BRIEFING button.
8. In the BRIEFING window, review the mission briefing.
9. Press the FLY button.
10. After all players are ready to start, remove the game from pause by pressing the [Pause] key.

Connecting to a Server (Client)

Below are step-by-step instructions for connecting to an already-existing game server:

1. Open the MULTIPLAYER window.
2. In the list of servers, select the desired server. If necessary, press the refresh button  to refresh the list of servers, or manually connect to a server by inputting its IP address.
3. Click the JOIN button.
4. In the SELECT ROLE window, select the unit for the game or remain in the list of spectators.
5. Click the BRIEFING button.
6. Familiarize yourself with the mission details.
7. Click on FLY to start the game.

Ending the Session

1. Press [Esc].
2. In the MISSION PAUSED window, click LEAVE SERVER or QUIT TO DESKTOP.



3. You may also end the session by clicking on DISCONNECT or EXIT in the SELECT ROLE window.

DCS World



EXIT

EXIT

Exit by pressing the EXIT button on the Main Menu page.

